Primordial Cosmology

Study Notes based on Patrick Peter & Jean-Philippe Uzan

Department of Physics / Cosmology Notes

Personal Study Notes
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Preface

These notes are personal study materials based primarily on Patrick Peter and Jean-Philippe Uzan's *Primordial Cosmology* (Oxford University Press). While they derive insights from multiple contemporary references, Peter and Uzan's comprehensive text serves as the primary roadmap and foundational framework for this project.

The central goal of these notes is to make the subject of general relativistic cosmology, early-universe thermodynamics, cosmic nucleosynthesis, cosmological perturbation theory, and CMB anisotropies more accessible, both for my own mastery and for any other students navigating this demanding field. To achieve this, every effort has been made to **derive all calculations explicitly**. By filling in intermediate steps, establishing gauge-invariant formulations, and clarifying technical nuances often omitted in standard texts, I hope to ensure the material is easily parsable and the underlying mathematical and physical narrative is completely transparent.

All vector figures and diagrams have been directly extracted from the original source material to maintain absolute fidelity to the textbook presentation.

I welcome feedback, corrections, or discussions regarding the content of these notes. Please feel free to reach out to me via email.

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Disclaimer: These are pedagogical study notes. While I have strived for accuracy, any errors in the calculations, typos, or physical interpretations remain entirely my own. I am actively revising and correcting the notes; the latest version can always be found on the project page at lunar-photon.github.io.

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Part I

Overview of the Theoretical Basis

Chapter 1

General relativity

Since gravity is the only long-range interaction that cannot be screened or neutralized, the physical properties and large-scale dynamics of the Universe are essentially determined by this force.

The goal of this chapter is to present the fundamental hypotheses, mathematical framework, and key formulas of general relativity that are necessary for the study of primordial cosmology:

- In Section 1.1, we trace the evolution of the mathematical concept of space-time from Newtonian physics and special relativity to general relativity.
- Section 1.2 summarizes the core definitions and tools of differential geometry.
- In Section 1.3, we derive the Einstein field equations governing the dynamics of space-time and the conservation equations for matter.
- Section 1.4 analyzes kinetics and geodesic motion in curved space-time.
- Section 1.5 establishes how Newtonian gravity is recovered in the weak-field limit.
- Section 1.6 summarizes the classical and modern experimental tests of general relativity.

1.1 Space-time and gravity

1.1.1 Absolute space and time of Newtonian physics

In Newtonian physics, space is described by an absolute, immutable three-dimensional Euclidean space $\mathbb{E}^3$. We can endow this space with an origin and three mutually orthogonal reference axes, thereby defining an absolute reference frame.

Time is likewise ideal and absolute: it flows uniformly, independently of the motion of any observer, playing the role of an external evolution parameter. For any event P, there exists an intuitive and absolute notion of simultaneity (defined as the set of all events occurring at the same time). If we consider a second event Q, three mutually exclusive causal possibilities arise:

1. One can in principle travel from P to Q; thus Q belongs to the future of P.
2. One can in principle travel from Q to P; thus Q belongs to the past of P.
3. It is impossible to travel between P and Q by any physical trajectory; the two events are therefore simultaneous.

This causal structure is illustrated in Figure 1.1.

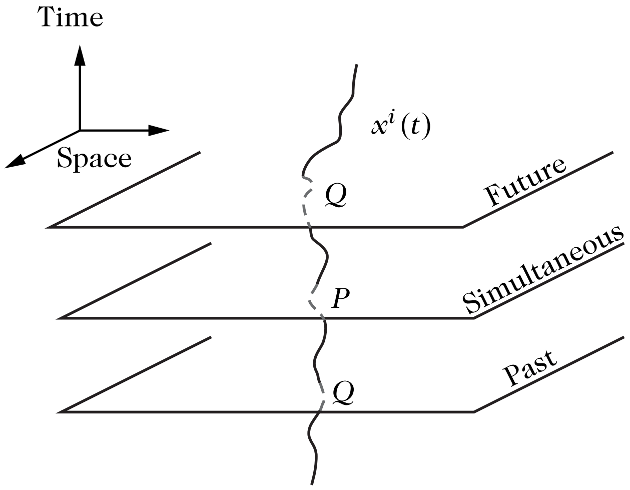


Figure 1.1: The causal structure of Newtonian space-time. If Q can be related to P by a trajectory $x^i(t)$ with finite speed ($\mathrm{d}x^i \,/\, \mathrm{d}t < \infty$), Q is in the past or the future of P, otherwise it is simultaneous. The set of all simultaneous events is a three-dimensional Euclidean space.

Because spatial geometry is assumed to be Euclidean, Pythagoras' theorem gives the distance $\mathrm{d}\ell$ between two neighboring points in Cartesian coordinates:

$$\mathrm{d}\ell^2 = (\mathrm{d}x^1)^2 + (\mathrm{d}x^2)^2 + (\mathrm{d}x^3)^2. \tag{1.1.1}$$

We can write this line element in compact form as:

$$\mathrm{d}\ell^2 = \sum_{i=1}^{3}(\mathrm{d}x^i)^2 = \sum_{i,j=1}^{3} \delta_{ij}\, \mathrm{d}x^i\, \mathrm{d}x^j \equiv \delta_{ij}\, \mathrm{d}x^i\, \mathrm{d}x^j\,, \tag{1.1.2}$$

where δ_{ij} is the Kronecker delta symbol ($\delta_{ij} = 1$ if $i = j$, and 0 otherwise), and Latin indices $i, j, \ldots$ run from 1 to 3. Here, we adopt the **Einstein summation convention**, where a summation over repeated upper and lower indices is implicitly understood. For instance, the scalar product of two vectors $\mathbf{T}$ and $\mathbf{V}$ is written as $\mathbf{T}\cdot\mathbf{V} \equiv T^iV^i = \delta_{ij}T^iV^j = T^1V^1 + T^2V^2 + T^3V^3$.

The trajectory of any physical body is described parametrically by $x^i(t)$. The time elapsed during travel from point A at t_A to point B at t_B is given by $t_B - t_A$, which is strictly independent of the spatial trajectory taken between A and B.

Physical laws do not depend on the use of Cartesian coordinates. If we introduce a general coordinate system (y^i) related to Cartesian coordinates (x^j) by $x^j(y^i)$, the displacement vector $\mathrm{d}x$ with components $\mathrm{d}x^j$ has coordinates $\mathrm{d}y^i = (\partial y^i/\partial x^j)\,\mathrm{d}x^j$. The distance between neighboring points in an arbitrary coordinate system takes the form:

$$\mathrm{d}\ell^2 = g_{ij}(y^k)\,\mathrm{d}y^i\,\mathrm{d}y^j\,, \qquad g_{ij}(y^k) = \frac{\partial x^m}{\partial y^i}\frac{\partial x^n}{\partial y^j}\delta_{mn}, \tag{1.1.3}$$

where g_{ij} is the metric tensor of space in the new coordinates.

For example, spherical coordinates (r, θ, ϕ) are defined by:

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta. \tag{1.1.4}$$

Applying Equation (1.1.3), the only non-vanishing components of the metric in spherical coordinates are:

$$g_{rr} = 1, \qquad g_{\theta\theta} = r^2, \qquad g_{\phi\phi} = r^2\sin^2\theta, \tag{1.1.5}$$

and the line element (1.1.2) takes the familiar form:

$$\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\;\mathrm{d}\phi^2\right). \tag{1.1.6}$$

Galilean group

The choice of spatial reference frame is arbitrary. It is instructive to examine the frame transformations that preserve the Euclidean form (1.1.1) of the line element $\mathrm{d}\ell^2$. This is equivalent to finding all coordinate transformations that keep the three spatial reference axes orthogonal. The set of all such transformations is given by rigid spatial rotations and translations of the origin:

$$x^i \longrightarrow y^i = \Lambda^i{}_j(t)x^j + T^i(t), \qquad \Lambda^i{}_j(t)\Lambda_k{}^j(t) = \delta^i_k. \tag{1.1.7}$$

The rotation matrix $\Lambda^i{}_j$ may depend on time, but it is rigid because it does not depend on position x^j, acting identically on all points at a given time t. $T^i(t)$ represents a time-dependent translation. The invariance of physical laws under Equation (1.1.7) reflects the isotropy and homogeneity of Euclidean space.

Among these rigid transformations, the **Galilean group** plays a central role. It constitutes the subgroup of time-independent rotations and translations linear in time, $T^i(t) = T^i_0 - v^i t$ (where T^i_0 and v^i are constant vectors), thereby preserving the equivalence among inertial reference frames. The laws of Newtonian mechanics are identical in all inertial frames, being invariant under the action of a Galilean transformation:

$$x^i \longrightarrow y^i = \Lambda^i{}_j x^j + T^i_0 - v^i t, \qquad \Lambda^i{}_j\Lambda_k{}^j = \delta^i_k. \tag{1.1.8}$$

This group is formed by 3 spatial rotations, 3 spatial translations, and 3 changes of inertial frame (boosts). Including the freedom to change the time origin ($t \to t + t_0$), the Galilean group is a **10-parameter Lie group** (3 Euler angles, 3 translation coordinates, 3 velocity components, and 1 time shift).

Trajectory of a massive body

The equations of motion of a body of mass m evolving in a gravitational potential ϕ can be derived by extremizing the action constructed from the Lagrangian:

$$L = \frac{1}{2}m_\mathrm{I}v^2 + m_\mathrm{G}\phi = \frac{1}{2}m_\mathrm{I}g_{ij}(x^k)\dot{x}^i\dot{x}^j + m_\mathrm{G}\phi(x^k), \tag{1.1.9}$$

where we have explicitly distinguished between the **inertial mass** m_I and the **gravitational mass** m_G.

The Euler–Lagrange equations,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{x}^i}\right) - \frac{\partial L}{\partial x^i} = 0, \tag{1.1.10}$$

take the following form in any arbitrary spatial coordinate system:

$$\ddot{x}^i + \Gamma^i_{jk}\dot{x}^j\dot{x}^k = -\frac{m_\mathrm{G}}{m_\mathrm{I}}g^{ij}\partial_j\phi, \tag{1.1.11}$$

where Γ^i_{jk} are the Christoffel connection coefficients:

$$\Gamma^i_{jk} = \frac{1}{2}g^{ip}\left(\partial_j g_{kp} + \partial_k g_{jp} - \partial_p g_{jk}\right). \tag{1.1.12}$$

In Cartesian coordinates, $g_{ij} = \delta_{ij}$ and $\Gamma^k_{ij} = 0$, so Equation (1.1.11) reduces to the standard classical equation of motion $\ddot{\mathbf{x}} = -\boldsymbol{\nabla}\phi$ when $m_\mathrm{I} = m_\mathrm{G}$. The connection coefficients Γ^k_{ij} represent the fictitious inertial forces (centrifugal, Coriolis, etc.) that arise when describing motion in a non-inertial or curvilinear coordinate system.

Importantly, Equation (1.1.11) is entirely independent of the mass of the body if $m_\mathrm{I} = m_\mathrm{G}$: all bodies fall in identical fashion in a given gravitational field, regardless of their mass or chemical composition. This is known as the **weak equivalence principle** or the **universality of free fall**.

In the Newtonian framework, the equality between inertial and gravitational mass appears fortuitous. Using pendulums of different compositions, Newton (1686) experimentally verified that:

$$\frac{|m_{\mathrm{G}} - m_{\mathrm{I}}|}{m_{\mathrm{G}} + m_{\mathrm{I}}} < 10^{-3}. \tag{1.1.13}$$

Newton regarded this equality as a central pillar of gravitation and highlighted it in the introduction of the *Principia*.

1.1.2 Space-time of special relativity

Maxwell's equations of electromagnetism are not invariant under the Galilean group (1.1.8). For example, the electrostatic force generated by a static charge distribution is not invariant under a Galilean boost, since a magnetic force appears in a moving reference frame. Furthermore, Maxwell's theory predicts that light is an electromagnetic wave propagating at a speed $c = 1/\sqrt{\epsilon_0\mu_0}$.

To preserve Galilean relativity, classical physicists postulated the existence of a mechanical medium—the *aether*—in which electromagnetic waves propagate. Maxwell's equations were assumed to be strictly valid only in this absolute rest frame. In any other inertial reference frame, the speed of light would be measured as $c \pm v$, which offered the theoretical possibility of detecting the absolute reference frame at the price of abandoning the principle of Galilean relativity.

Lorentz transformations

The Michelson–Morley experiment (1887) refuted the aether hypothesis, demonstrating that the speed of light is isotropic and identical in all inertial reference frames. In 1905, Einstein resolved the contradiction by reaffirming the principle of relativity: *all laws of nature take the same form in all inertial reference frames*. Consequently, Maxwell's equations and the speed of light c must be invariant across all inertial frames, implying that the transformation laws between inertial frames cannot be those of Galileo.

The correct transformation group is the **Lorentz group**. An observer in an inertial frame $\mathcal{S}$ locates events by Cartesian coordinates (x, y, z) and synchronized clocks measuring time t, labeling events as (t, x, y, z). A second observer in an inertial frame $\mathcal{S}'$ moving with velocity v along the Ox axis labels the same events as (t', x', y', z'). The two coordinate sets are related by the Lorentz boost:

$$ct' = \frac{ct - (v/c)x}{\sqrt{1 - v^2/c^2}}, \qquad x' = \frac{x - (v/c)ct}{\sqrt{1 - v^2/c^2}}, \qquad y' = y, \qquad z' = z. \tag{1.1.14}$$

These transformations reduce to Galilean transformations in the non-relativistic limit $v \ll c$, while keeping the speed of light c invariant in all inertial frames.

Minkowski space-time

Special relativity postulates that all inertial observers are physically equivalent; individual coordinate systems have no intrinsic meaning. The invariant quantity under a change of reference frame is given by a pseudo-Euclidean interval. The spatial line element (1.1.1) is replaced by the spacetime interval between two events:

$$\mathrm{d}s^2 = -(\mathrm{d}x^0)^2 + (\mathrm{d}x^1)^2 + (\mathrm{d}x^2)^2 + (\mathrm{d}x^3)^2 \equiv \eta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu\,, \tag{1.1.15}$$

with $x^0 \equiv ct$. Greek indices run from 0 to 3, and the Einstein summation convention applies. In Cartesian coordinates, $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$.

Unlike the Euclidean metric, $\mathrm{d}s^2$ can be positive, negative, or zero for distinct events. Space and time are unified into a four-dimensional continuum: **Minkowski space-time**.

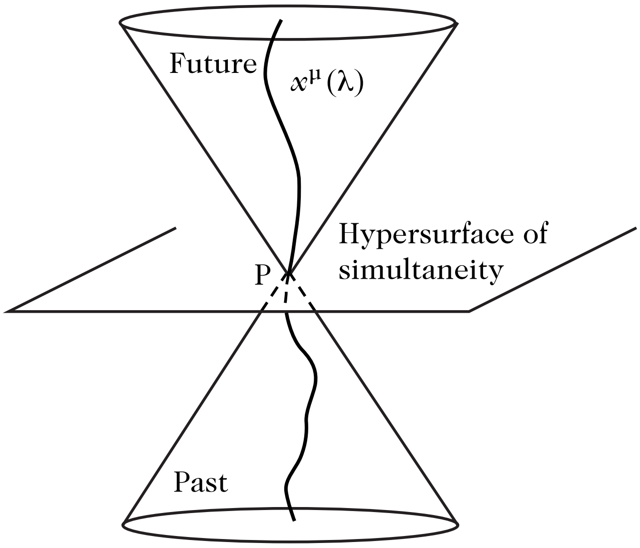


Figure 1.2: The causal structure of Minkowski space-time. The notion of simultaneity is replaced by the notion of a light cone defined by $\mathrm{d}s^2 = 0$. The worldline $x^\mu(\lambda)$ emanating from P is found at the interior of the light cone ($\mathrm{d}s^2 < 0$). All points exterior to the cone can never be joined by a physical trajectory leaving from P. They are separated from P by a space-like interval ($\mathrm{d}s^2 > 0$).

The set of points satisfying $\mathrm{d}s^2 = 0$ defines the **light cone**. For any point M, it separates space-time into events that can receive a light signal emitted at M (future light cone) and events that could have sent a signal received at M (past light cone), as shown in Figure 1.2. Because physical velocities are bounded by c:

- **Time-like interval ($\mathrm{d}s^2 < 0$):** Events lie inside the light cone and can be connected by physical trajectories. The **proper time** τ is defined by:
$$\mathrm{d}s^2 = -c^2\,\mathrm{d}\tau^2\,. \tag{1.1.16}$$
- **Space-like interval ($\mathrm{d}s^2 > 0$):** Events lie outside the light cone and cannot be causally connected.
- **Null / Light-like interval ($\mathrm{d}s^2 = 0$):** Connected only by light rays.

Proper time

The trajectory of a massive particle is represented parametrically by $x^\mu(\lambda)$, where λ is an arbitrary monotonically increasing parameter. The worldline can be reparameterized by proper time:
$$\tau = \frac{1}{c}\int \sqrt{-\eta_{\mu\nu}U^\mu U^\nu}\,\mathrm{d}\lambda\,, \tag{1.1.17}$$
where $U^\mu = \mathrm{d}x^\mu/\mathrm{d}\lambda$ is the tangent vector. In this parameterization, the four-velocity is $u^\mu = \mathrm{d}x^\mu/\mathrm{d}\tau$, satisfying:
$$u^\mu u_\mu = \eta_{\mu\nu}u^\mu u^\nu = -c^2. \tag{1.1.18}$$

Consider an observer $\mathcal{O}$ at rest in an inertial frame $\mathcal{S}$ ($\mathrm{d}x^i = 0$), whose proper time is $\mathrm{d}\tau_{\mathcal{O}}^2 = \mathrm{d}t^2$. A moving observer $\mathcal{O}'$ travels with velocity $v^i = \mathrm{d}x^i/\mathrm{d}t$ relative to $\mathcal{S}$. Its proper time is:

$$\mathrm{d}\tau^2_{\mathcal{O}'} = \mathrm{d}t^2 - \delta_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j\,/c^2 = \left(1 - \frac{v^2}{c^2}\right)\mathrm{d}t^2\,. \tag{1.1.19}$$

Eliminating coordinate time t, the proper times measured by the two observers are related by:

$$\mathrm{d}\tau_{\mathcal{O}'} = \sqrt{1 - \frac{v^2}{c^2}}\,\mathrm{d}\tau_{\mathcal{O}}\,. \tag{1.1.20}$$

Thus, the duration of a journey depends on the observer's worldline—a manifestation of relativistic **time dilation**.

Poincaré group

Just as in Newtonian space, the metric (1.1.15) can be expressed in any arbitrary coordinate system (y^μ) related to Minkowski coordinates (x^ν) by $x^\nu(y^\mu)$. The displacement vector transforms as $\mathrm{d}x^\alpha = (\partial x^\alpha/\partial y^\mu)\,\mathrm{d}y^\mu$, leading to:

$$\mathrm{d}s^2 = g_{\mu\nu}(y^\lambda)\,\mathrm{d}y^\mu\,\mathrm{d}y^\nu\,, \qquad g_{\mu\nu}(y^\lambda) = \frac{\partial x^\alpha}{\partial y^\mu}\frac{\partial x^\beta}{\partial y^\nu}\eta_{\alpha\beta}. \tag{1.1.21}$$

For example, in **Rindler coordinates**, related to Minkowski coordinates by:

$$ct = R\sinh T, \qquad x = R\cosh T, \qquad y = y', \qquad z = z', \tag{1.1.22}$$

the line element takes the form:

$$\mathrm{d}s^2 = -R^2\,\mathrm{d}T^2 + \mathrm{d}R^2 + \mathrm{d}y'^2 + \mathrm{d}z'^2\,. \tag{1.1.23}$$

The coordinate transformations that leave the Minkowski metric form-invariant constitute the **Poincaré group** (inhomogeneous Lorentz group):

$$x^\mu \longrightarrow y^\mu = \Lambda^\mu{}_\nu x^\nu + T^\mu, \qquad \Lambda^\mu{}_\nu \Lambda_\sigma{}^\nu = \delta^\mu_\sigma, \tag{1.1.24}$$

where $\Lambda^\mu{}_\nu$ is a constant Lorentz matrix and T^μ is a constant spacetime translation vector. Unlike the Galilean group, rotations cannot depend on time. The Poincaré group contains 3 rotations, 3 boosts, and 4 spacetime translations, possessing **10 degrees of freedom**.

Poincaré algebra

The Poincaré group can be constructed via infinitesimal transformations. For an infinitesimal translation $T^\mu = \delta x^\mu$, a scalar field transforms as $f(y^\mu) = (1 - i\delta x^\alpha P_\alpha)f(x^\mu)$. Expanding to first order identifies the translation generator:

$$P_\mu = i\partial_\mu. \tag{1.1.25}$$

Similarly, an infinitesimal Lorentz transformation is $y^\mu = x^\mu + \omega^\mu{}_\nu x^\nu$ with $\omega_{\nu\mu} = -\omega_{\mu\nu}$. Defining the generator $J_{\mu\nu}$ by $f(y^\mu) = (1 - \frac{i}{2}\omega^{\alpha\beta}J_{\alpha\beta})f(x^\mu)$, we obtain:

$$J_{\mu\nu} = i\left(x_\mu\partial_\nu - x_\nu\partial_\mu\right). \tag{1.1.26}$$

These generators satisfy the commutation relations of the **Poincaré Lie algebra**:

$$[P_\alpha, P_\beta] = 0, \tag{1.1.27}$$

$$[J_{\mu\nu}, P_\alpha] = i\left(\eta_{\mu\alpha}P_\nu - \eta_{\nu\alpha}P_\mu\right), \tag{1.1.28}$$

$$[J_{\mu\nu}, J_{\alpha\beta}] = i\left(\eta_{\mu\alpha}J_{\beta\nu} + \eta_{\mu\beta}J_{\alpha\nu} + \eta_{\nu\alpha}J_{\mu\beta} + \eta_{\nu\beta}J_{\alpha\mu}\right), \tag{1.1.29}$$

where $[A,B] \equiv AB - BA$. Equation (1.1.29) forms the independently closed Lorentz algebra $\mathfrak{so}(3,1)$.

Trajectory of a free body

In analogy with Newtonian mechanics, the trajectory of a massive free particle is obtained by extremizing the action:

$$L = \frac{1}{2} g_{\mu\nu}(x^\lambda)\dot{x}^\mu \dot{x}^\nu, \tag{1.1.30}$$

where a dot denotes differentiation with respect to λ. The Euler–Lagrange equations yield the 4D geodesic equation:

$$\dot{u}^\mu + \Gamma^\mu_{\nu\lambda} u^\nu u^\lambda = 0, \qquad \Gamma^\mu_{\nu\lambda} = \frac{1}{2} g^{\mu\sigma} \left(\partial_\nu g_{\lambda\sigma} + \partial_\lambda g_{\nu\sigma} - \partial_\sigma g_{\nu\lambda}\right), \tag{1.1.31}$$

with $u^\mu = \mathrm{d}x^\mu \,/\, \mathrm{d}\lambda$. In Minkowski Cartesian coordinates, an inertial observer follows a path of constant velocity:

$$\dot{u}^\mu = 0 \iff u^\nu \partial_\nu u^\mu = 0. \tag{1.1.32}$$

Note that in Equation (1.1.30), it is not possible to simply add a scalar gravitational potential term in a Lorentz-covariant manner.

1.1.3 General relativity and curved space-time

Special relativity successfully reconciles electrodynamics with relativity, but conflicts with Newtonian gravity. Newtonian gravity relies on instantaneous action at a distance ($\nabla^2\phi = 4\pi G_{\mathrm{N}}\rho$), which is incompatible with the relativistic causal speed limit c.

Equivalence principle

Einstein based his approach on the universality of free fall: all test bodies fall identically in a gravitational field regardless of mass or composition.

In an accelerated reference frame, objects appear subject to fictitious forces (inertia, Coriolis). These forces are indistinguishable from gravity because a gravitational field can be locally eliminated in a freely falling reference frame. An accelerated observer experiences the same physical laws as an observer in a gravitational field: one can locally "transform away" gravity by an appropriate choice of frame.

This elimination is possible only because the equivalence principle holds strictly ($m_{\mathrm{I}} = m_{\mathrm{G}}$). Einstein elevated this to a first principle. This property is unique to gravity: in electromagnetism, the acceleration of a particle in an electric field depends on q/m, which varies across particles, preventing any single frame acceleration from eliminating the electric force for all bodies simultaneously.

The **equivalence principle** ensures that one can always find locally a reference frame in which the force of gravity is eliminated.

Tidal forces and space-time curvature

Gravity can only be eliminated *locally*. Consider a body A orbiting the Earth at distance $\mathbf{r}_A$, with Newtonian acceleration $\mathbf{a}_A = -G_{\mathrm{N}}M_\oplus \mathbf{r}_A/r_A^3$. A neighboring body B at $\mathbf{r}_B = \mathbf{r}_A + \delta\mathbf{r}$ experiences a relative acceleration with respect to A:

$$\delta\mathbf{a}_{AB} = -\frac{G_{\mathrm{N}}M_\oplus}{r_A^3}\left[\delta\mathbf{r} - 3\frac{\mathbf{r}_A\cdot\delta\mathbf{r}}{r_A^2}\mathbf{r}_A\right] \tag{1.1.33}$$

to first order in $\delta\mathbf{r}$.

This relative acceleration cannot be cancelled out:

- If $\mathbf{r}_A \cdot \delta\mathbf{r} = 0$, body B approaches body A.

- If $\mathbf{r}_A \cdot \delta\mathbf{r} = \pm r_A \delta r$, body B recedes from body A.

Thus, an initially spherical droplet of test particles deforms into an elongated ellipsoid while conserving its volume ($\boldsymbol{\nabla} \cdot \mathbf{g} = 0$). This distortion characterizes **tidal forces**.

In flat space-time, only a single body can free itself from gravity and fictitious forces. In curved space-time, however, **all bodies are free and follow lines of extremal length (geodesics)**. Gravity has locally vanished for each body, but neighboring geodesics reveal relative acceleration (geodesic deviation), uncovering the curvature of space-time. Gravity is not an external force; it is the physical manifestation of **space-time curvature**.

Gravity as a manifestation of geometry

Einstein's equivalence principle forms the foundation of all metric theories of gravitation, including general relativity. It rests on three conditions:

1. **Weak Equivalence Principle (WEP)** (universality of free fall): The trajectory of an uncharged test body with negligible self-gravitational energy is independent of its internal structure and composition.

2. **Local Position Invariance (LPI)**: The outcome of any local non-gravitational experiment is independent of the spacetime location where it is performed.

3. **Local Lorentz Invariance (LLI)**: The outcome of any local non-gravitational experiment is independent of the velocity of the freely falling laboratory.

When the Einstein equivalence principle holds, gravitation is described by a **metric theory**, possessing three properties:

- Space-time geometry is described by a metric $g_{\mu\nu}$.

- Freely falling bodies follow geodesics of that metric.

- In a local freely falling frame, physical laws take the same form as in special relativity.

This implies that non-gravitational binding energies contribute equally to inertial and gravitational mass.

Generalizing this property to gravitational binding energy leads to the **Strong Equivalence Principle (SEP)**: gravitational binding energy contributes equally to inertial and gravitational mass, and external gravitational fields can be locally eliminated for all experiments, gravitational ones included. A theory satisfying SEP is necessarily non-linear, as the gravitational field itself carries energy and generates secondary gravitational curvature ("gravity gravitates"). General relativity satisfies SEP, and experimental tests of these principles are detailed in Section 1.6.

1.2 Elements of differential geometry

The heuristic argument of the preceding section demonstrates that the motion of a freely falling body (that is, a body acted upon by no force other than gravity) can be described by a geodesic in a four-dimensional curved space-time. Space-time is therefore modeled as a four-dimensional continuum, requiring four coordinates to locate each event, just as in special relativity.

In Newtonian mechanics and special relativity, this coordinate mapping is assumed to hold globally, so that the entire set of events is mapped bijectively to $\mathbb{R}^4$. In general relativity, we make no such restrictive assumption about the global topology of space-time before solving the field equations. This situation is directly analogous to describing a two-dimensional sphere: two numbers specify the location of any point, and locally the sphere resembles the Euclidean plane

$\mathbb{R}^2$, but globally it possesses a distinct topology and cannot be covered by a single non-singular coordinate system.

The space-time of general relativity is therefore modeled as a **four-dimensional differentiable manifold**—a space of four dimensions that locally resembles $\mathbb{R}^4$, but not necessarily globally.

1.2.1 Manifolds and tensors

Manifolds

To define a manifold rigorously, recall that an open set in $\mathbb{R}^n$ is defined as a finite union of open balls. An open ball of radius r centered at $\mathbf{y} = (y^1, \ldots, y^n)$ is the set of points $\mathbf{x} \in \mathbb{R}^n$ such that $\|\mathbf{x} - \mathbf{y}\| < r$.

A **differentiable manifold** $\mathcal{M}$ of dimension n is a topological space composed of local neighborhoods that locally resemble $\mathbb{R}^n$ and are smoothly glued together. More precisely, $\mathcal{M}$ is endowed with a collection $\{O_i\}$ of open sets satisfying three conditions:

1. The collection $\{O_i\}$ covers $\mathcal{M}$: for every point $p \in \mathcal{M}$, there exists at least one open set O_i containing p.

2. For each i, there exists a homeomorphism ψ_i mapping O_i onto an open subset $U_i \subset \mathbb{R}^n$. The continuous map ψ_i assigns to every point $p \in O_i$ an n-tuple of real numbers $(x^1, \ldots, x^n)$, called the *coordinates* of p in the chart (O_i, ψ_i).

3. If two open sets O_i and O_j overlap ($O_i \cap O_j \neq \emptyset$), the transition map
$$\psi_j \circ \psi_i^{-1} : \psi_i(O_i \cap O_j) \subset U_i \longrightarrow \psi_j(O_i \cap O_j) \subset U_j \tag{1.2.1}$$
is an infinitely differentiable ($\mathcal{C}^\infty$) function from $\mathbb{R}^n$ to $\mathbb{R}^n$.

The collection of charts $\{(O_i, \psi_i)\}$ is called an **atlas**. Familiar examples of manifolds include the Euclidean plane $\mathbb{R}^2$ (which requires only a single chart) and the 2-sphere S^2 (which requires at least two charts, e.g., via stereographic projection).

Vectors

In Newtonian physics, Euclidean space is a global vector space. In curved spaces (such as the surface of a sphere), this global vector space structure is lost, but it is recovered locally in the limit of infinitesimal displacements.

In $\mathbb{R}^n$, there is a natural bijective correspondence between vectors and directional derivative operators: every vector $\mathbf{y} = (y^1, \ldots, y^n)$ defines a directional derivative $y^\mu(\partial/\partial x^\mu)$ acting on smooth functions. On a manifold $\mathcal{M}$, let $\mathcal{F}$ denote the set of $\mathcal{C}^\infty$ scalar functions $f : \mathcal{M} \to \mathbb{R}$. A **tangent vector y** at a point $p \in \mathcal{M}$ is defined as a linear map $y : \mathcal{F} \to \mathbb{R}$ satisfying the **Leibniz product rule**:

1. Linearity: $y(af + bg) = a\, y(f) + b\, y(g)$ for all $f, g \in \mathcal{F}$ and all $a, b \in \mathbb{R}$,

2. Leibniz rule: $y(fg) = f(p)\, y(g) + g(p)\, y(f)$.

The set of all tangent vectors at point p forms an n-dimensional real vector space V_p (or $T_p\mathcal{M}$), called the **tangent space** at p. Given a coordinate system $\{x^\mu\}$, a coordinate basis $\{X_\mu\}$ of V_p is defined by:
$$X_\mu(f) \equiv \frac{\partial f}{\partial x^\mu}. \tag{1.2.2}$$

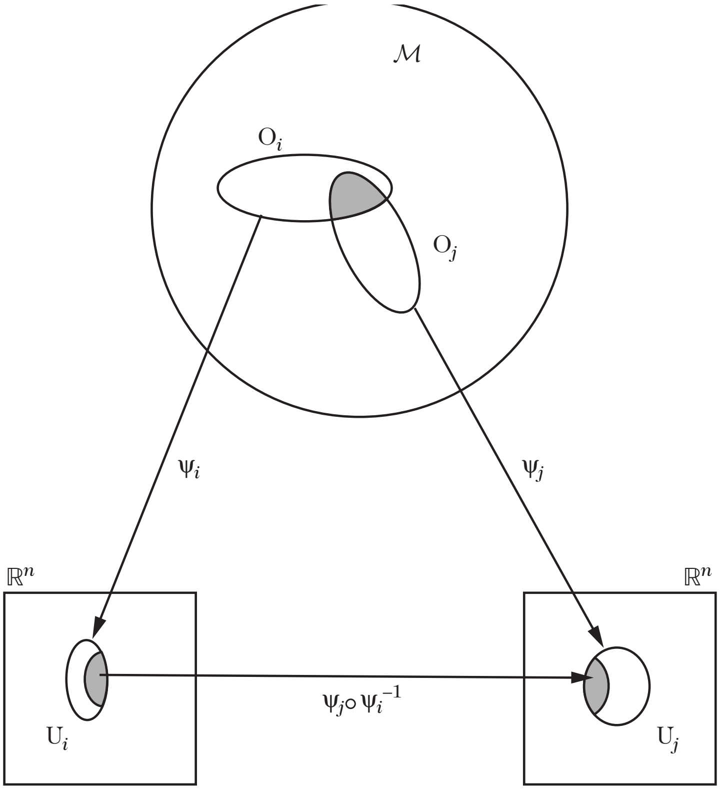


Figure 1.3: Illustration of the relationship between different charts of the same manifold. Locally, one can relate $\mathcal{M}$ and $\mathbb{R}^n$, but globally this is only possible if we use a collection of charts (an atlas) that cover all the manifold.

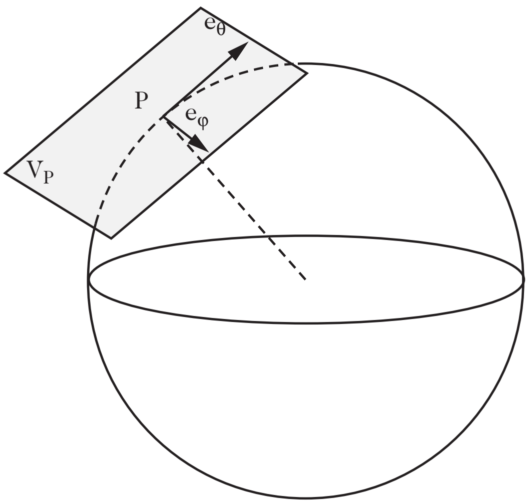


Figure 1.4: The tangent space of a two-dimensional sphere is a plane whose basis is given by $e_\theta = \partial_\theta$ and $e_\phi = \partial_\phi$.

We denote the basis vectors simply by $\partial_\mu \equiv \partial/\partial x^\mu$. Any tangent vector $\mathbf{y} \in V_p$ is expressed as a linear combination:

$$y = y^\mu \partial_\mu, \tag{1.2.3}$$

where y^μ are the contravariant components of the vector.

The tangent vector $\mathbf{t}$ to a parameterized curve $x^\mu(\tau)$ acting on a function $f \in \mathcal{F}$ is:

$$t(f) = \frac{\mathrm{d}}{\mathrm{d}\tau} f\left[x^\mu(\tau)\right] = \frac{\partial f}{\partial x^\mu}\frac{\mathrm{d}x^\mu(\tau)}{\mathrm{d}\tau}, \tag{1.2.4}$$

so that $t = t^\mu \partial_\mu$ with components $t^\mu = \mathrm{d}x^\mu(\tau)/\,\mathrm{d}\tau$.

Under a smooth coordinate transformation $x^\mu \to x'^\mu(x^\nu)$ with non-vanishing Jacobian determinant $\det(\partial x'^\mu/\partial x^\nu) \neq 0$, the chain rule implies $\partial_\nu f = (\partial_\nu x'^\mu)(\partial'_\mu f)$. Consequently, the coordinate basis vectors transform as:

$$\partial_\mu \longrightarrow \partial'_\mu = \frac{\partial x^\nu}{\partial x'^\mu}\partial_\nu, \tag{1.2.5}$$

and the vector components transform contravariantly:

$$y^\mu \longrightarrow y'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} y^\nu. \tag{1.2.6}$$

Tensors

The concept of a tensor generalizes vectors to multilinear objects under coordinate transformations.

Given the tangent space V_p, we construct its **dual space** V_p^* (the cotangent space $T_p^*\mathcal{M}$), defined as the vector space of all linear maps from V_p to $\mathbb{R}$. The elements of V_p^* are called **1-forms** (or covariant vectors). A dual basis $\{\mathrm{d}x^\nu\}$ for V_p^* is defined by its action on the coordinate basis vectors $\{\partial_\mu\}$:

$$\mathrm{d}x^\nu(\partial_\mu) = \delta^\nu_\mu. \tag{1.2.7}$$

The 1-form $\mathrm{d}x^\nu$ acts on an arbitrary vector $y = y^\mu\partial_\mu$ to extract its ν-th component: $\mathrm{d}x^\nu(y) = y^\nu$. Under a coordinate transformation, the basis 1-forms transform as:

$$\mathrm{d}x^\mu \longrightarrow \mathrm{d}x'^\mu = \frac{\partial x'^\mu}{\partial x^\nu}\,\mathrm{d}x^\nu\,. \tag{1.2.8}$$

A **tensor T of rank** (r, q) is a multilinear map:

$$T : \underbrace{V_p^* \times \cdots \times V_p^*}_{r \text{ copies}} \times \underbrace{V_p \times \cdots \times V_p}_{q \text{ copies}} \longrightarrow \mathbb{R}, \tag{1.2.9}$$

expanded in the tensor product basis as:

$$T = T^{\mu_1\ldots\mu_r}_{\nu_1\ldots\nu_q}\,\partial_{\mu_1}\otimes\cdots\otimes\partial_{\mu_r}\otimes \mathrm{d}x^{\nu_1}\otimes\cdots\otimes\mathrm{d}x^{\nu_q}\,. \tag{1.2.10}$$

Such a tensor is r times contravariant and q times covariant. In particular, a tensor of rank $(0,0)$ is an invariant scalar, a rank $(1,0)$ tensor is a contravariant vector, and a rank $(0,1)$ tensor is a 1-form.

Under a coordinate transformation, the components of a rank-(r,q) tensor transform as:

$$T'^{\mu_1\ldots\mu_r}_{\nu_1\ldots\nu_q} = \frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}}\cdots\frac{\partial x'^{\mu_r}}{\partial x^{\alpha_r}}\frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}}\cdots\frac{\partial x^{\beta_q}}{\partial x'^{\nu_q}}T^{\alpha_1\ldots\alpha_r}_{\beta_1\ldots\beta_q}\,. \tag{1.2.11}$$

A tangent space is attached to each point of the manifold; hence tensors defined at different points belong to different vector spaces. A **tensor field** assigns a tensor of rank (r,q) smoothly to each point of $\mathcal{M}$.

The Metric Tensor. To define distances on the manifold, we introduce the **metric tensor** g, which is a symmetric, non-degenerate tensor field of rank $(0,2)$. The spacetime interval between two infinitesimally separated events is:

$$\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu\,. \tag{1.2.12}$$

Because $g_{\mu\nu} = g_{\nu\mu}$, the metric in 4 dimensions possesses 10 independent components. Under a coordinate transformation, it transforms as:

$$g'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta}. \tag{1.2.13}$$

Taking the determinant of Equation (1.2.13), the metric determinant $g \equiv \det(g_{\mu\nu})$ transforms as a scalar density of weight -2:

$$g' = \left[\det\left(\frac{\partial x^\alpha}{\partial x'^\mu}\right)\right]^2 g. \tag{1.2.14}$$

Tensor calculus

Tensor algebraic operations include:

1. **Addition:** Tensors of identical rank (r,q) can be added linearly:

$$R^{\mu_1\ldots\mu_r}_{\nu_1\ldots\nu_q} = S^{\mu_1\ldots\mu_r}_{\nu_1\ldots\nu_q} + T^{\mu_1\ldots\mu_r}_{\nu_1\ldots\nu_q}. \tag{1.2.15}$$

2. **Outer Product:** The tensor product of a rank-(r,s) tensor with a rank-(p,q) tensor yields a tensor of rank $(r+p, s+q)$:

$$R^{\mu_1\ldots\mu_r\alpha_1\ldots\alpha_p}_{\nu_1\ldots\nu_s\beta_1\ldots\beta_q} = S^{\mu_1\ldots\mu_r}_{\nu_1\ldots\nu_s} T^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}. \tag{1.2.16}$$

3. **Contraction:** Setting one contravariant and one covariant index equal implies a summation over that index, reducing the rank from (r,q) to $(r-1,q-1)$:

$$R^{\mu_2\ldots\mu_r}_{\nu_2\ldots\nu_q} = S^{\mu_1\mu_2\ldots\mu_r}_{\mu_1\nu_2\ldots\nu_q} = S^{\alpha\mu_2\ldots\mu_r}_{\beta\nu_2\ldots\nu_q}\delta^\beta_\alpha. \tag{1.2.17}$$

   Complete contraction yields an invariant scalar. In particular, the scalar product of two vectors is $\mathbf{T}\cdot\mathbf{V} = T^\mu V_\mu = g_{\mu\nu}T^\mu V^\nu$. An object $T^{\mu_1\ldots\mu_r}_{\nu_1\ldots\nu_q}$ is a tensor if and only if its contraction with every arbitrary tensor $R^{\nu_1\ldots\nu_q}_{\mu_1\ldots\mu_r}$ produces an invariant scalar (the quotient law).

4. **Raising and Lowering Indices:** The metric $g_{\mu\nu}$ and its inverse $g^{\mu\nu}$ (satisfying $g^{\mu\alpha}g_{\alpha\nu} = \delta^\mu_\nu$) raise and lower indices:

$$R^{\mu_1\ldots\mu_r\mu}_{\nu_2\ldots\nu_q} = R^{\mu_1\ldots\mu_r}_{\nu_1\nu_2\ldots\nu_q}g^{\nu_1\mu}, \qquad T^\mu = g^{\mu\nu}T_\nu. \tag{1.2.18}$$

A tensor is symmetric if $T_{\mu\nu} = T_{\nu\mu}$ and antisymmetric if $T_{\mu\nu} = -T_{\nu\mu}$. Any rank-2 tensor can be decomposed into its symmetric and antisymmetric parts:

$$T_{(\mu\nu)} \equiv \frac{1}{2}\left(T_{\mu\nu} + T_{\nu\mu}\right), \qquad T_{[\mu\nu]} \equiv \frac{1}{2}\left(T_{\mu\nu} - T_{\nu\mu}\right). \tag{1.2.19}$$

For indices enclosed between vertical bars, the symmetrization excludes those indices: $T_{(\mu|\alpha\beta|\nu)} = \frac{1}{2}(T_{\mu\alpha\beta\nu} + T_{\nu\alpha\beta\mu})$. Complete symmetrization and antisymmetrization over n indices are defined by:

$$T_{(\mu_1\ldots\mu_n)} \equiv \frac{1}{n!}\sum_{\sigma\in\mathcal{S}_n} T_{\mu_{\sigma(1)}\cdots\mu_{\sigma(n)}}, \tag{1.2.20}$$

$$T_{[\mu_1\dots\mu_n]} \equiv \frac{1}{n!}\sum_{\sigma\in\mathcal{S}_n}\mathrm{sgn}(\sigma)T_{\mu_{\sigma(1)}\dots\mu_{\sigma(n)}}. \tag{1.2.21}$$

In 4 dimensions, any rank-2 tensor can be uniquely decomposed into an antisymmetric part, a traceless symmetric part, and a trace:

$$T_{\mu\nu} = T_{[\mu\nu]} + \left(T_{(\mu\nu)} - \frac{1}{4}T^{\alpha}{}_{\alpha}g_{\mu\nu}\right) + \frac{1}{4}T^{\alpha}{}_{\alpha}g_{\mu\nu}. \tag{1.2.22}$$

The Levi-Civita Tensor. The completely antisymmetric **Levi-Civita tensor** of rank n is defined by:

$$\epsilon_{\mu_1\mu_2\dots\mu_n} = \epsilon_{[\mu_1\mu_2\dots\mu_n]}, \tag{1.2.23}$$

with components $\epsilon_{0123} = \sqrt{-g}$ and $\epsilon^{0123} = -1/\sqrt{-g}$ in four dimensions. The contraction identities of the 4D Levi-Civita tensor are:

$$\epsilon^{\alpha\beta\delta\gamma}\epsilon_{\alpha\lambda\nu\mu} = -\delta^{\beta}_{\lambda}\delta^{\delta}_{\nu}\delta^{\gamma}_{\mu} - \delta^{\beta}_{\nu}\delta^{\delta}_{\mu}\delta^{\gamma}_{\lambda} - \delta^{\beta}_{\mu}\delta^{\delta}_{\lambda}\delta^{\gamma}_{\nu} + \delta^{\beta}_{\lambda}\delta^{\delta}_{\mu}\delta^{\gamma}_{\nu} + \delta^{\beta}_{\mu}\delta^{\delta}_{\nu}\delta^{\gamma}_{\lambda} + \delta^{\beta}_{\nu}\delta^{\delta}_{\lambda}\delta^{\gamma}_{\mu}. \tag{1.2.24}$$

Successive contractions yield:

$$\epsilon^{\alpha\beta\delta\gamma}\epsilon_{\alpha\beta\nu\mu} = -2\left(\delta^{\delta}_{\nu}\delta^{\gamma}_{\mu} - \delta^{\delta}_{\mu}\delta^{\gamma}_{\nu}\right), \tag{1.2.25}$$

$$\epsilon^{\alpha\beta\delta\gamma}\epsilon_{\alpha\beta\delta\mu} = -6\delta^{\gamma}_{\mu}, \tag{1.2.26}$$

$$\epsilon^{\alpha\beta\delta\gamma}\epsilon_{\alpha\beta\delta\gamma} = -24. \tag{1.2.27}$$

1.2.2 Geodesic equations and Christoffel symbols

We now return to the trajectory of a freely falling body moving in a curved space-time with metric $g_{\mu\nu}$. Such a trajectory minimizes the spacetime distance, governed by the action $S = \int \mathrm{d}s$. However, because $\mathrm{d}s^2$ can be zero or negative, it is mathematically advantageous to extremize the quadratic action:

$$S = \int L\,\mathrm{d}\lambda\,, \qquad L = g_{\mu\nu}(x^{\alpha})\dot{x}^{\mu}\dot{x}^{\nu}, \tag{1.2.28}$$

where $\dot{x}^{\mu} \equiv \mathrm{d}x^{\mu}\,/\,\mathrm{d}\lambda$. Extremizing this action yields identical trajectories to $S = \int \mathrm{d}s$ for timelike curves, while remaining well-defined and valid for **null (light-like) geodesics** ($\mathrm{d}s^2 = 0$).

The Euler–Lagrange equations take the form:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial \dot{x}^{\alpha}}\right) - \frac{\partial L}{\partial x^{\alpha}} = 0. \tag{1.2.29}$$

Evaluating the derivatives:

$$\frac{\partial L}{\partial x^{\alpha}} = \partial_{\alpha}g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu}, \tag{1.2.30}$$

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial \dot{x}^{\alpha}}\right) = \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(2g_{\mu\alpha}\dot{x}^{\mu}\right) = 2g_{\mu\alpha}\ddot{x}^{\mu} + 2\partial_{\beta}g_{\mu\alpha}\dot{x}^{\beta}\dot{x}^{\mu}. \tag{1.2.31}$$

Setting the difference to zero:

$$g_{\mu\alpha}\ddot{x}^{\mu} = \frac{1}{2}\left[\partial_{\alpha}g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} - \partial_{\beta}g_{\mu\alpha}\dot{x}^{\beta}\dot{x}^{\mu} - \partial_{\mu}g_{\beta\alpha}\dot{x}^{\beta}\dot{x}^{\mu}\right], \tag{1.2.32}$$

where we have symmetrized the term $\partial_{\beta}g_{\mu\alpha}\dot{x}^{\beta}\dot{x}^{\mu} = \frac{1}{2}(\partial_{\beta}g_{\mu\alpha} + \partial_{\mu}g_{\beta\alpha})\dot{x}^{\beta}\dot{x}^{\mu}$.

Introducing the **Christoffel symbols of the first kind**:

$$\Gamma_{\lambda\mu\nu} \equiv \frac{1}{2}\left(\partial_{\mu}g_{\lambda\nu} + \partial_{\nu}g_{\mu\lambda} - \partial_{\lambda}g_{\mu\nu}\right), \tag{1.2.33}$$

and the **Christoffel symbols of the second kind**:

$$\Gamma^{\alpha}_{\mu\nu} \equiv g^{\lambda\alpha}\Gamma_{\lambda\mu\nu} = \frac{1}{2}g^{\alpha\lambda}\left(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}\right), \tag{1.2.34}$$

contracting with $g^{\alpha\sigma}$ yields the **geodesic equation**:

$$\ddot{x}^\mu + \Gamma^\mu_{\nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0. \tag{1.2.35}$$

Because the metric is symmetric, the Christoffel symbols are symmetric in their lower indices:

$$\Gamma^\alpha_{\mu\nu} = \Gamma^\alpha_{\nu\mu}, \qquad \Gamma_{\alpha\mu\nu} = \Gamma_{\alpha\nu\mu}. \tag{1.2.36}$$

In four dimensions, there are $4 \times 10 = 40$ independent Christoffel symbols, matching the 40 partial derivatives of the metric $\partial_\alpha g_{\mu\nu}$. Inverting Equation (1.2.33) gives:

$$\partial_\alpha g_{\mu\nu} = g_{\lambda\mu}\Gamma^\lambda_{\alpha\nu} + g_{\lambda\nu}\Gamma^\lambda_{\alpha\mu}. \tag{1.2.37}$$

Under a general coordinate transformation $x^\mu \to x'^\mu(x^\nu)$, the Christoffel symbols transform as:

$$\Gamma'^{\alpha}_{\mu\nu} = \frac{\partial x'^\alpha}{\partial x^\beta}\frac{\partial x^\sigma}{\partial x'^\mu}\frac{\partial x^\rho}{\partial x'^\nu}\Gamma^\beta_{\sigma\rho} - \frac{\partial^2 x'^\alpha}{\partial x^\sigma \partial x^\rho}\frac{\partial x^\sigma}{\partial x'^\mu}\frac{\partial x^\rho}{\partial x'^\nu}. \tag{1.2.38}$$

Due to the second inhomogeneous term containing second derivatives $\partial^2 x'^\alpha/\partial x^\sigma \partial x^\rho$, **the Christoffel symbols do not form a tensor**.

Metric Determinant Relations. The determinant $g = \det(g_{\mu\nu})$ can be expanded in terms of matrix minors $m^{\mu\nu}$, where the inverse metric is $g^{\mu\nu} = m^{\mu\nu}/g$, yielding $\mathrm{d}g = m^{\mu\nu}\,\mathrm{d}g_{\mu\nu} = gg^{\mu\nu}\,\mathrm{d}g_{\mu\nu}$. Thus:

$$\partial_\alpha g = gg^{\mu\nu}\partial_\alpha g_{\mu\nu}. \tag{1.2.39}$$

Using this result and the contraction $\Gamma^\mu_{\mu\alpha} = \frac{1}{2}g^{\mu\lambda}\partial_\alpha g_{\mu\lambda}$, we obtain the contraction identity:

$$\Gamma^\mu_{\mu\alpha} = \partial_\alpha \ln\sqrt{-g} = \frac{1}{\sqrt{-g}}\partial_\alpha\sqrt{-g}. \tag{1.2.40}$$

Furthermore, differentiating the relation $g_{\mu\nu}g^{\mu\nu} = 4$ yields:

$$g_{\mu\nu}\partial_\alpha g^{\mu\nu} = -g^{\mu\nu}\partial_\alpha g_{\mu\nu}. \tag{1.2.41}$$

1.2.3 Covariant derivative, parallel transport, and Lie derivative

Covariant derivative

Physical laws are formulated as partial differential equations. Differentiating a tensor field must produce an object that is itself a valid tensor.

The ordinary partial derivative of an arbitrary tensor field $\partial_\alpha T^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q}$ is generally *not* a tensor (except for the derivative of a scalar field, $\partial_\alpha\phi$, which is a genuine 1-form). For a vector T^μ, differentiating its transformed component $T'^\mu = (\partial x'^\mu/\partial x^\nu)T^\nu$ gives:

$$\frac{\partial T'^\mu}{\partial x'^\alpha} = \frac{\partial x'^\mu}{\partial x^\nu}\frac{\partial x^\sigma}{\partial x'^\alpha}\frac{\partial T^\nu}{\partial x^\sigma} + \frac{\partial^2 x'^\mu}{\partial x^\sigma \partial x^\nu}\frac{\partial x^\sigma}{\partial x'^\alpha}T^\nu. \tag{1.2.42}$$

The non-tensorial second derivative term arises because the partial derivative compares the vector at x^α and $x^\alpha + \mathrm{d}x^\alpha$, where the local coordinate basis vectors point in different directions.

The **covariant derivative** ∇_μ corrects for this by parallel transporting the tensor before computing the difference. For contravariant and covariant vectors, the covariant derivative is defined by:

$$\nabla_\mu T^\nu = \partial_\mu T^\nu + \Gamma^\nu_{\mu\alpha}T^\alpha, \tag{1.2.43}$$

$$\nabla_\mu T_\nu = \partial_\mu T_\nu - \Gamma^\alpha_{\mu\nu} T_\alpha. \tag{1.2.44}$$

Throughout this text, we denote the covariant derivative interchangeably by $\nabla_\mu T^\nu \equiv T^\nu{}_{;\mu}$, and the ordinary partial derivative by $\partial_\mu T^\nu \equiv T^\nu{}_{,\mu}$.

For an arbitrary tensor of rank (p, q), the covariant derivative is:

$$\nabla_\alpha T^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q} = \partial_\alpha T^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q} + \sum_{i=1}^{p} \Gamma^{\mu_i}_{\alpha\lambda} T^{\mu_1\ldots\lambda\ldots\mu_p}_{\nu_1\ldots\nu_q} - \sum_{j=1}^{q} \Gamma^{\lambda}_{\alpha\nu_j} T^{\mu_1\ldots\mu_p}_{\nu_1\ldots\lambda\ldots\nu_q}. \tag{1.2.45}$$

Metric Compatibility. Computing the covariant derivative of the metric tensor:

$$\nabla_\alpha g_{\mu\nu} = \partial_\alpha g_{\mu\nu} - \Gamma^\sigma_{\alpha\mu} g_{\sigma\nu} - \Gamma^\sigma_{\alpha\nu} g_{\sigma\mu}. \tag{1.2.46}$$

Substituting Equation (1.2.37), we obtain the fundamental property of **metric compatibility**:

$$\nabla_\alpha g_{\mu\nu} = 0, \qquad \nabla_\alpha g^{\mu\nu} = 0. \tag{1.2.47}$$

The Levi-Civita connection defined by the Christoffel symbols is the unique torsion-free connection compatible with the metric.

Parallel transport

The covariant derivative defines the parallel transport of a tensor along a curve $x^\mu(\tau)$ with tangent four-velocity $u^\mu = \mathrm{d}x^\mu / \mathrm{d}\tau$. A vector T^μ is **parallel transported** along the curve if its absolute derivative along the curve vanishes:

$$\frac{\mathrm{D}T^\mu}{\mathrm{D}\tau} \equiv u^\nu \nabla_\nu T^\mu = \frac{\mathrm{d}T^\mu}{\mathrm{d}\tau} + \Gamma^\mu_{\nu\alpha} T^\alpha u^\nu = 0. \tag{1.2.48}$$

On a curved manifold, parallel transport is path-dependent: transporting a vector between two points along different routes results in different final vectors (Figure 1.5).

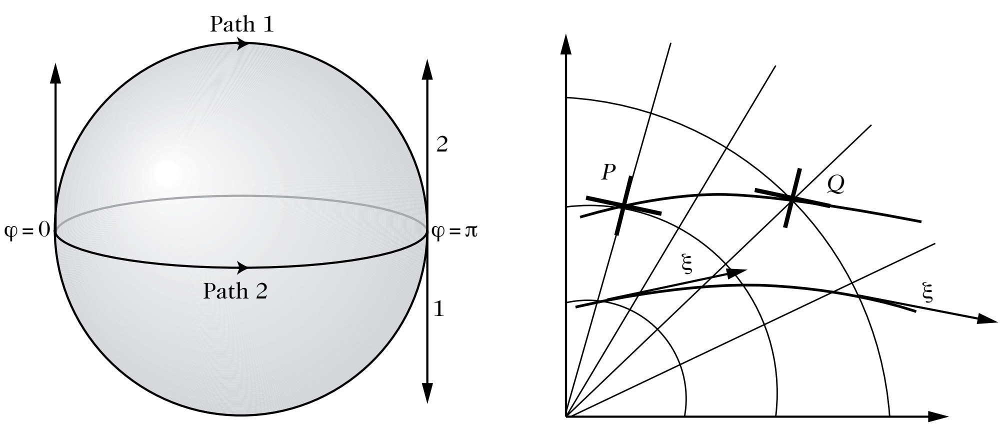


Figure 1.5: (Left): On a 2-dimensional sphere, a vector can be transported parallel to itself along a path 1 corresponding to a great circle until its antipodal point. We can reach the same point following the path 2 along the equator. The two vectors that we get do not coincide. (Right): The Lie derivative evaluates the change determined by an observer that goes from a point P to a point Q along the flow curves of the vector field ξ^μ transporting its local frame with it.

A curve is **auto-parallel** if its tangent vector is parallel-transported along itself:

$$\frac{\mathrm{D}u^\mu}{\mathrm{D}\tau} = u^\nu \nabla_\nu u^\mu = 0 \iff \frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\tau^2} + \Gamma^\mu_{\nu\lambda} \frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} \frac{\mathrm{d}x^\lambda}{\mathrm{d}\tau} = 0. \tag{1.2.49}$$

This is precisely the geodesic equation (1.2.35): **geodesics are the auto-parallel curves of the space-time metric**.

Differential operators

Using the contracted Christoffel symbol identity (1.2.40), we formulate standard covariant differential operators:

1. **Divergence of a Vector Field:**

$$\nabla_\mu T^\mu = \partial_\mu T^\mu + \Gamma^\mu_{\mu\rho} T^\rho = \partial_\mu T^\mu + T^\rho \partial_\rho \ln\sqrt{-g} = \frac{1}{\sqrt{-g}} \partial_\mu \left(\sqrt{-g} T^\mu\right). \tag{1.2.50}$$

2. **Divergence of an Antisymmetric Tensor:** For an antisymmetric tensor $F^{\mu\nu} = -F^{\nu\mu}$, $\Gamma^\mu_{\mu\rho} F^{\rho\nu} + \Gamma^\nu_{\mu\rho} F^{\mu\rho} = \Gamma^\mu_{\mu\rho} F^{\rho\nu}$ (since $\Gamma^\nu_{\mu\rho} F^{\mu\rho} = 0$ by symmetry-antisymmetry contraction), yielding:

$$\nabla_\mu F^{\mu\nu} = \frac{1}{\sqrt{-g}} \partial_\mu \left(\sqrt{-g} F^{\mu\nu}\right). \tag{1.2.51}$$

3. **The d'Alembertian Operator:** The covariant d'Alembertian is $\Box \equiv \nabla_\mu \nabla^\mu = g^{\mu\nu} \nabla_\mu \nabla_\nu$. Acting on a scalar field ϕ (where $\nabla_\nu \phi = \partial_\nu \phi$):

$$\Box\phi = \nabla_\mu \nabla^\mu \phi = \frac{1}{\sqrt{-g}} \partial_\mu \left(\sqrt{-g} g^{\mu\nu} \partial_\nu \phi\right). \tag{1.2.52}$$

In flat Minkowski space-time, this reduces to $\Box = -\partial_t^2 + \nabla^2$.

4. **Curl of a 1-form:** The covariant curl of a 1-form A_μ satisfies:

$$A_{\nu;\mu} - A_{\mu;\nu} = \partial_\mu A_\nu - \Gamma^\alpha_{\mu\nu} A_\alpha - \left(\partial_\nu A_\mu - \Gamma^\alpha_{\nu\mu} A_\alpha\right) = \partial_\mu A_\nu - \partial_\nu A_\mu = A_{\nu,\mu} - A_{\mu,\nu}, \tag{1.2.53}$$

due to the symmetry of the Christoffel symbols ($\Gamma^\alpha_{\mu\nu} = \Gamma^\alpha_{\nu\mu}$).

Lie derivative

Another fundamental method for constructing a derivative on a manifold is the **Lie derivative**. Consider a vector field ξ^μ. We determine how the components of a tensor field change for an observer transported along the integral flow curves of ξ^μ from a point $P(x^\alpha)$ to a neighboring point $Q(x^\alpha + \epsilon\xi^\alpha)$, carrying their coordinate frame with them (Figure 1.5, right).

Using the coordinate system defined at P at the new point Q corresponds to performing a local coordinate transformation at Q given by $x'^\mu = x^\mu - \epsilon\xi^\mu$. To first order in the infinitesimal parameter ϵ, the Jacobian matrices are:

$$\frac{\partial x'^\mu}{\partial x^\nu} = \delta^\mu_\nu - \epsilon\partial_\nu \xi^\mu, \qquad \frac{\partial x^\nu}{\partial x'^\mu} = \delta^\nu_\mu + \epsilon\partial_\nu \xi^\mu. \tag{1.2.54}$$

The components of the vector T^μ evaluated at Q in this transported frame are:

$$\begin{aligned} T'^\mu(Q) &= \frac{\partial x'^\mu}{\partial x^\nu} T^\nu(x^\alpha + \epsilon\xi^\alpha) \\ &= \left(\delta^\mu_\nu - \epsilon\partial_\nu \xi^\mu\right) \left[T^\nu(P) + \epsilon\xi^\alpha \partial_\alpha T^\nu(P)\right] \\ &= T^\mu(P) + \epsilon\left[\xi^\alpha \partial_\alpha T^\mu(P) - T^\nu(P)\partial_\nu \xi^\mu\right] + \mathcal{O}(\epsilon^2). \end{aligned} \tag{1.2.55}$$

The **Lie derivative** $\mathcal{L}_\xi T^\mu$ is defined by comparing this transported value with the original vector $T^\mu(P)$:

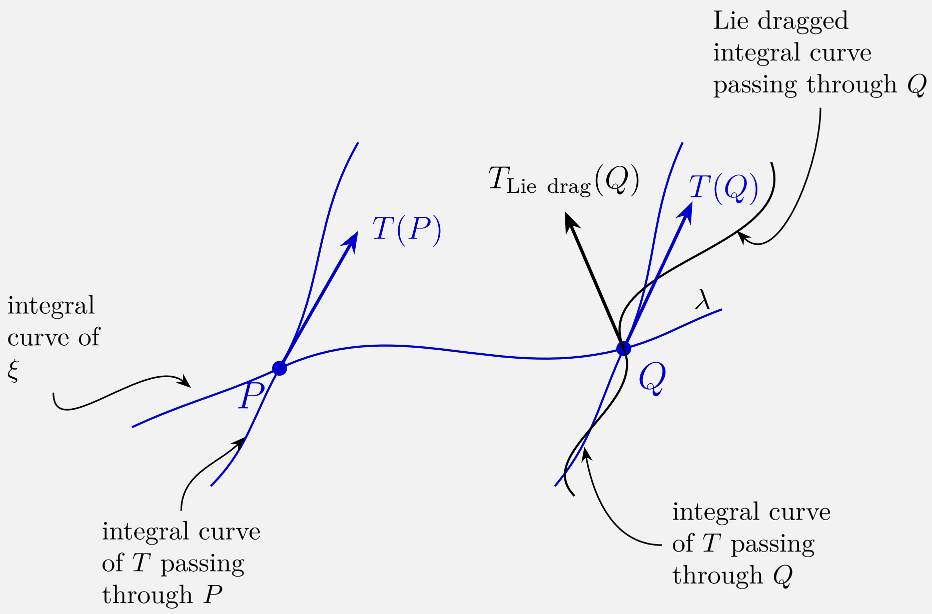
$$\mathcal{L}_\xi T^\mu \equiv \lim_{\epsilon\to 0} \frac{1}{\epsilon} \left[T'^\mu(Q) - T^\mu(P)\right] = \xi^\nu \partial_\nu T^\mu - T^\nu \partial_\nu \xi^\mu = [\xi, T]^\mu. \tag{1.2.56}$$

Generalizing to an arbitrary tensor field of rank (p, q), the Lie derivative is:

$$\mathcal{L}_\xi T^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q} = \xi^\alpha \partial_\alpha T^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q} - \sum_{i=1}^{p} T^{\mu_1\ldots\alpha\ldots\mu_p}_{\nu_1\ldots\nu_q} \partial_\alpha \xi^{\mu_i} + \sum_{j=1}^{q} T^{\mu_1\ldots\mu_p}_{\nu_1\ldots\beta\ldots\nu_q} \partial_{\nu_j} \xi^\beta. \tag{1.2.57}$$

Because the connection coefficients $\Gamma^\alpha_{\mu\nu}$ are symmetric, all partial derivatives in Equation (1.2.57) can be replaced by covariant derivatives ∇.

One intuitive way to derive the Lie derivative is with the notion of integral curves.



Let ξ be a vector field and consider its integral curves $x^\mu(\lambda)$. Along such a curve,

$$\xi = \frac{dx^\mu}{dt} \frac{\partial}{\partial x^\mu} = \xi^\mu \partial_\mu.$$

Let P and Q be two nearby points on an integral curve of ξ, with Q obtained from P by moving a parameter distance ϵ along the curve. Thus, to first order in ϵ,

$$x^\mu(Q) = x^\mu(P) + \epsilon\, \xi^\mu(P) + \mathcal{O}(\epsilon^2).$$

Now consider another vector field T, whose integral curves $x^\mu(\lambda)$ satisfy

$$T = \frac{dx^\mu}{d\lambda} \frac{\partial}{\partial x^\mu} = T^\mu \partial_\mu.$$

At P, the vector $T(P)$ is tangent to the integral curve of T passing through P. We now transport this vector from P to Q along the integral curve of ξ. The Lie-dragged vector at Q is

$$\begin{aligned} T_{\text{dragged}}(Q) &= \frac{d}{d\lambda}(x^\mu + \epsilon\, \xi)\Big|_P \frac{\partial}{\partial x^\mu}\Big|_Q \\ &= [T^\mu + \epsilon\, \xi^\nu \partial_\nu T^\mu]\Big|_P \frac{\partial}{\partial x^\mu}\Big|_Q + \mathcal{O}(\epsilon^2). \end{aligned}$$

We must express the coordinate basis at Q in terms of the basis at P. From

$$x^\mu(Q) = x^\mu(P) + \epsilon\, \xi^\mu(P),$$

we obtain the inverse transformation

$$x^\mu(P) = x^\mu(Q) - \epsilon\,\xi^\mu(Q) + \mathcal{O}(\epsilon^2).$$

Hence,

$$\begin{aligned}\frac{\partial}{\partial x^\mu}\bigg|_Q &= \frac{\partial x^\nu(P)}{\partial x^\mu(Q)}\frac{\partial}{\partial x^\nu}\bigg|_P \\ &= \Big[\delta^\nu_\mu - \epsilon\,\partial_\mu \xi^\nu\Big]_P\,\partial_\nu|_P + \mathcal{O}(\epsilon^2).\end{aligned}$$

Consequently,

$$\begin{aligned}T_{\text{dragged}}(Q) &= (T^\mu + \epsilon\,\xi^\nu\partial_\nu T^\mu)\left(\delta^\sigma_\mu - \epsilon\,\partial_\mu\xi^\sigma\right)\partial_\sigma + \mathcal{O}(\epsilon^2) \\ &= [T^\sigma + \epsilon\,\xi^\nu\partial_\nu T^\sigma - \epsilon\,T^\mu\partial_\mu\xi^\sigma]\,\partial_\sigma + \mathcal{O}(\epsilon^2) \\ &= [T^\sigma + \epsilon\,(\xi^\nu\partial_\nu T^\sigma - T^\nu\partial_\nu\xi^\sigma)]\,\partial_\sigma + \mathcal{O}(\epsilon^2).\end{aligned}$$

Recognising the Lie bracket,

$$[\xi, T]^\sigma = \xi^\nu\partial_\nu T^\sigma - T^\nu\partial_\nu\xi^\sigma,$$

we obtain

$$T_{\text{dragged}}(Q) = T(P) + \epsilon[\xi, T]_P + \mathcal{O}(\epsilon^2).$$

Thus the Lie derivative of T along ξ is

$$\begin{aligned}\pounds_\xi T &= \lim_{\epsilon\to 0}\frac{T_{\text{dragged}}(Q) - T(P)}{\epsilon} \\ &= [\xi, T].\end{aligned}$$

Hence,

$$\boxed{\pounds_\xi T = [\xi, T]}. \tag{1}$$

The Lie derivative plays a central role in characterising the spacetime symmetries of a Riemannian or pseudo-Riemannian manifold. If $\pounds_\xi T^{\mu_1\cdots\mu_p}_{\nu_1\ldots\nu_q} = 0$, the tensor field is strictly invariant under the continuous coordinate transformation generated by the flow of ξ^μ. In particular, for the metric tensor $g_{\mu\nu}$, applying Equation (1.2.57) yields:

$$\mathcal{L}_\xi g_{\mu\nu} = \nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu. \tag{1.2.58}$$

1.2.4 Curvature

The Riemann curvature tensor

As illustrated in Figure 1.5, parallel transporting a vector along a closed loop in a curved space does not return the vector to its initial orientation. This holonomy distinguishes curved geometry from flat Euclidean or Minkowski space.

Consider parallel transporting a vector T^μ around the infinitesimal quadrilateral $P_1P_2P_3P_4$ with coordinate displacements $\mathrm{d}x^\alpha$ and $\mathrm{d}x^\beta$ (Figure 1.6). From P_1 to P_2, parallel transport gives $\delta_{12}T^\mu = \mathrm{d}x^\alpha\,\nabla_\alpha T^\mu = \mathrm{d}T^\mu + \Gamma^\mu_{\alpha\sigma}T^\sigma\,\mathrm{d}x^\alpha$. Iterating along $P_2 \to P_3$ yields:

$$\delta_{123}T^\mu = \mathrm{d}x^\alpha\,\mathrm{d}x^\beta\,\nabla_\beta\nabla_\alpha T^\mu = \mathrm{d}^2T^\mu + 2\Gamma^\mu_{\alpha\nu}\,\mathrm{d}x^\alpha\,\mathrm{d}T^\nu + \left(\partial_\beta\Gamma^\mu_{\alpha\sigma} + \Gamma^\mu_{\beta\nu}\Gamma^\nu_{\alpha\sigma}\right)\mathrm{d}x^\alpha\,\mathrm{d}x^\beta\,T^\sigma. \tag{1.2.59}$$

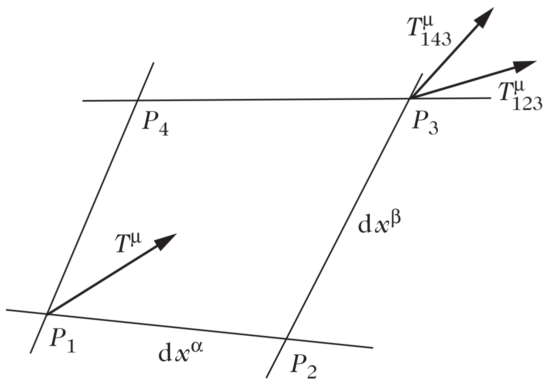


Figure 1.6: The vector T^{μ} is parallel transported along the path $P_1 \to P_3$ passing either through P_2 (yielding T^{μ}_{123}) or through P_4 (yielding T^{μ}_{143}). These two vectors coincide if and only if the Riemann curvature tensor vanishes.

Similarly, transporting along the alternative route $P_1 \to P_4 \to P_3$ gives $\delta_{143}T^{\mu} = \mathrm{d}x^{\alpha}\,\mathrm{d}x^{\beta}\,\nabla_{\alpha}\nabla_{\beta}T^{\mu}$. The net difference between the two paths depends only on the commutator of covariant derivatives:

$$(\nabla_{\alpha}\nabla_{\beta} - \nabla_{\beta}\nabla_{\alpha})\,T^{\mu} = R^{\mu}{}_{\nu\alpha\beta}T^{\nu}, \tag{1.2.60}$$

where $R^{\mu}{}_{\nu\alpha\beta}$ is the **Riemann curvature tensor**:

$$R^{\mu}{}_{\nu\alpha\beta} \equiv \partial_{\alpha}\Gamma^{\mu}_{\nu\beta} - \partial_{\beta}\Gamma^{\mu}_{\nu\alpha} + \Gamma^{\mu}_{\sigma\alpha}\Gamma^{\sigma}_{\nu\beta} - \Gamma^{\mu}_{\sigma\beta}\Gamma^{\sigma}_{\nu\alpha}, \tag{1.2.61}$$

and its fully covariant form is $R_{\sigma\nu\alpha\beta} \equiv g_{\sigma\mu}R^{\mu}{}_{\nu\alpha\beta}$.

Algebraic Symmetries of the Riemann Tensor. The Riemann tensor satisfies three fundamental algebraic symmetries:

1. **Antisymmetry in the first and second index pairs:**

$$R_{\alpha\mu\nu\sigma} = -R_{\mu\alpha\nu\sigma} = -R_{\alpha\mu\sigma\nu}. \tag{1.2.62}$$

2. **Pair-exchange symmetry:**

$$R_{\alpha\mu\nu\sigma} = R_{\nu\sigma\alpha\mu}. \tag{1.2.63}$$

3. **First Algebraic Bianchi Identity (Cyclic Sum):**

$$3R_{\alpha[\mu\nu\sigma]} \equiv R_{\alpha\mu\nu\sigma} + R_{\alpha\nu\sigma\mu} + R_{\alpha\sigma\mu\nu} = 0. \tag{1.2.64}$$

In an n-dimensional space, the first two symmetries (1.2.62) and (1.2.63) reduce the components of the Riemann tensor to those of a symmetric $\frac{n(n-1)}{2} \times \frac{n(n-1)}{2}$ matrix, giving $\frac{1}{2}\left[\frac{n(n-1)}{2}\right]\left[\frac{n(n-1)}{2}+1\right]$ components. The cyclic identity (1.2.64) imposes $\binom{n}{4}$ independent constraints. Consequently, the number of algebraically independent components of the Riemann tensor is:

$$N_{\text{Riem}} = \frac{n^2(n^2-1)}{12}. \tag{1.2.65}$$

In four spacetime dimensions ($n = 4$), $N_{\text{Riem}} = \frac{16\times 15}{12} = \mathbf{20}$ independent components.

The Differential Bianchi Identity. The covariant derivatives of the Riemann tensor satisfy the cyclic identity:

$$3R_{\alpha\mu[\nu\sigma;\beta]} \equiv R_{\alpha\mu\nu\sigma;\beta} + R_{\alpha\mu\sigma\beta;\nu} + R_{\alpha\mu\beta\nu;\sigma} = 0. \tag{1.2.66}$$

Ricci and Einstein tensors

Due to the algebraic symmetries of the Riemann tensor, only a single independent rank-2 trace can be formed by contraction. This defines the symmetric **Ricci tensor**:

$$R_{\mu\nu} \equiv R^{\alpha}{}_{\mu\alpha\nu} = -R^{\alpha}{}_{\mu\nu\alpha}. \tag{1.2.67}$$

Because $R_{\mu\nu} = R_{\nu\mu}$, the Ricci tensor in 4 dimensions possesses **10 independent components**.

Contraction of the Ricci tensor with the inverse metric yields the **Ricci scalar** (scalar curvature):

$$R \equiv g^{\mu\nu} R_{\mu\nu} = R^{\mu}{}_{\mu}. \tag{1.2.68}$$

Contraction of the differential Bianchi identity (1.2.66) with $g^{\alpha\beta}$ and $g^{\mu\sigma}$ yields:

$$\nabla^{\alpha} R_{\alpha\mu\nu\sigma} - \nabla_{\nu} R_{\mu\sigma} + \nabla_{\sigma} R_{\mu\nu} = 0, \tag{1.2.69}$$

$$\nabla^{\alpha} R_{\alpha\nu} - \frac{1}{2}\nabla_{\nu} R = 0 \iff \nabla_{\mu}\left(R^{\mu\nu} - \frac{1}{2} R g^{\mu\nu}\right) = 0. \tag{1.2.70}$$

This identifies the geometrically conserved **Einstein tensor**:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}, \qquad \nabla^{\mu} G_{\mu\nu} = 0. \tag{1.2.71}$$

Killing vector fields

A vector field ξ^{μ} satisfying the **Killing equation**,

$$\mathcal{L}_{\xi} g_{\mu\nu} = \nabla_{\mu}\xi_{\nu} + \nabla_{\nu}\xi_{\mu} = 0, \tag{1.2.72}$$

is called a **Killing vector field**. Its flow generates an isometry (a continuous spacetime symmetry leaving the metric invariant).

Differentiating the Killing equation and using the Ricci commutator identity (1.2.60) yields the integrability condition:

$$\nabla_{\alpha}\nabla_{\beta}\xi_{\nu} = R^{\mu}{}_{\alpha\beta\nu}\xi_{\mu}. \tag{1.2.73}$$

Thus, a Killing vector field across the entire manifold is uniquely specified by its value $\xi_{\nu}(p)$ (N parameters) and its antisymmetric first derivative $\nabla_{\mu}\xi_{\nu}(p)$ ($N(N-1)/2$ parameters) at a single point p. Consequently, an N-dimensional space can admit at most:

$$N_{\text{Killing}} \leq \frac{N(N+1)}{2} \tag{1.2.74}$$

linearly independent Killing vector fields. In four dimensions ($N = 4$), $N_{\text{Killing}} \leq 10$, matching the 10 parameters of the Poincaré group $\mathrm{ISO}(3,1)$.

Maximally Symmetric Spaces. A spacetime possessing the maximal number of $N(N+1)/2$ Killing vector fields is called a **maximally symmetric space**. For such spaces, isotropy and homogeneity require the Riemann tensor to take the constant-curvature form:

$$R_{\mu\nu\alpha\beta} = \frac{R}{N(N-1)}\left(g_{\mu\alpha}g_{\nu\beta} - g_{\mu\beta}g_{\nu\alpha}\right), \qquad R_{\mu\nu} = \frac{R}{N} g_{\mu\nu}, \tag{1.2.75}$$

where the Ricci scalar R is a spacetime constant. In 4 dimensions, there exist exactly three maximally symmetric spacetimes:

- **Minkowski space** ($R = 0$),
- **de Sitter space** (dS) ($R > 0$, positive constant curvature),
- **anti-de Sitter space** (AdS) ($R < 0$, negative constant curvature).

1.2.5 Covariant approach (1+3 / 3+1 splitting)

We consider a congruence of timelike worldlines representing a family of fundamental observers (or fluid elements of a cosmological medium). Each observer possesses a unit timelike 4-velocity $u^\mu = \mathrm{d}x^\mu \,/\, \mathrm{d}\tau$ normalized as (setting $c = 1$):

$$u^\mu u_\mu = -1. \tag{1.2.76}$$

3+1 decomposition

The 4-velocity u^μ introduces two mutually orthogonal projection tensors:

$$U^\mu{}_\nu \equiv -u^\mu u_\nu, \qquad \gamma_{\mu\nu} \equiv g_{\mu\nu} + u_\mu u_\nu, \tag{1.2.77}$$

projecting parallel to and orthogonal to u^μ, respectively. They satisfy:

$$U^\mu{}_\nu U^\nu{}_\alpha = U^\mu{}_\alpha, \qquad U^\mu{}_\mu = 1, \qquad U^\mu{}_\nu u^\nu = u^\mu, \tag{1.2.78}$$
$$\gamma^\mu{}_\nu \gamma^\nu{}_\alpha = \gamma^\mu{}_\alpha, \qquad \gamma^\mu{}_\mu = 3, \qquad \gamma^\mu{}_\nu u^\nu = 0. \tag{1.2.79}$$

The spacetime line element decomposes naturally into temporal and spatial components:

$$\mathrm{d}s^2 = -\left(u_\mu \,\mathrm{d}x^\mu\right)^2 + \gamma_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu\,. \tag{1.2.80}$$

Any 4-vector v^μ decomposes into parallel and orthogonal parts:

$$v^\mu = v^\nu U^\mu{}_\nu + v^\mu_\perp, \qquad v^\mu_\perp \equiv v^\nu \gamma^\mu{}_\nu. \tag{1.2.81}$$

For a particle of mass m with 4-momentum $p^\mu = mv^\mu$ ($p^\mu p_\mu = -m^2$), the energy measured by observer u^μ is:

$$E = -p_\mu u^\mu \implies E^2 - p^\mu_\perp p_{\mu\perp} = m^2. \tag{1.2.82}$$

If the particle is acted upon by a non-gravitational 4-force F^μ, its equation of motion is $ma^\mu = mv^\nu \nabla_\nu v^\mu = F^\mu$, with orthogonality condition $F^\mu v_\mu = 0$ (derived from $v^\mu v_\mu = -1$).

Spatial and time derivatives

The covariant derivative along the observer's worldline (proper time derivative) is defined by:

$$\dot{T}^{\alpha\beta\ldots}_{\mu\nu\ldots} \equiv u^\lambda \nabla_\lambda T^{\alpha\beta\ldots}_{\mu\nu\ldots}. \tag{1.2.83}$$

The spatial projected covariant derivative in the instantaneous orthogonal rest space is defined by:

$$\bar{\nabla}_\lambda T^{\alpha\beta\ldots}_{\mu\nu\ldots} \equiv \gamma^{\lambda_1}{}_\lambda \gamma^\alpha{}_{\alpha_1} \gamma^\beta{}_{\beta_1} \ldots \gamma^{\mu_1}{}_\mu \gamma^{\nu_1}{}_\nu \ldots \nabla_{\lambda_1} T^{\alpha_1\beta_1\ldots}_{\mu_1\nu_1\ldots}, \tag{1.2.84}$$

which is metric-compatible with the spatial metric ($\bar{\nabla}_\alpha \gamma_{\mu\nu} = 0$). For a scalar field ρ, the commutator $[\bar{\nabla}_\mu, \bar{\nabla}_\nu]\rho = u_{[\mu;\nu]}\dot{\rho}$ vanishes if and only if the flow is irrotational ($u_{[\mu;\nu]} = 0$).

Distortion of the geodesic flow (Fluid Kinematics)

The covariant gradient of the 4-velocity field u^μ can be decomposed into its irreducible kinematic parts:

$$\nabla_\mu u_\nu = -u_\mu \dot{u}_\nu + \frac{1}{3}\Theta\gamma_{\mu\nu} + \sigma_{\mu\nu} + \omega_{\mu\nu}. \tag{1.2.85}$$

- **Acceleration vector $\dot{u}^\nu \equiv u^\mu \nabla_\mu u^\nu$:** Vanishes if the fluid elements follow geodesics ($\dot{u}^\nu = 0$).
- **Expansion scalar $\Theta \equiv \nabla_\mu u^\mu$:** Quantifies the isotropic volume expansion rate of the flow.

- **Shear tensor** $\sigma_{\mu\nu} \equiv \left[\gamma^{\alpha}{}_{(\mu}\gamma^{\beta}{}_{\nu)} - \frac{1}{3}\Theta\gamma_{\mu\nu}\right]\nabla_{\alpha}u_{\beta}$**:** Symmetric ($\sigma_{\mu\nu} = \sigma_{\nu\mu}$), traceless ($\sigma^{\mu}{}_{\mu} = 0$), and purely spatial ($\sigma_{\mu\nu}u^{\mu} = 0$). It describes volume-preserving anisotropic shearing, with shear scalar $\sigma^2 \equiv \frac{1}{2}\sigma_{\mu\nu}\sigma^{\mu\nu}$.

- **Vorticity tensor** $\omega_{\mu\nu} \equiv \gamma^{\alpha}{}_{[\mu}\gamma^{\beta}{}_{\nu]}\nabla_{\alpha}u_{\beta}$**:** Antisymmetric ($\omega_{\mu\nu} = -\omega_{\nu\mu}$) and purely spatial ($\omega_{\mu\nu}u^{\mu} = 0$). It characterizes the local rotation of the flow, with vorticity vector $\omega^{\sigma} \equiv \frac{1}{2}u_{\alpha}\epsilon^{\alpha\sigma\mu\nu}\omega_{\mu\nu} \iff \omega_{\mu\nu} = u^{\beta}\epsilon_{\beta\sigma\mu\nu}\omega^{\sigma}$ and vorticity scalar $\omega^2 \equiv \frac{1}{2}\omega_{\mu\nu}\omega^{\mu\nu}$.

A representative length scale $S(\tau)$ of the fluid element is defined along each worldline by:

$$\frac{\dot{S}}{S} \equiv \frac{1}{3}\Theta, \tag{1.2.86}$$

such that the comoving volume of the fluid element scales as $V \propto S^3$.

The Raychaudhuri equation

Evaluating the proper time derivative of the expansion $\dot{\Theta} = u^{\nu}\nabla_{\nu}(\nabla_{\mu}u^{\mu})$ using the kinematic decomposition (1.2.85) and the Ricci identity (1.2.60) yields the **Raychaudhuri equation**:

$$\dot{\Theta} = -\frac{1}{3}\Theta^2 + \dot{u}^{\mu}\dot{u}_{\mu} + 2\left(\omega^2 - \sigma^2\right) + \bar{\nabla}_{\mu}\dot{u}^{\mu} - R_{\mu\nu}u^{\mu}u^{\nu}. \tag{1.2.87}$$

This fundamental equation governs the focusing of timelike geodesic congruences and gravitational collapse.

Singularity Theorems and Energy Conditions. For an irrotational ($\omega = 0$), geodesic ($\dot{u}^{\mu} = 0$) flow satisfying the **Strong Energy Condition** ($R_{\mu\nu}u^{\mu}u^{\nu} \geq 0$), Equation (1.2.87) reduces to the inequality:

$$\dot{\Theta} + \frac{1}{3}\Theta^2 \leq 0 \implies \frac{1}{\Theta(\tau)} \geq \frac{1}{\Theta_0} + \frac{\tau}{3}. \tag{1.2.88}$$

- If $\Theta_0 < 0$ (converging flow), $\Theta(\tau) \to -\infty$ within a finite proper time $\tau \leq 3/|\Theta_0|$, signaling the formation of a caustic or gravitational singularity.

- If $\Theta_0 > 0$ (expanding Universe), expressing the equation in terms of the scale factor $S(\tau)$ gives $\ddot{S}/S \leq 0$, proving the existence of an initial singularity in the past ($S \to 0$) at $\tau_0 \leq 3/\Theta_0$ (the Big Bang singularity).

Table 1.1: Standard Energy Conditions in General Relativity

Name	Mathematical Definition	Physical Meaning
Null (NEC)	$G_{\mu\nu}k^{\mu}k^{\nu} \geq 0 \quad (\forall k^{\mu}, k^{\mu}k_{\mu} = 0)$	Light rays are focused by matter
Weak (WEC)	$G_{\mu\nu}u^{\mu}u^{\nu} \geq 0 \quad (\forall u^{\mu}, u^{\mu}u_{\mu} = -1)$	Energy density is non-negative for any observer
Strong (SEC)	$R_{\mu\nu}u^{\mu}u^{\nu} \geq 0 \quad (\forall u^{\mu}, u^{\mu}u_{\mu} = -1)$	Gravity is universally attractive ($\ddot{S} < 0$)
Dominant (DEC)	Causal energy flux: $G^{\mu}{}_{\nu}u^{\nu}$ is time-like/null	Energy does not propagate faster than light

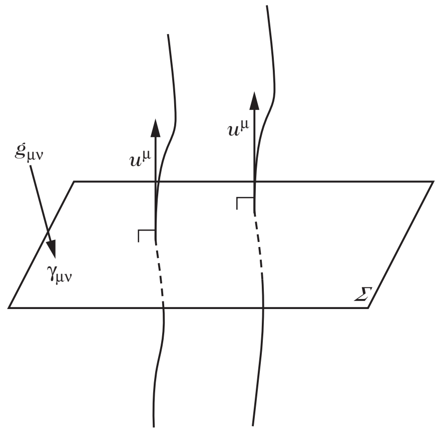


Figure 1.7: A vector flow u^μ defines an orthogonal surface at each point of the flow. A global notion of space and time exists if and only if the flow vorticity vanishes.

Case of vanishing vorticity ($\omega = 0$)

When the vorticity vanishes ($\omega = 0$), Frobenius' theorem guarantees that the instantaneous local rest spaces integrate globally into a foliation of spatial hypersurfaces Σ_t orthogonal to u^μ (Figure 1.7).

The **extrinsic curvature tensor** $K_{\mu\nu}$ measures the bending of Σ in spacetime $\mathcal{M}$:

$$K_{\mu\nu} \equiv \gamma^\alpha{}_\mu \nabla_\alpha u_\nu = \frac{1}{2}\mathcal{L}_u \gamma_{\mu\nu} = \frac{1}{3}\Theta\gamma_{\mu\nu} + \sigma_{\mu\nu}. \tag{1.2.89}$$

Projecting the spacetime Riemann tensor onto Σ yields the **Gauss relation**:

$$^{(3)}R^\alpha{}_{\beta\mu\nu} = \gamma^{\alpha'}{}_\alpha \gamma^{\beta'}{}_\beta \gamma^{\mu'}{}_\mu \gamma^\nu{}_{\nu'} R^{\alpha'}{}_{\beta'\mu'\nu'} - K^\alpha{}_\mu K^\beta{}_\nu + K^\beta{}_\mu K^\alpha{}_\nu. \tag{1.2.90}$$

Contracting and taking the trace yields the **scalar Gauss relation**:

$$^{(3)}R + K^2 - K^{\alpha\beta}K_{\alpha\beta} = R + 2R_{\mu\nu}u^\mu u^\nu = 2G_{\mu\nu}u^\mu u^\nu, \tag{1.2.91}$$

which, expressed in terms of kinematic variables, gives:

$$^{(3)}R = -\frac{2}{3}\Theta^2 + 2\sigma^2 + 2G_{\mu\nu}u^\mu u^\nu. \tag{1.2.92}$$

Projecting once along the normal u^μ yields the **Codazzi relation**:

$$\gamma^\mu{}_\rho u^\sigma \gamma^{\alpha'}{}_\alpha \gamma^{\beta'}{}_\beta R^\rho{}_{\sigma\alpha'\beta'} = \bar{\nabla}_\beta K^\mu{}_\alpha - \bar{\nabla}_\alpha K^\mu{}_\beta, \tag{1.2.93}$$

and its contracted form:

$$\gamma^\mu{}_\alpha u^\nu R_{\mu\nu} = \bar{\nabla}_\mu K^\mu{}_\alpha - \bar{\nabla}_\alpha K. \tag{1.2.94}$$

Decomposition of rank-2 tensors

Relative to the timelike flow u^μ, any symmetric rank-2 tensor $T_{\mu\nu}$ (such as the stress-energy tensor) decomposes uniquely into irreducible components:

$$T_{\mu\nu} = Au_\mu u_\nu + 2q_{(\mu}u_{\nu)} + B\gamma_{\mu\nu} + \pi_{\mu\nu}, \tag{1.2.95}$$

where:

- $A = T_{\mu\nu}u^\mu u^\nu$ is the energy density measured by observer u^μ,

- $q_\mu \equiv -\gamma^\alpha{}_\mu T_{\alpha\beta} u^\beta$ is the spatial energy flux vector ($q_\mu u^\mu = 0$),
- $B = \frac{1}{3} T_{\mu\nu}\gamma^{\mu\nu}$ is the isotropic pressure,
- $\pi_{\mu\nu} \equiv \left[\gamma^\alpha{}_{(\mu}\gamma^\beta{}_{\nu)} - \frac{1}{3}\gamma_{\mu\nu}\gamma^{\alpha\beta}\right] T_{\alpha\beta}$ is the symmetric, traceless, spatial anisotropic stress tensor ($\pi_{\mu\nu}u^\mu = 0, \pi^\mu{}_\mu = 0$).

1.3 Equations of motion

1.3.1 Einstein's equations

In analogy with electrodynamics, we seek second-order partial differential field equations for the metric tensor $g_{\mu\nu}$, derived from an action containing at most first derivatives of the metric in the Lagrangian.

The simplest action satisfying these criteria is the **Einstein–Hilbert action**:

$$S = \frac{1}{2\kappa}\int (R - 2\Lambda)\sqrt{-g}\,\mathrm{d}^4x + \int \mathcal{L}_{\mathrm{mat}}\sqrt{-g}\,\mathrm{d}^4x, \tag{1.3.1}$$

where κ is the gravitational coupling constant (to be determined by matching the Newtonian weak-field limit), Λ is the cosmological constant, and $\mathcal{L}_{\mathrm{mat}}$ is the Lagrangian density of matter fields.

Variation of the Einstein–Hilbert action

Varying the gravitational action with respect to the inverse metric $g^{\mu\nu}$:

$$\begin{aligned}\delta\int R\sqrt{-g}\,\mathrm{d}^4x &= \delta\int R_{\mu\nu}g^{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x \\ &= \int R_{\mu\nu}\delta g^{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x + \int R\delta\sqrt{-g}\,\mathrm{d}^4x + \int g^{\mu\nu}\delta R_{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x.\end{aligned} \tag{1.3.2}$$

Using the determinant variation identity $\delta g = -g g_{\mu\nu}\delta g^{\mu\nu}$, the second term gives:

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu} \implies \int R\delta\sqrt{-g}\,\mathrm{d}^4x = -\frac{1}{2}\int R g_{\mu\nu}\delta g^{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x. \tag{1.3.3}$$

For the third term, the variation of the Ricci tensor (Palatini identity) evaluated in a local Riemann normal coordinate frame (where $\Gamma^\alpha_{\mu\nu} = 0$) is:

$$\delta R_{\mu\nu} = \nabla_\sigma(\delta\Gamma^\sigma_{\mu\nu}) - \nabla_\mu(\delta\Gamma^\sigma_{\sigma\nu}). \tag{1.3.4}$$

Because the difference between two connections $\delta\Gamma^\sigma_{\mu\nu}$ is a genuine rank-$(1,2)$ tensor, this expression holds in any coordinate system. Contracting with $g^{\mu\nu}$ gives:

$$g^{\mu\nu}\delta R_{\mu\nu}\sqrt{-g} = \sqrt{-g}\nabla_\sigma\left(g^{\mu\nu}\delta\Gamma^\sigma_{\mu\nu} - g^{\mu\sigma}\delta\Gamma^\rho_{\rho\mu}\right) = \partial_\sigma\left[\sqrt{-g}\left(g^{\mu\nu}\delta\Gamma^\sigma_{\mu\nu} - g^{\mu\sigma}\delta\Gamma^\rho_{\rho\mu}\right)\right], \tag{1.3.5}$$

which is a pure boundary divergence that vanishes under standard boundary conditions ($\delta g^{\mu\nu}|_{\partial V} = 0$).

Combining these results, the variation of the gravitational action yields:

$$\delta\int (R - 2\Lambda)\sqrt{-g}\,\mathrm{d}^4x = \int (G_{\mu\nu} + \Lambda g_{\mu\nu})\,\delta g^{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x. \tag{1.3.6}$$

The stress-energy tensor

The variation of the matter action with respect to the metric defines the **stress-energy tensor** $T_{\mu\nu}$:

$$\delta \int \mathcal{L}_{\rm mat}\sqrt{-g}\,\mathrm{d}^4x \equiv -\frac{1}{2}\int T_{\mu\nu}\delta g^{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x, \tag{1.3.7}$$

yielding the definition:

$$T_{\mu\nu} \equiv -\frac{2}{\sqrt{-g}}\frac{\delta\left(\mathcal{L}_{\rm mat}\sqrt{-g}\right)}{\delta g^{\mu\nu}} = g_{\mu\nu}\mathcal{L}_{\rm mat} - 2\frac{\delta\mathcal{L}_{\rm mat}}{\delta g^{\mu\nu}}. \tag{1.3.8}$$

Einstein's field equations

Requiring the total action variation $\delta S = 0$ for arbitrary metric variations $\delta g^{\mu\nu}$ yields the **Einstein Field Equations**:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \tag{1.3.9}$$

where $\kappa \equiv 8\pi G_{\rm N}/c^4$ (or $8\pi G_{\rm N}$ in $c = 1$ units).

1.3.2 Conservation equations

The geometric Bianchi identities $\nabla_\nu G^{\mu\nu} = 0$, combined with metric compatibility $\nabla_\alpha g_{\mu\nu} = 0$ and Einstein's equations (1.3.9), require the covariant conservation of the total stress-energy tensor:

$$\nabla_\nu T^{\mu\nu} = 0. \tag{1.3.10}$$

This fundamental equation governs the dynamics and thermodynamics of all matter and radiation components in curved spacetime.

Covariant 1+3 projection of conservation equations

Decomposing an arbitrary symmetric stress-energy tensor relative to a fluid 4-velocity u^μ ($u^\mu u_\mu = -1$) as in Equation (1.2.95):

$$T_{\mu\nu} = \rho u_\mu u_\nu + P\gamma_{\mu\nu} + 2q_{(\mu}u_{\nu)} + \pi_{\mu\nu}, \tag{1.3.11}$$

where $\rho = T_{\mu\nu}u^\mu u^\nu$ is the energy density, $P = \frac{1}{3}T_{\mu\nu}\gamma^{\mu\nu}$ is the isotropic pressure, $q_\mu = -\gamma^\alpha{}_\mu T_{\alpha\beta}u^\beta$ is the energy flux ($q_\mu u^\mu = 0$), and $\pi_{\mu\nu}$ is the symmetric, traceless anisotropic stress tensor ($\pi_{\mu\nu}u^\mu = 0, \pi^\mu{}_\mu = 0$).

Projecting the conservation equation $\nabla_\mu T^{\mu\nu} = 0$ parallel to u_ν and orthogonally onto the spatial hypersurface via $\gamma^\lambda{}_\nu$ yields two exact dynamical equations:

1. **Energy Conservation Equation (Parallel Projection $u_\nu\nabla_\mu T^{\mu\nu} = 0$):**

$$\dot\rho + \bar\nabla_\mu q^\mu + (\rho + P)\Theta + 2q_\mu\dot u^\mu + \sigma^\mu{}_\nu\pi^\nu{}_\mu = 0. \tag{1.3.12}$$

2. **Momentum Conservation Equation (Orthogonal Projection $\gamma^\lambda{}_\nu\nabla_\mu T^{\mu\nu} = 0$):**

$$\gamma^\lambda{}_\nu\dot q^\nu + \bar\nabla^\lambda P + \bar\nabla_\alpha\pi^{\alpha\lambda} + \frac{4}{3}\Theta q^\lambda + \left(\sigma^\lambda{}_\alpha + \omega^\lambda{}_\alpha\right)q^\alpha + (\rho + P)\dot u^\lambda + \pi^{\alpha\lambda}\dot u_\alpha = 0. \tag{1.3.13}$$

These conservation laws are completely model-independent, valid for any matter content and any spacetime metric.

The perfect fluid

For a **perfect fluid**, dissipative effects vanish ($q^\mu = 0$, $\pi_{\mu\nu} = 0$), so that $T_{\mu\nu} = (\rho+P)u_\mu u_\nu + P g_{\mu\nu}$. Equations (1.3.12) and (1.3.13) simplify to:

$$\dot\rho + (\rho + P)\Theta = 0, \tag{1.3.14}$$
$$(\rho + P)\dot u^\mu + \bar\nabla^\mu P = 0. \tag{1.3.15}$$

Equation (1.3.14) is the **relativistic continuity equation**. Equation (1.3.15) is the **relativistic Euler equation**.

For **pressureless dust** ($P = 0$), the acceleration vanishes ($\dot u^\mu = 0$), proving that dust particles follow geodesics of spacetime.

For an irrotational fluid flow ($\omega = 0$), substituting Einstein's field equations (1.3.9) into the scalar Gauss relation (1.2.92) gives the **generalized Friedmann equation**:

$$^{(3)}R = 2\kappa\rho - \frac{2}{3}\Theta^2 + 2\sigma^2 + 2\Lambda. \tag{1.3.16}$$

In terms of the energy density ρ and pressure P, the standard energy conditions (Table 1.1) translate to:

- **Null Energy Condition (NEC):** $\rho + P \geq 0$.
- **Weak Energy Condition (WEC):** $\rho \geq 0$ and $\rho + P \geq 0$.
- **Strong Energy Condition (SEC):** $\rho + 3P \geq 0$ and $\rho + P \geq 0$.
- **Dominant Energy Condition (DEC):** $\rho \geq |P|$.

Electromagnetism in curved spacetime

Electromagnetism is described by the antisymmetric **Faraday tensor** $F_{\mu\nu} = -F_{\nu\mu}$, defined in terms of the electromagnetic 4-potential A_μ as:

$$F_{\mu\nu} \equiv \nabla_\mu A_\nu - \nabla_\nu A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{1.3.17}$$

The homogeneous Maxwell equations are geometric identities ($3F_{[\mu\nu;\alpha]} = 0$):

$$\nabla_\alpha F_{\mu\nu} + \nabla_\mu F_{\nu\alpha} + \nabla_\nu F_{\alpha\mu} = 0. \tag{1.3.18}$$

The electromagnetic action is:

$$S_{\text{em}} = -\frac{1}{4}\int F_{\mu\nu}F^{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x + \int A_\mu j^\mu \sqrt{-g}\,\mathrm{d}^4x, \tag{1.3.19}$$

yielding the inhomogeneous curved-spacetime Maxwell equations:

$$\nabla_\mu F^{\nu\mu} = j^\nu \iff \frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}F^{\nu\mu}\right) = j^\nu. \tag{1.3.20}$$

In the **Lorenz gauge** ($\nabla_\mu A^\mu = 0$), commuting covariant derivatives yields the wave equation:

$$\Box A^\mu - R^\mu{}_\nu A^\nu = -j^\mu. \tag{1.3.21}$$

The electromagnetic stress-energy tensor obtained from varying Equation (1.3.19) is:

$$T^{\mu\nu}_{\text{em}} = F^\mu{}_\lambda F^{\nu\lambda} - \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}. \tag{1.3.22}$$

Note that $T^{\mu\nu}_{\rm em}$ is identically traceless in 4 dimensions: $T^{\mu}{}_{\mu} = 0$.

An observer with 4-velocity u^μ measures electric and magnetic field vectors:

$$E^\mu \equiv F^{\mu\nu} u_\nu, \qquad B^\mu \equiv -\frac{1}{2}\epsilon^{\mu\nu\rho\sigma} u_\nu F_{\rho\sigma}, \tag{1.3.23}$$

satisfying $E^\mu u_\mu = 0$ and $B^\mu u_\mu = 0$. The Faraday tensor reconstructs as:

$$F_{\mu\nu} = 2u_{[\mu}E_{\nu]} - \epsilon_{\mu\nu\rho\sigma} u^\rho B^\sigma. \tag{1.3.24}$$

In terms of E^μ and B^μ, the electromagnetic stress-energy tensor decomposes as a fluid with:

$$\rho_{\rm em} = \frac{1}{2}\left(E^2 + B^2\right), \qquad P_{\rm em} = \frac{1}{3}\rho_{\rm em}, \qquad q^\mu = \epsilon^{\mu\nu\rho\sigma} u_\nu E_\rho B_\sigma, \tag{1.3.25}$$

and anisotropic stress tensor:

$$\pi_{\mu\nu} = \frac{1}{3}\left(E^2 + B^2\right)\gamma_{\mu\nu} - E_\mu E_\nu - B_\mu B_\nu. \tag{1.3.26}$$

The trajectory of a particle of charge e and mass m is governed by the curved-spacetime Lorentz force:

$$u^\mu \nabla_\mu u^\nu = \frac{e}{m} F^\nu{}_\mu u^\mu = \frac{e}{m} E^\nu. \tag{1.3.27}$$

Projecting Maxwell's equations along u^μ and $\gamma^\lambda{}_\mu$ yields the 1+3 relativistic Maxwell system:

$$\bar\nabla_\mu E^\mu - 2\omega_\mu B^\mu = \epsilon_e, \tag{1.3.28}$$

$$\bar\nabla_\mu B^\mu + 2\omega_\mu B^\mu = 0, \tag{1.3.29}$$

$$\gamma^\lambda{}_\mu \dot E^\mu - \epsilon^{\alpha\lambda\rho\sigma} u_\alpha \bar\nabla_\rho B_\sigma = -\gamma^\lambda{}_\nu j^\nu - \frac{2}{3}\Theta E^\lambda + \sigma^\lambda{}_\nu E^\nu + \epsilon^{\alpha\lambda\rho\sigma} u_\alpha \left(\dot u_\rho B_\sigma + \omega_\rho E_\sigma\right), \tag{1.3.30}$$

$$\gamma^\lambda{}_\mu \dot B^\mu + \epsilon^{\alpha\lambda\rho\sigma} u_\alpha \bar\nabla_\rho E_\sigma = -\frac{2}{3}\Theta B^\lambda + \sigma^\lambda{}_\nu B^\nu - \epsilon^{\alpha\lambda\rho\sigma} u_\alpha \left(\dot u_\rho E_\sigma - \omega_\rho B_\sigma\right), \tag{1.3.31}$$

where $\epsilon_e \equiv -j^\mu u_\mu$ is the comoving charge density.

Scalar fields

A minimally coupled real scalar field ϕ with self-interaction potential $V(\phi)$ has action:

$$S_\phi = -\int \left[\frac{1}{2} g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + V(\phi)\right]\sqrt{-g}\,\mathrm{d}^4x. \tag{1.3.32}$$

Varying with respect to ϕ yields the **Klein–Gordon equation**:

$$\Box\phi - V'(\phi) = 0 \iff \frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}g^{\mu\nu}\partial_\nu\phi\right) - V'(\phi) = 0. \tag{1.3.33}$$

Varying the action with respect to the metric gives the stress-energy tensor:

$$T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu}\left[\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi + V(\phi)\right]. \tag{1.3.34}$$

When $\partial_\mu\phi$ is timelike, the scalar field can be mapped directly to an effective perfect fluid with 4-velocity:

$$u_\mu = -\frac{\partial_\mu\phi}{\psi}, \qquad \psi \equiv \dot\phi = \sqrt{-\partial_\mu\phi\partial^\mu\phi}. \tag{1.3.35}$$

The effective energy density and isotropic pressure are:

$$\rho_\phi = \frac{1}{2}\dot\phi^2 + V(\phi), \qquad P_\phi = \frac{1}{2}\dot\phi^2 - V(\phi). \tag{1.3.36}$$

Notice that when potential energy dominates ($V(\phi) \gg \frac{1}{2}\dot\phi^2$), the equation of state approaches $P_\phi \approx -\rho_\phi$, driving cosmological inflation.

For this scalar fluid, the vorticity vanishes identically ($\omega_{\mu\nu} = 0$), and energy conservation $\dot\rho_\phi + \Theta(\rho_\phi + P_\phi) = 0$ is exactly equivalent to the Klein–Gordon equation:

$$\ddot\phi + \Theta\dot\phi + V'(\phi) = 0. \tag{1.3.37}$$

1.3.3 ADM Hamiltonian formulation of general relativity

The **Arnowitt–Deser–Misner (ADM)** 3+1 Hamiltonian formulation decomposes spacetime into a continuous family of spacelike hypersurfaces Σ_t parameterized by global time coordinate t (valid for globally hyperbolic spacetimes).

The ADM line element is written as:

$$\mathrm{d}s^2 = -N^2\,\mathrm{d}t^2 + \gamma_{ij}\left(N^i\,\mathrm{d}t + \mathrm{d}x^i\right)\left(N^j\,\mathrm{d}t + \mathrm{d}x^j\right), \tag{1.3.38}$$

where:

- $N(t, \mathbf{x})$ is the **lapse function**, measuring the rate of flow of proper time relative to coordinate time t normal to Σ_t,
- $N^i(t, \mathbf{x})$ is the **shift vector**, measuring the spatial coordinate drift between successive spatial hypersurfaces,
- $\gamma_{ij}(t, \mathbf{x})$ is the **induced 3-metric** of the spatial hypersurface Σ_t.

The unit timelike normal to Σ_t is $n_\alpha = (-N, 0, 0, 0)$ with $n^\alpha = \frac{1}{N}(1, -N^i)$, satisfying $n^\alpha n_\alpha = -1$. The 4-metric determinant satisfies:

$$\sqrt{-g} = N\sqrt{\gamma}, \qquad \gamma \equiv \det(\gamma_{ij}). \tag{1.3.39}$$

The extrinsic curvature of Σ_t is:

$$K_{ij} = \frac{1}{2N}\left(D_i N_j + D_j N_i - \dot{\gamma}_{ij}\right), \tag{1.3.40}$$

where D_i denotes the 3-dimensional spatial covariant derivative compatible with γ_{ij}.

In terms of ADM variables, the gravitational action becomes:

$$S_{\text{grav}} = \int \mathrm{d}t \int_{\Sigma_t} N\left({}^{(3)}R + K_{ij}K^{ij} - K^2\right)\sqrt{\gamma}\,\mathrm{d}^3x. \tag{1.3.41}$$

Because the Lagrangian density $\mathcal{L}_{\text{ADM}} = N({}^{(3)}R + K_{ij}K^{ij} - K^2)\sqrt{\gamma}$ contains no time derivatives of N or N^i, the lapse and shift are non-dynamical Lagrange multipliers. The only dynamical variables are the spatial metric components γ_{ij}.

The conjugate momentum tensor to γ_{ij} is:

$$\pi^{ij} \equiv \frac{\partial \mathcal{L}}{\partial \dot{\gamma}_{ij}} = \sqrt{\gamma}\left(K\gamma^{ij} - K^{ij}\right). \tag{1.3.42}$$

Performing a Legendre transformation $H = \int(\pi^{ij}\dot{\gamma}_{ij} - \mathcal{L})\mathrm{d}^3x$, the ADM Hamiltonian takes the form:

$$H = -\int_{\Sigma_t}\left(N\mathcal{C}_0 - 2N^i\mathcal{C}_i\right)\sqrt{\gamma}\,\mathrm{d}^3x, \tag{1.3.43}$$

where varying with respect to N and N^i produces the fundamental **ADM constraint equations**:

$$\mathcal{C}_0 \equiv {}^{(3)}R + K^2 - K_{ij}K^{ij} = 16\pi G_{\text{N}}\rho \qquad \text{(Hamiltonian / Energy constraint)}, \tag{1.3.44}$$

$$\mathcal{C}_i \equiv D_j K^j{}_i - D_i K = 8\pi G_{\text{N}} P_i \qquad \text{(Momentum constraint)}, \tag{1.3.45}$$

with energy density $\rho \equiv T_{\mu\nu}n^\mu n^\nu$ and momentum density $P_i \equiv -\gamma^\mu{}_i T_{\mu\nu} n^\nu$.

1.4 Geodesics in curved space-time

1.4.1 Conserved quantities along geodesics

Whenever a spacetime admits a continuous symmetry generated by a Killing vector field ξ^μ ($\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = 0$), there exists a strictly conserved quantity along any freely falling trajectory:

$$u^\nu \nabla_\nu \left(u^\mu \xi_\mu\right) = \xi_\mu \left(u^\nu \nabla_\nu u^\mu\right) + u^\mu u^\nu \nabla_\nu \xi_\mu = 0. \tag{1.4.1}$$

The first term vanishes because u^μ is geodesic ($u^\nu \nabla_\nu u^\mu = 0$), and the second vanishes by contracting the symmetric tensor $u^\mu u^\nu$ with the antisymmetric Killing tensor $\nabla_{(\mu}\xi_{\nu)} = 0$.

Similarly, for any conserved stress-energy tensor $\nabla_\mu T^{\mu\nu} = 0$, the current $J^\mu \equiv T^{\mu\nu}\xi_\nu$ is covariantly conserved:

$$\nabla_\mu \left(T^{\mu\nu}\xi_\nu\right) = \xi_\nu \nabla_\mu T^{\mu\nu} + T^{\mu\nu}\nabla_\mu \xi_\nu = 0. \tag{1.4.2}$$

The corresponding conserved charge integrated over a spatial volume V is:

$$Q \equiv \int_V T^{0\nu}\xi_\nu \sqrt{-g}\, \mathrm{d}^3 x, \qquad \frac{\mathrm{d}Q}{\mathrm{d}t} = 0. \tag{1.4.3}$$

1.4.2 Geodesic deviation and tidal forces

Consider a 1-parameter family of neighboring geodesics $x^\mu(\lambda, s)$, parameterized by affine parameter λ along the curves and labeled by s. The separation vector between adjacent geodesics is $n^\mu \equiv \frac{\partial x^\mu}{\partial s}$, satisfying the commutation relation:

$$u^\mu \nabla_\mu n^\nu = n^\mu \nabla_\mu u^\nu. \tag{1.4.4}$$

The relative acceleration between neighboring geodesics is $a^\mu \equiv \frac{\mathrm{D}^2 n^\mu}{\mathrm{D}\lambda^2} = u^\alpha \nabla_\alpha (u^\beta \nabla_\beta n^\mu)$. Using Equation (1.4.4) and the Ricci identity for the commutator of covariant derivatives, we obtain the **equation of geodesic deviation** (Jacobi equation):

$$\frac{\mathrm{D}^2 n^\mu}{\mathrm{D}\lambda^2} = R^\mu{}_{\nu\alpha\beta} u^\nu u^\alpha n^\beta = -R^\mu{}_{\nu\beta\alpha} u^\nu n^\beta u^\alpha. \tag{1.4.5}$$

This rigorous derivation confirms the physical insight that space-time curvature is directly detectable via the relative acceleration (tidal deviation) of freely falling test bodies.

Covariant kinematic deviation and the generalized Hubble law

Decomposing the separation vector as $n^\mu = \ell e^\mu$, with unit spatial direction $e^\mu e_\mu = 1$ and $e^\mu u_\mu = 0$, evaluating the rate of change of separation ℓ along the flow yields the **generalized Hubble law**:

$$\frac{\dot{\ell}}{\ell} = \frac{1}{3}\Theta + \sigma_{\mu\nu} e^\mu e^\nu, \tag{1.4.6}$$

while the orthogonal projection describes the drift in the apparent line-of-sight observation direction:

$$\gamma^\mu{}_\lambda \dot{e}^\lambda = \left[\sigma^\mu{}_\lambda - \left(\sigma_{\alpha\beta} e^\alpha e^\beta\right)\gamma^\mu{}_\lambda - \omega^\mu{}_\lambda\right] e^\lambda. \tag{1.4.7}$$

Geometric optics and cosmological redshift

In the high-frequency geometric optics approximation, an electromagnetic wave is represented as $A_\mu = C_\mu e^{i\phi}$. The wavevector $k_\mu \equiv \partial_\mu \phi$ satisfies:

$$k_\mu k^\mu = 0, \qquad k^\mu \nabla_\mu k^\nu = 0, \qquad C_\mu k^\mu = 0. \tag{1.4.8}$$

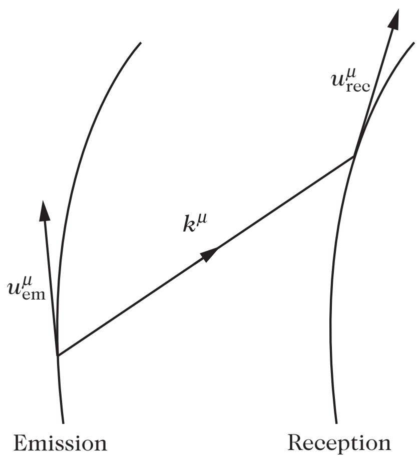


Figure 1.8: A photon emitted with wavevector k^μ by an observer with 4-velocity $u^\mu_{\rm em}$ travels along a null geodesic to an observer with 4-velocity $u^\mu_{\rm rec}$.

Light propagates along **null geodesics** of the spacetime metric.

An observer with 4-velocity u^μ measures a photon energy and frequency $E = \hbar\omega = -k_\mu u^\mu$. If a photon is emitted at spacetime event em and received at event rec (Figure 1.8), the **cosmological redshift** z is defined by:

$$1 + z \equiv \frac{\lambda_{\rm rec}}{\lambda_{\rm em}} = \frac{\omega_{\rm em}}{\omega_{\rm rec}} = \frac{(k_\mu u^\mu)_{\rm em}}{(k_\mu u^\mu)_{\rm rec}}. \tag{1.4.9}$$

Decomposing the null wavevector as $k^\mu = E(u^\mu + e^\mu) = \frac{\mathrm{d}x^\mu}{\mathrm{d}p}$ (where p is an affine parameter along the photon trajectory, $e^\mu e_\mu = 1$, and $e^\mu u_\mu = 0$), the rate of change of wavelength along the ray is:

$$\frac{1}{\lambda}\frac{\mathrm{d}\lambda}{\mathrm{d}p} = -\frac{1}{E}\frac{\mathrm{d}E}{\mathrm{d}p} = \left(\frac{1}{3}\Theta + \dot{u}_\mu e^\mu + \sigma_{\mu\nu}e^\mu e^\nu\right) E. \tag{1.4.10}$$

This exact formula reveals that cosmological redshift receives isotropic contributions from volume expansion Θ, kinematic Doppler effects from observer acceleration $\dot{u}_\mu$, and anisotropic shifts from shear $\sigma_{\mu\nu}$.

1.5 The weak-field regime

1.5.1 The Newtonian limit

To determine the gravitational coupling constant κ in Einstein's field equations (1.3.9), we consider the weak-field limit, where spacetime is a small perturbation around flat Minkowski spacetime:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1. \tag{1.5.1}$$

Linearized Einstein equations

To first order in the metric perturbation $h_{\mu\nu}$, the inverse metric is $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$, where indices are raised and lowered using the background Minkowski metric $\eta_{\mu\nu}$. The trace is $h \equiv h^\alpha{}_\alpha = \eta^{\alpha\beta}h_{\alpha\beta}$.

The linearized Christoffel symbols are:

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2}\eta^{\alpha\lambda}\left(\partial_\mu h_{\lambda\nu} + \partial_\nu h_{\lambda\mu} - \partial_\lambda h_{\mu\nu}\right). \tag{1.5.2}$$

The linearized Ricci tensor and scalar curvature are:

$$R_{\mu\nu} = \partial_\alpha \Gamma^\alpha_{\mu\nu} - \partial_\mu \Gamma^\alpha_{\alpha\nu} = \frac{1}{2}\left(2\partial^\lambda \partial_{(\mu} h_{\nu)\lambda} - \Box h_{\mu\nu} - \partial_\mu \partial_\nu h\right), \tag{1.5.3}$$

$$R = \partial_\alpha \partial_\beta h^{\alpha\beta} - \Box h, \tag{1.5.4}$$

where $\Box \equiv \eta^{\alpha\beta}\partial_\alpha\partial_\beta = -\partial_0^2 + \nabla^2$ is the flat-space d'Alembertian.

Introducing the **trace-reversed metric perturbation**:

$$\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2} h \eta_{\mu\nu} \iff \bar{h} = -h, \tag{1.5.5}$$

the linearized Einstein equations take the form:

$$\Box \bar{h}_{\mu\nu} + \eta_{\mu\nu}\partial_\alpha\partial_\beta \bar{h}^{\alpha\beta} - \partial^\lambda \partial_\mu \bar{h}_{\nu\lambda} - \partial^\lambda \partial_\nu \bar{h}_{\mu\lambda} = -2\kappa T_{\mu\nu}. \tag{1.5.6}$$

Gauge freedom and the harmonic gauge

Under an infinitesimal coordinate transformation $x^\mu \to x'^\mu = x^\mu + \xi^\mu(x)$, the metric perturbation transforms as:

$$h_{\mu\nu} \longrightarrow h'_{\mu\nu} = h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu \implies \bar{h}_{\mu\nu} \longrightarrow \bar{h}'_{\mu\nu} = \bar{h}_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu + \eta_{\mu\nu}\partial_\alpha \xi^\alpha. \tag{1.5.7}$$

We can always choose the gauge parameter ξ^μ satisfying $\Box \xi^\mu = \partial_\nu \bar{h}^{\mu\nu}$ to enforce the **harmonic gauge** (also known as the **de Donder gauge** or Lorenz gauge):

$$\partial^\nu \bar{h}_{\mu\nu} = 0. \tag{1.5.8}$$

In this gauge, the linearized Einstein field equations simplify to the decoupled wave equations:

$$\Box \bar{h}_{\mu\nu} = -2\kappa T_{\mu\nu}. \tag{1.5.9}$$

Retarded potential solutions

Equation (1.5.9) is solved using the standard retarded Green's function for the d'Alembertian operator:

$$\bar{h}_{\mu\nu}(\mathbf{x}, t) = \frac{\kappa}{2\pi} \int \frac{T_{\mu\nu}\left(\mathbf{x}', t - |\mathbf{x} - \mathbf{x}'|/c\right)}{|\mathbf{x} - \mathbf{x}'|} \mathrm{d}^3 x'. \tag{1.5.10}$$

The general solution is $\bar{h}_{\mu\nu} = \bar{h}^{(\mathrm{ret})}_{\mu\nu} + \bar{h}^{(\mathrm{hom})}_{\mu\nu}$, where $\bar{h}^{(\mathrm{hom})}_{\mu\nu}$ satisfies the homogeneous vacuum wave equation $\Box \bar{h}^{(\mathrm{hom})}_{\mu\nu} = 0$, describing freely propagating **gravitational waves**.

Weak-field geodesic motion and determination of κ

Consider the trajectory of a non-relativistic massive test body ($v \ll c$) in a weak, static gravitational field. The 4-velocity is $u^\mu = \gamma^{-1}(c, v^i)$, where $v^i \equiv \frac{\mathrm{d}x^i}{\mathrm{d}t}$ and $\gamma = \frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{1 - v^2/c^2}$.

To linear order in the metric perturbation $h_{\mu\nu}$ and velocity v/c, the spatial component of the geodesic equation (1.2.35) reduces to:

$$\frac{\mathrm{d}^2 x^i}{\mathrm{d}t^2} = -c^2 \left[\Gamma^i_{00} + \left(\Gamma^0_{00}\delta^i_j - \frac{2}{c}\Gamma^i_{0j}\right) v^j\right]. \tag{1.5.11}$$

For a static field ($\partial_0 h_{\mu\nu} = 0$), the only non-vanishing Christoffel symbol at leading order is $\Gamma^i_{00} = -\frac{1}{2}\delta^{ij}\partial_j h_{00}$, yielding:

$$\frac{\mathrm{d}^2 x^i}{\mathrm{d}t^2} = \frac{c^2}{2}\delta^{ij}\partial_j h_{00}. \tag{1.5.12}$$

Comparing this directly with Newton's second law in a gravitational potential Φ_N, $\frac{\mathrm{d}^2x^i}{\mathrm{d}t^2} = -\delta^{ij}\partial_j\Phi_\mathrm{N}$, we identify:

$$h_{00} = -\frac{2\Phi_\mathrm{N}}{c^2} \implies g_{00} = -\left(1 + \frac{2\Phi_\mathrm{N}}{c^2}\right), \qquad \bar{h}_{00} = -\frac{4\Phi_\mathrm{N}}{c^2}. \tag{1.5.13}$$

For a non-relativistic pressureless source ($P \ll \rho c^2$), the dominant stress-energy component is $T_{00} \approx \rho c^2$. In the static limit, the 00-component of the linearized Einstein equation (1.5.9) becomes:

$$\nabla^2\bar{h}_{00} = -2\kappa T_{00} \implies -\frac{4}{c^2}\nabla^2\Phi_\mathrm{N} = -2\kappa\rho c^2 \iff \nabla^2\Phi_\mathrm{N} = \frac{1}{2}\kappa c^4\rho. \tag{1.5.14}$$

Comparing with Newton's Poisson equation $\nabla^2\Phi_\mathrm{N} = 4\pi G_\mathrm{N}\rho$, we determine the fundamental coupling constant of general relativity:

$$\kappa = \frac{8\pi G_\mathrm{N}}{c^4}. \tag{1.5.15}$$

Gravito-electromagnetism

The linearized Einstein equations display a striking structural analogy with Maxwell's electrodynamics. Defining the gravito-magnetic vector potential $A_i \equiv c\,\bar{h}_{0i}$, the non-relativistic geodesic equation takes the form of a Lorentz force:

$$\mathbf{a} = -\boldsymbol{\nabla}\Phi_\mathrm{N} - \frac{1}{c}\frac{\partial\mathbf{A}}{\partial t} + \mathbf{v}\times(\boldsymbol{\nabla}\times\mathbf{A}) + \frac{3}{c^2}\frac{\partial\Phi_\mathrm{N}}{\partial t}\mathbf{v}. \tag{1.5.16}$$

Defining the gravito-electric field $\mathbf{g} \equiv -\boldsymbol{\nabla}\Phi_\mathrm{N} - \frac{1}{c}\partial_t\mathbf{A}$ and gravito-magnetic field $\mathbf{B}_\mathrm{g} \equiv \boldsymbol{\nabla}\times\mathbf{A}$, the linearized field equations mirror Maxwell's equations:

$$\boldsymbol{\nabla}\cdot\mathbf{B}_\mathrm{g} = 0, \qquad \boldsymbol{\nabla}\times\mathbf{g} = -\frac{1}{c}\frac{\partial\mathbf{B}_\mathrm{g}}{\partial t}, \tag{1.5.17}$$

$$\boldsymbol{\nabla}\cdot\mathbf{g} = -4\pi G_\mathrm{N}\rho, \qquad \boldsymbol{\nabla}\times\mathbf{B}_\mathrm{g} = -\frac{16\pi G_\mathrm{N}}{c^2}\mathbf{j}_m + \frac{1}{c}\frac{\partial\mathbf{g}}{\partial t}, \tag{1.5.18}$$

where $\mathbf{j}_m \equiv \rho\mathbf{v}$ is the mass-current density.

Table 1.2: Analogy between Electromagnetism and Linearized General Relativity

Property	**Electromagnetism**	**Linearized Gravity**
Potential Field	$A^\mu = (\Phi/c, \mathbf{A})$	$\bar{h}^{\mu\nu}$
Gauge Transformation	$A_\mu \to A_\mu + \partial_\mu\chi$	$\bar{h}_{\mu\nu} \to \bar{h}_{\mu\nu} - 2\partial_{(\mu}\xi_{\nu)} + \eta_{\mu\nu}\partial_\alpha\xi^\alpha$
Gauge Condition	$\partial_\mu A^\mu = 0$ (Lorenz)	$\partial_\mu\bar{h}^{\mu\nu} = 0$ (Harmonic / de Donder)
Wave Equations	$\Box A^\mu = -\mu_0 j^\mu$	$\Box\bar{h}^{\mu\nu} = -2\kappa T^{\mu\nu}$
Residual Gauge Invariance	$\Box\chi = 0$	$\Box\xi^\mu = 0$

Scalar-Vector-Tensor (SVT) decomposition

To analyze cosmological perturbations systematically, we decompose the metric perturbation $h_{\mu\nu}$ into its irreducible spatial components under spatial rotations SO(3):

$$h_{00} = -2A, \tag{1.5.19}$$

$$h_{0i} = \partial_i B + \bar{B}_i, \tag{1.5.20}$$

$$h_{ij} = 2C\delta_{ij} + 2\partial_i\partial_j E + 2\partial_{(i}\bar{E}_{j)} + \bar{E}_{ij}, \tag{1.5.21}$$

where $\bar{B}_i$ and $\bar{E}_i$ are transverse divergence-free vector fields ($\partial_i \bar{B}^i = 0, \partial_i \bar{E}^i = 0$), and $\bar{E}_{ij}$ is a symmetric, transverse, and traceless (TT) tensor field ($\partial^i \bar{E}_{ij} = 0, \bar{E}^i{}_i = 0$).

The 10 independent components of $h_{\mu\nu}$ decompose into:

- **4 scalar degrees of freedom:** A, B, C, E,
- **4 vector degrees of freedom (2 transverse vectors):** $\bar{B}_i, \bar{E}_i$,
- **2 tensor degrees of freedom (1 TT tensor):** $\bar{E}_{ij}$.

Under an infinitesimal coordinate gauge transformation $\xi^0 = -T$ and $\xi^i = -\partial^i L - \bar{L}^i$ (with $\partial_i \bar{L}^i = 0$), the metric perturbations transform as:

$$A \longrightarrow A + \dot{T}, \qquad B \longrightarrow B + \dot{L} - T, \qquad C \longrightarrow C, \qquad E \longrightarrow E + L, \tag{1.5.22}$$

$$\bar{B}_i \longrightarrow \bar{B}_i + \dot{\bar{L}}_i, \qquad \bar{E}_i \longrightarrow \bar{E}_i + \bar{L}_i, \qquad \bar{E}_{ij} \longrightarrow \bar{E}_{ij}. \tag{1.5.23}$$

The tensor perturbations $\bar{E}_{ij}$ are automatically **gauge-invariant**.

Constructing gauge-invariant combinations for the scalar and vector sectors:

$$\Phi \equiv A + \dot{B} - \ddot{E} \qquad \text{(Bardeen scalar potential)}, \tag{1.5.24}$$

$$\Psi \equiv -C \qquad \text{(Bardeen scalar potential)}, \tag{1.5.25}$$

$$\bar{\Phi}_i \equiv \dot{\bar{E}}_i - \bar{B}_i \qquad \text{(Gauge-invariant vector perturbation)}. \tag{1.5.26}$$

This leaves 2(scalar) + 2(vector) + 2(tensor) = **6** physical degrees of freedom after fixing the 4 coordinate gauge choices.

1.5.2 Gravitational waves in vacuum

In vacuum ($T_{\mu\nu} = 0$), the linearized Einstein field equations decouple into:

$$\nabla^2 \Psi = 0, \qquad \partial_i \dot{\Psi} = 0, \qquad \Phi - \Psi = 0 \implies \Phi = \Psi = 0, \tag{1.5.27}$$

$$\nabla^2 \bar{\Phi}_i = 0 \implies \bar{\Phi}_i = 0. \tag{1.5.28}$$

Thus, **no scalar or vector gravitational waves can propagate in flat spacetime**.

The only propagating gravitational degrees of freedom are the tensor modes, satisfying the transverse-traceless wave equation:

$$\Box \bar{E}_{ij} = 0, \qquad \bar{E}^i{}_i = 0, \qquad \partial^i \bar{E}_{ij} = 0. \tag{1.5.29}$$

In the **Transverse-Traceless (TT) gauge**, the only non-vanishing metric perturbations are $h_{ij}^{\mathrm{TT}} = \bar{E}_{ij}$.

Polarization modes of gravitational waves

For a monochromatic plane gravitational wave propagating in the $+z$ direction with frequency $\omega = ck$, the general solution in the TT gauge is a linear combination of two orthogonal polarization states:

$$h_{ij}^{\mathrm{TT}}(t, z) = E_+(ct - z) e_{ij}^+ + E_\times(ct - z) e_{ij}^\times, \tag{1.5.30}$$

with polarization tensors:

$$e_{ij}^+ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad e_{ij}^\times = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{1.5.31}$$

Effect on test particles

Consider a ring of test particles in the xy-plane at $z = 0$. The relative tidal acceleration governed by the geodesic deviation equation (1.4.5) is:

$$\frac{\mathrm{d}^2 n^i}{\mathrm{d}t^2} = \frac{1}{2}\frac{\partial^2 h_{ij}^{\mathrm{TT}}}{\partial t^2} n^j. \tag{1.5.32}$$

Integrating Equation (1.5.32) over one wave period $T = 2\pi/\omega$ shows that the $+$ polarization stretches the ring along the x-axis while compressing it along the y-axis (and vice versa half a cycle later), whereas the $\times$ polarization produces identical quadrupolar deformations rotated by 45° (Figure 1.9).

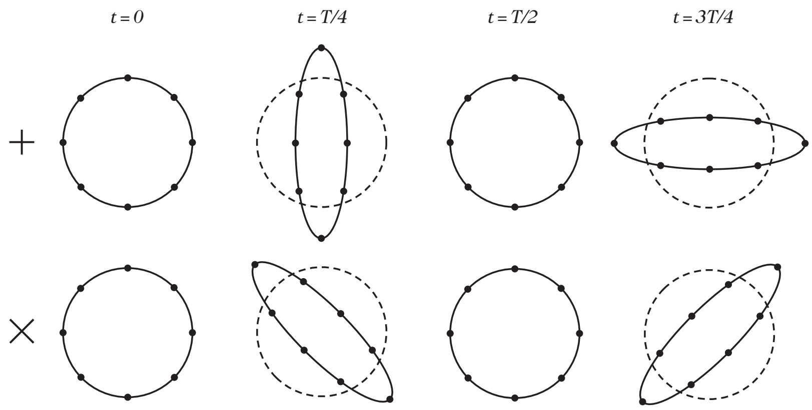


Figure 1.9: Deformation of a ring of test particles in the xy-plane caused by a gravitational plane wave propagating along the z-axis for the $+$ (top) and $\times$ (bottom) polarization states across four phases of a wave period T.

1.6 Experimental tests of general relativity

Because cosmological models rely on extrapolating general relativity to the entire observable Universe, it is vital to scrutinize its experimental foundations and empirical precision.

1.6.1 Tests of the Equivalence Principle

Universality of free fall (Weak Equivalence Principle)

The mass of any body receives contributions from different physical interactions (rest mass, electromagnetic binding energy, strong nuclear interactions, and gravitational self-energy). If interaction i with binding energy E_i couples differently to gravitational mass m_{G} than to inertial mass m_{I}:

$$m_{\mathrm{G}} = m_{\mathrm{I}} + \sum_i \eta_i \frac{E_i}{c^2}, \tag{1.6.1}$$

the acceleration in an external gravitational field $\mathbf{g}$ becomes $\mathbf{a} = \left(1 + \sum_i \eta_i \frac{E_i}{m_{\mathrm{I}} c^2}\right)\mathbf{g}$.

The difference in acceleration between two bodies A and B is quantified by the **Eötvös parameter** η:

$$\eta \equiv \frac{2|\mathbf{a}_A - \mathbf{a}_B|}{|\mathbf{a}_A + \mathbf{a}_B|} = \sum_i \eta_i \left[\left(\frac{E_i}{m_{\mathrm{I}} c^2}\right)_A - \left(\frac{E_i}{m_{\mathrm{I}} c^2}\right)_B\right]. \tag{1.6.2}$$

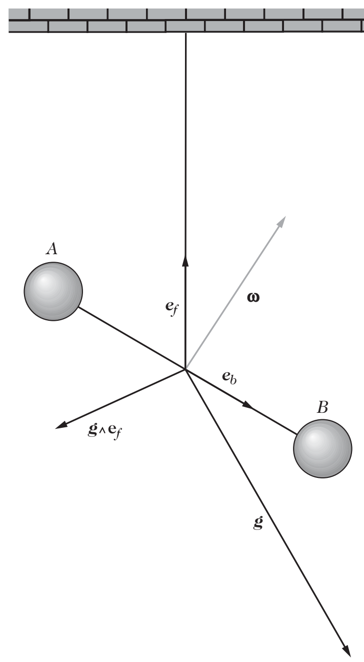


Figure 1.10: Principle of the Eötvös torsion balance. Two test bodies A and B of different compositions are suspended on a beam of length $2L$ supported by a torsion fiber $\mathbf{e}_f$. The balance experiences gravitational acceleration $\mathbf{g}$ and centrifugal acceleration $\mathbf{c} = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$.

1. **Free Pendulums:** The period $T = 2\pi\sqrt{\frac{m_\mathrm{I}}{m_\mathrm{G}}\frac{L}{g}}$ gives $\eta = 2\frac{|T_A - T_B|}{T_A + T_B}$.

2. **Eötvös Torsion Balance (Figure 1.10):** Two test bodies of different compositions are mounted at opposite ends of a beam suspended by a torsion fiber. The net gravitational force $\mathbf{F} = m_\mathrm{G}\mathbf{g} + m_\mathrm{I}\mathbf{c}$ creates a restoring torque on the fiber:

$$M = \eta \frac{L}{|\mathbf{g} + \mathbf{c}|} (\mathbf{e}_b \times \mathbf{c}) \cdot \mathbf{g}. \tag{1.6.3}$$

   Modern torsion-balance experiments (e.g., the Eöt-Wash group) constrain the equivalence principle to $\eta < 10^{-13}$.

Lunar Laser Ranging and the Strong Equivalence Principle (SEP). Laboratory test bodies have negligible gravitational binding energy ($E_\mathrm{grav}/mc^2 \sim 10^{-25}$). Testing the **Strong Equivalence Principle** requires astronomical bodies with significant self-gravity ($E_\mathrm{grav}/mc^2 \approx -4.6 \times 10^{-10}$ for Earth, -0.2×10^{-10} for the Moon).

In the gravitational field of the Sun, any SEP violation causes a differential acceleration (Nordtvedt effect) polarizing the lunar orbit toward the Sun:

$$\delta a \leq 2.6 \times 10^{-12} \eta\,\mathrm{m} \cdot \mathrm{s}^{-2} \implies \delta r_\mathrm{max} \leq -14\eta\,\mathrm{m}. \tag{1.6.4}$$

Lunar Laser Ranging (LLR), tracking retroreflectors on the Moon with sub-centimeter precision over three decades, confirms $\eta_\mathrm{SEP} = (-0.7 \pm 1.4) \times 10^{-13}$, verifying the SEP to high precision.

Local Lorentz Invariance (LLI)

The most sensitive tests of Local Lorentz Invariance are Hughes–Drever nuclear magnetic resonance experiments. Observing the $J = 3/2$ nuclear ground state of ^{7}Li in an external magnetic field, any spatial anisotropy in the inertial mass tensor $\delta m_\mathrm{I}^{ij} = \sum_A \delta_A \frac{E_A}{c^2}$ would cause an anomalous shift in the Zeeman level splittings. These experiments establish the stringent bound:

$$|\delta m^{ij} c^2| \lesssim 1.7 \times 10^{-16}\,\text{eV}. \tag{1.6.5}$$

Local Position Invariance (LPI)

Local Position Invariance requires that gravitational redshift depend solely on the potential difference $\Delta\Phi_\text{N}$:

$$z = (1 + \alpha_\text{LPI})\frac{\Delta\Phi_\text{N}}{c^2}. \tag{1.6.6}$$

The Pound–Rebka experiment (1960) using ^{57}Fe Mössbauer spectroscopy verified this to 10%, while the sub-orbital Gravity Probe A hydrogen maser rocket experiment (Vessot et al., 1980) confirmed $\alpha_\text{LPI} = (1.4 \pm 70) \times 10^{-5}$.

Fifth force tests and Yukawa modifications

Deviations from Newton's inverse-square law are parameterized by a Yukawa-type potential of characteristic interaction range λ:

$$V(r) = -\frac{G_\text{N} m_1 m_2}{r}\left(1 + \alpha_{12} \mathrm{e}^{-r/\lambda}\right). \tag{1.6.7}$$

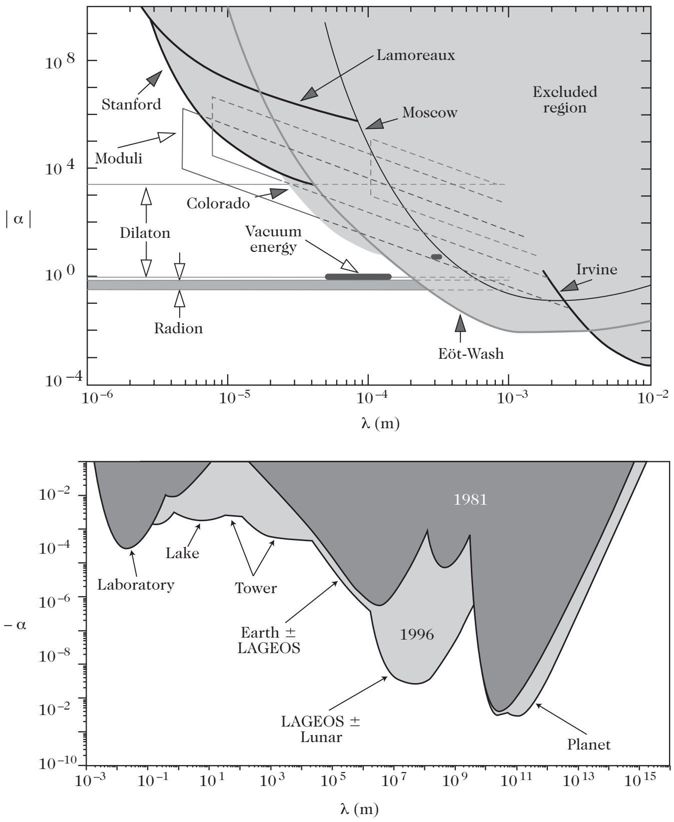


Figure 1.11: Experimental constraints on the strength $|\alpha|$ and range λ of a Yukawa fifth-force modification to Newton's law. Shaded regions are excluded by laboratory Cavendish/torsion balances, satellite geodesy (LAGEOS), lunar ranging, and planetary ephemerides.

Constraints on $(|\alpha|, \lambda)$ span from sub-millimeter scales in laboratory Casimir and torsion tests to Solar System scales (10^{11} m), as summarized in Figure 1.11.

1.6.2 Solar System tests and the Parameterized Post-Newtonian (PPN) framework

The Schwarzschild metric

The unique spherically symmetric, static vacuum solution to Einstein's equations outside a mass M is the **Schwarzschild metric**:

$$\mathrm{d}s^2 = -\left(1 - \frac{2G_\mathrm{N}M}{c^2 r}\right) c^2\,\mathrm{d}t^2 + \left(1 - \frac{2G_\mathrm{N}M}{c^2 r}\right)^{-1} \mathrm{d}r^2 + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\right). \tag{1.6.8}$$

Parameterized Post-Newtonian (PPN) expansion

In the weak-field regime ($\frac{G_\mathrm{N}M}{c^2 r} \ll 1$), we generalize the isotropic metric to the PPN framework:

$$\mathrm{d}s^2 = -\left[1 - \frac{2G_\mathrm{N}M}{c^2 r} + 2\left(\beta_\mathrm{PPN} - \gamma_\mathrm{PPN}\right)\left(\frac{G_\mathrm{N}M}{c^2 r}\right)^2\right] c^2\,\mathrm{d}t^2 + \left(1 + 2\gamma_\mathrm{PPN}\frac{G_\mathrm{N}M}{c^2 r}\right)\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2\,. \tag{1.6.9}$$

General relativity predicts $\gamma_\mathrm{PPN} = 1$ (spatial curvature produced per unit mass) and $\beta_\mathrm{PPN} = 1$ (non-linearity in the gravitational superposition).

Deflection of light by a massive body

For a light ray moving in the equatorial plane ($\theta = \pi/2$), time-translation and rotational isometries yield two constants of motion along the null geodesic $k^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}$:

$$\frac{\mathrm{d}ct}{\mathrm{d}\lambda} = E\,\mathrm{e}^{-\mu(r)}, \qquad r^2\frac{\mathrm{d}\phi}{\mathrm{d}\lambda} = L. \tag{1.6.10}$$

The null condition $k^\mu k_\mu = 0$ yields the trajectory equation:

$$\left(\frac{1}{r^2}\frac{\mathrm{d}r}{\mathrm{d}\phi}\right)^2 = \left(\frac{E}{L}\right)^2 \mathrm{e}^{-\nu(r)-\mu(r)} - \frac{\mathrm{e}^{-\nu(r)}}{r^2}. \tag{1.6.11}$$

Introducing the dimensionless variable $y \equiv \frac{G_\mathrm{N}M}{c^2 r}$ and impact parameter $b \equiv cL/E$, differentiating with respect to ϕ and expanding to first post-Newtonian order yields:

$$\frac{\mathrm{d}^2 y}{\mathrm{d}\phi^2} + y = 3\gamma_\mathrm{PPN} y^2 + \left(1 - \gamma_\mathrm{PPN}\right)\left(\frac{G_\mathrm{N}M}{bc^2}\right)^2. \tag{1.6.12}$$

To second order in $G_\mathrm{N}M/bc^2$, the trajectory is $y(\phi) = \frac{G_\mathrm{N}M}{bc^2}\sin\phi + \frac{G_\mathrm{N}M}{bc^2}(1 + \gamma_\mathrm{PPN}\cos^2\phi)$. The distance of closest approach to the deflecting body is $r_0 \simeq b\left(1 - \frac{G_\mathrm{N}M}{bc^2}\right)$ (Figure 1.12).

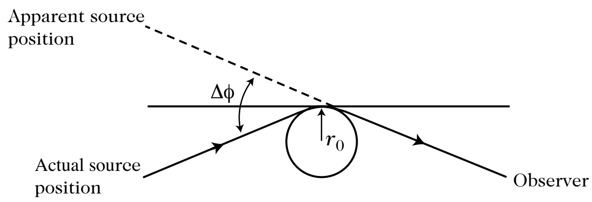


Figure 1.12: Geometry of light deflection by a central mass M. The apparent position of the source is displaced by the deflection angle $\Delta\phi$ relative to its actual position.

In the asymptotic limit $r \to \infty$, the total deflection angle $\Delta\phi = \phi_+ - \phi_-$ is:

$$\Delta\phi = 2\left(1 + \gamma_\mathrm{PPN}\right)\frac{G_\mathrm{N}M}{r_0 c^2}. \tag{1.6.13}$$

For a light ray grazing the solar limb ($r_0 = R_\odot$), general relativity ($\gamma_{\rm PPN} = 1$) predicts:

$$\Delta\phi = \frac{4G_{\rm N}M_\odot}{R_\odot c^2} = 1.751''. \tag{1.6.14}$$

Modern Very Long Baseline Interferometry (VLBI) measuring radio source deflections by the Sun confirms $|\gamma_{\rm PPN} - 1| < 4 \times 10^{-4}$.

The Shapiro time delay effect

Discovered by Irwin Shapiro in 1964, the **Shapiro effect** is a gravitational time delay experienced by electromagnetic signals traversing a gravitational potential well. From the null geodesic equation (1.6.11), the coordinate velocity of light is:

$$\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 = c^2 \mathrm{e}^{\mu(r)-\nu(r)} \left(1 - \mathrm{e}^{\mu(r)}\frac{b^2}{r^2}\right). \tag{1.6.15}$$

The round-trip travel time of a radar signal between Earth ($r_{\rm E}$) and a target planet or spacecraft ($r_{\rm p}$) grazing the Sun at closest approach r_0 is:

$$\Delta t = \frac{2}{c}\int_{r_0}^{r_{\rm E}} \frac{\mathrm{e}^{\frac{1}{2}[\nu(r)-\mu(r)]}}{\sqrt{1 - \mathrm{e}^{\mu(r)}\frac{b^2}{r^2}}}\,\mathrm{d}r + \frac{2}{c}\int_{r_0}^{r_{\rm P}} \frac{\mathrm{e}^{\frac{1}{2}[\nu(r)-\mu(r)]}}{\sqrt{1 - \mathrm{e}^{\mu(r)}\frac{b^2}{r^2}}}\,\mathrm{d}r\,. \tag{1.6.16}$$

Expanding in the PPN metric, the relativistic excess time delay is:

$$\Delta t_{\rm excess} = (1 + \gamma_{\rm PPN})\,\frac{2G_{\rm N}M_\odot}{c^3}\ln\left(\frac{4r_{\rm E}r_{\rm P}}{r_0^2}\right). \tag{1.6.17}$$

For the Cassini spacecraft solar occultation (2003), tracking Doppler shifts constrained the time delay rate to $|\gamma_{\rm PPN} - 1| = (2.1 \pm 2.3) \times 10^{-5}$.

Precession of the perihelion of Mercury

For a massive test body orbiting on a timelike geodesic ($u^\mu u_\mu = -c^2$), the radial energy equation in the equatorial plane is:

$$\mathrm{e}^{-\mu(r)}E^2 = c^2 + \mathrm{e}^{\nu(r)}\frac{L^2}{r^4}\left(\frac{\mathrm{d}r}{\mathrm{d}\phi}\right)^2 + \frac{L^2}{r^2}. \tag{1.6.18}$$

Setting $y \equiv 1/r$, the orbital equation to post-Newtonian order is:

$$\frac{\mathrm{d}^2y}{\mathrm{d}\phi^2} + y = \frac{G_{\rm N}M}{L^2} + (2 + 2\gamma_{\rm PPN} - \beta_{\rm PPN})\,\frac{G_{\rm N}M}{c^2}y^2. \tag{1.6.19}$$

The perturbative solution describes an ellipse whose perihelion precesses at each revolution by:

$$\Delta\phi = \frac{2\pi G_{\rm N}M}{a\,(1-e^2)\,c^2}\,(2 + 2\gamma_{\rm PPN} - \beta_{\rm PPN})\,, \tag{1.6.20}$$

where a is the semi-major axis and e is the orbital eccentricity.

For Mercury ($a = 5.79 \times 10^{10}\,\mathrm{m}, e = 0.2056$), general relativity ($\gamma_{\rm PPN} = \beta_{\rm PPN} = 1$) yields:

$$\Delta\phi_{\rm GR} = \frac{6\pi G_{\rm N}M_\odot}{a(1-e^2)c^2} = 42.98''/\text{century}, \tag{1.6.21}$$

in exquisite agreement with the observed anomalous residual of $(43.1 \pm 0.5)''$/century after subtracting planetary Newtonian perturbations.

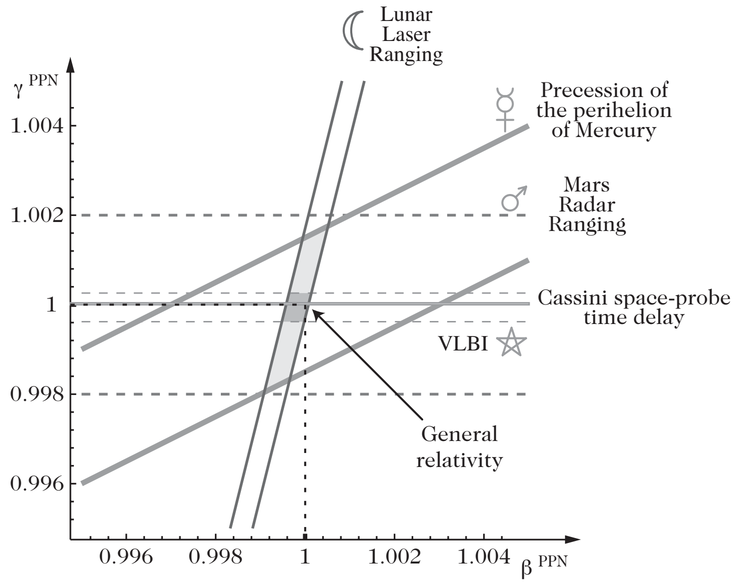


Figure 1.13: Summary of experimental constraints on the Parameterized Post-Newtonian parameters $(\beta_{\text{PPN}}, \gamma_{\text{PPN}})$ in the Solar System. The intersection of bounds from Mercury's perihelion precession, Lunar Laser Ranging, VLBI light deflection, and the Cassini radio time delay isolates general relativity ($\beta_{\text{PPN}} = \gamma_{\text{PPN}} = 1$) to high precision.

Table 1.3: Experimental Constraints on Gravitational Parameters in the Solar System

Parameter	**Bound / Measurement**	**Experimental Method**	**Source / System**
η (Weak WEP)	$< 10^{-3}$	Free pendulums	Newton (1686)
η (Weak WEP)	$< 5 \times 10^{-9}$	Torsion balance	Eötvös (1922)
η (Weak WEP)	$(-1.9 \pm 2.5) \times 10^{-12}$	Modern torsion balance	Be–Cu laboratory
η (Weak WEP)	$< 5.5 \times 10^{-13}$	Rotating torsion balance	Eöt-Wash group
η_{SEP} (Strong SEP)	$(-0.7 \pm 1.4) \times 10^{-13}$	Lunar Laser Ranging (LLR)	Earth–Moon in Sun field
α_{LPI} (Grav. Redshift)	$< 10^{-2}$	^{57}Fe Mössbauer effect	Pound–Rebka (1960)
α_{LPI} (Grav. Redshift)	$(1.4 \pm 70) \times 10^{-5}$	Suborbital H-maser clock	Gravity Probe A (1980)
$\lvert\delta m_{\text{I}}^{ij} c^2\rvert$ (LLI)	$\lesssim 1.7 \times 10^{-16}$ eV	Hughes–Drever NMR	^{7}Li nuclear Zeeman
$\lvert\gamma_{\text{PPN}} - 1\rvert$	$< 4 \times 10^{-4}$	Solar light deflection	VLBI astrometry
$\lvert\gamma_{\text{PPN}} - 1\rvert$	$(2.1 \pm 2.3) \times 10^{-5}$	Radar time delay	Cassini spacecraft
$\lvert 2\gamma - \beta - 1\rvert$	$< 3 \times 10^{-3}$	Perihelion advance	Mercury orbit
$\lvert 4\beta - \gamma - 3\rvert$	$(-0.7 \pm 1.0) \times 10^{-3}$	Nordtvedt effect	LLR tracking

1.6.3 Gravitational radiation and binary pulsars

The quadrupole radiation formula

In linearized general relativity, time-varying mass distributions generate gravitational radiation. In the far-zone radiation field ($r \gg \lambda_{\rm GW}$), the transverse-traceless metric perturbation is governed by the **Einstein quadrupole formula**:

$$h_{ij}^{\rm TT}(\mathbf{x},t) = \frac{2G_{\rm N}}{c^4 r}\mathcal{P}_{ijkl}\frac{{\rm d}^2 Q_{kl}}{{\rm d}t^2}\left(t - \frac{r}{c}\right), \tag{1.6.22}$$

where $\mathcal{P}_{ijkl}$ is the transverse-traceless projection operator, and $Q_{kl}(t)$ is the trace-free mass quadrupole moment of the source:

$$Q_{kl}(t) \equiv \int \rho(\mathbf{x},t)\left(x_k x_l - \frac{1}{3}r^2\delta_{kl}\right){\rm d}^3x. \tag{1.6.23}$$

Because total mass-energy and total linear momentum are conserved, there is **no monopole or dipole gravitational radiation** in general relativity.

The total gravitational energy loss rate (luminosity) is:

$$\frac{{\rm d}E}{{\rm d}t} = -\frac{G_{\rm N}}{5c^5}\dddot{Q}_{kl}\dddot{Q}_{kl}. \tag{1.6.24}$$

The Hulse–Taylor binary pulsar PSR B1913+16

Discovered in 1974 by Russell Hulse and Joseph Taylor, the relativistic binary pulsar PSR B1913+16 provided the first empirical confirmation of gravitational radiation. For two orbiting point masses m_1 and m_2 in an eccentric Keplerian orbit with semi-major axis a, period P, and eccentricity e, the gravitational radiation reaction reduces the orbital period at the rate:

$$\left\langle \frac{{\rm d}P}{{\rm d}t}\right\rangle = -\frac{192\pi}{5c^5}\left(\frac{2\pi G_{\rm N}}{P}\right)^{5/3}\frac{m_1 m_2}{(m_1+m_2)^{1/3}}\frac{1+\frac{73}{24}e^2+\frac{37}{96}e^4}{(1-e^2)^{7/2}}. \tag{1.6.25}$$

For PSR B1913+16 ($m_1 = 1.441\,M_\odot, m_2 = 1.387\,M_\odot, e = 0.6171, P \approx 7.75\,{\rm h}$), the theoretical prediction $\langle \dot{P}\rangle = -2.402\times 10^{-12}\,{\rm s/s}$ matches the measured orbital decay to within 0.2%, establishing that gravitational interactions propagate at the speed of light c.

1.6.4 Need for tests at astrophysical and cosmological scales

While general relativity is tested to parts in 10^5 within the Solar System and binary pulsar systems, its application to the Universe as a whole involves vast extrapolations across 15 orders of magnitude in length scale and requires careful empirical cross-checks:

1. **Galactic Rotation Curves and Dark Matter:** On galactic scales, flat rotation curves ($v \approx {\rm const} \implies \Phi_{\rm N} \propto \ln r$) indicate either the existence of non-baryonic **dark matter** halo distributions or infrared modifications of gravity.

2. **Galaxy Clusters and Gravitational Lensing:** Weak and strong gravitational lensing by clusters ($\sim 2\,{\rm Mpc}$) probes the metric potential $\Phi + \Psi$, while X-ray gas temperature profiles probe the Newtonian potential Φ. Concordance between the two confirms the relativistic relation $\Phi = \Psi$ up to dark matter modeling.

3. **Cosmic Acceleration and Dark Energy:** The observed accelerated cosmic expansion cannot be driven by ordinary positive-pressure matter within standard general relativity (as required by the Raychaudhuri equation). It necessitates either an exotic negative-pressure **dark energy** component (cosmological constant Λ, dynamical scalar fields / quintessence) or large-scale modifications of gravity ($f(R)$, scalar-tensor theories).

4. **Cosmic Tests of Fundamental Constants:** Cosmological bounds on the variation of fundamental constants ($\alpha_{\mathrm{EM}}, \mu = m_p/m_e, G_{\mathrm{N}}$) over gigayear timescales test Local Position Invariance across Hubble horizons.

These cosmological questions form the core subject matter of the remaining chapters of this book.

Part II

The Modern Standard Cosmological Model

Chapter 2

The cosmological solution of Friedmann–Lemaître

The first relativistic cosmological solution of Einstein's field equations was formulated by Albert Einstein in 1917, who introduced a cosmological constant Λ to construct a static, spatially closed universe. The general dynamical cosmological solutions were discovered independently by Alexandre Friedmann (1922, 1924) and Georges Lemaître (1927).

This chapter details the mathematical derivation of the Friedmann–Lemaître spacetime from the symmetries of homogeneity and isotropy, analyzes its kinematics and relativistic dynamics, and establishes the observational relationships foundational to modern primordial cosmology.

2.1 The Friedmann–Lemaître spacetime

2.1.1 Constructing a cosmological model

Within the framework of general relativity, the central objective is to identify which exact solution of Einstein's field equations provides an accurate, idealized description of our Universe on the largest observable scales.

This task is constrained by fundamental observational limitations:

1. **Uniqueness of the Universe:** We observe a single realization of the Universe and cannot perform statistical ensembles over independent universes.

2. **Spacetime Localization:** Astronomical observations are gathered exclusively along our **past light cone**, while geophysical and local experiments probe a narrow worldtube around our worldline.

3. **Observable Universe vs. Global Universe:** We must distinguish between the *observable Universe* (the interior of our past particle horizon from which causal signals have reached us) and the *global Universe*, whose geometry beyond our horizon cannot be directly probed without theoretical extrapolation.

The standard cosmological model rests on four foundational hypotheses:

H1 General Relativity as the Theory of Gravitation: Gravity is governed by Einstein's general relativity, and the Einstein Equivalence Principle ensures that local non-gravitational physical laws remain invariant throughout cosmic history.

H2 Cosmic Matter Content: On large scales, the Universe is filled with a statistical mixture of pressureless baryonic and dark matter (dust), relativistic radiation (photons and neutrinos), and dark energy / cosmological constant Λ.

H3 Uniformity Principles (Homogeneity and Isotropy): On scales larger than galaxy clusters and superclusters ($\gtrsim 100\,\mathrm{Mpc}$), spacetime possesses spatial homogeneity and spatial isotropy.

H4 Global Spatial Topology: For a globally hyperbolic spacetime $\mathcal{M} = \mathbb{R} \times \Sigma$, the spatial slices Σ possess a simply connected or multi-connected topology compatible with the local constant curvature.

2.1.2 The Cosmological Principle and the Copernican Principle

Astronomical surveys of distant galaxies and the extreme angular uniformity of the Cosmic Microwave Background (CMB, isotropic to $\Delta T/T \sim 10^{-5}$) demonstrate that the observable Universe is highly isotropic around our vantage point.

Two distinct interpretations arise:

1. **Non-Copernican Centered Geometry:** Our Galaxy resides at the geometric center of a spherically symmetric, radially inhomogeneous universe (Figure 2.1, right).

2. **The Copernican Principle:** Humans do not occupy a privileged or special location in the Universe. If the Universe is isotropic around us, and our position is typical, then the Universe must be **isotropic around every point** (Figure 2.1, left).

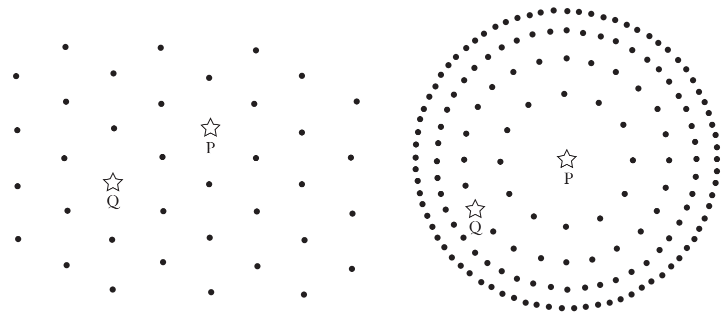


Figure 2.1: Point distributions illustrating uniformity: (Left) Statistically isotropic around every point, implying spatial homogeneity. (Right) Isotropic around a single privileged center P, where points P and Q are physically distinct. The Copernican principle excludes centered models.

The **Cosmological Principle** asserts that the Universe is both **spatially homogeneous** and **spatially isotropic** when averaged on scales greater than the largest bound structures ($\lambda_{\text{smooth}} \gtrsim 100\,\mathrm{Mpc}$).

2.1.3 Spacetime symmetries and the Robertson–Walker metric

Homogeneity and isotropy are distinct geometric properties:

- **Spatial Homogeneity:** Spacetime $\mathcal{M}$ admits a foliation by a 1-parameter family of spacelike hypersurfaces Σ_t such that for any two points $P, Q \in \Sigma_t$, there exists an isometry of the spatial metric mapping P to Q (Figure 2.2, left).

- **Spatial Isotropy:** There exists a congruence of timelike worldlines with unit tangent 4-velocity u^μ (the **fundamental / comoving observers**) such that at any point P, any unit spatial tangent vector $e_1^\mu \in T_P\Sigma_t$ can be rotated into any other unit vector $e_2^\mu \in T_P\Sigma_t$ by an isometry that leaves P and $u^\mu(P)$ fixed (Figure 2.2, right).

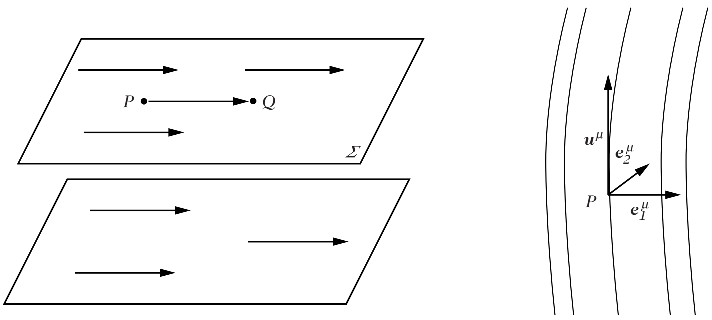


Figure 2.2: (Left) A spatially homogeneous spacetime foliated by spacelike slices Σ_t. (Right) Spatial isotropy ensures that any spatial tangent direction e_1^μ is isometric to e_2^μ about the fundamental observer u^μ.

Isotropy requires that the worldlines of fundamental observers be strictly orthogonal to the hypersurfaces of homogeneity Σ_t. If this were not the case, the non-vanishing projection of u^μ on Σ_t would single out a preferred spatial direction, violating isotropy.

Consequently, each spatial slice Σ_t is a 3-dimensional **maximally symmetric Riemannian space** of constant spatial curvature $K(t)$. The 3-dimensional Riemann curvature tensor of Σ_t satisfies:

$$ {}^{(3)}R_{ijkl} = \frac{K(t)}{a^2(t)} \left(\gamma_{ik}\gamma_{jl} - \gamma_{il}\gamma_{jk}\right), \tag{2.1.1} $$

where γ_{ij} is the dimensionless comoving spatial metric, $a(t)$ is the cosmic scale factor, and $K \in \{-1, 0, +1\}$ is the normalized curvature index:

1. **Spherical Geometry ($K = +1$):** Σ_t is a 3-sphere S^3 embedded in $\mathbb{R}^4$ via $x^2+y^2+z^2+w^2 = R^2(t)$:
$$ \mathrm{d}\ell^2_{(3)} = a^2(t)\left[\mathrm{d}\chi^2 + \sin^2\chi \left(\mathrm{d}\theta^2 + \sin^2\theta\, \mathrm{d}\phi^2\right)\right], \qquad \chi \in [0, \pi]. \tag{2.1.2} $$

2. **Flat Euclidean Geometry ($K = 0$):** $\Sigma_t = \mathbb{R}^3$:
$$ \mathrm{d}\ell^2_{(3)} = a^2(t)\left[\mathrm{d}\chi^2 + \chi^2 \left(\mathrm{d}\theta^2 + \sin^2\theta\, \mathrm{d}\phi^2\right)\right], \qquad \chi \in [0, \infty). \tag{2.1.3} $$

3. **Hyperbolic Geometry ($K = -1$):** Σ_t is a 3-hyperboloid H^3 embedded in Minkowski space $\mathbb{R}^{3,1}$ via $-w^2 + x^2 + y^2 + z^2 = -R^2(t)$:
$$ \mathrm{d}\ell^2_{(3)} = a^2(t)\left[\mathrm{d}\chi^2 + \sinh^2\chi \left(\mathrm{d}\theta^2 + \sin^2\theta\, \mathrm{d}\phi^2\right)\right], \qquad \chi \in [0, \infty). \tag{2.1.4} $$

The Friedmann–Lemaître–Robertson–Walker (FLRW) line element

Setting the proper time of fundamental observers to cosmic time t ($u^\mu\, \mathrm{d}x_\mu \equiv -\,\mathrm{d}t$), the 4-dimensional spacetime metric takes the universal form:

$$ \mathrm{d}s^2 = -\,\mathrm{d}t^2 + a^2(t)\gamma_{ij}(\mathbf{x})\, \mathrm{d}x^i\, \mathrm{d}x^j\, . \tag{2.1.5} $$

Introducing **conformal time** η, defined by:

$$ \mathrm{d}t = a(t)\, \mathrm{d}\eta \implies \eta(t) = \int_{t_0}^{t} \frac{\mathrm{d}t'}{a(t')}, \tag{2.1.6} $$

the line element becomes conformally related to a static spacetime:

$$ \mathrm{d}s^2 = a^2(\eta)\left[-\,\mathrm{d}\eta^2 + \gamma_{ij}\, \mathrm{d}x^i\, \mathrm{d}x^j\right]. \tag{2.1.7} $$

In comoving spherical coordinates (χ, θ, ϕ), the spatial line element is:

$$\mathrm{d}\sigma^2 \equiv \gamma_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j = \mathrm{d}\chi^2 + f_K^2(\chi)\,\mathrm{d}\Omega^2\,, \qquad \mathrm{d}\Omega^2 \equiv \mathrm{d}\theta^2 + \sin^2\theta\;\mathrm{d}\phi^2\,, \tag{2.1.8}$$

where the coordinate distance function $f_K(\chi)$ is:

$$f_K(\chi) = \begin{cases} \frac{1}{\sqrt{K}}\sin\left(\sqrt{K}\chi\right) & (K>0,\ \text{Closed}),\\ \chi & (K=0,\ \text{Flat}),\\ \frac{1}{\sqrt{-K}}\sinh\left(\sqrt{-K}\chi\right) & (K<0,\ \text{Open}). \end{cases} \tag{2.1.9}$$

Alternatively, defining the radial coordinate $r \equiv f_K(\chi)$, the metric takes the standard curvature form:

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + a^2(t)\left[\frac{\mathrm{d}r^2}{1-Kr^2} + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\;\mathrm{d}\phi^2\right)\right]. \tag{2.1.10}$$

Symmetry reduces the 10 metric functions of an arbitrary spacetime to a single dynamical function $a(t)$ and a discrete constant $K \in \{+1, 0, -1\}$.

Geometric tensors of the FLRW spacetime

From the metric (2.1.5), the non-vanishing Christoffel connection coefficients are:

$$\Gamma^0_{ij} = Ha^2\gamma_{ij}, \qquad \Gamma^i_{j0} = H\delta^i_j, \qquad \Gamma^i_{jk} = {}^{(3)}\Gamma^i_{jk}, \tag{2.1.11}$$

where $H(t) \equiv \frac{\dot{a}}{a} = \frac{c}{a}\frac{\mathrm{d}a}{\mathrm{d}(ct)}$ is the **Hubble expansion parameter**, and an overdot denotes differentiation with respect to cosmic time t.

The non-vanishing components of the Ricci tensor are:

$$R_{00} = -3\frac{\ddot{a}}{a}, \qquad R_{ij} = \left(2H^2 + \frac{\ddot{a}}{a} + \frac{2K}{a^2}\right)a^2\gamma_{ij}, \tag{2.1.12}$$

yielding the Ricci scalar curvature:

$$R = 6\left(H^2 + \frac{\ddot{a}}{a} + \frac{K}{a^2}\right) = 6\left[\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 + \frac{K}{a^2}\right]. \tag{2.1.13}$$

The non-vanishing components of the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ are:

$$G_{00} = 3\left(H^2 + \frac{K}{a^2}\right), \tag{2.1.14}$$

$$G_{ij} = -\left(H^2 + 2\frac{\ddot{a}}{a} + \frac{K}{a^2}\right)a^2\gamma_{ij}. \tag{2.1.15}$$

Killing and Conformal Killing vectors

A maximally symmetric 3-dimensional spatial section admits $\frac{3(3+1)}{2} = \mathbf{6}$ independent spatial Killing vector fields satisfying $\nabla_{(\mu}\xi_{\nu)} = 0$:

1. **3 Spatial Translations:**

$$P^0_{[r]} = 0, \qquad P^i_{[r]} = \sqrt{1-Kr^2}\,\delta^i_r \qquad (r = 1,2,3). \tag{2.1.16}$$

2. **3 Spatial Rotations:**

$$R^0_{[rs]} = 0, \qquad R^i_{[rs]} = \delta^i_r x_s - \delta^i_s x_r \qquad (r,s = 1,2,3). \tag{2.1.17}$$

Because the Universe expands ($\dot{a} \neq 0$), time-translation invariance is broken; there is no timelike Killing vector. However, FLRW spacetime possesses **4 conformal Killing vectors** satisfying $\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = \frac{1}{2}(\nabla_\alpha \xi^\alpha) g_{\mu\nu}$:

$$T^0 = \sqrt{1 - Kr^2}, \qquad T^k = -Hx^k \sqrt{1 - Kr^2}, \tag{2.1.18}$$

$$L^0_{[r]} = x_r, \qquad L^k_{[r]} = H \left[\frac{1}{2} \delta^k_r \left(r^2 - \eta^2 \right) - x^k x_r \right] \quad (K = 0). \tag{2.1.19}$$

2.2 Kinematics of the expanding Universe

2.2.1 Fundamental observers and the Hubble expansion law

In comoving coordinates $\mathbf{x}$, the worldline of a fundamental observer at rest with the expanding cosmological fluid is $\mathbf{x} = \text{constant}$. Their 4-velocity is:

$$u^\mu = \delta^\mu_0 = (1, 0, 0, 0). \tag{2.2.1}$$

For two comoving galaxies at comoving separation $\Delta\mathbf{x}$, their physical distance at cosmic time t is:

$$\mathbf{r}(t) = a(t) \Delta\mathbf{x}. \tag{2.2.2}$$

Differentiating with respect to cosmic time yields the **Hubble Law**:

$$\mathbf{v}_{\text{rec}} \equiv \dot{\mathbf{r}} = \dot{a}(t) \Delta\mathbf{x} = \frac{\dot{a}}{a} \mathbf{r} = H(t) \mathbf{r}. \tag{2.2.3}$$

The recession velocity $\mathbf{v}_{\text{rec}}$ is the kinematic expansion of space itself.

For a non-comoving object with coordinate motion $\dot{\mathbf{x}}$, the total physical velocity splits as:

$$\mathbf{v}_{\text{tot}} = \dot{\mathbf{r}} = H\mathbf{r} + a(t)\dot{\mathbf{x}} \equiv \mathbf{v}_{\text{rec}} + \mathbf{v}_{\text{pec}}, \tag{2.2.4}$$

where $\mathbf{v}_{\text{pec}} \equiv a(t)\dot{\mathbf{x}}$ is the **peculiar velocity** relative to the comoving Hubble flow.

2.2.2 Propagation of light and cosmological redshift

In geometric optics, light rays travel along null geodesics ($k^\mu k_\mu = 0, k^\mu \nabla_\mu k^\nu = 0$). Decomposing the wavevector relative to comoving observers:

$$k^\mu = E u^\mu + p^\mu, \qquad E = -k_\mu u^\mu = \hbar\omega, \qquad p^\mu u_\mu = 0, \tag{2.2.5}$$

where $E^2 = a^2(t) \gamma_{ij} p^i p^j = p^2$.

Projecting the geodesic equation $k^\nu \nabla_\nu k^\mu = 0$ along u_μ:

$$u_\mu k^\nu \nabla_\nu k^\mu = 0 \implies E\dot{E} + \Gamma^0_{ij} p^i p^j = 0 \implies E\dot{E} + Ha^2 \gamma_{ij} p^i p^j = 0. \tag{2.2.6}$$

Using $a^2 \gamma_{ij} p^i p^j = E^2$, we obtain the exact energy damping law:

$$E \frac{\mathrm{d}E}{\mathrm{d}t} = -\frac{\dot{a}}{a} p^2 \implies \frac{\dot{E}}{E} = -H = -\frac{\dot{a}}{a} \implies E(t) \propto \frac{1}{a(t)}. \tag{2.2.7}$$

Because photon frequency $\nu \propto E$ and wavelength $\lambda = c/\nu$, the wavelength stretches in direct proportion to the cosmic scale factor:

$$\lambda(t) \propto a(t). \tag{2.2.8}$$

A photon emitted at cosmic time t_{em} and observed at t_{obs} undergoes an achromatic **cosmological redshift** z:

$$1 + z \equiv \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} = \frac{\nu_{\text{em}}}{\nu_{\text{obs}}} = \frac{a(t_{\text{obs}})}{a(t_{\text{em}})}. \tag{2.2.9}$$

Setting $a(t_{\text{obs}}) \equiv 1$ today, the scale factor at redshift z is simply $a(z) = \frac{1}{1+z}$.

2.2.3 Dynamics of massive test particles

For a massive test particle with 4-velocity v^μ ($v^\mu v_\mu = -1$), decomposing relative to the fundamental observers gives:

$$v^\mu = \alpha u^\mu + p^\mu, \qquad u_\mu p^\mu = 0, \qquad \alpha = \sqrt{1+p^2}, \tag{2.2.10}$$

where $p^2 \equiv a^2\gamma_{ij}p^ip^j$.

Projecting the geodesic equation $v^\nu \nabla_\nu v^\mu = 0$ along u_μ yields:

$$\alpha\dot{\alpha} + Hp^2 = 0 \iff \alpha\dot{\alpha} + H\left(\alpha^2 - 1\right) = 0 \implies \alpha^2 - 1 \propto a^{-2}(t). \tag{2.2.11}$$

Consequently, the physical momentum and peculiar velocity of any freely moving particle decay as:

$$E^2 = p^2 \implies E\frac{\mathrm{d}E}{\mathrm{d}t} = p\frac{\mathrm{d}p}{\mathrm{d}t} \tag{2.2.12}$$

Then using $p\frac{\mathrm{d}p}{\mathrm{d}t} = -Hp^2$, we have

$$p(t) \propto \frac{1}{a(t)}, \qquad v_{\mathrm{pec}}(t) \propto \frac{1}{a(t)} \quad \text{(non-relativistic limit)}. \tag{2.2.13}$$

Any freely moving particle is decelerated by the cosmic expansion, coming to rest with respect to the comoving fundamental frame as $a \to \infty$.

2.3 Cosmological dynamics of the FLRW spacetime

2.3.1 Energy-momentum tensor of the cosmological fluid

The stress-energy tensor $T_{\mu\nu}$ is a symmetric rank-2 tensor. Under the requirement of spatial homogeneity and isotropy, its most general form relative to the fundamental comoving observers $u^\mu = (1,0,0,0)$ is that of a **perfect fluid**:

$$T_{\mu\nu} = \rho(t)u_\mu u_\nu + P(t)\gamma_{\mu\nu}(t) = \begin{pmatrix} \rho(t) & 0 & 0 & 0 \\ 0 & & & \\ 0 & & a^2(t)P(t)\gamma_{ij} & \\ 0 & & & \end{pmatrix}, \tag{2.3.1}$$

where $\rho(t) = T_{00}$ is the energy density and $P(t) = \frac{1}{3}T^i_{\ i}$ is the isotropic pressure, both depending exclusively on cosmic time t. Anisotropic stresses ($\pi_{\mu\nu} = 0$) and spatial energy fluxes ($q_\mu = 0$) vanish identically by isotropy.

2.3.2 The Friedmann equations and conservation laws

Substituting the Einstein tensor (2.1.14)–(2.1.15) and stress-energy tensor (2.3.1) into Einstein's field equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_{\mathrm{N}} T_{\mu\nu}$ yields the two foundational **Friedmann equations**:

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G_{\mathrm{N}}}{3}\rho - \frac{K}{a^2} + \frac{\Lambda}{3}, \tag{2.3.2}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G_{\mathrm{N}}}{3}(\rho + 3P) + \frac{\Lambda}{3}. \tag{2.3.3}$$

Covariant conservation of stress-energy $\nabla_\mu T^{\mu\nu} = 0$ reduces to the **fluid continuity equation**:

$$\dot{\rho} + 3H(\rho + P) = 0 \iff \frac{\mathrm{d}}{\mathrm{d}t}\left(\rho a^3\right) + P\frac{\mathrm{d}a^3}{\mathrm{d}t} = 0. \tag{2.3.4}$$

The continuity equation is the first law of thermodynamics ($\mathrm{d}U = -P\,\mathrm{d}V$) applied to a comoving volume $V \propto a^3$. Due to the Bianchi identities, only two of the three equations (2.3.2), (2.3.3), and (2.3.4) are functionally independent with (2.3.2) being the Hamiltonian constraint of the theory. Since only two of them are independent, we only solve (2.3.2) and (2.3.4).

2.3.3 Cosmological equations in conformal time

Working in terms of conformal time $\mathrm{d}\eta = \mathrm{d}t\,/a(t)$, derivatives satisfy $X' \equiv \frac{\mathrm{d}X}{\mathrm{d}\eta} = a\dot{X}$ and $X'' - \mathcal{H}X' = a^2\ddot{X}$, where $\mathcal{H} \equiv \frac{a'}{a} = aH$ is the **comoving Hubble expansion rate**. The Friedmann and continuity equations become:

$$\mathcal{H}^2 = \frac{8\pi G_\mathrm{N}}{3}\rho a^2 - K + \frac{\Lambda}{3}a^2, \tag{2.3.5}$$

$$\mathcal{H}' = -\frac{4\pi G_\mathrm{N}}{3}a^2\,(\rho + 3P) + \frac{\Lambda}{3}a^2, \tag{2.3.6}$$

$$\rho' + 3\mathcal{H}\,(\rho + P) = 0. \tag{2.3.7}$$

Since (2.3.7) is $\nabla_{(\mu}T^{\mu\nu} = 0$ in disguise. This equation holds true for each fluid component individually and hence for constant equation of state (EoS) fluid, eq. (2.3.10) holds for them individually.

For a power-law scale factor in cosmic time $a(t) \propto t^n$ ($n \neq 1$), the conformal time dependence is:

$$a(t) \propto t^n \iff a(\eta) \propto \eta^{\frac{n}{1-n}}. \tag{2.3.8}$$

2.3.4 Equations of state and cosmic components

To close the system of dynamical equations, we specify a barotropic **equation of state**:

$$P = w\rho, \qquad w \equiv \frac{P}{\rho} = \text{constant}. \tag{2.3.9}$$

Integrating the continuity equation (2.3.4) for constant w yields:

$$\frac{\dot{\rho}}{\rho} = -3(1+w)\frac{\dot{a}}{a} \implies \rho(a) = \rho_0\left(\frac{a}{a_0}\right)^{-3(1+w)}. \tag{2.3.10}$$

Note that during reheating we do not expect w to be constant and thus this form will not work during that era.

Standard cosmological components scale as:

- **Non-relativistic Matter / Dust ($w = 0$):** Baryons and cold dark matter (CDM) have negligible thermal pressure ($P \ll \rho c^2$), scaling as comoving particle dilution:

$$\rho_m(a) = \rho_{m0}\left(\frac{a}{a_0}\right)^{-3} = \rho_{m0}(1+z)^3. \tag{2.3.11}$$

- **Relativistic Radiation ($w = 1/3$):** Photons and relativistic neutrinos experience both number density dilution $\propto a^{-3}$ and individual photon energy redshift $\propto a^{-1}$:

$$\rho_r(a) = \rho_{r0}\left(\frac{a}{a_0}\right)^{-4} = \rho_{r0}(1+z)^4. \tag{2.3.12}$$

- **Cosmological Constant / Vacuum Energy ($w = -1$):** Constant energy density:

$$\rho_\Lambda(a) = \rho_{\Lambda 0} = \frac{\Lambda}{8\pi G_\mathrm{N}} = \text{constant.} \tag{2.3.13}$$

- **Spatial Curvature ($w_\mathrm{eff} = -1/3$):** Curvature acts as an effective fluid with $\rho_K \propto a^{-2}$.

For a general fluid, the adiabatic sound speed $c_s^2 \equiv \frac{\partial P}{\partial \rho}$ governs the time variation of the equation of state:

$$w' = -3\mathcal{H}\,(1+w)\left(c_s^2 - w\right). \tag{2.3.14}$$

2.3.5 Dimensionless density parameters and the cosmic sum rule

Dividing the first Friedmann equation (2.3.2) by $H^2(t)$, we define the **critical energy density**:

$$\rho_\mathrm{crit}(t) \equiv \frac{3H^2(t)}{8\pi G_\mathrm{N}}, \qquad \rho_\mathrm{crit,0} = \frac{3H_0^2}{8\pi G_\mathrm{N}} = 1.878 \times 10^{-26} h^2\,\mathrm{kg}\cdot\mathrm{m}^{-3}, \tag{2.3.15}$$

where $H_0 \equiv 100\,h\,\mathrm{km}\cdot\mathrm{s}^{-1}\cdot\mathrm{Mpc}^{-1}$.

We define dimensionless **density parameters**:

$$\Omega_X(t) \equiv \frac{\rho_X(t)}{\rho_\mathrm{crit}(t)} = \frac{8\pi G_\mathrm{N}\rho_X}{3H^2}, \qquad \Omega_\Lambda(t) \equiv \frac{\Lambda}{3H^2}, \qquad \Omega_K(t) \equiv -\frac{K}{H^2a^2}. \tag{2.3.16}$$

The first Friedmann equation (2.3.2) becomes the exact **cosmic sum rule** upon substituting (2.3.16):

$$\sum_X \Omega_X(t) + \Omega_\Lambda(t) + \Omega_K(t) = 1. \tag{2.3.17}$$

Using the solutions of the conservation equation, it follows that from (2.3.10)

$$\Omega_X = \Omega_{X0}\left(\frac{a}{a_0}\right)^{-3(1+w_{X0})}\left(\frac{H_0}{H}\right)^2 \tag{2.3.18}$$

Evaluating today (subscript 0), the normalized expansion rate $E(z) \equiv H(z)/H_0$ is:

$$E^2(z) = \Omega_{r0}(1+z)^4 + \Omega_{m0}(1+z)^3 + \Omega_{K0}(1+z)^2 + \Omega_{\Lambda 0}, \tag{2.3.19}$$

with spatial curvature parameter $\Omega_{K0} = 1 - \Omega_{m0} - \Omega_{r0} - \Omega_{\Lambda 0}$.

This form allows us to calculate the age of the universe with $\Omega_{m0} = 0.315$, $H_0 = 67.36 \times 1.02269 \times 10^{-3}\mathrm{Gyr}^{-1}$, $\Omega_{r0} = 9.22 \times 10^{-5}$, $\Omega_{\Lambda 0} = 0.685$ and $\Omega_{K0} = 0$,

$$\int_0^t dt = \int_0^{a_0} \frac{da}{aH(a)} = \int_0^{a_0} \frac{da}{aH_0E(a)} = 13.8\mathrm{Gyr} \tag{1}$$

```
import numpy as np
from scipy.integrate import quad

Omega_m0 = 0.315
Omega_r0 = 9.22e-5
Omega_Lambda0 = 0.685
Omega_K0 = 0.0

H0 = 67.36 * 1.02269e-3
```

```
def E(a):
        return np.sqrt(Omega_r0/a**4 + Omega_m0/a**3
                + Omega_K0/a**2 + Omega_Lambda0)

t0 = quad(lambda a: 1/(a*E(a)), 0, 1)[0]/H0

print(f"Age of the Universe = {t0:.4f} Gyr")
```

2.3.6 Exact analytical solutions of the Friedmann equations

Flat single-component universes ($K = 0, \Lambda = 0, w \neq -1$)

For a flat spacetime filled with a single fluid with equation of state w, $\rho \propto a^{-3(1+w)}$, the first Friedmann equation $\dot{a}/a \propto a^{-3(1+w)/2}$ integrates to:

$$a(t) = a_0 \left(\frac{t}{t_0}\right)^{\frac{2}{3(1+w)}}, \qquad a(\eta) = a_0 \left(\frac{\eta}{\eta_0}\right)^{\frac{2}{1+3w}} \quad (w \neq -1/3)\,. \tag{2.3.20}$$

- **Einstein–de Sitter Universe ($w = 0$, Dust):**

$$a(t) \propto t^{2/3}, \qquad a(\eta) \propto \eta^2, \qquad t_0 = \frac{2}{3H_0}. \tag{2.3.21}$$

- **Radiation-Dominated Universe ($w = 1/3$):**

$$a(t) \propto t^{1/2}, \qquad a(\eta) \propto \eta, \qquad t_0 = \frac{1}{2H_0}. \tag{2.3.22}$$

de Sitter spacetime ($K = 0, w = -1$)

When vacuum energy dominates ($\rho = \rho_\Lambda = \text{const}$), $H = H_\Lambda = \sqrt{\Lambda/3} = \text{const}$:

$$a(t) = a_0\, \mathrm{e}^{H_\Lambda t}, \qquad a(\eta) = -\frac{1}{H_\Lambda \eta} \quad (\eta < 0). \tag{2.3.23}$$

In global closed slicing ($K = +1$), de Sitter space is described by:

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + \frac{\cosh^2(H_\Lambda t)}{H_\Lambda^2}\left[\mathrm{d}\chi^2 + \sin^2\chi\, \mathrm{d}\Omega^2\right]. \tag{2.3.24}$$

Curved single-component universes ($K = \pm 1, w \neq -1/3$)

In conformal time, the Friedmann equation $\mathcal{H}^2 = \frac{\Omega_0}{|1-\Omega_0|}(a/a_0)^{-(1+3w)} - K$ integrates to the parametric solution:

$$\frac{a(\eta)}{a_0} = \left(\frac{\Omega_0}{|1-\Omega_0|}\right)^{\frac{1}{1+3w}} \begin{cases} \left[\sin\left(\frac{1+3w}{2}\eta\right)\right]^{\frac{2}{1+3w}} & (K = +1), \\ \left[\sinh\left(\frac{1+3w}{2}\eta\right)\right]^{\frac{2}{1+3w}} & (K = -1). \end{cases} \tag{2.3.25}$$

For a closed matter-dominated universe ($K = +1, w = 0$), this yields the classical **cycloid solution**:

$$a(\eta) = a_{\max} \sin^2\left(\frac{\eta}{2}\right) = \frac{a_{\max}}{2}\left(1 - \cos\eta\right), \qquad t(\eta) = \frac{a_{\max}}{2}\left(\eta - \sin\eta\right). \tag{2.3.26}$$

Flat ΛCDM **universe** $(K=0, w=0, \Lambda>0)$

For the standard cosmological model with cold matter and cosmological constant, the Friedmann equation integrates analytically to:

$$a(t) = a_0 \left(\frac{\Omega_{m0}}{\Omega_{\Lambda 0}}\right)^{1/3} \sinh^{2/3}\left(\frac{3}{2}\sqrt{\Omega_{\Lambda 0}} H_0 t\right). \tag{2.3.27}$$

At early times ($t \ll 1/H_0$), $a(t) \propto t^{2/3}$ (matter domination); at late times ($t \gg 1/H_0$), $a(t) \propto \mathrm{e}^{\sqrt{\Omega_{\Lambda 0}} H_0 t}$ (de Sitter acceleration).

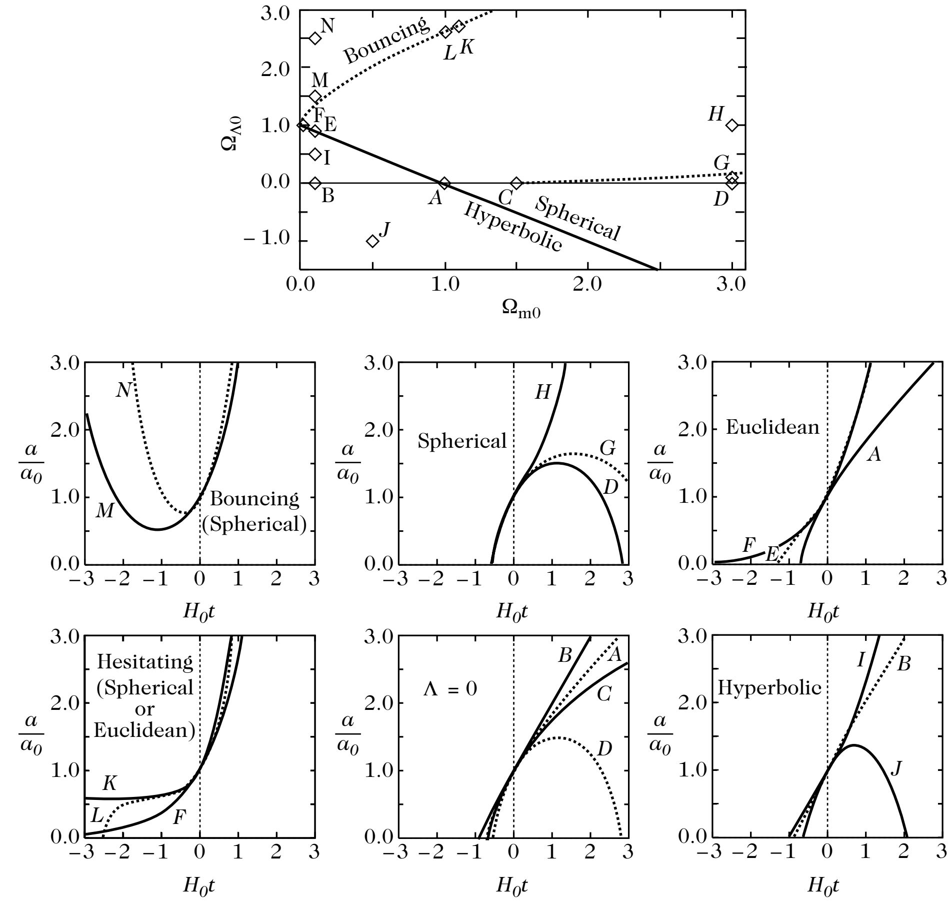


Figure 2.3: Evolution profiles of the cosmic scale factor $a(t)/a_0$ as a function of $H_0 t$ for various cosmological parameter combinations $(\Omega_{m0}, \Omega_{\Lambda 0}, K)$, illustrating Big Bang, Big Crunch, bouncing, hesitating, and de Sitter asymptotic regimes.

2.3.7 Dynamical system analysis in phase space

To analyze the global qualitative dynamics across all parameter regimes, we formulate the Friedmann equations as an autonomous dynamical system.

Defining the **deceleration parameter**:

$$q \equiv -\frac{\ddot{a}a}{\dot{a}^2} = -1 - \frac{\dot{H}}{H^2} = \frac{1}{2}(3\gamma - 2)(1 - \Omega_K) - \frac{3\gamma}{2}\Omega_\Lambda, \tag{2.3.28}$$

with $\gamma \equiv 1 + w$. Introducing the logarithmic e-fold time variable $p \equiv \ln(a/a_0)$, the derivative is $X' \equiv \frac{\mathrm{d}X}{\mathrm{d}p} = \frac{\dot{X}}{H}$.

where we used

$$\frac{\mathrm{d}}{\mathrm{d}\ln\left(\frac{a}{a_0}\right)} = \frac{\mathrm{d}t}{\mathrm{d}\ln\left(\frac{a}{a_0}\right)}\frac{\mathrm{d}}{\mathrm{d}t} = \frac{1}{\frac{\mathrm{d}\ln\left(\frac{a}{a_0}\right)}{\mathrm{d}t}}\frac{\mathrm{d}}{\mathrm{d}t} = \frac{1}{H}\frac{\mathrm{d}}{\mathrm{d}t} \tag{1}$$

The Friedmann equations transform into the 2-dimensional autonomous system:

$$\Omega'_\Lambda = 2\,(1+q)\,\Omega_\Lambda, \tag{2.3.29}$$
$$\Omega'_K = 2q\Omega_K, \tag{2.3.30}$$

with $\Omega_m = 1 - \Omega_\Lambda - \Omega_K$.

Fixed points and linear stability

Setting $\Omega'_\Lambda = 0$ and $\Omega'_K = 0$, we identify three fundamental equilibrium fixed points:

1. **Einstein–de Sitter (EdS) Point** $(\Omega_K, \Omega_\Lambda) = (0,0)$**:** Flat, matter-dominated universe $(\Omega_m = 1)$.
2. **de Sitter (dS) Point** $(\Omega_K, \Omega_\Lambda) = (0,1)$**:** Vacuum-dominated accelerating universe $(\Omega_\Lambda = 1)$.
3. **Milne (M) Point** $(\Omega_K, \Omega_\Lambda) = (1,0)$**:** Empty hyperbolic universe $(\Omega_K = 1)$.

Table 2.1: Linear Stability of Cosmological Fixed Points as a Function of $\gamma = 1 + w$

Fixed Point	$\gamma \in (-\infty, 0)$	$\gamma = 0$	$\gamma \in (0, 2/3)$	$\gamma = 2/3$	$\gamma \in (2/3, +\infty)$
Einstein–de Sitter (EdS)	Attractor	—	Saddle	—	Repeller
de Sitter (dS)	Saddle	Attractor	Attractor	Attractor	Attractor
Milne (M)	Saddle	Repeller	Repeller	Repeller	Saddle

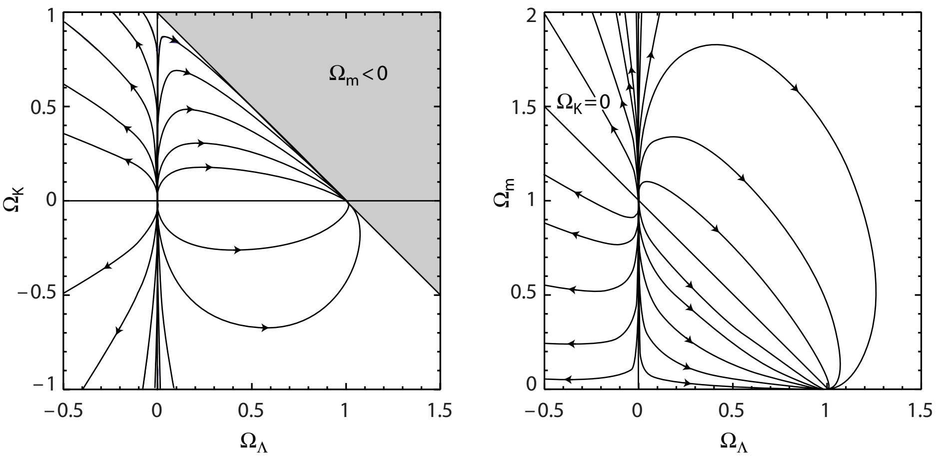


Figure 2.4: Phase space trajectories of the cosmological dynamical system in the $(\Omega_K, \Omega_\Lambda)$ plane (left) and $(\Omega_m, \Omega_\Lambda)$ plane (right) for dust $(\gamma = 1, w = 0)$. Arrows indicate the direction of cosmic time evolution.

Linear stability analysis (Table 2.1 and Figure 2.4) shows that for physical matter ($\gamma \geq 1$), the de Sitter point is the unique global **future attractor**, while the Einstein–de Sitter state is an **unstable repeller** (the flatness problem of standard Big Bang cosmology).

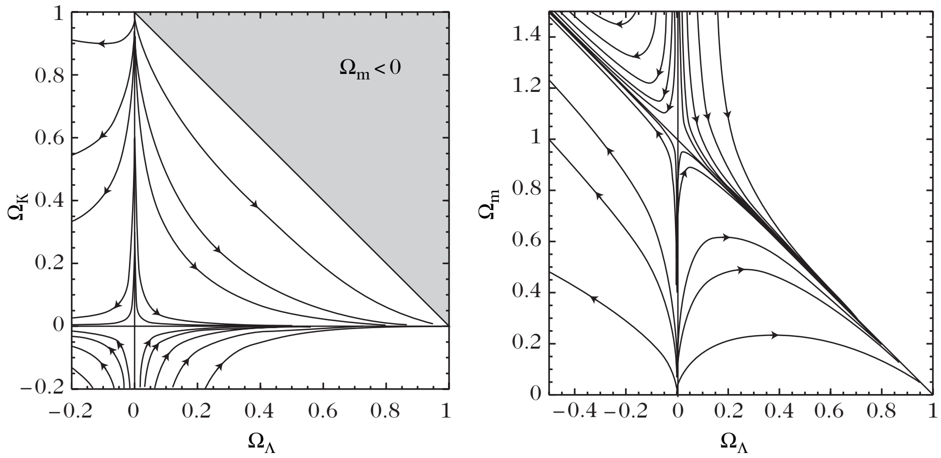


Figure 2.5: Phase space trajectories of the cosmological dynamical system in the $(\Omega_K, \Omega_\Lambda)$ plane (left) and $(\Omega_m, \Omega_\Lambda)$ plane (right) for radiation ($\gamma = 4/3, w = 1/3$).

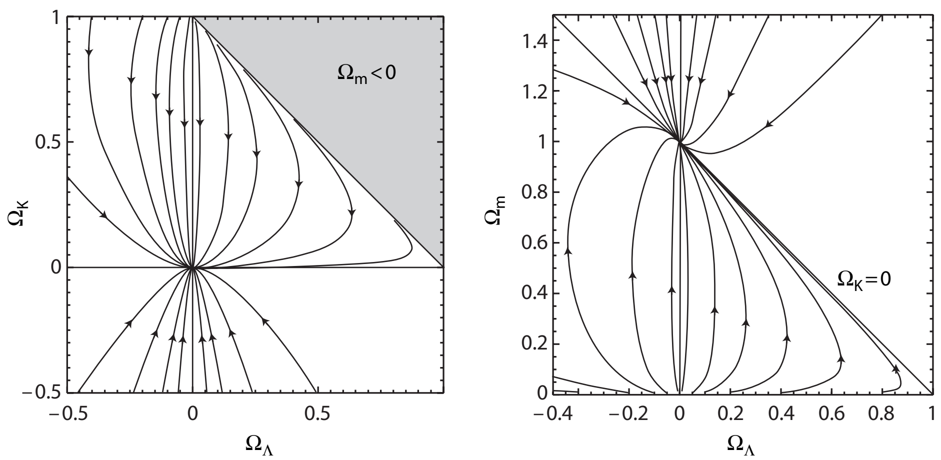


Figure 2.6: Phase space trajectories of the cosmological dynamical system in the $(\Omega_K, \Omega_\Lambda)$ plane (left) and $(\Omega_m, \Omega_\Lambda)$ plane (right) for a phantom / vacuum fluid ($\gamma = 0, w = -1$).

The analytical boundary in the $(\Omega_{m0}, \Omega_{\Lambda0})$ plane separating universes with eternal expansion from those collapsing into a Big Crunch is:

$$\Omega_{\Lambda0} = \begin{cases} 0 & (\Omega_{m0} \leq 1), \\ 1 - \Omega_{m0} + \frac{2}{3} \cos\left(\arctan \Xi_{m0}^{-1} - \pi\right) & (\Omega_{m0} \geq 1), \end{cases} \tag{2.3.31}$$

where $\Xi_{m0} \equiv \sqrt{|2\Omega_{m0} - 1|}$. Similarly, the boundary separating eternally expanding universes from bouncing cosmologies (which lack a past Big Bang singularity) is:

$$\Omega_{\Lambda 0} = \begin{cases} 1 - \Omega_{m0} + \frac{3}{2}\Omega_{m0}^{1/3}\left[(1-\Omega_{m0}+\Xi_{m0})^{1/3} + (1-\Omega_{m0}-\Xi_{m0})^{1/3}\right] & \left(0 \le \Omega_{m0} \le \frac{1}{2}\right), \\ 1 - \Omega_{m0} + \frac{2}{3}\cos\left(\arctan \Xi_{m0}^{-1} + \pi\right) & \left(\Omega_{m0} \ge \frac{1}{2}\right). \end{cases} \tag{2.3.32}$$

2.4 Cosmological timescales and distance measures

The characteristic distance and time scales of an expanding FLRW universe are set by the current Hubble expansion rate $H_0 \equiv 100\,h\,\mathrm{km}\cdot\mathrm{s}^{-1}\cdot\mathrm{Mpc}^{-1}$:

$$t_{H0} \equiv \frac{1}{H_0} = 9.778 \times 10^9\,h^{-1}\,\mathrm{yr} \approx 13.97\,\mathrm{Gyr} \quad (h \approx 0.70), \tag{2.4.1}$$

$$D_{H0} \equiv \frac{c}{H_0} = 2997.9\,h^{-1}\,\mathrm{Mpc} \approx 4280\,\mathrm{Mpc}. \tag{2.4.2}$$

The sphere of radius $D_H(t) \equiv c/H(t)$ is the **Hubble sphere**, outside of which recession velocities exceed the speed of light ($\mathbf{v}_{\mathrm{rec}} > c$).

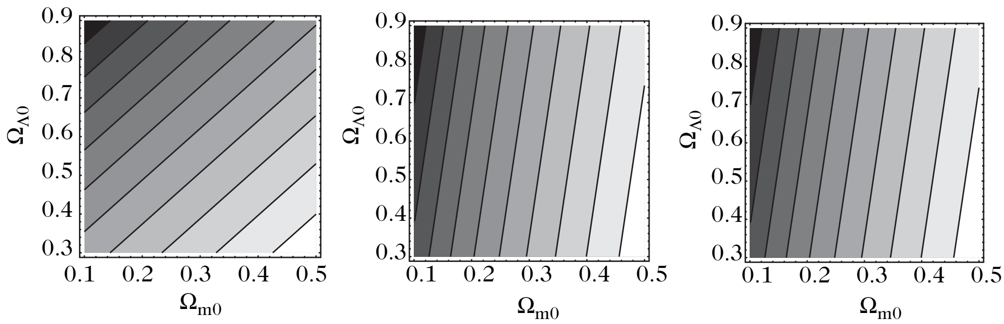


Figure 2.7: Isocontours of the normalized expansion rate function $E(z)$ in the $(\Omega_{m0}, \Omega_{\Lambda 0})$ plane at redshifts $z = 1$ (left), $z = 3$ (middle), and $z = 10$ (right). The degeneracy directions rotate with redshift, demonstrating how multi-epoch observations break cosmological parameter degeneracies.

2.4.1 Age of the Universe and look-back time

From the Hubble definition $H(a) = \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t} \implies \mathrm{d}t = \frac{\mathrm{d}a}{aH(a)} = \frac{\mathrm{d}a}{aH_0E(a)} = \frac{t_{H0}\,\mathrm{d}a}{aE(a)}$, the **cosmic age** of the Universe today is:

$$t_0 = t_{H0}\int_0^1 \frac{a_0}{a}\frac{\mathrm{d}a}{a_0}\frac{1}{E(a)} = t_{H0}\int_0^1 \frac{\mathrm{d}x}{xE(x)} = t_{H0}\int_0^\infty \frac{\mathrm{d}z}{(1+z)E(z)}. \tag{2.4.3}$$

The conformal age is:

$$d\eta = \frac{dt}{a} \implies \eta_0 = t_{H0}\int_0^1 \frac{\mathrm{d}x}{x^2E(x)} = t_{H0}\int_0^\infty \frac{\mathrm{d}z}{E(z)}. \tag{2.4.4}$$

For a light source observed at redshift z_*, the cosmic time at emission was:

$$t(z_*) = t_{H0}\int_{z_*}^\infty \frac{\mathrm{d}z}{(1+z)E(z)}. \tag{2.4.5}$$

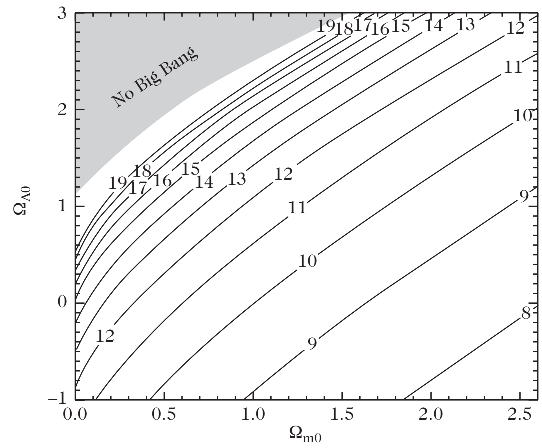


Figure 2.8: Isocontours of the present cosmic age t_0 (in units of 10^9 yr) in the $(\Omega_{m0}, \Omega_{\Lambda 0})$ plane. The shaded region denotes non-singular bouncing models lacking a past Big Bang.

The **look-back time** $\Delta t(z_*) \equiv t_0 - t(z_*)$ represents the elapsed time since the light departed the source:

$$\Delta t(z_*) = t_{H0} \int_0^{z_*} \frac{dz}{(1+z)E(z)}. \tag{2.4.6}$$

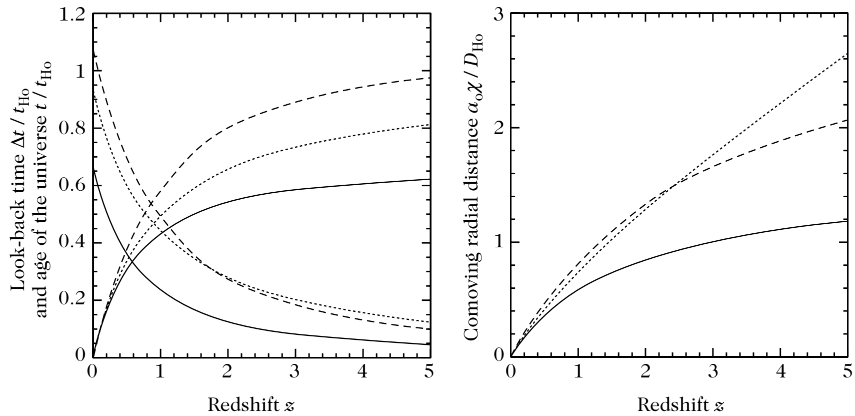


Figure 2.9: (Left) Cosmic age $t(z)/t_{H0}$ and look-back time $\Delta t(z)/t_{H0}$ as a function of redshift z for three cosmological models: Einstein–de Sitter ($\Omega_{m0} = 1, \Omega_{\Lambda 0} = 0$, solid), open ($\Omega_{m0} = 0.05, \Omega_{\Lambda 0} = 0$, dotted), and concordance flat ΛCDM ($\Omega_{m0} = 0.2, \Omega_{\Lambda 0} = 0.8$, dashed). (Right) Dimensionless comoving radial distance $a_0\chi(z)/D_{H0}$ for the same three models.

2.4.2 Comoving radial distance

Along a radial null geodesic arriving at the observer at $\chi = 0$ and $t = t_0$, the FLRW metric gives

$$ds^2 = -c^2\, dt^2 + a^2(t)\, d\chi^2 = 0 \quad \Longrightarrow \quad d\chi = \frac{c\, dt}{a(t)}. \tag{2.4.7}$$

Integrating from emission at t_* to observation at t_0, we obtain

$$a_0\chi(z_*) = ca_0 \int_{t_*}^{t_0} \frac{dt}{a(t)} = c \int_{t_*}^{t_0} \frac{a_0}{a(t)} \frac{d(\frac{a}{a_0})}{(\frac{a}{a_0})H(\frac{a}{a_0})} = \frac{c}{H_0} \int_{x_*}^{1} \frac{dx}{x^2 E(x)}. \tag{2.4.8}$$

Since $x = (1+z)^{-1}$, we have $\mathrm{d}x/x^2 = -\,\mathrm{d}z$, and hence

$$a_0\chi(z_*) = \frac{c}{H_0}\int_0^{z_*}\frac{\mathrm{d}z}{E(z)} = D_{H0}\int_0^{z_*}\frac{\mathrm{d}z}{E(z)}, \tag{2.4.9}$$

where $D_{H0} \equiv c/H_0$ is the present-day Hubble distance. In a flat, matter-dominated universe ($\Omega_{m0} = 1, \Omega_{\Lambda 0} = 0, E(z) = (1+z)^{3/2}$):

$$a_0\chi(z) = 2D_{H0}\left(1 - \frac{1}{\sqrt{1+z}}\right). \tag{2.4.10}$$

2.4.3 Angular diameter distance and reciprocity theorem

Comoving angular diameter

The comoving cross-sectional area of a source subtending a solid angle $\mathrm{d}\Omega_{\mathrm{obs}}$ at comoving coordinate distance χ is $\mathrm{d}S_{\mathrm{com}} = R^2_{\mathrm{ang}}(\chi)\,\mathrm{d}\Omega_{\mathrm{obs}}$, where:

$$R_{\mathrm{ang}}(z) = f_K\left[\chi(z)\right], \tag{2.4.11}$$

with $f_K(\chi)$ given by Eq. (2.1.9).

Physical angular diameter distance

The **angular diameter distance** $D_A(z)$ is defined as the ratio of the source's proper transverse physical diameter $\mathrm{d}\ell_{\mathrm{phys}}$ to its observed angular size $\mathrm{d}\theta_{\mathrm{obs}}$:

$$D_A(z) \equiv \frac{\mathrm{d}\ell_{\mathrm{phys}}}{\mathrm{d}\theta_{\mathrm{obs}}} = \sqrt{\frac{\mathrm{d}S_{\mathrm{phys}}}{\mathrm{d}\Omega_{\mathrm{obs}}}} = a(t_{\mathrm{em}})R_{\mathrm{ang}}(z) = \frac{a_0 f_K\left[\chi(z)\right]}{1+z}. \tag{2.4.12}$$

Because $1+z$ grows in the denominator, $D_A(z)$ is **non-monotonic**: it increases at low redshift, reaches a maximum at $z \approx 1.5$, and then decreases at higher redshifts (Figure 2.10, left). Beyond this turnover redshift, distant objects appear with larger angular sizes on the sky.

Etherington's reciprocity theorem

If light travels along null geodesics and photons are conserved, the angular distance r_s measured from the source to the observer relates to the observer-measured distance $r_o = D_A$ by **Etherington's reciprocity theorem**:

$$r_s = r_o\,(1+z) = D_A(z)\,(1+z)\,. \tag{2.4.13}$$

2.4.4 Luminosity distance and the distance modulus

The luminosity distance

The **luminosity distance** $D_L(z)$ is defined by the inverse-square law relating the intrinsic bolometric luminosity L_{source} to the observed energy flux $\mathcal{F}_{\mathrm{obs}}$:

$$\mathcal{F}_{\mathrm{obs}} \equiv \frac{L_{\mathrm{source}}}{4\pi D_L^2(z)}. \tag{2.4.14}$$

Two relativistic factors reduce the received bolometric flux:

1. **Photon Energy Redshift:** Individual photon energies are redshifted by $E_{\mathrm{obs}} = E_{\mathrm{em}}/(1+z)$.

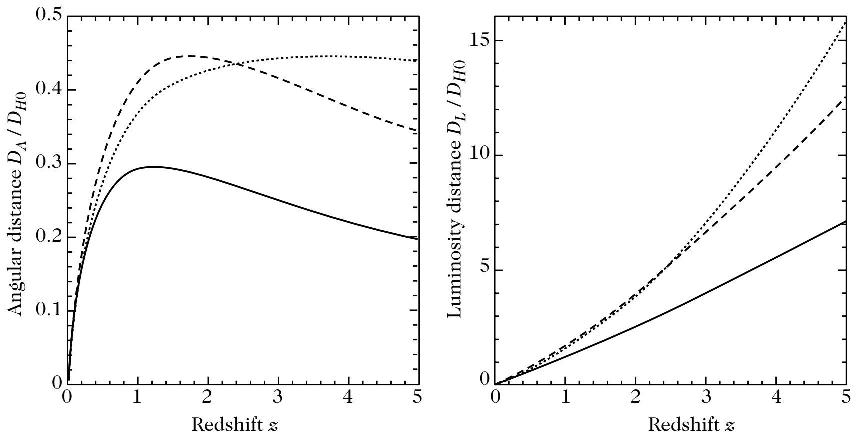


Figure 2.10: (Left) Angular diameter distance $D_A(z)/D_{H0}$ and (Right) Luminosity distance $D_L(z)/D_{H0}$ as a function of redshift z for three cosmological models: Einstein–de Sitter (solid), open universe (dotted), and flat ΛCDM (dashed).

2. **Time Dilation of Arrival Rate:** The arrival rate of photons is dilated by $\Delta t_{\rm obs} = \Delta t_{\rm em}(1+z)$.

Therefore, $L_{\rm obs} = \frac{\Delta E_{\rm obs}}{\Delta t_{\rm obs}} = \frac{\Delta E_{\rm em}}{\Delta t_{\rm em}} \left(\frac{a}{a_0}\right)^2 = L_{\rm source}/(1+z)^2$. Equating $\mathcal{F}_{\rm obs} = \frac{L_{\rm source}/(1+z)^2}{4\pi a_0^2 f_K^2(\chi)}$ with Eq. (2.4.14) yields:

$$D_L(z) = a_0\,(1+z)\,f_K\left[\chi(z)\right]. \tag{2.4.15}$$

Distance duality relation

Comparing Eq. (2.4.12) and Eq. (2.4.15), we obtain the exact **Etherington distance duality relation**:

$$D_L(z) = (1+z)^2\,D_A(z). \tag{2.4.16}$$

This relation is strictly geometric, depending only on Riemannian metric spacetime and photon conservation.

Differentiating $D_L(z)/(1+z)$ enables a direct model-independent reconstruction of the expansion rate $H(z)$:

$$\frac{1}{H(z)} = \frac{1}{c}\left[1 + \Omega_{K0}H_0^2\left(\frac{D_L(z)}{c(1+z)}\right)^2\right]^{-1/2}\frac{\mathrm{d}}{\mathrm{d}z}\left[\frac{D_L(z)}{1+z}\right]. \tag{2.4.17}$$

One can derive this by considering

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}z}\left(\frac{D_L}{1+z}\right) &= a_0\frac{\mathrm{d}f_K}{\mathrm{d}z} = a_0\frac{\mathrm{d}f_K}{\mathrm{d}\chi}\frac{\mathrm{d}\chi}{\mathrm{d}z} &(1)\\ &= \frac{\mathrm{d}f_K}{\mathrm{d}\chi}\frac{1}{H(z)} = \sqrt{1-Kf_K^2}\frac{1}{H(z)} &(2)\\ &= \sqrt{1-\frac{K}{a^2}\left(\frac{D_L}{1+z}\right)^2}\frac{1}{H(z)} &(3)\end{aligned}$$

which using (2.3.16) becomes:

$$\frac{1}{H(z)} = \frac{1}{c}\left[1 + \Omega_{K0}H_0^2\left(\frac{D_L(z)}{c(1+z)}\right)^2\right]^{-1/2}\frac{\mathrm{d}}{\mathrm{d}z}\left[\frac{D_L(z)}{1+z}\right]. \tag{4}$$

Distance modulus and astronomical magnitudes

We usually express luminosity distance in terms of the *distance modulus*, $m - M$. This quantity is defined so that it equals zero when a light source is located at a standard distance of 10 pc:

$$m - M = -2.5 \log \left[\frac{\phi(z)}{\phi(10 \text{ pc})} \right]. \tag{2.4.18}$$

[1]

Weber–Fechner Law

Astronomical magnitudes use a logarithmic scale based on the Weber–Fechner law of human perception. This psychophysical law states that the minimum perceptible change in a stimulus, $\mathrm{d}S$, is directly proportional to the stimulus intensity S, such that $\mathrm{d}S \propto S$. Integrating this relationship yields the mathematical expression:

$$\rho(S) - \rho(S_0) = \alpha \ln \left(\frac{S}{S_0} \right), \tag{1}$$

where S_0 is a reference stimulus level, $\rho(S)$ represents the perceived intensity (sensation), and α is an empirical proportionality constant.

Using Eq. (3.76) and noting that at a distance of 10 pc, the luminosity distance is simply 10 pc, we obtain $m - M = 5 \log[D_{\mathrm{L}}(z)/10 \text{ pc}]$.[2] Substituting $10 \text{ pc} = D_{\mathrm{H}_0}/(3 \times 10^8 h)$ allows us to simplify this relation to:

$$m - M = 25 + 5 \log \left[3000 \frac{D_{\mathrm{L}}}{D_{\mathrm{H}_0}} \right] - 5 \log h. \tag{2.4.19}$$

Here, m is the apparent magnitude, and the absolute magnitude M is defined relative to the Solar luminosity:

$$M - M_\odot = -2.5 \log_{10} \left(\frac{L/4\pi D_L^2}{L_\odot/4\pi D_L^2} \right) \implies M = -2.5 \log \left(\frac{L}{L_\odot} \right) + 4.76, \tag{2.4.20}$$

where $L_\odot \sim 3.8 \times 10^{26}$ W is the Sun's luminosity, and $M_\odot = 4.76$ is its absolute magnitude. Under this definition, brighter objects have lower (or more negative) magnitudes.

3.3.4.4 Magnitudes

The luminosities derived so far represent total power emitted across all wavelengths, defining the *bolometric magnitude*. Because measuring full-spectrum bolometric magnitudes directly is challenging, astronomers use optical filters to isolate specific frequency bands. The magnitude M_X within a given filter band X is defined as:

$$M_X = -2.5 \log \left(\frac{L_X}{L_{\odot X}} \right) + M_{\odot X}. \tag{2.4.21}$$

The standard astronomical filter bands and their corresponding properties are summarized below:

[1] The factor of -2.5 is a historical convention chosen to match the magnitude scale originally established by Hipparcos.

[2] For same source such that L_{source} cancels.

Band	Central Wavelength (nm)	Width (nm)	$M_\odot$
U (ultraviolet)	365	66	5.61
B (blue)	445	94	5.48
V (visible)	551	88	4.64
R (red)	658	138	4.42
I (infra-red)	806	149	4.08
J	1200	213	3.64
K	2190	390	3.28
Bolometric	—	∞	4.76

Light detected in a frequency band $\Delta\nu_X$ was emitted at a higher rest-frame frequency band $(1+z)\Delta\nu_X$ due to cosmic expansion. The observed flux per unit frequency is given by:

$$\phi_X = \frac{\mathrm{d}\phi}{\mathrm{d}\nu_0} = \frac{L_\nu d\nu_e}{4\pi D_L^2\,\mathrm{d}\nu_0} = \frac{L_\nu}{4\pi D_L^2}\frac{\mathrm{d}\nu_e}{\mathrm{d}\nu_o} \implies \phi_X(\nu) = (1+z)\frac{L_\nu[\nu(1+z)]}{4\pi D_\mathrm{L}^2}, \tag{2.4.22}$$

where L_ν is the source luminosity per unit frequency. Integrating over the bandwidth yields:

$$\phi_X(\nu) = \frac{(1+z)}{4\pi D_\mathrm{L}^2}\int_{\Delta\nu_X} L_\nu[\nu(1+z)]\,\mathrm{d}\nu. \tag{2.4.23}$$

Because spectral deformation vanishes when a source is placed at the reference distance of 10 pc, the distance modulus in band X ($m_X - M_X = -2.5\log[\phi_X/\phi_X(10\text{ pc})]$) takes the form:

$$m_X - M_X = 5\log\left(\frac{D_\mathrm{L}}{10\text{ pc}}\right) + K, \tag{2.4.24}$$

where the K-correction accounts for spectral distortion caused by cosmological expansion:

$$K = -2.5\log(1+z) - 2.5\log\left\{\frac{\int_{\Delta\nu_X} L_\nu[\nu(1+z)]\,\mathrm{d}\nu}{\int_{\Delta\nu_X} L_\nu[\nu]\,\mathrm{d}\nu}\right\}. \tag{2.4.25}$$

Additional physical effects, such as interstellar or intergalactic extinction, can further modify the apparent magnitude.

2.4.5 Cosmic volume elements and galaxy number counts

The proper volume element $\mathrm{d}V_\text{prop}$ subtended by a solid angle $\mathrm{d}\Omega$ in the redshift interval $[z, z+\mathrm{d}z]$ is the product of the transverse physical area $\mathrm{d}S_\text{phys} = D_A^2(z)\,\mathrm{d}\Omega$ and proper radial line element $c\ \mathrm{d}t = \frac{D_{H0}\,\mathrm{d}z}{(1+z)E(z)}$:

$$\frac{\mathrm{d}V_\text{prop}}{\mathrm{d}\Omega\,\mathrm{d}z} = D_A^2(z)\frac{c\ \mathrm{d}t}{\mathrm{d}z} = D_{H0}\frac{a_0^2 f_K^2\,[\chi(z)]}{(1+z)^3E(z)}. \tag{2.4.26}$$

For a population of galaxies with proper number density $n_\text{prop}(z)$, the differential number count is:

$$\frac{\mathrm{d}N}{\mathrm{d}\Omega\,\mathrm{d}z} = n_\text{prop}(z)\frac{\mathrm{d}V_\text{prop}}{\mathrm{d}\Omega\,\mathrm{d}z} = D_{H0}n_\text{prop}(z)\frac{a_0^2 f_K^2\,[\chi(z)]}{(1+z)^3E(z)}. \tag{2.4.27}$$

For non-evolving conserved galaxies with comoving number density n_com, $n_\text{prop}(z) = n_\text{com}(1+z)^3$, simplifying the count to $\frac{\mathrm{d}N}{\mathrm{d}\Omega\mathrm{d}z} = D_{H0}n_\text{com}\frac{a_0^2 f_K^2[\chi(z)]}{E(z)}$.

The cumulative **optical depth** $\tau(z)$ for a line of sight to intersect a galaxy of cross-sectional area $\sigma = \pi r_*^2$ up to redshift z is:

$$\tau(z) = \pi r_*^2 D_{H0}\int_0^z \frac{n_\text{prop}(z')}{(1+z')E(z')}\ \mathrm{d}z'\,. \tag{2.4.28}$$

2.5 Kinematic expansions at small redshifts ($z \ll 1$)

2.5.1 Taylor expansion of the scale factor and deceleration parameter

Taylor expanding the scale factor $a(t)$ around the present epoch t_0:

$$a(t) = a_0 \left[1 + H_0 (t - t_0) - \frac{1}{2} q_0 H_0^2 (t - t_0)^2 + \frac{1}{6} j_0 H_0^3 (t - t_0)^3 + \dots \right], \tag{2.5.1}$$

where $H_0 \equiv \frac{\dot{a}_0}{a_0}$ is the Hubble constant, and q_0 is the dimensionless **deceleration parameter**:

$$q_0 \equiv -\frac{\ddot{a}_0 a_0}{\dot{a}_0^2} = - \left. \frac{\ddot{a}}{a H^2} \right|_{t=t_0}. \tag{2.5.2}$$

The sign convention is historical: $q_0 > 0$ corresponds to decelerating expansion, while $q_0 < 0$ signifies cosmic acceleration.

From the second Friedmann equation (2.3.3), q_0 is directly linked to the present cosmic energy content:

$$q_0 = \frac{1}{2} \sum_X (1 + 3 w_X) \Omega_{X0} = \frac{1}{2} \Omega_{m0} + \Omega_{r0} - \Omega_{\Lambda 0}. \tag{2.5.3}$$

For a flat ΛCDM universe ($\Omega_{m0} \approx 0.3, \Omega_{\Lambda 0} \approx 0.7$), $q_0 = \frac{1}{2}(0.3) - 0.7 = -0.55 < 0$, confirming current cosmic acceleration.

2.5.2 Small-z expansions of cosmological observables

Expanding $E(z) = \sqrt{\Omega_{m0}(1+z)^3 + \Omega_{K0}(1+z)^2 + \Omega_{\Lambda 0}}$ to second order in z:

$$E(z) = 1 + (1 + q_0) z + \mathcal{O}\left(z^2\right). \tag{2.5.4}$$

Integrating the distance and time relations yields the classical low-redshift kinematic expansions:

$$\Delta t(z) = t_{H0} \left[z - \frac{1}{2} (q_0 + 2) z^2 + \mathcal{O}\left(z^3\right) \right], \tag{2.5.5}$$

$$a_0 \chi(z) = D_{H0} \left[z - \frac{1}{2} (1 + q_0) z^2 + \mathcal{O}\left(z^3\right) \right], \tag{2.5.6}$$

$$D_L(z) = D_{H0} \left[z + \frac{1}{2} (1 - q_0) z^2 + \mathcal{O}\left(z^3\right) \right], \tag{2.5.7}$$

$$D_A(z) = D_{H0} \left[z - \frac{1}{2} (3 + q_0) z^2 + \mathcal{O}\left(z^3\right) \right]. \tag{2.5.8}$$

The distance modulus at low redshift is:

$$\mu(z) = 25 + 5 \log_{10} \left(\frac{cz}{H_0 \, \mathrm{Mpc}} \right) + 1.086 (1 - q_0) z + \mathcal{O}\left(z^2\right), \tag{2.5.9}$$

providing the foundational formula for measuring H_0 and q_0 using Type Ia supernovae.

2.5.3 The cosmic redshift drift (Sandage–Loeb effect)

If an astronomical source is observed at epoch t_0 with redshift z and monitored over a time interval δt_0, the cosmic expansion induces a continuous secular **redshift drift**:

$$1 + z + \delta z = \frac{a(t_0 + \delta t_0)}{a(t_{\mathrm{em}} + \delta t_{\mathrm{em}})}. \tag{2.5.10}$$

Using $\delta t_{\text{em}} = \delta t_0/(1+z)$, the exact time rate of change of redshift is:

$$\frac{\delta z}{\delta t_0} = H_0\left[(1+z) - E(z)\right]. \tag{2.5.11}$$

At small redshift ($z \ll 1$), this reduces to:

$$\frac{\delta z}{\delta t_0} = -q_0 H_0 z + \mathcal{O}\left(z^2\right). \tag{2.5.12}$$

Though tiny ($\delta z/\delta t_0 \sim 10^{-8}\,\text{century}^{-1}$), measuring this spectroscopic drift provides a direct, non-geometric probe of cosmic acceleration free of source astrophysical evolution.

2.6 Cosmological horizons and global conformal structure

An **horizon** is a null or spacelike hypersurface that divides spacetime into causally accessible and inaccessible regions (Figures 2.11 and 2.12).

and 3.12.

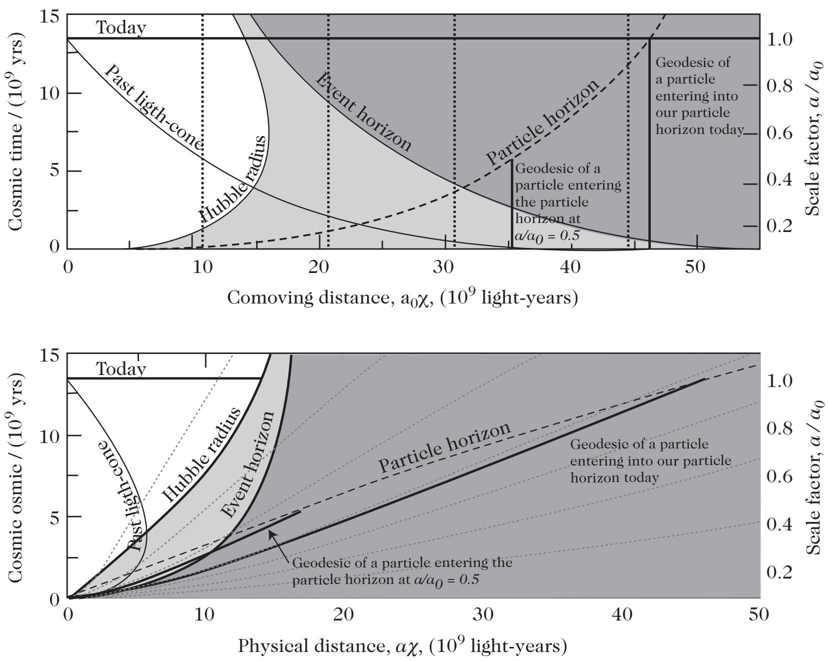


Figure 2.11: Particle horizon geometry in comoving coordinates (top) and physical coordinates (bottom) for a ΛCDM universe ($\Omega_{m0} = 0.3, \Omega_{\Lambda 0} = 0.7, h = 0.7$). The particle horizon $\chi = \varphi(t)$ expands monotonically, engulfing more comoving fundamental observers over cosmic time.

2.6.1 The cosmological event horizon

For a fundamental observer O, the **event horizon** is the boundary separating events that can ever emit a signal that reaches O from events that remain forever causally disconnected from O.

A necessary and sufficient condition for the existence of an event horizon is the convergence of the conformal time integral as $t \to \infty$:

$$\chi_{\text{event}}(t_0) = \int_{t_0}^{\infty} \frac{c\,\mathrm{d}t}{a(t)} < \infty. \tag{2.6.1}$$

Any photon emitted at epoch t_0 at comoving distance $\chi > \chi_{\text{event}}(t_0)$ will never reach the observer.

A photon on the $t = t_0$ hypersurface at $\chi = \chi_0$ reaches the observer at origin at future infinity. If $\chi_0 = \infty$ then all of $t = t_0$ hypersurface is observable otherwise the observer can only observe events on $t = t_0$ hypersurface from $\chi = 0$ to $\chi = \chi_0$.

For single-fluid power-law universes $a(t) \propto t^n$, the integral converges if and only if $n > 1$, which translates to:

$$w < -\frac{1}{3} \iff \rho + 3P < 0 \quad \text{(Violation of Strong Energy Condition)}. \tag{2.6.2}$$

In de Sitter spacetime ($a = e^{Ht}$), the physical event horizon radius is static:

$$R_{\text{event}}(t_0) = a(t_0)\chi_{\text{event}}(t_0) = \frac{c}{H} = \text{constant}. \tag{2.6.3}$$

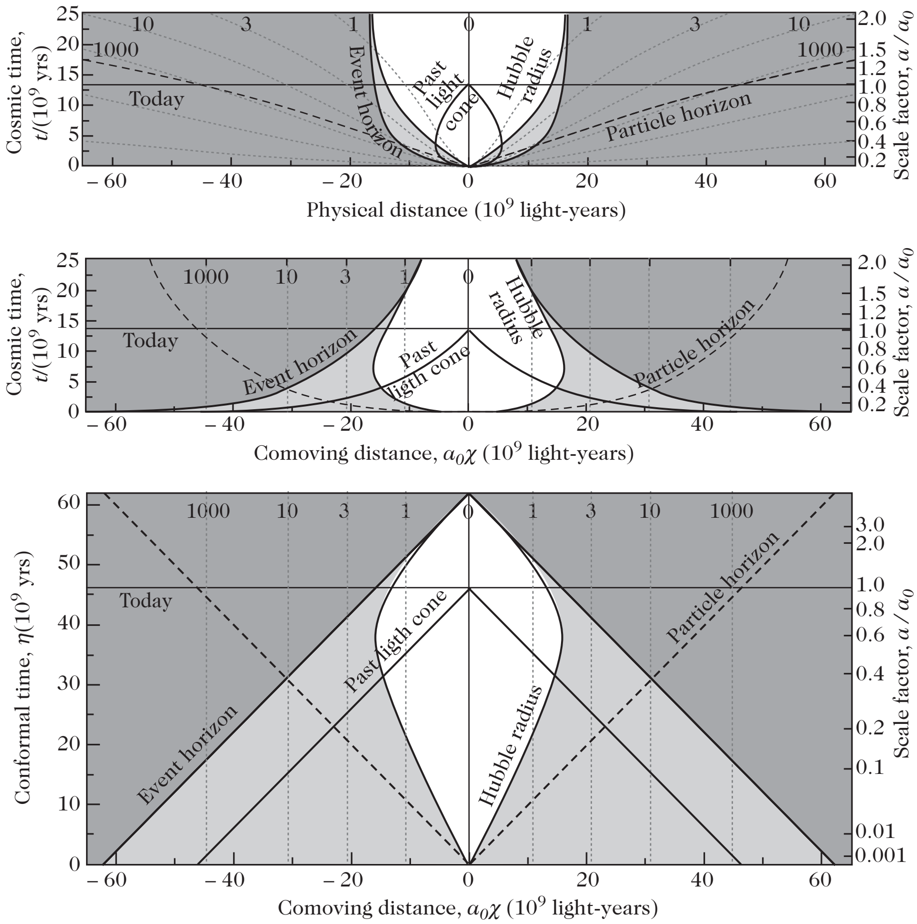


Figure 2.12: Spacetime diagrams for a concordance ΛCDM model ($\Omega_{m0} = 0.3, \Omega_{\Lambda 0} = 0.7, h = 0.7$) plotted in: (Top) physical coordinates $(a\chi, t)$, (Middle) comoving coordinates (χ, t), and (Bottom) conformal coordinates (χ, η). Shown are comoving galaxy worldlines (dotted), past light cone (bold line), event horizon (bold curve), Hubble sphere (solid line), and particle horizon (dashed line).

2.6.2 The cosmological particle horizon

The **particle horizon** at epoch t_0 is the 2-sphere dividing all particles into those that have been observable by O since the initial singularity ($t = 0$) and those that have not yet entered causal contact.

A particle horizon exists if and only if the past conformal time integral converges as $t \to 0$:

$$\chi_{\text{particle}}(t_0) = \int_0^{t_0} \frac{c\, \mathrm{d}t}{a(t)} < \infty. \tag{2.6.4}$$

For a power-law scale factor $a(t) \propto t^n$, a particle horizon exists if and only if $n < 1$:

$$w > -\frac{1}{3} \iff \rho + 3P > 0 \quad \text{(Standard decelerating cosmology)}. \tag{2.6.5}$$

In single-component power-law cosmologies, the event horizon and particle horizon are **mutually exclusive**:

- $w > -1/3$ (matter, radiation): Particle horizon exists; Event horizon does not.
- $w < -1/3$ (dark energy, de Sitter inflation): Event horizon exists; Particle horizon does not.

The physical diameter of the particle horizon at cosmic time t_2 for an initial emission epoch $t_1 \ll t_2$ is:

$$D_{\text{p.h.}}(t_2) = 2a(t_2) \int_0^{t_2} \frac{c\, \mathrm{d}t}{a(t)} = \frac{4}{1+3w} \frac{c}{H(t_2)} = \frac{4}{1+3w} D_H(t_2). \tag{2.6.6}$$

For matter domination ($w = 0$), $D_{\text{p.h.}}(t) = 4D_H(t) = 6ct$; for radiation domination ($w = 1/3$), $D_{\text{p.h.}}(t) = 2D_H(t) = 4ct$.

2.6.3 Global conformal structure and Penrose–Carter diagrams

To analyze the global asymptotic causal structure of cosmological spacetimes, we perform a conformal transformation $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$ mapping infinite physical spacetime onto a compact manifold $\widetilde{\mathcal{M}}$ with boundary $\partial\widetilde{\mathcal{M}} = \mathscr{I}$ (scri), where light cones remain at $\pm 45°$.

Minkowski spacetime as a conformal archetype

In advanced and retarded null coordinates $v = t + r$ and $u = t - r$, defining $V = \arctan v$ and $U = \arctan u$, followed by $T = V + U$ and $R = V - U$, the Minkowski metric is conformally embedded in the **Einstein static cylinder** $\mathbb{R} \times S^3$:

$$\mathrm{d}\bar{s}^2 = -\,\mathrm{d}T^2 + \mathrm{d}R^2 + \sin^2 R\, \mathrm{d}\Omega^2\,, \qquad -\pi < T \pm R < \pi, \quad R \geq 0. \tag{2.6.7}$$

The boundary consists of:

- $\mathscr{I}^+$ (Future Null Infinity) and $\mathscr{I}^-$ (Past Null Infinity): 3D null hypersurfaces where outgoing/ingoing null rays terminate and originate.
- i^+ (Future Timelike Infinity) and i^- (Past Timelike Infinity): Vertices where timelike geodesics end and begin.
- i^0 (Spatial Infinity): Endpoint of all spacelike geodesics.

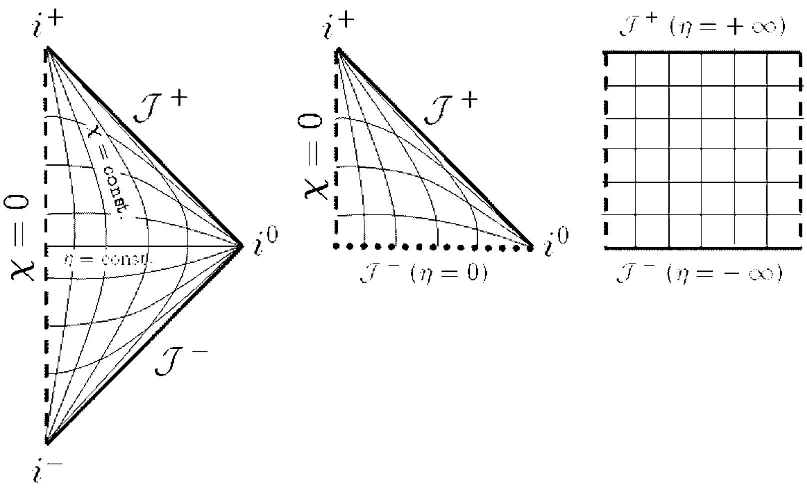


Figure 2.13: Conformal Penrose diagrams: (Left) Minkowski spacetime diamond, (Middle) Spatially flat FLRW universe with $\Lambda = 0, P > 0$ featuring an initial spacelike Big Bang singularity, and (Right) de Sitter spacetime slab with spacelike past and future conformal infinities $\mathscr{I}^{\pm}$.

Conformal diagrams of FLRW and de Sitter spacetimes

Because FLRW spacetimes are conformally related to static manifolds ($\mathrm{d}s^2 = a^2(\eta)[-\,\mathrm{d}\eta^2 + \mathrm{d}\sigma_K^2]$), their conformal diagrams correspond to sub-regions of the Einstein cylinder (Figures 2.13 and 2.14):

1. **Flat FLRW ($K = 0, \Lambda = 0, w \geq 0$):** Conformal time range $\eta \in (0, \infty)$ maps to the upper half of the Minkowski diamond ($T > 0$). The Big Bang at $\eta = 0$ is a **spacelike initial singularity boundary** (Figure 2.13, middle).

2. **de Sitter Spacetime:** The conformal time runs over $\eta \in (-\infty, 0)$ in flat coordinates, or $T \in (0, \pi)$ in global closed slicing, forming a square strip whose past and future boundaries $\mathscr{I}^{\pm}$ are **spacelike** (Figure 2.13, right).

3. **Closed FLRW ($K = +1$):** Conformal time spans $\eta \in (0, 2\pi)$ for cycloid dust collapse (spacelike Big Bang and Big Crunch), $\eta \in (0, \infty)$ for hesitating/Lemaître universes, and $\eta \in (-\infty, \infty)$ for non-singular bouncing models (Figure 2.14).

Theorem 2.1 (Conformal Horizon Criterion)**.** The existence of cosmological horizons is determined by the causal nature of the conformal boundaries:

$$\mathscr{I}^- \text{ is spacelike} \iff \text{Existence of a Particle Horizon,} \tag{2.6.8}$$

$$\mathscr{I}^+ \text{ is spacelike} \iff \text{Existence of an Event Horizon.} \tag{2.6.9}$$

2.7 Beyond the cosmological principle: anisotropic and inhomogeneous spacetimes

While the Friedmann–Lemaître model assumes maximal spatial symmetry (6 Killing vectors), testing the robustness of standard cosmological predictions requires examining exact solutions of Einstein's field equations with reduced symmetry groups.

If $\mathcal{J}^+$ is space-like, the worldline of any fundamental terminates on a point O of $\mathcal{J}^+$ and the past light cone of O divides the Universe into events that can be seen by the observer and events that he can never see; there is en event horizon. If $\mathcal{J}^+$ is null, all the worldlines will pass through the vertex i^+, so that $0 = i^+$ and its past light cone is $\mathcal{J}^+$ and there is no event horizon.

In conclusion,

$$\boxed{\begin{aligned} &\mathcal{J}^- \text{ space-like} \iff \text{existence of a particle horizon,} \\ &\mathcal{J}^+ \text{ space-like} \iff \text{existence of an event horizon.} \end{aligned}}$$

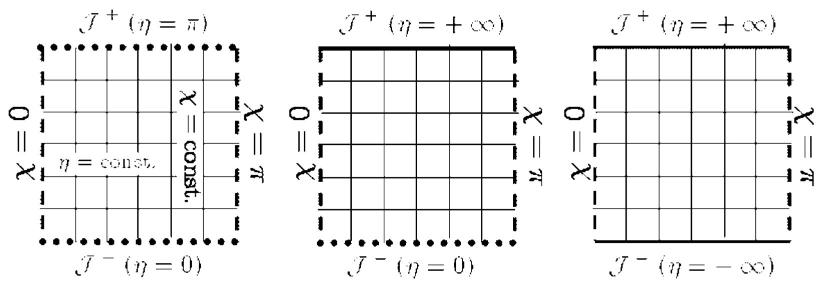


Figure 2.14: Conformal diagrams of FLRW spacetimes with spherical spatial sections ($K = +1$): (Left) Big Bang to Big Crunch cycloid ($\Lambda = 0$), (Middle) Hesitating Lemaître universe with asymptotic de Sitter future ($\Lambda > 0$), and (Right) Non-singular bouncing cosmology ($\Lambda > 0$). Dashed horizontal boundaries represent curvature singularities.

2.7.1 Classification of spacetime isometries and Lie algebras

Continuous symmetries are generated by Killing vector fields satisfying Killing's equation $\mathcal{L}_\xi g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = 0$.

The vector space of Killing fields forms a finite-dimensional Lie algebra of dimension $r \leq n(n+1)/2 = 10$ with Lie bracket defined by the commutator:

$$[\xi_a, \xi_b]^\alpha \equiv \xi_a^\beta \partial_\beta \xi_b^\alpha - \xi_b^\beta \partial_\beta \xi_a^\alpha = C^c{}_{ab} \xi_c^\alpha , \tag{2.7.1}$$

where $C^c{}_{ab} = -C^c{}_{ba}$ are the **structure constants** of the isometry group, satisfying the Jacobi identity:

$$C^e{}_{[ab} C^d{}_{c]e} = 0. \tag{2.7.2}$$

2.7.2 Classification of cosmological models

The isometry group of a 4-dimensional spacetime manifold $\mathcal{M}$ is characterized by:

- $s = \dim(\text{surfaces of transitivity / orbits}) \leq 4$,
- $q = \dim(\text{isotropy subgroup at a point}) \leq 3$,
- $r = s + q \leq 10$, the total dimension of the Lie algebra of Killing vector fields.

For physical cosmological fluids satisfying the null energy condition ($\rho + P > 0$), the 4-velocity u^μ is an invariant timelike vector, restricting spatial isotropy to subgroups of SO(3), so that $q \in \{0, 1, 3\}$.

Table 2.2: Classification of Cosmological Spacetimes by Homogeneity (s) and Isotropy (q)

Isotropy Group	**Spacetime Homogeneous** ($s = 4$)	**Spatially Homogeneous** ($s = 3$)	**Inhomogeneous** ($s = 2$)
$q = 6$ (Maximal 4D)	Minkowski, de Sitter, Anti-de Sitter	—	—
$q = 3$ (Maximal 3D)	Einstein static universe	**Friedmann–Lemaître (FLRW)**	—
$q = 1$ (Axial Symmetry)	LRS Bianchi types	LRS Bianchi, Kantowski–Sachs	**Lemaître–Tolman–Bondi**
$q = 0$ (Anisotropic)	—	**General Bianchi Types I–IX**	Szekeres models

2.7.3 Spatially inhomogeneous spacetimes

The Lemaître–Tolman–Bondi (LTB) metric

The simplest exact inhomogeneous cosmological solution is the **Lemaître–Tolman–Bondi (LTB)** spacetime ($s = 2, q = 1, r = 3$), describing a spherically symmetric, pressureless dust fluid ($P = 0$):

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + S^2(t,r)\,\mathrm{d}r^2 + R^2(t,r)\,\mathrm{d}\Omega^2 \,. \tag{2.7.3}$$

Einstein's field equations reduce to three coupled equations:

$$S(t,r) = \frac{R'(t,r)}{\sqrt{1 - K(r) r^2}}, \tag{2.7.4}$$

$$\dot{R}^2(t,r) = \frac{2G_\mathrm{N} M(r)}{R(t,r)} - K(r) r^2, \tag{2.7.5}$$

$$4\pi G_{\mathrm{N}}\rho(t,r) = \frac{M'(r)}{R^2(t,r)R'(t,r)}, \tag{2.7.6}$$

where $R' \equiv \frac{\partial R}{\partial r}$ and $\dot{R} \equiv \frac{\partial R}{\partial t}$.

The general solution is specified by three arbitrary radial functions:

1. $M(r)$: The gravitational mass enclosed within the comoving shell r.
2. $E(r) \equiv -\frac{1}{2}K(r)r^2$: The local spatial curvature profile determining whether shells are locally hyperbolic, flat, or elliptic.
3. $t_0(r)$: The inhomogeneous "bang time" function describing the local epoch at which $R(t_0(r), r) = 0$.

The standard FLRW metric is recovered when $R(t,r) = a(t)r$, $K(r) = K$, and $t_0(r) = 0$. LTB models are widely used to analyze nonlinear cosmic voids, local Hubble bubbles, and backreaction effects. In FLRW, one has rotational symmetry about every point which is broken here and we only have rotational symmetry about the origin.

The Swiss-cheese model

Inhomogeneous Swiss-cheese models are constructed by cutting spherical cavities out of a background FLRW universe and inserting concentric spherically symmetric Schwarzschild mass distributions. By Darmois–Israel junction conditions, matching across the boundary requires:

$$M_{\mathrm{Schw}} = \frac{4\pi}{3}\left[\rho a^3 R^3\right]_{\mathrm{FLRW}}. \tag{2.7.7}$$

Because the cavity mass equals the missing FLRW mass, Birkhoff's theorem ensures that the gravitational field outside the cavity is completely unperturbed: the local inhomogeneity is gravitationally screened.

2.7.4 Spatially homogeneous anisotropic spacetimes (Bianchi models)

Spatially homogeneous models ($s = 3$) possess a 3-dimensional isometry group acting transitively on spatial slices Σ_t. The Lie algebra structure constants $C^c{}_{ab}$ can be decomposed as:

$$C^c{}_{ab} = \varepsilon_{dab}n^{cd} + \delta^c_b A_a - \delta^c_a A_b, \tag{2.7.8}$$

where $n^{ab} = \mathrm{diag}(n_1, n_2, n_3)$ with $n_i \in \{0, \pm 1\}$, $A_b = (a, 0, 0)$, and the Jacobi identity requires $an_1 = 0$.

This classifies 3D Lie algebras into the **9 Bianchi types** (I through IX), partitioned into **Class A** ($a = 0$) and **Class B** ($a \neq 0$).

The Bianchi type I universe and shear dominance

The simplest anisotropic cosmology is the **Bianchi I** spacetime with flat spatial sections:

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + X^2(t)\,\mathrm{d}x^2 + Y^2(t)\,\mathrm{d}y^2 + Z^2(t)\,\mathrm{d}z^2 = -\,\mathrm{d}t^2 + S^2(t)\mathrm{e}^{2\beta_i(t)}\,\mathrm{d}x^i\,\mathrm{d}x^i\,, \tag{2.7.9}$$

where $S(t) \equiv (XYZ)^{1/3}$ is the mean scale factor and $\sum_{i=1}^3 \beta_i(t) = 0$.

The kinematic shear tensor $\sigma_{ij} = \frac{1}{2}\dot{\gamma}_{ij}$ evolves as:

$$\sigma^i{}_j = \frac{\Sigma^i{}_j}{S^3(t)}, \qquad \sigma^2 \equiv \frac{1}{2}\sigma_{ij}\sigma^{ij} = \frac{\Sigma^2}{2S^6(t)}, \tag{2.7.10}$$

where $\Sigma^i{}_j$ is a constant trace-free matrix.

Table 2.3: Bianchi Classification of Spatially Homogeneous Anisotropic Spacetimes

Class	**Bianchi Type**	a	n_1	n_2	n_3	**Isotropic FLRW Limit**
Class A ($a = 0$)	**I**	0	0	0	0	Flat FLRW ($K = 0$)
	II	0	0	0	1	—
	$\mathbf{VII}_0$	0	0	1	1	Flat FLRW ($K = 0$)
	$\mathbf{VI}_0$	0	0	1	−1	—
	IX	0	1	1	1	Closed FLRW ($K = +1$, Mixmaster)
	VIII	0	1	1	−1	—
Class B ($a \neq 0$)	**V**	1	0	0	0	Open FLRW ($K = -1$)
	IV	1	0	0	1	—
	III	1	0	1	−1	—
	$\mathbf{VII}_a$	a	0	1	1	Open FLRW ($K = -1$)
	$\mathbf{VI}_a$	a	0	1	−1	—

The generalized Friedmann equation is:

$$3\left(\frac{\dot{S}}{S}\right)^2 = 8\pi G_{\rm N}\rho(t) + \frac{\Sigma^2}{2S^6(t)}. \tag{2.7.11}$$

Because the shear energy density scales as $\rho_\sigma \propto S^{-6}$, it **strictly dominates all matter and radiation components** ($\rho_m \propto S^{-3}, \rho_r \propto S^{-4}$) as $S \to 0$ near the initial singularity:

$$\lim_{S\to 0} \frac{\rho_\sigma}{\rho_{\rm fluid}} \to \infty. \tag{2.7.12}$$

In the vacuum shear-dominated limit (the **Kasner metric**):

$$X(t) \propto t^{p_1}, \quad Y(t) \propto t^{p_2}, \quad Z(t) \propto t^{p_3}, \qquad \sum_{i=1}^{3} p_i = 1, \quad \sum_{i=1}^{3} p_i^2 = 1. \tag{2.7.13}$$

Except for the flat static case $(1, 0, 0)$, one index is always negative ($p_1 < 0 < p_2 \leq p_3$): space contracts along one spatial axis while expanding along the other two.

As the universe expands ($S \to \infty$), the shear decays rapidly ($\rho_\sigma \propto S^{-6} \to 0$), and the spacetime undergoes natural **isotropization**, converging asymptotically to the isotropic Friedmann–Lemaître solution.

Chapter 3

The Standard Cosmological Model: Observational Tests and Paradigm

The standard Big Bang model rests upon four empirical pillars established across observational astronomy and nuclear astrophysics:

1. **The Hubble Expansion of Spacetime:** The systematic recession velocity of galaxies proportional to distance ($v = H_0 d$).
2. **The Cosmic Microwave Background (CMB):** An isotropic, thermal blackbody radiation field at $T_0 = 2.7255\,\mathrm{K}$ predicted by George Gamow, Ralph Alpher, and Robert Herman (1948) and discovered by Arno Penzias and Robert Wilson (1965).
3. **Big Bang Nucleosynthesis (BBN):** The primordial synthesis of light elements ($^4\mathrm{He}, \mathrm{D}, ^3\mathrm{He}, ^7\mathrm{Li}$) during the first three minutes of cosmic time.
4. **The Growth of Cosmological Large-Scale Structure:** The gravitational amplification of primordial density perturbations into the cosmic web of galaxies and clusters.

This chapter details the primary observational methods that transformed Friedmann's mathematical solutions into the physical standard cosmological paradigm (ΛCDM).

3.1 The Hubble diagram and cosmic age constraints

3.1.1 The Hubble expansion law

In an expanding FLRW spacetime, the spectral lines of distant galaxies undergo a systematic cosmological redshift $z = \frac{\Delta\lambda}{\lambda_{\mathrm{em}}}$. At low redshifts ($z \ll 1$), the apparent recession velocity $v \approx cz$ relates to physical distance d via the **Hubble Law**:

$$v = H_0 d \iff z = \frac{H_0}{c} D_L(z), \tag{3.1.1}$$

where the slope directly yields the **Hubble constant** $H_0 \equiv 100\,h\,\mathrm{km \cdot s^{-1} \cdot Mpc^{-1}}$.

To determine H_0 reliably, astronomical observations must probe galaxies at distances where cosmic recession velocities dominate over local peculiar velocities ($v_{\mathrm{pec}} \approx 300$–$600\,\mathrm{km \cdot s^{-1}}$ induced by local gravitational clustering). Probing galaxies beyond $v \gtrsim 10^4\,\mathrm{km \cdot s^{-1}}$ ($z \gtrsim 0.03$) reduces peculiar velocity contamination below 3%.

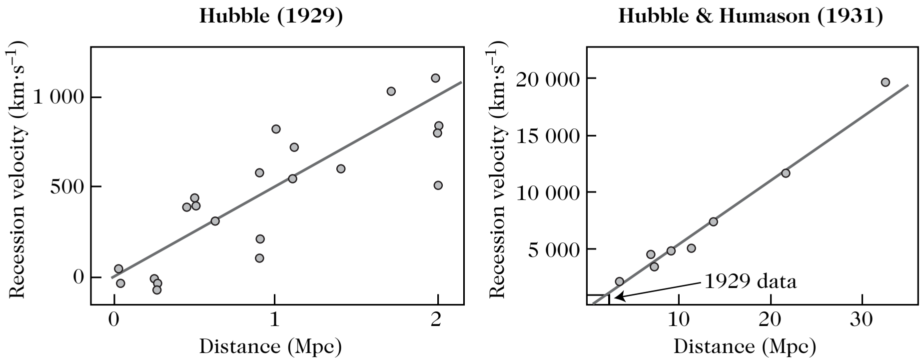


Figure 3.1: Historical Hubble diagrams: (Left) Edwin Hubble's landmark 1929 discovery using 18 galaxies out to $v \approx 1000\,\mathrm{km}\cdot\mathrm{s}^{-1}$, yielding an initial overestimate $H_0 \approx 500\,\mathrm{km}\cdot\mathrm{s}^{-1}\cdot\mathrm{Mpc}^{-1}$. (Right) Hubble and Humason's 1931 extension out to $v \approx 20\,000\,\mathrm{km}\cdot\mathrm{s}^{-1}$ confirming linearity across larger cosmic volumes.

3.1.2 The extragalactic distance ladder

Because direct geometric trigonometric parallaxes were historically limited to nearby stars ($\lesssim 100\,\mathrm{pc}$ prior to space astrometry like Gaia), determining extragalactic distances requires an overlapping **cosmic distance ladder**:

1. **Geometric Primary Calibrators:** Trigonometric parallax (Hipparcos, Gaia), detached eclipsing binaries in the LMC, and water megamasers in active galactic nuclei (e.g., NGC 4258).

2. **Pulsating Standard Candles:** Classical **Cepheid variable stars**, obeying the Henrietta Leavitt Period-Luminosity relation:

$$M_V = -\alpha \log_{10} P + \beta, \tag{3.1.2}$$

   and Tip of the Red Giant Branch (TRGB) stars.

3. **Secondary Standard Candles:** **Type Ia Supernovae (SNe Ia)**, Tully–Fisher relation for spiral galaxies ($L \propto v_{\mathrm{rot}}^4$), and the Fundamental Plane for elliptical galaxies, extending into the unperturbed Hubble flow ($z \sim 0.01$–0.1).

Table 3.1: Measurements of the Hubble Constant H_0 from the Cosmic Distance Ladder (HST Key Project)

Astronomical Method / Tracer	**Measured H_0 ($\mathrm{km}\cdot\mathrm{s}^{-1}\cdot\mathrm{Mpc}^{-1}$)**
Type Ia Supernovae (SN Ia)	$71 \pm 2_{\mathrm{stat}} \pm 6_{\mathrm{sys}}$
Tully–Fisher Relation	$71 \pm 3_{\mathrm{stat}} \pm 7_{\mathrm{sys}}$
Surface Brightness Fluctuations (SBF)	$70 \pm 5_{\mathrm{stat}} \pm 6_{\mathrm{sys}}$
Type II Supernovae (SN II)	$72 \pm 9_{\mathrm{stat}} \pm 7_{\mathrm{sys}}$
Fundamental Plane of Ellipticals	$82 \pm 2_{\mathrm{stat}} \pm 6_{\mathrm{sys}}$
HST Key Project Combined	$\mathbf{72 \pm 8}$

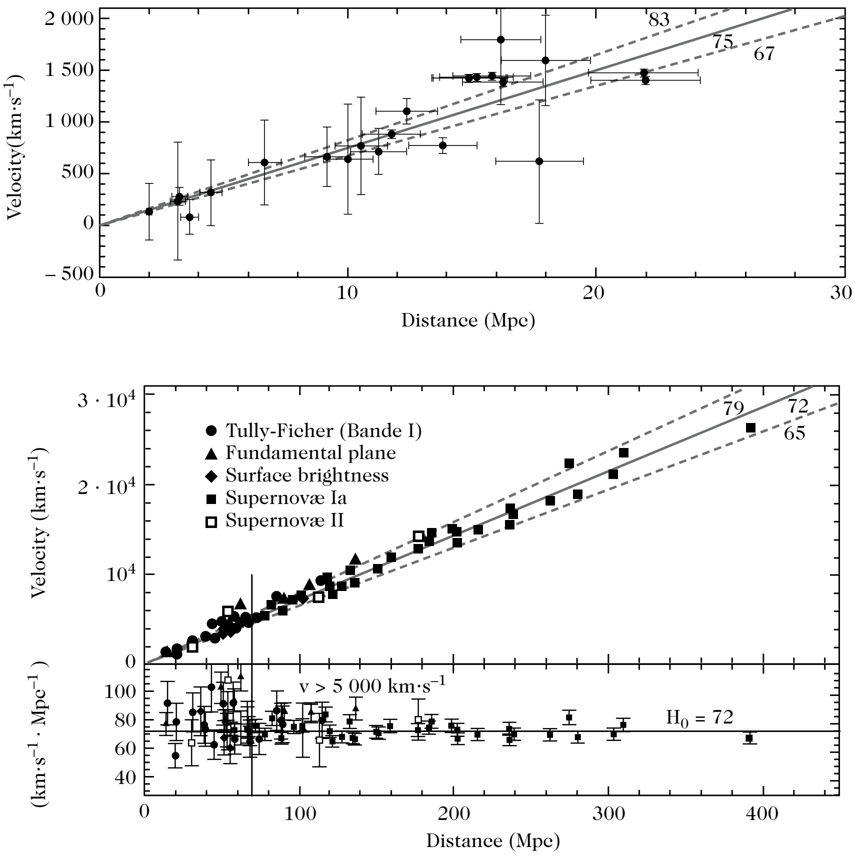


Figure 3.2: (Top) Hubble diagram for Cepheids obtained by the HST Key Project out to $\sim 20\,$Mpc. (Bottom) Combined Hubble diagram extending out to $\sim 300\,$Mpc utilizing secondary standard candles (Type Ia supernovae, Tully–Fisher, Fundamental Plane, surface brightness fluctuations, and Type II supernovae).

3.1.3 Physical, non-ladder methods for measuring H_0

In parallel with distance ladder methods, three independent physical techniques determine H_0 directly:

1. **Expanding Photosphere Method (EPM) of Type II SNe:** Combining the spectroscopic Doppler expansion velocity $v_{\rm phot}$ of the supernova shock front with photometric blackbody angular diameter $\theta = 2R_{\rm phot}/D_L$ yields $D_L = \frac{2v_{\rm phot}(t-t_{\rm exp})}{\theta}$.
2. **Gravitational Lensing Time Delays:** Measuring the differential arrival time Δt_{AB} between multiple images A and B of a time-variable gravitationally lensed quasar:
$$\Delta t_{AB} = \frac{1+z_L}{c}\frac{D_L D_S}{D_{LS}}\left[\frac{1}{2}\left(\boldsymbol{\theta}_A - \boldsymbol{\beta}\right)^2 - \psi(\boldsymbol{\theta}_A) - \frac{1}{2}\left(\boldsymbol{\theta}_B - \boldsymbol{\beta}\right)^2 + \psi(\boldsymbol{\theta}_B)\right] \propto H_0^{-1}. \quad (3.1.3)$$
3. **Sunyaev–Zel'dovich (SZ) Effect and X-Ray Cluster Emission:** Combining the redshift-independent CMB spectral distortion $\Delta T_{\rm SZ} \propto \int n_e T_e \, {\rm d}\ell$ with the X-ray bremsstrahlung surface brightness $S_X \propto D_A^{-1} \int n_e^2 T_e^{1/2} \, {\rm d}\ell$ isolates the physical cluster depth $\ell_{\rm cluster} = D_A\theta$, determining $D_A \propto H_0^{-1}$.

Table 3.2: Independent Physical Determinations of the Hubble Constant H_0

Physical Method	Value of H_0 ($\mathrm{km \cdot s^{-1} \cdot Mpc^{-1}}$)	Reference
SN II Expanding Photosphere	73 ± 15	Schmidt et al. (1994)
Gravitational Lensing Delay	$72 \pm 7_{\rm stat} \pm 15_{\rm sys}$	Tonry & Franx (1998)
Gravitational Lensing (B1608)	60 ± 20	Fassnacht et al. (2000)
Thermal Sunyaev–Zel'dovich	60 ± 4	Reese et al. (2002)
WMAP 1-Year Primary CMB	72 ± 5	Spergel et al. (2003)

3.2 Type Ia supernovae and the discovery of cosmic acceleration

3.2.1 Standardizing Type Ia supernovae

Type Ia supernovae originate from the thermonuclear disruption of a carbon-oxygen white dwarf accreting mass from a binary companion until approaching the Chandrasekhar limit $M_{\rm Ch} \approx 1.44\,M_\odot$. Because the exploding mass is uniform, the peak absolute magnitude is nearly universal ($M_B \approx -19.3\,$mag).

The residual intrinsic dispersion ($\sim 0.3\,$mag) is calibrated empirically using the **Phillips relation**: supernovae with broader, slower-decaying light curves are intrinsically more luminous at peak, standardizing distances to $\sim 6\%$ precision.

3.2.2 The 1998 discovery of cosmic acceleration

In 1998, the **Supernova Cosmology Project (SCP)** and the **High-Z Supernova Search Team (HZT)** independently measured the luminosity distances of high-redshift SNe Ia ($z \approx$ 0.2–0.8), finding that distant supernovae are systematically $\sim 0.25\,$mag fainter (more distant) than expected in a matter-dominated decelerating universe (Figure 3.3).

At second order in redshift, the luminosity distance is $D_L(z) = \frac{cz}{H_0}[1 + \frac{1}{2}(1-q_0)z + \mathcal{O}(z^2)]$. The supernova data established:

$$q_0 = \frac{1}{2}\Omega_{m0} - \Omega_{\Lambda 0} < 0 \quad \text{(at } 2.8\sigma \text{ confidence)}, \quad (3.2.1)$$

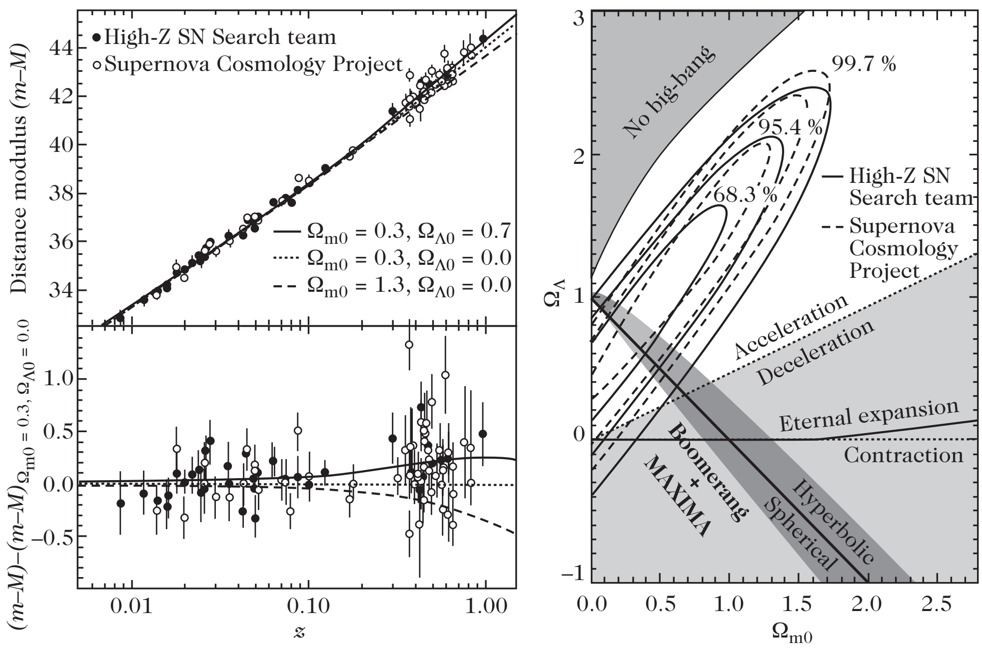


Figure 3.3: (Left) Hubble diagram of high-z Type Ia supernovae from SCP and HZT compared with FLRW models. The lower panel shows distance modulus residuals relative to an empty universe ($\Omega_{m0} = 0.3, \Omega_{\Lambda 0} = 0$). (Right) Joint statistical likelihood contours in the $(\Omega_{m0}, \Omega_{\Lambda 0})$ plane, demonstrating $q_0 < 0$ and isolating $\Omega_{\Lambda 0} \approx 0.7, \Omega_{m0} \approx 0.3$ when combined with CMB balloon data.

which translates to the linear constraint relation:

$$8\Omega_{m0} - 6\Omega_{\Lambda 0} \simeq -2 \pm 1. \tag{3.2.2}$$

Combined with CMB evidence for spatial flatness ($\Omega_{m0} + \Omega_{\Lambda 0} \approx 1$), the concordance cosmological parameters are:

$$\Omega_{\Lambda 0} \approx 0.70, \qquad \Omega_{m0} \approx 0.30. \tag{3.2.3}$$

3.2.3 Dark energy equation of state

Generalizing beyond a cosmological constant ($w = -1$), dark energy is described by an effective fluid with equation of state $P_X = w_X \rho_X$. Assuming constant w_X in a flat universe, SNe Ia constrain $w_X < -0.6$ (Figure 3.5, left).

For a dynamical dark energy component parameterized by the Chevallier–Polarski–Linder (CPL) ansatz:

$$w(a) = w_0 + w_a (1 - a) = w_0 + w_a \frac{z}{1+z}, \tag{3.2.4}$$

joint cosmological probes constrain:

$$w_0 = -0.98 \pm 0.19, \qquad w_a = 0.05^{+0.83}_{-0.65}, \tag{3.2.5}$$

consistent with a static cosmological constant ($w_0 = -1, w_a = 0$).

3.3 Cosmic age determinations and the age crisis resolution

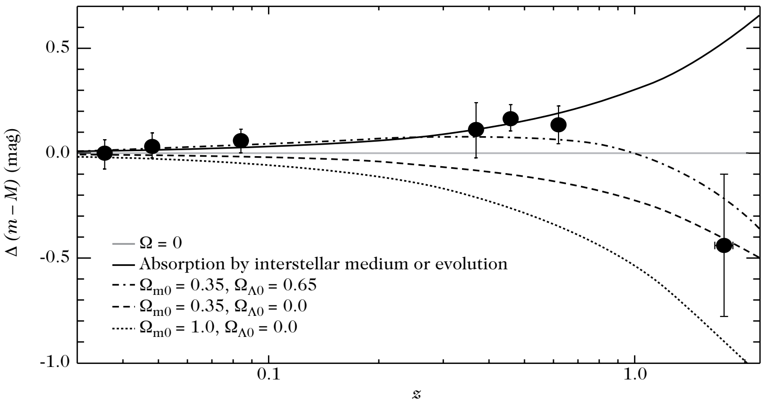


Figure 3.4: Residual Hubble diagram $\Delta(m - M)$ relative to an empty universe $\Omega = 0$. The detection of very high-redshift SNe Ia at $z \approx 1.7$ (HST GOODS survey) rules out monotonic astrophysical dimming (grey intergalactic dust or progenitor metallicity evolution) and confirms the transition from past deceleration ($z > 0.5$) to present acceleration ($z < 0.5$).

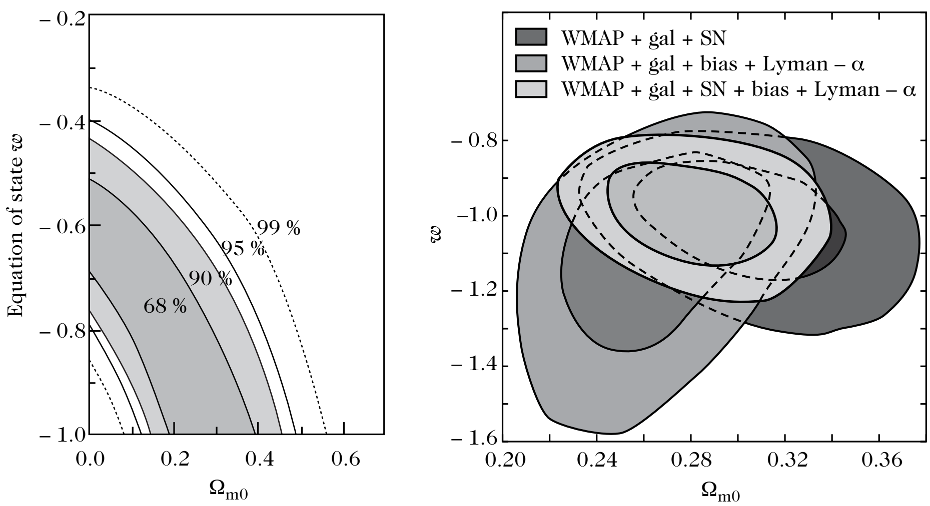


Figure 3.5: (Left) Joint constraints on constant dark energy equation of state w and Ω_{m0} in a flat universe. (Right) 68% and 95% confidence regions combining CMB (WMAP), SDSS galaxy clustering, SNe Ia, and Lyman-α forest data.

3.3.1 Astrophysical age indicators in the Milky Way

The age of the Universe t_u must strictly exceed the ages of its oldest stellar constituents ($t_u > t_{\rm star}$). Three independent astrophysical chronometers determine stellar ages:

1. **Nuclear Cosmochronology:** Dating ultra-metal-poor halo stars (e.g., CS 22892, CS 31082-0018) via the radioactive decay ratios of rapid neutron-capture (r-process) isotopes:
   - Thorium-232 (^{232}Th, half-life $\tau_{1/2} = 14.05\,$Gyr): $t_* = 15.2 \pm 3.7\,$Gyr.
   - Uranium-238 (^{238}U, half-life $\tau_{1/2} = 4.47\,$Gyr): $t_* = 12.5 \pm 3.0\,$Gyr.
2. **Cooling of Galactic White Dwarfs:** Modeling the fading luminosity function of degenerate white dwarfs in the solar neighborhood yields a Galactic disk age $t_{\rm disk} = 9.8^{+1.1}_{-0.8}\,$Gyr.
3. **Globular Cluster Main-Sequence Turnoff (MSTO):** The luminosity at the main-sequence turnoff in globular clusters is governed by core hydrogen exhaustion: $L_{\rm MSTO} \propto M_{\rm MSTO}^{3.5} \implies t_{\rm GC} \propto M_{\rm MSTO}/L_{\rm MSTO} \propto L_{\rm MSTO}^{-0.71}$. Absolute isochrone fitting yields $t_{\rm GC} \approx$ 11.5–14.0 Gyr.

Table 3.3: Summary of Astrophysical Age Constraints on the Universe

Chronometer / Method	**Stellar / Cluster Age (Gyr)**	**Reference**
Nucleochronology (^{232}Th, CS 22892)	15.2 ± 3.7	Sneden et al. (1996)
Nucleochronology (^{238}U, CS 31082)	12.5 ± 3.0	Cayrel et al. (2001)
Globular Clusters (MSTO Isochrones)	11.5 ± 1.3	Chaboyer et al. (1998)
Hipparcos-calibrated MSTO	11.8 ± 1.2	Gratton et al. (1997)
Eclipsing Binaries in Clusters	14.0 ± 1.2	Pont et al. (1998)
WMAP + Large-Scale Structure	$\mathbf{13.7 \pm 0.2}$	Spergel et al. (2003)

Accounting for an incubation time of $\Delta t_{\rm form} \approx 0.5$–$1.5\,$Gyr for the first stars and galaxies to form ($z \approx 6$–20), the empirical age of the Universe is bounded by:

$$9.6\,\text{Gyr} \le t_u \le 15.4\,\text{Gyr}. \tag{3.3.1}$$

3.3.2 Resolution of the cosmological age crisis

In a matter-dominated Einstein–de Sitter universe ($\Omega_m = 1, \Omega_\Lambda = 0$), the expansion age is:

$$t_0^{\rm EdS} = \frac{2}{3H_0} = 6.52\,h^{-1}\,\text{Gyr} \approx 9.3\,\text{Gyr} \quad (h \approx 0.70). \tag{3.3.2}$$

This conflicted with the ages of globular clusters ($t_{\rm GC} \ge 12\,$Gyr), creating the historic **cosmic age crisis**.

In a flat ΛCDM universe with dark energy, the accelerated expansion stretches the cosmic age to:

$$t_0^{\Lambda\rm CDM} = \frac{2}{3H_0\sqrt{\Omega_{\Lambda 0}}}\,\text{arcsinh}\left(\sqrt{\frac{\Omega_{\Lambda 0}}{\Omega_{m0}}}\right) \approx \frac{0.96}{H_0} \approx 13.8\,\text{Gyr}, \tag{3.3.3}$$

reconciling theoretical expansion dynamics with stellar nuclear chronometers.

3.4 Thermodynamics of the expanding cosmic plasma

3.4.1 The radiation-dominated early Universe

Because relativistic radiation energy density scales as $\rho_r \propto a^{-4}$ while non-relativistic matter scales as $\rho_m \propto a^{-3}$, the early Universe was radiation-dominated.

Equating $\rho_m(z_{\rm eq}) = \rho_r(z_{\rm eq})$ defines the **matter-radiation equality redshift**:

$$1 + z_{\rm eq} = \frac{\Omega_{m0}}{\Omega_{r0}} \simeq 3612\,\Theta_{2.7}^{-4}\left(\frac{\Omega_{m0}h^2}{0.15}\right), \tag{3.4.1}$$

where $\Theta_{2.7} \equiv T_{\rm CMB}/2.7255\,\mathrm{K}$.

The corresponding photon plasma temperature at matter-radiation equality is:

$$T_{\rm eq} = T_{\rm CMB}\,(1 + z_{\rm eq}) \simeq 5.65\,\Theta_{2.7}^{-3}\left(\Omega_{m0}h^2\right)\,\mathrm{eV} \approx 6.56 \times 10^4\,\mathrm{K}. \tag{3.4.2}$$

When the thermal energy exceeds particle rest masses ($k_{\rm B}T \gtrsim 2mc^2$), relativistic particle–antiparticle pairs ($e^+e^-, \mu^+\mu^-, q\bar{q}$) are produced in equilibrium with the photon bath, increasing the effective relativistic degrees of freedom.

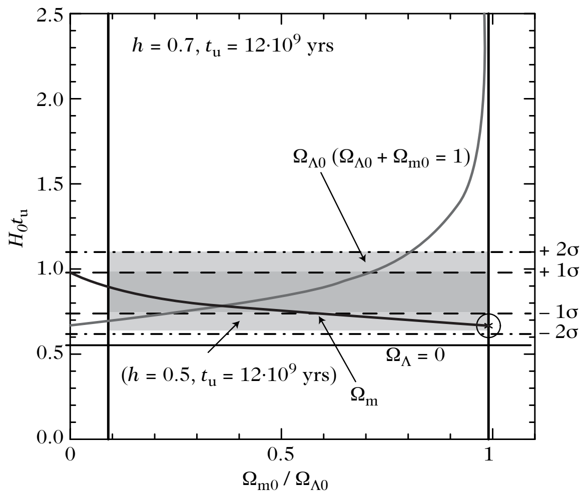


Figure 3.6: Comparison between the empirical constraint on H_0t_u (from stellar ages $t_u \approx 12\,$Gyr and $h \approx 0.7$, shaded bands) and the theoretical dynamical expansion age $H_0t_0(\Omega_{m0}, \Omega_{\Lambda 0})$. The dark curve denotes flat universes ($\Omega_{m0} + \Omega_{\Lambda 0} = 1$) vs. $\Omega_{\Lambda 0}$; the light curve denotes $\Omega_{\Lambda 0} = 0$ vs. Ω_{m0}. The Einstein–de Sitter point ($H_0t_0 = 2/3$) is excluded at $> 1.5\sigma$.

3.4.2 Equilibrium relativistic thermodynamics

Phase-space distribution functions

Under spatial homogeneity and isotropy, the phase-space distribution function $f_i(\mathbf{x}, \mathbf{p}, t)$ of particle species i depends only on the particle energy $E = \sqrt{p^2 + m_i^2}$ and cosmic time t through the local temperature $T_i(t)$.

For a gas in local thermodynamic equilibrium (kinetic and chemical), $f_i(E, T)$ is given by the **Fermi–Dirac** (+) or **Bose–Einstein** (−) distribution:

$$f_i(E, T) = \frac{1}{\exp\left[\frac{E-\mu_i}{T_i(t)}\right] \pm 1}, \tag{3.4.3}$$

where g_i is the internal spin/polarization degeneracy factor, and μ_i is the chemical potential.

The covariant definition of distribution function take the form of

$$f_i(U,T) = \frac{1}{\exp\left[\frac{-p_\mu U^\mu - \mu_i}{T_i(t)}\right] \pm 1}, \tag{1}$$

Thermal Decoupling and Freeze-Out Criteria

In the early Universe, cosmic expansion acts as a competitive dynamical process against particle interactions. A particle species i maintains local thermodynamic equilibrium (both thermal and chemical) with the background plasma if its interaction rate, $\Gamma_i \equiv n_i \langle \sigma v \rangle$, significantly exceeds the Hubble expansion rate $H(t) \equiv \dot{a}/a$:

$$\Gamma_i(T) \gg H(T) \iff \tau_{\text{coll}} \ll \tau_H \quad \text{(Thermal Equilibrium)}, \tag{3.4.4}$$

where $\tau_{\text{coll}} = \Gamma_i^{-1}$ is the mean free time between scattering events and $\tau_H = H^{-1}$ represents the characteristic timescale of space expansion.

Thermal Evolution and Cosmic Expansion. As the Universe undergoes adiabatic expansion, entropy conservation in a comoving volume ($S \propto g_{*S} T^3 a^3 = \text{const.}$) implies that the temperature of the cosmic plasma scales inversely with the scale factor:

$$T \propto \frac{1}{a(t)}. \tag{3.4.5}$$

Consequently, physical volume elements expand as $V \propto a^3 \propto T^{-3}$. During the radiation-dominated era, the total energy density is dominated by relativistic species ($\rho \approx \rho_{\text{rad}} = \frac{\pi^2}{30} g_* T^4$). The Friedmann equation then dictates the temperature dependence of the expansion rate:

$$H^2 = \frac{8\pi G}{3}\rho_{\text{rad}} \implies H(T) = \left(\frac{8\pi^3 g_*}{90}\right)^{1/2} \frac{T^2}{M_{\text{Pl}}} \propto T^2 \propto \frac{1}{a^2}, \tag{3.4.6}$$

where $M_{\text{Pl}} \equiv G^{-1/2} \approx 1.22 \times 10^{19}$ GeV is the Planck mass.

Interaction Rate Scaling. The interaction rate $\Gamma_i = n_i \langle \sigma v \rangle$ depends strongly on whether the species is relativistic ($T \gg m_i$) or non-relativistic ($T \ll m_i$). For a relativistic species, the number density scales purely with the thermal background as $n_i \propto T^3 \propto a^{-3}$. Considering an interaction cross-section that scales thermally as $\sigma \sim G^2 E^n \sim G^2 T^n$ (where G is the coupling constant with dimensions of mass^{-2}):

$$\Gamma_i(T) \sim n_i \sigma v \propto T^3 \cdot T^n = T^{n+3}. \tag{3.4.7}$$

Comparing the interaction rate to the Hubble parameter yields the dimensionless ratio governing equilibrium:

$$\frac{\Gamma_i(T)}{H(T)} \propto \frac{T^{n+3}}{T^2/M_{\text{Pl}}} \propto M_{\text{Pl}} T^{n+1} \propto a^{-(n+1)}. \tag{3.4.8}$$

Freeze-Out Condition. For weak interactions mediated by massive gauge bosons below the electroweak scale (such as standard light neutrino decoupling), $n = 2$ and the cross-section scales as $\sigma \sim G_{\text{F}}^2 T^2$, where G_{F} is the Fermi constant. Thus:

$$\frac{\Gamma}{H} \propto T^3 \propto \frac{1}{a^3}. \tag{3.4.9}$$

Because $\Gamma(T)$ falls faster than $H(T)$ as $a(t)$ grows, the interaction rate eventually drops below the expansion rate. The system undergoes **thermal freeze-out / decoupling** at a characteristic decoupling temperature T_D (or scale factor a_D):

$$\Gamma_i(T_D) \approx H(T_D) \implies T_D = \text{Decoupling Temperature.} \tag{3.4.10}$$

For $T < T_D$ (or $a > a_D$), $\Gamma_i < H$, meaning the time required for particles to interact becomes longer than the age of the Universe. The species drops out of chemical and thermal equilibrium, maintaining a frozen comoving number density ($Y_i \equiv n_i/s = \text{const.}$) and free-streaming along null or timelike geodesics.

Thermodynamic macroscopic integrals

Integrating over isotropic momentum space $\mathrm{d}^3p = 4\pi p^2\ \mathrm{d}p = 4\pi\sqrt{E^2 - m_i^2}E\ \mathrm{d}E$, the particle number density n_i, energy density ρ_i, and isotropic pressure P_i are:

$$n_i(T) = \frac{g_i}{(2\pi)^3}\int f_i(\mathbf{p})\,\mathrm{d}^3p = \frac{g_i}{2\pi^2}\int_{m_i}^{\infty}\frac{\sqrt{E^2 - m_i^2}E\ \mathrm{d}E}{\exp\left(\frac{E-\mu_i}{T}\right) \pm 1}, \tag{3.4.11}$$

$$\rho_i(T) = \frac{g_i}{(2\pi)^3}\int E(p)f_i(\mathbf{p})\,\mathrm{d}^3p = \frac{g_i}{2\pi^2}\int_{m_i}^{\infty}\frac{\sqrt{E^2 - m_i^2}E^2\ \mathrm{d}E}{\exp\left(\frac{E-\mu_i}{T}\right) \pm 1}, \tag{3.4.12}$$

$$P_i(T) = \frac{g_i}{(2\pi)^3}\int \frac{p^2}{3E}f_i(\mathbf{p})\,\mathrm{d}^3p = \frac{g_i}{6\pi^2}\int_{m_i}^{\infty}\frac{(E^2 - m_i^2)^{3/2}\ \mathrm{d}E}{\exp\left(\frac{E-\mu_i}{T}\right) \pm 1}. \tag{3.4.13}$$

Table 3.4: Asymptotic Limits of Thermodynamic Quantities for Relativistic and Non-Relativistic Species

Regime	**Statistics**	**Number Density** n	**Energy Density** ρ	**Pressure** P
Relativistic ($T \gg m, \lvert\mu\rvert \ll T$)	**Bosons** **Fermions**	$g\frac{\zeta(3)}{\pi^2}T^3$ $\frac{3}{4}g\frac{\zeta(3)}{\pi^2}T^3$	$g\frac{\pi^2}{30}T^4$ $\frac{7}{8}g\frac{\pi^2}{30}T^4$	$\frac{1}{3}\rho$ $\frac{1}{3}\rho$
Degenerate Fermi ($\mu \gg T \gg m$)	**Fermions**	$g\frac{\mu^3}{6\pi^2}$	$g\frac{\mu^4}{8\pi^2}$	$\frac{1}{3}\rho$
Non-Relativistic ($T \ll m$)	**Both**	$g\left(\frac{mT}{2\pi}\right)^{3/2}\mathrm{e}^{\frac{\mu-m}{T}}$	$mn + \frac{3}{2}nT \approx mn$	$nT \ll \rho$

Energy Density

We will now, explicitly find the energy density, pressure and number density in the early universe.

$$\begin{aligned}\rho^{\text{eq}} &= \sum_{\text{all species}}\frac{g_i}{(2\pi\hbar)^3}\int_0^{\infty}\frac{\sqrt{p^2c^2 + m^2c^4}}{\exp\left(\sqrt{p^2c^2+m^2c^4}-\mu/k_BT\right) \pm 1}4\pi p^2 dp \\ &= \sum_{\text{boson}}\frac{g_i}{2\pi^2\hbar^3}\int_0^{\infty}dp\frac{p^2\sqrt{p^2c^2 + m^2c^4}}{\exp\left(\sqrt{p^2c^2+m^2c^4}-\mu/k_BT\right) - 1}\end{aligned}$$

$$+ \sum_{\text{fermion}} \frac{g_i}{2\pi^2\hbar^3} \int_0^\infty dp \frac{p^2\sqrt{p^2c^2+m^2c^4}}{\exp\left(\sqrt{p^2c^2+m^2c^4}-\mu/k_BT\right)+1}$$

defining $x \equiv mc^2/k_BT$ and $\xi \equiv pc/k_BT$, we get

$$= \sum_{\text{boson}} \frac{g_i}{2\pi^2\hbar^3} \int_0^\infty \frac{k_BT}{c} d\xi \frac{\frac{k_B^2T^2}{c^2}\xi^2\sqrt{k_B^2T^2\xi^2+k_B^2T^2x^2}}{\exp\left(\sqrt{\xi^2+x^2}-\mu/k_BT\right)-1}$$

$$+ \sum_{\text{fermion}} \frac{g_i}{2\pi^2\hbar^3} \int_0^\infty \frac{k_BT}{c} d\xi \frac{\frac{k_B^2T^2}{c^2}\xi^2\sqrt{k_B^2T^2\xi^2+k_B^2T^2}}{\exp\left(\sqrt{\xi^2+x^2}-\mu/k_BT\right)+1}$$

$$= \frac{k_B^4T^4}{2\pi^2\hbar^3c^3}\left[\sum_{\text{boson}} g_i \int_0^\infty d\xi \frac{\xi^2\sqrt{\xi^2+x^2}}{\exp\left(\sqrt{\xi^2+x^2}-\mu/k_BT\right)-1}\right.$$

$$\left. + \sum_{\text{fermion}} g_i \int_0^\infty d\xi \frac{\xi^2\sqrt{\xi^2+x^2}}{\exp\left(\sqrt{\xi^2+x^2}-\mu/k_BT\right)+1}\right]$$

Ultra-Relativistic Limit In the early universe, $\mu \approx 0$. However, if we assume that $x \to 0$, then the integrals can be evaluated exactly and this condition is referred to as ultra-relativistic limit.

$$\begin{aligned}
\rho^{\text{eq}} &= \frac{k_B^4T^4}{2\pi^2\hbar^3c^3}\left[\sum_{\text{boson}} g_i \int_0^\infty d\xi \frac{\xi^3}{e^\xi-1} + \sum_{\text{fermion}} g_i \int_0^\infty d\xi \frac{\xi^3}{e^\xi+1}\right] \\
&= \frac{k_B^4T^4}{2\pi^2\hbar^3c^3}\left[\sum_{\text{boson}} g_i \int_0^\infty d\xi \frac{\xi^3}{e^\xi-1} + \sum_{\text{fermion}} g_i \int_0^\infty d\xi \frac{\xi^3}{e^\xi-1} - \sum_{\text{fermion}} g_i \int_0^\infty d\xi \frac{2\xi^3}{e^{2\xi}-1}\right] \\
&= \frac{k_B^4T^4}{2\pi^2\hbar^3c^3}\left\{\sum_{\text{boson}} g_i\zeta(3+1)\Gamma(3+1) + \sum_{\text{fermion}} g_i\zeta(3+1)\Gamma(3+1) - \frac{g_i}{8}\zeta(3+1)\Gamma(3+1)\right\} \\
&= \frac{k_B^4T^4}{2\pi^2\hbar^3c^3}\left\{\sum_{\text{boson}} g_i\frac{\pi^4}{15} + \sum_{\text{fermion}} g_i\left(1-\frac{1}{8}\right)\frac{\pi^4}{15}\right\} \\
&= \frac{k_B^4T^4}{2\pi^2\hbar^3c^3}\frac{\pi^4}{15}\underbrace{\left\{\sum_{\text{boson}} g_i + \frac{7}{8}\sum_{\text{fermion}} g_i\right\}}_{g^\star} \\
&= \frac{\pi^2k_B^4T^4}{30\hbar^3c^3}g^\star
\end{aligned}$$

where $g^\star$ is referred as effective degree of freedom. Note that the above derivation only holds if $k_BT_i >> m_ic^2$ for the species of particle under consideration therefore only relativistic bosons and fermions contribute to it. As the universe was cooling down due to adiabatic expansion. Temperature of universe would eventually fall below the energy scale associated with the mass of particle, in that condition the relativistic particle approximation would start getting violated. Thus those non-relativistic species of particle would stop contributing to the effective degree of freedom.

Non-Relativistic Limit Here, we have to note that, when we convert the integral from momentum space to integral over E via $E^2 = (mc^2)^2 + (Pc)^2$, the lower limit can't be set to $E = 0$ since, the smallest value of E is mc^2. Here we assume $\xi << x$, so $\xi/x << 1$.

$$\rho^{\text{eq}} = \sum_{\text{all species}} g_i \frac{k_B^4T^4}{2\pi^2\hbar^3c^3}\int_0^\infty d\xi \frac{\xi^2\sqrt{\xi^2+x^2}}{e^{\sqrt{\xi^2+x^2}}\pm 1}$$

$$
\begin{aligned}
&= \sum_{\text{all species}} g_i \frac{k_B^4 T^4}{2\pi^2\hbar^3 c^3} \int_0^\infty d\xi \frac{\xi^2 x\sqrt{1+\xi^2/x^2}}{e^{x(1+\xi^2/x^2)^{1/2}}} \\
&= \sum_{\text{all species}} g_i \frac{k_B^4 T^4}{2\pi^2\hbar^3 c^3} \left[\int_0^\infty d\xi \xi^2 x e^{-x-\xi^2/2x} + \int_0^\infty d\xi \frac{\xi^4}{2x} e^{-x-\xi^2/2x}\right] \\
&\approx \sum_{\text{all species}} g_i \frac{k_B^4 T^4}{2\pi^2\hbar^3 c^3} x e^{-x} \int_0^\infty d\xi \xi^2 e^{-\xi^2/2x} \\
&= \sum_{\text{all species}} g_i mc^2 \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3} e^{-mc^2/k_B T} \int_0^\infty d\xi \xi^2 e^{-\xi^2/2x} \qquad (\text{using } x = mc^2/k_B T) \\
&= \sum_{\text{all species}} mc^2 g_i \left(\frac{mk_B T}{2\pi\hbar^2}\right)^{3/2} e^{-mc^2/k_B T}
\end{aligned}
$$

The Number Density

The number density, we will define using $n_+ - n_-$, but first do the calculation without assuming the particle and anti-particle nature:

$$
\begin{aligned}
n_i^{\text{eq}} &= \frac{g_i}{(2\pi\hbar)^3} \int_0^\infty \frac{1}{\exp\left(\sqrt{p^2c^2+m^2c^4} - \mu/k_B T\right) \pm 1} 4\pi p^2 dp \\
&= \frac{g_i}{2\pi^2\hbar^3} \int_0^\infty dp \frac{p^2}{\exp\left(\sqrt{p^2c^2+m^2c^4} - \mu/k_B T\right) \pm 1}
\end{aligned}
$$

defining $x \equiv mc^2/k_B T$ and $\xi \equiv pc/k_B T$, we get

$$
\begin{aligned}
&= \frac{g_i}{2\pi^2\hbar^3} \int_0^\infty \frac{k_B T}{c} d\xi \frac{\frac{k_B^2 T^2}{c^2}\xi^2}{\exp\left(\sqrt{\xi^2+x^2} - \mu/k_B T\right) \pm 1} \\
&= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3} \int_0^\infty d\xi \frac{\xi^2}{\exp\left(\sqrt{\xi^2+x^2} - \mu/k_B T\right) \pm 1}
\end{aligned}
$$

Ultra-relativistic Limit In the early universe, $\mu \approx 0$. At temperatures above about 10^{16} K, all particles of the Standard Model are highly relativistic, so that their masses can be neglected. Therefore we assume that $x \to 0$, the integrals can then be evaluated exactly and this condition is referred to as ultra-relativistic limit.

$$
\begin{aligned}
n^{\text{eq}} &= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3} \int_0^\infty d\xi \frac{\xi^2}{\exp\left(\sqrt{\xi^2+x^2} - \mu/k_B T\right) \pm 1} \\
&= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3} \int_0^\infty d\xi \frac{\xi^2}{e^{\xi-\mu/k_B T} \pm 1}
\end{aligned}
$$

for $\mu/k_B T << 1$

$$
= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3} \begin{cases} \frac{3}{2}\zeta(3) & \text{for fermions} \\ \\ 2\zeta(3) & \text{for bosons} \end{cases}
$$

we will now, evaluate the net number density:

$$
n_+^{\text{eq}} - n_-^{\text{eq}} = g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3} \left[\int_0^\infty d\xi \frac{\xi^2}{e^{\xi-\mu/k_B T} \pm 1} - \int_0^\infty d\xi \frac{\xi^2}{e^{\xi+\mu/k_B T} \pm 1}\right]
$$

doing shift of variable $y = \xi - {}^{\mu}/_{k_BT}$ in first integral and $y = \xi + {}^{\mu}/_{k_BT}$ in the second integral, we have

$$
\begin{aligned}
&= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3}\left[\int_{-\mu/k_BT}^{\infty} dy \frac{(y+\mu/k_BT)^2}{e^y \pm 1} - \int_{\mu/k_BT}^{\infty} dy \frac{(y-\mu/k_BT)^2}{e^y \pm 1}\right] \\
&= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3}\left[\int_{-\mu/k_BT}^{\infty} dy \frac{(y+\mu/k_BT)^2}{e^y \pm 1} - \int_{\mu/k_BT}^{\infty} dy \frac{(y-\mu/k_BT)^2}{e^y \pm 1}\right] \\
&= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3}\left[\int_{-\mu/k_BT}^{0} dy \frac{(y+\mu/k_BT)^2}{e^y \pm 1} + \int_{0}^{\infty} dy \frac{(y+\mu/k_BT)^2}{e^y \pm 1}\right. \\
&\qquad \left. - \left\{\int_{0}^{\infty} dy \frac{(y-\mu/k_BT)^2}{e^y \pm 1} - \int_{0}^{\mu/k_BT} dy \frac{(y-\mu/k_BT)^2}{e^y \pm 1}\right\}\right] \\
&= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3}\left[\int_{-\mu/k_BT}^{0} dy \frac{(y+\mu/k_BT)^2}{e^y \pm 1} + \int_{0}^{\mu/k_BT} dy \frac{(y-\mu/k_BT)^2}{e^y \pm 1}\right. \\
&\qquad \left. + \int_{0}^{\infty} dy \frac{(y+\mu/k_BT)^2 - (y-\mu/k_BT)^2}{e^y \pm 1}\right]
\end{aligned}
$$

redefining $y \to -y$ in the first integral and using $a^2 - b^2 = (a+b)(a-b)$ in the last integral

$$
\begin{aligned}
&= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3}\left[\int_{0}^{\mu/k_BT} dy \frac{(-y+\mu/k_BT)^2}{e^{-y} \pm 1} + \int_{0}^{\mu/k_BT} dy \frac{(y-\mu/k_BT)^2}{e^y \pm 1}\right. \\
&\qquad \left. + \int_{0}^{\infty} dy \frac{(\mu/k_BT + \mu/k_BT)(y+y)}{e^y \pm 1}\right] \\
&= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3}\left[\int_{0}^{\mu/k_BT} dy (y-\mu/k_BT)^2 + 4\frac{\mu}{k_BT}\int_{0}^{\infty} dy \frac{y}{e^y \pm 1}\right] \\
&= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3}\left[\left.\frac{(y-\mu/k_BT)^3}{3}\right|_{y=0}^{y=\mu/k_BT} + 4\frac{\mu}{k_BT}\int_{0}^{\infty} dy \frac{y}{e^y \pm 1}\right] \\
&= g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3}\left[\frac{\mu^3}{3k_B^3T^3} + \mathcal{I}\right]
\end{aligned}
$$

We will now, explictely evaluate the last integral for fermion and boson. We use the following identity:

$$
\begin{aligned}
\frac{1}{e^y+1} &= \frac{1}{e^y-1} + \text{something} \\
\text{something} &= \frac{1}{e^y+1} - \frac{1}{e^y-1} \\
&= \frac{e^y - 1 - e^y - 1}{(e^y+1)(e^y-1)} \\
&= -\frac{2}{e^{2y}-1}
\end{aligned}
$$

hence,

$$
\frac{1}{e^y+1} = \frac{1}{e^y-1} - \frac{2}{e^{2y}-1}
$$

therefore,

$$
4\frac{\mu}{k_BT}\int_0^{\infty} dy \frac{y}{e^y+1} = 4\frac{\mu}{k_BT}\left\{\int_0^{\infty} dy \frac{y}{e^y-1} - \int_0^{\infty} dy \frac{2y}{e^{2y}-1}\right\}
$$

$$
\begin{aligned}
&= 4\frac{\mu}{k_BT}\left\{\Gamma(2)\zeta(2) - \frac{1}{2}\Gamma(2)\zeta(2)\right\} \\
&= 2\frac{\mu}{k_BT}\Gamma(2)\zeta(2) = \frac{\pi^2\mu}{3k_BT}
\end{aligned}
$$

and for bosons,

$$4\frac{\mu}{k_BT}\int_0^\infty dy\frac{y}{e^y-1} = 4\frac{\mu}{k_BT}\Gamma(2)\zeta(2) = \frac{2\pi^2\mu}{3k_BT}$$

Since, the matter surrounding us are mostly fermions, we get the net fermion density as

$$n^{\text{eq}}_{\text{fermion}} = g_i\frac{k_B^3T^3}{6\pi^2\hbar^3c^3}\left[\frac{\mu^3}{k_B^3T^3} + \frac{\pi^2\mu}{k_BT}\right]$$

and

$$n^{\text{eq}}_{\text{boson}} = g_i\frac{k_B^3T^3}{6\pi^2\hbar^3c^3}\left[\frac{\mu^3}{k_B^3T^3} + \frac{2\pi^2\mu}{k_BT}\right]$$

Non-Relativistic Limit Here we assume $\xi << x$, so ${}^{\xi}/_{x} << 1$.

$$
\begin{aligned}
n^{\text{eq}} &= g_i\frac{k_B^3T^3}{2\pi^2\hbar^3c^3}\int_0^\infty d\xi\frac{\xi^2}{e^{\sqrt{\xi^2+x^2}+\mu/k_BT}\pm 1} \\
&= g_i\frac{k_B^3T^3}{2\pi^2\hbar^3c^3}\int_0^\infty d\xi\frac{\xi^2}{e^{x(1+\xi^2/x^2)^{1/2}+\mu/k_BT}} \\
&= g_i\frac{k_B^3T^3}{2\pi^2\hbar^3c^3}\int_0^\infty d\xi\xi^2e^{-x-\xi^2/2x+\mu/k_BT} \\
&= g_i\frac{k_B^3T^3}{2\pi^2\hbar^3c^3}e^{-x+\mu/k_BT}\int_0^\infty d\xi\xi^2e^{-\xi^2/2x} \\
&= g_i\frac{k_B^3T^3}{2\pi^2\hbar^3c^3}\sqrt{\frac{\pi}{2}}x^{3/2}e^{-x+\mu/k_BT} \qquad\qquad (\text{where } x = mc^2/k_BT) \\
&= g_i\frac{k_B^3T^3}{2\pi^2\hbar^3c^3}\sqrt{\frac{\pi}{2}}\left(\frac{mc^2}{k_BT}\right)^{3/2}e^{-mc^2/k_BT+\mu/k_BT} \\
&= g_i\frac{1}{\hbar^3}\left(\frac{mk_BT}{2\pi}\right)^{3/2}e^{-mc^2-\mu/k_BT}
\end{aligned}
$$

Thus we arrive at, with $\mu = 0$

$$\boxed{n^{\text{eq}} = g_i\left(\frac{mk_BT}{2\pi\hbar^2}\right)^{3/2}e^{-mc^2/k_BT}}$$

Pressure

The quantity of interest is pressure:

$$
\begin{aligned}
P^{\text{eq}} &= \sum_{\text{all species}}\frac{g_i}{(2\pi\hbar)^3}\int\frac{|\vec{p}|^2c^2}{3E}f(\vec{p})d^3p \\
&= \sum_{\text{all species}}\frac{g_i}{2\pi^2\hbar^3}\int_0^\infty\frac{p^2c^2}{3\sqrt{p^2c^2+m^2c^4}}\frac{1}{e^{\sqrt{p^2c^2+m^2c^4}-\mu/k_BT}\pm 1}p^2dp
\end{aligned}
$$

defining $x \equiv {}^{mc^2}/_{k_BT}$ and $\xi = {}^{pc}/_{k_BT}$, we get

$$= \sum_{\text{all species}}\frac{g_i}{2\pi^2\hbar^3}\int_0^\infty\frac{k_B^2T^2\xi^2}{3k_BT\sqrt{\xi^2+x^2}}\frac{1}{\exp\left(\sqrt{\xi^2+x^2}-\mu/k_BT\right)\pm 1}\left(\frac{k_BT}{c}\xi\right)^2\frac{k_BT}{c}d\xi$$

$$= \sum_{\text{all species}} \frac{g_i}{2\pi^2\hbar^3} \int_0^\infty \frac{k_B T \xi^2}{3\sqrt{\xi^2 + x^2}} \frac{1}{\exp\left(\sqrt{\xi^2 + x^2} - \mu/k_B T\right) \pm 1} \left(\frac{k_B T}{c}\xi\right)^2 \frac{k_B T}{c} d\xi$$
$$= \frac{1}{3} \sum_{\text{all species}} g_i \frac{k_B^4 T^4}{2\pi^2\hbar^3 c^3} \int_0^\infty \frac{\xi^4}{\sqrt{\xi^2 + x^2}} \frac{1}{\exp\left(\sqrt{\xi^2 + x^2} - \mu/k_B T\right) \pm 1} d\xi$$

Ultra-relativistic limit In the ultra-relativistic limit $x \to 0$ and $\mu \approx 0$, we observe that

$$P^{\text{eq}} = \frac{1}{3} \sum_{\text{all species}} g_i \frac{k_B^4 T^4}{2\pi^2\hbar^3 c^3} \int_0^\infty \frac{\xi^4}{\sqrt{\xi^2}} \frac{1}{\exp\left(\sqrt{\xi^2} - \mu/k_B T\right) \pm 1} d\xi$$
$$= \frac{1}{3} \underbrace{\frac{k_B^4 T^4}{2\pi^2\hbar^3 c^3} \left[\sum_{\text{boson}} g_i \int_0^\infty d\xi \frac{\xi^3}{e^{\xi - \mu/k_B T} - 1} + \sum_{\text{fermion}} g_i \int_0^\infty d\xi \frac{\xi^3}{e^{\xi - \mu/k_B T} + 1}\right]}_{\rho^{\text{eq}}}$$
$$= \frac{1}{3} \rho^{\text{eq}}$$

Non-relativistic Limit In the limit $\xi << x \implies \xi/x << 1$

$$P^{\text{eq}} = \frac{1}{3} \sum_{\text{all species}} g_i \frac{k_B^4 T^4}{2\pi^2\hbar^3 c^3} \int_0^\infty \frac{\xi^4}{\sqrt{\xi^2 + x^2}} \frac{1}{\exp\left(\sqrt{\xi^2 + x^2} - \mu/k_B T\right) \pm 1} d\xi$$
$$= \frac{1}{3} \sum_{\text{all species}} g_i \frac{k_B^4 T^4}{2\pi^2\hbar^3 c^3} \int_0^\infty \frac{\xi^4}{x\sqrt{1 + \frac{\xi^2}{x^2}}} \frac{1}{\exp\left(x\sqrt{1 + \frac{\xi^2}{x^2}} - \mu/k_B T\right) \pm 1} d\xi$$
$$\approx \frac{1}{3} \sum_{\text{all species}} g_i \frac{k_B^4 T^4}{2\pi^2\hbar^3 c^3} \int_0^\infty \frac{\xi^4}{x + \frac{\xi^2}{2x}} \frac{1}{\exp\left(x + \frac{\xi^2}{2x} - \mu/k_B T\right) \pm 1} d\xi$$
$$= \frac{k_B T}{3} \left[\sum_{\text{all species}} g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3} \int_0^\infty \frac{\xi^4}{x + \frac{\xi^2}{2x}} \frac{1}{\exp\left(x + \frac{\xi^2}{2x} - \mu/k_B T\right) \pm 1} d\xi\right]$$
$$\approx \frac{k_B T}{3} \left[\sum_{\text{all species}} g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3} \int_0^\infty \frac{\xi^4}{x} e^{-x - \xi^2/2x + \mu/k_B T} d\xi\right]$$
$$= \frac{k_B T}{3} \left[\sum_{\text{all species}} g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3} \int_0^\infty \frac{\xi^4}{x} e^{-x - \xi^2/2x + \mu/k_B T} d\xi\right]$$
$$= \frac{k_B T}{3} \left[\sum_{\text{all species}} g_i \frac{k_B^3 T^3}{2\pi^2\hbar^3 c^3} \frac{1}{x} \times 3\sqrt{\frac{\pi}{2}} x^{5/2} e^{-x - \mu/k_B T}\right]$$
$$= k_B T \underbrace{\left[\sum_{\text{all species}} g_i \left(\frac{m k_B T}{2\pi\hbar^2}\right)^{3/2} e^{-mc^2 - \mu/k_B T}\right]}_{n^{\text{eq}}}$$
$$= n^{\text{eq}} k_B T$$

3.4.3 Using Maxwell-Boltzmann distribution

Now that we have seen how keeping the explicit $\hbar$ and c, makes the calculation look somewhat clustered, we will switch to natural units for this part. Following the previous section

Equilibrium Number density

The equilibrium number density is given by,

$$n^{\rm eq} = \frac{g_i}{(2\pi\hbar)^3}\int_0^\infty d^3p e^{-\frac{\sqrt{p^2c^2+m^2c^4}}{k_BT}} = \frac{g}{(2\pi)^3}\int d^3p\, e^{-\frac{\sqrt{p^2+m^2}}{T}},$$

where $p = |\mathbf{p}|$ and m is the mass of the particle. Since the integrand depends only on the magnitude of the momentum, it is convenient to switch to spherical coordinates in momentum space. In three dimensions, the volume element in spherical coordinates is

$$d^3p = p^2\, dp\, d\Omega = 4\pi p^2\, dp$$

Thus, the integral reduces to

$$n^{\rm eq} = \frac{g}{(2\pi)^3}4\pi\int_0^\infty p^2 e^{-\frac{\sqrt{p^2+m^2}}{T}}\, dp.$$

To proceed, we use the substitution

$$p = m\sinh t, \quad \text{so that} \quad dp = m\cosh t\, dt.$$

With this substitution, the energy becomes

$$\sqrt{p^2+m^2} = \sqrt{m^2\sinh^2 t + m^2} = m\cosh t,$$

and the factor p^2dp transforms as

$$p^2dp = (m\sinh t)^2\cdot(m\cosh t\, dt) = m^3\sinh^2 t\cosh t\, dt.$$

Substituting these expressions into the integral, we obtain

$$n^{\rm eq} = \frac{g}{2\pi^2}m^3\int_0^\infty \sinh^2 t\cosh t\, e^{-\frac{m\cosh t}{T}}\, dt.$$

Using the identity $\sinh^2 t = \cosh^2 t - 1$, we have

$$\sinh^2 t\cosh t = (\cosh^2 t - 1)\cosh t = \cosh^3 t - \cosh t.$$

Hence, the integral becomes

$$n^{\rm eq} = \frac{g}{2\pi^2}m^3\int_0^\infty (\cosh^3 t - \cosh t)\, e^{-\frac{m\cosh t}{T}}\, dt.$$

We can relate powers of $\cosh t$ to $\cosh(3t)$ using the identity

$$\cosh 3t = 4\cosh^3 t - 3\cosh t \quad\Rightarrow\quad \cosh^3 t - \cosh t = \frac{1}{4}(\cosh 3t - \cosh t).$$

Thus, the integral can be rewritten as

$$I = \frac{g}{2\pi^2}m^3\int_0^\infty \frac{1}{4}(\cosh 3t - \cosh t)\, e^{-\frac{m\cosh t}{T}}\, dt = I = \frac{g}{8\pi^2}m^3\int_0^\infty (\cosh 3t - \cosh t)\, e^{-\frac{m\cosh t}{T}}\, dt.$$

Next, we recall the integral representation of the modified Bessel function of the second kind:

$$K_\nu(z) = \int_0^\infty e^{-z\cosh t}\cosh(\nu t)\, dt, \quad \mathrm{Re}(z) > 0.$$

Comparing with our integral, we identify

$$\int_0^\infty \cosh 3t\, e^{-m\cosh t/T}\, dt = K_3\left(\frac{m}{T}\right), \quad \int_0^\infty \cosh t\, e^{-m\cosh t/T}\, dt = K_1\left(\frac{m}{T}\right).$$

Hence, we can write

$$I = \frac{g}{8\pi^2} m^3 \left[K_3\left(\frac{m}{T}\right) - K_1\left(\frac{m}{T}\right)\right].$$

Finally, using the recurrence relation for modified Bessel functions

$$K_{\nu+1}(z) - K_{\nu-1}(z) = \frac{2\nu}{z} K_\nu(z),$$

with $\nu = 2$, we get

$$K_3(z) - K_1(z) = \frac{4}{z} K_2(z).$$

Therefore, the final result of the integral is

$$n^{\text{eq}} = \frac{g}{8\pi^2} m^3 \cdot \frac{4T}{m} K_2\left(\frac{m}{T}\right) = \frac{g}{(2\pi)^3} 4\pi m^2 T\, K_2\left(\frac{m}{T}\right)$$

Equilibrium Energy Density

We consider a particle species of mass m and internal degrees of freedom g, in thermal equilibrium at temperature T, obeying Maxwell–Boltzmann statistics. The equilibrium energy density is defined as

$$\rho_{\text{eq}}(T) = g \int \frac{d^3p}{(2\pi)^3}\, E\, e^{-E/T}, \qquad E = \sqrt{p^2 + m^2}. \tag{1}$$

Using spherical symmetry in momentum space,

$$d^3p = 4\pi p^2 dp, \tag{2}$$

we obtain

$$\rho_{\text{eq}} = \frac{g}{2\pi^2} \int_0^\infty dp\, p^2 \sqrt{p^2 + m^2}\, e^{-\sqrt{p^2+m^2}/T}. \tag{3}$$

Introduce the change of variables

$$p = m \sinh u, \qquad E = \sqrt{p^2 + m^2} = m \cosh u. \tag{4}$$

Then

$$dp = m \cosh u\, du, \tag{5}$$

and

$$p^2 E\, dp = m^4 \sinh^2 u \cosh^2 u\, du. \tag{6}$$

Substituting into (3),

$$\rho_{\text{eq}} = \frac{gm^4}{2\pi^2} \int_0^\infty du\, \sinh^2 u \cosh^2 u\, e^{-m\cosh u/T}. \tag{7}$$

Using the identity

$$\sinh^2 u = \cosh^2 u - 1, \tag{8}$$

we find

$$\sinh^2 u \cosh^2 u = \cosh^4 u - \cosh^2 u. \tag{9}$$

Thus,

$$\rho_{\rm eq} = \frac{gm^4}{2\pi^2}\int_0^\infty du\,\left(\cosh^4 u - \cosh^2 u\right)e^{-m\cosh u/T}. \tag{10}$$

We use the identities

$$\cosh^2 u = \frac{1}{2}\big(\cosh 2u + 1\big), \tag{11}$$

$$\cosh^4 u = \frac{1}{8}\big(\cosh 4u + 4\cosh 2u + 3\big). \tag{12}$$

Subtracting,

$$\cosh^4 u - \cosh^2 u = \frac{1}{8}\cosh 4u + \frac{1}{4}\cosh 2u - \frac{1}{8}. \tag{13}$$

Substituting into (10),

$$\rho_{\rm eq} = \frac{gm^4}{16\pi^2}\int_0^\infty du\,\Big[\cosh 4u + 2\cosh 2u - 1\Big]e^{-m\cosh u/T}. \tag{14}$$

The modified Bessel function of the second kind is defined by

$$K_\nu(z) = \int_0^\infty du\; e^{-z\cosh u}\cosh(\nu u). \tag{15}$$

Using this definition, equation (14) becomes

$$\rho_{\rm eq} = \frac{gm^4}{16\pi^2}\left[K_4(x) + 2K_2(x) - K_0(x)\right], \qquad x \equiv \frac{m}{T}. \tag{16}$$

The modified Bessel functions satisfy the recursion relation

$$K_{\nu+1}(x) + K_{\nu-1}(x) = \frac{2\nu}{x}K_\nu(x). \tag{17}$$

Applying this relation:

$$K_4 + K_2 = \frac{6}{x}K_3, \tag{18}$$

$$K_2 + K_0 = \frac{2}{x}K_1. \tag{19}$$

Subtracting the second from the first,

$$K_4 - K_0 = \frac{6}{x}K_3 - \frac{2}{x}K_1. \tag{20}$$

Using the identity

$$K_3 - K_1 = \frac{4}{x}K_2, \tag{21}$$

we obtain

$$K_4 + 2K_2 - K_0 = \frac{4}{x}K_1 + \frac{12}{x^2}K_2. \tag{22}$$

Substituting into (16),

$$\rho_{\rm eq} = \frac{gm^4}{16\pi^2}\left(\frac{4}{x}K_1 + \frac{12}{x^2}K_2\right). \tag{23}$$

Restoring $x = m/T$,

$$\boxed{\rho_{\rm eq}(T) = \frac{gm^2T^2}{2\pi^2}\left[mK_1\!\left(\frac{m}{T}\right) + 3TK_2\!\left(\frac{m}{T}\right)\right]}. \tag{24}$$

Decay Rate and Relative Velocity

This derivation follows *Landau & Lifshitz, The Classical Theory of Fields*, Section 12, where we obtain a Lorentz-invariant expression for the relative velocity between two particles in special relativity. Let the four-momenta of the two particles be

$$p_1^\mu = (E_1, \mathbf{p}_1) = (\gamma_1 m_1, \gamma_1 m_1 \mathbf{v}_1), \qquad p_2^\mu = (E_2, \mathbf{p}_2) = (\gamma_2 m_2, \gamma_2 m_2 \mathbf{v}_2), \tag{25}$$

where natural units $c = 1$ are used and $\gamma_i = (1 - v_i^2)^{-1/2}$.
To determine the relative velocity between the two particles, we consider the scalar product of their four-momenta in the rest frame of the second particle. In this frame, the momentum of particle 2 is

$$p_2^\mu = (m_2, \mathbf{0}),$$

while particle 1 moves with velocity $\mathbf{v}_{\rm rel}$ and Lorentz factor $\gamma_{\rm rel} = (1 - v_{\rm rel}^2)^{-1/2}$. Thus,

$$p_1^\mu = (\gamma_{\rm rel} m_1, \gamma_{\rm rel} m_1 \mathbf{v}_{\rm rel}).$$

The Lorentz scalar product between the two momenta is

$$p_1 \cdot p_2 = E_1 E_2 - \mathbf{p}_1 \cdot \mathbf{p}_2 = \gamma_1 \gamma_2 m_1 m_2 (1 - \mathbf{v}_1 \cdot \mathbf{v}_2) = E_1 E_2 (1 - \mathbf{v}_1 \cdot \mathbf{v}_2) \tag{26}$$

where we substituted the definitions $E_i = \gamma_i m_i$ and $\mathbf{p}_i = \gamma_i m_i \mathbf{v}_i$. Upon squaring, we have

$$(p_1 \cdot p_2)^2 = (m_1 m_2 \gamma_1 \gamma_2)^2 (1 - \mathbf{v}_1 \cdot \mathbf{v}_2)^2. \tag{27}$$

Expanding the square of the scalar product, we find

$$(p_1 \cdot p_2)^2 = (m_1 m_2 \gamma_1 \gamma_2)^2 (1 - 2\mathbf{v}_1 \cdot \mathbf{v}_2 + (\mathbf{v}_1 \cdot \mathbf{v}_2)^2). \tag{28}$$

Recall the vector identity

$$(\mathbf{a} \times \mathbf{b})^2 = |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2,$$

We may write

$$(\mathbf{v}_1 \cdot \mathbf{v}_2)^2 = v_1^2 v_2^2 - (\mathbf{v}_1 \times \mathbf{v}_2)^2.$$

Substituting this, we obtain

$$(p_1 \cdot p_2)^2 = (m_1 m_2 \gamma_1 \gamma_2)^2 \left[1 - 2\mathbf{v}_1 \cdot \mathbf{v}_2 + v_1^2 v_2^2 - (\mathbf{v}_1 \times \mathbf{v}_2)^2\right]. \tag{29}$$

We also have

$$2\mathbf{v}_1 \cdot \mathbf{v}_2 = v_1^2 + v_2^2 - (\mathbf{v}_1 - \mathbf{v}_2)^2$$

Substituting this form back, we have

$$\begin{aligned}(p_1 \cdot p_2)^2 &= (m_1 m_2 \gamma_1 \gamma_2)^2 \left[(\mathbf{v}_1 - \mathbf{v}_2)^2 + 1 - v_1^2 - v_2^2 + v_1^2 v_2^2 - (\mathbf{v}_1 \times \mathbf{v}_2)^2\right] \\ &= (m_1 m_2 \gamma_1 \gamma_2)^2 \left[(\mathbf{v}_1 - \mathbf{v}_2)^2 + (1 - v_1^2)(1 - v_2^2) - (\mathbf{v}_1 \times \mathbf{v}_2)^2\right]\end{aligned}$$

Using $1 - v_i^2 = 1/\gamma_i^2$, this becomes

$$(p_1 \cdot p_2)^2 = (m_1 m_2 \gamma_1 \gamma_2)^2 \left[(\mathbf{v}_1 - \mathbf{v}_2)^2 + \frac{1}{\gamma_1^2 \gamma_2^2} - (\mathbf{v}_1 \times \mathbf{v}_2)^2\right]. \tag{30}$$

Subtracting $m_1^2 m_2^2$ from both sides gives

$$(p_1 \cdot p_2)^2 - m_1^2 m_2^2 = (m_1 m_2 \gamma_1 \gamma_2)^2 \left[(\mathbf{v}_1 - \mathbf{v}_2)^2 - (\mathbf{v}_1 \times \mathbf{v}_2)^2\right]. \tag{31}$$

Diving by (27), we get our expression for the magnitude of relative velocity $v_{\rm rel}$:

$$v_{\rm rel} = \frac{\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}}{p_1 \cdot p_2} = \frac{\sqrt{(\mathbf{v}_1 - \mathbf{v}_2)^2 - (\mathbf{v}_1 \times \mathbf{v}_2)^2}}{1 - \mathbf{v}_1 \cdot \mathbf{v}_2}. \tag{32}$$

This is the general Lorentz-invariant expression for the relative velocity between two particles moving with arbitrary velocities $\mathbf{v}_1$ and $\mathbf{v}_2$. We can express the above using the mandelstam variable

$$p_1 \cdot p_2 = \frac{s - m_1^2 - m_2^2}{2}$$

Then,

$$\begin{aligned} v_{\rm rel} &= \frac{\sqrt{\left(\frac{s-m_1^2-m_2^2}{2}\right)^2 - m_1^2 m_2^2}}{\frac{s-(m_1^2+m_2^2)}{2}} = \frac{\sqrt{(s - m_1^2 - m_2^2)^2 - 4m_1^2 m_2^2}}{[s - (m_1^2 + m_2^2)]} \\ &= \frac{\sqrt{[s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]}}{[s - (m_1^2 + m_2^2)]} \end{aligned} \tag{33}$$

For colinear motion, $\mathbf{v}_1 \parallel \mathbf{v}_2$, the cross product vanishes and the expression reduces to the familiar relativistic law of velocity addition,

$$v_{\rm rel} = \frac{|\mathbf{v}_1 - \mathbf{v}_2|}{1 - \mathbf{v}_1 \mathbf{v}_2} = \frac{E_1 E_2}{p_1 \cdot p_2} |\mathbf{v}_1 - \mathbf{v}_2| \tag{34}$$

3.4.4 Decay Rate

The thermal average of $\sigma v_{\rm rel}$ following Mirco Cannoni is

$$\langle \sigma |\mathbf{v}_1 - \mathbf{v}_2| \rangle = \frac{1}{(4\pi)^2\, T^2\, m_1^2 m_2^2 K_2(m_1/T) K_2(m_2/T)} \int d^3\mathbf{p}_1 d^3\mathbf{p}_2 \frac{(p_1 \cdot p_2)}{E_1 E_2} v_{\rm rel}\, \sigma(s)\, e^{-(E_1+E_2)/T}. \tag{35}$$

This is a complicated integral which can be simplified by transforming it into a single integral over the invariant mass squared $s = (p_1 + p_2)^2$. All calculations are performed in the rest frame of the heat bath, $u^\mu = (1, \mathbf{0})$. The integration measure could be simplified as:

$$d^3\mathbf{p}_i = p_i^2 dp_i\, d\Omega_i, \qquad E_i = \sqrt{p_i^2 + m_i^2}, \qquad i = 1, 2.$$

Let us perform the following transformation $\theta_1 \to \theta_1$ and $\theta_2 \to (\theta_2 - \theta_1) \equiv \theta$ which is equivalent to choosing $\mathbf{p}_1$ along the z-axis and allowing θ be the angle between $\mathbf{p}_1$ and $\mathbf{p}_2$. Then $d\Omega_1 = 4\pi$ and $d\Omega_2 = 2\pi\, d(\cos\theta)$, giving

$$d^3\mathbf{p}_1\, d^3\mathbf{p}_2 = 8\pi^2\, |\mathbf{p_1}|^2 |\mathbf{p_2}|^2 dp_1 dp_2\, d(\cos\theta).$$

Also, using $d\mathbf{p}_i = (E_i/\mathbf{p}_i)\, dE_i$, we have

$$\frac{d^3\mathbf{p}_1}{E_1} \frac{d^3\mathbf{p}_2}{E_2} = 8\pi^2\, |\mathbf{p_1}||\mathbf{p_2}| dE_1\, dE_2\, d(\cos\theta).$$

We recall

$$p_{1\mu} p^{2\mu} = E_1 E_2 - |\mathbf{p_1}||\mathbf{p_2}| \cos\theta, \qquad s = m_1^2 + m_2^2 + 2(E_1 E_2 - |\mathbf{p_1}||\mathbf{p_2}| \cos\theta).$$

Hence,

$$p_{1\mu}p^{2\mu} = \frac{s-(m_1^2+m_2^2)}{2}.$$

Let's perform the change of variable as suggested by P. Gondolo and G. Gelmini, Nucl. Phys. B 360

$$Y = E_1 + E_2, \qquad Z = E_1 - E_2, \qquad s = (p_1+p_2)^2.$$

The associated Jacobian for the transformation can be found as:

$$\frac{\partial(Y,Z,s)}{\partial(E_1,E_2,\cos\theta)} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ \dfrac{\partial s}{\partial E_1} & \dfrac{\partial s}{\partial E_2} & \dfrac{\partial s}{\partial \cos\theta} \end{pmatrix}.$$

The remaining derivatives can be computed from $s = m_1^2 + m_2^2 + 2(E_1E_2 - p_1p_2\cos\theta)$:

$$\frac{\partial s}{\partial E_1} = 2\left(E_2 - \cos\theta\,\frac{E_1|\mathbf{p_2}|}{|\mathbf{p_1}|}\right)$$

$$\frac{\partial s}{\partial E_2} = 2\left(E_1 - \cos\theta\,\frac{E_2|\mathbf{p_1}|}{|\mathbf{p_2}|}\right),$$

$$\frac{\partial s}{\partial \cos\theta} = 2|\mathbf{p_1}||\mathbf{p_2}|$$

Now the determinant:

$$\det\frac{\partial(Y,Z,s)}{\partial(E_1,E_2,\cos\theta)} = -4|\mathbf{p_1}||\mathbf{p_2}|$$

Hence,

$$dE_1\,dE_2\,d(\cos\theta) = \frac{1}{4|\mathbf{p_1}||\mathbf{p_2}|}\,dY\,dZ\,ds$$

Plugging this into the momentum-space measure:

$$\frac{d^3\mathbf{p}_1}{E_1}\frac{d^3\mathbf{p}_2}{E_2} = 8\pi^2|\mathbf{p_1}||\mathbf{p_2}|dE_1\,dE_2\,d(\cos\theta) = 2\pi^2\,dY\,dZ\,ds.$$

Inserting the above results into (35), we obtain

$$\langle\sigma|\mathbf{v}_1 - \mathbf{v}_2|\rangle = \frac{\pi^2}{(4\pi)^2T^2\,m_1^2m_2^2K_2(m_1/T)K_2(m_2/T)}\int dY\,dZ\,ds\,\sigma(s)\,[s-(m_1^2+m_2^2)]v_{\rm rel}\,e^{-Y/T}. \tag{36}$$

The variables Y and Z are restricted by kinematics: for fixed s, not every pair (Y,Z) is possible. The restriction arises from the condition $|\cos\theta| \le 1$. From $s = m_1^2 + m_2^2 + 2(E_1E_2 - p_1p_2\cos\theta)$ we isolate $\cos\theta$:

$$\cos\theta = \frac{2E_1E_2 + m_1^2 + m_2^2 - s}{2p_1p_2}.$$

Substitute $E_1 = (Y+Z)/2$, $E_2 = (Y-Z)/2$, and $p_i = \sqrt{E_i^2 - m_i^2}$. Then

$$\cos\theta = \frac{Y^2 - Z^2 + 2(m_1^2+m_2^2) - 2s}{4\sqrt{\left(\frac{Y+Z}{2}\right)^2 - m_1^2}\sqrt{\left(\frac{Y-Z}{2}\right)^2 - m_2^2}}.$$

The condition $|\cos\theta| \le 1$ determines the allowed range of Z for each (Y, s). Squaring and solving for Z gives the two roots $Z_\pm$ satisfying

$$Z_\pm = \frac{Y(m_1^2 - m_2^2) \pm 2\sqrt{s}\, p'(s)\, \sqrt{Y^2 - s}}{s},$$

where

$$p'(s) = \frac{\sqrt{[s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]}}{2\sqrt{s}} \tag{37}$$

Hence the integration interval in Z is $Z \in [Z_-, Z_+]$, and its length is

$$Z_+ - Z_- = \frac{4p'(s)}{\sqrt{s}}\sqrt{Y^2 - s}.$$

We note that the integrand in (36) is independent of Z, so the Z-integration is simply:

$$\int_{Z_-}^{Z_+} dZ\, e^{-Y/T} = e^{-Y/T}\,(Z_+ - Z_-) = e^{-Y/T}\, \frac{4p'(s)}{\sqrt{s}}\sqrt{Y^2 - s}.$$

Substitute this back into (36):

$$\langle \sigma v \rangle = \frac{1}{4T\, m_1^2 m_2^2 K_2(m_1/T) K_2(m_2/T)}$$
$$\int_{(m_1+m_2)^2}^{\infty} ds\, \sigma(s)\, [s - (m_1^2 + m_2^2)] v_{\rm rel}\, \frac{p'(s)}{\sqrt{s}} \int_{\sqrt{s}}^{\infty} dY\, e^{-Y/T}\sqrt{Y^2 - s}. \tag{38}$$

Let $y = Y/\sqrt{s}$, so $dY = \sqrt{s}\, dy$ and $Y^2 - s = s(y^2 - 1)$. Then

$$\int_{\sqrt{s}}^{\infty} dY\, e^{-Y/T}\sqrt{Y^2 - s} = s\int_1^{\infty} dy\, e^{-(\sqrt{s}/T)y}\sqrt{y^2 - 1}.$$

The standard integral representation of the modified Bessel function K_1 is

$$K_1(a) = a\int_1^{\infty} dy\, e^{-ay}\sqrt{y^2 - 1}, \qquad \mathrm{Re}\, a > 0.$$

Thus

$$\int_{\sqrt{s}}^{\infty} dY\, e^{-Y/T}\sqrt{Y^2 - s} = T\sqrt{s}\, K_1(\sqrt{s}/T)$$

Using above with (33) and (37), we find

$$\langle \sigma |\mathbf{v}_1 - \mathbf{v}_2| \rangle = \frac{1}{8T\, m_1^2 m_2^2 K_2(m_1/T) K_2(m_2/T)}$$
$$\times \int_{(m_1+m_2)^2}^{\infty} ds\, \frac{[s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]}{\sqrt{s}} \sigma(s)\, K_1(\sqrt{s}/T). \tag{39}$$

3.4.5 Effective relativistic degrees of freedom $g_*(T)$

Summing over all relativistic species ($m_i \ll T$), the total radiation energy density is parameterized by $g_*(T)$:

$$\rho_r(T) = g_*(T)\frac{\pi^2}{30}T^4, \tag{3.4.14}$$

where the **effective number of relativistic degrees of freedom** for energy is:

$$g_*(T) \equiv \sum_{i=\text{bosons}} g_i \left(\frac{T_i}{T}\right)^4 + \frac{7}{8} \sum_{i=\text{fermions}} g_i \left(\frac{T_i}{T}\right)^4. \tag{3.4.15}$$

The factor 7/8 originates from the integral ratio $\int_0^\infty \frac{x^3\,\mathrm{d}x}{\mathrm{e}^x+1} = \frac{7}{8}\int_0^\infty \frac{x^3\,\mathrm{d}x}{\mathrm{e}^x-1} = \frac{7}{8}\frac{\pi^4}{15}$.

In the radiation era ($H^2 = \frac{8\pi G_\mathrm{N}}{3}\rho_r$), the Hubble expansion rate and cosmic time are:

$$H(T) = \sqrt{\frac{8\pi G_\mathrm{N}}{3}\frac{\pi^2}{30}g_*(T)T^4} = \left(\frac{4\pi^3 G_\mathrm{N}}{45}g_*\right)^{1/2} T^2 \simeq 1.66\, g_*^{1/2}\frac{T^2}{M_\mathrm{Pl}}, \tag{3.4.16}$$

$$t(T) = \frac{1}{2H(T)} \simeq 0.301\, g_*^{-1/2}\frac{M_\mathrm{Pl}}{T^2} \simeq 2.42\, g_*^{-1/2}\left(\frac{T}{1\,\mathrm{MeV}}\right)^{-2}\ \mathrm{s}, \tag{3.4.17}$$

where $M_\mathrm{Pl} \equiv 1/\sqrt{G_\mathrm{N}} = 1.22 \times 10^{19}$ GeV is the Planck mass.

3.4.6 Chemical potentials and matter–antimatter asymmetry

Inelastic processes such as double Compton scattering ($e^- + \gamma \leftrightarrow e^- + 2\gamma$) and Bremsstrahlung ($e^- + p \leftrightarrow e^- + p + \gamma$) create and destroy photons freely, demanding:

$$\mu_\gamma = 0. \tag{3.4.18}$$

For any particle-antiparticle pair A and $\bar{A}$ in thermal equilibrium via rapid annihilation $A + \bar{A} \leftrightarrow 2\gamma$:

$$\mu_A + \mu_{\bar{A}} = 2\mu_\gamma = 0 \implies \mu_{\bar{A}} = -\mu_A. \tag{3.4.19}$$

In the relativistic regime ($T \gg m_A$), the net particle number asymmetry is:

$$n_A - n_{\bar{A}} = \frac{g_A T^3}{6\pi^2}\left[\pi^2\left(\frac{\mu_A}{T}\right) + \left(\frac{\mu_A}{T}\right)^3\right] \approx \frac{g_A T^3}{6}\left(\frac{\mu_A}{T}\right) \quad (\mu_A \ll T). \tag{3.4.20}$$

Cosmic electric charge neutrality ($n_p = n_e - n_{\bar{e}}$) and the empirical baryon-to-photon ratio $\eta_B \equiv n_B/n_\gamma \approx 6 \times 10^{-10}$ impose:

$$\frac{\mu_e}{T} \approx \frac{\mu_p}{T} \sim 10^{-9} \ll 1, \tag{3.4.21}$$

justifying setting $\mu_i \approx 0$ for all standard model species in the early Universe.

3.4.7 Comoving entropy conservation and $g_{*S}(T)$

From the thermodynamic relation $\mathrm{d}U = T\,\mathrm{d}S - P\,\mathrm{d}V + \mu\,\mathrm{d}N$, the relativistic **entropy density** is:

$$s \equiv \frac{\rho + P - \sum_i \mu_i n_i}{T} \approx \frac{\rho + P}{T}. \tag{3.4.22}$$

Combining with the fluid continuity equation $\mathrm{d}(\rho a^3) = -P\,\mathrm{d}a^3$, the total comoving entropy is strictly conserved:

$$\mathrm{d}(sa^3) = 0 \iff S \equiv s(T)a^3(t) = \text{constant}. \tag{3.4.23}$$

The entropy density is parameterized by $g_{*S}(T)$ (also denoted $q_*(T)$):

$$s(T) = g_{*S}(T)\frac{2\pi^2}{45}T^3, \tag{3.4.24}$$

where:

$$g_{*S}(T) \equiv \sum_{i=\text{bosons}} g_i \left(\frac{T_i}{T}\right)^3 + \frac{7}{8} \sum_{i=\text{fermions}} g_i \left(\frac{T_i}{T}\right)^3. \tag{3.4.25}$$

When all relativistic species share a common temperature $T_i = T$, $g_{*S}(T) = g_*(T)$.

The **comoving abundance (yield)** of any species i is defined by:

$$Y_i \equiv \frac{n_i}{s}, \tag{3.4.26}$$

which remains constant under cosmic expansion in the absence of creation/annihilation.

3.4.8 Thermal history: Neutrino decoupling and e^+e^- annihilation

Cosmic neutrino decoupling ($T \approx 1\,\mathrm{MeV}$)

Neutrinos (ν_e, ν_μ, ν_τ) are kept in thermal equilibrium with charged leptons via electroweak weak neutral- and charged-current interactions ($\nu\bar{\nu} \leftrightarrow e^+e^-$, $\nu e^- \leftrightarrow \nu e^-$). The weak interaction rate is:

$$\Gamma_{\text{weak}} = n_e \langle \sigma_{\text{weak}} v \rangle \approx G_{\mathrm{F}}^2 T^5. \tag{3.4.27}$$

Comparing with the Hubble rate $H \approx 1.66 g_*^{1/2} T^2 / M_{\mathrm{Pl}}$:

$$\frac{\Gamma_{\text{weak}}}{H} \approx \left(\frac{T}{1\,\mathrm{MeV}} \right)^3. \tag{3.4.28}$$

At $T_{\nu,D} \approx 1\,\mathrm{MeV}$ ($t \approx 1\,\mathrm{s}$), $\Gamma_{\text{weak}} \lesssim H$, and the three neutrino flavors decouple, free-streaming with temperature $T_\nu \propto a^{-1}$.

Electron-positron annihilation and photon reheating

Shortly after neutrino decoupling, the temperature drops below the electron mass ($T < m_e = 0.511\,\mathrm{MeV}$). Electrons and positrons annihilate ($e^+ + e^- \to 2\gamma$), but because neutrinos are already decoupled, the entropy of the $e^\pm$ pairs is dumped **exclusively into the photon bath**:

- Before annihilation ($T > m_e$): The electromagnetic plasma contains photons ($g_\gamma = 2$) and $e^\pm$ pairs ($g_e = 4$):

$$g_{*S,\text{em}}(T > m_e) = 2 + \frac{7}{8}(2+2) = \frac{11}{2}. \tag{3.4.29}$$

- After annihilation ($T < m_e$): Only photons remain relativistic ($g_\gamma = 2$):

$$g_{*S,\text{em}}(T < m_e) = 2. \tag{3.4.30}$$

By comoving entropy conservation, $S_{\text{em}} = g_{*S,\text{em}}(aT)^3 = \text{constant}$:

$$(aT_\gamma)^3_{\text{after}} = \frac{g_{*S}(T > m_e)}{g_{*S}(T < m_e)} (aT_\gamma)^3_{\text{before}} = \frac{11}{4} (aT_\nu)^3. \tag{3.4.31}$$

This yields the exact temperature ratio between photons and the Cosmic Neutrino Background (CNB):

$$T_\nu = \left(\frac{4}{11} \right)^{1/3} T_\gamma \approx 0.7137\, T_\gamma. \tag{3.4.32}$$

Given $T_{\gamma 0} = 2.7255\,\mathrm{K}$ today, the relic neutrino temperature is $T_{\nu 0} = 1.945\,\mathrm{K}$ ($k_{\mathrm{B}} T_{\nu 0} = 1.676 \times 10^{-4}\,\mathrm{eV}$), with total number density:

$$n_\nu = 3 \times \frac{3}{4} \left(\frac{4}{11} \right) n_\gamma = \frac{9}{11} n_\gamma \approx 336\,\mathrm{cm}^{-3} \quad (112\,\mathrm{cm}^{-3} \text{ per flavor}). \tag{3.4.33}$$

Present degrees of freedom g_{*0} and g_{*S0}

Evaluating today for photons and 3 relativistic neutrino species:

$$g_{*0} = 2 + 3 \times \frac{7}{8}(2)\left(\frac{T_\nu}{T_\gamma}\right)^4 = 2 + \frac{21}{4}\left(\frac{4}{11}\right)^{4/3} = 3.363, \tag{3.4.34}$$

$$g_{*S0} = 2 + 3 \times \frac{7}{8}(2)\left(\frac{T_\nu}{T_\gamma}\right)^3 = 2 + \frac{21}{4}\left(\frac{4}{11}\right) = \frac{43}{11} = 3.909. \tag{3.4.35}$$

Table 3.5: Particle Content and Effective Relativistic Degrees of Freedom $g_*(T)$ in the Standard Model

Temperature T	**Mass Threshold**	**Relativistic Particle Species**	$g_*(T)$
$T < m_e$	< 0.511 MeV	γ (+3 decoupled ν)	3.36
$m_e < T < T_{\nu,D}$	0.511–1.0 MeV	$\gamma, e^\pm, 3\nu$	10.75
$T_{\nu,D} < T < m_\mu$	1–106 MeV	$\gamma, e^\pm, \mu^\pm, 3\nu$	14.25
$m_\mu < T < m_\pi$	106–135 MeV	$\gamma, e^\pm, \mu^\pm, \pi^\pm, \pi^0, 3\nu$	17.25
$T_{\text{QCD}} < T < m_s$	150–400 MeV	$\gamma, e^\pm, \mu^\pm, 3\nu, u, d, 8g$ (QGP)	61.75
$m_s < T < m_c$	0.5–1.3 GeV	$+s, \bar{s}$	75.75
$m_c < T < m_\tau$	1.3–1.8 GeV	$+c, \bar{c}$	86.25
$m_\tau < T < m_b$	1.8–4.2 GeV	$+\tau^\pm$	92.25
$m_b < T < m_W$	4.2–80.4 GeV	$+b, \bar{b}$	96.25
$m_W < T < m_t$	80.4–173 GeV	$+W^\pm, Z^0$	106.75
$T > m_H, m_t$	> 173 GeV	Full Standard Model $(+t, \bar{t}, H^0)$	**106.75**
$T > T_{\text{GUT}}$ (Minimal SU(5))	$> 10^{15}$ GeV	Full GUT Multiplets	161.75
$T > T_{\text{SUSY}}$ (MSSM)	> 1 TeV	Full Supersymmetric Spectrum	228.75

3.4.9 Out-of-equilibrium relativistic Boltzmann equation

To compute the relic abundance of decoupled species (dark matter, light nuclei in BBN, and recombination), we formulate the **covariant Boltzmann transport equation**:

$$\hat{\mathcal{L}}[f] = \mathcal{C}[f], \tag{3.4.36}$$

where $\mathcal{C}[f]$ is the collision integral, and $\hat{\mathcal{L}}$ is the Liouville operator along the phase-space geodesic congruence:

$$\hat{\mathcal{L}}[f] = p^\alpha \frac{\partial f}{\partial x^\alpha} - \Gamma^\alpha_{\beta\gamma} p^\beta p^\gamma \frac{\partial f}{\partial p^\alpha}. \tag{3.4.37}$$

In a spatially homogeneous and isotropic FLRW metric, $f = f(E, t)$, reducing the Liouville operator to:

$$\hat{\mathcal{L}}[f] = E\frac{\partial f}{\partial t} - Hp^2 \frac{\partial f}{\partial E}. \tag{3.4.38}$$

Integrating over momentum space $\frac{g_i}{(2\pi)^3}\int \frac{\mathrm{d}^3p}{E}$, the left-hand side yields the macroscopic rate equation:

$$\frac{g_i}{(2\pi)^3}\int \hat{\mathcal{L}}[f]\frac{\mathrm{d}^3p}{E} = \dot{n}_i + 3Hn_i = \frac{1}{a^3}\frac{\mathrm{d}}{\mathrm{d}t}\left(n_i a^3\right). \tag{3.4.39}$$

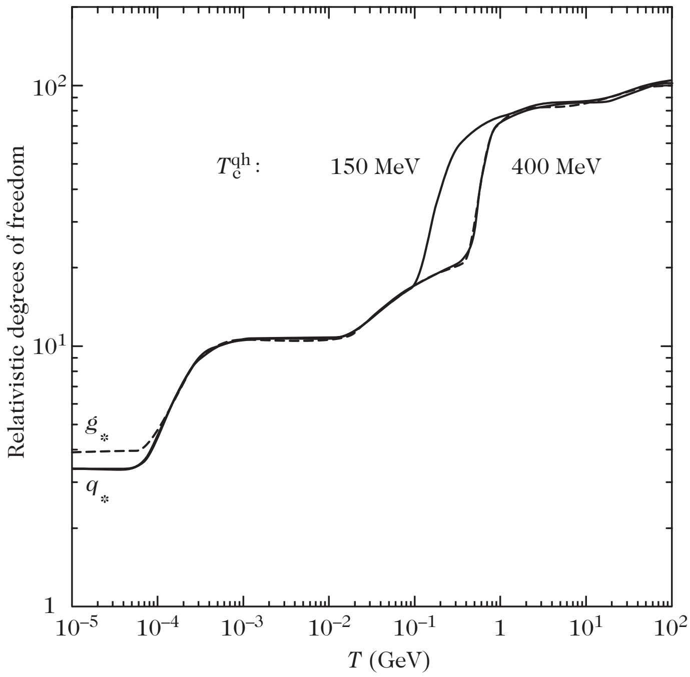


Figure 3.7: Evolution of the effective relativistic degrees of freedom $g_*(T)$ and $g_{*S}(T)$ as a function of temperature T in the Standard Model $SU(3)_c \times SU(2)_L \times U(1)_Y$, showing the electroweak, quark-hadron QCD ($T_c \approx 150$–$400\,\text{MeV}$), and e^+e^- annihilation steps.

Boltzmann Transport Equation

Boltzmann Equation, in theory, can be used to derive the distribution function and its time evolution. However, the problem is the need for boundary condition. Which means, we need to know the distribution at some time t. But, in the framework where the system begins to move away from the equilibrium, we already know the distribution at some time t from the equilibrium statistical mechanics. Thus, it can be used to inquire more about the system.

$$\frac{\partial f}{\partial t} + \frac{\vec{p}}{m} \cdot \nabla_{\vec{q}} f + \vec{F} \cdot \nabla_{\vec{p}} f = \left(\frac{\partial f}{\partial t}\right)_{\text{collision}}$$

Example

Let us consider Maxwell-Boltzmann distribution[a]:

$$f(\vec{p}, \vec{q}) = A e^{-\beta\left(\frac{|\vec{p}|^2}{2m} + U(\vec{q})\right)}, \text{ where } \beta = \frac{1}{k_B T}.$$

Since this distribution describes equilibrium distribution the collision term would vanish i.e., $\partial_t f = 0$, whereas the gradients are:

$$\nabla_{\vec{q}} f = -\beta f \nabla_{\vec{q}} U = \beta \vec{F} f, \qquad \nabla_{\vec{p}} f = -\beta \frac{\vec{p}}{m} f,$$

that is the sum of the two gradient term in the Boltzmann Transport Equation is zero. The density distribution under this distribution is not uniform, unless the potential is a constant, in which the distribution becomes Maxwell distribution

$$f(\vec{p}) = A e^{-\beta \frac{|\vec{p}|^2}{2m}},$$

that gives a non-zero gradient $\nabla_{\vec{p}} f = -\beta \frac{\vec{p}}{m} f$, but still satisfies the Boltzmann equation, since now $\vec{F} = -\nabla_{\vec{q}} U = 0$.

3.4.10 Derivation

The derivation that we will follow is from Kerson Huang's Statistical Mechanics as well as Donald McQuarrie's Stat Mech. We assume binary collision and ignore the impact of any external force on the collision. We will model the interaction taking place between the particles as sort of collision to get a better visualization into the problem.
In the absense of collision, all the molecule that start out at (r, v, t) will end up at $(r + vt, v + \delta v, t + dt)$. However in the presence of collision, some molecule may leave the stream. Let the number of $j-$molecule lost due to collision be represented by $\Gamma^-_{ji} dr dv_j dt$. Similary let the number of $j-$molecule that join the stream due to collision with $i-$molecule be represented by $\Gamma^+_{ji} dr dv_j dt$. Where i and j labels the species of molecule.

$$\left(\frac{\partial f_j}{\partial t}\right)_{\text{collision}} = \sum_i \Gamma^+_{ji} - \Gamma^-_{ji}$$

Consider a $12 \to 1'2'$ scattering process where we consider the '$2'$ as target and '$1'$ as projectile. The number of collision taking place in the time interval is given as

$$\text{Number of collision} = \underbrace{dN_{12}}_{\text{initial number of colliding particles}} \; \underbrace{dP_{12\to 1'2'}}_{\text{rate of collision}} \; \delta t$$

Where dN_{12} is the initial number of colliding pair $(\vec{p}_1, \vec{p}_2)$. We introduce the two-particle correlation function F which encodes their initial correlated distribution information by

$$dN_{12} = F(\vec{r}, \vec{p}_1, \vec{p}_2, t) d^3r d^3p_1 d^3p_2.$$

This can be rewritten in terms of cross-section σ, since a collision is just transition from the initial state to a set of final states via interaction. If we assume binary collision. Then,

$$\frac{\text{number of collision}}{\text{volume of phase space}} = \frac{\text{net change in the number of particle}}{\text{volume of phase space}} = (\Gamma^+ - \Gamma^-)\delta t$$

$$\begin{aligned}
\Gamma^+ \cancel{d^3r d^3p_1 \delta t} &= dN_{12} dP_{12\to 1'2'} \delta t \\
&= F(\vec{r}, \vec{p}_1, \vec{p}_2, t) \cancel{d^3r d^3p_1} d^3p_2 dP_{12\to 1'2'} \cancel{\delta t} \\
\Gamma^+ &= d^3p_2 dP_{12\to 1'2'} F(\vec{r}, \vec{p}_1, \vec{p}_2, t)
\end{aligned}$$

Thus, we have

$$\left(\frac{\partial f_1}{\partial t}\right)_{\text{collision}} = d^3p_2 \, \overbrace{|\vec{v}_1 - \vec{v}_2| d\sigma}^{\text{from transition rate for 1 particle}} \, (F_{1'2'} - F_{12})$$

Now, we will make a crucial assumption, also known as molecular chaos assumption or stosszahlansatz. This assumption is generally not true and implies that position and velocities are uncorrelation. It is the precisely why we apply the Boltzmann Equation for dilute gases.

$$\frac{\mathrm{d}f_1}{\mathrm{d}t} = \frac{\partial f_1}{\partial t} + \frac{\vec{p}_1}{m} \cdot \nabla_{\vec{q}} f_1 + \vec{F} \cdot \nabla_{\vec{p_1}} f_1 = d^3p_2 |\vec{v}_1 - \vec{v}_2| \overset{\text{integrated over all final state}}{d\sigma} \; (f_{1'} f_{2'} - f_1 f_2) \tag{1}$$

The above equation holds true in flat spacetime, however in curved spacetime the LHS of Boltzmann Transport Equation needs modification as follows:

$$
\begin{aligned}
\frac{\mathrm{d}f}{\mathrm{d}\lambda} &= \frac{\partial f}{\partial t}\frac{\mathrm{d}t}{\mathrm{d}\lambda} + \frac{\partial f}{\partial x^i}\frac{\mathrm{d}x^i}{\mathrm{d}\lambda} + \frac{\partial f}{\partial E}\frac{\mathrm{d}E}{\mathrm{d}\lambda} + \frac{\partial f}{\partial p^i}\frac{\mathrm{d}p^i}{\mathrm{d}\lambda} \\
&= p^0\frac{\partial f}{\partial t} + p^i\frac{\partial f}{\partial x^i} + \frac{\mathrm{d}p^\mu}{\mathrm{d}\lambda}\frac{\partial f}{\partial p^\mu} \\
&= p^\mu\frac{\partial f}{\partial x^\mu} + \frac{\mathrm{d}p^\mu}{\mathrm{d}\lambda}\frac{\partial f}{\partial p^\mu} \\
&= p^\mu\frac{\partial f}{\partial x^\mu} - \Gamma^\mu_{\alpha\beta}p^\alpha p^\beta\frac{\partial f}{\partial p^\mu} = \mathcal{L}[f]
\end{aligned}
$$

We have substituted $dp^\mu/d\lambda$ from the geodesic equation. Now, we will simplify the above in case of FLRW geometry.

$$
\begin{aligned}
\frac{\mathrm{d}f}{\mathrm{d}\lambda} &= p^\mu\frac{\partial f}{\partial x^\mu} + \frac{p^\mu}{p^\mu}\frac{\mathrm{d}p^\mu}{\mathrm{d}\lambda}\frac{\partial f}{\partial p^\mu} \\
&= p^\mu\frac{\partial f}{\partial x^\mu} + \frac{\mathrm{d}\ln p^\mu}{\mathrm{d}\lambda}\frac{\partial f}{\partial \ln p^\mu} \\
&= p^\mu\frac{\partial f}{\partial x^\mu} - \frac{\mathrm{d}\ln a}{\mathrm{d}\lambda}\frac{\partial f}{\partial \ln p^\mu} && \text{(used } p^\mu a = \text{constant)} \\
\frac{\mathrm{d}t}{\mathrm{d}\lambda}\frac{\mathrm{d}f}{\mathrm{d}t} &= p^\mu\frac{\partial f}{\partial x^\mu} - \frac{\mathrm{d}t}{\mathrm{d}\lambda}\frac{\mathrm{d}\ln a}{\mathrm{d}t}\frac{\partial f}{\partial \ln p^\mu}
\end{aligned}
$$

assuming $f \equiv f(t,p)$

$$
\begin{aligned}
p^0\frac{\mathrm{d}f}{\mathrm{d}t} &= p^0\frac{\partial f}{\partial t} - p^0\frac{\mathrm{d}\ln a}{\mathrm{d}t}\frac{\partial f}{\partial \ln p^\mu} = C[f] \\
\frac{\mathrm{d}f}{\mathrm{d}t} &= \frac{C[f]}{p^0}
\end{aligned}
$$

We can express the Boltzmann Equation simply as:[b]

$$
\frac{\mathrm{d}f}{\mathrm{d}\lambda} = \mathcal{L}[f] = C[f]
$$

Now, we will integrate f with respect to d^3p

$$
\begin{aligned}
g\int\frac{\mathrm{d}f}{\mathrm{d}t}\frac{d^3p}{(2\pi\hbar)^3} &= g\int\left[\frac{\partial f}{\partial t} - \frac{\mathrm{d}\ln a}{\mathrm{d}t}\frac{\partial f}{\partial \ln p^\mu}\right]\frac{d^3p}{(2\pi\hbar)^3} \\
&= \frac{\partial}{\partial t}\frac{g}{(2\pi\hbar)^3}\int f(t,p)d^3p - \frac{\mathrm{d}\ln a}{\mathrm{d}t}\frac{g}{(2\pi\hbar)^3}\int\frac{\partial f}{\partial \ln p^\mu}d^3p \\
&= \frac{\mathrm{d}n}{\mathrm{d}t} - \frac{\mathrm{d}\ln a}{\mathrm{d}t}\frac{g}{(2\pi\hbar)^3}\left\{ f(t,p)\times 4\pi p^3\Big|_{p=0}^{p=\infty} - \int 3f(t,p)d^3p\right\} \\
&= \frac{\mathrm{d}n}{\mathrm{d}t} + \frac{\mathrm{d}\ln a}{\mathrm{d}t}\times 3n = \frac{1}{a^3}\frac{\mathrm{d}na^3}{\mathrm{d}t} && (2)
\end{aligned}
$$

where we evaluated the second integral as follows:

$$
\begin{aligned}
\int\frac{\partial f}{\partial \ln p^\mu}d^3p &= \int\frac{\partial f}{\partial \ln p^\mu}4\pi p^2\frac{p}{p}dp \\
&= \int_0^\infty\frac{\partial f}{\partial \ln p^\mu}d\ln p\times 4\pi p^3 - \int_0^\infty f d(4\pi p^3) \\
&= f4\pi p^3\Big|_{p=0}^{p=\infty} - 3\int_0^\infty f4\pi p^2 dp
\end{aligned}
$$

3.4.11 Away from Equilibrium

We first note that, the number density given as

$$n = \frac{g}{(2\pi\hbar)^3}\int d^3p f(\vec{p}) \tag{3}$$

can be used in Boltzmann Transport Equation to express the evolution of number density:

$$\begin{aligned}
\frac{g}{(2\pi\hbar)^3}\int d^3p \frac{\mathrm{d}f(\vec{p})}{\mathrm{d}t} &= \overbrace{\frac{g}{(2\pi\hbar)^3}\int C[f]\frac{d^3p}{E}}^{\text{Lorentz Invariant}} = \tilde{C}[f] \\
\frac{1}{a^3}\frac{\mathrm{d}na^3}{\mathrm{d}t} &= \prod_{i=1}^{4}\int \frac{g_i d^3p_i}{(2\pi\hbar)^3 2E_i}\left|\tilde{\mathcal{M}}_{12\to34}\right|^2 (2\pi)^4\delta^4(p_1+p_2-p_3-p_4) \\
&\qquad \times \frac{1}{S}\left[f_3f_4(1\pm f_1)(1\pm f_2) - f_1f_2(1\pm f_3)(1\pm f_4)\right]
\end{aligned}$$

where $\tilde{C}$ is the new collision term which is directly defined in the second line. g is the internal degree of freedom and S is the appropriate symmetry factors for identical particles in the initial or final states, taking into account that there can be multiple particles of the same species in the final state. δ corresponds to the delta function, and $\left|\tilde{\mathcal{M}}_{12\to34}\right|$ is the amplitude of specific process, averaged over initial and final spins as well as internal degree of freedom.

$$\left|\tilde{\mathcal{M}}_{12\to34}\right| = \frac{1}{\prod_i g_i}\sum_{\text{spins } i}|\mathcal{M}_i|$$

The plus and minus sign in $1 \pm f_i$ corresponds to the particle species being bosons and fermions respectively. We now assume, the particles are approximately obeying Maxwell Boltzmann distribution for further simplification.

$$\frac{1}{e^{E-\mu/k_BT} \pm 1} \simeq e^{-E+\mu/k_BT}$$

Under this assumption, the Bose enhancement and Pauli blocking term $1 \pm f_i$ can be dropped upto first order in perturbation theory.

$$\begin{aligned}
f_3f_4(1\pm f_1)(1\pm f_2) - f_1f_2(1\pm f_3)(1\pm f_4) &\approx f_3f_4 - f_1f_2 \\
&= e^{-E_1+E_2/k_BT}\left\{e^{\mu_3+\mu_4/k_BT} - e^{\mu_1+\mu_2/k_BT}\right\}
\end{aligned}$$

where we used $E_1 + E_2 = E_3 + E_4$ for simplification. The above approximation, also leads to:

$$\left.\begin{aligned}
n_i &= g_i e^{\mu_i/k_BT}\int \tfrac{d^3p}{(2\pi\hbar)^3} e^{-E_i/k_BT} \\
n_i^{(0)} &= g_i \int \tfrac{d^3p}{(2\pi\hbar)^3} e^{-E_i/k_BT}
\end{aligned}\right\} \frac{n_i}{n_i^{(0)}} = e^{\mu_i/k_BT} \tag{4}$$

Let us define thermally averaged cross section as[c]

$$\begin{aligned}
\langle\sigma v\rangle &\equiv \frac{1}{n_1n_2}\prod_{i=1}^{4}\int\frac{d^3p_i}{(2\pi\hbar)^3 2E_i}\underbrace{e^{-E_1+\mu_1-E_2+\mu_2/k_BT}}_{\text{distribution function}}\left|\tilde{\mathcal{M}}\right|^2(2\pi)^4\delta(p_1+p_2-p_3-p_4) \\
&= \frac{1}{n_1^{(0)}n_2^{(0)}}\prod_{i=1}^{4}\int\frac{d^3p_i}{(2\pi\hbar)^3 2E_i}e^{-(E_1+E_2)/k_BT}\left|\tilde{\mathcal{M}}\right|^2(2\pi)^4\delta(p_1+p_2-p_3-p_4)
\end{aligned}$$

Then, finally:

$$\frac{1}{a^3}\frac{\mathrm{d}n_1a^3}{\mathrm{d}t} = \langle\sigma v\rangle\, n_1^{(0)} n_2^{(0)} \left\{ \frac{n_3n_4}{n_3^{(0)}n_4^{(0)}} - \frac{n_1n_2}{n_1^{(0)}n_2^{(0)}} \right\}$$

$$\frac{1}{a^3}\frac{\mathrm{d}n_1a^3}{\mathrm{d}\ln a}\frac{\mathrm{d}\ln a}{\mathrm{d}t} = -\overset{\text{thermally averaged cross section}}{\langle\sigma v\rangle}\, n_1 n_2 \left\{1 - \frac{n_3n_4}{n_1n_2}\left(\frac{n_1^{(0)}n_2^{(0)}}{n_3^{(0)}n_4^{(0)}}\right)\right\}$$

$$\frac{1}{n_1a^3}\frac{\mathrm{d}n_1a^3}{\mathrm{d}\ln a}\frac{\dot{a}}{a} = -n_2\,\langle\sigma v\rangle \left\{1 - \frac{n_3n_4}{n_1n_2}\left(\frac{n_1^{(0)}n_2^{(0)}}{n_3^{(0)}n_4^{(0)}}\right)\right\}$$

The Boltzmann Equation can be rewritten in more suggestive form:

$$\frac{1}{n_1a^3}\frac{\mathrm{d}n_1a^3}{\mathrm{d}x} = -\frac{\Gamma_1}{H}\left\{1 - \frac{n_3n_4}{n_1n_2}\left(\frac{n_1n_2}{n_3n_4}\right)_{eq}\right\}$$

where $\Gamma_1 \equiv n_2\,\langle\sigma v\rangle$ is the interaction rate and H is the expansion rate. and the second factor just tells us how close we are to equilibrium. Lets try to understand the solution of this general equation a bit better which will reveal a few concepts like decouping and freeze-out that will be very useful later on. If $\Gamma_1 >> H$ then the prefactor on the right hand side is huge and the system is efficiently driven towards chemical equlibrium. This is not hard to see from the form of the equation and is also easy to understand physically: if the interaction rate is large relative to the expansion rate of the Universe then we have many interactions in the time it takes the Universe to double in size and we are easily able to maintain equilibrium $n_i \sim n_i^{\mathrm{eq}}$ even though the number density of each species gets smaller and smaller. However this cannot last forever. Remember that the equilibrium distribution for massive partices $n_i \propto e^{-mc^2/k_BT}$ gets exponensialy suppressed when the temperature goes down (so if equilibrium had lasted we would have no more massive particles left in the Universe today). Thus the equilibrium we have must eventually fail. Once Γ_1 drops below H then the interactions are no longer efficient enough to maintain equilibrium and we say that it decouples. After it decouples the interaction is no longer relevant, so knowing when ($\Gamma_1 \sim H$) this happens already gives us important information. If a particle species has no more efficient interactions that ties it to the rest of the primordial plasma then it can no longer stay in equilibrium with it and will evolve on its own like all the other stuff wasn't there (and if anything happens in the plasma that changes, say, its temperature then that change will not be reflected in stuff that has decoupled).
As we reach $\Gamma_1 << H$ the right hand side is practically zero and in this regime the equation reduces to $d(n_1a^3)dx \approx 0 \implies n_1 \propto 1/a^3$ which is just normal volume dilution in an expanding Universe (i.e. the number of particles in a co-moving volume stays constant, its the physical volume that gets bigger $\propto a^3$) which is what we would expect if there was no interactions at all. When this happens for a massive particle species we say that it **freezes out**. The evolution is exponential decay of n_1a^3 until decoupling, $\Gamma_1 \sim H$, and then it flattens out us and leaves us with a freeze-out abundance $n_1a^3 \approx$constant. What this freeze-out abundance really is typucally requires numerical solutions of the Boltzmann equation (though we shall see an example of dark matter production where we can do some analytical estimates). We will encounter freeze-out in two main situations 1) when we look at recombination we see that not all of the free electrons and protons will end up in atoms 2) freeze-out is one of the most common production mechanisms for producing dark matter (for example WIMP's).

[a]It can be derived for dilute gases using Boltzmann Transport Equation, see chapter 4 of Kerson Huang's Statistical Mechanics

[b]better derivation for collision term can be found in "Boltzmann equation and its cosmological applications" by Seishi Enomoto

[c]The thermal averaging will come from the integral over p_1 and p_2 whereas p_3 and p_4 will be stored inside cross-section.

Collision integral for $2 \leftrightarrow 2$ scatterings

The fundamental challenge in solving the Boltzmann equation lies in evaluating the collision functional $\mathcal{C}[f_i]$. Consider a two-body scattering and annihilation process:

$$i + j \longleftrightarrow k + l. \tag{3.4.40}$$

The macroscopic collision term C_i decomposes into creation $(k + l \to i + j)$ and destruction $(i + j \to k + l)$ components:

$$C_i = C_{kl\to ij} - C_{ij\to kl}, \tag{3.4.41}$$

where the destruction rate per unit volume is:

$$\begin{aligned} C_{ij\to kl} = (2\pi)^4 \int \frac{g_i \mathrm{d}^4 p_i}{(2\pi)^3}\, \delta(E_i^2 - \mathbf{p}_i^2 - m_i^2) \ldots \frac{g_l \mathrm{d}^4 p_l}{(2\pi)^3}\, \delta(E_l^2 - \mathbf{p}_l^2 - m_l^2) \\ \times\, \delta^{(4)}(p_i + p_j - p_k - p_l)\, |\mathcal{M}|^2_{ij\to kl}\, f_i f_j (1 \pm f_k)(1 \pm f_l). \end{aligned} \tag{3.4.42}$$

Here, the $+$ sign corresponds to Bose enhancement and the $-$ sign to Fermi blocking for the final-state particles k and l. The matrix element $|\mathcal{M}|^2_{ij\to kl}$ is the spin-averaged quantum transition amplitude, and the four-dimensional Dirac delta enforces energy-momentum conservation $p_i + p_j = p_k + p_l$.

Derivation of the collision integral
Consider the reaction

$$i + j \leftrightarrow k + l.$$

The collision term in the Boltzmann equation measures the *net* change in the number of particles of species i due to this reaction. There are therefore two contributions:

$$\text{net rate} = \text{rate of } k + l \to i + j - \text{rate of } i + j \to k + l.$$

The derivation naturally separates into three parts. First, we determine how the occupation numbers of the quantum states enter the transition probability. Second, we determine the transition rate associated with a given set of momenta. Finally, we sum over all available momentum and internal states to obtain the collision integral.
We use natural units, $\hbar = c = k_B = 1$, and work in a local inertial frame. The four species are taken to be distinct, so no additional identical-particle symmetry factors are required. Correlations between distinct momentum and internal-state modes are neglected. The occupation number N_a refers to a single quantum mode, while

$$f_a(t, \mathbf{p}_a) \equiv \langle N_a \rangle$$

denotes its mean occupation. The quantity n_a will denote the corresponding particle number density.

1. Occupation-number factors
The first question is: given a particular set of occupation numbers, what is the amplitude for the transition $i + j \to k + l$?

The operator responsible for this transition has the structure

$$a_k^\dagger a_l^\dagger a_j a_i.$$

The two annihilation operators remove the incoming particles, whereas the two creation operators add the outgoing particles.
For a bosonic or fermionic mode, define

$$\eta_a = \begin{cases} +1, & \text{boson,} \\ -1, & \text{fermion.} \end{cases}$$

The canonical (anti)commutation relations imply

$$a_a a_a^\dagger = 1 + \eta_a a_a^\dagger a_a.$$

Since

$$\widehat{N}_a = a_a^\dagger a_a,$$

we have

$$\begin{aligned} \|a_a \left|N_a\right\rangle\|^2 &= \left\langle N_a\right| a_a^\dagger a_a \left|N_a\right\rangle = N_a, \\ \|a_a^\dagger \left|N_a\right\rangle\|^2 &= \left\langle N_a\right| a_a a_a^\dagger \left|N_a\right\rangle = 1 + \eta_a N_a. \end{aligned}$$

Thus

$$a_a \left|N_a\right\rangle = \sqrt{N_a} \left|N_a - 1\right\rangle,$$

whereas

$$a_a^\dagger \left|N_a\right\rangle = \sqrt{1 + \eta_a N_a} \left|N_a + 1\right\rangle.$$

The physical interpretation is important. An incoming particle can be removed with an amplitude proportional to $\sqrt{N_a}$, since there are N_a particles available in the mode. The creation of an outgoing particle gives the factor $\sqrt{1 + \eta_a N_a}$. For bosons this becomes $\sqrt{1 + N_a}$, giving Bose enhancement. For fermions it becomes $\sqrt{1 - N_a}$, which vanishes when the state is already occupied and therefore gives Pauli blocking.
For the complete transition,

$$\begin{aligned} &a_k^\dagger a_l^\dagger a_j a_i \left|\ldots, N_i, \ldots, N_j, \ldots, N_k, \ldots, N_l, \ldots\right\rangle \\ &\quad = \sqrt{N_i N_j (1 + \eta_k N_k)(1 + \eta_l N_l)} \\ &\qquad \times \left|\ldots, N_i - 1, \ldots, N_j - 1, \ldots, N_k + 1, \ldots, N_l + 1, \ldots\right\rangle, \end{aligned}$$

up to an irrelevant phase and the appropriate ordering convention for fermionic operators. Consequently, if $\mathcal{A}_0$ denotes the transition amplitude with the occupation-number dependence removed, then

$$|\mathcal{A}_{\{N\}}|^2 = |\mathcal{A}_0|^2 N_i N_j (1 + \eta_k N_k)(1 + \eta_l N_l).$$

Thus the statistical factors appearing in the collision integral are not inserted by hand. They follow directly from the matrix elements of the creation and annihilation operators.

2. Statistical averaging
A macroscopic gas is not described by one particular occupation-number state. Instead, we average over an ensemble described by a density operator ρ:

$$\langle O \rangle \equiv \mathrm{Tr}(\rho O).$$

In particular,

$$f_a = \langle a_a^\dagger a_a \rangle = \langle N_a \rangle.$$

Using the operator identity above,

$$\langle a_a a_a^\dagger \rangle = 1 + \eta_a f_a.$$

If correlations between the four participating modes are neglected, the occupation-number factor therefore factorizes:

$$\begin{aligned}&\langle N_i N_j (1+\eta_k N_k)(1+\eta_l N_l)\rangle \\ &\quad \simeq f_i f_j (1+\eta_k f_k)(1+\eta_l f_l).\end{aligned}$$

Thus the forward reaction is weighted by

$$f_i f_j (1+\eta_k f_k)(1+\eta_l f_l),$$

while the inverse reaction is weighted by

$$f_k f_l (1+\eta_i f_i)(1+\eta_j f_j).$$

No assumption of thermal equilibrium has been made. In equilibrium one would have

$$f_a^{\mathrm{eq}}(E_a) = \frac{1}{\exp[(E_a-\mu_a)/T] - \eta_a},$$

but the collision term is valid for arbitrary nonequilibrium distributions $f_a(t, \mathbf{p}_a)$.

3. Continuum normalization

We now derive the rate directly in the continuum, without introducing a finite box. This is the cleanest way to obtain the final form of the collision integral.

Use relativistically normalized one-particle states,

$$\langle \mathbf{p}, r | \mathbf{p}', r' \rangle = 2E(2\pi)^3 \delta^{(3)}(\mathbf{p}-\mathbf{p}')\delta_{rr'},$$

where

$$E = \sqrt{\mathbf{p}^2 + m^2}.$$

For fixed momenta and internal states, the S-matrix element is written as

$$\langle f|S|i\rangle = \mathrm{i}(2\pi)^4 \delta^{(4)}(q)\,\mathcal{M}_{ij\to kl},$$

with

$$q = p_i + p_j - p_k - p_l.$$

The invariant amplitude $\mathcal{M}$ contains the microscopic dynamics of the scattering process. The remaining factors arise from the normalization of states and from the statistical occupation of those states.

For a fixed incoming pair, the probability per unit time to produce particles with momenta in $\mathrm{d}^3 p_k$ and $\mathrm{d}^3 p_l$ is proportional to the final-state phase-space measure:

$$\begin{aligned}\mathrm{d}\Gamma_{ij\to kl} &= (2\pi)^4 \delta^{(4)}(q) |\mathcal{M}_{ij\to kl}|^2 \\ &\quad \times \frac{\mathrm{d}^3 p_k}{(2\pi)^3\, 2E_k} \frac{\mathrm{d}^3 p_l}{(2\pi)^3\, 2E_l}.\end{aligned}$$

To describe a gas, rather than one particular incoming pair, we must sum over the available incoming particles. This introduces the same invariant phase-space measure for each incoming particle:

$$d\Pi_a \equiv \frac{d^3p_a}{(2\pi)^3\, 2E_a}.$$

Hence the destruction rate per unit volume for $i + j \to k + l$ is

$$C_{ij\to kl} = (2\pi)^4 \int d\Pi_i d\Pi_j d\Pi_k d\Pi_l\, \delta^{(4)}(q)|\mathcal{M}_{ij\to kl}|^2 \\ \times f_i f_j (1+\eta_k f_k)(1+\eta_l f_l).$$

The delta function enforces energy-momentum conservation, the matrix element describes the microscopic interaction, and the occupation factors determine how many particles are available initially and how the quantum statistics of the final states modify the transition.

4. The invariant phase-space measure

The one-particle measure

$$d\Pi_a = \frac{d^3p_a}{(2\pi)^3\, 2E_a}$$

is Lorentz invariant. Its appearance follows from the relativistic normalization of one-particle states.

The factor $2E_a$ comes from the normalization

$$\langle \mathbf{p}|\mathbf{p}'\rangle = 2E(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{p}'),$$

while $(2\pi)^3$ is the conventional momentum-space normalization. Thus the invariant phase-space measure is simply the continuum form of the quantum-state normalization.

5. Internal-state degeneracies

If a species has g_a internal states, such as spin or color states, the internal-state labels must also be summed. Explicitly, one has

$$\sum_{r_i,r_j,r_k,r_l} |\mathcal{M}_{r_i r_j\to r_k r_l}|^2.$$

For an unpolarized gas, it is conventional to define a matrix element averaged over the initial internal states and summed over the final ones:

$$\overline{|\mathcal{M}|^2} \equiv \frac{1}{g_i g_j} \sum_{r_i,r_j,r_k,r_l} |\mathcal{M}_{r_i r_j\to r_k r_l}|^2.$$

The important point is that the convention must be used consistently: the internal-state sums should not be counted both in $|\mathcal{M}|^2$ and again through the phase-space factors.

6. Gain and loss

We can now combine the forward and inverse reactions. The forward reaction

$$i + j \to k + l$$

removes particles of species i and therefore gives a loss term:

$$C^{\text{loss}}_{ij\to kl} = (2\pi)^4 \int d\Pi_i d\Pi_j d\Pi_k d\Pi_l\, \delta^{(4)}(q)|\mathcal{M}|^2 \\ \times f_i f_j (1+\eta_k f_k)(1+\eta_l f_l).$$

The inverse reaction

$$k + l \to i + j$$

creates particles of species i and therefore gives a gain term:

$$\begin{aligned} C^{\text{gain}}_{kl\to ij} = (2\pi)^4 \int & \mathrm{d}\Pi_i \mathrm{d}\Pi_j \mathrm{d}\Pi_k \mathrm{d}\Pi_l \, \delta^{(4)}(q) |\mathcal{M}|^2 \\ & \times f_k f_l (1+\eta_i f_i)(1+\eta_j f_j). \end{aligned}$$

Assuming microscopic reversibility, the same invariant matrix element may be used for the forward and inverse reactions. Hence

$$\begin{aligned} C_i &= C^{\text{gain}}_{kl\to ij} - C^{\text{loss}}_{ij\to kl} \\ &= (2\pi)^4 \int \mathrm{d}\Pi_i \mathrm{d}\Pi_j \mathrm{d}\Pi_k \mathrm{d}\Pi_l \, \delta^{(4)}(q) |\mathcal{M}|^2 \\ &\quad \times \Big[f_k f_l (1+\eta_i f_i)(1+\eta_j f_j) \\ &\qquad\quad - f_i f_j (1+\eta_k f_k)(1+\eta_l f_l) \Big]. \end{aligned}$$

This is the desired collision integral.
Expanding the phase-space measures gives

$$\begin{aligned} C_i = (2\pi)^4 \int & \prod_{a=i,j,k,l} \frac{\mathrm{d}^3 p_a}{(2\pi)^3\, 2E_a} \\ & \times \delta^{(4)}(p_i + p_j - p_k - p_l) |\mathcal{M}|^2 \\ & \times \Big[f_k f_l (1+\eta_i f_i)(1+\eta_j f_j) \\ & \qquad - f_i f_j (1+\eta_k f_k)(1+\eta_l f_l) \Big]. \end{aligned}$$

Writing

$$1 + \eta_a f_a = \begin{cases} 1 + f_a, & \text{bosons}, \\ 1 - f_a, & \text{fermions}, \end{cases}$$

gives the familiar quantum-statistical form of the collision integral.
Finally, the collision term enters the Boltzmann equation as

$$\dot{n}_i + 3H n_i = C_i.$$

The term $3Hn_i$ describes dilution of the number density due to the expansion of the universe, while C_i describes the local creation and destruction of particles through microscopic reactions.
In the dilute classical limit,

$$f_a \ll 1,$$

the Bose-enhancement and Pauli-blocking factors approach unity, and the statistical part of the collision term becomes

$$f_k f_l - f_i f_j.$$

The usual classical Boltzmann collision term is therefore recovered.
For reactions involving identical particles, additional symmetry and particle-counting factors must be included. These factors are distinct from the quantum-statistical factors derived above.

Assuming $\hat{\mathcal{C}}\hat{\mathcal{P}}$ invariance (or $\hat{\mathcal{T}}$ invariance / microscopic reversibility), the forward and backward transition amplitudes are identical:

$$|\mathcal{M}|^2_{ij\to kl} = |\mathcal{M}|^2_{kl\to ij} \equiv |\mathcal{M}|^2. \tag{3.4.43}$$

Using the on-shell identity $\mathrm{d}^4p\,\delta(E^2 - \mathbf{p}^2 - m^2) = \frac{\mathrm{d}^3p}{2E}$, the total net collision rate takes the compact form:

$$\begin{aligned} C_i = (2\pi)^4 \int \delta^{(4)}(p_i + p_j - p_k - p_l) \left(\prod_{a=i,j,k,l} \frac{g_a \mathrm{d}^3 p_a}{2(2\pi)^3 E_a} \right) |\mathcal{M}|^2 \\ \times \left[f_k f_l (1 \pm f_i)(1 \pm f_j) - f_i f_j (1 \pm f_k)(1 \pm f_l) \right]. \end{aligned} \tag{3.4.44}$$

The Boltzmann equation $\dot{n}_i + 3Hn_i = C_i$ thus captures three physical effects:

1. **Hubble dilution** ($3Hn_i$), reducing particle density due to volume expansion;
2. **Particle creation** ($C_{kl\to ij}$), populating state i via inverse reactions;
3. **Particle destruction** ($C_{ij\to kl}$), depleting state i via forward annihilations.

Maxwell–Boltzmann approximation and chemical equilibrium

In most cosmological applications (such as freeze-out and nucleosynthesis), particle energies satisfy $E_a - \mu_a \gg T$, allowing quantum statistical factors to be neglected:

$$1 \pm f_a \approx 1, \qquad f_a(p_a) \approx \mathrm{e}^{(\mu_a - E_a)/T}. \tag{3.4.45}$$

Under this approximation, the actual particle number density n_a relates to its fictitious zero-chemical-potential equilibrium density $\bar{n}_a \equiv n_a(\mu_a = 0)$ via:

$$n_a = \mathrm{e}^{\mu_a/T}\bar{n}_a. \tag{3.4.46}$$

Energy conservation $E_k + E_l = E_i + E_j$ enables the phase-space distribution product to factorize as:

$$\begin{aligned} f_k f_l - f_i f_j &= \mathrm{e}^{-(E_k+E_l)/T} \left[\mathrm{e}^{(\mu_k+\mu_l)/T} - \mathrm{e}^{(\mu_i+\mu_j)/T} \right] \\ &= \mathrm{e}^{-(E_i+E_j)/T} \left[\frac{n_k n_l}{\bar{n}_k \bar{n}_l} - \frac{n_i n_j}{\bar{n}_i \bar{n}_j} \right]. \end{aligned} \tag{3.4.47}$$

Substituting Eq. (3.4.47) into Eq. (3.4.44) yields the **integrated Boltzmann rate equation**:

$$\dot{n}_i + 3Hn_i = -\langle \sigma v \rangle \left(n_i n_j - \frac{\bar{n}_i \bar{n}_j}{\bar{n}_k \bar{n}_l} n_k n_l \right), \tag{3.4.48}$$

where the **thermally averaged cross section** $\langle \sigma v \rangle$ is defined covariantly by:

$$\bar{n}_i \bar{n}_j \langle \sigma v \rangle \equiv \int \frac{\mathrm{d}^3 p_i}{2E_i} \frac{\mathrm{d}^3 p_j}{2E_j} \frac{\mathrm{d}^3 p_k}{2E_k} \frac{\mathrm{d}^3 p_l}{2E_l} \frac{\delta^{(4)}(p_i + p_j - p_k - p_l)}{(2\pi)^8} |\mathcal{M}|^2 \, \mathrm{e}^{-(E_i+E_j)/T}. \tag{3.4.49}$$

When the reaction rate per particle $\Gamma \sim n_j \langle \sigma v \rangle$ exceeds the expansion rate H ($\Gamma \gg H$), the bracketed term in Eq. (3.4.48) must vanish, establishing **chemical equilibrium**:

$$\frac{n_i n_j}{\bar{n}_i \bar{n}_j} = \frac{n_k n_l}{\bar{n}_k \bar{n}_l} \iff \mu_i + \mu_j = \mu_k + \mu_l. \tag{3.4.50}$$

As the Universe expands and cools, H eventually overtakes Γ, breaking chemical equilibrium and freezing out particle abundances.

3.4.12 Frozen interactions and thermal relic freeze-out

Consider a stable massive particle species X and its antiparticle $\bar{X}$ undergoing pair annihilation into light Standard Model bath particles $l, \bar{l}$:

$$X + \bar{X} \longleftrightarrow l + \bar{l}. \tag{3.4.51}$$

Because the light particles $l, \bar{l}$ remain tightly coupled to the thermal plasma at zero chemical potential ($\mu_l = \mu_{\bar{l}} = 0$), their distribution satisfies $f_l f_{\bar{l}} = \mathrm{e}^{-(E_l+E_{\bar{l}})/T} = \mathrm{e}^{-(E_X+E_{\bar{X}})/T} = \bar{f}_X \bar{f}_{\bar{X}}$. For a symmetric relic ($n_X = n_{\bar{X}}$), the rate equation simplifies to:

$$\dot{n}_X + 3Hn_X = -\langle\sigma v\rangle \left(n_X^2 - \bar{n}_X^2\right). \tag{3.4.52}$$

Consider a process $\chi\bar{\chi} \to f\bar{f}$ with χ and $\bar{\chi}$ being the same in the non-relativistic limit. It implies that, for χ chemical potential should vanish i.e. $\mu = 0$.

$$\begin{aligned}
\tilde{C} &= \prod_{i=1}^{4} \int \frac{g_i d^3 p_i}{(2\pi\hbar)^3 2E_i} \left|\tilde{\mathcal{M}}_{12\to34}\right|^2 (2\pi)^4 \delta^4(p_1 + p_2 - p_3 - p_4) \left(\frac{1}{2} \times 2\right) (f_3 f_4 - f_1 f_2) \\
&= \prod_{i=1}^{2} \int \frac{g_i d^3 p_i}{(2\pi\hbar)^3 2\cancel{E_i}} \sigma v \cancel{2E_1 2E_2} \times (f_3 f_4 - f_1 f_2) \\
&= \prod_{i=1}^{2} \int \frac{g_i d^3 p_i}{(2\pi\hbar)^3} (f_3 f_4 - f_1 f_2) \times \sigma v
\end{aligned}$$

now, let us consider $f_3 f_4 = f_f f_{\bar{f}} = (f_\chi f_{\bar{\chi}})_{eq}$ and $f_1 f_2 = f_\chi f_{\bar{\chi}}$ and thermal averaging of cross-section.

$$\begin{aligned}
\frac{1}{a^3} \frac{\mathrm{d} n_\chi a^3}{\mathrm{d}t} &= \prod_{i=1}^{2} \int \frac{g_i d^3 p_i}{(2\pi\hbar)^3} (f_f f_{\bar{f}} - f_\chi f_{\bar{\chi}}) \times \sigma v \\
&= \prod_{i=1}^{2} \int \frac{g_i d^3 p_i}{(2\pi\hbar)^3} (f_\chi f_{\bar{\chi}})_{eq} \times \sigma v - \prod_{i=1}^{2} \int \frac{g_i d^3 p_i}{(2\pi\hbar)^3} f_\chi f_{\bar{\chi}} \times \sigma v \\
\frac{\mathrm{d} n_\chi}{\mathrm{d}t} + 3Hn_\chi &= \langle\sigma v\rangle \left(n^2_{\chi\ \text{equilibrium}} - n_\chi^2\right)
\end{aligned}$$

where we have assumed that the $(f_f)_{eq}$ and $(f_{\bar{f}})_{eq}$ are the distribution function of χ and $\bar{\chi}$, if they were in thermal equilibrium. In the last step we used the definition of thermally averaged cross section from where the n_χ terms came into the expression.

To eliminate the Hubble dilution term, we express the abundance in terms of the **comoving yield** (number of particles per comoving entropy volume):[1]

$$Y_X \equiv \frac{n_X}{s}, \qquad \bar{Y}_X \equiv \frac{\bar{n}_X}{s}. \tag{3.4.53}$$

Using comoving entropy conservation $\frac{\mathrm{d}(sa^3)}{\mathrm{d}t} = 0 \implies \dot{n}_X + 3Hn_X = s\dot{Y}_X$, we get:

$$\dot{Y}_X = \frac{\dot{n}_X}{s} - \frac{n_X \dot{s}}{s^2} = \frac{1}{s}\left[\dot{n}_X - \frac{n_X}{s}(-3Hs)\right] = -\langle\sigma v\rangle s \left(Y_X^2 - \bar{Y}_X^2\right). \tag{3.4.54}$$

[1]Since $n_X \propto \frac{1}{a^3}$ and $s \propto \frac{1}{a^3}$, we find that Y does not get diluted.

where in the last step we used (3.4.52) and (3.4.53). Introducing the dimensionless inverse temperature parameter:

$$x \equiv \frac{m_X}{T}, \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = m_X \frac{\mathrm{d}(1/T)}{\mathrm{d}t} = -\frac{m_X}{T}\frac{\dot{T}}{T} = Hx, \tag{3.4.55}$$

the evolution equation becomes:

$$\frac{\mathrm{d}Y_X}{\mathrm{d}x} = -\frac{\langle\sigma v\rangle s}{Hx}\left(Y_X^2 - \bar{Y}_X^2\right). \tag{3.4.56}$$

During the radiation-dominated era, the expansion rate and entropy density are:

$$H(T) = \sqrt{\frac{4\pi^3 g_*}{45}}\frac{T^2}{M_{\mathrm{Pl}}} = \sqrt{\frac{4\pi^3 g_*}{45}}\frac{m_X^2}{M_{\mathrm{Pl}}x^2}, \qquad s(T) = \frac{2\pi^2}{45}q_*(T)\frac{m_X^3}{x^3}, \tag{3.4.57}$$

where $q_* \equiv g_{*S}$ denotes the entropy relativistic degrees of freedom. Defining the dimensionless coupling parameter:

$$\lambda \equiv \langle\sigma v\rangle\sqrt{\frac{\pi}{45}}\frac{q_*}{\sqrt{g_*}}m_X M_{\mathrm{Pl}}, \tag{3.4.58}$$

and the deviation from equilibrium $\Delta \equiv Y_X - \bar{Y}_X$, Eq. (3.4.56) takes the standard Lee–Weinberg form:

$$\frac{\mathrm{d}\Delta}{\mathrm{d}x} = -\frac{\mathrm{d}\bar{Y}_X}{\mathrm{d}x} - \lambda x^{-2}\Delta\left(\Delta + 2\bar{Y}_X\right). \tag{3.4.59}$$

3.4.13 Asymptotic solutions: Cold and hot relics

This is interaction based mechanism for producing dark matter. Observeration does not tell us which mechanism was used for the production, only the final abundance Y.

Cold thermal relics ($x_f \gg 1$)

A **cold relic** freezes out when it is non-relativistic ($T_f \ll m_X$, i.e., $x_f \gtrsim 20$). The non-relativistic equilibrium yield is:

$$\bar{Y}_X(x) = \frac{\bar{n}_X}{s} = \frac{g_X\left(\frac{m_X T}{2\pi}\right)^{3/2}\mathrm{e}^{-m_X/T}}{\frac{2\pi^2}{45}q_* T^3} = \frac{45}{2\pi^4}\sqrt{\frac{\pi}{8}}\frac{g_X}{q_*}x^{3/2}\mathrm{e}^{-x}. \tag{3.4.60}$$

The thermally averaged cross section is parameterized by a velocity expansion $\langle\sigma v\rangle = \sigma_0 f(x) = \sigma_0 x^{-n}$, where $n = 0$ corresponds to s-wave annihilation and $n = 1$ to p-wave annihilation. Defining:

$$\lambda_0 \equiv \sigma_0\sqrt{\frac{\pi}{45}}\frac{q_*}{\sqrt{g_*}}m_X M_{\mathrm{Pl}}, \tag{3.4.61}$$

we analyze the two asymptotic regimes of Eq. (3.4.59):

1. **Pre-decoupling regime ($1 < x \ll x_f$):** The abundance tracks equilibrium ($Y_X \approx \bar{Y}_X$, $\Delta \ll \bar{Y}_X$). Setting $\frac{\mathrm{d}\Delta}{\mathrm{d}x} \approx 0$ yields:

$$\frac{\mathrm{d}\Delta}{\mathrm{d}x} = -\frac{\mathrm{d}\bar{Y}}{\mathrm{d}x} - \lambda\underbrace{\frac{\Delta^2}{x^2}}_{\approx 0} - \frac{2\lambda\Delta\bar{Y}}{x^2} \approx 0 \tag{3.4.62}$$

using (3.4.58) and (3.4.61), we can write

$$\Delta \simeq -\frac{x^2}{2\lambda_0 f(x)}\frac{\mathrm{d}\ln\bar{Y}_X}{\mathrm{d}x} \simeq \frac{x^2}{2\lambda_0 f(x)}, \tag{3.4.63}$$

since $\frac{\mathrm{d}\ln\bar{Y}_X}{\mathrm{d}x} = -1 + \frac{3}{2x} \approx -1$ for $x \gg 1$.

2. **Post-decoupling regime ($x \gg x_f$):** The equilibrium yield drops exponentially ($\bar{Y}_X \to 0$), leaving $Y_X \approx \Delta$. The Boltzmann equation reduces to:

$$\frac{\mathrm{d}\Delta}{\mathrm{d}x} \simeq -\lambda_0 \frac{f(x)}{x^2}\Delta^2 \implies \Delta_\infty^{-1} - \Delta_f^{-1} = \lambda_0 \int_{x_f}^{\infty} \frac{f(x)}{x^2}\mathrm{d}x. \tag{3.4.64}$$

Since $\Delta_f^{-1} \ll \Delta_\infty^{-1}$, the asymptotic relic yield is:

$$Y_\infty \simeq \left[\lambda_0 \int_{x_f}^{\infty} \frac{f(x)}{x^2}\mathrm{d}x\right]^{-1} = \frac{(n+1)x_f^{n+1}}{\lambda_0}. \tag{3.4.65}$$

Freeze-out temperature determination: The freeze-out point x_f is defined where $\Delta(x_f) = c\bar{Y}_X(x_f)$ with $c \sim \mathcal{O}(1)$ (numerical calibration gives $c(c+2) \approx 1$). Equating Eq. (3.4.63) and Eq. (3.4.60) yields the transcendental equation:

$$x_f^{1/2}\mathrm{e}^{x_f} f(x_f) = \frac{45}{\pi^4}\sqrt{\frac{\pi}{8}}\frac{g_X}{q_*}c\lambda_0. \tag{3.4.66}$$

Equivalently, using the intuitive condition $\Gamma(x_f) = H(x_f)$ gives:

$$\frac{g_X}{\sqrt{g_*}} x_f^{1/2}\mathrm{e}^{-x_f} f(x_f)\sigma_0 m_X M_{\mathrm{Pl}} = 4\pi^3\sqrt{\frac{2}{45}}. \tag{3.4.67}$$

Iterative solution gives $x_f \approx \ln(m_X M_{\mathrm{Pl}}\sigma_0) \approx 20$–$30$ over a vast range of particle masses ($10\,\mathrm{GeV} \le m_X \le 10\,\mathrm{TeV}$).

Present relic density and the WIMP miracle: The present relic number density $n_{X0} = s_0 Y_\infty$ is:

$$n_{X0} = 1.09 \times 10^4 \frac{\sqrt{g_*}}{q_*}\left(\frac{q_{*0}}{3.91}\right)\Theta_{2.7}^3 \left[\int_{x_f}^{\infty} \frac{f(x)}{x^2}\mathrm{d}x\right]^{-1} \frac{1}{m_X M_{\mathrm{Pl}}\sigma_0}\,\mathrm{cm}^{-3}. \tag{3.4.68}$$

The cosmological density parameter $\Omega_{X0} \equiv \rho_{X0}/\rho_{\mathrm{crit},0} = m_X n_{X0}/\rho_{\mathrm{crit},0}$ evaluates to:

$$\Omega_{X0}h^2 \simeq 0.31\left(\frac{g_*(x_f)}{100}\right)^{-1/2}(n+1)x_f^{n+1}\left(\frac{q_{*0}}{3.91}\right)\Theta_{2.7}^3\left(\frac{\sigma_0}{10^{-38}\,\mathrm{cm}^2}\right)^{-1}. \tag{3.4.69}$$

Remarkably, a typical electroweak-scale cross section $\sigma_0 \approx 10^{-38}\,\mathrm{cm}^2 \approx 1\,\mathrm{pb} \sim \alpha_W^2/m_{\mathrm{weak}}^2$ naturally yields $\Omega_{X0}h^2 \approx 0.12$, matching the observed cold dark matter density. This striking coincidence is known as the **WIMP miracle**.

Hot thermal relics ($x_f \ll 1$)

A **hot relic** decouples while still highly relativistic ($T_f \gg m_X$, $x_f \ll 1$). Its equilibrium comoving yield is constant:

$$\bar{Y}_X = \frac{45\zeta(3)}{2\pi^4}\varepsilon_{\mathrm{FB}}\frac{g_X}{q_*}, \tag{3.4.70}$$

where $\varepsilon_{\mathrm{FB}} = 1$ for bosons and $3/4$ for fermions. The asymptotic relic yield simply equals the equilibrium yield at decoupling:

$$Y_\infty = \bar{Y}_X(x_f) = \frac{45\zeta(3)}{2\pi^4}\varepsilon_{\mathrm{FB}}\frac{g_X}{q_*(x_f)}. \tag{3.4.71}$$

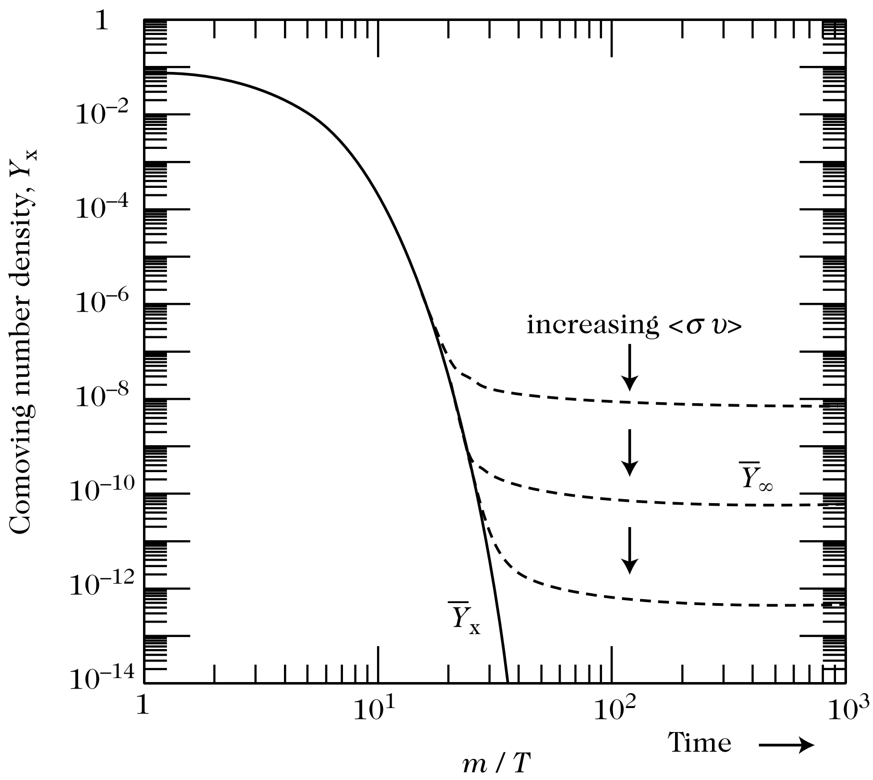


Figure 3.8: Numerical integration of the Boltzmann equation (3.4.59) for comoving abundance $Y_X(x)$ as a function of $x = m/T$. The solid curves display freeze-out departures from the equilibrium curve $\bar{Y}_X(x)$ for increasing annihilation cross sections $\langle \sigma v \rangle$, showing that larger cross sections maintain equilibrium longer and result in smaller residual relic abundances $Y_\infty \propto 1/\langle \sigma v \rangle$.

Assuming subsequent expansion remains adiabatic, today's number density is:

$$n_{X0} = s_0 Y_\infty = 776.8 \left(\frac{q_{*0}}{3.91}\right) \frac{\varepsilon_{\rm FB} g_X}{q_*(x_f)} \Theta_{2.7}^3 \,{\rm cm}^{-3}. \tag{3.4.72}$$

If the particle is non-relativistic today ($m_X > T_0 \approx 1.7 \times 10^{-4}\,$eV), its energy density is $\rho_{X0} = m_X n_{X0} = m_X s_0 Y_\infty$, yielding:

$$\Omega_{X0} h^2 = 0.739 \left(\frac{q_{*0}}{3.91}\right) \left(\frac{m_X}{10\,{\rm eV}}\right) \frac{\varepsilon_{\rm FB} g_X}{q_*(x_f)}. \tag{3.4.73}$$

The requirement $\Omega_{X0} h^2 \leq 1$ establishes the Lee–Weinberg / Cowsik–McClelland mass upper bound:

$$m_X \leq 13.5 \,\frac{q_*(x_f)}{\varepsilon_{\rm FB} g_X}\,{\rm eV}. \tag{3.4.74}$$

Light neutrinos and warm relics: For standard left-handed neutrinos decoupling at $T_{\nu,D} \approx 1\,$MeV ($g_X = 2$, $\varepsilon_{\rm FB} = 3/4$, $q_*(T_{\nu,D}) = 10.75$), Eq. (3.4.73) yields the celebrated **Gershtein–Zeldovich / Cowsik–McClelland formula**:

$$\Omega_{\nu 0} h^2 = \frac{\sum m_\nu}{91.5\,{\rm eV}}. \tag{3.4.75}$$

For particles that decouple much earlier (e.g., at the electroweak scale $T \sim 300\,$GeV with $q_* \approx 106.75$), the relic density is suppressed by entropy injection from subsequent particle annihilations:

$$\Omega_{X0} h^2 = \frac{m_X}{910\,{\rm eV}}. \tag{3.4.76}$$

Such species have lower present temperatures and abundances relative to CMB photons and are designated as **warm relics**.

3.5 Primordial Big-Bang Nucleosynthesis (BBN)

At temperatures $T \sim 100$–$0.01\,$MeV, the Universe comprises a relativistic thermal bath of photons (γ), electrons and positrons ($e^\pm$), and three families of decoupled neutrinos (ν_e, ν_μ, ν_τ). Non-relativistic baryons (protons p and neutrons n) contribute negligibly to the overall energy density ($\rho_b \ll \rho_r$).

The interconversion between neutrons and protons is driven by charged-current weak interactions:

$$\nu_e + n \longleftrightarrow p + e^-, \tag{3.5.1}$$
$$e^+ + n \longleftrightarrow p + \bar\nu_e, \tag{3.5.2}$$
$$n \longleftrightarrow p + e^- + \bar\nu_e. \tag{3.5.3}$$

Simultaneously, electron-positron electromagnetic annihilation maintains thermal contact:

$$e^- + e^+ \longleftrightarrow \gamma + \gamma. \tag{3.5.4}$$

3.5.1 Chronology of primordial nucleosynthesis

The synthesis of light elements proceeds across three distinct physical epochs:

1. **Thermal and Chemical Equilibrium ($T \gg 1\,\mathrm{MeV}$, $t \ll 1\,\mathrm{s}$):** Radiation dominates the expansion with $g_* = 10.75$. Weak interaction rates exceed the Hubble expansion rate ($\Gamma_w \gg H$), maintaining the neutron-to-proton ratio at its equilibrium value:

$$\left(\frac{n}{p}\right)_{\mathrm{eq}} = \mathrm{e}^{-Q/T} \approx 1, \qquad Q \equiv m_n - m_p \simeq 1.293\,\mathrm{MeV}. \tag{3.5.5}$$

Nuclear abundances follow Nuclear Statistical Equilibrium (NSE).

2. **Weak Interaction Freeze-out ($T \sim 1$–$0.7\,\mathrm{MeV}$, $t \sim 1\,\mathrm{s}$):** As the temperature drops below $T_f \approx 0.8\,\mathrm{MeV}$, the weak reaction rate $\Gamma_w \propto G_F^2 T^5$ falls below the Hubble rate $H \propto \sqrt{g_*}T^2/M_{\mathrm{Pl}}$. The $n \leftrightarrow p$ reactions freeze out, locking the neutron-to-proton ratio at:

$$\left(\frac{n}{p}\right)_f = \mathrm{e}^{-Q/T_f} \approx \frac{1}{5}\text{–}\frac{1}{6}. \tag{3.5.6}$$

Below T_f, free neutrons undergo spontaneous beta decay $n \to p + e^- + \bar{\nu}_e$ with measured mean lifetime $\tau_n \approx 887\,\mathrm{s}$.

3. **The Deuterium Bottleneck and Nuclear Combustion ($T \sim 0.7$–$0.05\,\mathrm{MeV}$, $t \approx 100$–$500\,\mathrm{s}$):** Electrons and positrons annihilate at $T \sim m_e \approx 0.511\,\mathrm{MeV}$, transferring entropy to photons and reducing g_* to 3.36. All nuclear synthesis proceeds through the two-body formation of deuterium:

$$p + n \longleftrightarrow \mathrm{D} + \gamma. \tag{3.5.7}$$

Because of the enormous photon-to-baryon ratio ($n_\gamma/n_b \sim 10^9$), high-energy Wien-tail photons efficiently photodissociate newly formed deuterium until the temperature falls to $T_{\mathrm{nuc}} \approx 0.08\,\mathrm{MeV}$ ($t_{\mathrm{nuc}} \approx 300$–$500\,\mathrm{s}$).

At T_{nuc}, the neutron-to-proton ratio has declined via beta decay to:

$$\left(\frac{n}{p}\right)_{\mathrm{nuc}} = \left(\frac{n}{p}\right)_f \mathrm{e}^{-t_{\mathrm{nuc}}/\tau_n} \approx \frac{1}{7}. \tag{3.5.8}$$

Once the bottleneck breaks, rapid two-body nuclear reactions fuse virtually all available neutrons into exceptionally tightly bound ^{4}He nuclei ($B_{^4\mathrm{He}} \approx 28.3\,\mathrm{MeV}$). The primordial helium-4 mass fraction Y_p is determined by simple stoichiometry:

$$Y_p \equiv \frac{4n_{^4\mathrm{He}}}{n_b} = \frac{4(n_n/2)}{n_n + n_p} = \frac{2(n/p)_{\mathrm{nuc}}}{1 + (n/p)_{\mathrm{nuc}}} \approx \frac{2/7}{1 + 1/7} = \frac{2}{8} \approx 0.25. \tag{3.5.9}$$

Because there are no stable nuclear isotopes at mass numbers $A = 5$ (^{5}He,^{5}Li) and $A = 8$ (^{8}Be), and because the Coulomb barriers rise steeply while the density and temperature rapidly drop, nuclear synthesis ceases after producing trace abundances of D, ^{3}He, and ^{7}Li. BBN effectively terminates at $T \approx 0.05\,\mathrm{MeV}$ ($t \approx 530\,\mathrm{s}$).

3.5.2 Key physical parameters governing BBN

Primordial nucleosynthesis is a non-equilibrium process governed by four key cosmological and microphysical parameters:

- **Relativistic degrees of freedom g_*:** Sets the expansion rate $H(T) \propto \sqrt{g_*}T^2/M_{\mathrm{Pl}}$. Equating $\Gamma_w \sim G_F^2 T_f^5 \sim H(T_f)$ yields $T_f \propto g_*^{1/6}$. An extra neutrino species ($\Delta N_\nu > 0$) increases g_*, accelerating expansion, raising T_f, boosting $(n/p)_f$, and thereby producing more ^{4}He.

- **Neutron lifetime τ_n:** Governs the weak reaction rates and the beta-decay survival factor $e^{-t_{\rm nuc}/\tau_n}$. The decay rate is:

$$\tau_n^{-1} = \frac{1.636}{2\pi^3} G_F^2 (1 + 3g_A^2) m_e^5 = (887 \pm 2\,\mathrm{s})^{-1}, \tag{3.5.10}$$

  where $g_A \approx 1.27$ is the axial-vector coupling constant.

- **Baryon-to-photon ratio η:** Defined as $\eta \equiv n_b/n_\gamma \sim (5\text{–}6) \times 10^{-10}$. A higher η triggers earlier deuterium synthesis (higher $T_{\rm nuc}$), shortening the beta-decay window, slightly increasing Y_p, and drastically depleting residual D/H and ^{3}He/H.

- **Fundamental constants:** G_N (gravity), G_F (weak force), α (fine-structure constant, setting Coulomb barriers and Q), and m_e.

3.5.3 Initial state: Nuclear statistical equilibrium

At $T \gg 1\,\mathrm{MeV}$, non-relativistic nuclei ${}^A_Z\mathrm{X}$ (with Z protons and $A - Z$ neutrons) are held in thermodynamic equilibrium by strong and electromagnetic reactions. Their Maxwell–Boltzmann number density is:

$$n_A = g_A \left(\frac{m_A T}{2\pi}\right)^{3/2} e^{-(m_A - \mu_A)/T}, \tag{3.5.11}$$

where $g_A = 2J_A + 1$ is the nuclear spin degeneracy.

Chemical equilibrium under nuclear reactions imposes:

$$\mu_A = Z\mu_p + (A - Z)\mu_n. \tag{3.5.12}$$

Neglecting the small mass difference $Q \approx 1.293\,\mathrm{MeV}$ in non-exponential prefactors, the free proton and neutron number densities are ($g_p = g_n = 2$):

$$n_{p,n} = 2\left(\frac{m_N T}{2\pi}\right)^{3/2} e^{-(m_{p,n} - \mu_{p,n})/T}, \qquad m_N \equiv \frac{m_p + m_n}{2}. \tag{3.5.13}$$

Expressing the chemical potential factor in Eq. (3.5.11) in terms of nucleon densities gives:

$$\begin{aligned} e^{-(m_A - \mu_A)/T} &= \left(e^{\mu_p/T}\right)^Z \left(e^{\mu_n/T}\right)^{A-Z} e^{-m_A/T} \\ &= 2^{-A} n_p^Z n_n^{A-Z} \left(\frac{2\pi}{m_N T}\right)^{3A/2} e^{B_A/T}, \end{aligned} \tag{3.5.14}$$

where B_A is the total nuclear binding energy:

$$B_A \equiv Z m_p + (A - Z) m_n - m_A. \tag{3.5.15}$$

Defining the **nuclear mass fraction** $X_A \equiv A n_A / n_b$ (with total baryon density $n_b = n_n + n_p + \sum A n_A$) and substituting the photon density $n_\gamma = \frac{2\zeta(3)}{\pi^2} T^3$, we derive the **nuclear Saha equation**:

$$X_A = F(A) \left(\frac{T}{m_N}\right)^{3(A-1)/2} \eta^{A-1} X_p^Z X_n^{A-Z} e^{B_A/T}, \tag{3.5.16}$$

where the dimensionless geometrical prefactor is:

$$F(A) \equiv g_A A^{5/2} \left[\frac{\zeta(3)}{\sqrt{\pi}} 2^{(3A-5)/(2A-2)}\right]^{A-1} = g_A A^{5/2} 2^{(3A-5)/2} \pi^{-(A-1)/2} [\zeta(3)]^{A-1}. \tag{3.5.17}$$

The entropy barrier and threshold synthesis temperatures

Equation (3.5.16) illustrates the profound competition between binding energy $e^{B_A/T}$ and cosmological entropy $\eta^{A-1} \sim (5\times10^{-10})^{A-1}$. Although nuclear binding energetically favors fusion at $T \sim B_A \approx 1$–30 MeV, the vast entropy (trillions of photons per nucleon) photodisintegrates any compound nuclei, maintaining nucleons almost entirely as free protons and neutrons.

The characteristic threshold temperature T_A at which $X_A \sim 1$ (with $X_p \sim X_n \sim 1$) is estimated by setting the exponent in Eq. (3.5.16) to unity:

$$T_A \approx \frac{B_A/(A-1)}{-\frac{3}{2}\ln(T_A/m_N) - \ln\eta}. \tag{3.5.18}$$

Table 3.6: Nuclear Binding Energies and Equilibrium Threshold Synthesis Temperatures ($\eta \approx 5 \times 10^{-10}$)

Nucleus	**Deuterium** (D)	**Tritium** (T)	**Helium-3** (^{3}He)	**Helium-4** (^{4}He)
Binding Energy B_A	2.22 MeV	6.92 MeV	7.72 MeV	28.3 MeV
Threshold Temp. T_A	0.066 MeV	0.10 MeV	0.11 MeV	0.28 MeV

Even though ^{4}He has a high threshold temperature $T_{^4\mathrm{He}} \approx 0.28$ MeV, its synthesis requires prior formation of deuterium via $p + n \to \mathrm{D} + \gamma$. Because $T_\mathrm{D} \approx 0.066$ MeV, the whole nucleosynthesis chain is blocked until $T \lesssim 0.08$ MeV—the famous **deuterium bottleneck**.

Equilibrium neutron abundance

Weak interactions (3.5.1)–(3.5.3) enforce chemical potential equilibrium:

$$\mu_n + \mu_\nu = \mu_p + \mu_e \implies \mu_n - \mu_p = \mu_e - \mu_\nu. \tag{3.5.19}$$

From Eq. (3.5.13), the equilibrium neutron-to-proton ratio is:

$$\left(\frac{n}{p}\right)_\mathrm{eq} \equiv \frac{n_n}{n_p} = \exp\left[-\frac{Q}{T} + \frac{\mu_e - \mu_\nu}{T}\right]. \tag{3.5.20}$$

Cosmic electrical neutrality requires $\mu_e/T \sim 10^{-10}$–10^{-8}. Assuming negligible neutrino chemical potential ($\mu_\nu \ll T$), the equilibrium ratio reduces to:

$$\left(\frac{n}{p}\right)_\mathrm{eq} = e^{-Q/T}. \tag{3.5.21}$$

At temperatures $T > 0.1$ MeV, complex nuclear abundances are negligible ($X_A \ll 1$), so the total nucleon density is $n_b = n_n + n_p$. The equilibrium free neutron mass fraction is therefore:

$$X_{n,\mathrm{eq}} \equiv \frac{n_n}{n_n + n_p} = \frac{1}{1 + e^{Q/T}}. \tag{3.5.22}$$

In the relativistic limit $T \gg Q$, $X_{n,\mathrm{eq}} = X_{p,\mathrm{eq}} = \frac{1}{2}$. At $T = 1$ MeV, the equilibrium deuterium fraction is already suppressed to $X_\mathrm{D} \sim 6 \times 10^{-12}$.

3.5.4 Freeze-out of the weak interaction

As the temperature falls below 1 MeV, the rates of the weak processes (3.5.1)–(3.5.3) drop precipitously. For $m_e < T < Q$, the total weak reaction rate per nucleon is computed from Fermi's four-fermion theory:

$$\Gamma_w(T) = \frac{7\pi}{60}(1+3g_A^2)G_F^2T^5. \tag{3.5.23}$$

Comparing the weak interaction rate $\Gamma_w(T)$ with the Hubble expansion rate $H(T) = \sqrt{\frac{4\pi^3 g_*}{45}}\frac{T^2}{M_{\rm Pl}}$ for $g_* = 10.75$, we obtain the dimensionless ratio:

$$\frac{\Gamma_w}{H} \simeq \left(\frac{T}{0.8\,\mathrm{MeV}}\right)^3, \tag{3.5.24}$$

which establishes the **weak interaction freeze-out temperature**:

$$T_f \approx 0.8\,\mathrm{MeV}, \qquad t_f \approx 1.15\,\mathrm{s}. \tag{3.5.25}$$

Neutron rate equation and analytic freeze-out solution

The macroscopic Boltzmann rate equation for the neutron mass fraction $X_n \equiv n_n/n_b$ is:

$$\dot{X}_n = \lambda_{pn}(1-X_n) - \lambda_{np}X_n, \tag{3.5.26}$$

where λ_{np} is the total conversion rate $n \to p$ and λ_{pn} is the reverse rate $p \to n$. Microscopic reversibility enforces:

$$\lambda_{np} = \lambda_{pn}\mathrm{e}^{Q/T}. \tag{3.5.27}$$

The differential equation (3.5.26) admits the exact integral solution:

$$X_n(t) = \int_{t_i}^{t} I(t,t')\,\lambda_{pn}(t')\,\mathrm{d}t' + I(t,t_i)\,X_n(t_i), \tag{3.5.28}$$

with the integrating kernel defined by:

$$I(t,t') \equiv \exp\left[-\int_{t'}^{t}(\lambda_{pn}(u)+\lambda_{np}(u))\,\mathrm{d}u\right]. \tag{3.5.29}$$

Because reaction rates diverge at early times ($t_i \to 0$), the memory of initial conditions $I(t,0)X_n(0) \to 0$ vanishes completely. Integrating Eq. (3.5.28) by parts yields:

$$\begin{aligned} X_n(t) &= \int_0^t I(t,t')\,\lambda_{pn}(t')\,\mathrm{d}t' \\ &\simeq X_{n,\mathrm{eq}}(t) - \int_0^t I(t,t')\,\frac{\mathrm{d}X_{n,\mathrm{eq}}(t')}{\mathrm{d}t'}\,\mathrm{d}t' \\ &\simeq X_{n,\mathrm{eq}}(t)\left[1+\frac{H}{\lambda_{pn}+\lambda_{np}}\frac{\mathrm{d}\ln X_{n,\mathrm{eq}}}{\mathrm{d}\ln T}\right]. \end{aligned} \tag{3.5.30}$$

At the freeze-out temperature $T_f \approx 0.8\,\mathrm{MeV}$, the neutron fraction locks to:

$$X_n(T_f) \simeq X_{n,\mathrm{eq}}(T_f) = \left[1+\exp\left(\frac{1.293\,\mathrm{MeV}}{0.8\,\mathrm{MeV}}\right)\right]^{-1} \approx 0.165 \approx \frac{1}{6}, \tag{3.5.31}$$

corresponding to a neutron-to-proton ratio:

$$\left(\frac{n}{p}\right)_{T_f} \approx 0.198 \approx \frac{1}{5}. \tag{3.5.32}$$

Remark. The Cosmological Coincidence $Q \sim T_f$: The ratio $Q/T_f \approx 1.293/0.8 \sim \mathcal{O}(1)$ is a profound cosmic coincidence. The neutron-proton mass difference $Q = (m_d - m_u) - \Delta E_{\mathrm{EM}}$ is determined entirely by strong and electromagnetic QCD/QED physics, whereas $T_f \propto g_*^{1/6}(G_N^{1/2}/G_F^2)^{1/3}$ is dictated by the electroweak Fermi constant G_F and gravitational Newton constant G_N. Had $Q \ll T_f$, we would have $n/p \approx 1$, resulting in 100% helium conversion and no primordial hydrogen for water or long-lived stellar fusion. Conversely, had $Q \gg T_f$, all neutrons would have converted into protons before freeze-out ($n/p \to 0$), leaving a universe of pure hydrogen with no light nuclei.

Explicit phase-space weak rates and post-freeze-out evolution

The three fundamental weak conversion channels have explicit phase-space rates:

$$\lambda(n\nu_e \to pe^-) = A\int_0^\infty \mathrm{d}q_\nu\, q_\nu^2 q_e\,(E_\nu + Q)(1 - f_e)f_\nu, \tag{3.5.33}$$

$$\lambda(ne^+ \to p\bar{\nu}_e) = A\int_0^\infty \mathrm{d}q_e\, q_e^2 q_\nu\,(E_e + Q)(1 - f_\nu)f_e, \tag{3.5.34}$$

$$\lambda(n \to pe^-\bar{\nu}_e) = A\int_0^{\sqrt{Q^2-m_e^2}} \mathrm{d}q_e\, q_e^2 q_\nu\,(E_e + Q)(1 - f_\nu)(1 - f_e). \tag{3.5.35}$$

For $T > m_e$, Pauli blocking is negligible ($1 - f \approx 1$) and $m_e \approx 0$, evaluating the leading integrals to:

$$\lambda(n\nu_e \to pe^-) = \lambda(ne^+ \to p\bar{\nu}_e) = AT^3\left(24T^2 + 12TQ + 2Q^2\right). \tag{3.5.36}$$

The experimental neutron lifetime $\tau_n = (887 \pm 2)\,\mathrm{s}$ fixes the normalization constant A:

$$\tau_n^{-1} = \frac{AQ^5}{6}\sqrt{1-\tilde{m}^2}\left[\frac{1}{5}\left(\frac{1}{6} - \frac{3}{4}\tilde{m}^2 - \frac{2}{3}\tilde{m}^4\right) + \frac{\tilde{m}^4}{4}\,\mathrm{arcosh}\left(\frac{1}{\tilde{m}}\right)\right] \simeq 0.0158\,AQ^5, \tag{3.5.37}$$

where $\tilde{m} \equiv m_e/Q \approx 0.395$.

Introducing the dimensionless variable $x \equiv Q/T$, the total $n \to p$ reaction rate becomes:

$$\lambda_{np} \simeq \frac{253}{\tau_n x^5}\left(12 + 6x + x^2\right). \tag{3.5.38}$$

The kernel $I(x, x') = \exp[K(x) - K(x')]$ is parameterized by:

$$\begin{aligned} K(x) &= b\int_x^\infty \frac{1 + \mathrm{e}^{-u}}{u^4}\left(12 + 6u + u^2\right)\mathrm{d}u \\ &= b\left[\left(\frac{4}{x^3} + \frac{3}{x^2} + \frac{1}{x}\right) + \left(\frac{4}{x^3} + \frac{1}{x^2}\right)\mathrm{e}^{-x}\right], \end{aligned} \tag{3.5.39}$$

where $b \equiv \frac{253}{\tau_n Q^2}\sqrt{\frac{45}{4\pi^3 g_*}}M_{\mathrm{Pl}} \approx 0.252$. Integrating the Boltzmann equation yields the asymptotic pre-decay neutron freeze-out abundance:

$$X_n(\infty) = 0.150, \tag{3.5.40}$$

which is attained at $T \approx 0.25\,\mathrm{MeV}$ ($x \approx 5$, $t \approx 20\,\mathrm{s}$).

3.5.5 Synthesis of light elements and the nuclear reaction network

Light nuclei are synthesized through a multi-step reaction network involving two-body collisions, illustrated in Figure 3.9. Direct multi-body reactions (such as $2n + 2p \to^4$ He) and weak fusion ($p + p \to \mathrm{D} + e^+ + \nu_e$) have negligible rates in the dilute cosmic plasma.

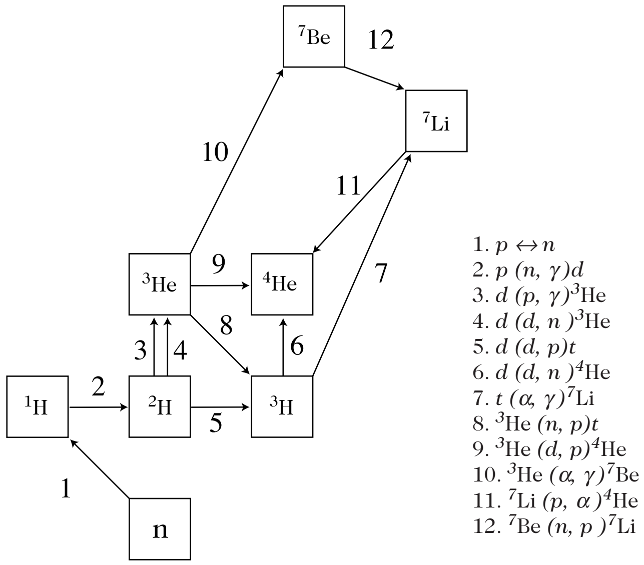


Figure 3.9: Simplified nuclear reaction network of Big-Bang Nucleosynthesis (BBN) connecting protons (p), neutrons (n), deuterium (D or ^{2}H), tritium (T or ^{3}H), helium-3 (^{3}He), helium-4 (^{4}He), lithium-7 (^{7}Li), and beryllium-7 (^{7}Be). Reactions: (1) $p \leftrightarrow n$, (2) $p(n,\gamma)$D, (3) D$(p,\gamma)^3$He, (4) D(D$,n)^3$He, (5) D(D$,p)$T, (6) D(D$,n)^4$He, (7) T$(\alpha,\gamma)^7$Li, (8) ^{3}He(n,p)T, (9) ^{3}He(D$,p)^4$He, (10) ^{3}He$(\alpha,\gamma)^7$Be, (11) ^{7}Li$(p,\alpha)^4$He, (12) ^{7}Be$(n,p)^7$Li.

Deuterium synthesis and the bottleneck breakdown

The gateway to all subsequent nucleosynthesis is the radiative capture:

$$p + n \longleftrightarrow \mathrm{D} + \gamma. \tag{3.5.41}$$

The forward reaction rate per neutron is $\lambda_{\mathrm{D}} = 4.55 \times 10^{-20} n_p \,\mathrm{cm}^3 \cdot \mathrm{s}^{-1}$. Because the binding energy $B_{\mathrm{D}} = 2.22\,\mathrm{MeV}$ is small, high-energy CMB photons efficiently photodisintegrate deuterium until the temperature drops to $T_{\mathrm{nuc}} \approx 0.066$–$0.086\,\mathrm{MeV}$.

At T_{nuc}, electrons and positrons have annihilated ($g_* = 3.36$), setting the cosmic time to $t_{\mathrm{nuc}} \approx 180$–$303\,\mathrm{s}$. During the delay $t_f \to t_{\mathrm{nuc}}$, free neutrons undergo beta decay $n \to p + e^- + \bar{\nu}_e$:

$$X_n(T_{\mathrm{nuc}}) = X_n(\infty)\,\mathrm{e}^{-t_{\mathrm{nuc}}/\tau_n} \approx 0.117\text{–}0.136, \tag{3.5.42}$$

yielding an effective neutron-to-proton ratio at nucleosynthesis:

$$\left(\frac{n}{p}\right)_{\mathrm{nuc}} \approx 0.133\text{–}0.145 \approx \frac{1}{7}. \tag{3.5.43}$$

In equilibrium, the deuterium-to-nucleon ratio ($g_{\mathrm{D}} = 3$) satisfies:

$$\frac{X_{\mathrm{D}}}{X_n X_p} = \frac{24\zeta(3)}{\sqrt{\pi}}\eta\left(\frac{T}{m_N}\right)^{3/2} \exp\left(\frac{B_{\mathrm{D}}}{T}\right). \tag{3.5.44}$$

Helium-4 production and stoichiometric abundance

Once deuterium forms in sufficient quantity, the rapid reactions (3)–(6), (8), and (9) burn nearly all available neutrons into ^{4}He ($B_{^4\mathrm{He}} \approx 28.3\,\mathrm{MeV}$), as shown in Figure 3.10.

The primordial helium-4 mass fraction $Y_p \equiv \rho(^4\mathrm{He})/\rho_b = 4n_{^4\mathrm{He}}/n_b$ is:

$$Y_p = \frac{2X_n(T_{\mathrm{nuc}})}{1 + X_n(T_{\mathrm{nuc}})} = \frac{2(n/p)_{\mathrm{nuc}}}{1 + (n/p)_{\mathrm{nuc}}} \approx 0.24\text{–}0.25. \tag{3.5.45}$$

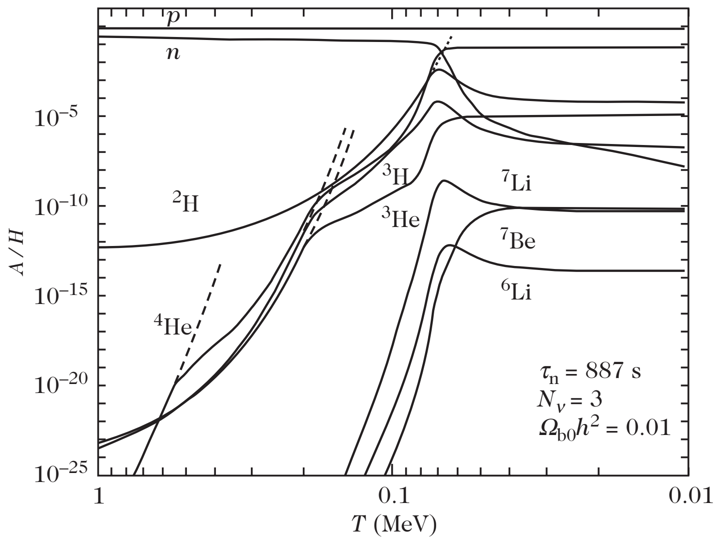


Figure 3.10: Dynamical evolution of light element mass fractions $(p, n, \mathrm{D}, \mathrm{T}, {}^3\mathrm{He}, {}^4\mathrm{He}, {}^6\mathrm{Li}, {}^7\mathrm{Li}, {}^7\mathrm{Be})$ as a function of temperature $T\,(\mathrm{MeV})$ during Big-Bang Nucleosynthesis ($\tau_n = 887\,\mathrm{s}, N_\nu = 3, \Omega_{b0}h^2 = 0.01$). Dashed lines represent Nuclear Statistical Equilibrium (NSE) predictions, highlighting the dramatic non-equilibrium departure below $T \sim 0.1\,\mathrm{MeV}$.

Neutrino degeneracy and parameter sensitivity: If neutrinos possess a non-zero chemical potential $\xi \equiv \mu_{\nu_e}/T$, the reaction rate $n + \nu_e \to p + e^-$ shifts detailed balance to $\lambda_{np} = \lambda_{pn}\exp(Q/T + \xi)$, modifying the freeze-out abundance to $X_n(\infty, \xi) = X_n(\infty, 0)\mathrm{e}^{-\xi}$.

Accurate numerical integrations across the standard range $3 \times 10^{-10} < \eta < 10^{-9}$ yield the linearized sensitivity formula:

$$Y_p = 0.245 + 0.014(N_\nu - 3) + 0.0002(\tau_n - 887\,\mathrm{s}) + 0.009\ln\left(\frac{\eta}{5 \times 10^{-10}}\right) - 0.25\,\xi. \tag{3.5.46}$$

Trace residual elements: Deuterium, Helium-3, and Lithium-7

As nucleosynthesis concludes at $t \sim 1000\,\mathrm{s}$ ($T \lesssim 0.03\,\mathrm{MeV}$), tiny residual fractions of intermediate isotopes survive unburned:

- **Deuterium (D/H $\sim 10^{-5}$):** Highly sensitive to the expansion rate and baryon density. Because deuterium burns into helium via two-body reactions, higher η results in more efficient destruction, making D/H a sensitive cosmic "baryometer".
- **Helium-3 (^{3}He/H $\sim 10^{-5}$):** Produced via $\mathrm{D}(p,\gamma)^3\mathrm{He}$ and tritium beta decay ($\mathrm{T} \to^3 \mathrm{He} + e^- + \bar{\nu}_e$, $\tau_{1/2} \approx 12.3\,\mathrm{yr}$).
- **Lithium-7 (^{7}Li/H $\sim 10^{-10}$):** Produced through two competing physical channels:
  1. At low baryon density ($\eta < 3\times10^{-10}$): $^4\mathrm{He}+\mathrm{T} \to^7 \mathrm{Li}+\gamma$, destroyed by $^7\mathrm{Li}+p \to 2\,^4\mathrm{He}$;
  2. At high baryon density ($\eta > 3 \times 10^{-10}$): $^4\mathrm{He} +^3 \mathrm{He} \to^7 \mathrm{Be} + \gamma$, followed much later by electron capture $^7\mathrm{Be} + e^- \to^7 \mathrm{Li} + \nu_e$.

  The transition between these two production mechanisms generates the characteristic "valley" in the ^{7}Li/H abundance curve shown in Figure 3.11.

In the observationally relevant window $10^{-10} < \eta < 10^{-9}$, semi-analytic power-law fits (with $\eta_{5.5} \equiv \eta/(5.5 \times 10^{-10})$) give:

$$\left(\frac{\mathrm{D}}{\mathrm{H}}\right)_p = 3.6 \times 10^{-5\pm0.06}\, \eta_{5.5}^{-1.6}, \tag{3.5.47}$$

$$\left(\frac{{}^3\mathrm{He}}{\mathrm{H}}\right)_p = 1.2 \times 10^{-5\pm0.06}\, \eta_{5.5}^{-0.63}, \tag{3.5.48}$$

$$\left(\frac{{}^7\mathrm{Li}}{\mathrm{H}}\right)_p = 1.2 \times 10^{-11\pm0.2} \left[\eta_{5.5}^{-2.38} + 21.7\, \eta_{5.5}^{2.38}\right]. \tag{3.5.49}$$

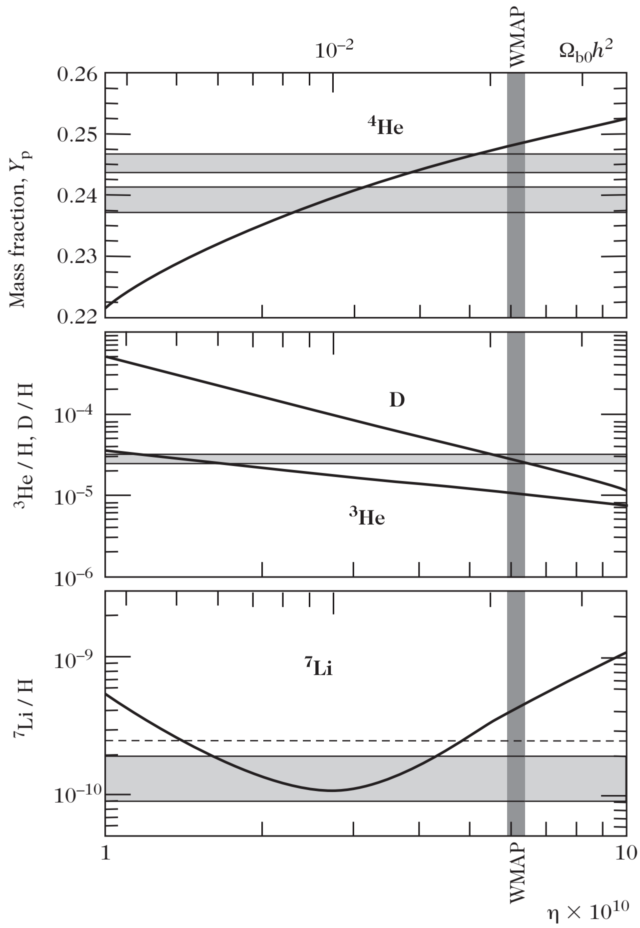


Figure 3.11: Theoretical primordial abundances of ^{4}He (mass fraction Y_p), D/H, ^{3}He/H, and ^{7}Li/H as a function of the baryon-to-photon ratio η and physical baryon density $\Omega_{b0}h^2$. The horizontal hatched bands depict spectroscopic astrophysical measurements in low-metallicity environments. The vertical shaded band denotes the independent cosmic microwave background constraint from WMAP.

3.5.6 Observational status and cosmological constraints

Measuring primordial abundances requires identifying astrophysical environments pristine enough to have undergone minimal stellar processing:

- **Helium-4:** Measured via optical emission lines in low-metallicity extragalactic H II regions of Blue Compact Dwarf (BCD) galaxies. Extrapolating to zero metallicity $(\mathrm{O/H}, \mathrm{N/H} \to 0)$

yields $Y_p = 0.2421 \pm 0.0021$ (Izotov & Thuan 2004) and $Y_p = 0.2443 \pm 0.0015$ (Luridiana et al. 2003).

- **Deuterium:** Because stars exclusively destroy deuterium (astration) at $T > 6 \times 10^5$ K, pristine D/H is measured in high-redshift ($z \sim 2$–4), low-metallicity Lyman-α absorption systems along lines of sight to distant quasars (QSOs), yielding $\mathrm{D/H} = (2.78^{+0.44}_{-0.38}) \times 10^{-5}$.
- **Lithium-7:** Measured in the warm atmospheres of metal-poor Population II Galactic halo dwarf stars (the *Spite plateau*), yielding $(\mathrm{Li/H})_{\mathrm{obs}} = (1.23^{+0.68}_{-0.32}) \times 10^{-10}$.
- **Helium-3:** Observed via the 8.665 GHz hyperfine transition in Galactic H II regions, giving $^3\mathrm{He/H} = (1\text{–}2) \times 10^{-5}$.

Cosmological baryon density and the Primordial Lithium Problem

Pure spectroscopic determinations of primordial abundances under standard BBN ($N_\nu = 3$) infer the cosmological baryon density:

$$\eta = (5.15 \pm 1.75) \times 10^{-10} \iff \Omega_{b0} h^2 = 0.018 \pm 0.006. \tag{3.5.50}$$

Precision measurements of the acoustic peaks in CMB temperature anisotropies (WMAP and Planck) determine the baryon density independently of nuclear astrophysics:

$$\Omega_{b0} h^2 = 0.0224 \pm 0.0009 \implies \eta = (6.14 \pm 0.25) \times 10^{-10}. \tag{3.5.51}$$

Table 3.7: Comparison of Standard BBN Abundance Predictions at CMB Baryon Density ($\Omega_{b0} h^2 = 0.0224$) with Observational Determinations

Isotope / Ratio	**Standard BBN Prediction (CMB η)**	**Spectroscopic Observation**
Helium-4 (Y_p)	0.2479 ± 0.0004	0.2421 ± 0.0021 (BCD H II regions)
Deuterium (D/H)	$(2.60^{+0.19}_{-0.17}) \times 10^{-5}$	$(2.78^{+0.44}_{-0.38}) \times 10^{-5}$ (QSO Ly-α absorbers)
Helium-3 (^{3}He/H)	$(1.04 \pm 0.04) \times 10^{-5}$	$(1\text{–}2) \times 10^{-5}$ (Galactic H II regions)
Lithium-7 (^{7}Li/H)	$(4.15^{+0.49}_{-0.45}) \times 10^{-10}$	$(1.23^{+0.68}_{-0.32}) \times 10^{-10}$ (Spite plateau halo stars)
Lithium-6 (^{6}Li/H)	$(1.2 \pm 0.08) \times 10^{-14}$	$\leq 10^{-11}$

While Y_p and D/H exhibit spectacular agreement between BBN theory and CMB cosmology, the predicted ^{7}Li abundance exceeds halo star observations by a factor of ~ 3–4. This discrepancy—the **Cosmological Lithium Problem**—spurs ongoing investigations into stellar turbulent diffusion and atmospheric depletion, resonant nuclear cross sections, and non-standard dark matter decays during the BBN epoch.

3.6 The Cosmic Microwave Background Radiation

Prior to neutral atom formation, the Universe is a fully ionized plasma where free electrons and photons are tightly coupled via Thomson scattering ($\gamma + e^- \leftrightarrow \gamma + e^-$). As the temperature drops below the binding energy of hydrogen, protons and electrons recombine into neutral hydrogen (H). Thomson scattering ceases, freeing photons to propagate unhindered across the cosmos as the **Cosmic Microwave Background (CMB)**.

3.6.1 Historical prediction and discovery

In 1948, Ralph Alpher and Robert Herman predicted the existence and temperature of the CMB from Gamow's hot Big-Bang model. To produce $Y_p \approx 0.25$, nucleosynthesis must take place at $T_{\rm BBN} \sim 10^9\,{\rm K}$ when the baryon density is $n_{b,\rm BBN} \sim 10^{18}\,{\rm cm}^{-3}$. Comparing with the present baryon density $n_{b0} \sim 10^{-7}\,{\rm cm}^{-3}$:

$$1 + z_{\rm BBN} = \left(\frac{n_{b,\rm BBN}}{n_{b0}}\right)^{1/3} \sim 2 \times 10^8 \implies T_{\rm CMB} = \frac{T_{\rm BBN}}{1 + z_{\rm BBN}} \sim 5\,{\rm K}. \tag{3.6.1}$$

In 1964, Arno Penzias and Robert Wilson serendipitously discovered this isotropic microwave radiation at Bell Laboratories, and Robert Dicke, Jim Peebles, Peter Roll, and David Wilkinson correctly interpreted it as the fossil thermal echo of the Big Bang at $T_0 = (3.5 \pm 1.0)\,{\rm K}$.

3.6.2 Thermodynamic recombination and the Saha equation

The photoionization-recombination equilibrium reaction is:

$$p + e^- \longleftrightarrow {\rm H} + \gamma. \tag{3.6.2}$$

Assuming cosmic electrical neutrality ($n_e = n_p$), we define the **fractional ionization**:

$$X_e \equiv \frac{n_e}{n_p + n_{\rm H}} = \frac{n_e}{n_b}, \qquad n_e = n_p = X_e n_b, \quad n_{\rm H} = (1 - X_e)n_b. \tag{3.6.3}$$

Applying chemical equilibrium ($\mu_p + \mu_e = \mu_{\rm H}$) to non-relativistic Maxwell–Boltzmann distributions yields the **Saha ionization equation**:

$$\frac{X_e^2}{1 - X_e} = \frac{1}{n_b}\frac{n_e n_p}{n_H} = \frac{1}{n_b}\left(\frac{m_e T}{2\pi}\right)^{3/2} {\rm e}^{-E_I/T}, \tag{3.6.4}$$

where we used (3.5.11) with $\frac{m_e m_p}{m_e + m_p} = \frac{m_e}{\frac{m_e}{m_p} + 1} \approx m_e$ and $E_I = m_e + m_p - m_{\rm H} = 13.6\,{\rm eV}$ is the ground-state hydrogen ionization energy.

Substituting $T(z) = T_0(1 + z) = 2.348 \times 10^{-4}(1 + z)\,{\rm eV}$ (for $T_0 = 2.7255\,{\rm K}$) and $n_b(z) = \eta n_{\gamma 0}(1 + z)^3 = 2.47 \times 10^{-21}\Omega_{b0}h^2(1 + z)^3\,{\rm GeV}^3$, Eq. (3.6.4) takes the explicit logarithmic form:

$$\log_{10}\left(\frac{X_e^2}{1 - X_e}\right) = 20.98 - \log_{10}\left[\Omega_{b0}h^2(1 + z)^{3/2}\right] - \frac{25163}{1 + z}. \tag{3.6.5}$$

Recombination redshifts and the photon-to-baryon delay

Because the photon-to-baryon ratio $\eta^{-1} \sim 10^9$ is enormous, recombination does not occur at $T \sim E_I \approx 13.6\,{\rm eV} \approx 1.5 \times 10^5\,{\rm K}$ (where the right-hand side of Eq. (3.6.4) is $\sim 10^{15}$, keeping $X_e \approx 1$). Instead, recombination is delayed until $T \ll E_I$, when the photon Wien tail no longer contains enough energetic ionizing photons. Taking the logarithm of the Saha equation yields:

$$\ln\left(\frac{X_{\rm e}^2}{1 - X_{\rm e}}\right) = \frac{3}{2}\ln\left(\frac{m_{\rm e}T}{2\pi}\right) - \frac{E_{\rm I}}{T} - \ln\left[\eta\frac{2}{\pi^2}\zeta(3)T^3\right], \tag{3.6.6}$$

from which we can isolate the ratio $E_{\rm I}/T$:

$$\frac{E_{\rm I}}{T} = \frac{3}{2}\ln\left(\frac{m_{\rm e}}{2\pi T}\right) - \ln\eta - \ln\left[\frac{2}{\pi^2}\zeta(3)\frac{X_{\rm e}^2}{1 - X_{\rm e}}\right]. \tag{3.6.7}$$

Since the final logarithmic term evaluated at $X_{\rm e} \sim 0.5$ is of order $\mathcal{O}(1)$ and negligible compared to the large value of $\ln(m_{\rm e}/T) \sim 10^2$ and $\ln\eta \sim -21$ (from (3.5.50)), solving this transcendental relation self-consistently yields the recombination temperature estimate:

$$T_{\rm rec} \sim 3500\ {\rm K}. \tag{3.6.8}$$

From Eq. (3.6.5), standard ionization thresholds correspond to:

- $X_e = 0.50 \implies z_{\rm rec} \approx 1376 \quad (T \approx 3750\,{\rm K} \approx 0.32\,{\rm eV})$;
- $X_e = 0.10 \implies z_{\rm rec} \approx 1258 \quad (T \approx 3430\,{\rm K} \approx 0.30\,{\rm eV})$;
- $X_e = 0.01 \implies z_{\rm rec} \approx 1138 \quad (T \approx 3100\,{\rm K} \approx 0.27\,{\rm eV})$.

3.6.3 Photon decoupling and the surface of last scattering

The Thomson scattering rate of photons on free electrons is:

$$\Gamma_{\rm T}(z) = n_e(z)\sigma_{\rm T} = 1.495 \times 10^{-31}\, X_e(z) \left(\frac{\Omega_{b0}h^2}{0.02}\right) (1+z)^3\,{\rm cm}^{-1}, \tag{3.6.9}$$

where $\sigma_{\rm T} = \frac{8\pi}{3}\left(\frac{\alpha}{m_e}\right)^2 \approx 6.65 \times 10^{-25}\,{\rm cm}^2$.

In a matter-and-radiation universe, the expansion rate is:

$$H^2(z) = \Omega_{m0} H_0^2 (1+z)^3 \left(1 + \frac{1+z}{1+z_{\rm eq}}\right). \tag{3.6.10}$$

The condition for photon decoupling $\Gamma_{\rm T}(z_{\rm dec}) = H(z_{\rm dec})$ leads to:

$$(1+z_{\rm dec})^{3/2} = \frac{280.01}{X_e(\infty)} \left(\frac{\Omega_{b0}h^2}{0.02}\right)^{-1} \left(\frac{\Omega_{m0}h^2}{0.15}\right)^{1/2} \left(1 + \frac{1+z_{\rm dec}}{1+z_{\rm eq}}\right)^{1/2}. \tag{3.6.11}$$

When the freeze-out residual ionization fraction $X_e(\infty) \approx 7 \times 10^{-3}$ is inserted, Eq. (3.6.11) determines the **photon decoupling redshift**:

$$z_{\rm dec} \approx 1100, \qquad T_{\rm dec} \approx 3000\,{\rm K} \approx 0.26\,{\rm eV}, \qquad t_{\rm dec} \approx 380\,000\,{\rm yr}. \tag{3.6.12}$$

3.6.4 Recombination dynamics and the Peebles three-level atom

The equilibrium Saha equation (3.6.4) assumes that photoionization and radiative recombination occur in instantaneous thermodynamic equilibrium. However, the microscopic physics of recombination in an expanding cosmos is governed by non-equilibrium atomic kinetics.

Consider the Boltzmann rate equation for the electron number density $n_e = n_b X_e$. Using $n_b a^3 = {\rm const} \implies \dot{n}_e + 3Hn_e = n_b \dot{X}_e$ and $n_{\rm H} = n_b(1 - X_e)$, the kinetic rate equation is:

$$\dot{X}_e = C_r \left[\beta(1-X_e) - \alpha^{(2)} n_b X_e^2\right], \tag{3.6.13}$$

where $\alpha^{(2)}(T)$ is the recombination coefficient to all excited states ($n \geq 2$), and $\beta(T)$ is the effective photoionization rate from these excited states:

$$\beta(T) = \left(\frac{m_e T}{2\pi}\right)^{3/2} \alpha^{(2)}(T) \exp\left(-\frac{E_I}{T}\right). \tag{3.6.14}$$

The recombination rate to excited states is computed quantum mechanically from the Milne relation:

$$\alpha^{(2)}(T) = \frac{64\sqrt{\pi}}{\sqrt{27}} \frac{e^4}{m_e^2} \left(\frac{T}{E_I}\right)^{-1/2} \phi_2(T), \qquad \phi_2(T) \simeq 0.448 \ln\left(\frac{E_I}{T}\right), \tag{3.6.15}$$

which is accurate to better than 1% for $T < 6000\,{\rm K}$.

The Lyman-α and two-photon decay bottlenecks

Direct recombination of an electron and proton to the ground state $1s$ produces a photon with energy $h\nu \geq 13.6\,\text{eV}$. In the dense primordial plasma, this ionizing photon is immediately re-absorbed by a neutral hydrogen atom ($h\nu + \text{H}(1s) \to p + e^-$), resulting in *zero net recombination*.

Therefore, recombination can only succeed through radiative capture into excited states ($n \geq 2$), followed by radiative cascade down to $n = 2$ ($2s$ and $2p$). From the $n = 2$ shell, transition to the ground state $1s$ is governed by two bottleneck processes:

1. **The $2s \to 1s$ Two-Photon Decay:** Single-photon electric dipole transitions ($2s \to 1s$) are forbidden by angular momentum conservation ($\Delta l = 0$). De-excitation occurs exclusively via the simultaneous emission of two photons with combined energy $E_{2s} - E_{1s} = 10.2\,\text{eV}$. This is a slow quantum second-order process with rate:

$$\Lambda_{2s} \approx 8.227\,\text{s}^{-1}. \tag{3.6.16}$$

   Because each emitted photon has $h\nu < 10.2\,\text{eV}$, neither can excite another ground-state atom to $n = 2$, permanently locking the atom into the ground state.

2. **Cosmological Redshifting of Lyman-α Photons ($2p \to 1s$):** Resonant Lyman-α photons emitted during $2p \to 1s$ decay have an optical depth $\tau_{\text{Ly}\alpha} \sim 10^7$. They undergo millions of resonant scatterings before escaping only when cosmic expansion redshifts their frequency below the resonant absorption profile:

$$\Lambda_\alpha = \frac{8\pi H}{\lambda_\alpha^3 n_{1s}} = \frac{8\pi H}{\lambda_\alpha^3 (1 - X_e) n_b}, \qquad \lambda_\alpha = \frac{4}{3E_I} = 1.216 \times 10^{-5}\,\text{cm}. \tag{3.6.17}$$

Combining these two channels defines the **Peebles reduction factor** C_r:

$$C_r = \frac{\Lambda_\alpha + \Lambda_{2s}}{\Lambda_\alpha + \Lambda_{2s} + \beta^{(2)}}, \qquad \beta^{(2)} \equiv \beta \exp\left(\frac{h\nu_\alpha}{T}\right). \tag{3.6.18}$$

Table 3.8: Comparison of Fractional Ionization $X_e(z)$ Predicted by the Equilibrium Saha Equation vs Peebles Multi-Level Recombination Kinetics ($\Omega_{b0}h^2 = 1$)

Redshift z	**Saha Equilibrium**	**Peebles Equation (3.6.13)**	**Full Multi-Level (Seager et al.)**
1667	9.43×10^{-1}	9.20×10^{-1}	9.14×10^{-1}
1480	3.80×10^{-1}	4.00×10^{-1}	4.00×10^{-1}
1296	2.60×10^{-2}	7.20×10^{-2}	7.15×10^{-2}
1111	8.80×10^{-4}	9.80×10^{-3}	9.68×10^{-3}
926	5.28×10^{-6}	9.20×10^{-4}	9.01×10^{-4}
741	2.40×10^{-9}	1.23×10^{-4}	1.20×10^{-4}

In the redshift range $1500 > z > 800$, Eq. (3.6.13) simplifies to:

$$\frac{\mathrm{d}X_e}{\mathrm{d}z} = -61.72\, \frac{\Omega_{b0}h^2}{\sqrt{\Omega_{m0}h^2}}\, f(z) \left[1 + 2.26 \times 10^4\, z\, \mathrm{e}^{-14486/(z\Theta_{2.7})}\right]^{-1} X_e^2, \tag{3.6.19}$$

with $f(z) \equiv (1 + 1.45z/z_{\text{eq}})^{-1/2}$. An accurate analytical solution is:

$$X_e(z) \simeq 2.74 \times 10^{-3}\, \mathrm{e}^{14.486(1-\Theta_{2.7}^{-1})}\, \frac{\sqrt{\Omega_{m0}h^2}}{\Omega_{b0}h^2} \left(\frac{z}{1000}\right)^{12.75}. \tag{3.6.20}$$

3.6.5 The optical depth and the Last Scattering Surface

The Thomson scattering optical depth from redshift z to the present observer ($z = 0$) is:

$$\tau(z) = \int_0^{\chi(z)} n_e \sigma_{\mathrm{T}}\, \mathrm{d}\chi \simeq 0.42\, \mathrm{e}^{14.486(1-\Theta_{2.7}^{-1})} \left(\frac{z}{1000}\right)^{14.25}. \tag{3.6.21}$$

Remarkably, the dependence on Ω_{m0} and Ω_{b0} cancels out to leading order in $\tau(z)$.

The probability distribution that a CMB photon underwent its final scattering between redshift z and $z + \mathrm{d}z$ is quantified by the **visibility function**:

$$g(z) \equiv \mathrm{e}^{-\tau(z)} \frac{\mathrm{d}\tau}{\mathrm{d}z}. \tag{3.6.22}$$

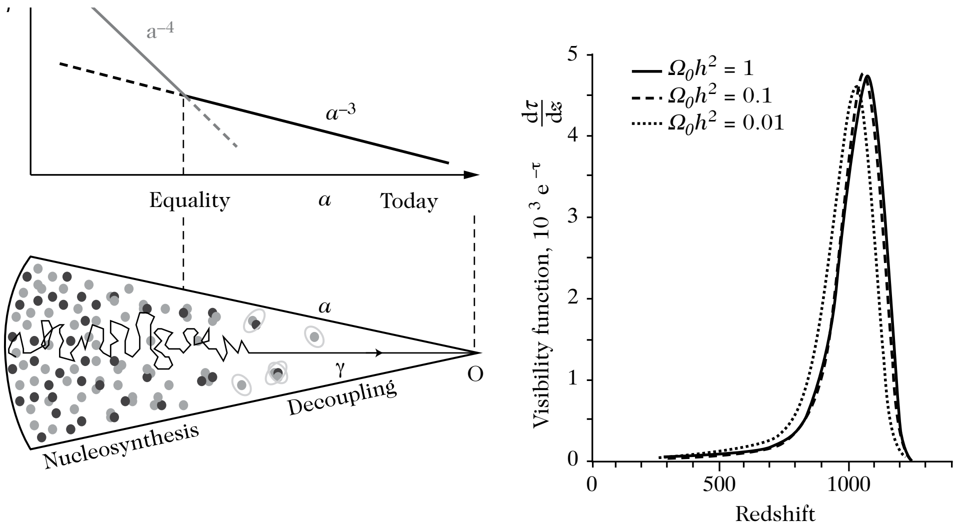


Figure 3.12: (Left): Thermal history of cosmic energy density $\rho(a)$ illustrating the matter-radiation equality, nucleosynthesis, and recombination epochs. (Right): The CMB visibility function $g(z) = \mathrm{e}^{-\tau} \frac{\mathrm{d}\tau}{\mathrm{d}z}$ plotted across redshift, showing a sharp peak that defines the Last Scattering Surface (LSS) at $z_{\mathrm{dec}} \approx 1089$.

The visibility function is sharply peaked (Figure 3.12), defining the **Last Scattering Surface (LSS)**:

$$z_{\mathrm{dec}} = 1089 \pm 1, \qquad \Delta z_{\mathrm{dec}} = 195 \pm 2, \qquad T_{\mathrm{dec}} \approx 2970\,\mathrm{K}. \tag{3.6.23}$$

The Last Scattering Surface represents a spherical shell centered on the observer at comoving distance $\chi(z_{\mathrm{dec}}) \approx 14\,\mathrm{Gpc}$, with finite physical thickness corresponding to $\Delta z_{\mathrm{dec}} \approx 195$.

3.6.6 Observed properties of the CMB spectrum

Precision blackbody spectrum and cosmological energy densities

The Far-Infrared Absolute Spectrophotometer (FIRAS) experiment aboard the NASA COBE satellite measured the absolute spectrum of the CMB across the frequency band 60–600 GHz (1–20 cm^{-1}), establishing an exquisitely precise Planck blackbody distribution:

$$I(\nu, T_0) = \frac{2h\nu^3}{c^2} \frac{1}{\mathrm{e}^{h\nu/k_{\mathrm{B}}T_0} - 1}, \tag{3.6.24}$$

with sky-averaged monopole temperature:

$$T_0 = 2.7255 \pm 0.0006\,\mathrm{K} \quad \text{(often quoted as } 2.725 \pm 0.001\,\mathrm{K} \text{ at } 2\sigma). \tag{3.6.25}$$

From this measured temperature, the present radiation quantities evaluate to:

$$n_{\gamma 0} = \frac{2\zeta(3)}{\pi^2} T_0^3 = 410.44\,\Theta_{2.7}^3\,\mathrm{cm}^{-3}, \tag{3.6.26}$$

$$\rho_{\gamma 0} = \frac{\pi^2}{15} T_0^4 = 4.6408 \times 10^{-34}\,\Theta_{2.7}^4\,\mathrm{g}\cdot\mathrm{cm}^{-3} = 0.26032\,\mathrm{eV}\cdot\mathrm{cm}^{-3}, \tag{3.6.27}$$

$$s_0 = \frac{2\pi^2}{45} q_{*0} T_0^3 = 2889.2 \left(\frac{q_{*0}}{3.91}\right) \Theta_{2.7}^3\,\mathrm{cm}^{-3} = 7.039 \left(\frac{q_{*0}}{3.91}\right) n_{\gamma 0}, \tag{3.6.28}$$

$$\Omega_{\gamma 0} h^2 = 2.4697 \times 10^{-5}\,\Theta_{2.7}^4, \tag{3.6.29}$$

$$\Omega_{r0} h^2 = \Omega_{\gamma 0} h^2 \left[1 + 3 \times \frac{7}{8}\left(\frac{4}{11}\right)^{4/3}\right] = 4.148 \times 10^{-5}\,\Theta_{2.7}^4. \tag{3.6.30}$$

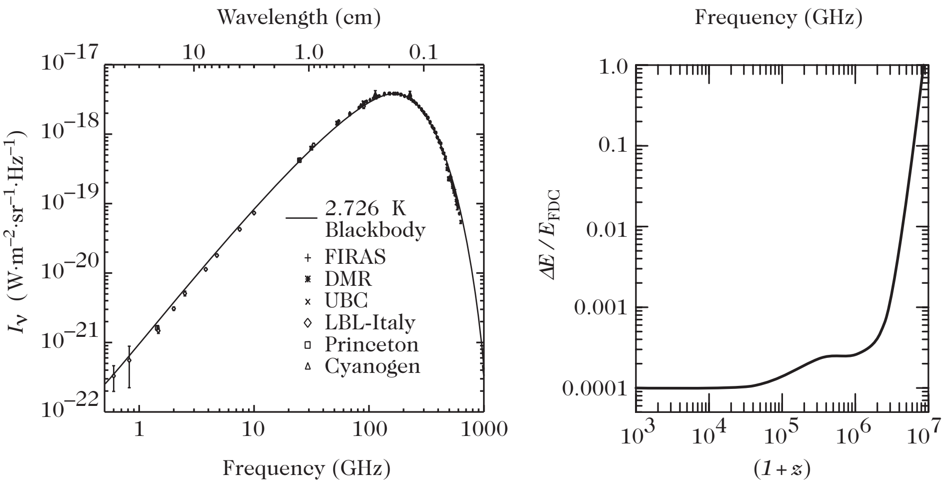


Figure 3.13: (Left): Comparison between the measured CMB spectrum and a perfect blackbody at $T_0 = 2.725\,\mathrm{K}$. (Right): Upper bounds on fractional energy injection $\Delta E/E_{\mathrm{FDC}}$ as a function of redshift $(1+z)$ from FIRAS spectral distortion constraints.

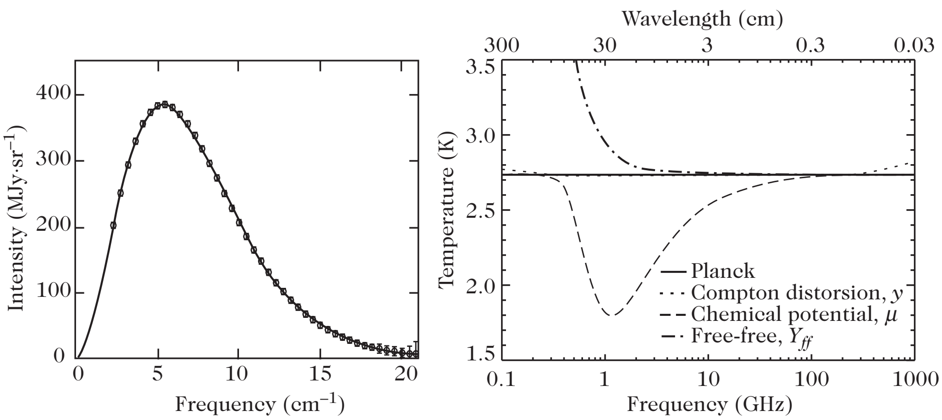


Figure 3.14: (Left): Spectrum of the cosmic microwave background measured by FIRAS/COBE (error bars multiplied by 200). (Right): Characteristic spectral distortions (Compton y, chemical potential μ, and free-free Y_{ff}).

Spectral distortions of the Planck distribution

Any energy injection into the cosmic plasma at early epochs alters the thermal equilibrium between matter and radiation, generating three distinct types of spectral distortions:

1. **Compton y-Distortions ($z \lesssim 10^5$):** At late times, hot electrons ($T_e \gg T_\gamma$) Compton upscatter low-energy photons to higher energies via the Sunyaev–Zel'dovich effect, depleting the Rayleigh–Jeans tail and enhancing the Wien tail. The distortion is parameterized by the Compton optical depth:
$$y \equiv \int \sigma_\mathrm{T} n_e \frac{k_\mathrm{B}(T_e - T_\gamma)}{m_e c^2}\, \mathrm{d}t. \tag{3.6.31}$$

2. **Chemical Potential μ-Distortions ($10^5 \lesssim z \lesssim 2\times10^6$):** In this redshift window, elastic Compton scattering $\gamma + e^- \leftrightarrow \gamma + e^-$ is fast enough to redistribute photon energies into a Bose–Einstein kinetic distribution, but photon-number-changing processes (Bremsstrahlung $e^- + p \leftrightarrow e^- + p + \gamma$ and double Compton $\gamma + e^- \leftrightarrow \gamma + \gamma + e^-$) are frozen out. The spectrum relaxes to a Bose–Einstein distribution with non-zero chemical potential:
$$I(\nu) \propto \frac{\nu^3}{\exp\left(\frac{h\nu-\mu}{k_\mathrm{B}T}\right) - 1}, \qquad \tilde{\mu} \equiv \frac{\mu}{k_\mathrm{B}T_\gamma}. \tag{3.6.32}$$

3. **Free-Free Distortions Y_ff ($z \lesssim 10^5$):** Emission and absorption of photons by thermal Bremsstrahlung at very low frequencies generates a distortion parameterized by:
$$Y_\mathrm{ff} \equiv \int \kappa\left(1 - \frac{T_\gamma}{T_e}\right) \mathrm{d}t, \qquad \kappa \equiv \frac{8\sqrt{\pi}\bar{g}_\mathrm{ff}}{3\sqrt{6}} \left(\frac{e^2}{4\pi\varepsilon_0}\right)^3 \frac{n_e^2}{m_e T_\gamma^3 \sqrt{m_e T_e}}. \tag{3.6.33}$$

FIRAS analysis places stringent 2σ upper bounds on all three distortion parameters:
$$|y| < 1.9\times10^{-5}, \qquad |\tilde{\mu}| < 9\times10^{-5}, \qquad |Y_\mathrm{ff}| < 1.5\times10^{-5}. \tag{3.6.34}$$

3.6.7 Kinematic dipole and galactic foregrounds

The kinematic dipole

The dominant angular anisotropy of the CMB is a dipole ($\ell = 1$) induced by the Doppler effect of our peculiar motion relative to the CMB rest frame:
$$T(\theta) = \frac{T_0}{\gamma(1 - \vec{\beta}\cdot\hat{n})} = \frac{T_0\sqrt{1 - v^2/c^2}}{1 - \frac{v}{c}\cos\theta} \simeq T_0\left[1 + \frac{v}{c}\cos\theta + \frac{1}{2}\left(\frac{v}{c}\right)^2 \cos 2\theta + \mathcal{O}\left(\frac{v^3}{c^3}\right)\right]. \tag{3.6.35}$$

We define the temperature to absorb the doppler shift of velocity
$$\nu_\mathrm{obs} = \nu\gamma(1 - \vec{\beta}\cdot\hat{n}) \tag{1}$$

FIRAS measured a dipole amplitude $\delta T = (3.346 \pm 0.017)\,\mathrm{mK}$, corresponding to a Solar System barycentric velocity:
$$v_\odot = 368 \pm 2\,\mathrm{km\cdot s^{-1}} \quad \text{towards } (l, b) = (263.85^\circ \pm 0.10^\circ, 48.25^\circ \pm 0.04^\circ). \tag{3.6.36}$$

Correcting for galactic rotation, the Local Group moves at:
$$v_\mathrm{LG} = 627 \pm 22\,\mathrm{km\cdot s^{-1}} \quad \text{towards } (l, b) = (276^\circ \pm 3^\circ, 30^\circ \pm 3^\circ). \tag{3.6.37}$$

After having subtracted these effects, some temperature anisotropies remain, with elative amplitude $\sim 10^{-5}$, which correspond to temperature fluctuations of the order of 30 μK. These anisotropies correspond to anisotropies in the cosmic microwave ackground at the time of recombination and redshifted by the cosmic expansion. hapter 6 is dedicated to the study of these temperature anisotropies.

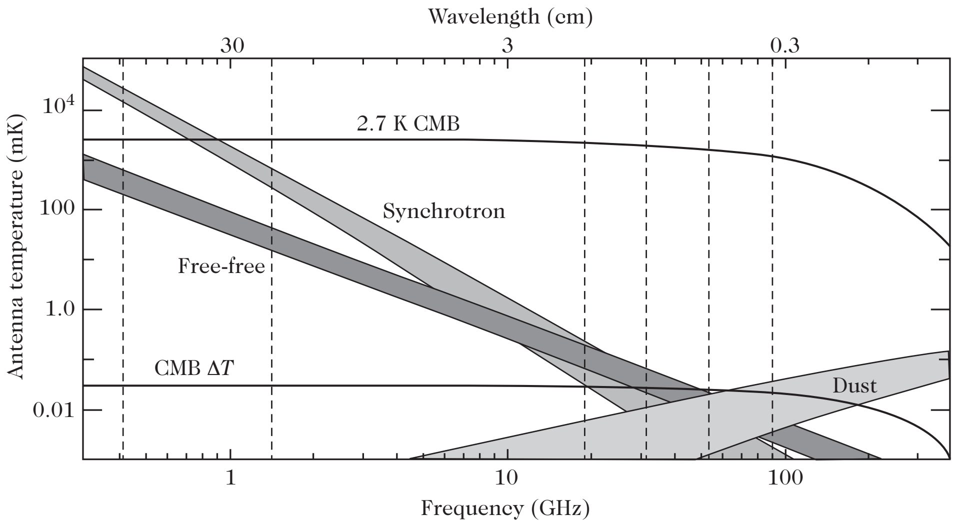


Figure 3.15: Frequency spectra of Galactic foreground emissions (synchrotron, free-free Bremsstrahlung, and thermal dust) compared with CMB primary temperature fluctuations $\Delta T \approx 30\,\mu$K. The minimum foreground contamination window occurs around 70–100 GHz.

3.6.8 Direct spectroscopic confirmation of cosmic cooling $T(z) = T_0(1+z)$

General relativity predicts that adiabatic expansion cools the photon temperature linearly with redshift:

$$T(z) = T_0(1+z) = 2.7255(1+z)\,\mathrm{K}. \tag{3.6.38}$$

This fundamental prediction has been directly verified observationally by measuring the excitation temperatures of atomic neutral carbon (C I) and carbon monoxide (CO) fine-structure and rotational levels in high-redshift absorption clouds illuminated by background quasars.

At redshift $z = 2.33771$, Srianand et al. (2000) measured $6.0\,\mathrm{K} \le T \le 14.0\,\mathrm{K}$, in striking agreement with the theoretical prediction $T(2.33771) = 9.1\,\mathrm{K}$ (Figure 3.16).

3.7 The Standard Cosmological Paradigm: Summary and Open Problems

3.7.1 Summary of cosmological parameters and thermal history

The Standard Cosmological Model (ΛCDM) is underpinned by three foundational observational pillars:

1. **Hubble Cosmic Expansion:** Directly established through supernovae, Cepheids, and distance-redshift relations.

2. **Big-Bang Nucleosynthesis:** The primordial origin and abundance of light isotopes (D,^{3} He,^{4} He,^{7} Li).

3. **Cosmic Microwave Background:** The pristine 2.7255 K blackbody relic bath and its primary anisotropies.

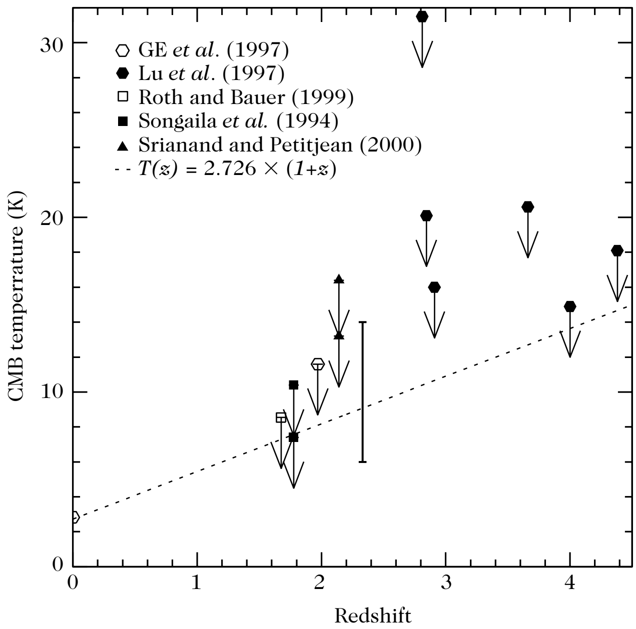


Figure 3.16: Spectroscopic measurements of the CMB temperature $T(z)$ as a function of redshift from quasar absorption systems (C I and CO molecular transitions). The dashed line represents the standard adiabatic expansion prediction $T(z) = 2.7255(1 + z)\,\mathrm{K}$.

The concordant cosmological parameters and thermal history milestones are summarized in Tables 3.9 and 3.10.

Table 3.9: Fundamental Cosmological Parameters of the Concordance ΛCDM Model

Parameter	**Physical Meaning**	**Observed Value**	**Reference / Method**
h	Dimensionless Hubble constant	0.72 ± 0.05	Section 3.1.1
Ω_{K0}	Spatial curvature parameter	-0.0049 ± 0.013	CMB + SNe Ia
$\Omega_{c0}h^2$	Cold dark matter density	0.113 ± 0.009	CMB + LSS
$\Omega_{b0}h^2$	Baryon density (BBN)	0.018 ± 0.006	Primordial D/H
$\Omega_{b0}h^2$	Baryon density (CMB)	0.0224 ± 0.0009	WMAP / Planck
$\Omega_{\gamma 0}h^2$	Photon energy density	$2.4697 \times 10^{-5}\,\Theta_{2.7}^4$	FIRAS Monopole
$\Omega_{\nu 0}h^2$	Massless neutrino density (3ν)	$1.68 \times 10^{-5}\,\Theta_{2.7}^4$	Relic Neutrinos
$\Omega_{r0}h^2$	Total radiation density	$4.148 \times 10^{-5}\,\Theta_{2.7}^4$	$\gamma + 3\nu$
$\Omega_{\Lambda 0}$	Dark energy density parameter	0.72 ± 0.03	SNe Ia + CMB
w	Dark energy equation of state	$w < -0.6$ ($w_0 = -1$)	SNe Ia ($z < 1.5$)

3.7.2 Classic problems of the standard Big-Bang paradigm

Despite its triumphs, the standard Big-Bang model cannot explain its own initial conditions and is plagued by five fundamental theoretical deficiencies:

Table 3.10: Key Physical Milestones in the Thermal History of the Universe

Milestone	**Cosmic Event**	**Redshift / Scale**	**Temperature T**
η	Baryon-to-photon ratio	$(6.14 \pm 0.25) \times 10^{-10}$	—
T_0	Present CMB temperature	$z = 0$	$2.7255 \pm 0.0006\,\mathrm{K}$
z_{eq}	Matter-radiation equality	$z_{\mathrm{eq}} \approx 3612$	$T_{\mathrm{eq}} \approx 0.85\,\mathrm{eV} \approx 9800\,\mathrm{K}$
z_{rec}	Recombination ($X_e = 0.5$)	$z_{\mathrm{rec}} \approx 1376$	$T_{\mathrm{rec}} \approx 3750\,\mathrm{K} \approx 0.32\,\mathrm{eV}$
z_{dec}	Photon decoupling (LSS peak)	$z_{\mathrm{dec}} = 1089 \pm 1$	$T_{\mathrm{dec}} \approx 2970\,\mathrm{K} \approx 0.26\,\mathrm{eV}$
Δz_{dec}	Last scattering surface thickness	$\Delta z_{\mathrm{dec}} = 195 \pm 2$	$\Delta T \approx 530\,\mathrm{K}$
z_{BBN}	Primordial nucleosynthesis	$z \sim 10^8$–10^9	$T \sim 0.08$–$1.0\,\mathrm{MeV}$
$z_{\nu,D}$	Neutrino decoupling	$z \sim 10^{10}$	$T \sim 1.0\,\mathrm{MeV}$

The Flatness / Curvature Problem

From the Friedmann equation, the curvature density parameter $\Omega_K(t) \equiv -\frac{K}{a^2H^2} = 1 - \Omega(t)$ evolves according to:

$$\frac{\mathrm{d}\Omega_K}{\mathrm{d}\ln a} = (1 + 3w)\,(1 - \Omega_K)\,\Omega_K, \tag{3.7.1}$$

where $w \equiv P/\rho$ is the cosmic equation of state. For constant w, Eq. (3.7.1) integrates to:

$$\Omega_K(z) = \frac{\Omega_{K0}}{(1 - \Omega_{K0})(1 + z)^{1+3w} + \Omega_{K0}}. \tag{3.7.2}$$

In standard cosmology, the expansion is dominated by ordinary matter ($w = 0$) and radiation ($w = 1/3$), satisfying the Strong Energy Condition $1 + 3w > 0$. Consequently:

$$|1 - \Omega(t)| \propto \frac{1}{\dot{a}^2} \propto \begin{cases} a^2 \propto t & \text{(Radiation era)} \\ a \propto t^{2/3} & \text{(Matter era)} \end{cases} \tag{3.7.3}$$

The flat fixed point $\Omega = 1$ ($\Omega_K = 0$) is dynamically *unstable* under forward cosmic expansion: any slight deviation from flatness grows rapidly with time.

Observational bounds today require $|\Omega_{K0}| < 0.01$. Tracing this constraint back through cosmological history requires severe fine-tuning:

$$|1 - \Omega(z_{\mathrm{eq}})| < 3 \times 10^{-5} \quad \text{(at matter-radiation equality } z_{\mathrm{eq}} \approx 3600), \tag{3.7.4}$$

$$|1 - \Omega(T_{\mathrm{BBN}})| < 10^{-16} \quad \text{(at nucleosynthesis } T \sim 1\,\mathrm{MeV}), \tag{3.7.5}$$

$$|1 - \Omega(T_{\mathrm{GUT}})| < 10^{-52} \quad \text{(at the GUT scale } T \sim 10^{16}\,\mathrm{GeV}), \tag{3.7.6}$$

$$|1 - \Omega(t_{\mathrm{Pl}})| < 10^{-60} \quad \text{(at the Planck epoch } t_{\mathrm{Pl}} \sim 10^{-43}\,\mathrm{s}). \tag{3.7.7}$$

The standard Big-Bang model provides no mechanism to explain why the initial curvature was fine-tuned to 60 decimal places.

Remark. Dynamical Resolution by Accelerated Expansion: Equation (3.7.1) demonstrates that $\Omega_K = 0$ becomes an attractive fixed point if the early Universe undergoes a phase of accelerated expansion driven by an effective fluid with $w < -1/3$ (i.e., $\ddot{a} > 0$), driving $\Omega(t) \to 1$ exponentially.

The Horizon / Causality Problem

In conformal time $\eta = \int \frac{\mathrm{d}t}{a}$, the maximum comoving distance a light signal could have traveled since the Big Bang ($t = 0$) defines the **particle horizon**:

$$\chi_{\mathrm{hor}}(t) = \int_0^t \frac{c\,\mathrm{d}t'}{a(t')} = \eta(t) - \eta(0). \tag{3.7.8}$$

In a radiation- or matter-dominated universe, $\eta(0) = 0$, so $\chi_{\mathrm{hor}}(t) = \eta(t) \propto t^{1/2}$ or $t^{1/3}$.

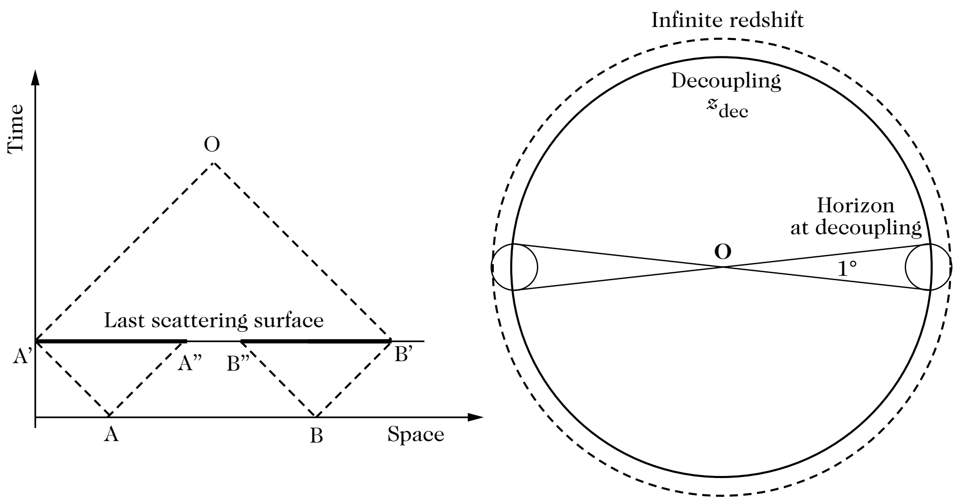


Figure 3.17: (Left): Spacetime conformal diagram showing the past light cone of the observer O intersecting the Last Scattering Surface (LSS) on a sphere of diameter $A'B'$. The causally connected regions ($A'A''$, ...) at decoupling are determined by the future light cones from the Big Bang singularity. (Right): Geometry of causally disconnected patches on the CMB sky.

As depicted in Figure 3.17, the comoving radius of a causally connected region at photon decoupling is $\chi_{\mathrm{hor}}(z_{\mathrm{dec}}) = \eta_{\mathrm{dec}}$, whereas the comoving distance from the observer to the Last Scattering Surface is $\chi(z_{\mathrm{dec}}) = \eta_0 - \eta_{\mathrm{dec}}$. The angular size subtended by a causal horizon patch on the present CMB sky is:

$$\theta_{\mathrm{hor}} \simeq \frac{\eta_{\mathrm{dec}}}{\eta_0 - \eta_{\mathrm{dec}}} \simeq \frac{1}{\sqrt{1 + z_{\mathrm{dec}}}} \approx 1^\circ. \tag{3.7.9}$$

The entire celestial sphere encompasses approximately:

$$N_{\mathrm{causal}} \sim \left(\frac{\eta_0 - \eta_{\mathrm{dec}}}{\eta_{\mathrm{dec}}}\right)^3 \approx 8(1 + z_{\mathrm{dec}})^{3/2} \sim 10^5\text{–}10^6 \tag{3.7.10}$$

mutually causally disconnected regions. Extrapolated back to the Planck epoch, the currently observable Universe contains:

$$N_{\mathrm{Planck}} = \frac{\frac{4}{3}\pi D_H^3}{\frac{4}{3}\pi\left(\ell_P \frac{a}{a_0}\right)^3} \sim \left(\frac{1 + z_{\mathrm{Pl}}}{\ell_{\mathrm{Pl}} H_0}\right)^3 \sim 10^{87} \tag{3.7.11}$$

independent causal volumes. The standard model offers no causal physical explanation for why these 10^{87} disconnected regions share the identical temperature $T_0 = 2.7255\,\mathrm{K}$ to one part in 10^5. The key question becomes: How was the thermal equilibrium established between causally disconnected regions?

The Origin of Inhomogeneities and the Trans-Planckian Problem

In standard cosmology, physical perturbation wavelengths scale as $\lambda_{\text{phys}}(t) \propto a(t)$, while the causal Hubble radius scales as $d_H(t) \equiv H^{-1} \propto t \propto a^2$ (radiation) or $a^{3/2}$ (matter).

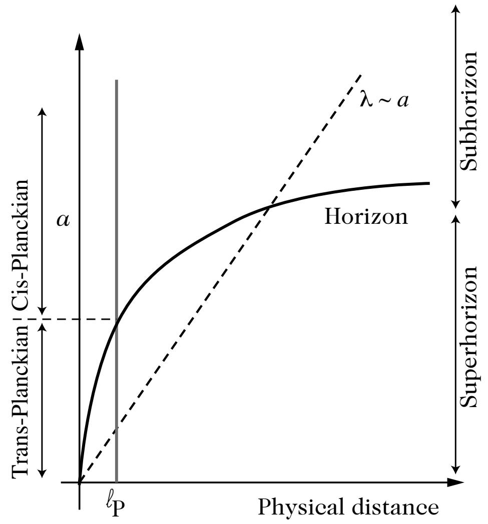


Figure 3.18: Evolution of physical perturbation wavelengths $\lambda_{\text{phys}} \propto a$ compared to the causal Hubble horizon H^{-1} in standard Big-Bang expansion. All cosmological scales observed today originated with super-horizon wavelengths ($\lambda > H^{-1}$) and sub-Planckian sizes ($\lambda < \ell_{\text{Pl}}$) at early times, defining the horizon and trans-Planckian problems.

As shown in Figure 3.18:

1. **Super-Horizon Initial Perturbations:** All large-scale structure modes and CMB multipoles ($\ell \lesssim 100$) had wavelengths $\lambda_{\text{phys}} > H^{-1}$ larger than the causal horizon at early epochs. In standard FLRW cosmology, no local causal process (such as sub-horizon thermal fluctuations) could have generated coherent, phase-synchronized perturbations across these super-horizon scales.

2. **The Trans-Planckian Problem:** Tracing perturbation wavelengths back towards the initial singularity ($a \to 0$), all physical lengths become smaller than the Planck length ($\lambda_{\text{phys}} < \ell_{\text{Pl}} \equiv \sqrt{\hbar G/c^3} \approx 1.6 \times 10^{-35}$ m), where classical spacetime geometry breaks down and quantum gravity effects dominate.

The Relic / Magnetic Monopole Problem

Grand Unified Theories (GUTs) predict that the simple gauge group G undergoes spontaneous symmetry breaking at $T_{\text{GUT}} \sim 10^{16}$ GeV to the Standard Model group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. By the Kibble mechanism, topological defects form whenever the vacuum manifold $\mathcal{M} = G/H$ has non-trivial homotopy groups.

For simple groups G, the second homotopy group $\pi_2(G/H) \neq 0$ implies the inevitable production of stable 't Hooft–Polyakov **magnetic monopoles** with mass $M_{\text{mono}} \sim T_{\text{GUT}}/\alpha_{\text{GUT}} \sim 10^{16}$ GeV. Because the Higgs field orientation is uncorrelated across distances greater than the causal horizon $d_H(t_{\text{GUT}}) \sim 2t_{\text{GUT}}$, at least one monopole forms per causal horizon volume:

$$n_{\text{mono}}(t_{\text{GUT}}) \sim \frac{1}{(2t_{\text{GUT}})^3} \sim T_{\text{GUT}}^6/M_{\text{Pl}}^3. \tag{3.7.12}$$

Because monopoles are massive and non-relativistic, their energy density dilutes as $\rho_{\rm mono} \propto a^{-3}$ while radiation dilutes as a^{-4}. Their present relic abundance evaluates to:

$$\Omega_{\rm mono,0} h^2 \sim 10^{17} \left(\frac{T_{\rm GUT}}{10^{16}\,{\rm GeV}} \right)^3 \left(\frac{M_{\rm mono}}{10^{16}\,{\rm GeV}} \right) \gg 1. \tag{3.7.13}$$

This would overclose the Universe by 17 orders of magnitude and induce recollapse within a few tens of thousands of years.

The Dark Sector, Cosmological Constant, and Coincidence Problems

Cosmological observations require that 95% of the cosmic energy budget resides in an unknown dark sector:

$$\Omega_{\gamma 0} : \Omega_{b0} : \Omega_{c0} : \Omega_{\Lambda 0} \sim 10^{-4} : 0.05 : 0.25 : 0.70. \tag{3.7.14}$$

This structure poses three deep theoretical puzzles:

- **The Cosmological Constant Problem:** Quantum vacuum zero-point energy density evaluates theoretically to $\rho_{\rm vac}^{\rm EW} \sim (1\,{\rm TeV})^4$ or $\rho_{\rm vac}^{\rm Pl} \sim (10^{19}\,{\rm GeV})^4$, conflicting with the observed value $\rho_\Lambda \sim (2.3 \times 10^{-3}\,{\rm eV})^4$ by 60 to 120 orders of magnitude.
- **The Coincidence Problem:** While $\rho_r \propto a^{-4}$, $\rho_m \propto a^{-3}$, and $\rho_\Lambda = {\rm const}$, we live in the brief cosmic epoch where matter and dark energy densities are of comparable order ($\Omega_m \sim \Omega_\Lambda$).
- **Baryogenesis:** Explaining the asymmetry $\eta = (n_b - n_{\bar b})/n_\gamma \approx 6 \times 10^{-10}$ requires satisfying Sakharov's three conditions (B-violation, C and CP violation, out-of-equilibrium dynamics) beyond the minimal Standard Model.

3.7.3 Synthesis and transition to the inflationary paradigm

The Standard Big-Bang model provides an impeccably verified description of the thermal history of the Universe from $T \sim 100\,{\rm MeV}$ ($t \sim 10^{-4}\,{\rm s}$) down to the present day. However, its profound initial condition paradoxes (flatness, horizons, super-horizon perturbations, and monopole overproduction) demonstrate that standard FLRW expansion must be preceded by a primordial epoch of accelerated expansion—**Cosmic Inflation**—which forms the foundation of modern early-universe cosmology.

Part III

The Inhomogeneous Universe

Chapter 4

The Inhomogeneous Universe: Perturbations and Their Evolution

Spacetimes of the Friedmann–Lemaître type serve as idealized cosmological backgrounds that provide an accurate description of the cosmos solely when averaged over very large scales. In reality, the observed Universe exhibits prominent hierarchical structures, including galaxies, clusters, and extensive filaments. During the 1980s, it was recognized that this cosmic web originates naturally from the gravitational amplification of tiny primordial density perturbations within a pressureless fluid, namely cold dark matter (Refs. [1–3]).

The central objective of this chapter is to examine spacetimes whose spatial geometry remains in the perturbative vicinity of an idealized Friedmann–Lemaître metric, and to delineate the gravitational evolution of density perturbations within an expanding cosmic medium:

- In Section 4.1, we review the classical Newtonian treatment of fluid perturbations in both static and expanding backgrounds.
- Section 4.2 formulates the full general-relativistic, gauge-invariant cosmological perturbation theory, which forms the theoretical bedrock of modern structure formation.
- Section 4.3 develops explicit dynamical solutions to these perturbation equations across distinct cosmological epochs.
- Section 4.4 analyzes the characteristic physical scales and the formal mathematical structure of the matter power spectrum.
- Section 4.5 reviews the primary observational probes of cosmological large-scale structure.

4.1 Newtonian perturbations

Prior to constructing the complete relativistic theory of cosmic perturbations, it is instructive to explore the simpler regime of fluid perturbations governed by Newtonian gravity in an expanding medium. As demonstrated in Section 5.4, this Newtonian formulation is fully sufficient for describing the late-time clustering of matter on sub-Hubble scales ($k \gg aH$) during the matter-dominated epoch (see Ref. [4] for an extensive review of Newtonian gravitational dynamics).

4.1.1 Case of a static space

In a flat Euclidean background space, fluid dynamics is governed by the standard hydrodynamic equations of continuity and momentum conservation (Euler equation) (Ref. [10]):

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{v}) = 0, \tag{4.1.1}$$

$$\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v} \cdot \boldsymbol{\nabla})\boldsymbol{v} = -\frac{1}{\rho}\boldsymbol{\nabla} P - \boldsymbol{\nabla}\Phi, \tag{4.1.2}$$

where ρ denotes the fluid mass density, P is the isotropic thermal pressure, $\boldsymbol{v}$ represents the velocity vector field, and Φ is the Newtonian gravitational potential.

Without gravity

Neglecting gravitational interactions ($\Phi = 0$) and decomposing the fluid density and pressure into uniform static background values and small fluctuations, $\rho = \bar{\rho} + \delta\rho$ and $P = \bar{P} + \delta P$, the linear perturbation of Eqs. (4.1.1) and (4.1.2) yields a wave equation:

$$\frac{\partial^2 \delta\rho}{\partial t^2} - \Delta \delta P = 0. \tag{4.1.3}$$

We define the compressibility coefficient as $\chi_a \equiv \rho^{-1} \left(\partial P/\partial \rho\right)_a$, where the index a designates either isothermal (T) or adiabatic (specific entropy S) conditions. When heat conduction operates much faster than the perturbation period, the flow remains isothermal; conversely, when the timescale is insufficient for thermal exchange (as on cosmological wavelengths), the flow is adiabatic. Under adiabatic conditions, Eq. (4.1.3) takes the boxed canonical wave form:

$$\boxed{\frac{\partial^2 \delta\rho}{\partial t^2} - c_{\mathrm{S}}^2 \Delta \delta\rho = 0, \qquad c_{\mathrm{S}}^2 = \rho \chi_S = \left(\frac{\partial P}{\partial \rho}\right)_S,} \tag{4.1.4}$$

showing that acoustic disturbances propagate at constant amplitude with the adiabatic sound speed c_{S}.

Effect of gravity

We now restore gravitational interactions. The gravitational potential is determined by the Poisson equation:

$$\Delta\Phi = 4\pi G_{\mathrm{N}} \delta\rho, \tag{4.1.5}$$

which introduces a gravitational self-attraction source term into the wave equation:

$$\frac{\partial^2 \delta\rho}{\partial t^2} - c_{\mathrm{S}}^2 \Delta \delta\rho = 4\pi G_{\mathrm{N}} \bar{\rho}\, \delta\rho. \tag{4.1.6}$$

Decomposing the density fluctuation into Fourier spatial plane waves, $\delta\rho \propto \exp[i(\omega t - \boldsymbol{k} \cdot \boldsymbol{x})]$, yields the fundamental dispersion relation:

$$\boxed{\omega^2 = \frac{4\pi^2 c_{\mathrm{S}}^2}{\lambda_{\mathrm{J}}^2} \left(\frac{\lambda_{\mathrm{J}}^2}{\lambda^2} - 1\right), \qquad \lambda_{\mathrm{J}} \equiv c_{\mathrm{S}} \sqrt{\frac{\pi}{G_{\mathrm{N}} \bar{\rho}}}.} \tag{4.1.7}$$

Here λ_{J} defines the **Jeans length**. For perturbation scales exceeding this threshold ($\lambda > \lambda_{\mathrm{J}}$), the squared frequency becomes negative ($\omega^2 < 0$), driving an exponential gravitational instability:

$$\delta\rho \propto \exp\left[\sqrt{4\pi G_{\mathrm{N}} \bar{\rho} \left(1 - \frac{\lambda_{\mathrm{J}}^2}{\lambda^2}\right)}\, t\right].$$

Physically, the Jeans length demarcates the scale where the acoustic crossing timescale $t_{\mathrm{sound}} \sim \lambda/c_{\mathrm{S}}$ equals the gravitational free-fall collapse time $t_{\mathrm{grav}} \sim (G_{\mathrm{N}} \bar{\rho})^{-1/2}$. On small scales ($\lambda < \lambda_{\mathrm{J}}$), acoustic pressure gradients react faster than gravity and support stable oscillations; on scales exceeding λ_{J}, pressure support fails, enabling gravitational collapse.

Energetic aspects

From an energetic standpoint, each fluid parcel can be modeled as a coupled harmonic oscillator with Lagrangian displacement $\boldsymbol{s}$ from its equilibrium position. The velocity is $\dot{\boldsymbol{s}}$, and mass conservation implies $\delta\rho = -\boldsymbol{\nabla}\cdot\boldsymbol{s}$. Integrating the dynamical equation (4.1.7) over a volume V yields the mechanical energy conservation law:

$$\langle E_M \rangle = \langle E_K \rangle + \langle E_E \rangle + \langle E_G \rangle \,, \tag{4.1.8}$$

where $\langle E_K \rangle = \frac{1}{2}\langle \dot{\boldsymbol{s}}^2 \rangle$ is the average kinetic energy density, $\langle E_E \rangle = \frac{1}{2}c_{\mathrm{S}}^2 \langle (\boldsymbol{\nabla}\cdot\boldsymbol{s})^2 \rangle$ is the elastic compressive energy density, and $\langle E_G \rangle = -2\pi G_{\mathrm{N}}\bar{\rho}\langle \boldsymbol{s}^2 \rangle$ is the self-gravitational potential energy density. The kinetic contribution reads:

$$\frac{1}{2}\left\langle \dot{\boldsymbol{s}}^2 \right\rangle = \frac{1}{2}\omega^2 \left\langle \boldsymbol{s}^2 \right\rangle, \qquad \omega^2 = c_{\mathrm{S}}^2\left(k^2 - k_{\mathrm{J}}^2\right), \qquad k_{\mathrm{J}} \equiv \frac{2\pi}{\lambda_{\mathrm{J}}}.$$

Consequently, the volume-averaged total mechanical energy evaluates to:

$$\langle E_M \rangle = \frac{1}{2}\left\langle \boldsymbol{s}^2 \right\rangle \left[\omega^2 + c_{\mathrm{S}}^2\left(k^2 - k_{\mathrm{J}}^2\right)\right].$$

By virtue of equipartition enforced by the dispersion relation, the total average energy density simplifies to:

$$\langle E_M \rangle = \left\langle \boldsymbol{s}^2 \right\rangle c_{\mathrm{S}}^2\left(k^2 - k_{\mathrm{J}}^2\right), \tag{4.1.9}$$

which is strictly negative on scales larger than the Jeans length ($k < k_{\mathrm{J}}$), proving that gravitationally unstable perturbations correspond to bound physical states.

4.1.2 Case of an expanding space

In an expanding cosmological background, physical spatial positions $\boldsymbol{r}(t)$ are related to comoving coordinates $\boldsymbol{x}$ via the cosmic scale factor $a(t)$:

$$\boldsymbol{r}(t) = a(t)\boldsymbol{x}. \tag{4.1.10}$$

Differentiating with respect to cosmic time t yields the total physical velocity:

$$\boldsymbol{v}(t) = \dot{\boldsymbol{r}} = H\boldsymbol{r} + \boldsymbol{u}, \tag{4.1.11}$$

where $H \equiv \dot{a}/a$ is the Hubble expansion parameter and $\boldsymbol{u} \equiv a\dot{\boldsymbol{x}}$ represents the peculiar velocity field relative to the cosmological Hubble flow.

Eulerian formulation

Transforming spatial and temporal derivatives to comoving coordinates via $\boldsymbol{\nabla}_{\boldsymbol{r}} = a^{-1}\boldsymbol{\nabla}_{\boldsymbol{x}}$ and $\frac{\partial}{\partial t}\Big|_{\boldsymbol{r}} = \frac{\partial}{\partial t}\Big|_{\boldsymbol{x}} - H\boldsymbol{x}\cdot\boldsymbol{\nabla}_{\boldsymbol{x}}$, and using $\boldsymbol{\nabla}_{\boldsymbol{x}}\cdot\boldsymbol{x} = 3$, the hydrodynamic equations take the comoving forms:

$$\dot{\rho}(\boldsymbol{x},t) + 3H\rho(\boldsymbol{x},t) + \frac{1}{a}\boldsymbol{\nabla}_{\boldsymbol{x}}\cdot\left[\rho(\boldsymbol{x},t)\boldsymbol{u}(\boldsymbol{x},t)\right] = 0, \tag{4.1.12}$$

and

$$\dot{\boldsymbol{u}}(\boldsymbol{x},t) + H\boldsymbol{u}(\boldsymbol{x},t) + \frac{1}{a}\left(\boldsymbol{u}\cdot\boldsymbol{\nabla}_{\boldsymbol{x}}\right)\boldsymbol{u}(\boldsymbol{x},t) = -\frac{1}{a}\boldsymbol{\nabla}_{\boldsymbol{x}}\Phi(\boldsymbol{x},t) - \frac{1}{a\rho}\boldsymbol{\nabla}_{\boldsymbol{x}}P(\boldsymbol{x},t). \tag{4.1.13}$$

Defining the fractional density contrast $\delta(\boldsymbol{x},t)$ relative to the background density $\bar{\rho}(t)$:

$$\rho(\boldsymbol{x},t) = \bar{\rho}(t)\left[1 + \delta(\boldsymbol{x},t)\right], \tag{4.1.14}$$

the continuity equation (4.1.12) becomes:

$$\dot{\delta} + \frac{1}{a}\boldsymbol{\nabla}\cdot[(1+\delta)\boldsymbol{u}] = 0. \tag{4.1.15}$$

Taking the time derivative of Eq. (4.1.15) and combining it with the divergence of the Euler equation (4.1.13) produces the exact non-linear evolution equation for the density contrast:

$$\ddot{\delta} + 2H\dot{\delta} = \frac{1}{\rho a^2}\Delta P + \frac{1}{a^2}\boldsymbol{\nabla}\cdot[(1+\delta)\boldsymbol{\nabla}\Phi] + \frac{1}{a^2}\partial_i\partial_j\left[(1+\delta)u^i u^j\right]. \tag{4.1.16}$$

Importantly, Eq. (4.1.16) involves no small-amplitude approximation and is valid even in the non-linear clustering regime. The gravitational potential satisfies the comoving Poisson equation:

$$\Delta\Phi = 4\pi G_{\mathrm{N}}\bar{\rho}\, a^2(t)\delta, \tag{4.1.17}$$

whose formal solution is given by:

$$\Phi(\boldsymbol{x},t) = -G_{\mathrm{N}}\bar{\rho}\, a^2 \int \frac{\mathrm{d}^3\boldsymbol{x}'\,\delta(\boldsymbol{x}',t)}{|\boldsymbol{x}-\boldsymbol{x}'|}. \tag{4.1.18}$$

Lagrangian description

In the Lagrangian framework, we follow fluid parcels along trajectories defined by the displacement field $\boldsymbol{X}(\boldsymbol{q},t)$, linking initial Lagrangian coordinates $\boldsymbol{q}$ to Eulerian positions $\boldsymbol{x}(t)$:

$$\boldsymbol{x}(t) = \boldsymbol{q} + \boldsymbol{X}(\boldsymbol{q},t).$$

Particle trajectories satisfy the equation of motion:

$$\ddot{\boldsymbol{x}} + 2H\dot{\boldsymbol{x}} = -\frac{1}{a^2}\boldsymbol{\nabla}_{\boldsymbol{x}}\Phi,$$

which, using conformal time η ($\mathrm{d}t = a\,\mathrm{d}\eta$, $\mathcal{H} \equiv a'/a$), takes the form:

$$\boldsymbol{x}'' + \mathcal{H}\boldsymbol{x}' = -\boldsymbol{\nabla}_{\boldsymbol{x}}\Phi.$$

Taking the divergence of this relation, using Poisson's equation, and invoking mass conservation $\bar{\rho}\mathrm{d}^3\boldsymbol{q} = \bar{\rho}[1+\delta(\boldsymbol{x},t)]\mathrm{d}^3\boldsymbol{x}$, we obtain the equation governing the displacement field:

$$J(\boldsymbol{q},t)\boldsymbol{\nabla}_{\boldsymbol{x}}\cdot\left[\boldsymbol{X}'' + \mathcal{H}\boldsymbol{X}'\right] = \frac{3}{2}\Omega_{\mathrm{m}}\mathcal{H}^2\left[J(\boldsymbol{q},t)-1\right], \tag{4.1.19}$$

where the overdensity is related to the Jacobian determinant:

$$1+\delta(\boldsymbol{x},t) = \frac{1}{\det\left(\delta_{ij}+X_{i,j}\right)} \equiv \frac{1}{J(\boldsymbol{q},t)},$$

with $X_{i,j} \equiv \partial X_i/\partial q_j$. Eulerian and Lagrangian gradients are connected via $(\boldsymbol{\nabla}_{\boldsymbol{x}})_i = (\delta_{ij} + X_{i,j})^{-1}(\boldsymbol{\nabla}_{\boldsymbol{q}})_j$. While highly effective in non-linear perturbation theory (Refs. [11, 12]), this Lagrangian mapping breaks down upon shell crossing ($J \to 0$), where multistreaming produces formal density singularities.

Linearized systems

In the linear regime, we assume small departures from homogeneity under the criteria:

$$\delta \ll 1, \qquad \left(\frac{ut}{d}\right)^2 \ll \delta,$$

where u represents the peculiar velocity and d is the perturbation coherence length. Linearizing Eq. (4.1.16) yields the cosmological Jeans equation:

$$\ddot{\delta} + 2H\dot{\delta} = \frac{c_{\mathrm{S}}^2}{a^2}\Delta\delta + 4\pi G_{\mathrm{N}}\bar{\rho}\,\delta. \tag{4.1.20}$$

Unlike a static universe, the expanding Jeans length is time-dependent through $\bar{\rho}(t) \propto a^{-3}$. Hence, a perturbation of fixed comoving scale can transition from an oscillating acoustic regime to gravitational instability as expansion lowers the Jeans scale.

Growth of density perturbations

For pressureless cold dark matter ($c_{\mathrm{S}} = 0$), the spatial and temporal dependencies decouple, admitting separable solutions:

$$\delta(\boldsymbol{x}, t) = D_+(t)\varepsilon_+(\boldsymbol{x}) + D_-(t)\varepsilon_-(\boldsymbol{x}),$$

where $D_+(t)$ is the growing mode, $D_-(t)$ is the decaying mode, and $\varepsilon_\pm(\boldsymbol{x})$ define the initial seed fluctuation field. The growth factors satisfy:

$$\ddot{D} + 2H(t)\dot{D} - \frac{3}{2}H^2(t)\Omega_{\mathrm{m}}(t)D = 0, \tag{4.1.21}$$

where $4\pi G_{\mathrm{N}}\bar{\rho}_{\mathrm{m}}(t) = \frac{3}{2}H^2(t)\Omega_{\mathrm{m}}(t)$. In an Einstein–de Sitter spacetime ($\Omega_{\mathrm{m}} = 1, a \propto t^{2/3}$), analytical integration yields power-law growth:

$$D_+(t) \propto t^{2/3} \propto a(t), \qquad D_-(t) \propto t^{-1} \propto a^{-3/2}(t). \tag{4.1.22}$$

Using the scale factor a as the evolution variable, Eq. (4.1.21) becomes:

$$\frac{\mathrm{d}^2 D}{\mathrm{d}a^2} + \left(\frac{1}{H}\frac{\mathrm{d}H}{\mathrm{d}a} + \frac{3}{a}\right)\frac{\mathrm{d}D}{\mathrm{d}a} - \frac{3}{2}\frac{\Omega_{\mathrm{m}0}}{a^5}\left(\frac{H}{H_0}\right)^{-2} D = 0. \tag{4.1.23}$$

One exact solution is $D_- \propto H(a)$. The second independent growing mode follows via reduction of order:

$$D_+(a) = \frac{5}{2}\frac{H(a)}{H_0}\Omega_{\mathrm{m}0}\int_0^a \frac{\mathrm{d}a'}{[a'E(a')]^3}, \tag{4.1.24}$$

where $E(a) \equiv H(a)/H_0$. For an open universe without dark energy ($\Lambda = 0, \Omega_{\mathrm{m}} < 1$), both modes admit closed forms:

$$D_- = \sqrt{\frac{1+x}{x^3}}, \qquad D_+ = 1 + \frac{3}{x} + 3D_-(x)\ln\left[\sqrt{1+x} - \sqrt{x}\right], \tag{4.1.25}$$

with $x \equiv \Omega_{\mathrm{m}}^{-1} - 1$. As $\Omega_{\mathrm{m}} \to 0$ ($x \gg 1$), $D_+ \to 1$, demonstrating that spatial curvature halts perturbation growth.

For a flat ΛCDM model, evaluating Eq. (4.1.24) requires numerical quadrature (Fig. 4.1), well approximated by the Lahav et al. fitting formula (Ref. [13]):

$$D_+ \simeq \frac{5}{2}\frac{a\Omega_{\mathrm{m}}}{\Omega_{\mathrm{m}}^{4/7} - \Omega_\Lambda + \left(1 + \frac{1}{2}\Omega_{\mathrm{m}}\right)\left(1 + \frac{1}{70}\Omega_\Lambda\right)}. \tag{4.1.26}$$

Analytically, for $K = 0$, D_+ can be expressed via Gauss's hypergeometric function (Ref. [14]):

$$D_+ = {}_2F_1\left[1, \frac{1}{3}; \frac{11}{6}; -\sinh^2\left(\frac{3\alpha t}{2}\right)\right]\sinh^{2/3}\left(\frac{3\alpha t}{2}\right),$$

where $\alpha \equiv H_0\sqrt{\Omega_{\Lambda 0}} = \sqrt{\Lambda/3}$.

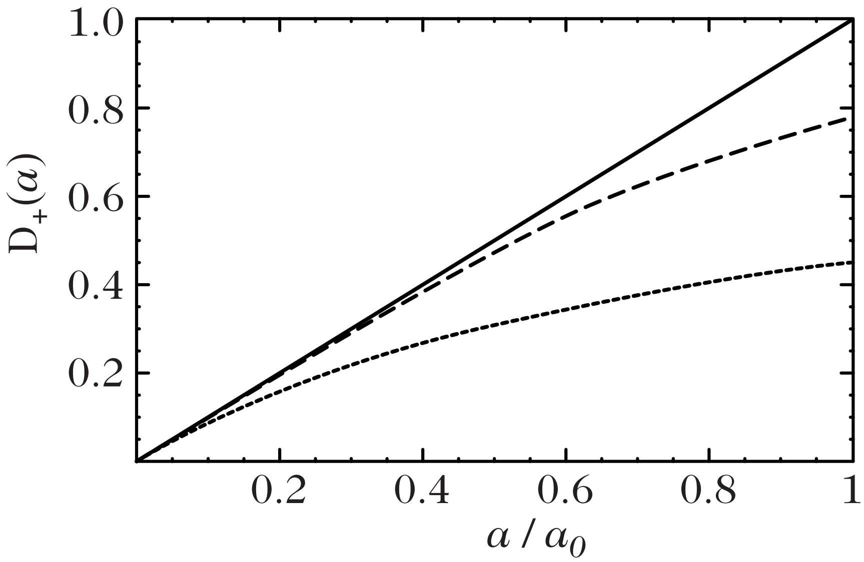


Figure 4.1: Evolution of the growing mode of the density contrast, $D_+(a)$, as a function of the scale factor, a, for $\Omega_{\mathrm{m}0} = 1$ [solid line], $(\Omega_{\mathrm{m}0}, \Omega_{\Lambda 0}) = (0.3, 0.7)$ [dashed line] and $(\Omega_{\mathrm{m}0}, \Omega_{\Lambda 0}) = (0.5, 0)$ [dotted line].

The Hubble parameter from the growth function

Equation (4.1.23) can be inverted into an equation for the normalized expansion rate $E^2(a)$:

$$\frac{\mathrm{d}E^2}{\mathrm{d}a} + 2\left[\frac{3}{a} + \frac{\mathrm{d}^2 D}{\mathrm{d}a^2}\left(\frac{\mathrm{d}D}{\mathrm{d}a}\right)^{-1}\right]E^2 - 3\frac{\Omega_{\mathrm{m}0}}{a^5}D\left(\frac{\mathrm{d}D}{\mathrm{d}a}\right)^{-1} = 0.$$

Integrating this first-order differential equation yields:

$$E^2(z) = -3\Omega_{\mathrm{m}0}(1+z)^2\left(\frac{\mathrm{d}D}{\mathrm{d}z}\right)^{-2}\int_z^\infty \frac{D}{1+z}\frac{\mathrm{d}D}{\mathrm{d}z}\,\mathrm{d}z\,, \tag{4.1.27}$$

as a function of redshift $1 + z = a_0/a$. This consistency relation links cosmological expansion directly to structure growth in ΛCDM.

Density–velocity relation

Because $\delta \propto D(t)$, the continuity equation (4.1.15) relates the peculiar velocity divergence directly to the overdensity:

$$\theta(\boldsymbol{x}, t) \equiv \frac{1}{aH}\boldsymbol{\nabla}\cdot\boldsymbol{u}(\boldsymbol{x}, t) = -f(\Omega_{\mathrm{m}0}, \Omega_{\Lambda 0})\delta, \tag{4.1.28}$$

where the dimensionless growth rate f is defined as:

$$f(\Omega_{\mathrm{m}0}, \Omega_{\Lambda 0}) \equiv \frac{\mathrm{d}\ln D(a)}{\mathrm{d}\ln a} \simeq \Omega_{\mathrm{m}0}^{0.6} + \frac{\Omega_{\Lambda 0}}{70}\left(1 + \frac{1}{2}\Omega_{\mathrm{m}0}\right). \tag{4.1.29}$$

Inverting the divergence, the general peculiar velocity field decomposes into potential and solenoidal components:

$$\boldsymbol{u} = -\frac{2}{3}\frac{f_+(\Omega_{\mathrm{m}0}, \Omega_{\Lambda 0})}{aH\Omega_{\mathrm{m}}}\boldsymbol{\nabla}\Phi + \boldsymbol{\nabla}\times\boldsymbol{U}, \tag{4.1.30}$$

where $\boldsymbol{U}$ is an arbitrary solenoidal vector potential. In an Einstein–de Sitter universe:

$$f_+ = 1, \quad \boldsymbol{u}_+ = -t^{1/3}\boldsymbol{\nabla}\Phi, \qquad f_- = -\frac{3}{2}, \quad \boldsymbol{u}_- = t^{-1/3}\boldsymbol{\nabla}\Phi. \tag{4.1.31}$$

Cosmic peculiar velocity surveys (Ref. [15]) directly measure these flow fields, providing sensitive tests of gravity on cosmological scales.

Gravitational potential

In the linear regime, the gravitational potential scales as $\Phi \propto D_+(t)/a(t)$. Thus, in an Einstein–de Sitter spacetime ($D_+ \propto a$), Φ remains strictly constant in time. Using the σ_8 normalization on $8\,h^{-1}$Mpc scales, Poisson's equation gives:

$$\sigma_\Phi = \frac{3}{2}\Omega_{\mathrm{m0}} \left(\frac{8\,\mathrm{Mpc}}{H_0^{-1}}\right)^2 \sigma_8 \sim 10^{-5}.$$

This smallness guarantees that the metric potential Φ remains perturbative ($\Phi \ll 1$) even when density clustering has become non-linear ($\delta \gg 1$), and characterizes the typical potential well depth across all scales under inflationary scale invariance.

4.1.3 Predictions and observables

The cosmic web observed today stems from the gravitational amplification of primordial fluctuations. To rigorously connect early-universe predictions with large-scale observations, a statistical formulation is essential.

The need for a statistical approach

Because individual primordial seed perturbations $\varepsilon_\pm(\boldsymbol{x})$ are inaccessible, theoretical models predict the statistical properties of the random field ensemble rather than deterministic local amplitudes. The observable Universe is interpreted as a single realization drawn from a statistical cosmic distribution.

Statistical description

Assuming statistical homogeneity (translational invariance of all joint probabilities) and statistical isotropy (rotational invariance), the two-point spatial correlation function:

$$\xi(\boldsymbol{x}, \boldsymbol{r}) = \langle \delta(\boldsymbol{x})\delta(\boldsymbol{x}+\boldsymbol{r})\rangle \tag{4.1.32}$$

depends purely on the separation distance $r \equiv |\boldsymbol{r}|$:

$$\xi(\boldsymbol{x}, \boldsymbol{r}) = \xi(r).$$

Expanding the real density contrast into spatial Fourier modes:

$$\delta(\boldsymbol{r}) = \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^{3/2}}\, \delta_{\boldsymbol{k}}\, e^{i\boldsymbol{k}\cdot\boldsymbol{r}}, \tag{4.1.33}$$

the Fourier coefficients satisfy the reality condition:

$$\delta^*_{\boldsymbol{k}} = \delta_{-\boldsymbol{k}}. \tag{4.1.34}$$

The ensemble expectation value evaluates to:

$$\langle \delta_{\boldsymbol{k}}\delta_{\boldsymbol{k}'}\rangle = \int \frac{\mathrm{d}^3\boldsymbol{x}}{(2\pi)^{3/2}} \frac{\mathrm{d}^3\boldsymbol{r}}{(2\pi)^{3/2}}\, \langle \delta(\boldsymbol{x})\delta(\boldsymbol{x}+\boldsymbol{r})\rangle\, e^{-i(\boldsymbol{k}+\boldsymbol{k}')\cdot\boldsymbol{x} - i\boldsymbol{k}'\cdot\boldsymbol{r}}, \tag{4.1.35}$$

which, after integrating over $\boldsymbol{x}$, defines the matter power spectrum $P_\delta(k)$:

$$\langle \delta_{\boldsymbol{k}}\delta_{\boldsymbol{k}'}\rangle = \delta^{(3)}(\boldsymbol{k}+\boldsymbol{k}')P_\delta(k). \tag{4.1.36}$$

The power spectrum is the Fourier transform of the real-space correlation function:

$$P_\delta(k) = \int \mathrm{d}^3\boldsymbol{r}\, \xi(r)\, e^{i\boldsymbol{k}\cdot\boldsymbol{r}}, \tag{4.1.37}$$

which inverts to the boxed canonical relation:

$$\boxed{\xi(r) = \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3}\, P_\delta(k)\, e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} = \int_0^\infty \frac{\mathrm{d}k}{2\pi^2}\, k^2 P_\delta(k)\, \frac{\sin kr}{kr}.} \tag{4.1.38}$$

Link with observable quantities and smoothing

Because astronomers observe a finite survey volume V of a single cosmic realization, the ensemble average is approximated operationally by the spatial volume average:

$$\xi_{\mathrm{obs}}(\boldsymbol{r}) = \frac{1}{V} \int_V \mathrm{d}^3\boldsymbol{r}'\, \delta_{\mathrm{obs}}(\boldsymbol{r}')\delta_{\mathrm{obs}}(\boldsymbol{r}+\boldsymbol{r}'). \tag{4.1.39}$$

For any statistical observable $\mathcal{O}$, an empirical estimator $\hat{\mathcal{O}}$ is characterized by its cosmic distribution function $\mathcal{P}(\hat{\mathcal{O}})$, with expectation value:

$$\left\langle \hat{\mathcal{O}} \right\rangle = \int \hat{\mathcal{O}}\, \mathcal{P}(\hat{\mathcal{O}})\, \mathrm{d}\hat{\mathcal{O}}\,.$$

Non-linear estimators typically carry a non-zero cosmic bias $b_{\mathcal{O}}$:

$$b_{\mathcal{O}} \equiv \frac{\left\langle \hat{\mathcal{O}} \right\rangle - \mathcal{O}}{\mathcal{O}},$$

which vanishes only as the survey volume approaches infinity. Robust estimators are designed to minimize both cosmic bias and cosmic variance.

In observational practice, this theoretical framework represents an idealization. Real galaxy catalogs record discrete counts of astronomical objects across a finite survey volume V, from which an underlying continuous density field must be reconstructed. The operationally inferred quantity is the locally smoothed density contrast $\delta_{\mathrm{obs},R}(\boldsymbol{x})$, obtained by convolving the raw field with a spatial window (or filter) function W_R:

$$\delta_{\mathrm{obs},R}(\boldsymbol{x}) = \int_V \mathrm{d}^3\boldsymbol{y}\, \delta_{\mathrm{obs}}(\boldsymbol{y})\, W_R(\boldsymbol{x}-\boldsymbol{y}), \tag{4.1.40}$$

which defines the observational window function W_R. To preserve total mass and characterize a physical smoothing scale R, the window function is subject to the normalization conditions:

$$\int W_R(\boldsymbol{x})\, \mathrm{d}^3\boldsymbol{x} = 1, \tag{4.1.41}$$

and

$$\int W_R(\boldsymbol{x})\, x^2\, \mathrm{d}^3\boldsymbol{x} = R^2. \tag{4.1.42}$$

Canonical choices for W_R include the spherical top-hat filter and the Gaussian smoothing kernel. Denoting the spatial Fourier transform of the window function by $\widehat{W}_R(\boldsymbol{k})$, the convolution theorem establishes that the Fourier modes of the smoothed field are simply:

$$\delta_R(\boldsymbol{k}) = \delta_{\boldsymbol{k}}\, \widehat{W}_R(\boldsymbol{k}). \tag{4.1.43}$$

The uncertainty in estimating cosmological parameters from finite galaxy catalogs arises from three primary sources (see Refs. [12, 16] for comprehensive analyses):

1. **Finite-volume effects:** A survey encompassing total volume V contains only approximately $N \sim V/L^3$ statistically independent patches of characteristic coherence size L. Consequently, the global background density cannot be determined with infinite precision. These sampling fluctuations scale with the volume integral of the two-point correlation function over the survey footprint and are universally designated as *cosmic variance*.

2. **Edge effects:** The finite, complex geometric boundaries of an observational catalog distort the spatial sampling, since estimators assign uneven statistical weights to regions situated near survey borders.

3. **Discreteness (shot-noise) effects:** Whereas theoretical cosmological fields are smooth and continuous, the observable universe is traced by point-like discrete sources (e.g., individual galaxies). The resulting Poisson-like shot noise decays as an inverse power of the total number density of observed tracers.

For any general physical field $Q(\boldsymbol{r})$, one seeks to determine the probability distribution function (PDF) $p(Q, R)\,\mathrm{d}Q$ of the filtered field $Q_R(\boldsymbol{r})$ (focusing on the one-point distribution for conciseness). This PDF quantifies the likelihood that the smoothed field assumes a value in the interval $[Q, Q + \mathrm{d}Q]$. For instance, if Q represents galaxy counts within a spherical top-hat of radius R, $p(Q, R)$ dictates the probability of detecting N galaxies within that volume. Standard inflationary cosmology posits that primordial fluctuations originate as a Gaussian random field:

$$p(Q_i, R) = \frac{1}{\sqrt{2\pi}\sigma_i(R)} \exp\left(-\frac{Q_{iR}^2}{2\sigma_i^2(R)}\right), \tag{4.1.44}$$

where $\sigma_i^2(R) \equiv \langle Q_R^2 \rangle$ represents the initial variance. Because a Gaussian distribution is entirely specified by its second central moment, comparing the primordial state with the late-time smoothed field Q_R requires accounting for two primary dynamical phenomena:

- Scale-dependent linear growth, whereby gravitational evolution modifies the initial power spectrum through a scale-dependent transfer function $T(k)$;

- Non-linear gravitational collapse, which drives departures from Gaussianity in the statistical distribution of fluctuations.

A third practical complication stems from the fact that telescopes register solely luminous baryonic tracers rather than dark matter directly. Ensuring that visible structures accurately reflect the underlying total gravitational matter field leads to the concept of galaxy bias (see Refs. [12, 17–19]).

The subsequent sections of this chapter investigate the relativistic evolution of linear perturbations and the resulting transfer functions, after first examining the onset of non-linear behavior within the Newtonian framework.

4.1.4 Towards the non-linear regime

The Newtonian formulation offers valuable insight into the initial stages of non-linear gravitational clustering (for an exhaustive review of perturbative and non-linear dynamics, see Ref. [12]).

Regardless of its initial infinitesimal amplitude, the density contrast δ undergoes continuous gravitational growth until it eventually approaches order unity ($\delta \sim 1$). The cosmic epoch at which linear theory breaks down depends on both the primordial amplitude and the cosmological growth factor $D(t)$. Observationally, the normalization of the present-day matter power spectrum is conventionally parameterized by the root-mean-square density fluctuation in a sphere of radius $R_8 = 8\,h^{-1}$Mpc, filtered with a spherical top-hat:

$$\sigma_8^2 \equiv \left\langle \left(\frac{3}{4\pi R_8^3} \int_{|\boldsymbol{x}| \leq R_8} \delta(\boldsymbol{x})\, \mathrm{d}^3\boldsymbol{x} \right)^2 \right\rangle \sim 1. \tag{4.1.45}$$

The characteristic scale R_8 thus marks the physical boundary below which non-linearities dominate and the linear approximation is rendered invalid.

To describe the onset of non-linear evolution, the density contrast is expanded in a perturbative series:

$$\delta = \delta^{(1)} + \delta^{(2)} + \dots,$$

where each successive term $\delta^{(n)}$ is of order $\mathcal{O}(\varepsilon^n)$ in terms of the initial seed perturbation field $\varepsilon(\boldsymbol{x}) \ll 1$. The first-order term reproduces the linear growing-mode solution established earlier:

$$\delta^{(1)} = D(t)\,\varepsilon(\boldsymbol{x}), \tag{4.1.46}$$

while at second order, the non-linear continuity and Euler equations yield the governing dynamical equation for $\delta^{(2)}$:

$$\ddot{\delta}^{(2)} + 2H\dot{\delta}^{(2)} = 4\pi G_{\mathrm{N}}\rho \left[\delta^{(1)}\right]^2 + \frac{1}{a^2}\boldsymbol{\nabla}\delta^{(1)} \cdot \boldsymbol{\nabla}\Phi + \frac{1}{a^2}\partial_i\partial_j \left[u^i_{(1)}u^j_{(1)}\right]. \tag{4.1.47}$$

Substituting the first-order solutions into the right-hand side source terms, this inhomogeneous differential equation integrates to (Ref. [20]):

$$\delta^{(2)} = D^2(t)\left\{\frac{2}{3}\left[1+\kappa(t)\right]\varepsilon^2(\boldsymbol{x}) + \boldsymbol{\nabla}\varepsilon \cdot \boldsymbol{\nabla}\Phi + \left[\frac{1}{2} - \kappa(t)\right]\sigma^2\right\}, \tag{4.1.48}$$

with the shear tensor and its scalar contraction defined by:

$$\sigma^2 = \sigma_{ij}\sigma^{ij}, \qquad \sigma_{ij} = \partial_i\partial_j\Phi - \frac{1}{3}\delta_{ij}\Delta\Phi.$$

The dimensionless function $\kappa(t)$ evolves very weakly with cosmic time and is accurately approximated by (Ref. [21]):

$$\kappa \simeq \frac{3}{14}\,\Omega_{\mathrm{m}}^{-2/63},$$

in a flat Universe with vanishing cosmological constant.

Application: Skewness

A direct diagnostic of non-linear gravitational dynamics is provided by the third central moment (or skewness) of the cosmic density field:

$$\left\langle \delta^3 \right\rangle = \left\langle \left(\delta^{(1)}\right)^3 \right\rangle + \left\langle \left(\delta^{(1)}\right)^2 \delta^{(2)} \right\rangle + \mathcal{O}(\varepsilon^5).$$

For strictly Gaussian primordial initial conditions, all odd moments vanish identically, so that $\left\langle (\delta^{(1)})^3 \right\rangle = 0$. Consequently, the emergence of a non-zero third moment serves as a direct indicator of non-linear gravitational clustering. Applying spatial filtering at scale R, the lowest-order non-vanishing contribution to the smoothed third moment evaluates to:

$$\left\langle \delta_R^3 \right\rangle = D^2(t)\int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3}\frac{\mathrm{d}^3\boldsymbol{k}'}{(2\pi)^3}\, P_\delta(k)P_\delta(k')\widehat{W}_R(k)\widehat{W}_R(k')\widehat{W}_R(|\boldsymbol{k}+\boldsymbol{k}'|)\, J(\boldsymbol{k},\boldsymbol{k}'), \tag{4.1.49}$$

where the non-linear coupling kernel is given by:

$$J(\boldsymbol{k},\boldsymbol{k}') = \frac{1}{2} + \kappa + \frac{k'}{k}\left(\frac{\boldsymbol{k}\cdot\boldsymbol{k}'}{kk'}\right) + \left(\frac{1}{2} - \kappa\right)\left(\frac{\boldsymbol{k}\cdot\boldsymbol{k}'}{kk'}\right)^2.$$

In the limit of an unfiltered field ($R \to 0$) and assuming a power-law spectrum $P(k) \propto k^n$, one obtains the reduced skewness parameter $S_3(0)$ (Ref. [21]):

$$S_3(0) \equiv \frac{\langle \delta^3 \rangle}{\langle \delta^2 \rangle^2} = 4 + 4\kappa - (n+3) \simeq \frac{34}{17}\left(\Omega_{\mathrm{m}}^{-2/63} - 1\right) - (n+3). \tag{4.1.50}$$

When the density field is smoothed with a spherical top-hat window of radius R, this relation receives a scale-dependent correction (Ref. [22]):

$$S_3(R) = S_3(0) + \frac{\mathrm{d}\ln\sigma_\delta^2(R)}{\mathrm{d}\ln R}, \tag{4.1.51}$$

where $\sigma_\delta(R)$ is the variance of the density field smoothed at scale R. Observational measurements of $S_3(R)$ across galaxy surveys (e.g., Ref. [23]) confirm that large-scale cosmic structure is fundamentally consistent with having grown out of Gaussian primordial seeds.

4.2 Gauge invariant cosmological perturbation theory

While Newtonian theory provides a faithful description of sub-Hubble perturbations during the late-time matter era, a rigorous general-relativistic framework is indispensable for tracking super-Hubble modes and causal horizons across the early Universe. The formulation developed in this section constitutes one of the foundations of modern theoretical cosmology, culminating in a relativistic generalization of the fluid equations (4.1.1), (4.1.2), and (4.1.5). This is needed as we do not merely want to study the dynamics of test particle but we wish to incorporate their backreaction into the picture as well.

Two distinct formalisms exist for treating relativistic cosmological perturbations. Here we adopt the coordinate-based approach in which an arbitrary parameterization of metric and fluid perturbations is introduced, from which gauge-invariant combinations are systematically constructed. Pioneered by Bardeen (Refs. [24, 25]), this formulation is the standard workhorse of modern cosmology and is reviewed extensively in Refs. [26–29]. An alternative, complementary approach employs the 1+3 covariant and gauge-invariant fluid formalism (Refs. [30–32]). Both routes yield identical physical predictions.

4.2.1 Perturbed space-time

Metric of the perturbed space-time

At linear order, the most general perturbed line element around a spatially flat or curved Friedmann–Lemaître geometry is parameterized in conformal time η by:

$$\mathrm{d}s^2 = a^2(\eta)\left[-(1+2A)\,\mathrm{d}\eta^2 + 2B_i\,\mathrm{d}x^i\,\mathrm{d}\eta + (\gamma_{ij}+h_{ij})\,\mathrm{d}x^i\,\mathrm{d}x^j\right], \tag{4.2.1}$$

where $A(\eta,\boldsymbol{x})$, $B_i(\eta,\boldsymbol{x})$, and $h_{ij}(\eta,\boldsymbol{x})$ represent small metric perturbations to be determined from the linearized Einstein equations. Since $\delta T_{\mu\nu} \neq 0$, we can have off shell scalar graviton degree of freedom. Decomposing the full physical metric into the background Friedmann–Lemaître tensor $\bar{g}_{\mu\nu}$ and its perturbation:

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \delta g_{\mu\nu}, \tag{4.2.2}$$

the inverse metric to linear order is given by:

$$g^{\mu\nu} = \bar{g}^{\mu\nu} + \delta g^{\mu\nu}, \qquad \delta g^{\mu\nu} = -\bar{g}^{\mu\sigma}\bar{g}^{\nu\lambda}\delta g_{\sigma\lambda}. \tag{4.2.3}$$

For the sake of perturbation theory, we always expland the full metric around some fixed background irrespective of the choice of coordinate. The choice of coordinate manifests itself as transformation law for the perturbation or as gauge invariance.

We denote the trace of the spatial metric perturbations with respect to the background 3-metric γ_{ij} by:

$$h \equiv h_{ij}\gamma^{ij},$$

and ∇_μ denotes the full four-dimensional covariant derivative. The complete explicit expressions for the perturbed Christoffel symbols, Riemann, Ricci, and Einstein tensors, and Ricci scalar are cataloged in Appendix C.

SVT decomposition

When we perform coordinate transformation on the full metric which happens to be the isometry of the background metric, we observe that the perturbation transforms as tensor under these. As such one can decompose the perturbation into the irreducible representation (fundamental building blocks) of this isometry group. Following Bardeen (Ref. [24]), we perform a scalar-vector-tensor (SVT) decomposition of all spatial perturbation fields, exactly as introduced in the geometric foundations of Chapter 1. This decomposition generalizes the Helmholtz theorem of vector calculus, which partitions any vector field into the gradient of a scalar potential and a divergence-free (solenoidal) vector:

$$B_i = D_i B + \bar{B}_i \qquad \text{with} \qquad D_i \bar{B}^i = 0. \tag{4.2.4}$$

Here D_i is the covariant derivative associated with spatil part of background metric γ_{ij}. For a velocity field, B represents the longitudinal scalar potential while $\bar{B}_i$ encodes rotational vorticity. Thus, the 3 components of the vector have been split into 1 scalar mode (B) and 2 vector modes ($\bar{B}_i$).

Similarly, any symmetric rank-2 spatial tensor field h_{ij} can be decomposed into scalar, vector, and trace-free transverse tensor components:

$$h_{ij} = 2C\gamma_{ij} + 2D_i D_j E + 2D_{(i}\bar{E}_{j)} + 2\bar{E}_{ij} \qquad \text{with} \qquad D_i \bar{E}^{ij} = 0, \quad \bar{E}^i{}_i = 0. \tag{4.2.5}$$

Equation (4.2.3) implies that the indices of these quantities are raised and lowered with respect to the unperturbed spatial metric γ_{ij} (for example, $\bar{B}^i = \gamma^{ij}\bar{B}_j$). We adopt the convention in which any 'barred' quantity is divergenceless (and traceless if it has two indices). The 6 components of h_{ij} have thus been split into 2 scalar (C and E), 2 vector ($\bar{E}_j$), and 2 tensor ($\bar{E}_{ij}$) components.

Note that the scalar potentials B and E are not uniquely defined without boundary conditions. In fact, B is a solution of $\Delta B = D^i B_i$ (the Laplacian being defined by $\Delta \equiv D_i D^i$), whose solution is unique only if some boundary conditions are imposed.

In total, the 10 degrees of freedom of the metric have thus been decomposed as:

- 4 scalars: A, B, C, and E, corresponding to 4 degrees of freedom;
- 2 vectors: $\bar{B}_i$ and $\bar{E}_i$, corresponding to $2 \times (3 - 1) = 4$ degrees of freedom;
- 1 tensor: $\bar{E}_{ij}$, corresponding to $6 - 1 - 3 = 2$ degrees of freedom.

The advantages of this decomposition lie in the fact that, to leading order, the three types of perturbations are decoupled and can thus be studied separately.

The gauge problem

To introduce the perturbed space-time metric, we have made the hypothesis (4.2.2) that the metric $g_{\mu\nu}$ was 'close' to that of the Friedmann–Lemaître background $\bar{g}_{\mu\nu}$.

In field theory, space-time is usually fixed and once the system of coordinates is chosen, one can define the perturbation of any quantity $Q(\boldsymbol{x}, t)$ (pullback from the perturbed manifold to background manifold) as:

$$\delta Q(\boldsymbol{x}, t) \equiv \phi^*(Q(\boldsymbol{x}, t)) - \bar{Q}(\boldsymbol{x}, t) = Q(\boldsymbol{x}, t) - \bar{Q}(\boldsymbol{x}, t),$$

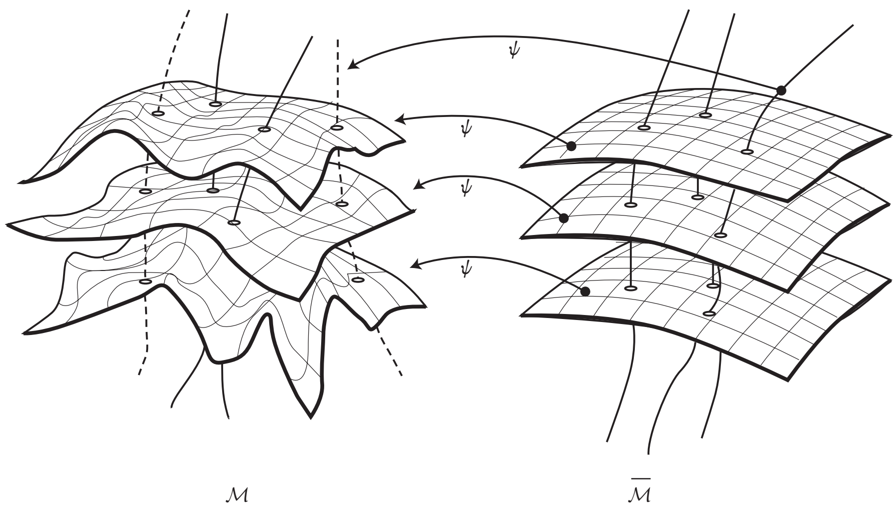


Figure 4.2: Any perturbed quantity is defined via a mapping between the Friedmann–Lemaître space-time, $\bar{\mathcal{M}}$, and the perturbed space-time $\mathcal{M}$.

where $\bar{Q}(\boldsymbol{x}, t)$ represents the unperturbed configuration. At any point of space-time, the quantity $Q(\boldsymbol{x}, t)$ can be compared with $\bar{Q}(\boldsymbol{x}, t)$ at the same point.

In general relativity, there exists an important difference with this standard approach. Indeed, we would like to compare two different space-times and make the hypothesis that $(\mathcal{M}, g_{\mu\nu})$ is 'close' to a Friedmann–Lemaître Universe $(\bar{\mathcal{M}}, \bar{g}_{\mu\nu})$. In making this comparison, there is some arbitrary freedom related to the way of identifying the points of these two space-times. Indeed, an isomorphism ψ should be introduced to relate the points of the two space-times (see Fig. 4.2). There is actually no 'natural' identification between the two spaces, and there is therefore a freedom of choice in the identification of the points in each space.[1] This implies that there exist some unphysical degrees of freedom related to the choice of the coordinate systems on the two manifolds.

> What we're doing in perturbation theory is that we are building a discription of the perturbed spacetime by using a set of tensor fields, i.e. the perturbation, which live on the background manifold. Since we use pullback on the full metric to define the perturbation on background manifold. The non uniqueness of such map manifests itself as the gauge invariance of the problem and the full metric ends up becoming a tensor field on the background manifold.

The decomposition (4.2.1) implicitly assumes that a system of coordinates has been chosen in the perturbed space. Any change of coordinates will modify the form of the metric coefficients. To illustrate the effect of such a change of coordinates, let us consider an unperturbed Friedmann–Lemaître space-time, as in Fig. 4.2, and let us perform an arbitrary space- and time-dependent change of coordinates $x^i \to y^i = x^i - \xi^i(x^j, \eta)$. We then get:

$$\mathrm{d}s^2 = a^2(\eta)\left[-\mathrm{d}\eta^2 + 2\xi_i'\,\mathrm{d}y^i\,\mathrm{d}\eta + (\gamma_{ij} + 2D_{(i}\xi_{j)})\,\mathrm{d}y^i\,\mathrm{d}y^j\right].$$

There appear two artificial terms $B_i = \xi_i'$ and $E_i = \xi_i$ that do not correspond to metric fluctuations, since this space is still a Friedmann–Lemaître space, but to a choice of the system of coordinates. The dependence of the quantities that are intrinsic to the manifold in which the

[1] Since the perturbation depends on both physical and unphysical dof.

perturbations evolve should thus be separated from the artificial ones, related to the arbitrary choice of a particular coordinate system.

Gauge invariant variables

In order to extract the physical degrees of freedom, let us consider an active transformation of the coordinate system defined by an arbitrary vector field ξ. The coordinates of any point change according to:

$$x^\mu \to x^\mu - \xi^\mu, \tag{4.2.6}$$

where the displacement ξ^μ is decomposed into 2 scalar degrees of freedom (T and L) and 2 vector degrees of freedom ($\bar{L}^i$, which is divergenceless $D_i\bar{L}^i = 0$) as:

$$\xi^0 = T, \qquad \xi^i = L^i = D^i L + \bar{L}^i. \tag{4.2.7}$$

Under this change of coordinates, the metric transforms as:

$$g_{\mu\nu} \to g_{\mu\nu} + \pounds_\xi g_{\mu\nu}.$$

For first order perturbation theory, we always expand the metric about the same fixed background metric.

$$\begin{aligned} g'_{\mu\nu} &= g_{\mu\nu} + \pounds_\xi g_{\mu\nu} && (1)\\ \bar{g}_{\mu\nu} + \delta g'_{\mu\nu} &= \bar{g}_{\mu\nu} + \delta g_{\mu\nu} + \pounds_\xi(\bar{g}_{\mu\nu} + \delta g'_{\mu\nu}) && (2)\\ \delta g'_{\mu\nu} &= \delta g_{\mu\nu} + \pounds_\xi \bar{g}_{\mu\nu} + \mathcal{O}\left(\xi^2\right) && (3) \end{aligned}$$

To first order in the perturbations, this is:

$$\delta g_{\mu\nu} \to \delta g_{\mu\nu} + \pounds_\xi \bar{g}_{\mu\nu} = \delta g_{\mu\nu} + 2\nabla_{(\mu}\xi_{\nu)},$$ [2]

Rather than evaluating the covariant derivative of the arbitrary vector on the background manifold, one can utilise the following to quickly evaluate how the perturbation transforms:

$$\begin{aligned} \pounds_\xi \bar{g}_{\mu\nu} &= \xi^\sigma \nabla_\sigma \bar{g}_{\mu\nu} + \bar{g}_{\sigma\nu}(\nabla_\mu \xi^\sigma) + \bar{g}_{\mu\sigma}(\nabla_\nu \xi^\sigma) && (1)\\ &= \xi^\sigma \partial_\sigma \bar{g}_{\mu\nu} + \bar{g}_{\sigma\nu}(\partial_\mu \xi^\sigma) + \bar{g}_{\mu\sigma}(\partial_\nu \xi^\sigma) && (2) \end{aligned}$$

which in turn implies that the metric perturbation variables (4.2.1) transform as:

$$\begin{aligned} A &\to A + T' + \mathcal{H}T, && (4.2.9)\\ B_i &\to B_i - D_i T + L'_i, \\ h_{ij} &\to h_{ij} + D_i L_j + D_j L_i + 2\mathcal{H}T\gamma_{ij}. && (4.2.10) \end{aligned}$$

[2]For completeness, we give the expressions of the Lie derivatives. They are easily obtained from the expressions of the Christoffel symbols of the Friedmann–Lemaître space-time and from $\xi_\mu = a^2(-T, L_i)$. They are given by

$$\begin{aligned} \pounds_\xi \bar{g}_{00} &= \xi^\sigma \partial_\sigma \bar{g}_{00} + \bar{g}_{\sigma 0}\partial_0 \xi^\sigma + \bar{g}_{0\sigma}\partial_0\xi^\sigma = T\,\partial_0(-a^2) + 2(-a^2)T' = -2aa'T - 2a^2T' = -2a^2(T' + \mathcal{H}T),\\ \pounds_\xi \bar{g}_{0i} &= \xi^\sigma \partial_\sigma \bar{g}_{0i} + \bar{g}_{\sigma i}\partial_0\xi^\sigma + \bar{g}_{0\sigma}\partial_i\xi^\sigma = a^2\gamma_{ji}\partial_0 L^j - a^2\partial_i T = a^2(L'_i - \partial_i T),\\ \pounds_\xi \bar{g}_{ij} &= \xi^\sigma \partial_\sigma \bar{g}_{ij} + \bar{g}_{\sigma j}\partial_i\xi^\sigma + \bar{g}_{i\sigma}\partial_j\xi^\sigma = T(a^2\gamma_{ij})' + a^2\gamma_{kj}\partial_i L^k + a^2\gamma_{ik}\partial_j L^k\\ &= 2a^2\mathcal{H}T\gamma_{ij} + a^2(D_iL_j + D_jL_i) = a^2(D_iL_j + D_jL_i + 2\mathcal{H}T\gamma_{ij}). \end{aligned}$$

Using the scalar-vector-tensor decomposition (4.2.4) and (4.2.5), we get:

$$\boxed{\begin{array}{lll} A \to A + T' + \mathcal{H}T, & \bar{B}^i \to \bar{B}^i + \bar{L}^{i\prime}, & \bar{E}_{ij} \to \bar{E}_{ij}, \\ B \to B - T + L', & \bar{E}^i \to \bar{E}^i + \bar{L}^i, & \\ C \to C + \mathcal{H}T, & & \\ E \to E + L, & & \end{array}} \tag{4.2.11}$$

respectively, for the S, V and T modes. These transformations are similar to the gauge transformations encountered, for instance, in electromagnetism. Therefore, one may reduce the arbitrariness by imposing some additional conditions on the variables: in other words, and again as in electromagnetism, one needs to fix the gauge before doing any actually relevant calculation.

As stressed in the previous section, it is interesting to define gauge-independent quantities. To do so, notice that some combinations of the previous quantities do not depend on L^i and T, for instance:

$$\Psi \equiv -C - \mathcal{H}(B - E'), \tag{4.2.12}$$
$$\Phi \equiv A + \mathcal{H}(B - E') + (B - E')', \tag{4.2.13}$$
$$\bar{\Phi}^i \equiv \bar{E}^{i\prime} - \bar{B}^i, \tag{4.2.14}$$
$$\bar{E}^{ij}. \tag{4.2.15}$$

We initially had 10 perturbation variables $(A, B, C, E, \bar{E}^i, \bar{B}^i, \bar{E}^{ij})$ and 4 gauge degrees of freedom $(T, L, \bar{L}^i)$ that can be absorbed into the definition of the gauge invariant variables, then described by $10 - 4 = 6$ degrees of freedom parameterized, for instance, by $\Psi, \Phi, \bar{\Phi}^i$ and $\bar{E}^{ij}$.[3] These quantities can be considered as the 'real' space-time perturbations in the sense that they cannot be removed by any change of coordinates.

Other gauge invariant quantities could have been constructed, such as, for instance:

$$X = A - C - \left(\frac{C}{\mathcal{H}}\right)', \tag{4.2.16}$$

but they can always be expressed in terms of the two functions Φ and Ψ. For instance, in the case of X defined just above:

$$X = \Psi + \Phi + \left(\frac{\Psi}{\mathcal{H}}\right)', \tag{4.2.17}$$

as can straightforwardly be checked.

Stewart–Walker lemma

Just as the metric transforms as (4.2.8) under a change of coordinates of the form (4.2.6), any scalar quantity δQ transforms as:

$$\delta Q \to \delta Q + \pounds_\xi \bar{Q},$$

to first order in perturbations. Since each vector field ξ generates a gauge transformation, we can conclude that the only gauge invariant quantities are the ones for which:

$$\pounds_\xi \bar{Q} = 0 \qquad \forall \xi.$$

This result is known under the name of the **Stewart–Walker lemma** (Ref. [33]).

Note an important corollary of this result. Since all relativistic equations are covariant by definition, they can always be written in the form $Q = 0$, where Q is a tensor field. It is thus always possible, to first order in perturbations, to write all the relevant equations in terms of gauge invariant quantities only. Such a treatment would ensure that no gauge, i.e. unphysical, degree of freedom is misleadingly used.

[3] Except $\bar{E}^{ij}$, rest are off shell graviton degree of freedom. It only happens that at linear order one is able to isolate all the graviton degrees of freedom whereas at higher order they are mixed with each other in complicated fashion.

4.2.2 Description of matter

In this section, we introduce various quantities that come into the description of a perfect fluid for which the energy-momentum tensor in an unperturbed space is of the form:

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + P g_{\mu\nu},$$

as was seen in Chapter 3.

The energy-momentum tensor of a perturbed fluid

The energy-momentum tensor of this perturbed fluid takes the general form:

$$\delta T_{\mu\nu} = (\delta\rho + \delta P)\bar{u}_\mu \bar{u}_\nu + \delta P\, \bar{g}_{\mu\nu} + 2(\rho + P)\bar{u}_{(\mu}\delta u_{\nu)} + P\, \delta g_{\mu\nu} + a^2 P \pi_{\mu\nu}, \tag{4.2.18}$$

where $u^\mu = \bar{u}^\mu + \delta u^\mu$ is the four-velocity of a comoving observer, satisfying $u_\mu u^\mu = -1$. The normalization condition of $\bar{u}^\mu$ (to zeroth order) provides the solution:

$$\bar{u}^\mu = a^{-1}\delta^\mu_0, \qquad \bar{u}_\mu = -a\delta^0_\mu,$$

and since the norm of $\bar{u}^\mu + \delta u^\mu$ should also be equal to -1, we infer that $2\bar{u}^\mu \delta u_\mu + \delta g_{\mu\nu}\bar{u}^\mu \bar{u}^\nu = 0$, and thus that $\delta u_0 = -Aa$. We then write $\delta u^i \equiv v^i/a$, so that:

$$\delta u^\mu = a^{-1}(-A, v^i), \qquad \delta u_\mu = a(-A, v_k + B_k), \tag{4.2.19}$$

and we decompose v_i into a scalar and a vector part according to:

$$v_i = D_i v + \bar{v}_i. \tag{4.2.20}$$

In the decomposition (4.2.18), $\pi_{\mu\nu}$ is the anisotropic stress tensor and characterizes the difference between the perturbed fluid and a perfect fluid. It can be chosen to be traceless ($g^{\mu\nu}\pi_{\mu\nu} = 0$), since its trace can be absorbed into a redefinition of the isotropic pressure, δP. It is also symmetric, since the energy-momentum tensor is, and can be chosen orthogonal to u^μ, i.e. $u^\mu \pi_{\mu\nu} = 0$. This implies that, without lack of generality, one can set $\pi_{00} = \pi_{0i} = 0$.

The anisotropic stress tensor then consists only of a spatial part, which can be decomposed into scalar, vector and tensor parts as:

$$\pi_{ij} = \Delta_{ij}\bar{\pi} + D_{(i}\bar{\pi}_{j)} + \bar{\pi}_{ij}, \tag{4.2.21}$$

> This is part of the pressure/stress that depends on the direction which is suppressed by scattering/collision.

where the operator Δ_{ij} is defined as:

$$\Delta_{ij} \equiv D_i D_j - \frac{1}{3}\gamma_{ij}\Delta. \tag{4.2.22}$$

We infer from (4.2.19), (4.2.20) and (4.2.21) that the components of the perturbation of the energy-momentum tensor (4.2.18) are:

$$\begin{aligned}
\delta T_{00} &= (\delta\rho + \delta P)(-a)^2 + \delta P(-a^2) + 2(\rho + P)(-a)(-aA) + P(-2a^2 A) + a^2 P \times 0 \\
&= \rho a^2(\delta + 2A), \qquad (4.2.23)\\
\delta T_{0i} &= (\rho + P)(-a)\left[a(D_i v + \bar{v}_i)\right] + (\rho + P)(aD_i B + a\bar{B}_i)(-a) + P\left[a^2(D_i B + \bar{B}_i)\right]
\end{aligned}$$

$$= -\rho a^2 \left[(1+w)(D_i v + \bar{v}_i) + D_i B + \bar{B}_i\right], \tag{4.2.24}$$

$$\delta T_{ij} = \underbrace{\delta(\rho+P)u_i u_j + (\rho+P)(u_i\delta u_j + \delta u_i u_j)}_{=0} + \delta P\, \bar{g}_{ij} + P\,\delta g_{ij} + a^2 P\pi_{ij}$$

$$= Pa^2\left[h_{ij} + \frac{\delta P}{P}\gamma_{ij} + \pi_{ij}\right], \tag{4.2.25}$$

where we recall that $\delta \equiv \delta\rho/\bar{\rho}$ is the density contrast.

It will be useful to introduce the entropy perturbation, Γ, defined by the relation:

$$\delta P = c_{\rm s}^2\delta\rho + P\Gamma, \tag{4.2.26}$$

which takes the form:

$$w\Gamma = \frac{1}{\rho}\left(\delta P - c_{\rm s}^2\delta\rho\right). \tag{4.2.27}$$

For an adiabatic perturbation, $\delta P/\delta\rho = c_{\rm s}^2$ and hence $\Gamma = 0$ and we introduce the notation used in the literature:

$$\delta P_{\rm nad} = P\Gamma, \tag{4.2.28}$$

where the index 'nad' stands for non-adiabatic.

Gauge invariant quantities

Upon performing a change of coordinates of the form (4.2.6), scalar and vectors transform as:

$$\delta Q \to \delta Q + \pounds_\xi \bar{Q}, \qquad \pounds_\xi \bar{Q} = \xi^\alpha\partial_\alpha\bar{Q} = T\bar{Q}',$$

and

$$\delta u^\mu \to \delta u^\mu + \pounds_\xi \bar{u}^\mu, \qquad \pounds_\xi \bar{u}^\mu = \xi^\alpha\partial_\alpha\bar{u}^\mu - \bar{u}^\alpha\partial_\alpha\xi^\mu,$$

so that it follows that, after SVT decomposition, $\delta P, \delta\rho, v$ and $\bar{v}_i$ transform as:

$$\boxed{\begin{array}{ll} \delta\rho \to \delta\rho + \rho' T, & v \to v - L', \\ \delta P \to \delta P + P' T, & \bar{v}_i \to \bar{v}_i - \bar{L}_i'. \end{array}} \tag{4.2.29}$$

The quantities $\bar{\pi}, \bar{\pi}_i$ and $\bar{\pi}_{ij}$ are gauge invariant from the Stewart–Walker lemma, since the unperturbed part of the anisotropic stress tensor must vanish to be compatible with the perfect fluid hypothesis imposed by the symmetries of the Friedmann–Lemaître space-time. One can also check that, as expected again from the Stewart–Walker lemma, Γ is gauge invariant since $\Gamma \to \Gamma + (P' - c_{\rm s}^2\rho')T = \Gamma$.

As in the case of the metric perturbations, one can likewise define the gauge invariant quantities associated with the quantities (4.2.29). Different choices are possible, and we define:

$$\delta^{\rm N} = \delta + \frac{\rho'}{\rho}(B - E'), \tag{4.2.30}$$

$$\delta^{\rm F} = \delta - \frac{\rho'}{\rho}\frac{C}{\mathcal{H}}, \tag{4.2.31}$$

$$\delta^{\rm C} = \delta + \frac{\rho'}{\rho}(v + B), \tag{4.2.32}$$

$$V = v + E', \tag{4.2.33}$$

$$\overline{V}_i = \bar{v}_i + \bar{B}_i, \tag{4.2.34}$$

$$\overline{W}_i = \bar{v}_i + \bar{E}_i'. \tag{4.2.35}$$

Even though these are gauge invariant density contrast but they are not the same object. They all converge to δ in the subhorizon limit. But relativistically speaking they correspond to perturbations measured in different hypersurfaces.

The pressure perturbations $\delta P^{\mathrm{N}}, \delta P^{\mathrm{F}}$ and δP^{C} are defined in an identical way. The different gauge invariant variables are related to each other and one can check that:

$$\delta^{\mathrm{F}} = \delta^{\mathrm{N}} + \frac{\rho'}{\rho}\frac{\Psi}{\mathcal{H}}, \qquad \delta^{\mathrm{C}} = \delta^{\mathrm{N}} + \frac{\rho'}{\rho}V, \qquad \overline{W}_i = \overline{V}_i + \bar{\Phi}_i. \tag{4.2.36}$$

In all these relations, we recall that $\delta = \delta\rho/\rho$ is the density contrast.

4.2.3 Choosing a gauge

From the Stewart–Walker lemma, we know that any perturbation equation can be written in terms of gauge invariant variables only. Indeed, such an equation always takes the form $Q = 0$ and is, by construction, linear in the first-order perturbation variables. One method for obtaining these gauge invariant equations is thus to write the equations in an arbitrary gauge and to regroup the terms to make the gauge invariant variables appear.

A more direct and physically transparent approach is to choose a specific coordinate system from the very beginning such that the metric and matter perturbations in that gauge reduce identically to gauge invariant variables. For instance, fixing the longitudinal gauge by the constraints $E = 0$ and $\overline{B}_i = 0$ directly implies $C = -\Psi$, $A = \Phi$, and so on. The governing perturbation equations are then derived immediately in terms of gauge invariant quantities. Once these dynamical relations are established, one can readily transform to any other gauge or reinstate the original gauge-dependent perturbation variables via the transformation relations (4.2.11).

A crucial conceptual point deserves special emphasis. Although perturbative calculations can be performed in any gauge of choice, connecting theoretical predictions with observational measurements generally requires a gauge transformation to the coordinate system of the observer. Gauge invariant quantities do not possess an intrinsic physical interpretation in the abstract; rather, their physical significance becomes apparent only within a gauge where the metric or fluid variables coincide with the gauge invariant definitions. For example, the gauge invariant variable δ^{N} can be directly interpreted as a fractional density fluctuation only when evaluated in the reference frame of an observer tied to the longitudinal gauge.

Furthermore, by the equivalence principle, all coordinate systems are locally indistinguishable. Consequently, deviations among different gauge representations manifest only on large spatial scales—more precisely, on scales exceeding a characteristic physical threshold set by the cosmological Hubble radius $\mathcal{H}^{-1}$, as we will see shortly.

Finally, a note of caution regarding perturbation theory: if a supposedly small perturbation variable becomes large in a given gauge while remaining small in another, this does not necessarily signify a physical breakdown of linear perturbation theory, nor does it prove that the perturbations remain linear. Instead, it typically indicates that the coordinate displacement relating the two slicings is no longer infinitesimal. In such circumstances, a higher-order perturbation calculation is required: if the second-order corrections remain small relative to the first order across all gauges, the pathology was merely an artifact of the initial coordinate system. Such checks are indispensable before drawing definitive physical conclusions.

Before deriving the general-relativistic evolution equations, we summarize the properties of several widely used cosmological gauges.

Newtonian or longitudinal gauge

The Newtonian (or longitudinal) gauge is defined by demanding that the scalar sector of the spatial metric perturbations is strictly diagonal, namely:

$$B = 0 \quad \text{and} \quad E = 0. \tag{4.2.37}$$

Under these two constraints, the vector sector cannot be simultaneously diagonalized; one conventionally eliminates the remaining vector gauge freedom by imposing additional 2 constraint:

$$\overline{B}_i = 0. \tag{4.2.38}$$

This fixes the gauge completely ($2 + 2 = 4$ constraints). Starting from an arbitrary coordinate system, one reaches the longitudinal gauge via the gauge transformation (4.2.11) with parameters:

$$T = B - E', \quad L = -E \quad \text{and} \quad \overline{L}'_i = -\overline{B}_i. \tag{4.2.39}$$

In this coordinate system, the cosmological expansion appears isotropic and $T, L, \bar{L}_i$ are completely fixed. The scalar potential Ψ characterizes the intrinsic spatial curvature perturbation on constant-time hypersurfaces [see (C.31)], while Φ plays the role of the relativistic gravitational potential entering the Poisson equation. On sub-Hubble scales, Φ reduces to the classical Newtonian gravitational potential, which motivates the name of the gauge.

In the longitudinal gauge, the remaining perturbed quantities reduce to the corresponding gauge invariant variables:

$$A = \Phi, \quad C = -\Psi, \quad \delta = \delta^{\mathrm{N}}, \quad \delta P = \delta P^{\mathrm{N}}, \quad v = V \quad \text{and} \quad \bar{v}_i = \overline{V}_i. \tag{4.2.40}$$

Flat-slicing gauge

The flat-slicing (or spatially flat) gauge is defined by requiring that the scalar curvature perturbation of the spatial hypersurfaces vanishes identically. From the expression for the spatial Ricci scalar, this condition imposes:

$$C = 0, \quad E = 0 \quad \text{and} \quad \overline{E}_i = 0. \tag{4.2.41}$$

This is total of $1 + 1 + 2 = 4$ constraints. As with the longitudinal gauge, the flat-slicing gauge is completely fixed. The gauge parameters required to pass from an arbitrary gauge to this frame are:

$$T = -\frac{C}{\mathcal{H}}, \quad L = -E \quad \text{and} \quad \overline{L}_i = -\overline{E}_i. \tag{4.2.42}$$

In this gauge, the remaining perturbed quantities are related to the gauge invariant variables by:

$$A = X, \quad E' = -\frac{\Psi}{\mathcal{H}}, \quad \delta = \delta^{\mathrm{F}}, \quad \delta P = \delta P^{\mathrm{F}}, \quad v = V \quad \text{and} \quad \bar{v}_i = \overline{W}_i. \tag{4.2.43}$$

A key property of this gauge is that the fluid conservation equations take an identical mathematical form to their Newtonian counterparts.

Synchronous or Gauss gauges

Synchronous gauges are defined by the coordinate conditions:

$$\delta g_{0\mu} = 0 \implies A = 0 \quad \text{and} \quad B_i = 0. \tag{4.2.44}$$

This is still $1 + 3 = 4$ constraints but as we will see some residual gauge symmetry survives. In such coordinates, only the spatial metric components are perturbed. The worldlines $x^i = \text{const.}$ are timelike geodesics orthogonal to constant-time hypersurfaces, and the proper time of comoving observers corresponds directly to cosmic coordinate time. This system therefore represents a foliation of space-time by freely falling observers.

While this geometric picture makes synchronous coordinates intuitive and widely used, the gauge suffers from notorious ambiguities: it is not uniquely fixed. Residual coordinate transformations of the form $t \to t' = f(t)$ or $x^i \to y^i = f^i(x^j)$, involving time or space separately, leave the constraints (4.2.44) invariant. The transformation from an arbitrary gauge to a synchronous gauge is given by:

$$T = -\frac{1}{a}\int Aa\,\mathrm{d}\eta + \frac{C_T}{a}, \quad L = \int (T - B)\,\mathrm{d}\eta + C_L \quad \text{and} \quad \overline{L}^i = -\int \overline{B}^i\,\mathrm{d}\eta + \overline{C}^i_L, \tag{4.2.45}$$

where C_T and $C^i_L = (C_L, \overline{C}^i_L)$ are four arbitrary integration functions that depend solely on the spatial coordinates x^i. Physically, C_T reflects the freedom in choosing the local origin of proper time for each observer, whereas C^i_L corresponds to an arbitrary relabeling of spatial coordinates on constant-time slices. This remaining freedom produces unphysical “gauge modes” in the solutions of the perturbation equations, which can be mistakenly interpreted as physical instabilities. Furthermore, because the gauge is not completely fixed, variables evaluated in synchronous gauge cannot always be uniquely translated into gauge invariant observables.

In the synchronous gauge, the residual spatial metric perturbations are conventionally parameterized by:

$$h \equiv 6C + 2\Delta E \quad \text{and} \quad \eta = -C. \tag{4.2.46}$$

Comoving gauge

The comoving gauge is defined such that the coordinate frame follows the overall bulk motion of the cosmic fluid, setting the momentum flux to zero:

$$\delta T^0{}_i = 0 \implies \rho(1+w)[D_i(v+B) + \bar{v}_i + \bar{B}_i] = 0. \tag{4.2.47}$$

In the scalar sector, one may impose $v + B = 0$ to fix the temporal gauge freedom associated with T. However, the comoving gauge condition is not sufficient to completely fix the gauge, since the spatial gauge freedom associated with L_i remains unfixed. An additional condition is therefore required to fix L_i. In the vector sector, however, $\bar{v}_i + \bar{B}_i$ is gauge invariant and hence cannot be set to zero by a gauge transformation. Thus, imposing $\bar{v}_i + \bar{B}_i = 0$ is an additional physical restriction, corresponding to a fluid with vanishing vorticity, rather than a gauge choice.

In a cosmological medium consisting of multiple components, a comoving gauge can be defined with respect to any individual fluid species (e.g., baryons or radiation) or with respect to the total matter fluid. Unless specified otherwise, “comoving gauge” will refer to the frame comoving with the total energy-momentum tensor.

The condition (4.2.47) demands $v_i + B_i = 0$. While one can always find coordinates in which the scalar velocity vanishes (since for potential flows, Frobenius’s theorem prevents flow lines from crossing, allowing coordinates to follow fluid elements), this is not possible for the vector component corresponding to vorticity, which cannot be eliminated by coordinate choices. Therefore, the comoving gauge must be supplemented by:

$$E = 0 \quad \text{and} \quad \overline{E}_i = 0, \tag{4.2.48}$$

to completely fix L and L_i. This gauge shares an important property with the flat-slicing gauge: the relativistic Poisson equation takes a form identical to that in the Newtonian framework.

4.2.4 Einstein equations: derivation

The linearized Einstein field equations governing cosmological perturbations take the form:

$$\delta G^{\mu}{}_{\nu} = \kappa\, \delta T^{\mu}{}_{\nu}.$$

All explicit expressions for the perturbed geometric tensors are collected in Appendix C. In what follows, we examine tensor modes first, proceed to vector modes, and conclude with the scalar perturbations.

Decomposition of 3-dimensional vectors and tensors

Vectors In Fourier space, any 3-vector field V_i can be uniquely decomposed into longitudinal and transverse parts:

$$V_i = k_i V + \overline{V}_i, \quad \text{with} \quad k^i \overline{V}_i = 0, \tag{4.2.49}$$

such that $\overline{V}_i$ lies in the 2-dimensional subspace orthogonal to the wavevector k_i. Thus, V_i is partitioned into one scalar degree of freedom (V) and two transverse vector modes ($\overline{V}_i$). Choosing an orthonormal basis $\{e_i^1, e_i^2\}$ for the 2D plane perpendicular to k_i, satisfying:

$$e_i^a k_j \gamma^{ij} = 0, \quad e_i^a e_j^b \gamma^{ij} = \delta^{ab} \qquad (a, b \in \{1, 2\}),$$

which is defined up to a rotation around the wavevector k_i, the transverse vector can be expanded as:

$$\overline{V}_i(k_i, \eta) = \sum_{a=1,2} V_a(\hat{k}_i, \eta)\, e_i^a(\hat{k}_i). \tag{4.2.50}$$

This decomposition defines two degrees of freedom V_a, which depend on $\hat{k}_i = k_i/k$. The transverse projector onto this subspace is:

$$P_{ij} \equiv e_i^1 e_j^1 + e_i^2 e_j^2 = \gamma_{ij} - \hat{k}_i \hat{k}_j, \tag{4.2.51}$$

which satisfies $P^i{}_j P^j{}_k = P^i{}_k$, $P^i{}_j k^j = 0$, and $P^{ij}\gamma_{ij} = 2$. Consequently, any vector field can be decomposed as:

$$V_i = (\hat{k}^j V_j)\hat{k}_i + P^j{}_i V_j, \tag{4.2.52}$$

providing a direct scalar-vector projection.

Tensors Similarly, any 3-dimensional symmetric rank-2 tensor field V_{ij} admits an irreducible SVT decomposition:

$$V_{ij} = T\gamma_{ij} + \Delta_{ij} S + 2D_{(i}\overline{V}_{j)} + 2\overline{V}_{ij}, \tag{4.2.53}$$

where $\Delta_{ij} \equiv D_i D_j - \frac{1}{3}\Delta\gamma_{ij}$, and:

$$D_i \overline{V}^i = 0, \quad \overline{V}^i{}_i = 0 = D_i \overline{V}^{ij}. \tag{4.2.54}$$

The symmetric tensor $\overline{V}_{ij}$ is transverse and trace-free, possessing two independent physical degrees of freedom corresponding to the two polarization states of gravitational waves:

$$\overline{V}_{ij}(k_i, \eta) = \sum_{\lambda=+,\times} V_\lambda(k_i, \eta)\, \varepsilon_{ij}^{\lambda}(\hat{k}_i), \tag{4.2.55}$$

where the polarization tensors are defined by:

$$\varepsilon_{ij}^{\lambda} = \frac{e_i^1 e_j^1 - e_i^2 e_j^2}{\sqrt{2}}\delta_+^{\lambda} + \frac{e_i^1 e_j^2 + e_i^2 e_j^1}{\sqrt{2}}\delta_\times^{\lambda}. \tag{4.2.56}$$

These polarization tensors are traceless ($\varepsilon^{\lambda}_{ij}\gamma^{ij} = 0$), transverse ($\varepsilon^{\lambda}_{ij}k^i = 0$), and mutually orthonormal ($\varepsilon^{\lambda}_{ij}\varepsilon^{ij}_{\mu} = \delta^{\lambda}_{\mu}$).

Introducing the transverse-traceless projection operator:

$$\Lambda^{ab}_{ij} = P^a{}_i P^b{}_j - \frac{1}{2}P_{ij}P^{ab},$$

and the trace-free longitudinal projector:

$$T^j{}_i = \hat{k}_i\hat{k}^j - \frac{1}{3}\delta^j{}_i,$$

the full SVT components of V_{ij} are extracted according to:

$$V_{ij} = \left(\frac{1}{3}V_{ab}\gamma^{ab}\right)\gamma_{ij} + \left(\frac{3}{2}V_{ab}T^{ab}\right)T_{ij} + 2\hat{k}_{(i}\left[P^a{}_{j)}\hat{k}^b V_{ab}\right] + \Lambda^{ab}_{ij}V_{ab}. \tag{4.2.57}$$

In this decomposition, V_{ij} is separated into two scalar degrees of freedom (T and S), two vector degrees of freedom ($\overline{V}_i$), and two tensor degrees of freedom ($\overline{V}_{ij}$). Any vector equation $V_i = 0$ can therefore be projected along $\hat{k}_i$ (scalar) and $P^i{}_j$ (vector), while any tensor equation $V_{ij} = 0$ can be projected along γ_{ij} (trace scalar), T^{ij} (longitudinal scalar), $P^i{}_l\hat{k}^j$ (vector), and Λ^{ij}_{ab} (tensor).

4.2.5 Einstein equations: SVT decomposition

Tensor modes

Tensor perturbations are gauge invariant by construction. Projecting the perturbed Einstein equations with the transverse-traceless operator,

$$\Lambda^{ij}_{kl}\,\delta G_{ij} = \kappa\,\Lambda^{ij}_{kl}\,\delta T_{ij},$$

yields a single dynamical equation governing gravitational waves:

$$\boxed{\overline{E}''_{kl} + 2\mathcal{H}\overline{E}'_{kl} + (2K - \Delta)\overline{E}_{kl} = \kappa a^2 P\overline{\pi}_{kl}.} \tag{4.2.58}$$

Apart from the spatial curvature term $2K$ and the Hubble friction damping $2\mathcal{H}\overline{E}'_{kl}$ arising from cosmic expansion, this equation is the exact general-relativistic analogue of the gravitational-wave equation (1.132) derived in Chapter 1 in flat space-time.

Using the cosmological density parameter $\Omega \equiv \kappa\rho a^2/(3\mathcal{H}^2)$ defined in (3.36), this wave equation can be recast as:

$$\overline{E}''_{kl} + 2\mathcal{H}\overline{E}'_{kl} + (2K - \Delta)\overline{E}_{kl} = 3\mathcal{H}^2\Omega w\overline{\pi}_{kl}. \tag{4.2.59}$$

It is also convenient to isolate each polarization mode by introducing the canonical Mukhanov-type field variable:

$$u_{\mathrm{T}} = a\overline{E}_{\lambda} \quad \text{and} \quad \overline{E}_{ij} = \sum_{\lambda}\overline{E}_{\lambda}\varepsilon^{\lambda}_{ij}, \tag{4.2.60}$$

which transforms Eq. (4.2.58) into:

$$u''_{\mathrm{T}} + \left(2K - \Delta - \frac{a''}{a}\right)u_{\mathrm{T}} = \kappa a^3 P\overline{\pi}_{\lambda}. \tag{4.2.61}$$

We can write

$$u_{\mathrm{T}}'' - -\frac{a''}{a}u_{\mathrm{T}} = \kappa a^3 P\overline{\pi}_\lambda - (2K - \Delta)\, u_{\mathrm{T}}. \tag{1}$$

Then it becomes obvious that $u = a(\eta)$ is one solution of the LHS and second solution can be found by the method of reduction by considering $u = y(\eta)a(\eta)$, which leads to

$$u' = y'a + ya' \tag{2}$$
$$u'' = y''a + 2y'a' + ya'' \tag{3}$$

Plugging it back to EoM, we get

$$u'' - \frac{a''}{a}u = 0 \implies (y''a + 2y'a' + ya'') - \frac{a''}{a}(ya) = 0 \tag{4}$$
$$\frac{y''}{y} = -\frac{2a''}{a} \tag{5}$$
$$\ln(y') = \ln\left(\frac{1}{a^2}\right) \tag{6}$$

which gives

$$y(\eta) = \int^\eta \frac{d\eta}{a^2} \tag{7}$$

For a spatially flat Universe ($K = 0$), the homogeneous left-hand side is mathematically equivalent to the equation of a field propagating in a time-dependent effective potential $V_{\text{eff}}(\eta) = a''/a$. This has the same structure as a 1D time-independent Schrödinger equation with potential $V(x)$, a correspondence that will prove especially fruitful when analyzing the quantum generation of gravitational waves during inflation (Chapter 8).

Vector modes

The vector sector of the Einstein equations provides two relations from the $(0i)$ and (ij) components, respectively. Using the perturbed Ricci tensor and the vector components of the energy-momentum tensor (4.2.24) and (4.2.25), one obtains:

$$\boxed{(\Delta + 2K)\overline{\Phi}_i = -2\kappa\rho a^2(1+w)\overline{V}_i,} \tag{4.2.62}$$

$$\boxed{\overline{\Phi}_i' + 2\mathcal{H}\overline{\Phi}_i = \kappa P a^2 \overline{\pi}_i.} \tag{4.2.63}$$

In terms of the cosmological density parameter Ω, these relations become:

$$(\Delta + 2K)\overline{\Phi}_i = -6\mathcal{H}^2\Omega(1+w)\overline{V}_i, \tag{4.2.64}$$
$$\overline{\Phi}_i' + 2\mathcal{H}\overline{\Phi}_i = 3\mathcal{H}^2\Omega w\overline{\pi}_i. \tag{4.2.65}$$

Scalar modes

To extract the Einstein equations for scalar perturbations, we evaluate four independent combinations:

- $\delta G^0{}_0 + 3\mathcal{H}D^{-1i}\delta G^0{}_i$,
- the traceless part of $\delta G^i{}_j$,
- $\delta G^0{}_i$,

- $\delta G^i{}_i + 3c_{\rm s}^2 \delta G^0{}_0$.

The first two yield two algebraic constraint equations:

$$(\Delta + 3K)\Psi = \frac{\kappa}{2} a^2 \rho\, \delta^{\rm C} = \frac{3}{2}(\mathcal{H}^2 + K)\delta^C, \tag{4.2.66}$$

$$\Psi - \Phi = \kappa a^2 P \bar{\pi}. \tag{4.2.67}$$

Equation (4.2.66) is Hamiltonian constraint and it reproduces the familiar Poisson equation when formulated in terms of the comoving density contrast $\delta^{\rm C}$. While equation (4.2.67) being momentum constraint shows that the two metric potentials Φ and Ψ are identical whenever the scalar anisotropic stress vanishes ($\bar{\pi} = 0$), which is the case for perfect fluids.

The remaining two combinations give dynamical evolution equations:

$$\Psi' + \mathcal{H}\Phi = -\frac{\kappa}{2} a^2 \rho (1 + w) V, \tag{4.2.68}$$

$$\begin{aligned} \Psi'' + 3\mathcal{H}(1 + c_{\rm s}^2)\Psi' + \left[2\mathcal{H}' + (\mathcal{H}^2 - K)(1 + 3c_{\rm s}^2)\right]\Psi - c_{\rm s}^2 \Delta\Psi = \\ - (\mathcal{H}^2 + 2\mathcal{H}' + K)\left[\frac{1}{2}\Gamma + (3\mathcal{H}^2 + 2\mathcal{H}')\bar{\pi} + \mathcal{H}\bar{\pi}' + \frac{1}{3}\Delta\bar{\pi}\right] \\ - 9c_{\rm s}^2 \mathcal{H}^2(\mathcal{H}^2 + K)\bar{\pi}, \end{aligned} \tag{4.2.69}$$

where in Eq. (4.2.69) we have eliminated Φ in favor of Ψ and $\bar{\pi}$ using (4.2.67).

Equivalent forms

It is instructive to express the scalar Einstein equations using other gauge invariant density contrasts. In particular, the Poisson equation (4.2.66) takes the following forms when written in terms of $\delta^{\rm N}$ or $\delta^{\rm F}$ using (4.2.36) and (2.3.7) with $P = w\rho$:

$$(\Delta + 3K)\Psi = \frac{\kappa}{2} a^2 \rho \left[\delta^{\rm N} - 3\mathcal{H}(1 + w)V\right], \tag{4.2.70}$$

$$\Delta\Psi - 3\mathcal{H}^2 X = \frac{\kappa}{2} a^2 \rho\, \delta^{\rm F}, \tag{4.2.71}$$

To derive (4.2.66) in flat slicing, we need to do some work. Let us start from

$$(\Delta + 3K)\Psi = \frac{3}{2}(\mathcal{H}^2 + K)\delta^{\rm C}, \tag{1}$$

where we have used the background Friedmann equation

$$\kappa a^2 \rho = 3(\mathcal{H}^2 + K). \tag{2}$$

The gauge-invariant density perturbations in the flat and comoving slicings are related to the Newtonian-gauge density perturbation by

$$\delta^{\rm F} = \delta^{\rm N} + \frac{\rho'}{\rho}\frac{\Psi}{\mathcal{H}}, \qquad \delta^{\rm C} = \delta^{\rm N} + \frac{\rho'}{\rho} V. \tag{3}$$

Hence

$$\delta^{\rm C} - \delta^{\rm F} = \frac{\rho'}{\rho}\left(V - \frac{\Psi}{\mathcal{H}}\right). \tag{4}$$

For a constant equation of state, $P = w\rho$, the background conservation equation gives

$$\frac{\rho'}{\rho} = -3\mathcal{H}(1 + w), \tag{5}$$

and therefore

$$\delta^{\mathrm{C}} = \delta^{\mathrm{F}} - 3\mathcal{H}(1+w)V + 3(1+w)\Psi. \tag{6}$$

The scalar $0i$ Einstein equation in Newtonian gauge is

$$\Psi' + \mathcal{H}\Phi = -\frac{\kappa}{2}a^2(\rho + P)V. \tag{7}$$

Using $P = w\rho$ together with (2), this gives

$$V = -\frac{2(\Psi' + \mathcal{H}\Phi)}{3(1+w)(\mathcal{H}^2 + K)}. \tag{8}$$

Substituting this into (6), we obtain

$$\delta^{\mathrm{C}} = \delta^{\mathrm{F}} + \frac{2\mathcal{H}(\Psi' + \mathcal{H}\Phi)}{\mathcal{H}^2 + K} + 3(1+w)\Psi. \tag{9}$$

Substituting (9) into the comoving constraint (1) gives

$$(\Delta + 3K)\Psi = \frac{3}{2}(\mathcal{H}^2 + K)\delta^{\mathrm{F}} + 3\mathcal{H}(\Psi' + \mathcal{H}\Phi) + \frac{9}{2}(1+w)(\mathcal{H}^2 + K)\Psi. \tag{10}$$

The background acceleration equation is

$$\mathcal{H}' = -\frac{1}{2}(1+3w)(\mathcal{H}^2 + K), \tag{11}$$

which implies

$$\mathcal{H}^2 - \mathcal{H}' + K = \frac{3}{2}(1+w)(\mathcal{H}^2 + K). \tag{12}$$

Hence

$$\frac{9}{2}(1+w)(\mathcal{H}^2 + K) = 3(\mathcal{H}^2 - \mathcal{H}' + K), \tag{13}$$

and (10) becomes

$$(\Delta + 3K)\Psi = \frac{3}{2}(\mathcal{H}^2 + K)\delta^{\mathrm{F}} + 3\mathcal{H}\Psi' + 3\mathcal{H}^2\Phi + 3(\mathcal{H}^2 - \mathcal{H}' + K)\Psi. \tag{14}$$

Now define

$$X = \Phi + \Psi + \left(\frac{\Psi}{\mathcal{H}}\right)'. \tag{15}$$

Expanding the derivative,

$$\mathcal{H}^2 X = \mathcal{H}^2\Phi + \mathcal{H}^2\Psi + \mathcal{H}\Psi' - \mathcal{H}'\Psi, \tag{16}$$

so that

$$3\mathcal{H}^2 X = 3\mathcal{H}^2\Phi + 3\mathcal{H}\Psi' + 3(\mathcal{H}^2 - \mathcal{H}')\Psi. \tag{17}$$

Therefore (14) becomes

$$(\Delta + 3K)\Psi = \frac{3}{2}(\mathcal{H}^2 + K)\delta^{\mathrm{F}} + 3\mathcal{H}^2 X + 3K\Psi. \tag{18}$$

The $3K\Psi$ terms cancel from both sides, yielding

$$\boxed{\Delta\Psi - 3\mathcal{H}^2 X = \frac{3}{2}(\mathcal{H}^2 + K)\delta^{\mathrm{F}}}. \tag{19}$$

Finally, using (2),

$$\boxed{\Delta\Psi - 3\mathcal{H}^2 X = \frac{\kappa}{2}a^2\rho\,\delta^{\mathrm{F}}}. \tag{20}$$

while the second-order evolution equation (4.2.69) can be expressed entirely through the single variable X:

$$\mathcal{H}X' + (\mathcal{H}^2 + 2\mathcal{H}')X = \frac{\kappa}{2}a^2\rho\left(w\Gamma + c_{\mathrm{s}}^2\delta^{\mathrm{F}} + \frac{2}{3}w\Delta\bar{\pi}\right). \tag{4.2.72}$$

4.2.6 Perturbed conservation equation for a fluid

The conservation of the energy-momentum tensor, expanded to linear order, reads:

$$\delta(\nabla_\mu T^\mu{}_\nu) = 0.$$

Explicit geometric connection coefficients and contracted Christoffel symbols are provided in Appendix C. Because perfect fluids do not support transverse-traceless shear stresses at linear order, there are no tensor conservation equations; the dynamics separates into vector and scalar sectors.

Vector modes

For the vector perturbations, the conservation law yields a single differential equation:

$$\boxed{\overline{V}_i' + \mathcal{H}(1 - 3c_{\mathrm{s}}^2)\overline{V}_i = -\frac{1}{2}\frac{w}{1+w}(\Delta + 2K)\overline{\pi}_i.} \tag{4.2.73}$$

This first-order equation is readily solvable in closed form in most cosmological scenarios of interest.

Scalar modes

Projecting the 4-dimensional conservation equation along timelike and spacelike directions generates two equations analogous to the continuity and Euler equations, respectively. In the longitudinal (Newtonian) gauge, these read:

$$\boxed{\delta^{\mathrm{N}\prime} + 3\mathcal{H}(c_{\mathrm{s}}^2 - w)\delta^{\mathrm{N}} = -(1+w)(\Delta V - 3\Psi') - 3\mathcal{H}w\Gamma,} \tag{4.2.74}$$

Consider the Newtonian-gauge metric

$$\mathrm{d}s^2 = a^2(\eta)\left[(1+2\Phi)\,\mathrm{d}\eta^2 - (1-2\Psi)\delta_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j\right], \tag{1}$$

where

$$\mathcal{H} \equiv \frac{a'}{a}. \tag{2}$$

We decompose the energy density and pressure as

$$\rho = \bar{\rho} + \delta\rho, \qquad P = \bar{P} + \delta P, \tag{3}$$

and define the Newtonian-gauge density contrast by

$$\delta^{\mathrm{N}} \equiv \frac{\delta\rho}{\bar{\rho}}. \tag{4}$$

The energy-momentum tensor is

$$T^\mu{}_\nu = (\rho + P)u^\mu u_\nu - P\delta^\mu_\nu + \Pi^\mu{}_\nu, \tag{5}$$

where $\Pi^\mu{}_\nu$ denotes the anisotropic stress. To first order,

$$T^0{}_0 = \bar\rho + \delta\rho, \tag{6}$$
$$T^i{}_0 = (\bar\rho + \bar P)V^i, \tag{7}$$
$$T^0{}_i = -(\bar\rho + \bar P)V_i, \tag{8}$$
$$T^i{}_j = -(\bar P + \delta P)\delta^i_j + \Pi^i{}_j, \tag{9}$$

with

$$\Pi^i{}_i = 0. \tag{10}$$

We start from the covariant conservation equation

$$\nabla_\mu T^\mu{}_\nu = 0. \tag{11}$$

For a mixed tensor,

$$\nabla_\mu T^\mu{}_\nu = \partial_\mu T^\mu{}_\nu + \Gamma^\mu_{\mu\lambda} T^\lambda{}_\nu - \Gamma^\lambda_{\mu\nu} T^\mu{}_\lambda. \tag{12}$$

Taking $\nu = 0$ gives

$$\nabla_\mu T^\mu{}_0 = \partial_\mu T^\mu{}_0 + \Gamma^\mu_{\mu\lambda} T^\lambda{}_0 - \Gamma^\lambda_{\mu 0} T^\mu{}_\lambda = 0. \tag{13}$$

Expanding the repeated indices into temporal and spatial parts,

$$\begin{aligned} 0 = {} & \partial_0 T^0{}_0 + \partial_i T^i{}_0 + \Gamma^\mu_{\mu 0} T^0{}_0 + \Gamma^\mu_{\mu i} T^i{}_0 \\ & - \Gamma^0_{00} T^0{}_0 - \Gamma^0_{i0} T^i{}_0 - \Gamma^j_{00} T^0{}_j - \Gamma^j_{i0} T^i{}_j. \end{aligned} \tag{14}$$

The Christoffel symbols required below are

$$\Gamma^0_{00} = \mathcal{H} + \Phi', \tag{15}$$
$$\Gamma^0_{i0} = \partial_i \Phi, \tag{16}$$
$$\Gamma^j_{00} = \partial^j \Phi, \tag{17}$$
$$\Gamma^j_{i0} = (\mathcal{H} - \Psi')\delta^j_i. \tag{18}$$

Furthermore,

$$\Gamma^\mu_{\mu 0} = \Gamma^0_{00} + \Gamma^i_{i0} = 4\mathcal{H} + \Phi' - 3\Psi'. \tag{19}$$

We now evaluate the eight terms in Eq. (14) separately.

First term:

$$\partial_0 T^0{}_0 = \partial_0(\bar\rho + \delta\rho) = \bar\rho' + \delta\rho'. \tag{20}$$

Second term: Using the homogeneity of the background,

$$\partial_i T^i{}_0 = (\bar\rho + \bar P)\partial_i V^i. \tag{21}$$

Defining

$$\Delta V \equiv \partial_i V^i, \tag{22}$$

we have

$$\partial_i T^i{}_0 = (\bar\rho + \bar P)\Delta V. \tag{23}$$

Third term:

$$\begin{aligned} \Gamma^\mu_{\mu 0} T^0{}_0 &= (4\mathcal{H} + \Phi' - 3\Psi')(\bar\rho + \delta\rho) \\ &= 4\mathcal{H}\bar\rho + 4\mathcal{H}\delta\rho + \bar\rho\Phi' - 3\bar\rho\Psi', \end{aligned} \tag{24}$$

where products of perturbations have been discarded.

Fourth term: Since $\Gamma^{\mu}_{\mu i}$ has no background contribution, it is already first order, while $T^i{}_0$ is also first order. Hence

$$\Gamma^{\mu}_{\mu i} T^i{}_0 = \mathcal{O}(2) = 0 \tag{25}$$

at linear order.

Fifth term:

$$\begin{aligned} -\Gamma^0_{00} T^0{}_0 &= -(\mathcal{H} + \Phi')(\bar{\rho} + \delta\rho) \\ &= -\mathcal{H}\bar{\rho} - \mathcal{H}\delta\rho - \bar{\rho}\Phi'. \end{aligned} \tag{26}$$

Sixth term: Since both $\Gamma^0_{i0} = \partial_i \Phi$ and $T^i{}_0 = (\bar{\rho} + \bar{P})V^i$ are first order,

$$-\Gamma^0_{i0} T^i{}_0 = \mathcal{O}(2) = 0. \tag{27}$$

Seventh term: Similarly, $\Gamma^j_{00} = \partial^j \Phi$ and $T^0{}_j = -(\bar{\rho} + \bar{P})V_j$ are both first order, so

$$-\Gamma^j_{00} T^0{}_j = \mathcal{O}(2) = 0. \tag{28}$$

Eighth term: Using

$$T^i{}_j = -(\bar{P} + \delta P)\delta^i_j + \Pi^i{}_j, \tag{29}$$

we obtain

$$\begin{aligned} -\Gamma^j_{i0} T^i{}_j &= -(\mathcal{H} - \Psi')\delta^j_i \left[-(\bar{P} + \delta P)\delta^i_j + \Pi^i{}_j \right] \\ &= (\mathcal{H} - \Psi') \left[3(\bar{P} + \delta P) - \Pi^i{}_i \right] \\ &= 3\mathcal{H}\bar{P} + 3\mathcal{H}\delta P - 3\bar{P}\Psi', \end{aligned} \tag{30}$$

where $\Pi^i{}_i = 0$ has been used.
Substituting all eight contributions into Eq. (14), we obtain

$$\begin{aligned} 0 = {} & \bar{\rho}' + \delta\rho' + (\bar{\rho} + \bar{P})\Delta V \\ & + 4\mathcal{H}\bar{\rho} + 4\mathcal{H}\delta\rho + \bar{\rho}\Phi' - 3\bar{\rho}\Psi' \\ & - \mathcal{H}\bar{\rho} - \mathcal{H}\delta\rho - \bar{\rho}\Phi' \\ & + 3\mathcal{H}\bar{P} + 3\mathcal{H}\delta P - 3\bar{P}\Psi'. \end{aligned} \tag{31}$$

The background terms give

$$\boxed{\bar{\rho}' + 3\mathcal{H}(\bar{\rho} + \bar{P}) = 0,} \tag{32}$$

while the first-order terms give

$$\delta\rho' + 3\mathcal{H}(\delta\rho + \delta P) + (\bar{\rho} + \bar{P})(\Delta V - 3\Psi') = 0. \tag{33}$$

Hence

$$\boxed{\delta\rho' + 3\mathcal{H}(\delta\rho + \delta P) = -(\bar{\rho} + \bar{P})(\Delta V - 3\Psi').} \tag{34}$$

Now define

$$w \equiv \frac{\bar{P}}{\bar{\rho}}, \qquad \delta^{\mathrm{N}} \equiv \frac{\delta\rho}{\bar{\rho}}. \tag{35}$$

Since

$$\delta\rho = \bar{\rho}\delta^{\mathrm{N}}, \tag{36}$$

we have

$$\delta\rho' = \bar{\rho}'\delta^{\rm N} + \bar{\rho}\delta^{\rm N\prime}. \tag{37}$$

Using Eq. (32),

$$\frac{\bar{\rho}'}{\bar{\rho}} = -3\mathcal{H}(1+w), \tag{38}$$

so Eq. (34) becomes

$$\boxed{\delta^{\rm N\prime} - 3\mathcal{H}w\delta^{\rm N} + 3\mathcal{H}\frac{\delta P}{\bar{\rho}} = -(1+w)(\Delta V - 3\Psi').} \tag{39}$$

Finally, define the adiabatic sound speed

$$c_s^2 \equiv \frac{\bar{P}'}{\bar{\rho}'}, \tag{40}$$

and the entropy perturbation Γ through

$$\boxed{\delta P = c_s^2\delta\rho + \bar{P}\Gamma.} \tag{41}$$

Since

$$\delta\rho = \bar{\rho}\delta^{\rm N}, \qquad \bar{P} = w\bar{\rho}, \tag{42}$$

we have

$$\frac{\delta P}{\bar{\rho}} = c_s^2\delta^{\rm N} + w\Gamma. \tag{43}$$

Substituting this into Eq. (39), we finally obtain

$$\boxed{\delta^{\rm N\prime} + 3\mathcal{H}(c_s^2 - w)\delta^{\rm N} = -(1+w)(\Delta V - 3\Psi') - 3\mathcal{H}w\Gamma.} \tag{44}$$

$$\boxed{V' + \mathcal{H}(1-3c_{\rm s}^2)V = -\Phi - \frac{c_{\rm s}^2}{1+w}\delta^{\rm N} - \frac{w}{1+w}\left[\Gamma + \frac{2}{3}(\Delta + 3K)\bar{\pi}\right].} \tag{4.2.75}$$

Using the background identity for w' [Eq. (C.19)], the continuity equation (4.2.74) can be recast into the compact form:

$$\left(\frac{\delta^{\rm N}}{1+w}\right)' = -(\Delta V - 3\Psi') - 3\mathcal{H}\frac{w}{1+w}\Gamma. \tag{4.2.76}$$

In terms of the flat-slicing and comoving density contrasts $\delta^{\rm F}$ and $\delta^{\rm C}$, these equations become:

$$\delta^{\rm F\prime} + 3\mathcal{H}(c_{\rm s}^2 - w)\delta^{\rm F} = -(1+w)\Delta V - 3\mathcal{H}w\Gamma, \tag{4.2.77}$$

$$V' + \mathcal{H}(1-3c_{\rm s}^2)V = -\Phi - 3c_{\rm s}^2\Psi - \frac{c_{\rm s}^2}{1+w}\delta^{\rm F} - \frac{w}{1+w}\left[\Gamma + \frac{2}{3}(\Delta + 3K)\bar{\pi}\right], \tag{4.2.78}$$

$$V' + \mathcal{H}V = -\Phi - \frac{c_{\rm s}^2}{1+w}\delta^{\rm C} - \frac{w}{1+w}\left[\Gamma + \frac{2}{3}(\Delta + 3K)\bar{\pi}\right]. \tag{4.2.79}$$

4.2.7 Interpretation of the perturbation equations

At first glance, the relativistic fluid equations resemble their Newtonian counterparts. Indeed, for a static background ($\mathcal{H} = 0$), non-relativistic pressureless dust ($w = c_{\rm s}^2 = 0$, $\bar{\pi} = 0$), and purely adiabatic perturbations ($\Gamma = 0$), identifying δ with $\delta^{\rm F}$ immediately recovers the classical relations (4.1.1) and (4.1.2). However, general relativity introduces five profound structural differences:

1. **Mode Diversity:** Relativistic perturbation theory encompasses vector and tensor degrees of freedom alongside scalar modes. In the linear regime around a Friedmann–Lemaître background, these sectors decouple completely.

2. **Dual Gravitational Potentials:** The scalar sector requires two independent potentials, Φ and Ψ. In the Newtonian limit and on sub-Hubble scales, they converge to a single potential provided anisotropic stresses are negligible. Notably, the relativistic Poisson equation aligns most closely with its Newtonian form in the comoving gauge, yet it involves the spatial curvature potential Ψ rather than the acceleration potential Φ.

3. **Super-Hubble Deviations and Gauge Ambiguity:** On scales larger than the Hubble radius, the Poisson equation deviates significantly from Newtonian gravity, and different gauge-invariant definitions of density contrast ($\delta^{\mathrm{N}}, \delta^{\mathrm{F}}, \delta^{\mathrm{C}}$) diverge markedly from one another.

4. **Relativistic Inertial Enthalpy:** The continuity equation contains an explicit factor of $(1+w)$, reflecting the fact that in relativity, energy flux (which includes pressure and enthalpy) rather than rest-mass flux governs gravitational dynamics.

5. **Cosmic Expansion Damping:** The Euler equation features an expansion damping term $-\mathcal{H}(1-3c_{\mathrm{s}}^2)V$. For non-relativistic matter ($c_{\mathrm{s}}^2 \approx 0$), this term forces peculiar velocities to decay as a^{-1}, causing the velocity field to align asymptotically with the Hubble flow. Conversely, for ultra-relativistic radiation ($c_{\mathrm{s}}^2 = 1/3$), this damping vanishes identically, which can be understood from the conformal invariance of null geodesics. (A rigorous kinetic treatment of radiation will be developed in Chapter 6, where velocity appears as the dipole moment of the photon distribution.)

4.3 Evolution

The perturbation equations established above can be integrated analytically in several fundamental regimes. In general, it is highly advantageous to expand spatial perturbations into Fourier modes (Appendix B) and examine the time evolution of each mode independently. Because spatial derivatives enter the linearized field equations solely through the Laplacian Δ, Fourier transformation maps the partial differential equations into decoupled ordinary differential equations for each wavenumber k.

In analyzing mode evolution, one distinguishes two distinct physical regimes based on weather $\mathcal{H}$ is increasing function of time or not:

$k \ll \mathcal{H}$	**super-Hubble mode**	wavelength larger than Hubble radius,
$k \gg \mathcal{H}$	**sub-Hubble mode**	wavelength smaller than Hubble radius.

In the cosmological literature, super-Hubble and sub-Hubble modes are often casually referred to as “super-horizon” and “sub-horizon” modes. This terminology originates historically from the standard Big Bang model (Chapter 4), where the Hubble radius $\mathcal{H}^{-1}$ is numerically comparable to the particle horizon. However, the two concepts are fundamentally distinct, especially in cosmological models featuring an inflationary epoch (Chapter 8), whose primary purpose is precisely to decouple the Hubble radius from the causal particle horizon. The suppression of spatial gradients on super-Hubble scales is not an enforcement of causal disconnectivity, but rather reflects the dynamical dominance of cosmic expansion damping ($\mathcal{H}^{-1}$) over the characteristic oscillation timescale of the mode ($\sim k^{-1}$).

So the modes which are superhubble has time period so large that they have not had enough time to oscillate even once and hence they behave as static gravitational potential.

Consequently, the super-Hubble behavior of growing modes cannot be deduced merely from heuristic causality arguments, contrary to frequent claims that perturbations are causally "frozen" outside the horizon. When the wavelength of a given mode is much larger than the hubble horizon, they are effectively frozen but when the wavelength is just barely outside the hubble horizon, it feels the edges and its not quite frozen yet.

4.3.1 Vector and tensor modes

Vector modes

In the absence of anisotropic vector stresses ($\bar{\pi}_i = 0$), the Einstein equation (4.2.63) yields:

$$\bar{\Phi} + 2\frac{a'}{a}\bar{\Phi} = 0 \implies \frac{\bar{\Phi}'}{\Phi} = -2\frac{a'}{a} \implies \overline{\Phi}_i \propto a^{-2}, \tag{4.3.1}$$

while the fluid conservation equation (4.2.73) implies:

$$\overline{V}^i \propto a^{-(1-3c_s^2)}. \tag{4.3.2}$$

Thus, cosmic expansion rapidly extinguishes vector metric perturbations. For a non-relativistic cosmic fluid ($c_s^2 = 0$), fluid vorticity decays as a^{-1}. As a result, primordial vector perturbations are dynamically washed out and play virtually no role in large-scale structure formation, unless continuously replenished by active sources such as cosmic defects or primordial magnetic fields. This happens because vector mode has no constant or growing part in the solution unlike the scalar and tensor perturbation if $\bar{\pi}_i = 0$.

Tensor modes

Assuming negligible anisotropic shear ($\bar{\pi}_{ij} = 0$) in a spatially flat Universe ($K = 0$), we project the tensor wave equation (4.2.58) into Fourier space. For a background expanding as a power law in conformal time, $a(\eta) \propto \eta^\nu$, Eq. (4.2.58) reduces to:

$$\frac{\mathrm{d}^2\overline{E}_{ij}}{\mathrm{d}x^2} + \frac{2\nu}{x}\frac{\mathrm{d}\overline{E}_{ij}}{\mathrm{d}x} + \overline{E}_{ij} = 0, \tag{4.3.3}$$

where $x \equiv k\eta$. The general solution can be expressed in terms of cylindrical Bessel functions:

$$\overline{E}_{ij} = x^{1/2-\nu}\left[A_{ij}J_{\nu-\frac{1}{2}}(x) + B_{ij}N_{\nu-\frac{1}{2}}(x)\right], \tag{4.3.4}$$

where A_{ij} and B_{ij} are constant transverse-traceless polarization tensors. Retaining only the regular solution as $x \to 0$ (the Neumann mode N diverges at the origin, representing a decaying transient that rapidly vanishes as time progresses), we obtain:

$$\overline{E}_{ij} = x^{1/2-\nu}J_{\nu-\frac{1}{2}}(x)A_{ij} = x^{1-\nu}j_{\nu-1}(x)A_{ij}, \tag{4.3.5}$$

where $j_n(x)$ denotes the spherical Bessel functions. Evaluating this expression for the matter- and radiation-dominated epochs yields:

	Matter	**Radiation**
$a(\eta)$	η^2	η
$\overline{E}_{ij}(\eta)$	$3j_1(k\eta)/(k\eta)$	$j_0(k\eta)$

The tensor perturbation amplitude $\overline{E}_{ij}$ remains constant on super-Hubble scales ($k\eta < 1$), where the physical wavelength exceeds the Hubble radius. Once the mode crosses inside the Hubble radius ($k\eta > 1$), it transitions into a freely propagating gravitational wave characterized by damped oscillations (see Fig. 5.3).

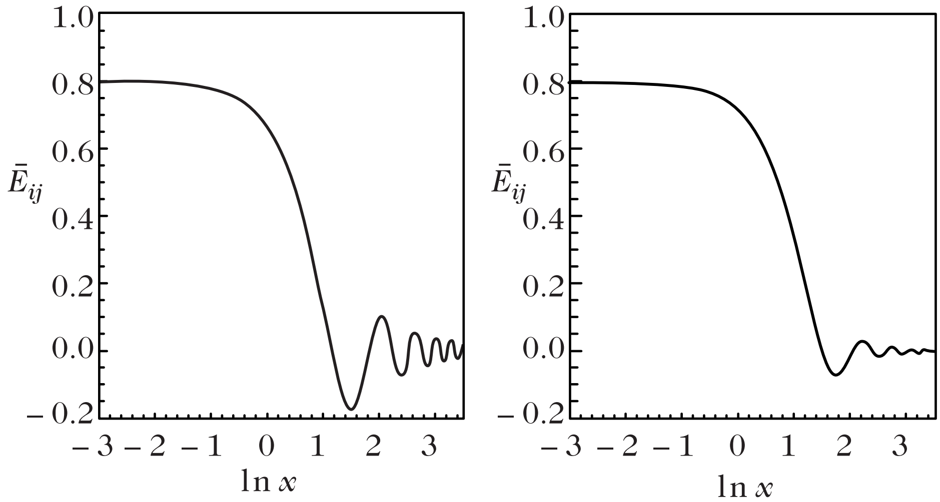


Figure 4.3: Evolution of gravitational waves in a radiation-dominated Universe ($w = 1/3, \nu = 1$, left) and pressureless matter ($w = 0, \nu = 2$, right) as a function of $\ln x = \ln(k\eta)$. The mode amplitude remains constant as long as the wavelength is super-Hubble ($x < 1$) and develops damped oscillatory behaviour as soon as it becomes sub-Hubble ($x > 1$).

Tensor modes in a Universe dominated by a mixture of matter and radiation

Let us now investigate a cosmological background driven by a simultaneous mixture of pressureless matter ($w = 0, \nu = 2$) and relativistic radiation ($w = 1/3, \nu = 1$). To parameterize the background expansion conveniently, we define the normalized scale factor:

$$y \equiv \frac{a}{a_{\rm eq}}, \tag{4.3.6}$$

The radiation and matter scales differently in the FLRW universe

$$\rho^m \propto a^{-3} \quad \Longrightarrow \quad \frac{\rho^m}{\rho^m_{\rm eq}} = \left(\frac{a}{a_{\rm eq}}\right)^{-3}, \tag{1}$$

$$\rho^r \propto a^{-4} \quad \Longrightarrow \quad \frac{\rho^r}{\rho^r_{\rm eq}} = \left(\frac{a}{a_{\rm eq}}\right)^{-4}. \tag{2}$$

But, at the matter–radiation equality, we have

$$\rho^m_{\rm eq} = \rho^r_{\rm eq}.$$

Therefore,

$$\frac{\rho^m}{\rho^r} = \frac{\rho^m_{\rm eq}}{\rho^r_{\rm eq}}\left(\frac{a}{a_{\rm eq}}\right) = \frac{a}{a_{\rm eq}}. \tag{3}$$

where $a_{\rm eq}$ designates the scale factor at the epoch of radiation-matter equality, defined by $\bar{\rho}_{\rm m}(a_{\rm eq}) = \bar{\rho}_{\rm r}(a_{\rm eq})$. The parameter y tracks the matter-to-radiation density ratio, $y = \rho_{\rm m}/\rho_{\rm r}$, and the effective equation-of-state parameter of the two-component mixture is:

$$w = \frac{1}{3}\frac{1}{1+y}. \tag{4.3.7}$$

This can be shown from a small calculation such as following

$$P = P_m + P_r = 0 + P_r = \frac{1}{3}\rho_r = \frac{1}{3}\frac{\rho_r}{\rho_m + \rho_r}\rho = \frac{1}{3}\frac{1}{1+y}\rho \tag{1}$$

Therefore the EoS of radiation+matter fluid is not constant but changing with time.

The dynamical wave equation (4.2.58) governing gravitational waves can then be recast using y as the independent evolution variable:

$$\frac{\mathrm{d}^2\overline{E}_{ij}}{\mathrm{d}y^2} + \frac{4+5y}{2y(1+y)}\frac{\mathrm{d}\overline{E}_{ij}}{\mathrm{d}y} + \left(\frac{k}{k_{\mathrm{eq}}}\right)^2 \frac{2}{1+y}\overline{E}_{ij} = 0. \tag{4.3.8}$$

To establish this differential equation, we have used the chain rule for derivatives with respect to conformal time, $\overline{E}'_{ij} = y' \,\mathrm{d}\overline{E}_{ij}\big/\mathrm{d}y$, along with the conformal Hubble parameter $\mathcal{H} = y'/y$. In terms of the variable y, the Friedmann equation reads:

$$\mathcal{H}^2 = \mathcal{H}_{\mathrm{eq}}^2 \frac{1+y}{2y^2}, \tag{4.3.9}$$

where $\mathcal{H}_{\mathrm{eq}}$ denotes the conformal Hubble parameter evaluated at equality ($y = 1$). We have also introduced the characteristic comoving wavenumber k_{eq}, representing the mode that enters the Hubble horizon at equality:

$$k_{\mathrm{eq}} \equiv \mathcal{H}_{\mathrm{eq}} = a_{\mathrm{eq}} H_{\mathrm{eq}}. \tag{4.3.10}$$

Equation (4.3.8) can be integrated numerically, and the resulting trajectories for representative modes k/k_{eq} are displayed in Fig. 4.4. As expected, modes entering the horizon earlier undergo more prolonged oscillatory damping, leading to greater suppression of their asymptotic amplitudes.

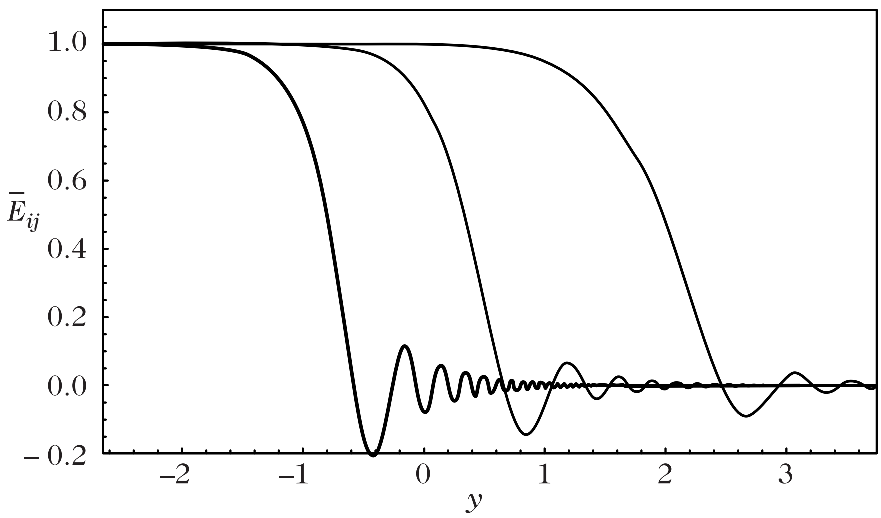


Figure 4.4: Evolution of gravitational waves in a Universe dominated by a mixture of matter and radiation in terms of $y = a/a_{\mathrm{eq}}$ ($y = 1$ at equality) for modes with $k/k_{\mathrm{eq}} = 10$, 1, and 0.1.

General solutions for long-wavelength modes

In a spatially flat Universe ($K = 0$), the gravitational wave equation (4.2.58) can be expressed as a formal Volterra integral equation:

$$u_\mathrm{T}(\eta, k) = a(\eta)\left[A_1(k) + A_2(k)\int^\eta \frac{\mathrm{d}\tilde{\eta}}{a^2(\tilde{\eta})} + k^2\int^\eta \frac{\mathrm{d}\tilde{\eta}}{a^2(\tilde{\eta})}\int^{\tilde{\eta}} \mathrm{d}\bar{\eta}\, a^2(\bar{\eta}) u_\mathrm{T}(\bar{\eta}, k)\right]. \tag{4.3.11}$$

For super-Hubble, long-wavelength modes satisfying the condition $k^2 \ll a''/a$, this integral equation truncates to:

$$u_\mathrm{T}(\eta, k) = a(\eta)A_1(k) + A_2(k)a(\eta)\int^\eta \frac{\mathrm{d}\tilde{\eta}}{a^2(\tilde{\eta})} + \mathcal{O}\left(\frac{k^2 a}{a''}\right), \tag{4.3.12}$$

This was done using the method of successive approximation

$$u(\eta) = \underbrace{a(\eta)}_{y_1} + \underbrace{a(\eta)\int \frac{d\eta}{a^2}}_{y_2} + k^2 \int d\eta' G(\eta, \eta') u(\eta') \tag{1}$$

where

$$G(\eta, \eta') = y_1(\eta) y_2(\eta')\Theta(\eta - \eta') + y_1(\eta') y_2(\eta)\Theta(\eta' - \eta) \tag{2}$$

where the two integration constants $A_1(k)$ and $A_2(k)$ are functions of wavenumber fixed by initial conditions. In analogy with quantum mechanical scattering in a one-dimensional barrier, Eq. (4.3.12) corresponds to the leading Born approximation for wave propagation through an effective potential.

4.3.2 Evolution of the gravitational potential

We now turn our attention to the dynamical evolution of the longitudinal gravitational potential. Neglecting anisotropic stress contributions ($\bar{\pi} = 0$), the scalar evolution equation (4.2.69) takes the form:

$$\Phi'' + 3\mathcal{H}\left(1 + c_\mathrm{s}^2\right)\Phi' + \left[2\mathcal{H}' + \left(\mathcal{H}^2 - K\right)\left(1 + 3c_\mathrm{s}^2\right)\right]\Phi - c_\mathrm{s}^2\Delta\Phi = \frac{\kappa}{2}a^2 P\Gamma. \tag{4.3.13}$$

It can be derived from (4.2.69) with $\bar{\pi} = 0$ and $\kappa P a^2 = -(\mathcal{H}^2 + 2\mathcal{H}' + K)$.

Another formulation of the evolution equation

A convenient reduced form of Eq. (4.3.13) can be established by introducing the background auxiliary function:

$$\theta \equiv \frac{1}{a}\left(\frac{\rho}{\rho + P}\right)^{1/2}\left(1 - \frac{3K}{\kappa\rho a^2}\right)^{1/2} = \frac{1}{a}\sqrt{\frac{\kappa\rho a^2 - 3K}{\kappa a^2(\rho + P)}}, \tag{4.3.14}$$

which, using the Friedmann equations, can be rewritten as:

$$\theta = \frac{\mathcal{H}}{a}\left[\frac{2}{3}\left(\mathcal{H}^2 - \mathcal{H}' + K\right)\right]^{-1/2}. \tag{4.3.15}$$

Derived from (4.3.14) using

$$\kappa\rho a^2 = 3(\mathcal{H}^2 + K) \qquad \kappa P a^2 = -(\mathcal{H}^2 + 2\mathcal{H}' + K) \tag{1}$$

Defining the rescaled Mukhanov–Sasaki type variable:

$$u_{\rm S} \equiv \frac{2}{3}\frac{a^2\theta}{\mathcal{H}}\Phi, \tag{4.3.16}$$

a lengthy but direct calculation transforms Eq. (4.3.13) into the canonical wave equation:

$$u_{\rm S}'' - \left(\frac{\theta''}{\theta} + c_{\rm s}^2\Delta\right)u_{\rm S} = \frac{\kappa}{3}\frac{\theta}{\mathcal{H}}a^4 P\Gamma. \tag{4.3.17}$$

Equation (4.3.17) shares structural similarity with the tensor mode equation (4.2.61) and can be analyzed using identical mathematical techniques.

We first collect the background relations that will be needed. The Friedmann equations can be written as

$$\kappa\rho a^2 = 3(\mathcal{H}^2 + K), \qquad \kappa P a^2 = -(\mathcal{H}^2 + 2\mathcal{H}' + K), \tag{1}$$

while energy conservation gives

$$\rho' = -3\mathcal{H}(\rho + P). \tag{2}$$

For an adiabatic background,

$$c_s^2 = \frac{P'}{\rho'}. \tag{3}$$

Subtracting the two equations in (1),

$$\mathcal{H}^2 - \mathcal{H}' + K = \frac{\kappa}{2}a^2(\rho + P), \tag{4}$$

allows the definition of θ to be written in the simpler form

$$\theta = \frac{\sqrt{3}\,\mathcal{H}}{a^2\sqrt{\kappa(\rho + P)}}. \tag{5}$$

This form makes its logarithmic derivative particularly simple. Since

$$\rho' + P' = -3\mathcal{H}(1 + c_s^2)(\rho + P), \tag{1}$$

we have

$$\frac{1}{2}\frac{(\rho + P)'}{\rho + P} = -\frac{3}{2}\mathcal{H}(1 + c_s^2). \tag{6}$$

Consequently,

$$\frac{\theta'}{\theta} = \frac{\mathcal{H}'}{\mathcal{H}} - 2\mathcal{H} + \frac{3}{2}\mathcal{H}(1 + c_s^2), \tag{2}$$

or

$$\frac{\theta'}{\theta} = \frac{\mathcal{H}'}{\mathcal{H}} - \frac{\mathcal{H}}{2} + \frac{3}{2}\mathcal{H}c_s^2. \tag{7}$$

Now use the definition of $w = \frac{2}{3}\frac{a^2\theta}{\mathcal{H}}$. Its logarithmic derivative is

$$\frac{w'}{w} = 2\mathcal{H} + \frac{\theta'}{\theta} - \frac{\mathcal{H}'}{\mathcal{H}}. \tag{3}$$

Using (7),

$$\boxed{\frac{w'}{w} = \frac{3}{2}\mathcal{H}(1+c_s^2)}. \tag{8}$$

Thus

$$2\frac{w'}{w} = 3\mathcal{H}(1+c_s^2),$$

which is precisely the coefficient multiplying Φ' in (4.3.13). We can therefore substitute

$$\Phi = \frac{u_S}{w}$$

into the evolution equation. The relevant derivative combination is

$$\Phi'' + 2\frac{w'}{w}\Phi' = \frac{1}{w}\left(u_S'' - \frac{w''}{w}u_S\right). \tag{9}$$

Since w depends only on time, one can write

$$\Delta\Phi = \frac{1}{w}\Delta u_S.$$

Hence, we get

$$u_S'' - \left[\frac{w''}{w} - 2\mathcal{H}' - (\mathcal{H}^2 - K)(1+3c_s^2)\right]u_S - c_s^2\Delta u_S = w\,\frac{\kappa}{2}a^2 P\Gamma. \tag{10}$$

The remaining task is to show that the quantity in square brackets is θ''/θ. From (8),

$$\frac{w''}{w} = \left(\frac{w'}{w}\right)' + \left(\frac{w'}{w}\right)^2, \tag{4}$$

so that

$$\frac{w''}{w} = \frac{3}{2}\mathcal{H}'(1+c_s^2) + \frac{3}{2}\mathcal{H}(c_s^2)' + \frac{9}{4}\mathcal{H}^2(1+c_s^2)^2. \tag{11}$$

For θ, it is convenient to use

$$\frac{\theta''}{\theta} = \left(\frac{\theta'}{\theta}\right)' + \left(\frac{\theta'}{\theta}\right)^2. \tag{12}$$

Substituting (7) into this expression introduces $\mathcal{H}''$. The background equations allow this derivative to be eliminated. Differentiating the Raychaudhuri equation and using (2) and (3) gives

$$\mathcal{H}'' = \mathcal{H}\left[(1+3c_s^2)(\mathcal{H}^2 + K) + \mathcal{H}'(1-3c_s^2)\right]. \tag{13}$$

Using this in (12), one finds

$$\frac{\theta''}{\theta} = \frac{5}{4}\mathcal{H}^2 + \frac{3}{2}\mathcal{H}^2 c_s^2 + \frac{9}{4}\mathcal{H}^2(c_s^2)^2 + K + 3Kc_s^2 + \frac{3c_s^2 - 1}{2}\mathcal{H}' + \frac{3}{2}\mathcal{H}(c_s^2)'. \tag{14}$$

On the other hand, substituting (11) into the coefficient appearing in (10) gives

$$\begin{aligned}
&\frac{w''}{w} - 2\mathcal{H}' - (\mathcal{H}^2 - K)(1+3c_s^2) \\
&\quad = \frac{3}{2}\mathcal{H}'(1+c_s^2) + \frac{3}{2}\mathcal{H}(c_s^2)' + \frac{9}{4}\mathcal{H}^2(1+c_s^2)^2 \\
&\qquad - 2\mathcal{H}' - (\mathcal{H}^2 - K)(1+3c_s^2)
\end{aligned}$$

$$= \frac{5}{4}\mathcal{H}^2 + \frac{3}{2}\mathcal{H}^2 c_s^2 + \frac{9}{4}\mathcal{H}^2 (c_s^2)^2 + K + 3K c_s^2$$
$$+ \frac{3c_s^2 - 1}{2}\mathcal{H}' + \frac{3}{2}\mathcal{H}(c_s^2)'. \tag{15}$$

Comparing (14) and (15), we obtain

$$\boxed{\frac{w''}{w} - 2\mathcal{H}' - (\mathcal{H}^2 - K)(1 + 3c_s^2) = \frac{\theta''}{\theta}}. \tag{16}$$

Finally, the source term follows directly from the definition of w:

$$w\,\frac{\kappa}{2}a^2 P\Gamma = \frac{2a^2\theta}{3\mathcal{H}}\frac{\kappa}{2}a^2 P\Gamma$$
$$= \frac{\kappa\theta}{3\mathcal{H}}a^4 P\Gamma. \tag{17}$$

Putting everything together, equation (10) becomes

$$\boxed{u_S'' - \left(\frac{\theta''}{\theta} + c_s^2\Delta\right)u_S = \frac{\kappa\theta}{3\mathcal{H}}a^4 P\Gamma}. \tag{18}$$

This is the desired canonical form.

Integral solution for long-wavelength modes

For super-Hubble modes where the spatial gradient term $c_{\mathrm{s}}^2\Delta$ is negligible, the homogeneous part of Eq. (4.3.17) admits $u_{\mathrm{S}} = \theta$ as an obvious particular solution. By reduction of order, the general solution in Fourier space is:

$$u_{\mathrm{S}}(k,\eta) = A(k)\theta(\eta) + B(k)\theta(\eta)\int^{\eta}\frac{\mathrm{d}\eta'}{\theta^2(\eta')} + \mathcal{O}\left(\frac{k^2 c_{\mathrm{s}}^2\theta}{\theta''}\right). \tag{4.3.18}$$

This solution holds for all modes during the matter-dominated era whenever non-adiabatic entropy perturbations Γ are negligible. In direct analogy with the tensor mode solution (4.3.12), Eq. (4.3.18) is recovered simply by replacing the scale factor $a(\eta)$ with $\theta(\eta)$.

First integral

For adiabatic perturbations on super-Hubble scales, Eq. (4.3.13) possesses an exact first integral. Introducing the gauge-invariant curvature perturbation on comoving slicing:

$$\zeta \equiv \Phi + \frac{2}{3\mathcal{H}}\frac{\Phi' + \mathcal{H}\Phi}{1 + w}, \tag{4.3.19}$$

the dynamical equation (4.3.13) takes the compact form:

$$\zeta' = \frac{2}{3}\frac{\mathcal{H}}{1+w}\left[\frac{K}{\mathcal{H}^2}\left(\frac{1+3w}{2\mathcal{H}}\Phi' + 3(c_{\mathrm{s}}^2 - w)\Phi\right) + \frac{\kappa}{2}\frac{a^2}{\mathcal{H}^2}P\Gamma - c_{\mathrm{s}}^2\left(\frac{k}{\mathcal{H}}\right)^2\Phi\right], \tag{4.3.20}$$

where background thermodynamic identities have been employed to evaluate w' and $\mathcal{H}'$. In the regime where spatial curvature is negligible ($K/\mathcal{H}^2 \ll 1$), for adiabatic fluctuations ($\Gamma = 0$) on super-Hubble scales ($k/\mathcal{H} \ll 1$), the quantity ζ is strictly conserved:

$$\zeta' = 0. \tag{4.3.21}$$

When non-adiabatic processes are present, Eq. (4.3.20) simplifies to:

$$\zeta' = \frac{\mathcal{H}}{\rho + P}\delta P_{\text{nad}}, \tag{4.3.22}$$

confirming that non-adiabatic entropy production drives departures from the conservation of ζ. Geometrically, ζ corresponds to the intrinsic three-dimensional Ricci scalar curvature evaluated on comoving spatial hypersurfaces.

Equivalently, the curvature perturbation evaluated in the flat-slicing gauge was introduced by Bardeen, Steinhardt, and Turner:

$$\zeta_{\text{BST}} \equiv -C + \frac{1}{3}\frac{\delta\rho}{\rho + P} = -C + \frac{1}{3}\frac{\delta}{1 + w}. \tag{4.3.23}$$

Choosing uniform density gauge

$$\delta\rho^U = \delta\rho + \rho' T \implies T = -\frac{\delta\rho}{\rho'} \tag{1}$$

Then

$$\zeta_{\text{BST}} = -C - \mathcal{H}T \tag{2}$$

Expressed in terms of the longitudinal potential Φ, this gauge-invariant variable is:

$$\zeta_{\text{BST}} = \Phi - \frac{2}{3}\frac{\mathcal{H}}{(1 + w)(\mathcal{H}^2 + K)}\left[\Phi' + \left(1 - \frac{K}{\mathcal{H}^2} + \frac{1}{3}\left(\frac{k}{\mathcal{H}}\right)^2\right)\mathcal{H}\Phi\right]. \tag{4.3.24}$$

Differentiating with respect to conformal time yields:

$$\zeta'_{\text{BST}} = -\frac{2}{3}\frac{\mathcal{H}^2}{(1 + w)(\mathcal{H}^2 + K)}\left[\frac{1}{3}\left(\frac{k}{\mathcal{H}}\right)^2(\Phi' + \mathcal{H}\Phi) + \frac{\kappa}{2}\frac{a^2}{\mathcal{H}^2}P\Gamma\right]. \tag{4.3.25}$$

Unlike Eq. (4.3.20), the right-hand side of Eq. (4.3.25) contains no explicit dependence on the background spatial curvature K. Consequently, ζ_{BST} remains exactly conserved for adiabatic ($\Gamma = 0$) super-Hubble modes ($k/\mathcal{H} \ll 1$), provided the decaying transient mode has vanished so that $\Phi' + \mathcal{H}\Phi$ does not diverge as $k/\mathcal{H} \to 0$.

On sub-Hubble scales, the comoving and flat-slicing curvature variables diverge according to the gradient relation:

$$\zeta = \zeta_{\text{BST}} - \frac{1}{3}\frac{\Delta\Phi}{\mathcal{H}' - \mathcal{H}^2}. \tag{4.3.26}$$

Remarkably, the conservation of ζ_{BST} on super-Hubble scales does not rely on the specific dynamics of Einstein's field equations, but follows solely from local energy-momentum conservation $\nabla_\mu T^{\mu\nu} = 0$ (Ref. [37]), ensuring its universal validity across generalized theories of gravity.

Effect of a variation in the cosmic fluid equation of state

To illustrate the utility of these conserved quantities, consider the evolution of the gravitational potential across a sudden change in the background equation of state. Suppose w changes from w_1 for $\eta < \eta_*$ (epoch 1) to w_2 for $\eta > \eta_*$ (epoch 2) in a flat, adiabatic background ($K = 0, \Gamma = 0$). During each epoch, the cosmic scale factor evolves as $a \propto t^{p_i} \propto \eta^{n_i}$, where $n_i = p_i/(1 - p_i)$ and $1 + w_i = 2/(3p_i)$.

Evaluating the long-wavelength general solution (4.3.18) yields:

$$\Phi = \frac{\mathcal{H}}{a^2}\left[A_- + A_+ \int a^2(1+w)\,\mathrm{d}\eta\right], \tag{4.3.27}$$

With $a(\eta) \propto \eta^{\frac{2}{1+3w}}$, we find $\mathcal{H} = \frac{a'}{a} = \frac{2}{1+3w}\frac{1}{\eta}$

$$\begin{aligned}
\Phi(\eta) &= \frac{\mathcal{H}}{a^2}\left[A_- + A_+ \int a^2(1+w)\,\mathrm{d}\eta\right] && (1)\\
&= \frac{2}{1+3w}\eta^{-\frac{5+3w}{1+3w}}A_- + \frac{2(1+w)}{5+3w}A_+ && (2)\\
&\approx \frac{2(1+w)}{5+3w}A_+ && (3)
\end{aligned}$$

where A_+ and A_- represent the amplitudes of the growing and decaying modes, respectively. Assuming the decaying mode during epoch 1 has had sufficient time to dissipate, the potential before the transition is dominated by the constant mode, $\Phi(\eta < \eta_*) \sim A_+(1+w_1)p_1/(p_1+1)|_{w=\frac{1}{3}} = 2A_+/[3(1+p_1)]$. Long after the transition, once the newly generated decaying transient has also died away, the potential settles to the constant plateau $\Phi(\eta \gg \eta_*) \sim A_+(1+w_2)p_2/(p_2+1)|_{w=0} = 2A_+/[3(1+p_2)]$. The net transition ratio is therefore:[4]

$$\frac{\Phi(\eta \gg \eta_*)}{\Phi(\eta < \eta_*)} = \frac{1+p_1}{1+p_2}. \tag{4.3.28}$$

For the cosmologically critical transition from radiation domination ($p_1 = 1/2$) to matter domination ($p_2 = 2/3$), Eq. (4.3.28) yields:

$$\frac{\Phi(\eta \gg \eta_*)}{\Phi(\eta < \eta_*)} = \frac{9}{10}, \tag{4.3.29}$$

for super-Hubble modes. The identical $9/10$ suppression factor follows immediately from the conservation of ζ across the transition, since for the dominant growing mode $\zeta \sim \{1 + 2/[3(1+w)]\}\Phi$ in both epochs. The same analytical result can also be established by matching the exact background two-fluid scale factor.

4.3.3 Scalar modes in the adiabatic regime

Throughout the following analysis, we assume negligible anisotropic shear stress ($\bar{\pi} = 0$) and vanishing entropy perturbations ($\Gamma = 0$). In this regime, the two gauge-invariant gravitational potentials coincide ($\Phi = \Psi$). We therefore focus on solving the master potential equation (4.3.13) under adiabatic conditions ($\Gamma = 0$), a regime that describes cosmic perturbation dynamics with exceptional accuracy throughout most of cosmic history.

Universe dominated by a fluid with equation of state $w \neq 0$

Consider a flat Universe ($K = 0$) dominated by a single fluid possessing a constant equation of state $w \neq 0$. Under adiabatic conditions, the sound speed is constant and equal to $c_s^2 = w$. The scale factor expands as a power law in conformal time, $a \propto \eta^\nu$ with $\nu \equiv 2/(1+3w)$. Defining $x \equiv k\eta$ and introducing the auxiliary variable $f \equiv x^\nu \Phi$, the Fourier-space potential equation (4.3.13) becomes:

$$\frac{\mathrm{d}^2 f}{\mathrm{d}x^2} + \frac{2}{x}\frac{\mathrm{d}f}{\mathrm{d}x} + \left(w - \frac{\nu(\nu+1)}{x^2}\right) f = 0. \tag{4.3.30}$$

[4] we ignore the decaying mode on both sides of the transition for computing this ratio.

The general analytical solution to Eq. (4.3.30) can be expressed in terms of spherical Bessel functions:

$$\Phi = -\frac{3\nu^2}{2}x^{-\nu}\left[Aj_\nu(c_\mathrm{s}x) + Bn_\nu(c_\mathrm{s}x)\right] \equiv -\frac{3\nu^2}{2}x^{-\nu}\mathcal{Z}_\nu(c_\mathrm{s}x), \tag{4.3.31}$$

where j_ν and n_ν denote the spherical Bessel functions of the first and second kind, and $\mathcal{Z}_\nu$ defines their linear combination. The comoving Poisson equation (4.1.17) then yields the comoving density contrast:

$$\delta^\mathrm{C} = x^{2-\nu}\mathcal{Z}_\nu(c_\mathrm{s}x), \tag{4.3.32}$$

while the velocity-potential relation (4.2.68) determines the peculiar velocity:

$$kV = -\frac{3}{4}x^{1-\nu}\left[\mathcal{Z}_\nu(c_\mathrm{s}x) - \frac{c_\mathrm{s}x}{\nu+1}\mathcal{Z}_{\nu-1}(c_\mathrm{s}x)\right]. \tag{4.3.33}$$

Using the asymptotic limits of the spherical Bessel functions, the asymptotic behavior of the perturbation fields in the super-acoustic and sub-acoustic regimes is summarized below:

	$c_\mathrm{s}x \ll 1$	$c_\mathrm{s}x \gg 1$
Φ	$\Phi_+ + \Phi_- x^{-1-2\nu}$	$\Phi_+ x^{-(1+\nu)}\cos(c_\mathrm{s}x - \alpha_\nu)$
$-\delta^\mathrm{C}$	$\frac{2}{3\nu^2}\left(\Phi_+ x^2 - \Phi_- x^{1-2\nu}\right)$	$\frac{2}{3\nu^2}\Phi_+ x^{1-\nu}\cos(c_\mathrm{s}x - \alpha_\nu)$
$-kV$	$\frac{2}{3\nu(1+w)}\left[\Phi_+ x - \Phi_-\left(1+\nu^{-1}\right)x^{-2\nu}\right]$	$\frac{2}{3\nu(1+w)}\Phi_+ x^{1-\nu}\cos(c_\mathrm{s}x - \alpha_\nu)$

Here the acoustic phase shift is $\alpha_\nu \equiv \frac{\pi}{2}(\nu+1)$. On sub-acoustic scales ($c_\mathrm{s}x \gg 1$), the density perturbation δ^C undergoes acoustic oscillations with an amplitude that experiences power-law damping whenever $-1/2 < w < 1/3$.

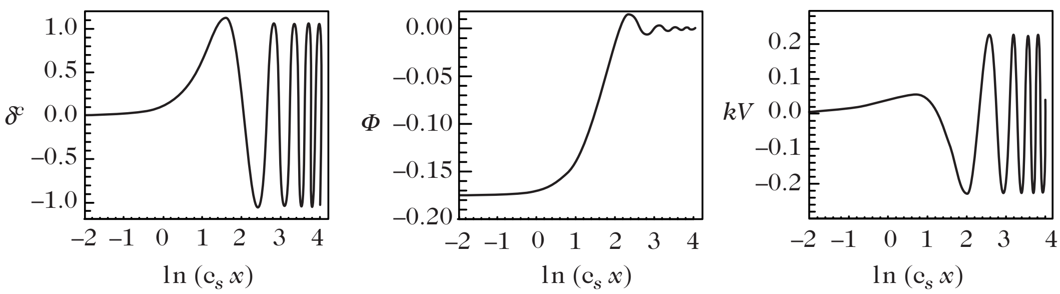


Figure 4.5: Evolution of the density contrast (left), gravitational potential (middle), and velocity perturbation (right) for scalar modes of a radiation fluid ($w = c_\mathrm{s}^2 = 1/3$) in a radiation-dominated Universe ($\nu = 1$) as a function of $c_\mathrm{s}x = c_\mathrm{s}k\eta$. The mode amplitude remains constant outside the sound horizon ($c_\mathrm{s}x < 1$) and initiates damped acoustic oscillations once inside ($c_\mathrm{s}x > 1$).

Radiation domination ($w = 1/3, \nu = 1$) represents the marginal threshold where the density oscillation amplitude remains constant without damping (see Fig. 4.5). Outside the acoustic horizon ($c_\mathrm{s}x \ll 1$), the growing mode of the gravitational potential remains strictly constant.

Furthermore, on super-Hubble scales during the radiation era, the comoving density fluctuation of the radiation fluid grows quadratically with the scale factor:

$$\delta_\mathrm{r}^\mathrm{C} \propto a^2. \tag{4.3.34}$$

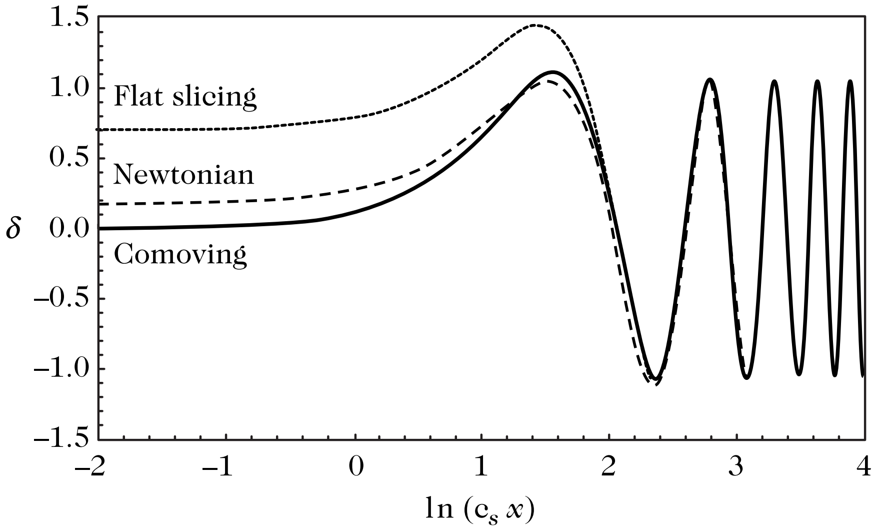


Figure 4.6: Evolution of the density contrasts $\delta^{\rm C}$ (comoving, solid line), $\delta^{\rm N}$ (Newtonian/longitudinal, dashed line), and $\delta^{\rm F}$ (flat slicing, dotted line) as a function of $c_{\rm s}x = c_{\rm s}k\eta$. These density contrast $\delta^{\rm C}$, $\delta^{\rm N}$, and $\delta^{\rm F}$ are gauge invariant but not the same physical object. Fundamentally they measure perturbation on different hypersurfaces. On super-acoustic scales ($c_{\rm s}x < 1$), substantial disparities exist among the three gauge definitions, which rapidly converge once modes penetrate the acoustic horizon ($c_{\rm s}x > 1$), illustrating the gauge ambiguity inherent to super-Hubble density perturbations. This happens because on subhorizon scale the general relativistic effects become unimportant and the newtonian description becomes valid. In this limit, all issue of gauge choice becomes irrelevant as all sensible gauges approach each other.

Matter fluid in a matter-dominated Universe

In the case of a dust-dominated cosmos ($w = c_{\rm s}^2 = 0$ and $\nu = 2$), Eq. (4.3.30) reduces to:

$$\frac{{\rm d}^2 f}{{\rm d}x^2} + \frac{2}{x}\frac{{\rm d}f}{{\rm d}x} - \frac{6}{x^2} f = 0. \tag{4.3.35}$$

The general solution provides the explicit dynamics of the potential, density contrast, and peculiar velocity:

$$\Phi = \Phi_+ + \Phi_- x^{-5}, \qquad \delta^{\rm C} = -\frac{1}{6}x^2\Phi, \qquad \text{and} \qquad -kV = \frac{1}{3}\Phi_+ x - \frac{1}{2}\Phi_- x^{-4}. \tag{4.3.36}$$

Since $a \propto \eta^2 \propto x^2$, the comoving density contrast of matter grows strictly linearly with the scale factor:

$$\delta_{\rm m}^{\rm C} \propto a, \tag{4.3.37}$$

independent of the perturbation wavelength. This reproduces the conclusion obtained from the Newtonian sub-Hubble analysis [Eq. (4.1.22)].

The constancy of the gravitational potential during matter domination and the modest linear growth rate $\delta \propto a$ prompted Landau and Lifshitz in 1946 to deduce that gravitational instability in an expanding FLRW background could not account for the observed cosmic structure (Ref. [38]). Indeed, initial thermal fluctuations could only grow by a factor of $\approx 10^4$ during the matter-dominated epoch, an amplification far too modest to generate galaxies and galaxy clusters today. This historical impasse underscored the necessity of a physical mechanism—such as primordial cosmic inflation—capable of generating seed perturbations of order 10^{-4} at the threshold of the matter-dominated era.

Effect of curvature

Because spatial curvature scales dynamically as a^{-2}, its energy density can dominate only at late cosmic times, well after radiation has become negligible. In a Universe governed by matter and spatial curvature, integrating Eq. (4.3.13) analytically for an open spatial geometry ($K=-1$) yields:

$$\Phi(k,\eta) = A(k)\frac{\sinh\eta}{\sinh^6(\eta/2)} + B(k)\frac{\sinh\eta(\sinh\eta - 3\eta) + 8\sinh^2(\eta/2)}{\sinh^6(\eta/2)}. \tag{4.3.38}$$

Once curvature dominates the cosmic dynamics ($\eta > 1$), the potential decays exponentially as $\Phi \sim B(k)e^{-\eta}$, and Poisson's equation (4.1.17) reveals that the density contrast freezes out to a constant plateau:

$$\delta_{\mathrm{m}}^{\mathrm{C}} = -\frac{8k^2|1-\Omega_0|}{\Omega_0}\sinh^2\left(\frac{\eta}{2}\right)\Phi \sim -\frac{8k^2B(k)|1-\Omega_0|}{\Omega_0}. \tag{4.3.39}$$

Thus, negative spatial curvature halts further gravitational clustering.

For a closed spatial geometry ($K=+1$), the general analytical solution to Eq. (4.3.13) is:

$$\Phi(k,\eta) = A(k)\frac{\sin\eta}{\sin^6(\eta/2)} + B(k)\frac{\sin\eta(\sin\eta - 3\eta) + 8\sin^2(\eta/2)}{\sin^6(\eta/2)}. \tag{4.3.40}$$

As $\eta \to \pi - \varepsilon$, the cosmic expansion nears its maximal turnaround radius, where $\mathcal{H} \to 0$ and $\Phi \sim 8B(k)$. In the subsequent recollapsing phase, both perturbation modes diverge rapidly, showing that cosmic contraction exponentially amplifies gravitational collapse. Immediately preceding the final big crunch at $\eta = 2\pi - \varepsilon$, the potential behaves as $\Phi \sim 2[6\pi B(k) - A(k)]/\sin^5(\eta/2)$, and the density contrast diverges as:

$$\delta_{\mathrm{m}}^{\mathrm{C}} \sim -2k^2\frac{6\pi B(k) - A(k)}{\sin^3(\eta/2)}\frac{|1-\Omega_0|}{\Omega_0}. \tag{4.3.41}$$

Figure 4.7 illustrates the contrasting evolution of $\delta_{\mathrm{m}}^{\mathrm{C}}$ in spherical and hyperbolic universes.

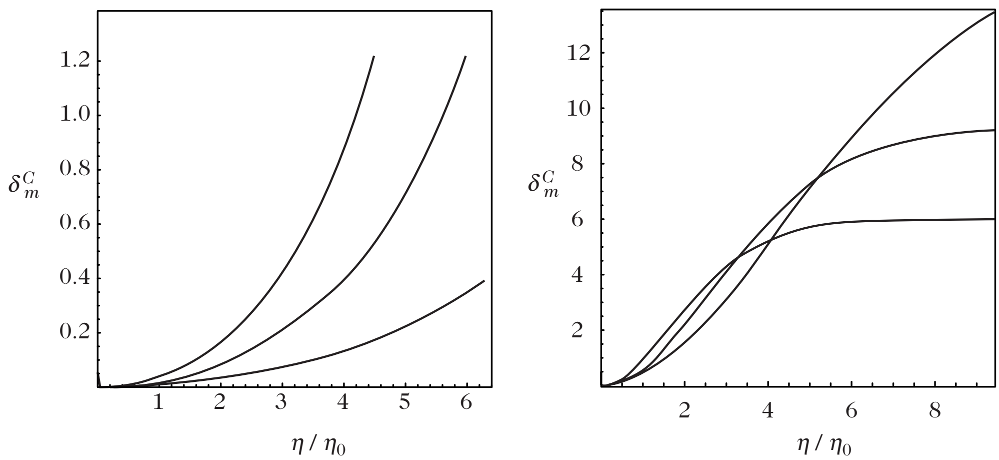


Figure 4.7: Evolution of the density contrast $\delta_{\mathrm{m}}^{\mathrm{C}}$ in a Universe with spherical spatial sections (left panel, for $\Omega_0 = 1.2, 1.3, 1.4$) and hyperbolic spatial sections (right panel, for $\Omega_0 = 0.2, 0.3, 0.4$) as a function of conformal time η/η_0.

Effect of the cosmological constant

A cosmological constant Λ also comes to dominate only at late cosmological times when radiation is negligible. Introducing the normalized variable $y_\Lambda \equiv a/a_\Lambda = (\Lambda/\kappa\rho)^{1/3}$, the transition from matter domination to cosmological constant domination occurs at the redshift:

$$1 + z_\Lambda = \left(\frac{\Omega_{\Lambda 0}}{\Omega_{\mathrm{m}0}}\right)^{1/3},$$

and the Friedmann equation takes the form:

$$\mathcal{H}^2 = \frac{\mathcal{H}_\Lambda^2}{2}\left(\frac{1}{y_\Lambda} + y_\Lambda^2\right).$$

Equation (4.3.13) then transforms into:

$$\ddot{\Phi} + \frac{1}{2y_\Lambda}\frac{7 + 10y_\Lambda^3}{1 + y_\Lambda^3}\dot{\Phi} + \frac{3y_\Lambda}{y_\Lambda^3 + 1}\Phi = 0. \tag{4.3.42}$$

The corresponding matter density contrast is given by:

$$\delta_{\mathrm{m}}^{\mathrm{C}} = -\frac{2}{3}\left(\frac{k}{\mathcal{H}_\Lambda}\right)^2 y_\Lambda \Phi. \tag{4.3.43}$$

Once dark energy dominates ($y_\Lambda \gg 1$), the potential decays as $\Phi \propto 1/y_\Lambda$, causing the density contrast to freeze out to a constant plateau (Fig. 4.8). In a Λ-dominated Universe, structure growth ceases near redshift $z \sim z_\Lambda$.

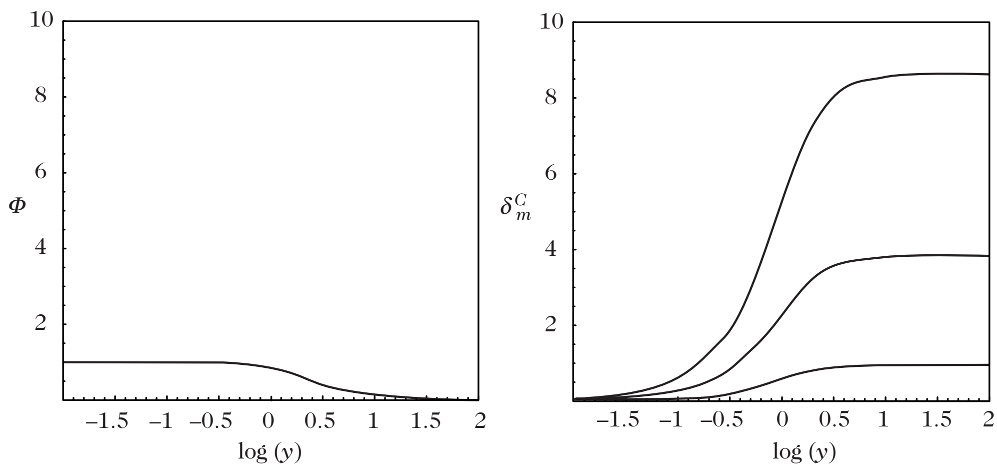


Figure 4.8: Evolution of the gravitational potential (left) and the density contrast (right) in a Universe dominated by a cosmological constant. The right panel displays the three modes $k/\mathcal{H}_\Lambda = 1, 2, 3$.

Case of a Universe dominated by a mixture of matter and radiation

We finally consider a spatially flat Universe ($K = 0$) dominated by a cosmic mixture of pressureless matter ($w = 0$) and radiation ($w = 1/3$). Adopting the equality-normalized scale factor $y = a/a_{\mathrm{eq}}$ defined in Eq. (4.3.6), the equation-of-state parameter and the adiabatic sound speed are:

$$w = \frac{1}{3}(1 + y)^{-1} \qquad \text{and} \qquad c_{\mathrm{s}}^2 = \frac{1}{3}\left(1 + \frac{3}{4}y\right)^{-1}. \tag{4.3.44}$$

Using the relations $\Phi' = \mathcal{H}y\,\mathrm{d}\Phi/\mathrm{d}y$, $\Phi'' = \mathcal{H}^2y^2\,\mathrm{d}^2\Phi/\mathrm{d}y^2 + y''\,\mathrm{d}\Phi/\mathrm{d}y$, and the Friedmann equation (4.3.9), the master potential equation (4.3.13) becomes:

$$\frac{\mathrm{d}^2\Phi}{\mathrm{d}y^2}+\frac{1}{2y}\left(7-\frac{1}{1+y}+\frac{8}{4+3y}\right)\frac{\mathrm{d}\Phi}{\mathrm{d}y}+\frac{1}{y(1+y)(4+3y)}\Phi+\frac{8}{3}\frac{1}{4+3y}\frac{1}{1+y}\left(\frac{k}{k_{\mathrm{eq}}}\right)^2\Phi=-\frac{3}{2y^2}w\Gamma, \tag{4.3.45}$$

where k_{eq} is defined in Eq. (4.3.10). For adiabatic initial conditions ($\Gamma = 0$), numerical integration of Eq. (4.3.45) yields the potential trajectories shown in Fig. 4.9.

In the presence of entropy perturbations, the dynamical behavior of Γ must be supplied to solve Eq. (4.3.45). On large wavelengths, Γ remains nearly constant in time, rendering the adiabatic solutions of Fig. 4.9 a solid leading-order approximation.

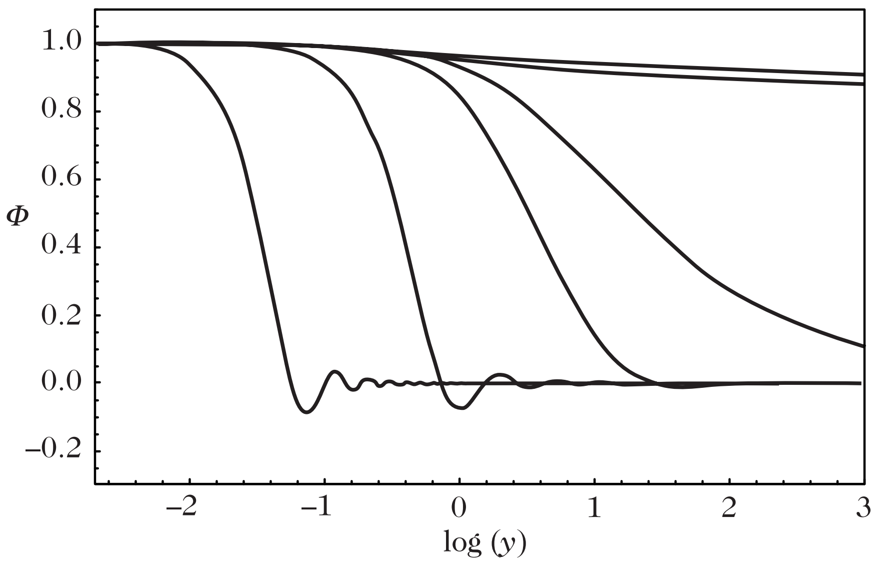


Figure 4.9: Evolution of the gravitational potential in a Universe dominated by a mixture of matter and radiation, assuming $\Gamma = 0$, for modes $k/k_{\mathrm{eq}} = 10^{-3}, 0.1, 1, 2, 10, 100$. (This should be compared with Fig. 5.12, where the dynamical evolution of the entropy perturbation is explicitly accounted for.)

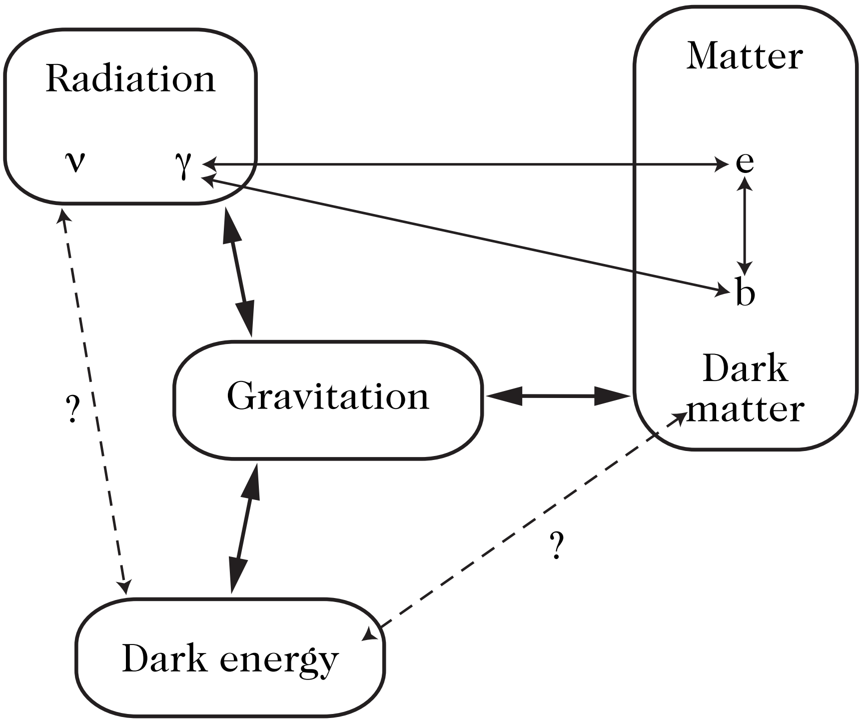
In the long-wavelength regime ($k/k_{\mathrm{eq}} \ll 1$), Eq. (4.3.45) can be solved in closed analytical form. Introducing the auxiliary field $\varphi \equiv y^3\Phi/\sqrt{1+y}$, one finds $\mathrm{d}\Phi/\mathrm{d}y \propto y^2(4+3y)/(1+y)^{3/2}$, which can be integrated directly.

$$\Phi = \frac{\Phi_0}{10y^3}\left(16\sqrt{1+y}+9y^3+2y^2-8y-16\right). \tag{4.3.46}$$

For super-Hubble perturbations, this reproduces the asymptotic ratio $\Phi(y \gg 1)/\Phi_0 = 9/10$, which represents a particular instance of the general result (4.3.29).

4.3.4 Mixture of several fluids

In realistic cosmological settings, multiple physical fluids coexist—such as baryons, cold dark matter, photons, and neutrinos. These different cosmological fluids interact with one another via gravitation, but they may also experience non-gravitational couplings. Figure 4.10 summarizes the principal fluid components that should be considered alongside their mutual interactions.



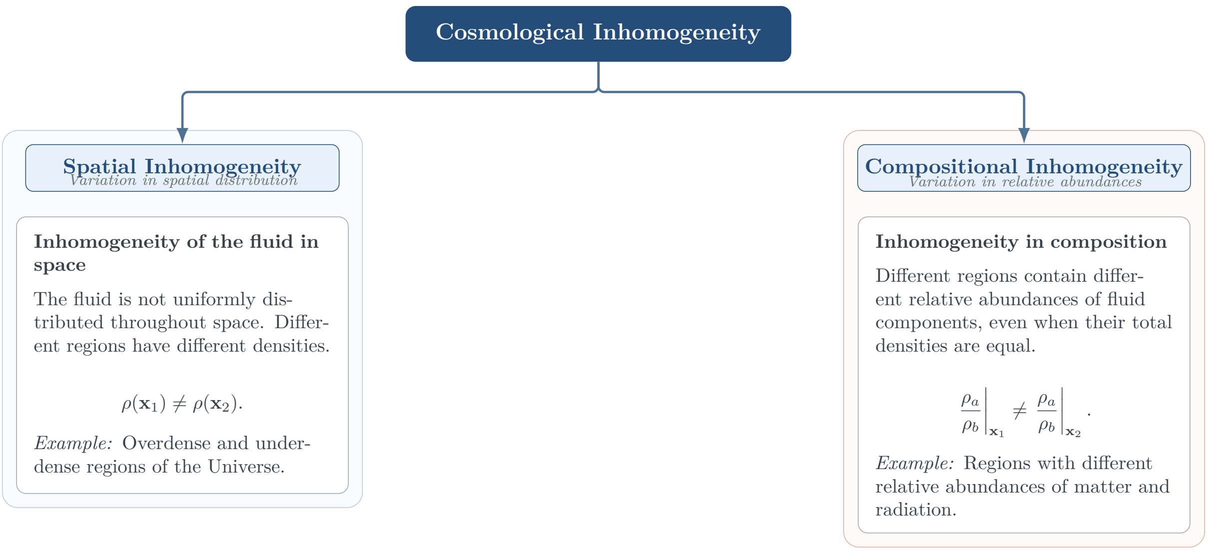


Figure 4.10: The cosmological fluid components and their interactions. Baryons and electrons are tightly coupled through the Coulomb interaction, while photons couple to charged matter via Compton scattering. Dark matter is assumed to interact with ordinary baryonic matter solely through gravity. It is not excluded that dark energy may interact with ordinary matter or with dark matter through an as yet unknown physical mechanism.

Description of the interactions

In the most general coupled multi-fluid system, conservation of the energy-momentum tensor holds only for the aggregate fluid mixture, whereas the individual energy-momentum tensors satisfy non-conserved balance equations of the form:

$$\nabla_\mu T_a^{\mu\nu} = Q_a^\nu, \tag{4.3.47}$$

where Q_a^μ denotes the external four-force and energy exchange acting on component a.

> This becomes important to take into account the interaction and decay rate energy transfer betweeen fluids and fields.

These interaction terms must satisfy the constraint:

$$\sum_a Q_a^\mu = 0, \tag{4.3.48}$$

which is simply the cosmological manifestation of Newton's third law of action and reaction. This condition follows directly from the contracted Bianchi identities ensuring total energy-momentum conservation $\nabla_\mu T^{\mu\nu} = 0$.

At background order, the spatial homogeneity and isotropy of the spacetime imply that the interaction four-vector has vanishing spatial components:

$$Q_\mu^a = (-aQ_a, \mathbf{0}), \tag{4.3.49}$$

so that $Q_\mu^a Q_\nu^a \bar{g}^{\mu\nu} = -Q_a^2$. Notice that $(\bar{g}_{\mu\nu} + \bar{u}_\mu \bar{u}_\nu) Q_a^\mu = 0$, meaning that background spacetime symmetries preclude the existence of any net spatial force between the fluids at the unperturbed background level. Taking energy transfer into account, the background continuity equation for component a generalizes to:

$$\rho_a' + 3\mathcal{H}(1 + w_a)\rho_a = aQ_a, \tag{4.3.50}$$

> If Q_a varues wutg position, it can lead to violation of (4.3.70) and generate isocurvature perturbation.

which yields the dynamical evolution of the equation-of-state parameter:

$$w_a' = -\left[3\mathcal{H}(1 + w_a) - \frac{Q_a}{\rho_a}\right](c_a^2 - w_a), \tag{4.3.51}$$

where $w_a \equiv P_a/\rho_a$ and $c_a^2 \equiv P_a'/\rho_a'$ denote the equation of state and the adiabatic sound speed of component a, respectively.

At linear order in cosmological perturbations, Q_a^μ can be decomposed with respect to the fluid four-velocity u_a^μ as:

$$Q_a^\mu = Q_a u_a^\mu + F_a^\mu, \qquad \text{with} \qquad F_a^\mu u_{a\mu} = 0. \tag{4.3.52}$$

We define $af_a^\mu \equiv (\delta_\nu^\mu + u^\mu u_\nu) F_a^\nu$, ensuring $f_a^0 = 0$, while the spatial part f_a^i decomposes into scalar and vector modes according to $f_a^i = \bar{f}_a^i + D^i f_a$. Under an infinitesimal gauge transformation (4.2.11), these quantities transform as:

$$\delta Q_a \to \delta Q_a + Q_a' T, \tag{4.3.53}$$

and f_a^i is inherently gauge invariant because it vanishes in the unperturbed background. We can therefore construct the gauge-invariant perturbation:

$$\delta Q_a^{\mathrm{N}} \equiv \delta Q_a + Q_a'(B - E'), \tag{4.3.54}$$

which is gauge invariant. With these notations, the continuity equation (4.2.74) and the Euler equation (4.2.75) for component a generalize as:

$$\left(\frac{\delta_a^{\mathrm{N}}}{1+w_a}\right)' = -(\Delta V_a - 3\Psi') - 3\mathcal{H}\frac{w_a}{1+w_a}\Gamma_a + \frac{a}{\rho_a + P_a}\left[\delta Q_a^{\mathrm{N}} + Q_a(\Phi - \delta_a^{\mathrm{N}})\right], \tag{4.3.55}$$

and

$$V_a' + \mathcal{H}V_a = -\Phi - \frac{c_a^2}{1+w_a}\delta_a^{\mathrm{C}} + \frac{w_a}{1+w_a}\left[\Gamma_a + \frac{2}{3}(\Delta + 3K)\bar{\pi}_a\right] + \frac{a}{\rho_a + P_a}(f_a - Q_a c_a^2 V_a). \tag{4.3.56}$$

System of fluids interacting gravitationally

We consider the simplified scenario where a set of fluids interact solely through gravity. In this case, $Q_a^\mu = 0$ for each individual fluid, and the total energy-momentum tensor is the simple sum:

$$T^{\mu\nu} = \sum_a T_a^{\mu\nu}.$$

> We have assumed that there is no energy or momentum transfer between different fuilds otherwise there would be collision term present corresponding to each kind of interaction coming from Boltzmann transport equation.

Each energy-momentum tensor can be decomposed as in Eq. (4.2.18), so that the aggregate density, pressure, and velocity field are defined by:

$$\rho = \sum_a \rho_a, \qquad P = \sum_a P_a, \qquad (\rho + P)v^i = \sum_a (\rho_a + P_a)v_a^i. \tag{4.3.57}$$

We have:

$$\Omega w = \sum_a \Omega_a w_a, \tag{4.3.58}$$

and

$$\Omega c_{\mathrm{s}}^2 = \sum_a \frac{1+w_a}{1+w}\Omega_a c_a^2. \tag{4.3.59}$$

For simplicity, let us examine a system of two fluids. It is described by two density contrasts, δ_a and δ_b, and two velocity fields, v_a^i and v_b^i. It is convenient to transform these degrees of freedom into a set of variables, δ and v, characterizing the total fluid, and a set of variables, S_{ab} and V_{ab}^i, describing their relative properties. The definitions (4.3.57) imply that:

$$\Omega\delta = \sum_a \Omega_a \delta_a \qquad \text{and} \qquad \Omega(1+w)v = \sum_a \Omega_a(1+w_a)v_a, \tag{4.3.60}$$

for the scalar modes. These total quantities satisfy the continuity equation (4.2.74) and the Euler equation (4.2.75). The total entropy perturbation Γ is defined as before by relation (4.2.26), so that for such a mixture of fluids, it takes the form:

$$\Omega w \Gamma = \sum_a w_a \Omega_a \Gamma_a + \sum_a \left(c_{\mathrm{s}}^2 - c_a^2\right)\Omega_a \delta_a. \tag{4.3.61}$$

The first term corresponds to the entropy contribution from the individual components of the mixture, while the second term, which does not vanish even if $\Gamma_a = 0$ for each component, represents the entropy of mixing. One can check that this second contribution is gauge invariant.

The Einstein field equations take the general form obtained previously, but for the quantities describing the total fluid. In particular, both (4.2.66) and (4.2.67) can be rewritten as:

$$(\Delta + 3K)\Psi = \frac{3}{2}\mathcal{H}^2 \sum_a \Omega_a \delta_a^{\mathrm{C}}, \tag{4.3.62}$$

$$\Psi - \Phi = 3\mathcal{H}^2 \sum_a \Omega_a w_a \bar{\pi}_a. \tag{4.3.63}$$

Now, S_{ab} and V_{ab} are defined, respectively, by:

$$S_{ab} \equiv \frac{\delta_a}{1+w_a} - \frac{\delta_b}{1+w_b} \qquad \text{and} \qquad V_{ab} \equiv v_a - v_b, \tag{4.3.64}$$

which are, by construction, gauge invariant.

> This can be thought of as relative weighted density contrast. It tells us how the spatial variation in entropy per particle between different species changes with position.

Using the relations (4.3.60), we readily obtain the inverse relation:

$$\left(\frac{\Omega_b}{1+w_a} + \frac{\Omega_a}{1+w_b}\right)\delta_a = \frac{\Omega}{1+w_b}\delta + \Omega_b S_{ab}, \tag{4.3.65}$$

and the analogous expression for δ_b is obtained by interchanging the indices a and b.

The evolution equations for the relative quantities S_{ab} and V_{ab} are obtained by combining the continuity equation in the form (4.2.76) and the Euler equation (4.2.79) for each fluid:

$$S'_{ab} = -\Delta V_{ab} - 3\mathcal{H}\Gamma_{ab}, \tag{4.3.66}$$

$$\begin{aligned} V'_{ab} = {} & -\mathcal{H}V_{ab} - \left(c_a^2 - c_b^2\right)\frac{\delta^{\mathrm{C}}}{1+w} - \left[c_a^2(1+w_b)\frac{\Omega_b}{\Omega} + c_b^2(1+w_a)\frac{\Omega_a}{\Omega}\right]\frac{S_{ab}}{1+w} \\ & - \left[\frac{2}{3}(\Delta + 3K)\bar{\pi}_{ab} + \Gamma_{ab}\right], \end{aligned} \tag{4.3.67}$$

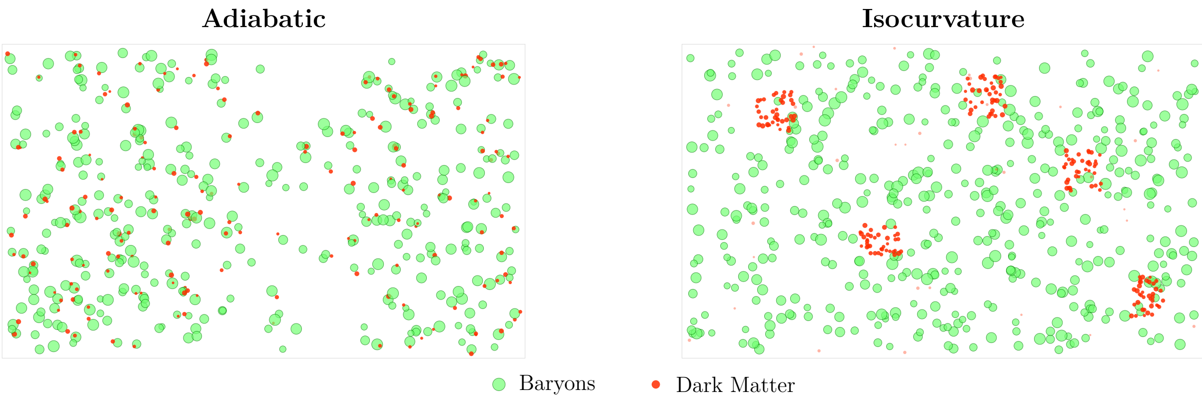
with the definitions:

$$\Gamma_{ab} \equiv \frac{w_a}{1+w_a}\Gamma_a - \frac{w_b}{1+w_b}\Gamma_b, \tag{4.3.68}$$

$$\bar{\pi}_{ab} \equiv \frac{w_a}{1+w_a}\bar{\pi}_a - \frac{w_b}{1+w_b}\bar{\pi}_b. \tag{4.3.69}$$

To completely solve the evolution of a system of two interacting fluids, we thus need to solve the evolution equation (4.3.13) for the gravitational potential, where Γ is now given by relation (4.3.61), together with the pair of equations (4.3.66) and (4.3.67); the relation (4.3.62) then gives δ^{C}.

Adiabatic and isocurvature modes



Two different types of initial conditions can be distinguished for a system of two fluids: we can require that the entropy perturbation vanishes initially (adiabatic initial conditions), or, on the contrary, arrange that the gravitational potential initially vanishes, i.e. 'balance' the density perturbations so that $\delta^{\mathrm{C}} = 0$ (isocurvature initial conditions). We thus define the two limiting regimes:

- **adiabatic**: it is assumed that $\Gamma = S_{ab} = 0$, from which we deduce that:
$$\frac{\delta_a}{1+w_a} = \frac{\delta_b}{1+w_b}, \tag{4.3.70}$$
- **isocurvature**: in this case, we impose that $\Psi = 0$ and thus we have:
$$\delta^{\mathrm{C}} = 0, \qquad \Omega_a \delta_a^{\mathrm{C}} + \Omega_b \delta_b^{\mathrm{C}} = 0. \tag{4.3.71}$$

Since the evolution equations are linear, any system can always be decomposed into a linear combination of an initially adiabatic system and an initially isocurvature system.

Mixture of two perfect fluids

The discussion in the previous section is very general. In practice, we are interested in simpler special cases. Let us consider that of a mixture of two perfect fluids ($\Gamma_a = 0$, $\bar{\pi}_a = 0$) with purely gravitational interactions. Using (4.3.58) and (4.3.59), the relation (4.3.61) reduces to:

$$w\Gamma = -\frac{\Omega_a \Omega_b (1+w_a)(1+w_b)}{\Omega^2 (1+w)} \left(c_a^2 - c_b^2\right) S_{ab},$$

so that a closed system can be written for δ^{C} and S_{ab}, namely:

$$\boxed{\begin{aligned}
&\Phi'' + 3\mathcal{H}\left(1+c_{\mathrm{s}}^2\right)\Phi' + \left[2\mathcal{H}' + \left(\mathcal{H}^2 - K\right)\left(1+3c_{\mathrm{s}}^2\right)\right]\Phi - c_{\mathrm{s}}^2 \Delta\Phi = \\
&\qquad\qquad \frac{3}{2}\mathcal{H}^2 \frac{\Omega_a \Omega_b (1+w_a)(1+w_b)}{\Omega(1+w)}\left(c_a^2 - c_b^2\right) S_{ab}, && (4.3.72)\\
&\qquad\qquad (\Delta + 3K)\Phi = \frac{3}{2}\mathcal{H}^2 \Omega \delta^{\mathrm{C}}, && (4.3.73)\\
&S_{ab}'' + \mathcal{H} S_{ab}' - \frac{\Omega_a(1+w_a)c_b^2 + \Omega_b(1+w_b)c_a^2}{\Omega(1+w)}\Delta S_{ab} = \left(c_a^2 - c_b^2\right)\frac{\Delta\delta^{\mathrm{C}}}{1+w}. && (4.3.74)
\end{aligned}}$$

This system reduces to a system of two coupled second-order differential equations if the gravitational potential is eliminated from (4.3.72) by differentiating (4.3.73).

Isocurvature perturbations from non-interacting perfect fluids

Consider two non-interacting perfect fluids with constant equations of state,

$$p_a = w_a \rho_a, \qquad p_b = w_b \rho_b.$$

Since the fluids are separately conserved,

$$\dot{\rho}_i + 3H(\rho_i + p_i) = 0, \qquad i = a, b.$$

Thus,

$$\rho_i \propto a^{-3(1+w_i)}.$$

In particular,

$$\rho_a = \rho_{a*}\left(\frac{a}{a_*}\right)^{-3(1+w_a)}, \qquad \rho_b = \rho_{b*}\left(\frac{a}{a_*}\right)^{-3(1+w_b)}. \tag{1}$$

The curvature perturbation associated with each fluid is

$$\zeta_i = -\psi - H\frac{\delta\rho_i}{\dot{\rho}_i}.$$

Using the continuity equation,

$$\dot{\rho}_i = -3H(1+w_i)\rho_i,$$

we obtain

$$\zeta_i = -\psi + \frac{1}{3(1+w_i)}\frac{\delta\rho_i}{\rho_i}.$$

The relative entropy, or isocurvature, perturbation between the two fluids is therefore

$$\begin{aligned} S_{ab} &\equiv 3(\zeta_a - \zeta_b) && (2)\\ &= \frac{1}{1+w_a}\frac{\delta\rho_a}{\rho_a} - \frac{1}{1+w_b}\frac{\delta\rho_b}{\rho_b}. && (3) \end{aligned}$$

Thus,

$$\boxed{S_{ab} = \frac{\delta_a}{1+w_a} - \frac{\delta_b}{1+w_b}}, \qquad \delta_i \equiv \frac{\delta\rho_i}{\rho_i}.$$

Now consider the dilution laws themselves. Taking a perturbation of

$$\rho_i \propto a^{-3(1+w_i)}$$

gives, for a common scale factor,

$$\frac{\delta\rho_i}{\rho_i} = \frac{\delta\rho_{i*}}{\rho_{i*}}.$$

Therefore the dilution law changes the background densities, but the common expansion does not by itself generate a new relative density perturbation.
For the ratio of the two densities, however,

$$\frac{\rho_a}{\rho_b} = \left(\frac{\rho_{a*}}{\rho_{b*}}\right)\left(\frac{a}{a_*}\right)^{-3(w_a-w_b)}.$$

Hence, if $w_a \neq w_b$, the *background composition* evolves with the expansion. For example,

$$\rho_m \propto a^{-3}, \qquad \rho_r \propto a^{-4}, \qquad \frac{\rho_m}{\rho_r} \propto a.$$

Nevertheless, the common dilution factor cancels when considering the spatial variation of the ratio:

$$\frac{\delta(\rho_a/\rho_b)}{\rho_a/\rho_b} = \frac{\delta\rho_a}{\rho_a} - \frac{\delta\rho_b}{\rho_b}.$$

Consequently, different dilution rates do not, by themselves, generate an isocurvature perturbation from initially adiabatic conditions. If initially

$$S_{ab} = 0,$$

then, for separately conserved barotropic fluids on super-Hubble scales,

$$\dot{\zeta}_a \simeq 0, \qquad \dot{\zeta}_b \simeq 0,$$

and hence

$$\boxed{\dot{S}_{ab} \simeq 0}.$$

Thus the dilution law determines how the individual components evolve with the expansion and how their *background relative abundance* changes, but a non-zero isocurvature perturbation requires a relative perturbation between the components to be present or a physical mechanism capable of generating one.

4.4 Power spectrum of density fluctuations

We will now be interested in the power spectrum of dark-matter density fluctuations. For this, and for the sake of simplicity, we assume that all matter is dark (i.e. $\Omega_{b0} \ll \Omega_{c0} \simeq \Omega_{m0} \simeq 1$). Moreover, we restrict ourselves to a flat Universe ($K = 0$ and $\Omega = 1$) with no cosmological constant. The Universe thus contains, by construction, only dark matter and radiation. We choose the normalization $a_0 = 1$.

As before, we introduce the scale factor y normalized at equality [see Eq. (4.3.6)] so that the Hubble law takes the form (4.3.9). The value of y today is:

$$y_0 = \frac{a_0}{a_{\rm eq}} = 1 + z_{\rm eq} \simeq 2.4 \times 10^4\, h^2$$

[see (4.8)]. The Friedmann equation evaluated today makes it possible to compute the numerical value of the mode entering the Hubble radius at the time of matter–radiation equality (4.3.10), namely:

$$k_{\rm eq} = H_0\sqrt{2\frac{\Omega_{\rm m0}a_0}{a_{\rm eq}}} = 0.072\,\Omega_{\rm m0}h^2\,{\rm Mpc}^{-1}, \tag{4.4.1}$$

corresponding to a wavelength:

$$k_{\rm eq}^{-1} = \mathcal{H}_{\rm eq}^{-1} = \frac{14}{\Omega_{\rm m0}h^2}\,{\rm Mpc}. \tag{4.4.2}$$

In general, a given mode k enters the Hubble radius when $k = \mathcal{H}$, and the Friedmann equation (4.3.9) allows us to determine that this occurs at a redshift:

$$y_*\left(\bar{k}\right) = \frac{1+\sqrt{1+8\bar{k}}}{4\bar{k}^2}, \qquad 1 + z_*\left(\bar{k}\right) = \frac{y_0}{y_*\left(\bar{k}\right)}, \tag{4.4.3}$$

where we have introduced the reduced wavenumber:

$$\bar{k} \equiv \frac{k}{k_{\rm eq}} \tag{4.4.4}$$

for further notational convenience.

4.4.1 Two equivalent approaches

We can follow two approaches to study this physical system. The first is to consider the four equations of evolution for the density contrasts and the velocity perturbations of each fluid, and the Poisson equation. The second is to use the results of the previous section for the total density contrast and the entropy perturbations. We will use one or the other formulation depending on the situation.

Evolution of the density contrasts

Our system is composed of a mixture of radiation and matter that only interacts gravitationally, so that the system of equations is deduced from (4.2.74) and (4.2.75). In Fourier modes, they reduce to:

$$\delta_{\mathrm{m}}^{\mathrm{N}\prime} = k^2 V_{\mathrm{m}} + 3\Phi', \tag{4.4.5}$$

$$\delta_{\mathrm{r}}^{\mathrm{N}\prime} = \frac{4}{3}k^2 V_{\mathrm{r}} + 4\Phi', \tag{4.4.6}$$

$$V_{\mathrm{m}}' = -\mathcal{H}V_{\mathrm{m}} - \Phi, \tag{4.4.7}$$

$$V_{\mathrm{r}}' = -\Phi - \frac{1}{4}\delta_{\mathrm{r}}^{\mathrm{N}}, \tag{4.4.8}$$

where we have used that $\Psi = \Phi$ since $\bar{\pi} = 0$. Another equation should be added to determine the gravitational potential and we choose to use the Poisson equation:[5]

$$-k^2\Phi = \frac{3}{2}\mathcal{H}^2\left[\Omega_{\mathrm{m}}\delta_{\mathrm{m}}^{\mathrm{N}} + \Omega_{\mathrm{r}}\delta_{\mathrm{r}}^{\mathrm{N}} - 3\mathcal{H}\left(\Omega_{\mathrm{m}}V_{\mathrm{m}} + \frac{4}{3}\Omega_{\mathrm{r}}V_{\mathrm{r}}\right)\right], \tag{4.4.9}$$

or (4.2.68) in the form:[6]

$$\Phi' + \mathcal{H}\Phi = -\frac{3\mathcal{H}^2}{2}\left(\Omega_{\mathrm{m}}V_{\mathrm{m}} + \frac{4}{3}\Omega_{\mathrm{r}}V_{\mathrm{r}}\right). \tag{4.4.10}$$

These two equations can also be used in a combined way to express Φ in terms of $\delta_{\mathrm{m}}^{\mathrm{N}}$ and $\delta_{\mathrm{r}}^{\mathrm{N}}$ only.

Evolution of the gravitational potential and of the entropy

Using the expressions (4.3.44) to express c_{s}^2 and w, the system (4.3.72)–(4.3.74) reduces to:

$$\ddot{\Phi} + \left(7 - \frac{1}{1+y} + \frac{8}{4+3y}\right)\frac{\dot{\Phi}}{2y} + \frac{\Phi}{y(1+y)(4+3y)} = \frac{2}{(4+3y)y^2}\left(\delta^{\mathrm{C}} - \frac{yS}{1+y}\right), \tag{4.4.11}$$

$$\delta^{\mathrm{C}} = -\frac{4}{3}\left(\frac{k}{k_{\mathrm{eq}}}\right)^2\frac{y^2}{1+y}\Phi, \tag{4.4.12}$$

$$\ddot{S} + \frac{3y+2}{2y(1+y)}\dot{S} = \frac{2}{4+3y}\left(\frac{k}{k_{\mathrm{eq}}}\right)^2\left(\delta^{\mathrm{C}} - \frac{y}{1+y}S\right), \tag{4.4.13}$$

with $S = \delta_{\mathrm{m}} - \frac{3}{4}\delta_{\mathrm{r}}$. We have also assumed that for a Universe with Euclidean spatial sections and vanishing cosmological constant ($K = \Lambda = 0$), the matter and radiation density parameters can be rewritten simply in terms of the rescaled scale factor y as:

$$\Omega_{\mathrm{m}} = \frac{y}{1+y} \qquad \text{and} \qquad \Omega_{\mathrm{r}} = \frac{1}{1+y}. \tag{4.4.14}$$

[5] Starting from (4.3.73) for $K = 0$ with (4.2.36) and $\frac{\rho'}{\rho} = -3\mathcal{H}^2(1+w)$

[6] Starting from (4.2.68) with $\frac{\rho'}{\rho} = -3\mathcal{H}^2(1+w)$ abd replace ΩV by $\sum_a \Omega_a V_a$

This system generalizes (4.3.45) to the case where the entropy perturbation is no longer neglected. The density contrast $\delta^{\rm N}$ is then given by:

$$\delta^{\rm N} = \delta^{\rm C} - 2\left(\Phi + y\dot{\Phi}\right), \tag{4.4.15}$$

and the density contrasts of each fluid are:

$$\delta_{\rm m} = \frac{3(1+y)\delta/4 + S}{1+3y/4} \qquad \text{and} \qquad \delta_{\rm r} = \frac{(1+y)\delta - yS}{1+3y/4}. \tag{4.4.16}$$

Types of initial conditions

To solve the systems (4.4.5)–(4.4.8) or (4.4.11)–(4.4.13), initial conditions for the perturbations need to be imposed (see, for example, Ref. [39]). These initial conditions are fixed deep within the radiation era ($y_i \ll 1$) and for super-Hubble modes ($k/\mathcal{H}_i \ll 1$). As seen earlier, we can distinguish two types of initial conditions:

- **adiabatic**: the super-Hubble modes are assumed constant (in other words the decaying mode has had enough time to decay) and, by definition, we have:

$$\Phi = \Phi_i, \qquad \Phi' = 0, \qquad S = 0, \qquad S' = 0,$$

which fixes everything for the system (4.4.11)–(4.4.13). The Poisson equation (4.3.73) implies that:

$$\delta_i^{\rm C} = -\frac{2}{3}x_i^2\Phi, \qquad \delta_{{\rm r},i}^{\rm C} = \delta_i^{\rm C}, \qquad \delta_{{\rm m},i}^{\rm C} = \frac{3}{4}\delta_i^{\rm C}, \tag{4.4.17}$$

where $x_i \equiv k/\mathcal{H}_i$. Equation (4.4.10) then implies that:

$$kV_i = -\frac{1}{2}x_i\Phi, \qquad V_{{\rm r},i} = V_{{\rm m},i} = V_i, \tag{4.4.18}$$

One can derive it by noting that

$$\left.\begin{aligned}\Omega_m &= \tfrac{y}{1+y} \to 0\\ \Omega_r &= \tfrac{1}{1+y} \to 1\end{aligned}\right\} \mathcal{H}\Phi = -\frac{3}{2}\mathcal{H}^2\left(0 + \frac{4}{3}V_r\right) \tag{1}$$

and (4.4.15) that:[7]

$$\delta_i^{\rm N} = -2\Phi_i, \qquad \delta_{{\rm r},i}^{\rm N} = \delta_i^{\rm N}, \qquad \delta_{{\rm m},i}^{\rm N} = \frac{3}{4}\delta_i^{\rm N}, \tag{4.4.19}$$

which completely fixes the initial conditions for the system (4.4.5)–(4.4.8).

- **isocurvature**: we still assume the super-Hubble modes to be constant and the definition yields:

$$\Phi = 0, \qquad \Phi' = 0, \qquad S = S_i, \qquad S' = 0.$$

The Poisson equation implies that:

$$\delta_i^{\rm C} = 0, \qquad \delta_{{\rm r},i}^{\rm C} = -y_iS_i, \qquad \delta_{{\rm m},i}^{\rm C} = S_i, \tag{4.4.20}$$

so that (4.4.10) then implies:

$$V_{{\rm r},i} = V_{{\rm m},i} = V_i = 0, \tag{4.4.21}$$

while (4.4.15) provides:

$$\delta_i^{\rm N} = \delta_i^{\rm C}, \tag{4.4.22}$$

which again completely fixes the initial conditions for the system (4.4.11)–(4.4.13).

[7]For superhorizon modes which sets the initial condition we have both $x \ll 1$ and $S = 0$ in the adiabatic limit, hence we find $\frac{\delta_i}{1+w_i} = \text{const}$

4.4.2 Different regimes

The behaviour of the solutions of these systems mainly depends on the value of the wavenumber k that determines the time at which the mode becomes sub-Hubble.

In order to have an idea of the orders of magnitudes, the following Table gives the values of y and of the redshift at the time a given mode becomes sub-Hubble for several selected values of k:

k (h^{-1} Mpc)	10^3	10^2	10	1	0.1	10^{-2}	10^{-3}
y_*	5×10^{-5}	5×10^{-4}	5×10^{-3}	5.1×10^{-2}	6.4×10^{-1}	26	2.5×10^3
z_*	4.75×10^8	4.75×10^7	4.74×10^6	4.63×10^5	3.7×10^4	9.05×10^2	8.4

We therefore have four kinds of behaviour to take into account, depending on whether the mode is (i) super-Hubble today, (ii) has become sub-Hubble during the matter-dominated era, or (iii) during the radiation-dominated era, or (iv) whether it has always been sub-Hubble.

Super-Hubble modes

Let us first consider long-wavelength modes. In practice, one can neglect all the terms that involve a Laplacian, which are proportional to k^2. Equation (4.3.66) then tells us that the entropy remains constant. Actually, we can also reach this conclusion by noting that (4.4.12) implies that $\delta^{\mathrm{C}} \propto k^2\Phi$. In the set of equations (4.4.11)–(4.4.13), the terms $k^2\Phi$ and k^2S are negligible in comparison to, respectively, $\ddot{\Phi}$ and $\ddot{S}$. We then infer that in (4.4.13), $k^4\Phi/k_{\mathrm{eq}}^2 \ll \ddot{S}$, which implies that S is constant.

Note then that using the transformation (4.2.29) to introduce $\delta_{\mathrm{r}}^{\mathrm{C}}$ which is of the order of $\delta^{\mathrm{C}} \propto k^2\Phi$ in the radiation era, the two equations (4.4.7) and (4.4.8) imply that:

$$V'_{\mathrm{rm}} = -\mathcal{H}V_{\mathrm{rm}} - \frac{1}{4}\delta^{\mathrm{C}},$$

and therefore that $V_{\mathrm{rm}} \propto a^{-1}$ during the radiation era.

In the case of adiabatic initial conditions, $S=0$ and the solution can be obtained analytically in the form (4.3.46). Deep enough into the radiation era, both the gravitational potential and the total density contrast are constant. In particular,

$$\delta^{\mathrm{N}}(y) = -2\Phi(y), \qquad \delta_{\mathrm{r}}^{\mathrm{N}}(y) = \frac{4}{3}\delta_{\mathrm{m}}^{\mathrm{N}}(y),$$

whatever y. In the radiation era, we obtain:

$$\delta_{\mathrm{r}}^{\mathrm{N}} = -2\Phi(y\ll1), \qquad \delta_{\mathrm{m}}^{\mathrm{N}}(y) = -\frac{3}{2}\Phi(y\ll1),$$

and in the matter era ($y\gg1$):

$$\Phi(y\gg1) = \frac{9}{10}\Phi(y\ll1), \qquad \delta_{\mathrm{r}}^{\mathrm{N}}(y\gg1) = -\frac{12}{5}\Phi(y\ll1), \qquad \delta_{\mathrm{m}}^{\mathrm{N}}(y\gg1) = -\frac{9}{5}\Phi(y\ll1). \tag{4.4.23}$$

To illustrate the effect of the initial conditions, let us now consider some isocurvature initial conditions and let us assume that initially $S=S_i\neq0$ and $\Phi=0$. As we have seen S remains constant. Deep in the radiation era ($y\ll1$), $\delta_{\mathrm{r}}^{\mathrm{N}} \sim \delta^{\mathrm{N}} = -2\Phi = 0$ and thus $\delta_{\mathrm{m}}^{\mathrm{N}} = S_i$. Note then that for super-Hubble modes, since Φ and S are constant, $\Phi=-2S$ is a particular solution of (4.4.11). The general solution satisfies the initial condition $\Phi_i=0$ and is thus obtained by adding this particular solution to the general solution (4.3.46) where the constant Φ_0 should be taken equal to $\Phi_0 = 2S_i$:

$$\Phi = \frac{2S_i}{10y^3}\left(16\sqrt{1+y} + 9y^3 + 2y^2 - 8y - 16\right) - 2S_i. \tag{4.4.24}$$

In the matter-dominated era ($y\gg1$), $\delta_{\mathrm{m}}^{\mathrm{N}} \sim \delta^{\mathrm{N}} = -2\Phi$ and $\Phi(y\gg1) = -S_i/5$. We conclude that, for $y\gg1$:

$$\Phi = -\frac{1}{5}S_i, \qquad \delta^{\mathrm{N}} \sim \delta_{\mathrm{m}}^{\mathrm{N}} = \frac{2}{5}S_i, \qquad \delta_{\mathrm{r}}^{\mathrm{N}} = -\frac{4}{5}S_i. \tag{4.4.25}$$

Modes entering the Hubble radius in the matter-dominated era

As can be seen in Fig. 4.11, even though the mode is sub-Hubble from a redshift of the order of $z_* \sim 8.5$, it is always constant. Actually, to establish this result, we had to assume that the Laplacian term was negligible in the evolution equation for the gravitational potential. This is legitimate as long as $c_s^2 k^2 \ll \mathcal{H}^2$, i.e. if:

$$\frac{2y^2}{3(1+y)(1+3y/4)}\bar{k}^2 \ll 1. \tag{4.4.26}$$

For the modes represented in Fig. 4.11, $\bar{k} = 10^{-3}$ and this ratio is still equal to 1.7×10^{-4} today. One can check that this ratio is of order unity for $k \sim k_{\rm eq}$ so that for all the modes entering the Hubble radius in the matter-dominated era, the gravitational potential remains constant, put aside the small variation due to the jump of the equation of state from 1/3 to 0 (see Fig. 5.12). The Poisson equation then tells us that:

$$\delta_{\rm m}^{\rm C} \propto a. \tag{4.4.27}$$

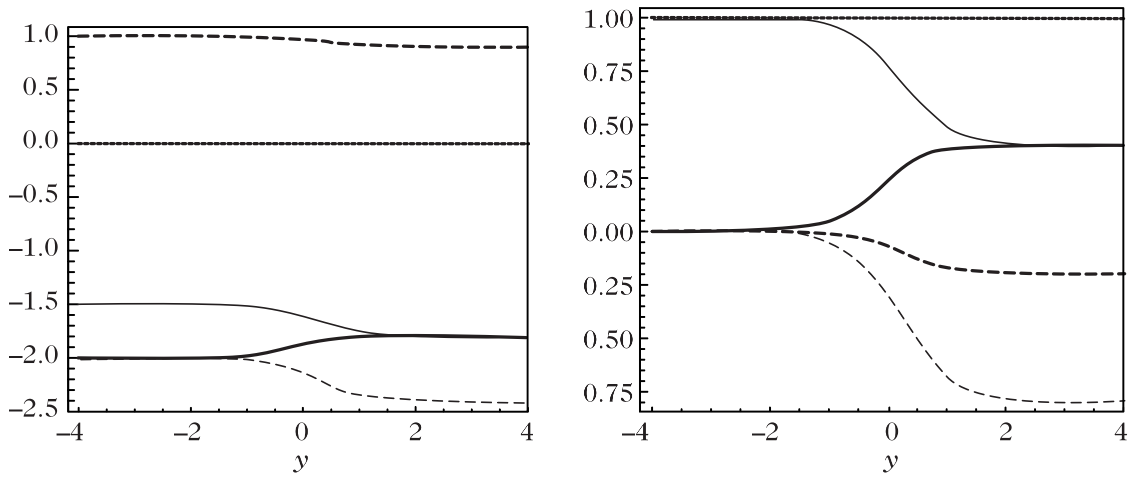


Figure 4.11: Evolution of the total density contrast $\delta^{\rm N}$ (thick solid line), the entropy S (thick dotted line) and gravitational potential (thick dashed line) as well as the two contrasts $\delta_{\rm r}^{\rm N}$ (thin dashed line) and $\delta_{\rm m}^{\rm N}$ (thin solid line) for a mode $\bar{k} = 10^{-3}$ assuming either adiabatic initial conditions ($\Phi_i = 1, S_i = 0$) [left] or isocurvature initial conditions ($\Phi_i = 0, S_i = -1$) [right].

Modes entering the Hubble radius in the radiation-dominated era

When the Universe is dominated by radiation, the gravitational potential is determined by the density fluctuations of radiation. It is therefore given by the general solution (4.3.31) and its evolution is represented in Fig. 4.5. Thus, as soon as the modes becomes sub-Hubble during the radiation-dominated era, the gravitational potential is subject to damped oscillations and tends rapidly to 0.

The dark-matter fluid density perturbations are then obtained by solving their evolution equations with the gravitational potential determined by the solution (4.3.31),

$$\delta_{\rm m}^{\rm N\prime\prime} + \mathcal{H}\delta_{\rm m}^{\rm N\prime} = 3\Phi'' + 3\mathcal{H}\Phi' - k^2\Phi, \tag{4.4.28}$$

where the Hubble parameter is given by $\mathcal{H} = \eta^{-1}$ deep enough into the radiation era. The solution of this equation is of the form

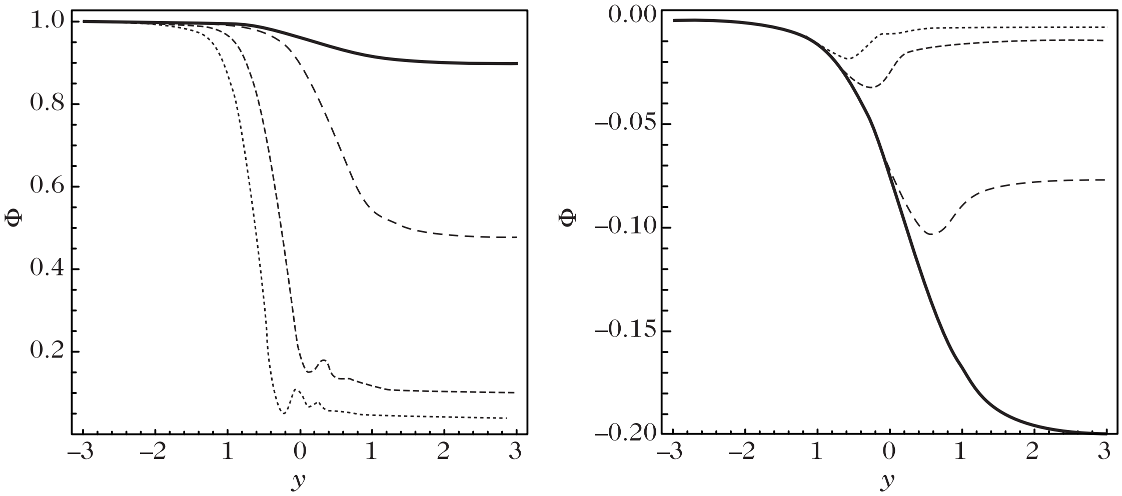


Figure 4.12: Evolution of the gravitational potential for $k = 10^{-2}$, 0.1, 0.5, 1 Mpc^{-1} for adiabatic (left) and isocurvature (right) initial conditions. These curves should be compared with Fig. 4.9, for which entropy fluctuations were neglected.

$$\delta_{\rm m}^{\rm N} = A + B\ln(k\eta) + \delta_{\rm part}^{\rm N}, \tag{4.4.29}$$

where the particular integral is obtained in the form

$$\delta_{\rm part}^{\rm N}(k,\eta) = \int_0^\eta {\rm d}\eta' \left[3\Phi''(k,\eta') + 3\mathcal{H}\Phi'(k,\eta') - k^2\Phi(k,\eta')\right]\eta' \ln\left(\frac{\eta'}{\eta}\right). \tag{4.4.30}$$

When $y \ll 1$, $\delta^{\rm N} \sim \delta_{\rm r}^{\rm N} \sim 2\Phi_{\rm i}$. Assuming adiabatic initial conditions ($\delta_{\rm m}^{\rm N} = \frac{3}{4}\delta_{\rm r}^{\rm N}$), one finds $\delta_{\rm m}^{\rm N} \sim \frac{3}{2}\Phi_{\rm i}$. The contribution from the particular solution turns out to be negligible on super-Hubble scales, so that one is left with $A = \frac{3}{2}\Phi_{\rm i}$ and $B = 0$.

As established by the analytical solution (4.3.31), the gravitational potential decays rapidly as soon as the perturbation enters the Hubble radius, remaining nearly constant beforehand. The integrand in Eq. (4.4.30) varies significantly only around $\eta' \sim 1/k$, so that the upper limit of the integral can be extended to infinity for sub-Hubble modes. Consequently, the integral evaluates to

$$\delta_{\rm part}^{\rm N}(k,\eta) = A + B\ln(k\eta), \tag{4.4.31}$$

where the constants

$$A = \int_0^\infty f(k\eta')k\eta' \ln(k\eta')\,{\rm d}(k\eta') \quad \text{and} \quad B = -\int_0^\infty f(k\eta')k\eta'\,{\rm d}(k\eta') \tag{4.4.32}$$

are governed by the auxiliary function $f(u)$, given explicitly by

$$f(u) = 9\Phi_{\rm i}u^{-5}\left[u(2u^2-27)\cos\frac{u}{\sqrt{3}} + \sqrt{3}(27-5u^2)\sin\frac{u}{\sqrt{3}}\right],$$

which is obtained by inserting the analytic potential solution (4.3.31). Numerically, one obtains $A \simeq -6\Phi_{\rm i}$ and $B \simeq 9\Phi_{\rm i}$, so that the general solution settles into the asymptotic form

$$\delta_{\rm m}^{\rm N}(k,\eta) \simeq \Phi_{\rm i}\left[-4.5 + 9\ln(k\eta)\right]. \tag{4.4.33}$$

Determining the exact numerical coefficients with high precision would of course necessitate a more detailed matching calculation. In summary, the dark matter perturbation evolves as

$$\delta_{\rm m}^{\rm N} \sim \delta_{\rm m}^{\rm C} \propto \ln a \quad (k\eta \gg 1); \qquad \delta_{\rm m}^{\rm N} \propto a^0 \quad (k\eta \ll 1) \tag{4.4.34}$$

during the radiation-dominated era (see Fig. 4.13 for a numerical integration).

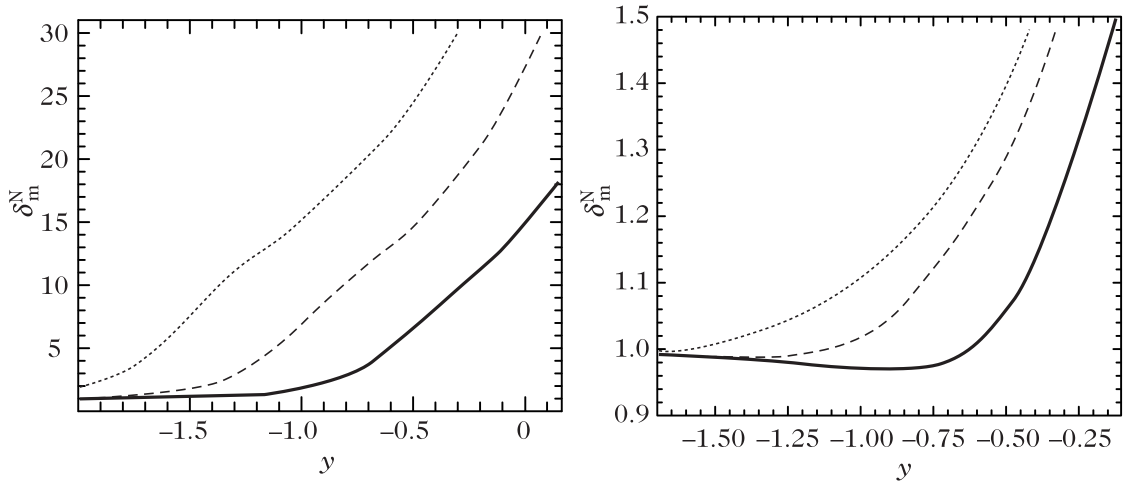


Figure 4.13: Evolution of the matter density contrast δ_m^N for $k = 1\,h\,\mathrm{Mpc}^{-1}$ (solid line), $k = 5\,h\,\mathrm{Mpc}^{-1}$ (dashed line) and $k = 10\,h\,\mathrm{Mpc}^{-1}$ (dotted line) for adiabatic (left) and isocurvature (right) initial conditions. These modes become sub-Hubble during the radiation-dominated era.

Sub-Hubble modes

Let us now focus on modes that were already sub-Hubble during the radiation era and examine how dark-matter perturbations respond across the radiation–matter transition. For these short-wavelength modes, we have seen (Fig. 4.6) that the density contrast is practically identical across all standard gauges.

For any mode that enters the Hubble radius during the radiation era, radiation perturbations δ_r remain approximately constant, whereas the dark matter perturbation grows logarithmically as $\delta_m \sim \ln a$. Even though ρ_m is initially small compared to ρ_r, the contribution of dark matter density fluctuations to the Poisson equation eventually outgrows that of radiation because

$$\frac{\rho_m \delta_m}{\rho_r \delta_r} \sim y \ln y.$$

We therefore concentrate on the regime where $\delta_r \ll \delta_m$. The total density contrast is then given by

$$\delta = \frac{y}{1+y}\delta_m,$$

and the Poisson equation takes the form

$$k^2\Phi = -\frac{3}{4y}k_{eq}^2\delta_m, \tag{4.4.35}$$

as long as velocity perturbations can be neglected. We now employ the two fluid equations (4.4.5)–(4.4.7) expressed in terms of the variable y. Differentiating Eq. (4.4.5) and eliminating V_m through Eq. (4.4.7) together with $\dot{V}_m$ from the continuity relation, one obtains a second-order differential equation for δ_m involving Φ and its derivatives, which can then be eliminated using Eq. (4.4.35).

To simplify this governing equation, observe that we are interested in short-wavelength modes satisfying $k/\mathcal{H} \gg 1$, which implies $k_{eq}^2 \ll k^2 y$. Dropping these negligible terms leads to the equation, known in the literature as the *Mészáros equation* [40],

$$\ddot{\delta}_m + \frac{2+3y}{2y(y+1)}\dot{\delta}_m - \frac{3}{2y(y+1)}\delta_m = 0. \tag{4.4.36}$$

This differential equation admits an affine solution, and the second independent solution can be found by variation of parameters. One can verify that two linearly independent solutions are

$$D_+(y) = y + \frac{2}{3} \quad \text{and} \quad D_-(y) = D_+(y) \ln\left(\frac{\sqrt{1+y}+1}{\sqrt{1+y}-1}\right) - 2\sqrt{1+y}. \tag{4.4.37}$$

In the matter-dominated era ($y \gg 1$), these correspond respectively to a growing and a decaying mode, with asymptotic behaviours

$$D_+(y) \propto y \quad \text{and} \quad D_-(y) \propto y^{-3/2}.$$

This solution can be continuously matched to the earlier sub-Hubble radiation-era solution (4.4.29)–(4.4.31) at the time when the mode enters the Hubble radius at $y_*(k)$ (see Fig. 4.14 for a numerical integration).

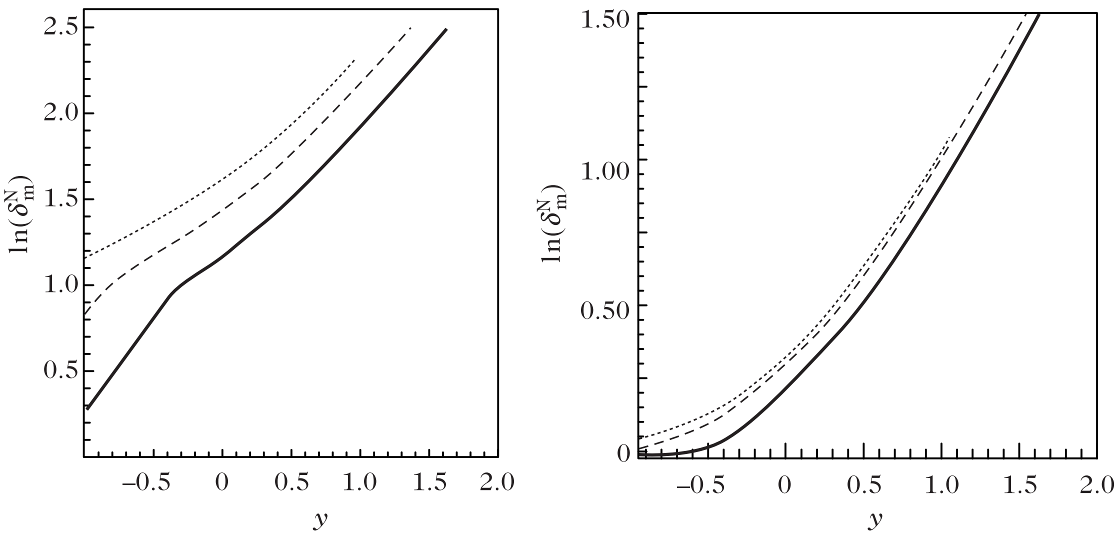


Figure 4.14: Evolution of the matter-density contrast δ_m^N for $k = 1\,h\,\mathrm{Mpc}^{-1}$ (solid line), $k = 5\,h\,\mathrm{Mpc}^{-1}$ (dashed line) and $k = 10\,h\,\mathrm{Mpc}^{-1}$ (dotted line) for adiabatic (left) and isocurvature (right) initial conditions.

Summary

In the simplified scenario of a Universe containing solely dark matter and radiation interacting gravitationally, the preceding analysis shows that the evolution of Φ and δ_m depends crucially on the value of the wavenumber k and on the cosmological epoch during which the mode crosses inside the Hubble horizon.

However, this basic analysis leaves out four significant physical effects:

1. Cosmic matter contains baryons that are tightly coupled to photons via Compton scattering. We describe this coupling and its consequences for cosmological perturbations in Chapter 6.

2. Radiation has been modeled under a simple fluid approximation. This description is rather coarse and must be generalized using a kinetic framework wherein radiation is governed by the Boltzmann equation. Such a kinetic approach yields an infinite hierarchy of moment equations, of which the fluid equations represent merely the first two moments. This is treated in detail in Chapter 6, dedicated to the cosmic microwave background.

3. The late-time evolution of cosmological perturbations can be influenced by spatial curvature or a cosmological constant. These effects were demonstrated in Figs. 4.7 and 4.8.

4. Because of the presence of relativistic neutrinos, $\Phi \neq \Psi$, which modifies, for example, the ratio 9/10 for the evolution of the gravitational potential on super-Hubble scales.

Transfer function

This simplified investigation nevertheless enables us to grasp the qualitative structure of the large-scale matter power spectrum. To formalize this, one defines the transfer function

$$T(k,a) = \frac{\delta(k,a)D_+(a_\mathrm{i})}{\delta(k,a=a_\mathrm{i})D_+(a)}, \tag{4.4.38}$$

where a_i represents an initial time and $D_+(a)$ is the linear growth factor (4.1.24). Note that a_i is arbitrary provided it corresponds to an era before any scale of cosmological interest has crossed the Hubble radius. This transfer function quantifies the modification of the initial matter power spectrum throughout cosmic history and depends on the cosmological parameters and matter content of the Universe. The normalization factor D_+ is included by convention because during the matter era

$$\Delta\Phi = \frac{3}{2}H_0^2\Omega_{\mathrm{m}0}\frac{D_+(a)}{a}\frac{\delta}{D_+(a)},$$

and the ratio $D_+(a)/a$ remains approximately equal to unity during matter domination (see Fig. 4.1). The definition (4.4.38) implies that

$$\Phi(k,a) = T(k,a)\Phi(k,a_\mathrm{i}).$$

The modes observable today were initially super-Hubble, and theoretical models of the primordial Universe allow us to predict the initial matter power spectrum on these scales. Consequently, the gauge in which the primordial density contrast $\delta(k,a=a_\mathrm{i})$ is evaluated must be explicitly specified. In practice, since Φ remains constant for these super-Hubble modes, the identical transfer function is obtained if one employs δ^C in the definition (4.4.38), because both density contrasts are related to the gravitational potential by the same Poisson equation. We therefore have

$$P_\Phi(k,a_0) = P_\Phi(k,a_\mathrm{i})T^2(k,a_0), \tag{4.4.39}$$

and

$$P_\delta(k,a_0) = P_\delta(k,a_\mathrm{i})T^2(k,a_0)\left[\frac{D_+(a_0)}{D_+(a_\mathrm{i})}\right]^2, \tag{4.4.40}$$

since $P_\delta(k,a_\mathrm{i}) = k^4 P_\Phi(k,a_\mathrm{i})$.

The qualitative behavior of the transfer function can be directly deduced from our previous analysis (see Fig. 4.15). Indeed, setting aside the logarithmic enhancement during the radiation era, modes that entered the Hubble radius during this epoch are effectively frozen and experience minimal growth between horizon crossing and matter–radiation equality. Their relative amplitude compared to long-wavelength modes ($k < k_\mathrm{eq}$) is therefore suppressed by a factor of $(k_\mathrm{eq}/k)^2$. One thus expects the limiting behaviors

$$T(k,a_0) \sim 1 \quad (k \ll k_\mathrm{eq}), \qquad T(k,a_0) \sim \left(\frac{k_\mathrm{eq}}{k}\right)^2 \quad (k \gg k_\mathrm{eq}). \tag{4.4.41}$$

This analytic estimate should be compared with Fig. 4.16, which is obtained from a complete numerical computation incorporating baryons and their coupling to radiation. Note also that the precise shape of the transfer function depends on the nature of the initial cosmological perturbations (adiabatic or isocurvature). Various analytic fitting formulas have been developed in the literature [19]. For example, for a cold dark matter model, one has

$$T(q) = \frac{\ln(1+2.34q)}{2.34q}\left[1+3.89q+(16.1q)^2+(5.46q)^3+(6.71q)^4\right]^{-1/4}, \tag{4.4.42}$$

$$T(q) = (5.6q)^2 \left\{1 + \left[15q + (0.9q)^{3/2} + (5.6q)^2\right]^{1.24}\right\}^{-1/1.24}, \tag{4.4.43}$$

with the dimensionless wavenumber $q = k/(\Gamma\,\mathrm{Mpc}^{-1})$ and shape parameter $\Gamma = \Omega_{\mathrm{m}0}h^2$, valid for adiabatic and isocurvature initial conditions, respectively.

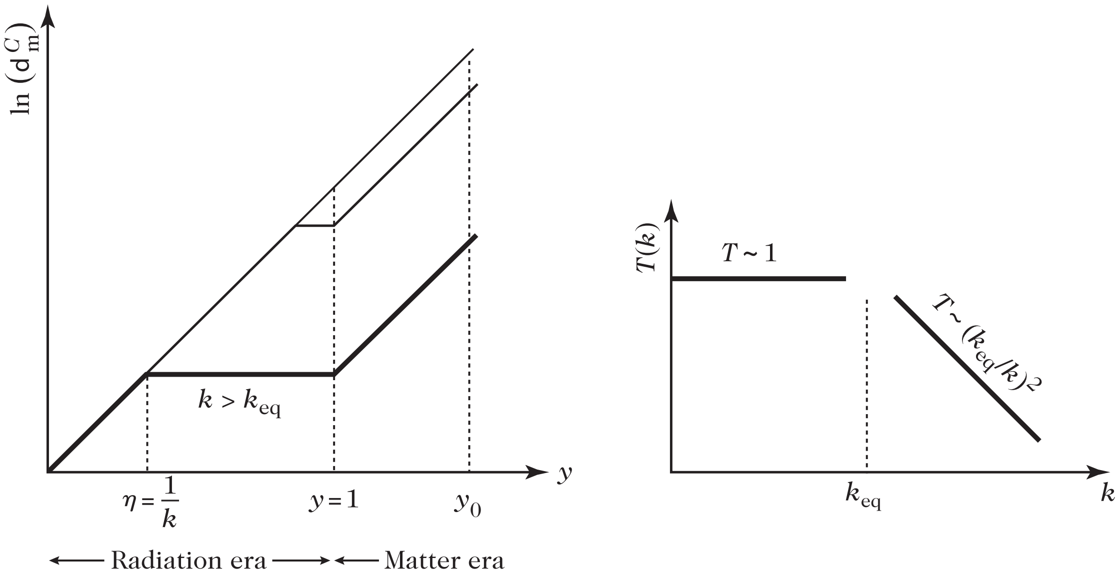


Figure 4.15: (left): Modes that become sub-Hubble during the radiation-dominated era ($k > k_{\mathrm{eq}}$) remain nearly constant (neglecting logarithmic growth) from $\eta \sim 1/k$ to $\eta \sim 1/k_{\mathrm{eq}}$. (right): The amplitude of the density contrast on these scales suffers from a suppression factor of order $(k_{\mathrm{eq}}/k)^2$. For $k > k_{\mathrm{eq}}$ one therefore expects the transfer function to scale as $T \propto (k_{\mathrm{eq}}/k)^2$, while remaining of order unity for $k < k_{\mathrm{eq}}$.

4.4.3 Some refinements

To conclude this chapter, we record the complete set of cosmological perturbation equations, formulated in the fluid approximation, for a Universe comprising photons, neutrinos, cold dark matter, and baryons. The reader is encouraged to implement this coupled system numerically to reproduce the curves presented throughout this chapter and to gain an appreciation for the various physical mechanisms omitted in the introductory presentation.

Dark matter

Cold dark matter is governed by the standard conservation and Euler equations employed throughout this chapter,

$$\delta_{\mathrm{c}}^{\mathrm{N}\prime} = k^2 V_{\mathrm{c}} + 3\Psi', \tag{4.4.44}$$

$$V_{\mathrm{c}}' = -\mathcal{H}V_{\mathrm{c}} - \Phi, \tag{4.4.45}$$

where we have explicitly retained the distinction between the two gravitational potentials Φ and Ψ.

Baryons

Electrons and atomic nuclei share the same density perturbation because electromagnetic interactions enforce local charge neutrality to extraordinary precision. Their continuity equation is identical to that of dark matter, but their Euler equation is modified due to Thomson scattering,

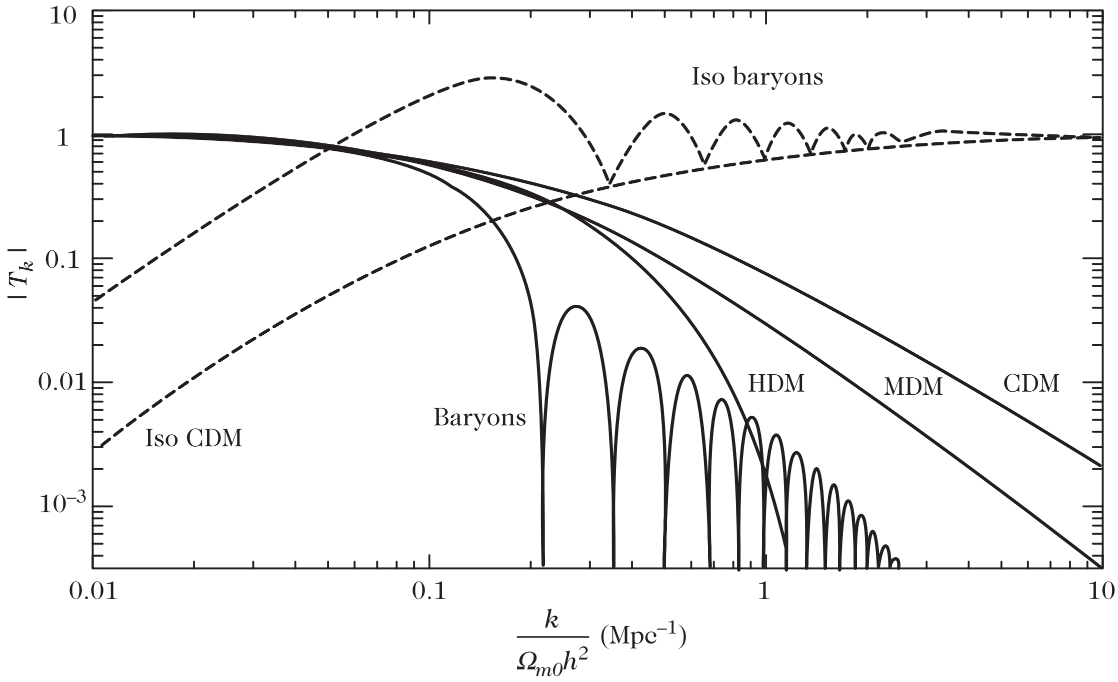


Figure 4.16: Transfer function for different cosmological dark-matter scenarios. The overall shape of this transfer function can be understood directly from the analytical considerations presented above, cf. (4.4.41). From Ref. [8].

which couples them dynamically to the photon bath. The momentum exchange between baryons and radiation must therefore be accounted for. As shown in Ref. [41], the interaction force entering the baryon Euler equation is

$$\frac{1}{\rho_\mathrm{b} + P_\mathrm{b}} f_\mathrm{b} = \frac{4}{3}\frac{\rho_\gamma}{\rho_\mathrm{b}} n_\mathrm{e}\sigma_\mathrm{T}(V_\gamma - V_\mathrm{b}), \tag{4.4.46}$$

where σ_T denotes the Thomson scattering cross-section and n_e is the number density of free electrons. The baryon sound speed does not vanish and can be computed by treating the cosmic gas as a monoatomic mixture of mean molecular weight μ (accounting for electrons, protons, and helium nuclei) at temperature T_b,

$$c_\mathrm{b}^2 = \frac{k_\mathrm{B} T_\mathrm{b}}{\mu}\left(1 - \frac{1}{3}\frac{\mathrm{d}\ln T_\mathrm{b}}{\mathrm{d}\ln a}\right), \tag{4.4.47}$$

neglecting the time variation of μ (in reality, $\dot{\mu}$ is significant only during recombination, when baryons contribute negligibly to the pressure of the combined photon–baryon plasma, making this approximation highly reliable). To determine the temperature evolution equation, one begins with the first law of thermodynamics (see Ref. [42]) $\delta Q = \frac{3}{2}\delta(P_\mathrm{b}/\rho_\mathrm{b}) + P_\mathrm{b}\delta(1/\rho_\mathrm{b})$, with the heating rate given by $Q' = 4(\rho_\gamma/\rho_\mathrm{b}) a n_\mathrm{e}\sigma_\mathrm{T}(T_\gamma - T_\mathrm{b})$. One concludes that the evolution of the baryon temperature is governed by

$$T_\mathrm{b}' = -\mathcal{H}T_\mathrm{b} + \frac{8}{3}\frac{\mu}{m_\mathrm{e}}\frac{\rho_\gamma}{\rho_\mathrm{b}} a n_\mathrm{e}\sigma_\mathrm{T}(T_\gamma - T_\mathrm{b}). \tag{4.4.48}$$

Since $P_\mathrm{b} \ll \rho_\mathrm{b}$, one can safely neglect w_b and c_b^2 throughout the perturbation equations, with the exception of the acoustic gradient term $c_\mathrm{b}^2\Delta\delta_\mathrm{b}^\mathrm{N}$. One eventually arrives at

$$\delta_\mathrm{b}^{\mathrm{N}\prime} = k^2 V_\mathrm{b} + 3\Psi', \tag{4.4.49}$$

$$V_\mathrm{b}' = -\mathcal{H}V_\mathrm{b} - \Phi - c_\mathrm{b}^2\delta_\mathrm{b}^\mathrm{N} + \frac{4}{3}\frac{\rho_\gamma}{\rho_\mathrm{b}} a n_\mathrm{e}\sigma_\mathrm{T}(T_\gamma - T_\mathrm{b}). \tag{4.4.50}$$

Photons

As will be detailed in the following chapter, radiation is rigorously described using the relativistic Boltzmann equation, which expands the photon and neutrino propagation equations into infinite multipole hierarchies. Here, we content ourselves with refining our fluid description by incorporating an anisotropic pressure term, π_γ, alongside the coupling to the baryon fluid:

$$\delta_\gamma^{\mathrm{N}\prime} = \frac{4}{3}k^2 V_\gamma + 4\Psi', \tag{4.4.51}$$

$$V_\gamma' = -\frac{1}{4}\delta_\gamma^{\mathrm{N}} - \Phi + k^2\sigma_\gamma + an_{\mathrm{e}}\sigma_{\mathrm{T}}(T_\gamma - T_{\mathrm{b}}). \tag{4.4.52}$$

Here, the coupling term represents the momentum transfer from the baryon fluid, and we have defined the anisotropic shear stress $\sigma_\gamma \equiv \pi_\gamma/6$. At this stage, we accept without proof the dynamical equation for the photon shear stress,

$$\sigma_\gamma' = -\frac{4}{15}V_\gamma - \frac{9}{5}an_{\mathrm{e}}\sigma_{\mathrm{T}}\sigma_\gamma. \tag{4.4.53}$$

In reality, Eq. (4.4.52) receives contributions from higher multipoles in the Boltzmann hierarchy, so that this fluid representation is merely a low-order truncation.

Neutrinos

As noted above, neutrinos must also be properly described via a Boltzmann equation. They satisfy equations formally identical to those for photons, except for the absence of collisional coupling with the baryon fluid. We thus have

$$\delta_\nu^{\mathrm{N}\prime} = \frac{4}{3}k^2 V_\nu + 4\Psi', \tag{4.4.54}$$

$$V_\nu' = -\frac{1}{4}\delta_\nu^{\mathrm{N}} - \Phi + k^2\sigma_\nu, \tag{4.4.55}$$

with $\sigma_\nu \equiv \pi_\nu/6$. As in the photon case, the dynamical evolution equation for the neutrino shear stress is given without proof by

$$\sigma_\nu' = -\frac{4}{15}V_\nu. \tag{4.4.56}$$

The more general scenario involving massive neutrinos is described in detail in Ref. [44].

Einstein equations

The perturbed Einstein equations reduce to

$$\Psi - \Phi = 6\frac{\mathcal{H}^2}{k^2}(\Omega_\gamma\sigma_\gamma + \Omega_\nu\sigma_\nu) \tag{4.4.57}$$

and

$$-k^2\Psi = \frac{3}{2}\mathcal{H}^2\left[\sum_i \Omega_i\delta_i^{\mathrm{N}} - \mathcal{H}\sum_i(1+w_i)\Omega_i V_i\right]. \tag{4.4.58}$$

Initial conditions

Just as in the preceding sections, initial conditions must be specified for super-Hubble modes deep in the radiation era. For adiabatic initial conditions, one obtains (see, for instance, Ref. [39])

$$\delta_\gamma^{\mathrm{N}}(\eta_{\mathrm{i}}) = \delta_\nu^{\mathrm{N}}(\eta_{\mathrm{i}}) = \frac{4}{3}\delta_{\mathrm{b}}^{\mathrm{N}}(\eta_{\mathrm{i}}) = \frac{4}{3}\delta_{\mathrm{c}}^{\mathrm{N}}(\eta_{\mathrm{i}}) \quad \text{and} \quad V_\gamma(\eta_{\mathrm{i}}) = V_\nu(\eta_{\mathrm{i}}) = V_{\mathrm{b}}(\eta_{\mathrm{i}}) = V_{\mathrm{c}}(\eta_{\mathrm{i}}). \tag{4.4.59}$$

As before, on super-Hubble scales we have

$$\delta_\gamma^{\rm N}(\eta_{\rm i}) = -2\Phi(\eta_{\rm i}), \qquad kV_\gamma(\eta_{\rm i}) = -\frac{1}{2}k\eta_{\rm i}\Phi(\eta_{\rm i}), \tag{4.4.60}$$

from which one deduces

$$\sigma_\nu(\eta_{\rm i}) = \frac{1}{15}(k\eta_{\rm i})^2\Phi, \qquad \sigma_\gamma(\eta_{\rm i}) = 0, \tag{4.4.61}$$

and

$$\Psi = \left(1 + \frac{2}{5}R_\nu\right)\Phi, \tag{4.4.62}$$

with the neutrino energy fraction defined by $R_\nu \equiv \rho_\nu/(\rho_\nu + \rho_\gamma)$.

Tight-coupling approximation

Note that prior to decoupling, $V_{\rm b} = V_\gamma$ because Compton scattering couples both species extraordinarily tightly. Following decoupling, the Universe neutralizes and both fluids evolve independently. This coupled system can therefore be solved in the "tight-coupling" approximation, wherein one transitions between the two regimes almost instantaneously (see, for instance, Ref. [43]). Prior to decoupling, one has

$$V_{\rm b} = V_\gamma = V, \qquad V' = -\frac{3\Omega_{\rm b}}{4\Omega_\gamma + 3\Omega_{\rm b}}\mathcal{H}V - \frac{\Omega_\gamma}{4\Omega_\gamma + 3\Omega_{\rm b}}\delta_\gamma^{\rm N} - \Phi. \tag{4.4.63}$$

Conclusions

The system of equations detailed in this section provides a markedly more realistic portrait of cosmological perturbation evolution. While remaining within a fluid description, it accounts for departures from ideal fluid behavior in photons and neutrinos via their anisotropic shear stresses. Consequently, the two gravitational potentials are no longer identical. The collisional coupling between baryons and photons is incorporated, although its rigorous derivation demands a relativistic kinetic approach.

This kinetic description of radiation will be systematically developed in the next chapter, where we will not only derive the unproved equations but also supply their full kinetic versions. In addition, one must describe the detailed ionization and recombination history to determine the time evolution of the free-electron density and move beyond the tight-coupling hypothesis. Solving such a multi-component system yields the matter transfer functions of Fig. 4.16 with excellent precision across a wide range of cosmological models.

4.5 The large-scale structure of the Universe

Up to this point, this chapter has concentrated predominantly on the linear cosmological regime. In Section 4.1.4 we touched briefly upon the weakly non-linear regime. We now turn to what observations reveal regarding the matter power spectrum.

4.5.1 Observing the large-scale structure

Galaxy catalogues

Observational reconstruction of the matter power spectrum is primarily achieved through the assembly of large galaxy redshift surveys. In recent years, two extensive surveys—the Sloan Digital Sky Survey (SDSS) [45] and the 2-degree Field Galaxy Redshift Survey (2dFGRS) [46]—have dramatically expanded both the volume of observed celestial sources and the total sky coverage. SDSS surveys roughly a quarter of the celestial sphere, determining the positions and luminosities of about 100 million celestial sources and measuring distances to more than one million galaxies; 2dFGRS has measured the spatial positions and redshifts of more than 250,000 galaxies. These surveys probe the cosmic web up to redshifts $z \sim 0.3$. A representative slice of the large-scale galaxy distribution mapped by 2dFGRS is displayed in Fig. 4.17.

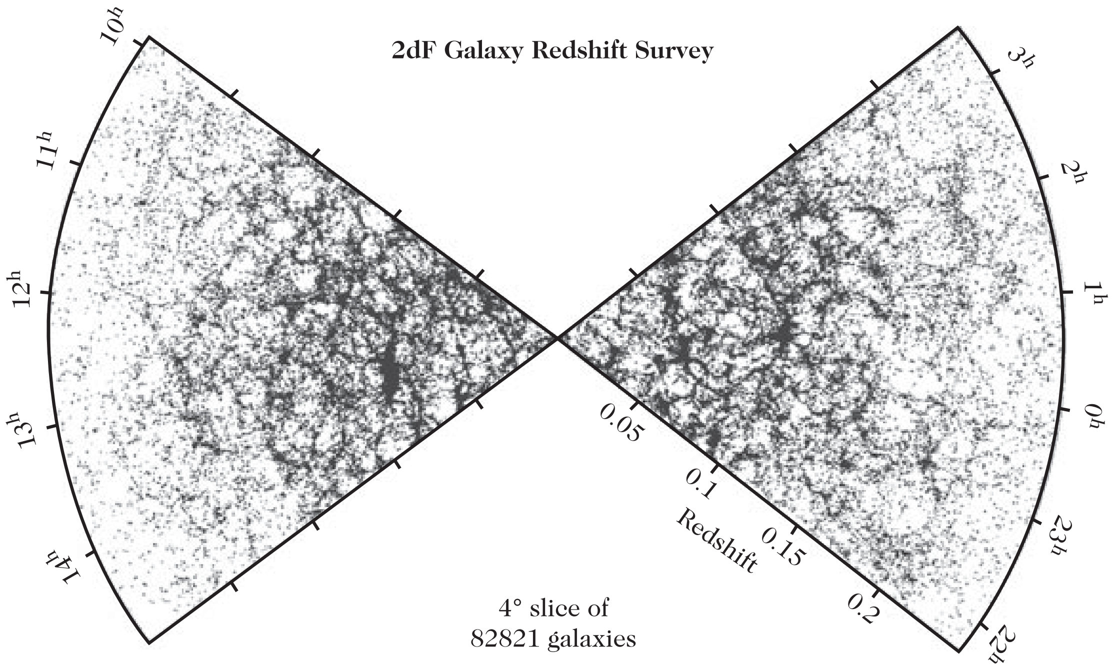


Figure 4.17: The distribution of galaxies across a 4°-wide slice from the 2dFGRS survey catalogue. Such observations of the large-scale structure of the Universe provide the empirical foundation for measurements of the matter power spectrum.

The relationship between these observational surveys and the underlying three-dimensional dark-matter power spectrum is not completely straightforward. Indeed, astronomical observations detect only luminous (baryonic) matter. To extract quantitative conclusions regarding the total matter distribution, one must ensure that luminous matter serves as a faithful tracer of the dark-matter field. This entails that the density contrast of luminous matter is proportional to that of dark matter. The proportionality factor, b, is termed the *bias parameter* [12, 18],

$$\delta_{\text{luminous}} = b\,\delta_{\text{c}}. \tag{4.5.1}$$

The validity of this linear bias relation, the possibility that b depends on spatial scale, and related observational nuances have been investigated extensively through numerical N-body simulations and observations [12]. Another practical difficulty stems from the fact that observations yield the sky position and redshift of astronomical sources with high precision. In redshift space, the inferred radial distance is perturbed by the peculiar velocity of each galaxy along the line of sight. Consequently, on small scales, a spherical gravitationally collapsing structure appears distorted and elongated along the radial line of sight into an ellipse (the "finger-of-God" effect). Such redshift-space distortions, arising from peculiar motions, must be carefully corrected.

Constraints on the power spectrum

Large galaxy redshift surveys permit the reconstruction of the matter power spectrum over approximately two decades in spatial scale. On larger scales, the linear primordial spectrum can be extracted from the angular anisotropies of the cosmic microwave background (Chapter 6).

On smaller scales, complementary observational techniques, such as weak gravitational lensing (Chapter 7) and the analysis of Lyman-α forest absorption in quasar spectra (which traces the distribution of diffuse intergalactic gas), make it possible to extend power spectrum measurements by nearly two additional orders of magnitude in wavelength. Importantly, the use of weak gravitational lensing offers the distinct advantage of circumventing the galaxy bias parameter altogether, since lensing directly measures the spatial distribution of the gravitational potential (see Chapter 7 for details).

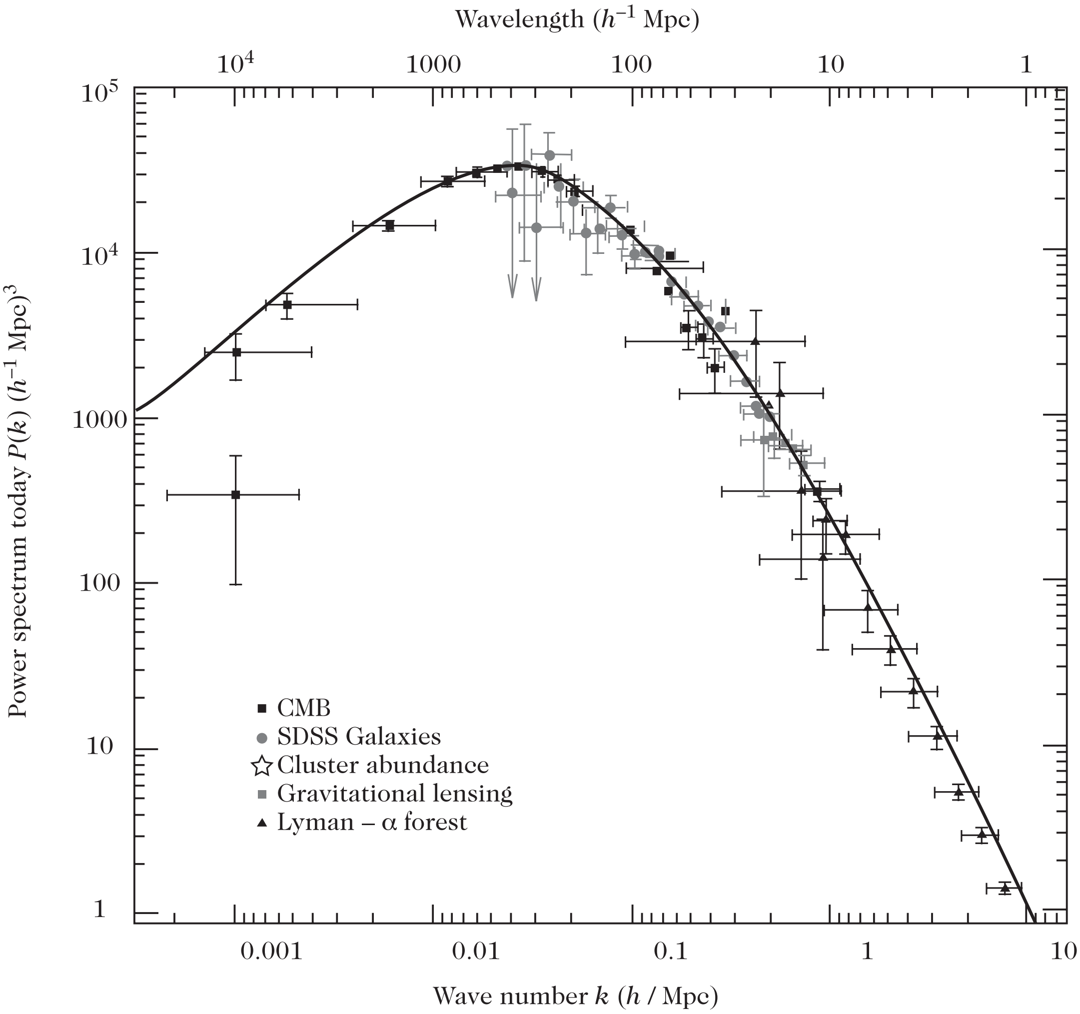


Figure 4.18: The matter power spectrum compared with various observational constraints, ranging from linear scales (cosmic microwave background) to non-linear scales (galaxy surveys, here SDSS), cluster abundances, gravitational lensing, and Lyman-α forests. From (4.4.42) we get the transfer function which can be used $P_m = P_\delta T^2(k) D^2(z)$, where $P_\delta \propto k$.

The code to produce the fig. 4.18

```
import numpy as np
import matplotlib.pyplot as plt

# =============================================================
# Cosmological parameters
# =============================================================
Om0 = 0.315
Or0 = 9.22e-5
```

```
Ol0 = 1.0 - Om0 - Or0

h = 0.674
H0 = 100.0 * h / 299792.458     # H0 in Mpc^{-1}

ns = 0.965
As = 2.1e-9
kpivot = 0.05                   # Mpc^{-1}

# ============================================================
# Primordial curvature power spectrum
# ============================================================
def Delta_zeta2(k):
        return As * (k / kpivot)**(ns - 1.0)


# ============================================================
# BBKS transfer function
#
# k_h   : k in h/Mpc
# k     : k in Mpc^{-1}
# ============================================================
def T_BBKS(k_h):
        q = k_h / (Om0 * h)
        numerator = np.log(1.0 + 2.34*q)
        denominator = 2.34*q * (1.0+ 3.89*q+ (16.1*q)**2
                + (5.46*q)**3 + (6.71*q)**4)**0.25

        return numerator / denominator


# ============================================================
# Wavenumber
# ============================================================
k_h = np.logspace(-4, 1, 1000)     # h/Mpc
k = h * k_h                         # Mpc^{-1}


# ============================================================
# Present-day matter transfer from curvature perturbation
# ============================================================
T = T_BBKS(k_h)

M = (2.0 * k**2 / (5.0 * Om0 * H0**2)) * T


# ============================================================
# Present-day dimensionless matter power spectrum
# ============================================================
Delta_m2 = M**2 * Delta_zeta2(k)


# ============================================================
# Present-day matter power spectrum
#
# P_m(k) = 2 pi^2 / k^3 * Delta_m^2(k)
# ============================================================
Pm = (2.0 * np.pi**2 / k**3) * Delta_m2


# ============================================================
# Plot
```

```
# ============================================================
plt.figure(figsize=(8, 5))

plt.loglog(k_h, Pm)

plt.xlabel(r"$k\,[h\,{\rm Mpc}^{-1}]$")
plt.ylabel(r"$P_m(k,0)\,[{\rm Mpc}^3]$")
plt.title("Present-day matter power spectrum")

plt.grid(True, which="both", alpha=0.3)
plt.tight_layout()
plt.show()
```

In conclusion, the cosmological matter power spectrum can be observationally reconstructed across approximately five decades in spatial scale. Figure 4.18 provides an overview of the observational state of the art and the power spectrum reconstructions obtained via these diverse astrophysical methods. It should be emphasized, however, that while impressive at first glance, this composite curve cannot be directly read off from observational raw data. Indeed, at small scales ($k \gtrsim 0.1\,h/\mathrm{Mpc}$), observations probe the power spectrum well into the non-linear regime. The linear reconstruction therefore relies on an analytical or numerical "mapping" between the linear and non-linear regimes (see Fig. 4.19). Furthermore, the inferral of the power spectrum from the Lyman-α forest involves non-trivial modeling of the intergalactic medium and should be interpreted with caution until confirmed by higher-resolution data.

Baryon acoustic oscillations

As established in earlier sections, baryons and radiation are tightly coupled prior to recombination. As a result, the combined photon–baryon plasma undergoes acoustic oscillations: radiation pressure resists gravitational compression, establishing standing sound waves in the photon fluid (see Fig. 4.5). The efficient momentum exchange mediated by Thomson scattering forces the baryon fluid to oscillate in acoustic phase with the photons.

As will be explored in Chapter 6, the imprint of these primordial acoustic sound waves on the cosmic microwave background has been measured with remarkable precision. These acoustic oscillations are defined by a characteristic physical scale corresponding to the sound horizon at decoupling.

Importantly, these acoustic sound waves are also imprinted onto the spatial distribution of galaxies. Because they reside directly within the baryon component, which constitutes only a fraction $\Omega_\mathrm{b}/\Omega_\mathrm{m} \sim 0.15$ of the total matter density, the resulting baryon acoustic oscillations (BAO) in the galaxy distribution are suppressed in amplitude relative to the dark matter background. The BAO peak was observationally detected in Ref. [48] using the two-point spatial correlation function of luminous red galaxies in the SDSS survey, manifesting as a distinct bump at a comoving separation of approximately $100\,h^{-1}$Mpc at redshift $z \sim 0.35$.

4.5.2 Structures in the non-linear regime

The dynamical equations governing the matter fluid are fundamentally non-linear, and our detailed analysis thus far has solved them only within the linearized approximation. To bring this chapter to a close, we review the principal physical paradigms and semi-analytical approaches used to investigate the non-linear regime, extending the preliminary treatment given in Section 4.1.4.

Hierarchical model of structure formation

To understand the non-linear evolution of cosmic structure, one must ascertain when a perturbation of a given comoving wavelength enters the non-linear regime ($\delta \sim 1$). If the matter density

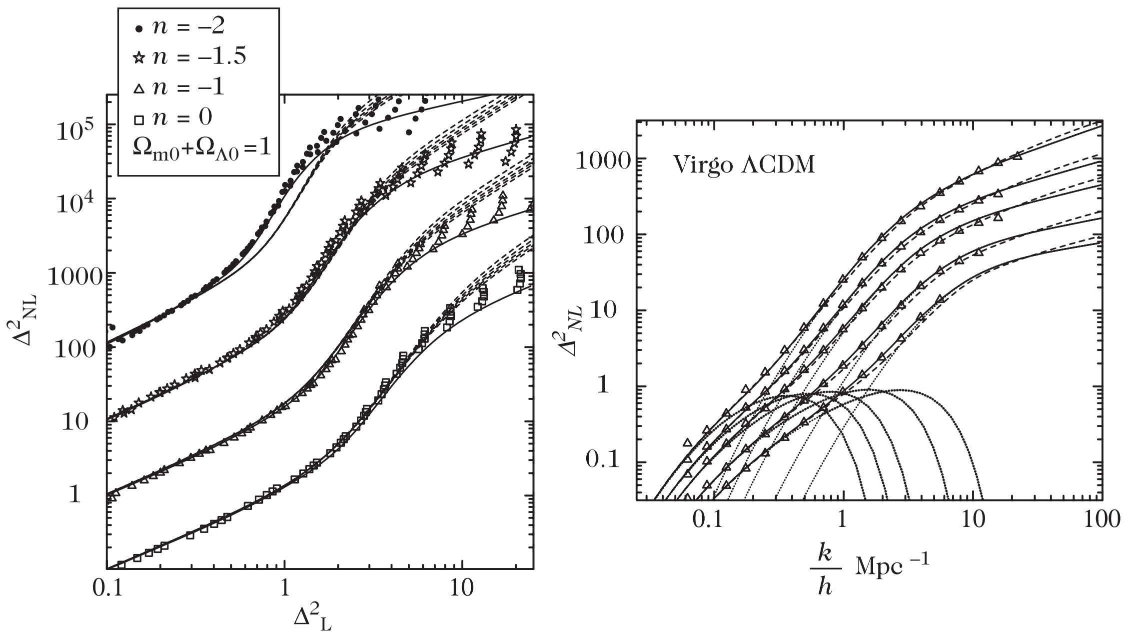


Figure 4.19: (left): Comparison between the analytic mapping developed by Ref. [51] and cosmological N-body numerical simulations [53]. For each spectral index, five epochs are plotted; from bottom to top, $1/(1+z) = 0.6$, 0.7, 0.8, 0.9, 1. (right): The non-linear power spectrum across five epochs, $z = 0$, 0.5, 1.0, 2.0, 3.0 from bottom to top. The solid curves represent the mapping of Ref. [53], whereas the dotted curves correspond to that of Ref. [52]. From Ref. [53].

contrast had remained strictly constant outside the horizon during the radiation era, all Fourier modes would have entered the non-linear regime simultaneously, since perturbation growth during matter domination is scale-independent ($D_+ \propto a$). However, sub-Hubble matter perturbations experience weak logarithmic amplification during radiation domination (see Figs. 4.13 and 4.14). Consequently, modes entering the horizon earlier undergo greater cumulative amplification prior to equality, causing smaller scales to reach the non-linear threshold first. Smaller structures therefore collapse earlier in cosmic history, establishing a *hierarchical* (or "bottom-up") scenario of structure formation.

At the epoch of matter–radiation equality, perturbation amplitudes on typical scales are roughly $\delta \sim 10^{-5}$. Short-wavelength perturbations enter the non-linear regime when $\delta_c(k,t) \sim 1$, initiating gravitational collapse into bound objects. If the Universe at that epoch had been smoothed on scales larger than these collapsing regions, it would have appeared remarkably smooth and homogeneous, so that the continuous linear density field remains valid on large scales.

As cosmic time advances, progressively larger scales cross into the non-linear regime. First, small bound halos (protogalaxies) collapse; these subsequently merge hierarchically to form massive galaxies, groups, and clusters. Today, the characteristic scale of non-linearity is approximately 20–40 Mpc, which exceeds the characteristic virial radius of galaxy clusters (~ 2 Mpc) by only one order of magnitude. Consequently, galaxy clusters are not distributed uniformly in space; instead, they trace an intricate network of filaments and sheets enclosing vast underdense cosmic voids whose typical sizes match the non-linearity scale. On these intermediate scales, the cosmic web exhibits a sponge-like morphology, while recovering statistical homogeneity when averaged over yet larger cosmological volumes.

Spherical collapse model

An insightful analytical model of non-linear gravitational collapse can be formulated by studying an idealized spherical overdense region in an expanding background. By virtue of Gauss's theorem (or Birkhoff's theorem in general relativity), the interior dynamical evolution of a spherically symmetric mass distribution depends solely on the total enclosed mass within that radius, remaining completely decoupled from the exterior universe.

In a matter-dominated Einstein–de Sitter background, the trajectory of a spherical shell of physical radius $r(t)$ enclosing mass M follows the parametric cycloid equations:

$$r = A(1 - \cos\theta), \quad \text{and} \quad t = B(\theta - \sin\theta), \tag{4.5.2}$$

where Newton's second law, $\ddot{r} = -G_{\rm N}M/r^2$, enforces the consistency relation $A^3 = G_{\rm N}MB^2$. The sphere expands, reaches a maximum turnaround radius at $\theta = \pi$, and subsequently collapses toward a point singularity ($r \to 0$) at $\theta = 2\pi$, corresponding to the collapse time $t_{\rm c} \equiv t|_{\theta=2\pi} = 2\pi B$. The mean interior density within the sphere is $\rho_{\rm int} = 3M/(4\pi r^3)$, while the background matter density in the surrounding Einstein–de Sitter spacetime is $\rho_{\rm ext} = (6\pi G_{\rm N} t^2)^{-1}$. The non-linear density contrast is given by the density ratio $\delta = \rho_{\rm int}/\rho_{\rm ext} - 1$.

In the early stages of collapse ($\theta \ll 1$), Taylor expanding the cycloid solutions (4.5.2) yields

$$r(t) \simeq \frac{A}{2}\left(\frac{6t}{B}\right)^{2/3}\left[1 - \frac{1}{20}\left(\frac{6t}{B}\right)^{2/3}\right].$$

At early times, the interior density contrast grows as $\delta \propto t^{2/3}$, exactly reproducing the growing linear solution (4.1.22) in an Einstein–de Sitter Universe.

Comparing the non-linear evolution of the sphere with the extrapolated linear perturbation solution in a flat $\Omega_{\rm m} = 1$ background reveals three critical evolutionary epochs:

1. *Turnaround and decoupling from expansion.* The overdense sphere represents a gravitationally bound system. Its physical radius reaches maximum expansion at $\theta = \pi$ ($t_{\rm ta} = \pi B = t_{\rm c}/2$), where it completely turns around and decouples from the cosmological expansion. At turnaround, the true physical density contrast reaches

$$1 + \delta_{\rm ta} = \frac{\rho_{\rm int}(t_{\rm ta})}{\rho_{\rm ext}(t_{\rm ta})} = \frac{9\pi^2}{16} \simeq 5.55 \qquad \Longrightarrow \qquad \delta_{\rm ta} \simeq 4.55,$$

   whereas the linear perturbation theory extrapolated to the same epoch predicts only $\delta_{\rm lin}(t_{\rm ta}) = \frac{3}{20}(6\pi)^{2/3} \simeq 1.06$.

2. *Gravitational collapse.* In the absence of pressure support or angular momentum, the idealized spherical shell collapses to a formal singularity at $\theta = 2\pi$ ($t = t_{\rm c}$), where $\delta \to \infty$. Extrapolating linear perturbation theory to the collapse epoch $t_{\rm c}$ yields the celebrated linear collapse threshold:

$$\delta_{\rm c} \equiv \delta_{\rm lin}(t_{\rm c}) = \frac{3}{20}(12\pi)^{2/3} \simeq 1.686 \approx 1.69.$$

3. *Virialization.* In any realistic astrophysical collapse, shell crossing, violent relaxation, and phase mixing convert the bulk kinetic energy of infalling matter into random isotropic velocity dispersion. The sphere does not collapse to zero volume; instead, by the virial theorem ($2K + U = 0$), it settles into a stable, virialized equilibrium configuration at radius $r_{\rm vir} = r_{\rm ta}/2$. Taking into account the cosmic background expansion up to $t_{\rm c}$, the mean overdensity of the resulting virialized dark matter halo relative to the background is

$$\Delta_{\rm vir} \equiv \frac{\rho_{\rm int}(t_{\rm c})}{\rho_{\rm ext}(t_{\rm c})} = 18\pi^2 \simeq 178 \approx 200.$$

This analytical model highlights the failure of linear perturbation theory once $\delta \gtrsim 1$. A rigorous relativistic description of spherically symmetric collapse in an expanding Universe is provided by the exact Lemaître–Tolman–Bondi solutions of Einstein's field equations [49] (as introduced in Chapter 3).

The non-linear power spectrum

To map the non-linear regime with quantitative precision, numerical N-body cosmological simulations are required. These simulations numerically integrate the non-linear Newtonian equations of motion (4.1.15)–(4.1.16) coupled to Poisson's equation (4.1.5) for millions to billions of gravitating particles. Comprehensive reviews of simulation algorithms, boundary conditions, and astrophysical implementations are provided in Ref. [12]; various modern N-body cosmological codes [50] are publicly available.

Extensive analysis of N-body simulations has produced phenomenological scaling relations (known as "mappings") [51–53] that relate the non-linear matter power spectrum directly to the linear power spectrum. These transformations provide powerful tools for predicting non-linear clustering without running costly simulations for every set of cosmological parameters.

A cornerstone concept is the *stable clustering hypothesis* [54], which posits that once high-density regions virialize and decouple from the Hubble flow, they maintain a fixed physical size and constant mean physical density. Consequently, the physical two-point correlation function within virialized objects scales inversely with the mean background density, $\xi(r,t) \propto 1/\bar{\rho}(t) \propto a^3$. For a scale-free primordial power spectrum of the form $P(k) \propto k^n$ in a flat universe, stable clustering implies that the non-linear two-point correlation function assumes a power-law form [54]:

$$\xi_{\rm nl}(r,t) \propto r^{-\gamma}, \quad \text{with} \quad \gamma = \frac{3(3+n)}{5+n}. \tag{4.5.3}$$

Thus, even deep in the non-linear regime, the correlation slope retains memory of the primordial spectral index n.

Building upon this insight, Hamilton et al. [51] proposed that non-linear clustering can be expressed through a universal mapping function $f_{\rm nl}$. The volume-averaged non-linear correlation function,

$$\bar{\xi}_{\rm nl}(x) = \frac{3}{x^3}\int_0^x y^2 \xi_{\rm nl}(y)\,{\rm d}y\,,$$

measured in simulations can be parameterized as a universal function of the linear correlation function $\bar{\xi}_{\rm lin}(\ell)$. The scale mapping between the non-linear radius x and the corresponding linear Lagrangian scale ℓ is obtained from mass conservation within a collapsing spherical halo. Prior to shell crossing, the enclosed mass is conserved:

$$m(<r) = \frac{4\pi}{3} r^3 \rho(<r) = \frac{4\pi}{3}\ell^3\rho(<\ell) = m_0(<\ell),$$

where $m_0(<\ell)$ is the mass initially enclosed within comoving radius ℓ. Identifying $1+\bar{\xi}_{\rm nl}$ with the local density amplification factor gives the transformation

$$x^3\left[1+\bar{\xi}_{\rm nl}(x,t)\right] = \ell^3,$$

relating the non-linear physical scale x to the linear Lagrangian scale ℓ. In terms of this transformed scale, one postulates $\bar{\xi}_{\rm nl}(x,t) = f_{\rm nl}\left[\bar{\xi}_{\rm lin}(\ell,t)\right]$.

This mapping translates naturally into Fourier space. Defining the dimensionless variance per logarithmic wavenumber interval as

$$\Delta^2(k) = 4\pi k^3 P(k), \tag{4.5.4}$$

the non-linear power spectrum $\Delta^2_{\rm nl}(k_{\rm nl})$ is related to the linear power spectrum $\Delta^2_{\rm lin}(k)$ via

$$\Delta^2_{\rm nl}(k_{\rm nl}) = f_{\rm nl}\left[\Delta^2_{\rm lin}(k)\right], \quad \text{with} \quad k^3 = \frac{k^3_{\rm nl}}{1+\Delta^2_{\rm nl}(k_{\rm nl})}. \tag{4.5.5}$$

Under the stable clustering hypothesis, the universal function must exhibit the asymptotic behavior [51]

$$f_{\rm nl}(x) \propto g^{-3}(\Omega)x^{3/2} \quad (x \gg 1), \qquad g(\Omega) \equiv \frac{D_+(a)}{a}, \tag{4.5.6}$$

where $D_+(a)$ is the linear growth factor (4.1.24). Accurate empirical fitting formulas for $f_{\rm nl}$ have been calibrated against high-resolution N-body simulations across varied ΛCDM cosmologies [51–53], and the physical existence of such a scaling relation has been grounded theoretically in the context of hierarchical collapse [55].

Figure 4.20 displays the morphological evolution of cosmic structures obtained from numerical simulations, while Fig. 4.19 compares this analytical mapping against simulation results and illustrates its deformation of the matter power spectrum. These non-linear mappings prove essential when evaluating weak gravitational lensing by large-scale cosmic structures in Chapter 7.

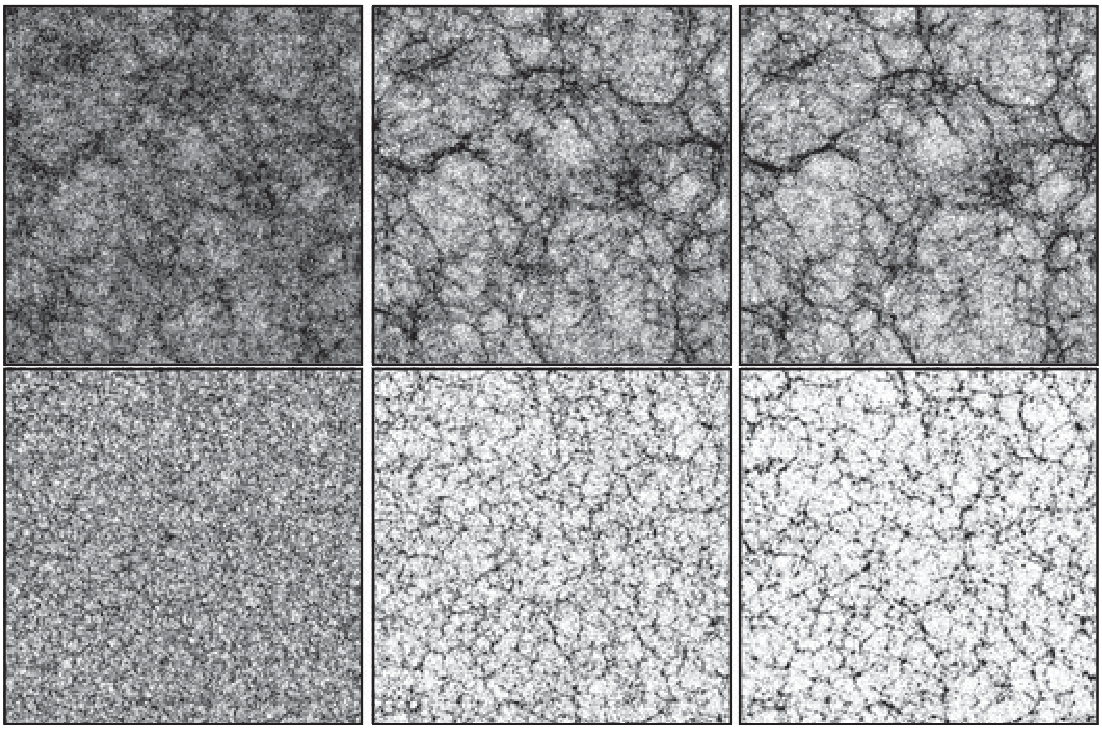


Figure 4.20: Cosmological N-body numerical simulation snapshots (top) for spectral index $n = -2$ at scale factors $1/(1+z) = 0.2$, 0.45, and 0.55 (from right to left) and (bottom) for $n = 0$ at $1/(1+z) = 0.25$, 0.63, and 0.83. The profound impact of the primordial spectral index on the morphology of large-scale structures—with prominent filamentary sheets for $n = -2$ and isolated, fragmented clumps for $n = 0$—is readily discernible. From Ref. [53].

Chapter 5

The cosmic microwave background

The primary objective of this chapter is to lay down the theoretical foundations governing the physics of the cosmic microwave background (CMB) and to elucidate the detailed derivations of the angular temperature anisotropies predicted across various cosmological frameworks.

We begin in Section 5.1 with an accessible, intuitive derivation of the angular power spectrum of temperature anisotropies, dissecting the distinct physical mechanisms that shape the observed multipoles. We establish the physical origin of CMB fluctuations and quantify how the temperature field mapped across the contemporary sky directly connects to primordial fluctuations imprinted on the last-scattering surface at decoupling. This foundational analysis equips us to analyze the qualitative and quantitative morphology of the angular power spectrum in Section 5.2.

In Section 6.3, we transition to the rigorous, modern kinetic description of the cosmic microwave background. Grounded in relativistic kinetic theory, this formulation necessitates solving the linearized relativistic Boltzmann equations for coupled photons, baryons, and collisionless components within an expanding spacetime. Finally, Section 6.5 investigates how cosmological parameters (including the respective matter energy fractions, the Hubble expansion rate, and baryonic content) alongside primordial initial conditions dictate the detailed structure and acoustic peaks of the CMB power spectrum. General foundational treatments and comprehensive reviews can be found in Refs. [1–4].

5.1 Origin of the cosmic microwave background anisotropies

Following the recombination and decoupling epoch, photons effectively cease to interact with matter and propagate freely along null geodesics to leading order. A transparent, physically intuitive picture of cosmic microwave background anisotropies can therefore be constructed by evaluating the total gravitational and kinematic redshift accumulated by a photon traversing a perturbed cosmological metric from the last-scattering surface to the present-day observer.

Within this streamlined framework, matter is treated in the hydrodynamic fluid approximation, and the last-scattering surface is idealized as an infinitely thin screen—corresponding to an instantaneous decoupling transition (as detailed in Section 4.4.1 of Chapter 4). This perspective clarifies the underlying gravitational and Doppler effects without sacrificing physical insight, while the complete kinetic treatment incorporating finite thickness and Thomson visibility will be fully developed in Section 6.3.

5.1.1 Sachs–Wolfe formula

The Sachs–Wolfe relation provides the foundational bridge connecting the observed energy of a CMB photon today to its local emission energy at the decoupling surface (Refs. [5, 6]).

Propagation of a photon in a perturbed space-time

Consider the propagation of a photon in a perturbed spacetime whose metric $g_{\mu\nu}$ takes the general form introduced in Eq. (5.52). The trajectory of the photon is defined by a null worldline $x^\mu(\lambda)$, parameterized by an affine parameter λ. The null tangent vector $k^\mu \equiv \mathrm{d}x^\mu / \mathrm{d}\lambda$ satisfies the fundamental null constraint and the geodesic equation (see Chapter 1):

$$k^\mu k_\mu = 0, \qquad k^\mu \nabla_\mu k^\nu = 0.$$

Recall that two conformally related spacetimes possess identical causal structures, meaning that their null paths coincide identically. Introducing the conformally rescaled metric $\hat{g}_{\mu\nu}$ defined by $g_{\mu\nu} = a^2 \hat{g}_{\mu\nu}$, if k^μ is the tangent vector to an affinely parameterized null geodesic of $g_{\mu\nu}$, then the vector $\hat{k}^\mu = a^2 k^\mu$ is the tangent vector to a null geodesic of $\hat{g}_{\mu\nu}$, satisfying

$$\hat{g}_{\mu\nu}\hat{k}^\mu \hat{k}^\nu = 0, \qquad \hat{k}^\mu \hat{\nabla}_\mu \hat{k}^\nu = 0,$$

where $\hat{\nabla}_\mu$ denotes the Levi-Civita connection compatible with $\hat{g}_{\mu\nu}$.

We decompose the conformal tangent four-vector $\hat{k}^\mu$ as

$$\hat{k}^\mu = E\left(1 + M,\, \hat{e}^i + \delta\hat{e}^i\right), \tag{5.1.1}$$

where E is an unperturbed constant,[1] while $(M, \delta\hat{e}^i)$ represent first-order perturbation fields. The null constraint $\hat{g}_{\mu\nu}\hat{k}^\mu \hat{k}^\nu = 0$ reduces the four perturbations to three independent degrees of freedom:

$$\gamma_{ij}\hat{e}^i\hat{e}^j = 1, \qquad \hat{e}_j \delta\hat{e}^j = A + M - B_i\hat{e}^i - \frac{1}{2}h_{ij}\hat{e}^i\hat{e}^j. \tag{5.1.2}$$

Starting from $\hat{k}^\mu \hat{k}_\mu = 0$, we find

$$\begin{aligned} 0 &= -(1+2A)[E(1+M)]^2 + 2B_iE^2(\hat{e}^i + \delta\hat{e}^i)(1+M) + (\gamma_{ij} + h_{ij})E^2(\hat{e}^i + \delta\hat{e}^i)(\hat{e}^j + \delta\hat{e}^j) \\ &= -\cancel{1} - 2A - 2M - 4AM + 2B_i(\hat{e}^i + \delta\hat{e}^i + \hat{e}^iM + \delta\hat{e}^iM) \\ &\quad + \cancel{\gamma_{ij}\hat{e}^i\hat{e}^j} + 2\gamma_{ij}\hat{e}^i\delta\hat{e}^j + \gamma_{ij}\delta\hat{e}^i\delta\hat{e}^j + h_{ij}\hat{e}^i\hat{e}^j + 2h_{ij}\hat{e}^i\delta\hat{e}^j + h_{ij}\delta\hat{e}^i\delta\hat{e}^j \end{aligned} \tag{1}$$

At first order, we find

$$\hat{e}_j\delta\hat{e}^j = A + M - B_i\hat{e}^i - \frac{1}{2}h_{ij}\hat{e}^i\hat{e}^j \tag{2}$$

To first order in metric and kinematic fluctuations, the geodesic equation for $\hat{k}^\mu$ simplifies to

$$\frac{\mathrm{d}\delta\hat{k}^\nu}{\mathrm{d}s} \equiv \hat{k}^\mu\partial_\mu\delta\hat{k}^\nu = -\delta\hat{\Gamma}^\nu_{\mu\rho}\hat{k}^\mu\hat{k}^\rho. \tag{5.1.3}$$

Evaluating the timelike component ($\nu = 0$) using the decomposition (5.1.1) yields the evolution equation for the energy perturbation M:

$$\frac{\mathrm{d}M}{\mathrm{d}s} = -A' - 2\hat{e}^i\partial_iA - \frac{1}{2}h'_{ij}\hat{e}^i\hat{e}^j + D_iB_j\hat{e}^i\hat{e}^j, \tag{5.1.4}$$

where we have defined the total derivative along the background path as $\mathrm{d}M / \mathrm{d}s \equiv \hat{e}^i\partial_iM + \partial_0M$.

[1] As an elementary consistency check, note that in an unperturbed Friedmann–Lemaître universe, the comoving observer four-velocity satisfies $u^0 \propto a^{-1}$, so that the physical photon energy scales as $k^\mu u_\mu = k^0u_0 = (\hat{k}^0/a^2)a \propto 1/a$, reflecting the cosmological redshift since $E = \hat{k}^0$ remains constant.

Sachs–Wolfe formula

As established in Chapter 1 [see Eq. (1.122)], the frequency and energy of a photon registered by an observer with four-velocity u^μ is given by $E = -k^\mu u_\mu$. Hence, the ratio of the photon energy detected today at η_0 to that emitted at $(\boldsymbol{x}_\mathrm{E}, \eta_\mathrm{E})$ is

$$\frac{E_0(\boldsymbol{e})}{E_\mathrm{E}(\boldsymbol{x}_\mathrm{E}, \eta_\mathrm{E})} = \frac{(k^\mu u_\mu)_0}{(k^\mu u_\mu)_\mathrm{E}}, \tag{5.1.5}$$

where u^μ is the baryon fluid four-velocity. In the unperturbed background, the emission and reception coordinates are linked by the light-cone trajectory

$$\boldsymbol{x}_\mathrm{E} = \boldsymbol{x}_0 + \boldsymbol{e}(\eta_0 - \eta_\mathrm{E}), \tag{5.1.6}$$

with $\boldsymbol{e} \equiv -\hat{\boldsymbol{e}}$ designating the line-of-sight unit vector pointing from the observer back toward the source. Throughout this analysis, we adopt the Born approximation, evaluating all perturbative expressions along this unperturbed background geodesic.

From Eq. (5.1.5), we deduce that the radiation temperature observed along direction $\boldsymbol{e}$ is connected to the emission temperature by $f(\eta_0) = f(\eta_\mathrm{E})$

$$\frac{E_0}{T_0} = \frac{E_\mathrm{E}}{T_\mathrm{E}} \implies \frac{T_0(\boldsymbol{e})}{T_\mathrm{E}(\boldsymbol{x}_\mathrm{E}, \eta_\mathrm{E})} = \frac{a(\eta_\mathrm{E})}{a(\eta_0)} \left\{1 + \left[M + A + e^i(v_{\mathrm{b}i} + B_i)\right]_\mathrm{E}^0\right\}, \tag{5.1.7}$$

we used

$$(k^\mu)_0 = \frac{E}{a^2(\eta_0)}(1 + M, \hat{e}^i + \delta\hat{e}^i)_0 \tag{1}$$

$$(k^\mu)_\mathrm{E} = \frac{E}{a^2(\eta_\mathrm{E})}(1 + M, \hat{e}^i + \delta\hat{e}^i)_\mathrm{E} \tag{2}$$

Then,

$$\frac{E_0}{E_\mathrm{E}} = \frac{a(\eta_\mathrm{E})}{a(\eta_0)} \frac{(1 + M, -\hat{e}^i + \delta\hat{e}^i) \cdot (-1 - A, v_k + B_k)|_0}{(1 + M, -\hat{e}^i + \delta\hat{e}^i) \cdot (-1 - A, v_k + B_k)|_\mathrm{E}} \tag{3}$$

$$= \frac{a(\eta_\mathrm{E})}{a(\eta_0)} \frac{[-1 - M_0 - A_0 - \hat{e}^i(v_k + B_i)_0 + \mathcal{O}(\epsilon^2)]}{[-1 - M_\mathrm{E} - A_\mathrm{E} - \hat{e}^i(v_k + B_i)_\mathrm{E} + \mathcal{O}(\epsilon^2)]} \tag{4}$$

$$= \frac{a(\eta_\mathrm{E})}{a(\eta_0)} \left\{1 + \left[M + A + e^i(v_{\mathrm{b}i} + B_i)\right]_\mathrm{E}^0\right\} \tag{5}$$

where we substituted the perturbed four-velocity $u_\mu = a(-1 - A, v_k + B_k)$, with $v_{\mathrm{b}i}$ being the velocity perturbation of the baryon fluid (see Chapter 4).

Because the recombination epoch is dictated by the photon–electron interaction, governed by the interaction coefficient $\sigma_\mathrm{T} n_\mathrm{e}$ (see Chapter 4, Section 4.4.1), the last-scattering surface can be modeled as a hypersurface of constant electron number density, $n_\mathrm{e} = \mathrm{const}$. Owing to local density fluctuations, recombination does not transpire synchronously across all points in the Universe (as illustrated in Fig. 5.1). The conformal time of emission is accordingly decomposed as

$$\eta_\mathrm{E} = \bar{\eta}_\mathrm{E} + \delta\eta_\mathrm{E}.$$

The emission and observed temperature fields are decomposed into spatial averages plus fractional contrasts:

$$T_\mathrm{E}(\boldsymbol{x}_\mathrm{E}, \eta_\mathrm{E}) = \bar{T}_\mathrm{E}(\eta_\mathrm{E}) \left[1 + \Theta_\mathrm{E}(\boldsymbol{x}_\mathrm{E}, \eta_\mathrm{E})\right],$$

$$T_0(\boldsymbol{e}) = \bar{T}_0(\eta_0) \left[1 + \Theta_0(\boldsymbol{e})\right],$$

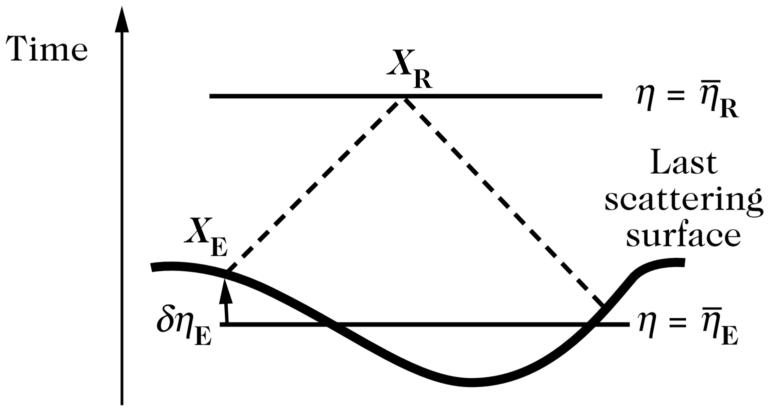


Figure 5.1: The last-scattering surface is a surface of constant electron number density, $n_\mathrm{e} = \mathrm{const.}$ Thus, it is not a constant-time hypersurface. The time of emission is related to the mean time by $\eta_\mathrm{E} = \bar{\eta}_\mathrm{E} + \delta\eta_\mathrm{E}$.

where $\bar{T}_\mathrm{E}$ is the spatial average across the emission hypersurface at time η_E, and $\Theta_0(\boldsymbol{e})$ denotes the temperature contrast observed in direction $\boldsymbol{e}$ compared to the average sky temperature $\bar{T}_0(\eta_0)$.

Substituting these two expansions into Eq. (5.1.7), we find that at lowest order in perturbations

$$\frac{\text{present day CMB temp.}}{\text{CMB temp. during emission}} = \frac{\bar{T}_0}{\bar{T}_\mathrm{E}} = \frac{a(\bar{\eta}_\mathrm{E})}{a(\eta_0)}, \tag{5.1.8}$$

which is simply the standard cosmological redshift scaling of blackbody radiation in a homogeneous and isotropic expanding spacetime (see Chapter 4).

Using the relation $\bar{T}_\mathrm{E}(\eta_\mathrm{E})a(\eta_\mathrm{E}) = \bar{T}_\mathrm{E}(\bar{\eta}_\mathrm{E})[1 + (T'/T)\delta\eta_\mathrm{E}]a(\bar{\eta}_\mathrm{E})[1 + (a'/a)\delta\eta_\mathrm{E}] = \bar{T}_\mathrm{E}(\bar{\eta}_\mathrm{E})a(\bar{\eta}_\mathrm{E})$, valid to first order in perturbations, we infer that the observed temperature contrast in the direction $\boldsymbol{e}$ is given by

$$\frac{T_0}{T_\mathrm{E}} = \frac{1 + \Theta_0(\mathbf{e})}{1 + \Theta_\mathrm{E}(\boldsymbol{x}_\mathrm{E}, \bar{\eta}_\mathrm{E})} \frac{a(\bar{\eta}_\mathrm{E})}{a(\eta_0)} \implies \Theta_0(\boldsymbol{e}) = \Theta_\mathrm{E}(\boldsymbol{x}_\mathrm{E}, \bar{\eta}_\mathrm{E}) + \Big[M + A + e^i(v_{\mathrm{b}i} + B_i)\Big]_\mathrm{E}^0. \tag{5.1.9}$$

Again, within the Born approximation, the bracketed first-order quantity may be evaluated either at $\bar{\eta}_\mathrm{E}$ or $\bar{\eta}_\mathrm{R}$.

To evaluate the intrinsic emission fluctuation Θ_E, we observe that the free electron density n_e is a function of the photon and baryon energy densities, ρ_γ and ρ_b. As evidenced by the Saha ionization equation (Ref. [6]), the effect of baryons can be neglected to a good approximation, so that the last-scattering surface coincides closely with the surface $\rho_\gamma = \mathrm{const}$ (see Chapter 4). Applying the Stefan–Boltzmann law $\rho_\gamma \propto T^4$, we obtain

$$\frac{\delta\rho_r}{\rho_r} = 4\frac{\delta T}{T} \implies \Theta_\mathrm{E}(\boldsymbol{x}_\mathrm{E}, \bar{\eta}_\mathrm{E}) = \frac{1}{4}\delta_\gamma(\boldsymbol{x}_\mathrm{E}, \bar{\eta}_\mathrm{E}). \tag{5.1.10}$$

To calculate the contribution $[M]_\mathrm{E}^0$, we integrate the geodesic equation (5.1.4) along the unperturbed ray:

$$[M]_\mathrm{E}^0 = -2[A]_\mathrm{E}^0 + \int_\mathrm{E}^0 \left(A' - \frac{1}{2}h'_{ij}e^i e^j + D_i B_j e^i e^j\right) \mathrm{d}s\,, \tag{5.1.11}$$

where the integration is performed along the unperturbed null path.

From (5.1.4)

$$\frac{\mathrm{d}M}{\mathrm{d}s} = A' - 2\frac{\mathrm{d}A}{\mathrm{d}s} - \frac{1}{2}h'_{ij}\hat{e}^i\hat{e}^j + D_iB_j\hat{e}^i\hat{e}^j, \tag{1}$$

using (4.2.5)

$$\frac{1}{2}h'_{ij} = C'\gamma_{ij} + D_iD_jE' + \frac{1}{2}(D_i\bar{E}'_j + D_j\bar{E}'_i) + \bar{E}'_{ij} \qquad \text{with} \qquad D_i\bar{E}^{ij} = 0, \quad \bar{E}^i{}_i = 0. \tag{2}$$

we find:

$$M|^0_{\mathrm{E}} = -2A|^0_{\mathrm{E}} + \int_{\mathrm{E}}^0 \left[A' - \left(C'\gamma_{ij} + D_iD_jE' + D_i\bar{E}'_j + \bar{E}'_{ij} + D_iB_j\right)\hat{e}^i\hat{e}^j\right] ds, \tag{3}$$

Combining Eqs. (5.1.9), (5.1.10), and (5.1.11), we deduce:

$$\begin{aligned}\Theta_0(\boldsymbol{e}) = &\left[\frac{1}{4}\delta_\gamma + A - e^i(v_{\mathrm{b}i} + B_i)\right](\boldsymbol{x}_{\mathrm{E}}, \eta_{\mathrm{E}}) \\ &+ \int_{\mathrm{E}}^0 \left[A' - C' - e^ie^j\left(D_iD_jE' + D_i\overline{E}'_j - D_iB_j + \overline{E}'_{ij}\right)\right]\mathrm{d}s + f(0),\end{aligned} \tag{5.1.12}$$

where $f(0)$ gathers terms evaluated today at the observer position. Rewriting this in terms of gauge-invariant cosmological perturbations, and utilizing the identity $e^jD_j(\overline{E}'_i - \overline{B}_i) = e^jD_j\overline{\Phi}_i = -\overline{\Phi}'_i + \mathrm{d}\overline{\Phi}_i/\,\mathrm{d}s$, this expression takes the form

$$\boxed{\begin{aligned}\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}) = &\left[\frac{1}{4}\delta^{\mathrm{N}}_\gamma + \Phi - e^i\left(D_iV_{\mathrm{b}} + \overline{V}_{\mathrm{b}i} + \overline{\Phi}_i\right)\right](\boldsymbol{x}_{\mathrm{E}}, \eta_{\mathrm{E}}) \\ &+ \int_{\mathrm{E}}^0 (\Phi' + \Psi')[\boldsymbol{x}(\eta), \eta]\mathrm{d}\eta - \int_{\mathrm{E}}^0 e^i\overline{\Phi}'_i[\boldsymbol{x}(\eta), \eta]\mathrm{d}\eta \\ &- \int_{\mathrm{E}}^0 e^ie^j\overline{E}'_{ij}[\boldsymbol{x}(\eta), \eta]\mathrm{d}\eta,\end{aligned}} \tag{5.1.13}$$

where we integrate along the unperturbed geodesic parameterized in conformal time as

$$\boldsymbol{x} = \boldsymbol{x}_0 + \boldsymbol{e}(\eta_0 - \eta), \tag{5.1.14}$$

so that the integral runs directly over η. We have intentionally dropped the local monopole shift $f(0)$, which depends solely on the observer location and cannot be measured observationally. Equation (5.1.13) is celebrated as the Sachs–Wolfe equation.

Interpretation

The general Sachs–Wolfe formula (5.1.13) decomposes naturally into the sum of three distinct physical contributions:

$$\Theta_{\mathrm{SW}} = \left(\frac{1}{4}\delta^{\mathrm{N}}_\gamma + \Phi\right)(\boldsymbol{x}_{\mathrm{E}}, \eta_{\mathrm{E}}) = \left(\frac{1}{4}\delta^{\mathrm{F}}_\gamma + \Phi + \Psi\right)(\boldsymbol{x}_{\mathrm{E}}, \eta_{\mathrm{E}}), \tag{5.1.15}$$

$$\Theta_{\mathrm{dop}} = -e^i\left(V_{\mathrm{b}i} + \overline{\Phi}_i\right)(\boldsymbol{x}_{\mathrm{E}}, \eta_{\mathrm{E}}), \tag{5.1.16}$$

$$\Theta_{\mathrm{ISW}} = \int_{\mathrm{E}}^0 \left[(\Phi' + \Psi') - e^i\overline{\Phi}'_i - e^ie^j\overline{E}'_{ij}\right]\mathrm{d}\eta. \tag{5.1.17}$$

The first term receives exclusively scalar contributions and is termed the ordinary Sachs–Wolfe effect; the second contains scalar and vector contributions representing the Doppler term; while the third term integrates across all three perturbation species along the photon path and is known as the integrated Sachs–Wolfe (ISW) term.

The ordinary Sachs–Wolfe term Θ_{SW} is evaluated locally at decoupling $(\boldsymbol{x}_{\mathrm{E}}, \bar{\eta}_{\mathrm{E}})$ and reflects two competing physical effects:

- the intrinsic density contrast of the radiation fluid, reflecting that overdense regions are hotter by virtue of the Stefan–Boltzmann law;
- the local gravitational potential, indicating that a photon emitted inside a gravitational potential well experiences an additional gravitational redshift (the Einstein effect).

The Doppler term Θ_{dop} arises purely from the relative kinematic velocity between the emitting plasma and the observing reference frame.

The integral term Θ_{ISW} depends upon the photon's propagation history between emission and reception, containing the conformal time derivatives of the gravitational potentials in the SVT decomposition. These terms receive significant contributions from:

- cosmic structures in the process of gravitational formation traversed by the photon: as a structure collapses, its gravitational potential deepens, breaking time symmetry as the photon enters and exits the well, imprinting a net frequency shift;
- late-time cosmic expansion in a universe with non-vanishing dark energy or spatial curvature, wherein the gravitational potentials decay in the recent cosmological epoch.

5.1.2 Angular power spectrum

The Sachs–Wolfe equation relates the observed directional temperature fluctuations to primordial cosmological perturbations. Thus, $\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e})$ constitutes a stochastic field on the celestial sphere, characterized statistically by its two-point angular correlation function:

$$C(\vartheta) = \langle \Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}_1)\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}_2)\rangle. \tag{5.1.18}$$

Statistical isotropy, guaranteed by the cosmological principle, dictates that this correlation function depends only on the separation angle ϑ between the observation unit vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$, with $\cos\vartheta = \boldsymbol{e}_1 \cdot \boldsymbol{e}_2$. It is accordingly convenient to expand $C(\vartheta)$ in the basis of Legendre polynomials:

$$C(\vartheta) = \sum_{\ell} \frac{2\ell+1}{4\pi} C_\ell P_\ell(\boldsymbol{e}_1 \cdot \boldsymbol{e}_2), \tag{5.1.19}$$

which defines the angular power spectrum C_ℓ. A multipole moment ℓ corresponds approximately to an angular aperture π/ℓ on the sky (see Appendix B), making C_ℓ a direct probe of the temperature fluctuation variance at that angular scale. For Gaussian temperature fluctuations, this spectrum completely specifies the statistical distribution.

Expanding $\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e})$ in spherical harmonics,

$$\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}) = \sum_{\ell\geq 1} \sum_{m=-\ell}^{\ell} a_{\ell m}(\boldsymbol{x}_0, \eta_0) Y_{\ell m}(\boldsymbol{e}), \tag{5.1.20}$$

the expansion multipole coefficients are given by

$$a_{\ell m}(\boldsymbol{x}_0, \eta_0) = \int \mathrm{d}^2\boldsymbol{e}\, \Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}) Y^*_{\ell m}(\boldsymbol{e}). \tag{5.1.21}$$

Using the addition theorem [Eq. (B.17)] to represent the Legendre polynomial in Eq. (5.1.19) and inverting using the orthogonality of spherical harmonics [Eq. (B.15)], we find

$$(2\ell+1)C_\ell = \sum_m \langle a_{\ell m}(\boldsymbol{x}_0, \eta_0) a^*_{\ell m}(\boldsymbol{x}_0, \eta_0)\rangle. \tag{5.1.22}$$

Since $a_{00} = 0$, and the $\ell = 1$ multipole is dominated by the Doppler effect arising from the motion of the Solar System with respect to the surface of last scattering, which cannot be cleanly separated from the observed signal, we restrict our analysis to $\ell \geq 2$.

We now proceed to compute this angular power spectrum as a function of the cosmological perturbations within the SVT decomposition. For each perturbation mode, we Fourier expand $\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e})$ as

$$\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}) = \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^{3/2}} \hat{\Theta}(\eta_0, \boldsymbol{k}, \boldsymbol{e}), \tag{5.1.23}$$

where the phase $\exp(\mathrm{i}\boldsymbol{k} \cdot \boldsymbol{x}_0)$ has been absorbed into the definition of $\hat{\Theta}$. It follows that

$$a_{\ell m}(\boldsymbol{x}_0, \eta_0) = \int \mathrm{d}^2\boldsymbol{e} \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^{3/2}} \hat{\Theta}(\eta_0, \boldsymbol{k}, \boldsymbol{e}) Y^*_{\ell m}(\boldsymbol{e}). \tag{5.1.24}$$

Henceforth, the explicit dependence on $(\boldsymbol{x}_0, \eta_0)$ in $a_{\ell m}$ is omitted.

Scalar modes

After substituting $\boldsymbol{x}_{\mathrm{E}}$ via Eq. (5.1.6), the scalar sector of Eq. (5.1.13) indicates that

$$\hat{\Theta}(\eta_0, \boldsymbol{k}, \boldsymbol{e}) = \left[\hat{\Theta}_{\mathrm{SW}}(\boldsymbol{k}) - \mathrm{i}k\mu\hat{V}_{\mathrm{b}}(\boldsymbol{k})\right] \mathrm{e}^{\mathrm{i}k\mu\Delta\eta_{\mathrm{E}}} + \int (\hat{\Phi}' + \hat{\Psi}')\mathrm{e}^{\mathrm{i}k\mu\Delta\eta}\mathrm{d}\eta, \tag{5.1.25}$$

with $\Delta\eta \equiv \eta_0 - \eta$, $\Delta\eta_{\mathrm{E}} \equiv \eta_0 - \bar{\eta}_{\mathrm{E}}$, and $\mu \equiv (\boldsymbol{k} \cdot \boldsymbol{e})/k$. The exponential factor $\exp(\mathrm{i}k\mu\Delta\eta)$ appears under the integral because the spatial Fourier transform contains $(\hat{\Phi}' + \hat{\Psi}') \exp[\mathrm{i}\boldsymbol{k} \cdot \boldsymbol{x}(\eta)]$, while Eq. (5.1.14) implies $\exp[\mathrm{i}\boldsymbol{k} \cdot \boldsymbol{x}(\eta)] = \exp(\mathrm{i}k\mu\Delta\eta) \exp(\mathrm{i}\boldsymbol{k} \cdot \boldsymbol{x}_0)$, with the constant phase absorbed into the Fourier amplitude [see Eq. (5.1.23)]. An identical mechanism accounts for the factor $\exp(\mathrm{i}k\mu\Delta\eta_{\mathrm{E}})$ in the first two terms.

Each term in Eq. (5.1.25) is a stochastic variable factorizing as $X(\boldsymbol{k}, \eta) = X(k, \eta)a(\boldsymbol{k})$, where $a(\boldsymbol{k})$ satisfies the random Gaussian property

$$\langle a(\boldsymbol{k})a^*(\boldsymbol{k}')\rangle = \delta^{(3)}(\boldsymbol{k} - \boldsymbol{k}'), \tag{5.1.26}$$

such that $\hat{\Theta}(\eta_0, \boldsymbol{k}, \boldsymbol{e}) = \hat{\Theta}(\eta_0, k, \mu)a(\boldsymbol{k})$ with

$$\hat{\Theta}(\eta_0, k, \mu) = \left[\hat{\Theta}_{\mathrm{SW}}(k) - \hat{V}_{\mathrm{b}}(k)\partial_{\Delta\eta_{\mathrm{E}}}\right] \mathrm{e}^{\mathrm{i}k\mu\Delta\eta_{\mathrm{E}}} + \int (\hat{\Phi}' + \hat{\Psi}')\mathrm{e}^{\mathrm{i}k\mu\Delta\eta}\mathrm{d}\eta. \tag{5.1.27}$$

Using the spherical harmonic plane-wave expansion [Eq. (B.21)] for the exponential factor, we obtain

$$\hat{\Theta}(\eta_0, \boldsymbol{k}, \mu) = 4\pi \sum_{\ell m} \hat{\Theta}^{(\mathrm{S})}_\ell(k) Y^*_{\ell m}(\hat{\boldsymbol{k}}) Y_{\ell m}(\boldsymbol{e}), \tag{5.1.28}$$

with the scalar multipole transfer function

$$\boxed{\hat{\Theta}^{(\mathrm{S})}_\ell(k) = \hat{\Theta}_{\mathrm{SW}}(k) j_\ell(k\Delta\eta_{\mathrm{E}}) - \frac{\hat{V}_{\mathrm{b}}(k)}{k} j'_\ell(k\Delta\eta_{\mathrm{E}}) + \int (\hat{\Phi}' + \hat{\Psi}') j_\ell(k\Delta\eta)\mathrm{d}\eta.} \tag{5.1.29}$$

Inserting these expressions into Eq. (5.1.22) and applying Eq. (5.1.26), we arrive at

$$\begin{aligned}(2\ell + 1)C^{\mathrm{S}}_\ell &= \sum_m \int \mathrm{d}^2\boldsymbol{e}_1 \mathrm{d}^2\boldsymbol{e}_2 \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3} \Theta(\eta_0, \boldsymbol{k}, \mu_1)\Theta^*(\eta_0, \boldsymbol{k}, \mu_2) Y_{\ell m}(\boldsymbol{e}_1) Y^*_{\ell m}(\boldsymbol{e}_2) \\ &= \frac{2}{\pi} \int k^2\, \mathrm{d}k \sum_{\ell_1 m_1 \ell_2 m_2 m} \hat{\Theta}^{(\mathrm{S})}_{\ell_1}(k) \hat{\Theta}^{(\mathrm{S})*}_{\ell_2}(k) \int \mathrm{d}^2\hat{\boldsymbol{k}} \mathrm{d}^2\boldsymbol{e}_1 \mathrm{d}^2\boldsymbol{e}_2\end{aligned}$$

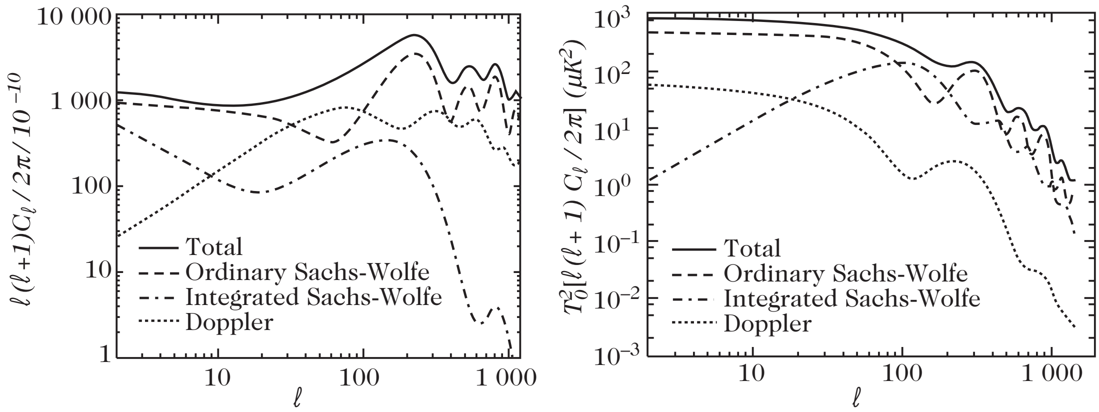


Figure 5.2: The angular power spectrum and its three contributions, ordinary Sachs–Wolfe, Doppler and integrated Sachs–Wolfe for adiabatic (left) and isocurvature (right) initial conditions and a scale invariant primordial spectrum ($n_S = 1$), as predicted by inflation (see Chapter 8).

$$\times Y_{\ell_1 m_1}(\hat{\boldsymbol{k}})Y^*_{\ell_2 m_2}(\hat{\boldsymbol{k}})Y_{\ell m}(\boldsymbol{e}_1)Y^*_{\ell_1 m_1}(\boldsymbol{e}_1)Y_{\ell_2 m_2}(\boldsymbol{e}_2)Y^*_{\ell m}(\boldsymbol{e}_2).$$

Integration over the unit wavevector $\hat{\boldsymbol{k}} = \boldsymbol{k}/k$ yields the factor $\delta_{\ell_1\ell_2}\delta_{m_1m_2}$, while integrating over $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ produces $\delta_{\ell_1\ell}\delta_{m_1m}$ and $\delta_{\ell\ell_2}\delta_{mm_2}$, respectively, yielding the boxed scalar angular power spectrum

$$\boxed{C_\ell^S = \frac{2}{\pi}\int \left|\hat{\Theta}_\ell^{(S)}(k)\right|^2 k^2\,\mathrm{d}k\,.} \tag{5.1.30}$$

Figure 5.2 decomposes the scalar angular power spectrum into its ordinary Sachs–Wolfe, Doppler, and integrated Sachs–Wolfe contributions.

Vector modes

We now perform the corresponding computation for the vector contributions. The Fourier expansion (5.1.23) yields (from (5.1.13))

$$\hat{\Theta}(\eta_0, \boldsymbol{k}, \boldsymbol{e}) = -e^i\left[\overline{V}_{bi}(\boldsymbol{k}) + \overline{\Phi}_i(\boldsymbol{k})\right]\mathrm{e}^{\mathrm{i}k\mu\Delta\eta_E} - \int e^i\overline{\Phi}_i'\mathrm{e}^{\mathrm{i}k\mu\Delta\eta}\mathrm{d}\eta. \tag{5.1.31}$$

Because vector perturbations are transverse, their Fourier coefficients possess no components longitudinal to the wavevector $\boldsymbol{k}$. They can thus be expanded in an orthonormal basis $(\boldsymbol{e}_1, \boldsymbol{e}_2)$ perpendicular to $\boldsymbol{k}$ as

$$\overline{X}_i(\boldsymbol{k},\eta) = \sum_{\lambda=\pm}\overline{X}_\lambda(k,\eta)e_i^\lambda(\boldsymbol{k})a_\lambda(\boldsymbol{k}), \tag{5.1.32}$$

with the complex circular polarization vectors $e_\pm^i \equiv e_1^i \pm \mathrm{i}e_2^i$, satisfying $e_\pm^i e_i = \sin\theta\exp(\pm\mathrm{i}\varphi)$. The two helicity states are statistically independent, with the random amplitudes satisfying

$$\langle a_\lambda(\boldsymbol{k})a^*_{\lambda'}(\boldsymbol{k}')\rangle = \delta^{(3)}(\boldsymbol{k}-\boldsymbol{k}')\delta_{\lambda\lambda'}. \tag{5.1.33}$$

Using Eq. (B.21) to expand the plane-wave exponential in spherical harmonics, we obtain

$$\hat{\Theta}(\eta_0, \boldsymbol{k}, \boldsymbol{e}) = 4\pi\sum_{\lambda=\pm}\sum_{\ell m}\left(\sin\vartheta\mathrm{e}^{\mathrm{i}\lambda\varphi}\right)\widetilde{\Theta}^{(V)}_{\lambda,\ell}(k)Y_{\ell m}(\boldsymbol{k})Y^*_{\ell m}(\boldsymbol{e})a_\lambda(\boldsymbol{k}), \tag{5.1.34}$$

with

$$\widetilde{\Theta}^{(V)}_{\lambda,\ell}(k) = -\left[\overline{V}_\lambda(k) + \overline{\Phi}_\lambda(k)\right]j_\ell(k\Delta\eta_E) - \int \overline{\Phi}'_\lambda(k)j_\ell(k\Delta\eta)\mathrm{d}\eta. \tag{5.1.35}$$

Inserting these expressions into Eq. (5.1.22), using the statistical orthogonality relation (5.1.33), and performing the integration over $\mathrm{d}^2\hat{\boldsymbol{k}}$, we obtain

$$(2\ell+1)C_\ell^{\mathrm{V}} = \frac{2}{\pi}\int k^2\,\mathrm{d}k \sum_{\ell_1,\lambda}\left|\widetilde{\Theta}^{(\mathrm{V})}_{\lambda,\ell_1}(k)\right|^2 \sum_{mm_1}\left|\int \mathrm{d}^2\boldsymbol{e}\,\sin\vartheta\mathrm{e}^{\mathrm{i}\lambda\varphi}Y_{\ell m}(\boldsymbol{e})Y^*_{\ell_1 m_1}(\boldsymbol{e})\right|^2 .$$

To calculate the integral over $\boldsymbol{e}$, we first note that $\sin\vartheta\mathrm{e}^{\mathrm{i}\lambda\varphi} = -\lambda 2\sqrt{2\pi/3}Y_{1\lambda}(\boldsymbol{e})$ and evaluate the triple spherical harmonic integral using Eq. (B.23). The summation over m and m_1 is then performed via the orthogonality relations of the Wigner $3j$-symbols [Eq. (B.25)]. This sum restricts the non-vanishing contributions strictly to $\ell_1 = \ell \pm 1$, causing the spherical Bessel functions j_{ℓ_1} appearing in $\widetilde{\Theta}^{(\mathrm{V})}_{\lambda,\ell_1}$ to recombine into the composite radial function [see Eq. (B.46)]:

$$j_\ell^{(11)}(x) = \sqrt{\frac{\ell(\ell+1)}{2}}\frac{j_\ell(x)}{x}. \tag{5.1.36}$$

We finally obtain the boxed vector angular power spectrum and transfer function:

$$\boxed{\begin{aligned} C_\ell^{\mathrm{V}} &= \frac{2}{\pi}\int\sum_\lambda\left|\hat{\Theta}^{(\mathrm{V})}_{\lambda,\ell}(k)\right|^2 k^2\,\mathrm{d}k\,, && (5.1.37)\\ \hat{\Theta}^{(\mathrm{V})}_{\lambda,\ell}(k) &= [\overline{V}_\lambda(k)+\overline{\Phi}_\lambda(k)]j_\ell^{(11)}(k\Delta\eta_{\mathrm{E}}) + \int\overline{\Phi}'_\lambda(k)j_\ell^{(11)}(k\Delta\eta)\mathrm{d}\eta. && (5.1.38)\end{aligned}}$$

Tensor modes

The computation for tensor modes proceeds along the same mathematical lines as that for vector modes. The Fourier expansion (5.1.23) yields

$$\hat{\Theta}(\eta_0,\boldsymbol{k},\boldsymbol{e}) = -\int e^j e^i \overline{E}'_{ij}\mathrm{e}^{\mathrm{i}k\mu\Delta\eta}\mathrm{d}\eta. \tag{5.1.39}$$

The transverse-traceless metric tensor $\overline{E}_{ij}$ is expanded as

$$\overline{E}_{ij}(\boldsymbol{k},\eta) = \sum_\lambda \overline{E}_\lambda(k,\eta)\varepsilon^\lambda_{ij}a_\lambda(\boldsymbol{k}), \tag{5.1.40}$$

where the random amplitude $a_\lambda(\boldsymbol{k})$ satisfies Eq. (5.1.33) and the polarization tensors are defined by

$$\varepsilon^\lambda_{ij} = e_i^+e_j^+\delta^\lambda_+ + e_i^-e_j^-\delta^\lambda_\times. \tag{5.1.41}$$

In particular, this implies that $\varepsilon^\lambda_{ij}e^ie^j = \sin^2\theta\exp(2\mathrm{i}\lambda\varphi)$.

Applying Eq. (B.21) to expand the plane-wave exponential factor, we find

$$\hat{\Theta}(\eta_0,\boldsymbol{k},\boldsymbol{e}) = 4\pi\sum_{\lambda=\pm}\sum_{\ell m}\left(\sin^2\vartheta\mathrm{e}^{2\mathrm{i}\lambda\varphi}\right)\widetilde{\Theta}^{(\mathrm{T})}_{\lambda,\ell}(k)Y_{\ell m}(\boldsymbol{k})Y^*_{\ell m}(\boldsymbol{e})a_\lambda(\boldsymbol{k}), \tag{5.1.42}$$

with

$$\widetilde{\Theta}^{(\mathrm{T})}_{\lambda,\ell}(k) = -\int\overline{E}'_\lambda(k)j_\ell(k\Delta\eta)\mathrm{d}\eta. \tag{5.1.43}$$

Inserting these expressions into Eq. (5.1.22), using Eq. (5.1.33), and performing the integration over $\mathrm{d}^2\hat{\boldsymbol{k}}$, we find that

$$(2\ell+1)C_\ell^{\mathrm{T}} = \frac{2}{\pi}\int k^2\,\mathrm{d}k\sum_{\ell_1,\lambda}\left|\widetilde{\Theta}^{(\mathrm{T})}_{\lambda,\ell_1}(k)\right|^2\sum_{mm_1}\left|\int\mathrm{d}^2\boldsymbol{e}\,\sin^2\vartheta\mathrm{e}^{2\mathrm{i}\lambda\varphi}Y_{\ell m}(\boldsymbol{e})Y^*_{\ell_1 m_1}(\boldsymbol{e})\right|^2 .$$

To evaluate the angular integral over $\boldsymbol{e}$, the procedure mirrors the vector derivation, using the identity $\sin^2\vartheta\mathrm{e}^{\pm 2\mathrm{i}\varphi} = 4\sqrt{2\pi/15}Y_{2\pm 2}(\boldsymbol{e})$. Summation over the Wigner $3j$-symbols restricts

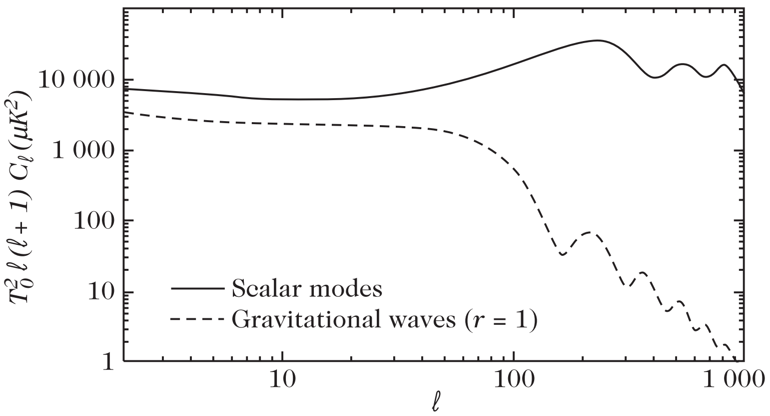


Figure 5.3: Angular power spectrum for the tensor modes, compared to that of scalars for a ratio $r = 1$ between the primordial spectra of the two modes.

surviving terms to $\ell_1 = \ell \pm 2$ and $\ell_1 = \ell$. Using the Bessel relation (B.47), the j_{ℓ_1} factors combine into the tensor radial function [see Eq. (B.46)]:

$$\tilde{j}_\ell^{(22)}(x) = \sqrt{\frac{(\ell+2)!}{(\ell-2)!}}\frac{j_\ell(x)}{x^2}. \tag{5.1.44}$$

We finally obtain the boxed tensor angular power spectrum and transfer function:

$$\boxed{C_\ell^{\mathrm{T}} = \frac{1}{\pi}\int \sum_\lambda \left|\hat{\Theta}_{\lambda,\ell}^{(\mathrm{T})}(k)\right|^2 k^2\,\mathrm{d}k\,,} \tag{5.1.45}$$

$$\boxed{\hat{\Theta}_{\lambda,\ell}^{(\mathrm{T})}(k) = -\int \overline{E}'_\lambda(k)\tilde{j}_\ell^{(22)}(k\Delta\eta)\mathrm{d}\eta.} \tag{5.1.46}$$

5.2 Properties of the angular power spectrum

The previous description allows us to understand the general shape of the power spectra of the scalar and tensor modes. The vector modes will be neglected in the rest of this chapter. Complementary references on the study of the properties of the temperature power spectrum can be found in Refs. [2, 7].

5.2.1 Large angular scales

On sufficiently large angular scales, corresponding to perturbation wavelengths that were super-Hubble at the recombination epoch, the gravitational potential remains effectively constant throughout the matter-dominated era (see Chapter 5). For such super-horizon modes, one can analytically evaluate the low-ℓ angular power spectrum directly in terms of the primordial power spectrum of metric fluctuations. This low-multipole regime of the angular spectrum constitutes the celebrated Sachs–Wolfe plateau.

Adiabatic perturbations

The primordial adiabatic initial conditions [Eq. (5.223)] dictate that initially $\delta_\gamma^{\mathrm{N}}(0) = -2\Psi(0)$ in the Newtonian gauge, meaning that radiation overdensities coincide precisely with potential wells, whilst the initial radiation velocity field vanishes, $V_\gamma(0) = 0$. On scales extending far beyond the Hubble radius at decoupling, causal causal transport prevents V_γ from undergoing appreciable dynamical growth. Equation (5.255) then demonstrates that the linear combination $\delta_\gamma^{\rm N} - 4\Psi$ is conserved. During the subsequent matter-dominated era, the fluid relations $\delta_\gamma^{\rm N}/4 = \delta_{\rm m}^{\rm N}/3 = -2\Phi/3$ [cf. Eqs. (5.220) and (5.227)] yield the net ordinary Sachs–Wolfe fluctuation:

$$\Theta_{\rm SW} = \frac{\delta_{\rm r}^{\rm N}}{4} + \Phi = \left(-\frac{2}{3} + 1\right)\Phi = \frac{1}{3}\Phi, \tag{5.2.1}$$

which represents the dominant contribution to large-scale temperature anisotropies, in full accordance with the numerical decomposition presented in Fig. 5.2.

Upon substituting (5.1.40) in (5.1.20) and using equation (5.1.30) subsequently implies that the scalar angular power spectrum is given by the convolution of the primordial potential spectrum with spherical Bessel functions:

$$C_\ell^{\rm S} = \frac{2}{\pi}\int j_\ell^2[k(\eta_0 - \eta_{\rm LSS})]\frac{1}{9}\mathcal{P}_\Phi(k)\frac{k^3\,{\rm d}k}{k}, \tag{5.2.2}$$

inasmuch as only the first term in Eq. (5.1.29) contributes significantly on super-horizon scales, with $\Delta\eta_{\rm E} = \eta_0 - \eta_{\rm LSS}$, where $\eta_{\rm LSS}$ denotes the conformal time of last scattering. Deep in the matter era, the gravitational potential is static and connects to the gauge-invariant comoving curvature perturbation through $\Phi = \frac{3}{5}\zeta$ for super-Hubble modes [Eq. (5.149)]. Consequently, $\mathcal{P}_\Phi = \frac{9}{25}\mathcal{P}_\zeta$. Defining the dimensionless power spectrum $\mathcal{P}_X \equiv k^3 P_X/2\pi^2$, we infer

$$C_\ell^{\rm S} = \pi\int A_{\rm S}^2(k) j_\ell^2[k(\eta_0 - \eta_{\rm LSS})]\frac{{\rm d}k}{k}, \tag{5.2.3}$$

for a primordial spectrum of the form predicted by inflation [Eq. (8.227)], with the normalization $A_{\rm S}^2 \equiv \frac{4}{25}\mathcal{P}_\zeta$ [Eq. (8.234)].

The integral in Eq. (5.2.3) is heavily weighted around modes satisfying $k\eta_0 \sim \ell$, so that one naturally expects $C_\ell \propto A_{\rm S}^2(k \sim \ell/\eta_0)$. More rigorously, for a power-law primordial spectrum (see Appendix B):

$$C_\ell^{\rm S} \simeq \pi A_{\rm S}^2(k_0)\left[\frac{\pi}{2}\frac{\Gamma(3-n_{\rm S})}{2^{3-n_{\rm S}}}\frac{\Gamma\left(\frac{\ell+n_{\rm S}-1}{2}\right)}{\Gamma^2\left(\frac{2-n_{\rm S}}{2}\right)\Gamma\left(\frac{\ell+5-n_{\rm S}}{2}\right)}\right], \tag{5.2.4}$$

Derived using (5.1.40) with $k = 2 - n_s$ in

$$\int_0^\infty j_\ell(t)j_\ell(t)t^{-k}dt = \frac{\pi}{2}\frac{\Gamma(k+1)\Gamma(\frac{2\ell+1-k}{2})}{2^{k+1}\Gamma^2\left(\frac{k+2}{2}\right)\Gamma\left(\frac{2\ell+k+3}{2}\right)} \tag{1}$$

adopting the pivot scale $k_* = k_0 = 1/\eta_0$ and invoking the approximation $\eta_{\rm LSS} \ll \eta_0$. For an exact scale-invariant Harrison–Zel'dovich spectrum ($n_{\rm S} = 1$), this reduces to

$$\ell(\ell+1)C_\ell^{\rm S} = \frac{\pi}{2}A_{\rm S}^2(k_0), \tag{5.2.5}$$

which defines the horizontal plateau of the angular power spectrum at low multipoles (Fig. 5.2). In the asymptotic limit of large ℓ, Stirling's formula $\Gamma(\ell + a) \propto \ell^a$ establishes that

$$\ell(\ell+1)C_\ell^{\rm S} \propto \ell^{n_{\rm S}-1}. \tag{5.2.6}$$

The primordial spectral index $n_{\rm S}$ therefore directly governs the tilt of the Sachs–Wolfe plateau. Furthermore, Eq. (5.2.1) shows that $\Theta_{\rm SW} = -\delta_\gamma^{\rm N}/6$, revealing that cold spots on large angular scales correspond physically to primordial overdense regions at the last-scattering surface due to the dominant gravitational redshift.

Isocurvature perturbations

For primordial isocurvature (entropy) perturbations, the initial conditions in the radiation era satisfy $\delta_\gamma^{\mathrm{N}}(0) = \Phi(0) = 0$ and $V_\gamma(0) = 0$ (Chapter 5). If $S(0)$ denotes the conserved primordial entropy fluctuation, then during the matter-dominated epoch the gravitational potential settles to $\Phi = -S/5$. On super-Hubble scales, the quantity $\delta_\gamma^{\mathrm{N}} - 4\Psi$ remains conserved, which implies $\delta_\gamma^{\mathrm{N}}/4 + \Phi = \Phi + \Psi \simeq 2\Phi$. The leading-order Sachs–Wolfe temperature fluctuation in the matter era is therefore

$$\Theta_{\mathrm{SW}} = 2\Phi = -\frac{2}{5}S; \tag{5.2.7}$$

see Eq. (5.229). Paralleling the derivation for adiabatic perturbations, we find

$$C_\ell^{\mathrm{S}} \simeq 36\pi A_{\mathrm{S}}^2(k_0) \left[\frac{\pi}{2} \frac{\Gamma(3-n_{\mathrm{S}})}{2^{3-n_{\mathrm{S}}}} \frac{\Gamma\left(\frac{\ell+n_{\mathrm{S}}-1}{2}\right)}{\Gamma^2\left(\frac{2-n_{\mathrm{S}}}{2}\right)\Gamma\left(\frac{\ell+5-n_{\mathrm{S}}}{2}\right)} \right]. \tag{5.2.8}$$

For a scale-invariant isocurvature spectrum ($n_{\mathrm{S}} = 1$), this produces an equivalent low-ℓ plateau:

$$\ell(\ell+1)C_\ell^{\mathrm{S}} = 36\pi A_{\mathrm{S}}^2(k_0). \tag{5.2.9}$$

5.2.2 Intermediate scales

The physical origin, morphological shape, and acoustic peak structure of the angular power spectrum at intermediate multipoles can be understood from first principles by analyzing the coupled photon–baryon plasma prior to recombination.

The linear evolution of photons tightly coupled to baryons is governed by the perturbation equations derived in Chapter 5 [Eqs. (5.253)–(5.256)]:

$$\delta_\gamma^{\mathrm{N}\prime} = \frac{4}{3}k^2 V_\gamma + 4\Psi', \qquad V_\gamma' = -\frac{1}{4}\delta_\gamma^{\mathrm{N}} - \Phi + \frac{1}{6}k^2\pi_\gamma + \tau'(V_{\mathrm{b}} - V_\gamma), \tag{5.2.10}$$

$$\delta_{\mathrm{b}}^{\mathrm{N}\prime} = k^2 V_{\mathrm{b}} + 3\Psi', \qquad V_{\mathrm{b}}' = -\mathcal{H}V_{\mathrm{b}} - \Phi + \frac{\tau'}{R}(V_\gamma - V_{\mathrm{b}}), \tag{5.2.11}$$

where $\tau' \equiv a n_{\mathrm{e}} \sigma_{\mathrm{T}}$ is the differential optical depth per unit conformal time. The residual radiation quadrupole π_γ is directly proportional to the photon velocity divergence:

$$\frac{k^2}{12}\pi_\gamma = -\frac{8}{45}\frac{k^2}{\tau'}V_\gamma. \tag{5.2.12}$$

This relation expresses the fact that an anisotropic stress quadrupole is locally generated by the shear gradient of the velocity field (look up (5.3.70)). As established in Eq. (6.231) below, incorporating the linear polarization of Thomson scattering fixes the numerical prefactor to $8/45$; if polarization is omitted, this coefficient simplifies to $2/15$.

Dynamics of the photon–baryon fluid

Consider perturbation modes whose spatial wavelengths are substantially larger than the photon mean free path, $k/\tau' \ll 1$. Expanding the photon and baryon Euler equations to leading order in k/τ' under negligible anisotropic stress yields the tight coupling conditions

$$V_{\mathrm{b}} = V_\gamma, \qquad \delta_\gamma' - \frac{4}{3}\delta_{\mathrm{b}}' = 0.$$

Even though photon-baryon anisotropic stress tensor can be ignored during radiation era. The anisotropic stress tensor from neutrino stays relevant fir precision cosmology. This is happening because the neutrinos are not undergoing collision and hence their velocity is no longer being isotropized.

The second relation guarantees that the entropy perturbation remains constant in time. In this tight-coupling regime, combining the two momentum equations eliminates the Thomson drag term:

$$[(1+R)V_\gamma]' = -\frac{1}{4}\delta_\gamma^{\rm N} - (1+R)\Phi. \tag{5.2.13}$$

Differentiating the photon continuity equation and substituting Eq. (5.2.13), we obtain the fundamental governing equation for the photon density contrast:

$$\delta_\gamma^{\rm N\prime\prime} + \frac{R'}{1+R}\delta_\gamma^{\rm N\prime} + k^2 c_{\rm s}^2 \delta_\gamma^{\rm N} = 4\mathcal{F}(\Phi,\Psi) = 4\left[\Psi'' + \frac{R'}{1+R}\Psi' - \frac{1}{3}k^2\Phi\right], \tag{5.2.14}$$

where the baryon-to-photon momentum density ratio R and the adiabatic sound speed $c_{\rm s}$ are defined by

$$R \equiv \frac{3}{4}\frac{\rho_{\rm b}}{\rho_\gamma}, \qquad c_{\rm s}^2 \equiv \frac{1}{3}\frac{1}{1+R}. \tag{5.2.15}$$

Equation (5.2.14) describes a forced, damped harmonic oscillator. The driving force $\mathcal{F}$ is governed by the gravitational potentials, which become dominated by cold dark matter as R increases, shifting the zero-point of the acoustic oscillations. Equation (5.2.14) can be recast in the self-adjoint form:

$$\left[(1+R)\delta_\gamma^{\rm N\prime}\right]' + \frac{1}{3}k^2\delta_\gamma^{\rm N} = 4\left\{[(1+R)\Psi']' - \frac{1}{3}k^2(1+R)\Phi\right\}. \tag{5.2.16}$$

Equation (5.2.16) lies at the heart of acoustic oscillations in the primordial plasma: radiation pressure supplies the restoring force against gravitational compression, driving sound waves at frequency $\omega_{\rm s} = kc_{\rm s}$. In the WKB limit, where the oscillation period is much shorter than the background evolution timescale ($\delta_\gamma^{\rm N\prime} \ll \omega_{\rm s}\delta_\gamma^{\rm N}$, while accounting for $\omega_{\rm s}'/\omega_{\rm s} = -R'/[2(1+R)]$), the homogeneous equation admits the two independent basis solutions

$$\delta_a = (1+R)^{-1/4}\cos(kr_{\rm s}), \qquad \delta_b = (1+R)^{-1/4}\sin(kr_{\rm s}), \tag{5.2.17}$$

where the sound horizon at conformal time η is given by

$$r_{\rm s}(\eta) \equiv \int_0^\eta c_{\rm s}{\rm d}\eta' = \int_0^\eta \frac{{\rm d}\eta'}{\sqrt{3(1+R)}}. \tag{5.2.18}$$

The Green's function associated with the driven oscillator is

$$\mathcal{G}(\eta,\eta') \equiv \frac{\delta_a(\eta')\delta_b(\eta) - \delta_a(\eta)\delta_b(\eta')}{\delta_a(\eta')\delta_b'(\eta') - \delta_a'(\eta')\delta_b(\eta')} = \frac{\sqrt{3}}{k}\frac{[1+R(\eta')]^{3/4}}{[1+R(\eta)]^{1/4}}\sin\left[kr_{\rm s}(\eta) - kr_{\rm s}(\eta')\right], \tag{5.2.19}$$

enabling the full inhomogeneous solution to be written as

$$\begin{aligned}[1+R(\eta)]^{1/4}\delta_\gamma^{\rm N} &= \delta_\gamma(0)\cos[kr_{\rm s}(\eta)] + \frac{\sqrt{3}}{k}\left[\delta_\gamma'(0) + \frac{1}{4}R'(0)\delta_\gamma(0)\right]\sin[kr_{\rm s}(\eta)] \\ &\quad + 4\frac{\sqrt{3}}{k}\int_0^\eta [1+R(\eta')]^{3/4}\sin\left[kr_{\rm s}(\eta) - kr_{\rm s}(\eta')\right]\mathcal{F}(\eta'){\rm d}\eta'.\end{aligned} \tag{5.2.20}$$

Sachs–Wolfe term

Assuming to leading order that the Bardeen potentials are quasi-static (as in the matter-dominated epoch) and neglecting the slow time evolution of R, the solution (5.2.20) simplifies to

$$\Theta_{\rm SW} = [\Theta_{\rm SW}(0) + R\Phi]\cos[kr_{\rm s}(\eta)] + \frac{\Theta'_{\rm SW}(0)}{kc_{\rm s}}\sin[kr_{\rm s}(\eta)] - R\Phi, \tag{5.2.21}$$

with $r_{\rm s} = c_{\rm s}\eta$. The Sachs–Wolfe term exhibits acoustic oscillations in conformal time representing sound waves propagating through the plasma. Crucially, the equilibrium center of oscillation is displaced from zero to $-R\Phi$ by the gravitational drag of the baryons. Over many periods, R and Φ evolve dynamically, causing the oscillation envelope of $\Theta_{\rm SW}$ to attenuate as $(1+R)^{-1/4}$. Figure 5.4 displays the temporal evolution of $\Theta_{\rm SW}$ for different modes prior to decoupling.

Doppler term

Neglecting the time derivatives of the metric potentials and R, the photon continuity equation sets $k^2V_\gamma/3 = \Theta'_{\rm SW} + (\Psi - \Phi)'$, leading in this approximation to

$$\frac{kV_\gamma}{3} = -c_{\rm s}[\Theta_{\rm SW}(0) + R\Phi]\sin[kr_{\rm s}(\eta)] + \frac{\Theta'_{\rm SW}(0)}{k}\cos[kr_{\rm s}(\eta)]. \tag{5.2.22}$$

Under tight coupling, $V_\gamma \simeq V_{\rm b}$, generating the kinematic Doppler anisotropy $\Theta_{\rm dop} \sim kV_\gamma/\sqrt{3}$. This term oscillates symmetrically about zero in temporal quadrature ($\pi/2$ out of phase) with $\Theta_{\rm SW}$, with an amplitude reduced by $\sqrt{3}c_{\rm s} = (1+R)^{-1/2}$. Consequently, the Doppler peaks naturally fill in the troughs of the Sachs–Wolfe power spectrum. In the pure radiation limit ($R \to 0$), the Doppler and Sachs–Wolfe amplitudes are comparable, whereas for realistic baryon fractions ($R > 0$), the Doppler contribution is suppressed relative to $\Theta_{\rm SW}$, preserving pronounced acoustic peaks (Fig. 5.2).

Positions of the acoustic peaks

The multipole locations of the acoustic peaks depend decisively on the primordial initial conditions. Inside the sound horizon, the Bardeen potentials are heavily suppressed. Accounting for photon diffusion damping via the factor $D(k,\eta) \equiv \exp(-k^2/k_{\rm D}^2)$ [Eq. (5.2.34)], the acoustic solution takes the form

$$\Theta_{\rm SW} = C_{\rm A}\frac{D(k,\eta)}{(1+R)^{1/4}}\cos(kr_{\rm s}) + C_{\rm I}\frac{D(k,\eta)}{(1+R)^{1/4}}\sin(kr_{\rm s}), \tag{5.2.23}$$

where $C_{\rm A}$ and $C_{\rm I}$ are determined by the initial conditions:

- **Adiabatic perturbations**: Here $\delta_\gamma^{\rm N}(0) = -2\Phi(0)$. Neglecting early velocity perturbations, $\delta_\gamma^{\rm N\prime} = 4\Phi'$, so that $\delta_\gamma^{\rm N}(\eta) = -2\Phi(0) + 4\Phi(\eta) - 4\Phi(0)$. When the perturbation crosses the Hubble radius, the potential decays, leaving $\delta_\gamma^{\rm N}(\eta) \simeq -6\Phi(0)$. At sound horizon entry, $\delta_\gamma^{\rm N}/4 = -3\Phi(0)/2$ and $\delta_\gamma^{\rm N\prime} = 0$, selecting $C_{\rm A} = -\frac{3}{2}\Phi(0)$ and $C_{\rm I} = 0$:

  $$\Theta_{\rm SW} = -\frac{3}{2}\Phi(0)\frac{D(k,\eta)}{(1+R)^{1/4}}\cos(kr_{\rm s}). \tag{5.2.24}$$

  The constant gravitational drive excites exclusively the cosine mode.

- **Isocurvature perturbations**: Here $\delta_\gamma^{\rm N}(0) = \Phi(0) = 0$. As shown in Eq. (5.220), $\delta_\gamma^{\rm N} \propto -aS$ while $S = S(0)$ remains constant. The potential then grows as $\Phi \propto -aS$ [Eq. (5.228)]. Since $a \propto \eta$ in the radiation era, $\mathcal{F} \propto \eta$ in Eq. (5.2.14), which excites the sine mode as $kr_{\rm s} \to 0$. Solving the dynamical equations yields

  $$C_{\rm A} = 0, \qquad C_{\rm I} = \frac{\sqrt{6}}{4}\frac{k_{\rm eq}}{k}S(0). \tag{5.2.25}$$

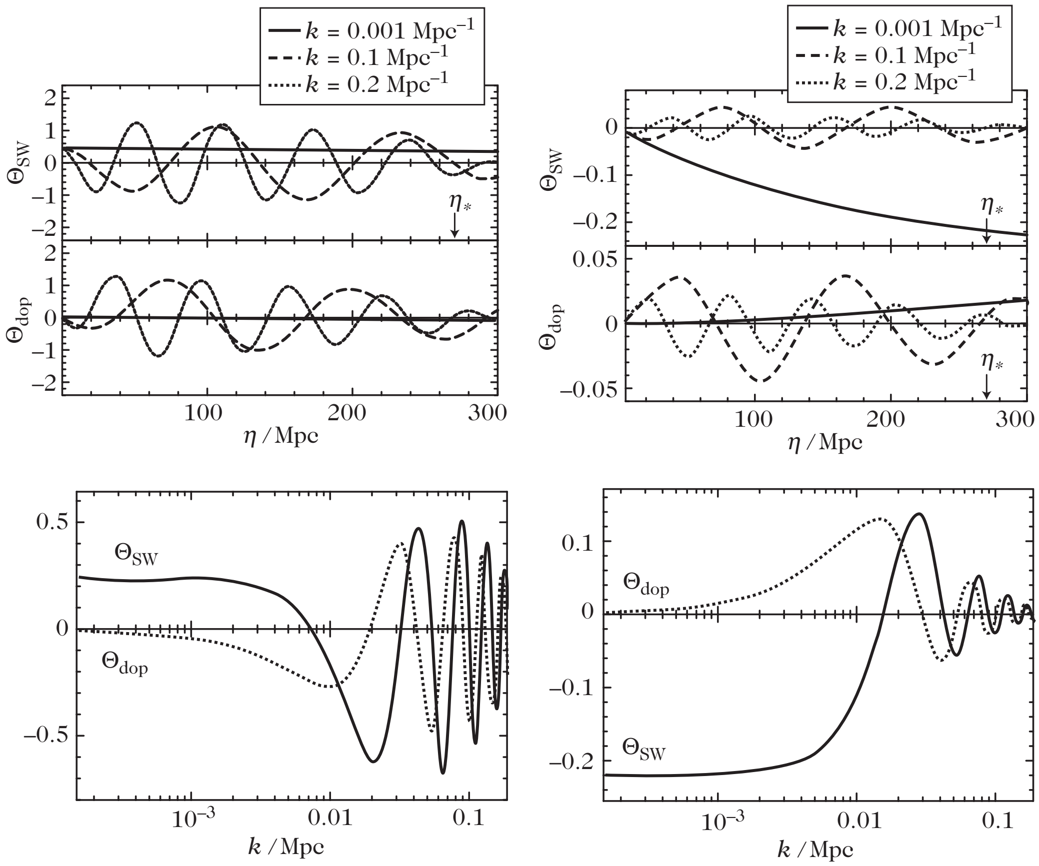


Figure 5.4: Evolution of $\Theta_{\rm SW} = \delta_\gamma^{\rm N}/4 + \Phi$ and of $\Theta_{\rm dop} \sim kV_\gamma/\sqrt{3}$ as a function of time for three modes with wavenumber $k = 10^{-3}$ (solid line), 10^{-1} (dashed) and $2 \times 10^{-1}\,\mathrm{Mpc}^{-1}$ (dotted). The contribution at the time of decoupling as a function of k is presented at the bottom. We compare the adiabatic (left) to the isocurvature (right) initial conditions.

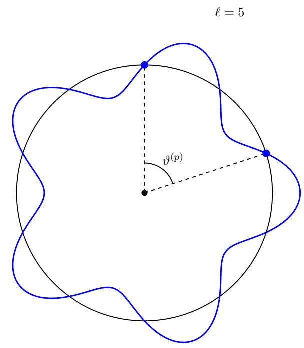
The factor k_{eq}/k reflects the growth of $\delta_\gamma^{\mathrm{N}} \propto -aS$ until the mode enters the Jeans scale at $a \sim k_{\mathrm{eq}}/k$.

At the decoupling epoch ($\eta = \eta_{\mathrm{LSS}}$), the temperature contrast $\Theta_{\mathrm{SW}}(k, \eta_{\mathrm{LSS}})$ achieves local extrema at wavenumbers satisfying

$$k^{(p)} r_{\mathrm{s}}(\eta_{\mathrm{LSS}}) = \begin{cases} p\pi, & \text{(adiabatic)} \\ \left(p - \frac{1}{2}\right)\pi, & \text{(isocurvature)} \end{cases} \tag{5.2.26}$$

for integers $p = 1, 2, \ldots$. The physical wavelength $\lambda^{(p)} = a(\eta_{\mathrm{LSS}})\pi/k^{(p)}$ subtends an observed angular scale $\vartheta^{(p)} \sim \lambda^{(p)}/D_{\mathrm{A}}(\eta_{\mathrm{LSS}})$, where D_{A} is the comoving angular diameter distance (given by (2.4.14) and (2.4.16)):

$$\begin{aligned} \vartheta^{(p)} &= \frac{\lambda^{(p)}}{D_A} = \frac{a_0}{1+z}\frac{\pi}{k^{(p)}}\frac{1}{D_A} \\ &= \frac{\pi}{k^{(p)} f_K(z_{\mathrm{LSS}})}. \end{aligned} \tag{5.2.27}$$



Relating the angular separation ϑ to multipole order $\ell \sim \pi/\vartheta$, the acoustic peaks are situated at

$$\ell^{(p)} = \frac{\pi}{\vartheta^{(p)}} = \frac{f_K(z_{\mathrm{LSS}})}{r_s(\eta_{\mathrm{LSS}})} \times k^{(p)} r_s(\eta_{\mathrm{LSS}}) = \frac{\pi f_K(z_{\mathrm{LSS}})}{r_{\mathrm{s}}(z_{\mathrm{LSS}})} \times \begin{cases} p, & \text{(adiabatic)} \\ p - \frac{1}{2}, & \text{(isocurvature)}. \end{cases} \tag{5.2.28}$$

The peak positions are therefore dictated primarily by the background cosmological geometry through z_{LSS} and $r_{\mathrm{s}}(z_{\mathrm{LSS}})$. Adiabatic models predict a harmonic series of peak locations in ratios $1 : 2 : 3 : \ldots$, whereas isocurvature models produce odd-harmonic ratios $1 : 3 : 5 : \ldots$. In isocurvature models, the $p = 1$ mode is heavily suppressed, placing the first observable peak at

$$\ell^{(1)} \sim \begin{cases} 220, & \text{(adiabatic)} \\ 330, & \text{(isocurvature)} \end{cases} \tag{5.2.29}$$

in a spatially flat universe with $\Omega_\Lambda = 0$. For an adiabatic model with arbitrary curvature, $\ell^{(1)} \sim 220/\sqrt{\Omega_0}$, providing an exquisitely sensitive geometric probe of spatial curvature. These acoustic signatures were first predicted by Sakharov in 1965 (Ref. [8]) and are widely known as Sakharov oscillations.

5.2.3 Small scales

On small spatial scales, k/τ' is no longer negligible, necessitating corrections beyond the tight-coupling approximation as the photon mean free path becomes comparable to the perturbation wavelength.

Silk damping

Consider perturbation modes that cross inside the Hubble radius during the radiation-dominated era. Because radiation dominates the cosmic energy density, the gravitational potentials decay rapidly during acoustic oscillations, allowing metric perturbations and gauge ambiguities to be neglected. Furthermore, because the oscillation period is much shorter than the expansion timescale, the expansion of the universe can be frozen ($R \approx \mathrm{const}$).

The baryon and radiation Euler equations [Eqs. (5.254) and (5.191)] reduce to

$$V_{\mathrm{b}}' = \frac{1}{R\tau'}(V_\gamma - V_{\mathrm{b}}), \qquad V_\gamma' = -\frac{1}{4}\delta_\gamma + \frac{1}{6}k^2\pi_\gamma - \tau'(V_\gamma - V_{\mathrm{b}}).$$

Because $\pi_\gamma \sim \mathcal{O}(1/\tau')$, expanding the velocity difference to order $(\tau')^0$ yields

$$\frac{R+1}{R}\tau'(V_\gamma - V_{\mathrm{b}}) = -\frac{1}{4}\delta_\gamma. \tag{5.2.30}$$

Combining both momentum equations into $(R+1)V_\gamma' + R(V_{\mathrm{b}}' - V_\gamma')$ and employing Eq. (5.2.30), we find

$$(R+1)V_\gamma' = -\frac{1}{4}\delta_\gamma + \frac{1}{6}k^2\pi_\gamma - \frac{1}{4}\frac{R^2}{1+R}\frac{\delta_\gamma'}{\tau'}. \tag{5.2.31}$$

Expressing π_γ via Eq. (5.2.12) and combining with the radiation continuity equation produces the damped wave equation:

$$\delta_\gamma'' + \frac{k^2 c_{\mathrm{s}}^2}{\tau'}\left[\frac{16}{15} + \frac{R^2}{1+R}\right]\delta_\gamma' + k^2 c_{\mathrm{s}}^2 \delta_\gamma = 0. \tag{5.2.32}$$

(If polarization is ignored, the factor 16/15 reverts to 4/5). The WKB solution to this equation exhibits exponential dissipation:

$$\delta_\gamma \propto \exp\left(-\frac{k^2}{k_{\mathrm{D}}^2}\right)\exp(\pm \mathrm{i} k r_{\mathrm{s}}), \tag{5.2.33}$$

where the Silk diffusion scale k_{D} is given by

$$k_{\mathrm{D}}^{-2} = \frac{1}{6}\int_0^\eta \frac{1}{1+R}\left[\frac{16}{15} + \frac{R^2}{1+R}\right]\frac{\mathrm{d}\eta'}{\tau'}. \tag{5.2.34}$$

The bracketed factor varies only between 16/15 (for $R = 0$) and 1 (as $R \to \infty$).

Physically, photons execute a random walk through Thomson scattering with mean free path $\lambda_{\mathrm{C}} \sim 1/\tau'$. In conformal time η, the accumulated diffusion distance is $\lambda_{\mathrm{D}} \sim \sqrt{N}\lambda_{\mathrm{C}}$ with $N \sim \eta/\lambda_{\mathrm{C}}$ steps, yielding $\lambda_{\mathrm{D}} \sim \sqrt{\eta/\tau'} \sim k_{\mathrm{D}}^{-1}$. In a flat universe without dark energy, assuming $z_{\mathrm{eq}} \sim z_{\mathrm{LSS}}$, the diffusion wavenumber satisfies

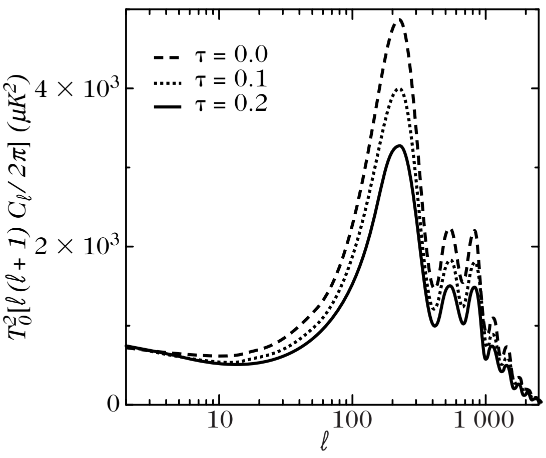
$$k_{\mathrm{D}}\eta_{\mathrm{LSS}} \sim (1 + z_{\mathrm{LSS}})\sqrt{6\frac{\sigma_{\mathrm{T}}}{m_{\mathrm{p}}}\sqrt{\frac{3H_0}{8\pi G_{\mathrm{N}}}}\Omega_{\mathrm{b}0}h}. \tag{5.2.35}$$

This photon diffusion effect (Silk damping, Ref. [9]) induces an exponential suppression of the angular power spectrum starting around $\ell \sim 1400\sqrt{\Omega_{\mathrm{b}0}h^2/0.019}$, well before the first acoustic peak, almost completely extinguishing power beyond $\ell \sim 2000$ (as demonstrated in Fig. 5.5).

Effect of the thickness of the last-scattering surface

Because recombination is not instantaneous, the last-scattering surface has a finite physical thickness. The visibility function $g(\eta)$ defines the probability distribution that a photon underwent its final Thomson scattering at conformal time η. It is related to the optical depth $\tau(\eta)$ by

$$\tau(\eta) = \int_\eta^{\eta_0} \tau'(u)\,\mathrm{d}u\,, \tag{5.2.36}$$



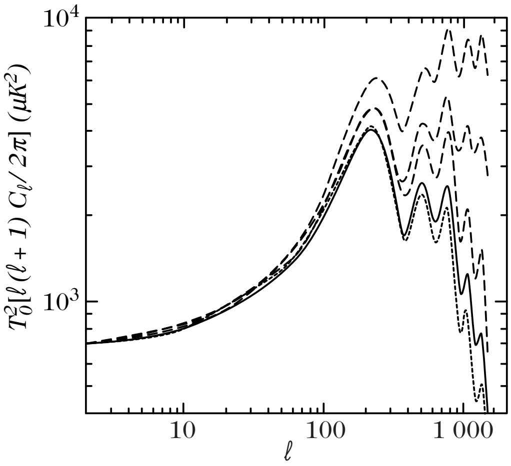


Figure 5.5: (left): Effect of the reionization on the temperature angular power spectrum. (right): The dashes represent the anisotropies of a standard cold dark matter model without taking into account the Silk damping and the thickness of the surface of last-scattering surface (top) or the thickness of the surface of last scattering alone (middle) or the Silk effect alone (bottom). The solid line represents the exact computation and the dotted line corresponds to a computation including the effect of the thickness of the last scattering surface and the Silk effect by an analytical correction deduced from our estimates.

Consider a monochromatic electromagnetic wave propagating through a dilute medium of scatterers. The electric field may be written as

$$\mathbf{E}(\mathbf{r},t) = \mathbf{E}_0 e^{i(\tilde{k}z-\omega t)}, \tag{1}$$

where the wave number is, in general, complex. Writing

$$\tilde{k} = k + i\kappa, \tag{2}$$

the electric field becomes

$$\mathbf{E}(\mathbf{r},t) = \mathbf{E}_0 e^{ikz} e^{-\kappa z} e^{-i\omega t}. \tag{3}$$

Since the intensity of an electromagnetic wave is proportional to the squared magnitude of the electric-field amplitude,

$$I(z) \propto |\mathbf{E}(z)|^2, \tag{4}$$

we obtain

$$I(z) = I_0 e^{-2\kappa z}. \tag{5}$$

For a dilute medium with number density n, the complex refractive index is related to the forward scattering amplitude $f(0)$ by

$$\tilde{n} - 1 = \frac{2\pi n}{k^2} f(0), \tag{6}$$

where $k = \omega/c$. Writing

$$\tilde{n} = n_r + i n_i, \tag{7}$$

and using

$$\tilde{k} = k\tilde{n}, \tag{8}$$

we find

$$\kappa = kn_i = \frac{2\pi n}{k} \operatorname{Im} f(0). \tag{9}$$

The optical theorem relates the imaginary part of the forward scattering amplitude to the total scattering cross section:

$$\sigma_{\text{tot}} = \frac{4\pi}{k} \operatorname{Im} f(0). \tag{10}$$

Consequently,

$$\kappa = \frac{n\sigma_{\text{tot}}}{2}, \tag{11}$$

Substituting this relation into Eq. (5) gives

$$I(z) = I_0 e^{-n\sigma_{\text{tot}} z}. \tag{12}$$

For a spatially varying medium, this generalizes to

$$I = I_0 \exp\left(-\int n\sigma_{\text{tot}} dl\right). \tag{13}$$

such that $\mathrm{d}\tau/\mathrm{d}\eta = -\tau'$. The visibility function is given by

$$g(\eta) \equiv \tau' \exp(-\tau), \tag{5.2.37}$$

normalized to $\int g(\eta)\mathrm{d}\eta = 1$.

The observed temperature field along direction $\boldsymbol{e}$ is therefore a visibility-weighted line-of-sight average:

$$\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}) = \int g(\eta)\left[\Theta_{\text{SW}} + \Theta_{\text{dop}}\right](\eta, [\eta_0 - \eta]\boldsymbol{e})\mathrm{d}\eta. \tag{5.2.38}$$

Moreover, gravitational potential variations only affect photons that have already decoupled. The decoupled fraction is given by

$$g_\gamma(\eta) = \int_{\eta_{\text{in}}}^{\eta} g(u)\,\mathrm{d}u\,, \tag{5.2.39}$$

which increases monotonically from 0 at early times to 1 today. The integrated Sachs–Wolfe contribution is accordingly modified to

$$\Theta_{\text{ISW}} = \int g_\gamma(\eta)[\Phi' + \Psi'](\eta, [\eta_0 - \eta]\boldsymbol{e})\mathrm{d}\eta. \tag{5.2.40}$$

In Fourier space, approximating the visibility function by a Gaussian centered on η_{LSS} with dispersion σ_{LSS}:

$$g(\eta) \sim \frac{1}{\sqrt{2\pi\sigma_{\text{LSS}}^2}} \exp\left[-\frac{1}{2}\frac{(\eta - \eta_{\text{LSS}})^2}{\sigma_{\text{LSS}}^2}\right],$$

its Fourier transform suppresses small-scale anisotropies according to

$$\hat{g}(k) \propto \exp\left(-\frac{1}{2}k^2\sigma_{\text{LSS}}^2\right). \tag{5.2.41}$$

The finite thickness of the last-scattering surface thus washes out temperature fluctuations on spatial scales smaller than σ_{LSS}, operating with comparable magnitude to Silk damping.

5.2.4 Other effects

Following decoupling, photons travel essentially unhindered, so that only gravitational field variations and late-time reionization can alter their temperature distribution.

Integrated Sachs–Wolfe

The temporal variation of the metric potentials and primordial gravitational waves contributes to the temperature contrast via Θ_{ISW} [Eq. (5.1.17)]. This term encompasses two distinct physical regimes:

- **Early Sachs–Wolfe effect**: The residual evolution of the gravitational potentials during the transition from radiation to matter domination leaves an imprint on modes entering the sound horizon between equality and decoupling. This effect enhances the power spectrum in the vicinity of the first acoustic peak. Because $\Theta_{\text{SW}} \sim -\Phi/2$ carries the same sign as $\Phi' < 0$ during this transition, Θ_{ISW} adds constructively to Θ_{SW}, boosting the height of the first peak (Fig. 5.2). For isocurvature modes, potential growth on scales larger than the sound horizon provides a substantial contribution below the first peak.

- **Late Sachs–Wolfe effect**: At low redshifts ($z \lesssim 1$), cosmic acceleration driven by dark energy (Λ) or non-zero spatial curvature reactivates the time evolution of the gravitational potentials. This effect operates primarily on very large angular scales, modifying the low-ℓ Sachs–Wolfe plateau.

Because Φ decays in a Universe dominated by dark energy or spatial curvature at low redshifts (Chapter 5), an additional net blueshift or redshift is imparted to photons traversing time-varying potential wells. This late integrated Sachs–Wolfe effect provides one of the few direct cosmological probes of dark energy through CMB anisotropies, detectable primarily via its cross-correlation with tracers of the large-scale gravitational potential in the local Universe (such as galaxy catalogs).

Furthermore, gravitational lensing by intervening large-scale structures deflects CMB photons along the line of sight. While the distortion induced in the temperature power spectrum is modest (smoothing the acoustic peaks and generating a small non-Gaussian signal at very high ℓ), weak gravitational lensing plays a central role in generating B-mode polarization from primordial E-modes (Chapter 7).

Reionization

Once the earliest generations of stars and quasars ignite, ultraviolet radiation reionizes the neutral intergalactic medium. Free electrons liberated during this epoch can Thomson-scatter CMB photons long after recombination. This rescattering tends to suppress primordial temperature anisotropies by averaging over multiple intersecting lines of sight arriving at each scattering center, while simultaneously generating secondary anisotropies through the motion of the scattering electrons.

Under the approximation of homogeneous reionization, the visibility function can be idealized as the sum of two Dirac delta distributions centered at decoupling (η_{LSS}) and reionization (η_{re}). Neglecting the integrated Sachs–Wolfe contribution, the observed temperature fluctuation along direction $\boldsymbol{e}$ is approximated by

$$\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}) \simeq (1-\mathrm{e}^{-\tau_{\text{re}}})(\Theta+\boldsymbol{e}\cdot\boldsymbol{V}_{\text{b}})[\boldsymbol{e}(\eta_0-\eta_{\text{re}}), \eta_{\text{re}}]+\mathrm{e}^{-\tau_{\text{re}}}(\Theta+\boldsymbol{e}\cdot\boldsymbol{V}_{\text{b}})[\boldsymbol{e}(\eta_0-\eta_{\text{LSS}}), \eta_{\text{LSS}}], \quad (5.2.42)$$

where τ_{re} denotes the optical depth from the reionization era to the present. The first term represents the line-of-sight averaging and the generation of secondary Doppler anisotropies due to the motion of scattering electrons at η_{re}. The second term corresponds to the unscattered fraction ($\mathrm{e}^{-\tau_{\text{re}}}$) of the primordial temperature anisotropy that originated at the primary last-scattering surface.

The quantity $\Theta[\boldsymbol{e}(\eta_0 - \eta_{\text{re}}), \eta_{\text{re}}]$ represents the directional average of $\Theta - \boldsymbol{e}' \cdot \boldsymbol{V}_{\text{b}}$ across the electron scattering surface at reionization. For long-wavelength modes satisfying $k(\eta_{\text{re}} - \eta_0) \ll 1$, this average reproduces $\Theta[\boldsymbol{e}(\eta_0 - \eta_{\text{LSS}}), \eta_{\text{LSS}}]$, while on sub-horizon scales it averages to zero.

Consequently, for super-Hubble modes at the reionization epoch, the observed temperature perturbation transforms as

$$\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}) \to \Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}) + (1 - \mathrm{e}^{-\tau_{\mathrm{re}}})\boldsymbol{e} \cdot \Delta \boldsymbol{V}_{\mathrm{b}},$$

where $\Delta \boldsymbol{V}_{\mathrm{b}}$ is the velocity differential of the electron fluid between reionization and the earlier scattering event. Because the Doppler contribution is negligible on super-Hubble scales, large-angle temperature anisotropies are left virtually unchanged. Conversely, on scales smaller than the horizon at reionization,

$$\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}) \to \mathrm{e}^{-\tau_{\mathrm{re}}}\Theta(\boldsymbol{x}_0, \eta_0, \boldsymbol{e}) + (1 - \mathrm{e}^{-\tau_{\mathrm{re}}})\boldsymbol{e} \cdot \boldsymbol{V}_{\mathrm{b}}(\eta_{\mathrm{re}}).$$

The net consequence is an exponential damping by $\mathrm{e}^{-\tau_{\mathrm{re}}}$ of the primordial acoustic peak amplitudes (as illustrated in Fig. 5.5). Cross-correlation measurements of CMB temperature and polarization by the WMAP mission constrain the optical depth to $\tau_{\mathrm{re}} \sim 0.17$, indicating a reionization redshift of $z_{\mathrm{re}} \sim 15$.

Sunyaev–Zel'dovich effect

Interactions between CMB photons and high-temperature astrophysical plasmas alter the photon energy distribution. In particular, the intracluster gas within massive galaxy clusters reaches temperatures of order $T_{\mathrm{e}} \sim 10^7 - 10^8$ K, emitting X-rays via thermal bremsstrahlung. Inverse Compton scattering between the free thermal electrons and the passing CMB photons shifts low-energy photons toward higher frequencies. This spectral distortion is known as the thermal Sunyaev–Zel'dovich (tSZ) effect and constitutes the dominant secondary anisotropy on arcminute angular scales.

To order of magnitude, the resulting temperature modification is given by $\Delta T_{\mathrm{SZ}} \sim L\overline{n_{\mathrm{e}}T_{\mathrm{e}}}$, where the overline denotes an average along the line of sight through the cluster, L is the physical path length, and T_{e}, n_{e} are the electron temperature and number density. By comparison, the thermal bremsstrahlung X-ray surface brightness of the cluster scales as

$$S_X \sim \frac{V}{4\pi D_{\mathrm{L}}^2}\overline{\left(n_{\mathrm{e}} n_{\mathrm{p}} T_{\mathrm{e}}^{1/2}\right)},$$

where $V = D_{\mathrm{A}}^2 \theta^2 L$ is the cluster volume subtended by an angular diameter θ, and D_{L}, D_{A} are the luminosity and angular diameter distances, respectively.

As a concrete example, consider an isothermal cluster described by a standard β-model electron density profile:

$$n_{\mathrm{e}}(r) = n_0 \left[1 + \left(\frac{r}{r_{\mathrm{c}}}\right)^2\right]^{-3\beta/2}, \tag{5.2.43}$$

for $0 < r < R_{\mathrm{cluster}}$ (and zero otherwise), where r_{c} is the cluster core radius. In the non-relativistic limit, the temperature decrement in the Rayleigh–Jeans regime is given by

$$\Delta T_{\mathrm{SZ}} = -2\frac{k_{\mathrm{B}} T_0}{m_{\mathrm{e}} c^2}\sigma_{\mathrm{T}} \int_{-L_{\max}}^{L_{\max}} n_{\mathrm{e}}\,\mathrm{d}\ell\,, \tag{5.2.44}$$

assuming that the electron gas is isothermal throughout the cluster. In the limit where the cluster integration boundary $R_{\mathrm{cluster}} \to \infty$, performing the line-of-sight integral yields

$$\Delta T_{\mathrm{SZ}} = -2\frac{k_{\mathrm{B}} T_0}{m_{\mathrm{e}} c^2}\sigma_{\mathrm{T}} n_0 r_{\mathrm{c}} B\left(\frac{3\beta - 1}{2}, \frac{1}{2}\right)\left[1 + \left(\frac{\theta}{\theta_{\mathrm{c}}}\right)^2\right]^{(1-3\beta)/2}, \tag{5.2.45}$$

where $B(x, y)$ is the Euler beta function and $\theta_{\mathrm{c}} = r_{\mathrm{c}}/D_{\mathrm{A}}$ is the angular core radius.

Unlike Thomson scattering—which preserves the blackbody spectral shape—the thermal Sunyaev–Zel'dovich effect generates a unique spectral signature characterized by a temperature decrease at long wavelengths (Rayleigh–Jeans region) and an increment at short wavelengths ($\nu > 217$ GHz). Because the surface brightness decrement is redshift-independent, the tSZ effect provides a powerful observational tool for detecting distant clusters and probing cosmic structure evolution (see Ref. [10] for an extensive review). Furthermore, if the cluster possesses a peculiar velocity $\boldsymbol{v}$ relative to the CMB rest frame, the Doppler effect introduces a secondary modification known as the kinetic Sunyaev–Zel'dovich (kSZ) effect, which is proportional to the line-of-sight radial component of the cluster velocity.

Gravitational waves

Deriving a tensor-mode counterpart to the scalar plateau [Eq. (5.2.4)] is more delicate because the tensor Sachs–Wolfe integral [Eq. (5.1.45)] depends explicitly on the time derivative of the gravitational wave amplitude E'_λ. During the matter-dominated era, the propagation of primordial gravitational waves is governed by Eq. (5.135), whose mode solution yields

$$E'_\lambda(k,\eta) = -3kE_\lambda(k,\eta_\mathrm{i})\frac{j_2(k\eta)}{k\eta}. \tag{5.2.46}$$

Substituting this result into Eq. (5.1.45) for an initial tensor power spectrum parametrized as $\mathcal{P}_\mathrm{T}(k) \propto k^{n_\mathrm{T}}$ (as predicted by standard slow-roll inflation, Eq. (8.234)) leads to the tensor angular power spectrum:

$$C_\ell^\mathrm{T} \propto \frac{(\ell+2)!}{(\ell-2)!}\int \frac{\mathrm{d}y_0}{y_0^{1-n_\mathrm{T}}}\left[\int_{y_0\eta_\mathrm{LSS}/\eta_0}^{y_0} \frac{j_\ell(y_0-y)}{(y_0-y)^2}\frac{j_2(y)}{y}\,\mathrm{d}y\right]^2, \tag{5.2.47}$$

where we have set the dimensionless integration variables $y = k\eta$ and $y_0 = k\eta_0$ with pivot scale $k_0 = 1/\eta_0$. The integrand is appreciable only when the arguments of the spherical Bessel functions simultaneously satisfy $y \sim 2$ and $y_0 - y \sim \ell$, as the Bessel functions rapidly decay away from these peaks. Given the domain of integration $y_0 \leq y\eta_0/\eta_\mathrm{LSS}$, both conditions can be met only for multipoles fulfilling

$$\ell \lesssim 2\frac{\eta_0}{\eta_\mathrm{LSS}} \sim 2\sqrt{z_\mathrm{LSS}+1} \sim 60. \tag{5.2.48}$$

This analytical estimate matches the numerical behavior depicted in Fig. 5.3, where C_ℓ^T drops off precipitously beyond $\ell \sim 100$. Consequently, the tensor spectrum C_ℓ^T is primarily sensitive to large angular scales and depends only weakly on background cosmological parameters.

To evaluate the integral in Eq. (5.2.47) analytically, we approximate the sharply peaked factor $j_2(y)/y$ by an effective delta function $j_2(y)/y \sim 2^{-1/2}\delta(y-2)$. Integrating over y under the assumption that $y_0 \gg 2$, the spectrum simplifies to

$$C_\ell^\mathrm{T} \propto \frac{(\ell+2)!}{(\ell-2)!}\int \frac{\mathrm{d}y_0}{y_0^{1-n_\mathrm{T}}}\left[\frac{j_\ell(y_0)}{y_0^2}\right]^2. \tag{5.2.49}$$

Evaluating the remaining integral over y_0 leads to the closed-form expression

$$C_\ell^\mathrm{T} \propto \frac{(\ell+2)!}{(\ell-2)!}\frac{\Gamma\left(\frac{\ell-2+n_\mathrm{T}}{2}\right)}{\Gamma\left(\frac{\ell+4+n_\mathrm{T}}{2}\right)}. \tag{5.2.50}$$

Applying Stirling's asymptotic expansion in an analogous manner to Eq. (5.2.4), we obtain the low-multipole scaling

$$\ell(\ell+1)C_\ell^\mathrm{T} \propto \ell^{n_\mathrm{T}} \tag{5.2.51}$$

for small multipoles ($\ell \gtrsim 2$). Thus, a scale-invariant primordial tensor background ($n_\mathrm{T} \simeq 0$) produces a flat quadrupole and low-ℓ plateau in $\ell(\ell+1)C_\ell^\mathrm{T}$.

5.3 Kinetic description

To achieve a rigorous and quantitatively accurate description of photon propagation, Thomson scattering, and the decoupling process from the baryonic plasma, one must move beyond the ideal fluid approximation and develop a relativistic kinetic framework. A kinetic approach is essential because photons possess a long mean free path in the transition regime and undergo directional and spectral evolution. Detailed presentations of relativistic kinetic theory can be found in Refs. [11–13], with foundational properties of distribution functions addressed in Ref. [14]. The application of the relativistic Boltzmann equation to cosmological perturbations is treated in Refs. [15–19].

5.3.1 Perturbed Boltzmann equation

Distribution function

As in Chapter 5, the state of the photon gas is described by a single-particle phase-space distribution function $f(x^\mu, p^\mu)$, which depends on the space-time position coordinates and 4-momentum components treated as independent variables. For particles of mass m, the distribution function is restricted to the mass shell

$$\mathcal{P}_m = \left\{ (x^\mu, p^\mu) \in TM \;\middle|\; p_\mu p^\mu = -m^2 \right\}, \tag{5.3.1}$$

defined on the tangent bundle $TM = \{(x^\mu, p^\mu) \mid x^\mu \in \mathcal{M}, p^\mu \in T_x\mathcal{M}\}$, where $\mathcal{M}$ is the four-dimensional space-time manifold and $T_x\mathcal{M}$ is the tangent space at point x.

To decompose the phase space, it is advantageous to introduce the unit timelike vector field normal to the spatial hypersurfaces of constant conformal time $\{\eta = \text{const.}\}$. For the linearly perturbed metric of Eq. (5.52), this normal vector and its dual are given by

$$N_\mu = -a(1 + A, \mathbf{0}), \qquad N^\mu = \frac{1}{a}(1 - A, -B^i), \tag{5.3.2}$$

to first order in cosmological perturbations. The photon 4-momentum k^μ (reserving p^μ for general massive particles and k^μ strictly for null geodesics) can then be parametrized as

$$k^\mu = \frac{E}{a}\left(1 - A,\; n^i - \left[B_j n^j + \frac{1}{2}h_{jk}n^j n^k\right] n^i\right), \tag{5.3.3}$$

where the unit spatial direction vector satisfies $\gamma_{ij}n^i n^j = 1$. This parameterization identically satisfies the null condition $k_\mu k^\mu = 0$ and identifies E as the physical photon energy measured by a comoving observer whose four-velocity is N^μ, since $N_\mu k^\mu = -E$.

In perturbation theory, the full distribution function is split into an unperturbed, isotropic and homogeneous background $\bar{f}(\eta, E)$ and a first-order perturbation $\delta f(\eta, x^i, E, n^j)$ depending on space-time position and direction:

$$\boxed{f(x^\mu, p_\mu) = f(\eta, x^i, E, n^j) = \bar{f}(\eta, E) + \delta f(\eta, x^i, E, n^j).} \tag{5.3.4}$$

The dynamical evolution of f is governed by the relativistic Boltzmann equation

$$\mathcal{L}[f] = C[f], \tag{5.3.5}$$

where $\mathcal{L} \equiv \mathrm{d}/\mathrm{d}\eta$ denotes the Liouville operator along the phase-space trajectory and $C[f]$ represents the collision term, implying that phase-space density is conserved in the absence of scattering interactions.

Collisionless Boltzmann equation

We first analyze the collisionless Liouville operator:

$$\mathcal{L}[f] = \frac{\mathrm{d}f}{\mathrm{d}\eta} = \left[k^\mu \frac{\partial f}{\partial x^\mu} + \frac{\mathrm{d}k^\mu}{\mathrm{d}\lambda}\frac{\partial f}{\partial k^\mu}\right]\frac{\mathrm{d}\lambda}{\mathrm{d}\eta}, \tag{5.3.6}$$

where λ is an affine parameter along the geodesic with tangent vector k^μ, so that $\mathrm{d}\eta/\mathrm{d}\lambda = k^0$. Utilizing the geodesic equation [Eq. (1.37)],

$$\frac{\mathrm{d}k^\mu}{\mathrm{d}\lambda} = -\Gamma^\mu_{\alpha\beta}k^\alpha k^\beta, \tag{5.3.7}$$

the Liouville operator takes the general form

$$\mathcal{L}[f] = \left[k^\mu \frac{\partial f}{\partial x^\mu} - \Gamma^\mu_{\alpha\beta}k^\alpha k^\beta \frac{\partial f}{\partial k^\mu}\right]\frac{1}{k^0}. \tag{5.3.8}$$

In terms of the coordinates (η, x^i, E, n^i) introduced in Eq. (5.3.4), the total derivative expands as

$$\mathcal{L}[f] = \frac{\partial f}{\partial \eta} + \frac{\mathrm{d}x^i}{\mathrm{d}\eta}\frac{\partial f}{\partial x^i} + \frac{\mathrm{d}E}{\mathrm{d}\eta}\frac{\partial f}{\partial E} + \frac{\mathrm{d}n^i}{\mathrm{d}\eta}\frac{\partial f}{\partial n^i}. \tag{5.3.9}$$

We evaluate each term in Eq. (5.3.9) systematically to linear order:

1. Because the directional derivative $\partial f/\partial n^i$ vanishes at zeroth order for an isotropic background, the rate $\mathrm{d}n^i/\mathrm{d}\eta$ is only required at the unperturbed level. To zeroth order, the geodesic equation gives

$$E' = -\mathcal{H}E, \qquad n^{i\prime} = -{}^{(3)}\Gamma^i_{jk}n^j n^k, \tag{5.3.10}$$

where ${}^{(3)}\Gamma^i_{jk}$ are the Christoffel symbols of the spatial metric γ_{ij} (Appendix A). Thus, the directional advection term evaluates to

$$\frac{\mathrm{d}n^i}{\mathrm{d}\eta}\frac{\partial f}{\partial n^i} = -{}^{(3)}\Gamma^i_{jk}n^j n^k \frac{\partial f}{\partial n^i}.$$

2. For the spatial gradient term, $\mathrm{d}x^i/\mathrm{d}\eta = k^i/k^0$. Since $\partial_i f$ is itself first order in perturbations, we can evaluate $\mathrm{d}x^i/\mathrm{d}\eta$ at background order, yielding $k^i/k^0 = n^i$, so that

$$\frac{\mathrm{d}x^i}{\mathrm{d}\eta}\frac{\partial f}{\partial x^i} = n^i \partial_i f.$$

3. To evaluate the energy evolution rate $\mathrm{d}E/\mathrm{d}\eta$, we examine the temporal component of the geodesic equation [Eq. (5.3.7)]:

$$\frac{\mathrm{d}k^0}{\mathrm{d}\eta} = -\Gamma^0_{\alpha\beta}\frac{k^\alpha k^\beta}{k^0}.$$

Recalling that $k^0 = \frac{E}{a}(1-A)$, its total derivative gives

$$\frac{\mathrm{d}}{\mathrm{d}\eta}\left(\frac{E}{a}\right) = \frac{E}{a}\left(A' + n^i\partial_i A\right) - \Gamma^0_{\alpha\beta}\frac{k^\alpha k^\beta}{E/a}(1+2A).$$

Substituting the perturbed Christoffel symbols [Eqs. (C.6) and (C.26)], we find

$$\frac{\mathrm{d}E}{\mathrm{d}\eta} = -E\left[\mathcal{H} + n^i\partial_i A + \left(\frac{1}{2}h'_{ij} - D_{(i}B_{j)}\right)n^i n^j\right].$$

Combining all four contributions yields the complete Liouville operator to first order in perturbations:

$$\boxed{\begin{aligned}\mathcal{L}[f] = f' + n^k\partial_k f - \left[\mathcal{H} + n^k\partial_k A + \left(\frac{1}{2}h'_{ij} - D_{(i}B_{j)}\right)n^i n^j\right] E\frac{\partial f}{\partial E} \\ - {}^{(3)}\Gamma^k_{ij} n^i n^j \frac{\partial f}{\partial n^k},\end{aligned}} \tag{5.3.11}$$

which can equivalently be written in compact form as

$$\boxed{\mathcal{L}[f] = f' + n^k\partial_k f - \left[\mathcal{H} + n^k\partial_k A + \left(\frac{1}{2}h'_{ij} - D_{(i}B_{j)}\right)n^i n^j\right] E\frac{\partial f}{\partial E} - {}^{(3)}\Gamma^k_{ij} n^i n^j \frac{\partial f}{\partial n^k}.} \tag{5.3.12}$$

Decomposing the distribution function according to Eq. (5.3.4) into background and perturbation components, we obtain the corresponding evolution equations:

$$\mathcal{L}[\bar{f}] = \bar{f}' - \mathcal{H}E\frac{\partial \bar{f}}{\partial E}, \tag{5.3.13}$$

$$\begin{aligned}\mathcal{L}[\delta f] = \delta f' + n^k\partial_k \delta f - \mathcal{H}E\frac{\partial \delta f}{\partial E} - \left[n^k\partial_k A + \left(\frac{1}{2}h'_{ij} - D_{(i}B_{j)}\right)n^i n^j\right] E\frac{\partial \bar{f}}{\partial E} \\ - {}^{(3)}\Gamma^k_{ij} n^i n^j \frac{\partial \delta f}{\partial n^k}.\end{aligned} \tag{5.3.14}$$

At the background level, the unperturbed photon distribution is a blackbody Bose–Einstein distribution (Chapter 4):

$$\bar{f}(E,\eta) = \left[\exp\left(\frac{E}{T(\eta)}\right) - 1\right]^{-1} = \bar{f}\left(\frac{E}{T}\right). \tag{5.3.15}$$

The background Liouville equation $\mathcal{L}[\bar{f}] = 0$ then guarantees that the background temperature redshifts as $T(\eta) \propto 1/a$, in complete harmony with the geodesic energy redshifting of individual photons.

At linear order, we consider perturbed distribution functions that maintain a local Planckian spectral shape with a direction-dependent temperature $T(\eta, x^i, n^j) = \bar{T}(\eta)[1 + \theta(\eta, x^i, n^j)]$, namely $f = \bar{f}(E/T)$. Taylor-expanding to first order in the fractional temperature perturbation θ, we obtain

$$f(\eta, x^i, E, n^j) = \bar{f}(\eta, E) - E\frac{\partial \bar{f}}{\partial E}\theta(\eta, x^i, n^j). \tag{5.3.16}$$

The temperature perturbation $\theta(\eta, x^i, n^j)$ is both inhomogeneous (dependent on position x^i) and anisotropic (dependent on direction n^j). This ansatz implicitly assumes that θ is energy-independent—a hypothesis justified by the energy independence of Thomson scattering in the Thomson regime, which introduces no spectral distortions.

Macroscopic quantities

Macroscopic fluid quantities are obtained as velocity moments of the phase-space distribution function. To carry out momentum integrations covariantly, we construct an orthonormal tetrad (vierbein) $(e^a_\mu)_{a=0\ldots3}$ representing the local Minkowski rest frame:

$$e^0_\mu = N_\mu, \qquad e^i_\mu e^j_\nu g^{\mu\nu} = \delta^{ij}, \qquad e^i_\mu N^\mu = 0.$$

Any null 4-momentum can then be projected onto the tetrad basis as $k^a = k^\mu e^a_\mu$. The energy-momentum tensor is defined by integrating over the mass shell in momentum space:

$$T^{\mu\nu}(x) = \int_{\mathcal{P}_m(x)} k^\mu k^\nu f(x^\alpha, k^\beta)\frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3 E_k}, \tag{5.3.17}$$

where $k^2 = \delta_{ab}k^a k^b$ and $E_k = \sqrt{k^2 + m^2} = k$ for massless photons. Absorbing the conventional factor $(2\pi)^3$ into the definition of the distribution function, the radiation energy-momentum tensor becomes

$$T^{\mu\nu}(x) = \int_{\mathcal{P}_m(x)} k^\mu k^\nu f(x^\alpha, k^\beta) E \,\mathrm{d}E \,\mathrm{d}^2\boldsymbol{n}. \tag{5.3.18}$$

Comparing this expression with the fluid decomposition of Eq. (5.69), we extract the radiation energy density, pressure, and anisotropic stress tensor:

$$\rho = \int (k^\mu u_\mu)^2 f E \,\mathrm{d}E \,\mathrm{d}^2\boldsymbol{n}, \tag{5.3.19}$$

$$P = \frac{1}{3}\int (k^\mu k^\nu \perp_{\mu\nu}) f E \,\mathrm{d}E \,\mathrm{d}^2\boldsymbol{n}, \tag{5.3.20}$$

$$\Pi^{\mu\nu} = \int k^\alpha k^\beta \left[\perp^\mu_\alpha \perp^\nu_\beta - \frac{1}{3} \perp_{\alpha\beta}\perp^{\mu\nu}\right] f E \,\mathrm{d}E \,\mathrm{d}^2\boldsymbol{n}, \tag{5.3.21}$$

where $\perp_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$ is the spatial projector orthogonal to the fluid four-velocity u^μ. The four-velocity is defined by Eq. (5.70) as $u_\mu = a(-1 - A, v_k + B_k)$. Because photons are null ($k_\mu k^\mu = 0$), contracting Eq. (5.3.18) yields $T^\mu_\mu = 0$, immediately recovering the radiation equation of state $P = \rho/3$.

Evaluating the contraction $k^\mu u_\mu$ to first order:

$$k^\mu u_\mu = -E\left[1 - n^i(v_i + B_i)\right].$$

Since the angular integral of the background distribution $\bar{f}n^i$ vanishes by isotropy, we obtain the perturbed macroscopic fluid quantities:

$$\delta\rho = \int \delta f E^3 \,\mathrm{d}E \,\mathrm{d}^2\boldsymbol{n}, \tag{5.3.22}$$

$$(\rho + P)(v^i + B^i) = \int n^i \delta f E^3 \,\mathrm{d}E \,\mathrm{d}^2\boldsymbol{n}, \tag{5.3.23}$$

$$\Pi^{ij} = P\pi^{ij} = \int n^{ij} \delta f E^3 \,\mathrm{d}E \,\mathrm{d}^2\boldsymbol{n}, \tag{5.3.24}$$

where the trace-free symmetric tensor product of the direction vector is defined by

$$n^{ij} \equiv n^i n^j - \frac{1}{3}\gamma^{ij}. \tag{5.3.25}$$

Brightness and temperature

The radiation brightness (or specific intensity integrated over frequency) I is obtained by integrating the distribution function weighted by E^3 over energy:

$$I(\eta, x^i, n^j) \equiv 4\pi \int E^3 f(\eta, x^i, E, n^j) \,\mathrm{d}E\,. \tag{5.3.26}$$

The brightness represents the directional energy density per unit solid angle propagating along direction n^i at space-time position (η, x^j). Splitting into background and perturbation, $I = \bar{I} + \delta I$:

$$\bar{I}(\eta) \equiv 4\pi \int E^3 \bar{f}(\eta, E) \,\mathrm{d}E\,, \qquad \delta I(\eta, x^i, n^j) \equiv 4\pi \int E^3 \delta f(\eta, x^i, E, n^j) \,\mathrm{d}E\,. \tag{5.3.27}$$

From Eq. (5.3.19), the total photon energy density ρ_γ is precisely the angular monopole of the brightness:

$$\rho_\gamma(\eta, x^j) = \int \frac{\mathrm{d}^2\boldsymbol{n}}{4\pi} I(\eta, x^i, n^j). \tag{5.3.28}$$

The background equation (5.3.13) immediately implies

$$\bar{I}' + 4\mathcal{H}\bar{I} = 0, \qquad \bar{\rho}'_\gamma + 4\mathcal{H}\bar{\rho}_\gamma = 0, \tag{5.3.29}$$

with $\bar{I} = \bar{\rho}_\gamma$, in agreement with background spatial isotropy.

Similarly, we define the energy-integrated collision operator acting on brightness perturbations:

$$C[\delta I] = 4\pi \int C[\delta f] E^3 \, \mathrm{d}E \,. \tag{5.3.30}$$

Integrating the first-order Liouville equation (5.3.14) over energy weighted by $4\pi E^3 \, \mathrm{d}E$ yields the brightness Boltzmann equation:

$$\delta I' + n^k \partial_k \delta I + 4\mathcal{H}\delta I + 4 \left[n^k \partial_k A + \left(\frac{1}{2} h'_{ij} - D_{(i}B_{j)} \right) n^i n^j \right] \bar{I} - {}^{(3)}\Gamma^k_{ij} n^i n^j \frac{\partial \delta I}{\partial n^k} = C[\delta I]. \tag{5.3.31}$$

The fractional temperature perturbation is defined from the brightness fluctuation via

$$\theta(\eta, x^i, n^i) = \frac{1}{4} \frac{\delta I}{\bar{I}}, \tag{5.3.32}$$

in direct correspondence with the ansatz of Eq. (5.3.16). The angular monopole of θ is directly proportional to the photon density contrast δ_γ:

$$\theta_0(\eta, x^i) = \int \theta(\eta, x^i, n^i) \frac{\mathrm{d}^2 \boldsymbol{n}}{4\pi} = \frac{1}{4} \delta_\gamma. \tag{5.3.33}$$

Using the background scaling $\bar{I}' = -4\mathcal{H}\bar{I}$ [Eq. (5.3.29)], the evolution equation for θ follows from Eq. (5.3.31):

$$\boxed{\theta' + n^k \partial_k \theta - {}^{(3)}\Gamma^k_{ij} n^i n^j \frac{\partial \theta}{\partial n^k} + \left[n^k \partial_k A + \left(\frac{1}{2} h'_{ij} - D_{(i}B_{j)} \right) n^i n^j \right] = C[\theta],} \tag{5.3.34}$$

where the normalized collision term is defined by $C[\theta] = C[\delta I]/(4\bar{I})$. This equation can alternatively be derived by inserting the temperature ansatz (5.3.16) directly into Eq. (5.3.14). Because the Thomson collision operator is energy-independent, the two formulations are identical.

Multipole expansion

In analogy with Eq. (5.3.28), the macroscopic fluid perturbations emerge as the low-order angular moments of the brightness fluctuation:

$$(\rho + P)(v^i + B^i) = \int \delta I n^i \frac{\mathrm{d}^2 \boldsymbol{n}}{4\pi}, \qquad \Pi^{ij} = \int n^{ij} \delta I \frac{\mathrm{d}^2 \boldsymbol{n}}{4\pi}. \tag{5.3.35}$$

To convert the directional partial differential equation (5.3.31) into a hierarchy of spatial tensor equations, we expand the brightness perturbation in symmetric trace-free tensor moments:

$$\delta I(\eta, x^i, n^j) = \sum_{\ell=0}^{\infty} \left(n^{i_1} \dots n^{i_\ell} \right) \mathcal{I}_{i_1 \dots i_\ell}(\eta, x^i). \tag{5.3.36}$$

Gauge transformations in linear perturbation theory affect only the scalar and vector metric perturbations; hence, only the monopole $\mathcal{I}_0$ and dipole $\mathcal{I}_{i_1}$ transform under a change of coordinates, while all higher multipoles ($\ell \geq 2$) are automatically gauge invariant. To make the decomposition unique, the moment tensors $\mathcal{I}_{i_1 \dots i_\ell}$ are chosen to be completely symmetric and trace-free (STF). Under these conditions, an STF tensor of rank ℓ in three spatial dimensions possesses exactly $2\ell+1$ independent degrees of freedom:

$$\frac{(\ell+1)(\ell+2)}{2} - \frac{\ell(\ell-1)}{2} = 2\ell+1.$$

Only the first three multipoles ($\ell = 0, 1, 2$) represent fluid quantities that couple directly into Einstein's field equations. A completely parallel STF multipole expansion can be applied to the temperature perturbation field $\theta(\eta, x^i, n^j)$, defining the moment tensors $\theta_{i_1\ldots i_\ell}(\eta, x^i)$.

The expansion (5.3.36) substantially streamlines the Boltzmann equation: in the advection term $n^k\partial_k \delta I$, the spatial derivative acts exclusively on the tensor coefficients $\mathcal{I}_{i_1\ldots i_\ell}$, enabling it to be combined with the spatial connection coefficients ${}^{(3)}\Gamma^k_{ij}$ into a spatial covariant derivative D_k. Eq. (5.3.34) then adopts the concise form:

$$\theta' + \sum_{\ell=0}^{\infty} n^k \left(n^{i_1}\ldots n^{i_\ell}\right) D_k\theta_{i_1\ldots i_\ell} + \left[n^k\partial_k A + \left(\frac{1}{2}h'_{ij} - D_{(i}B_{j)}\right) n^i n^j\right] = C[\theta]. \tag{5.3.37}$$

Because the metric perturbation terms depend on direction only up to quadratic order in n^i, they enter exclusively into the monopole and dipole evolution equations, matching the fluid equations of motion.

The scalar-vector-tensor (SVT) decomposition established in Chapter 5 extends naturally to arbitrary rank-ℓ tensors. Recall that any symmetric, trace-free rank-2 tensor (such as the anisotropic stress π_{ij}) can be decomposed into scalar, vector, and tensor components as

$$\pi_{ij} = \Delta_{ij}\pi + D_{(i}\pi_{j)} + \bar{\pi}_{ij},$$

with the individual projections defined by

$$\pi^{(\mathrm{S})}_{ij} = [D_{ij}\pi]^{\mathrm{STF}}, \qquad \pi^{(\mathrm{V})}_{ij} = [D_i\pi_j]^{\mathrm{STF}}, \qquad \pi^{(\mathrm{T})}_{ij} = [\bar{\pi}_{ij}]^{\mathrm{STF}},$$

where STF denotes the extraction of the symmetric trace-free part and where $\pi^{(\mathrm{V})}_{ij}$ and $\bar{\pi}_{ij}$ are constrained to be transverse.

For a general tensor moment of rank ℓ, the component of order m decomposes as

$$\theta^{(m)}_{i_1\ldots i_\ell} = \left[D_{i_1\ldots i_{\ell-m}}\theta^{(m)}_{\ell, i_{\ell-m+1}\ldots i_\ell}\right]^{\mathrm{STF}}. \tag{5.3.38}$$

The indices $|m| = 0, 1, 2$ correspond to scalar (S), vector (V), and tensor (T) perturbations, respectively. Decomposing into Fourier modes $\hat{\theta}(\boldsymbol{k})$ with orientation $\boldsymbol{e}_3 = \hat{\boldsymbol{k}} = \boldsymbol{k}/k$, the scalar sector of $\hat{\theta}_{i_1\ldots i_\ell}$ is proportional to $k_{i_1}\ldots k_{i_\ell}\hat{\theta}^{(0)}_\ell$. Contracting this tensor with the STF product of direction vectors $n^{i_1}\ldots n^{i_\ell}$ yields the Legendre polynomial $P_\ell(\hat{\boldsymbol{k}}\cdot\boldsymbol{n})$, so that

$$n^{i_1}\ldots n^{i_\ell}\hat{\theta}^{(0)}_{i_1\ldots i_\ell} \propto Y_{\ell 0}\hat{\theta}^{(0)}_\ell.$$

Vector modes possess two transverse polarization degrees of freedom orthogonal to $\boldsymbol{k}$ [cf. Eq. (5.1.32)]. Choosing the helicity basis vectors $\boldsymbol{e}^{\pm} = \boldsymbol{e}_1 \pm \boldsymbol{e}_2$, the vector portion of $\hat{\theta}_{i_1\ldots i_\ell}$ is proportional to the STF part of $k_{i_1}\ldots k_{i_{\ell-1}}e^{\pm}_{i_\ell}\hat{\theta}^{(\pm 1)}_\ell$. Contraction with $n^{i_1}\ldots n^{i_\ell}$ then leads to spherical harmonics of order $m = \pm 1$:

$$n^{i_1}\ldots n^{i_\ell}\hat{\theta}^{(\pm 1)}_{i_1\ldots i_\ell} \propto Y_{\ell,\pm 1}\hat{\theta}^{(\pm 1)}_\ell.$$

Generalizing this construction to all azimuthal indices m, the Fourier moment $\hat{\theta}_{i_1\ldots i_\ell}$ expands as

$$\hat{\theta}_{i_1\ldots i_\ell} = \sum_{m=-\ell}^{\ell} \hat{\theta}^{(m)}_{i_1\ldots i_\ell}, \tag{5.3.39}$$

where

$$\hat{\theta}^{(m)}_{i_1\ldots i_\ell} \propto \left[k_{i_1}\ldots k_{i_{\ell-m}}\boldsymbol{e}^{\pm}_{i_{\ell-m+1}}\otimes\cdots\otimes\boldsymbol{e}^{\pm}_{i_\ell}\right]^{\mathrm{STF}}\hat{\theta}^{(m)}_\ell, \tag{5.3.40}$$

taking the $+$ sign for $m \geq 0$ and the $-$ sign for $m < 0$. Projecting onto the directional line of sight yields (see Chapter 5 of Ref. [3] and Ref. [20] for explicit tensor properties):

$$n^{i_1}\ldots n^{i_\ell}\hat{\theta}^{(m)}_{i_1\ldots i_\ell} \propto Y_{\ell m}\hat{\theta}^{(m)}_\ell. \tag{5.3.41}$$

Each perturbation type carries 2 degrees of freedom (except for the scalar mode $m = 0$, which possesses only 1). For each multipole ℓ, there are thus $2\ell + 1$ independent degrees of freedom, precisely spanning the spherical harmonics $Y_{\ell m}$ on the celestial sphere.

This construction demonstrates that the radiation temperature perturbation $\theta(\eta, \boldsymbol{x}, \boldsymbol{n})$ at spatial position $\boldsymbol{x}$ propagating along direction $\boldsymbol{n}$ can be expanded in the complete basis functions $G_{\ell m}(\boldsymbol{x}, \boldsymbol{n})$ defined by

$$G_{\ell m}(\boldsymbol{x}, \boldsymbol{n}) = (-\mathrm{i})^\ell\sqrt{\frac{4\pi}{2\ell+1}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}Y_{\ell m}(\boldsymbol{n}), \tag{5.3.42}$$

yielding the modal expansion

$$\theta(\eta, \boldsymbol{x}, \boldsymbol{n}) = \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^{3/2}}\sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell}\hat{\theta}^{(m)}_\ell(\eta, \boldsymbol{k})G_{\ell m}(\boldsymbol{x}, \boldsymbol{n}). \tag{5.3.43}$$

At linear order, cosmological perturbations of the matter and space-time geometry only excite the low-order azimuthal modes $m = 0$ (scalar), $m = \pm 1$ (vector), and $m = \pm 2$ (tensor).

Relation with the SVT decomposition

Expanding scalar plane-wave modes according to

$$Q^{(0)} = \exp(\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}), \qquad Q^{(0)}_i = \partial_i Q^{(0)}, \qquad Q^{(0)}_{ij} = -\left[\partial_i\partial_j - \frac{1}{3}\delta_{ij}\Delta\right]Q^{(0)},$$

the corresponding angular projections evaluate to

$$Q^{(0)} = G_{00}, \qquad n^iQ^{(0)}_i = -kG_{10}, \qquad n^in^jQ^{(0)}_{ij} \propto k^2G_{20}. \tag{5.3.44}$$

Similarly, the vector modes are expanded as

$$Q^{(\pm1)}_i = -\frac{\mathrm{i}}{\sqrt{2}}e^{\pm}_i Q^{(0)}, \qquad Q^{(\pm1)}_{ij} = \partial_{(i}Q^{(\pm1)}_{j)},$$

so that contracting with the line-of-sight direction yields

$$n^iQ^{(\pm1)}_i = G_{1,\pm1}, \qquad n^in^jQ^{(\pm1)}_{ij} \propto -kG_{2,\pm1}. \tag{5.3.45}$$

This decomposition directly parallels Eq. (5.1.32) introduced earlier. Finally, the transverse-traceless tensor modes are expanded as

$$Q^{(\pm2)}_{ij} = -\sqrt{\frac{3}{8}}e^{\pm}_i\otimes e^{\pm}_j Q^{(0)},$$

leading to the angular projection

$$n^in^jQ^{(\pm2)}_{ij} = G_{2,\pm2}. \tag{5.3.46}$$

Note that this harmonic convention differs from the basis defined in Eq. (5.1.40) by a normalising prefactor of $\sqrt{3/8}$.

Collision term

We now incorporate the Thomson scattering of photons off free electrons into the relativistic Boltzmann equation. The collision term describes the net rate of change of the distribution function along the geodesic:

$$C[f]=\left(\frac{\mathrm{d}f}{\mathrm{d}\eta}\right)_{\mathrm{coll}}=\frac{\mathrm{d}f^{+}}{\mathrm{d}\eta}-\frac{\mathrm{d}f^{-}}{\mathrm{d}\eta}, \tag{5.3.47}$$

where $\mathrm{d}f^{+}/\mathrm{d}\eta$ denotes scattering into direction $\boldsymbol{n}$ (gain term) and $\mathrm{d}f^{-}/\mathrm{d}\eta$ represents scattering out of direction $\boldsymbol{n}$ (loss term). In the local rest frame of the baryon-electron plasma, these two rates are given by

$$\frac{\mathrm{d}f^{+}}{\mathrm{d}\eta}=\tau'\int f(\boldsymbol{n}')\omega(\boldsymbol{n},\boldsymbol{n}')\frac{\mathrm{d}^2\boldsymbol{n}'}{4\pi},\qquad \frac{\mathrm{d}f^{-}}{\mathrm{d}\eta}=\tau' f(\boldsymbol{n}), \tag{5.3.48}$$

where the differential optical depth is defined as

$$\tau'\equiv a\sigma_{\mathrm{T}}n_{\mathrm{e}}, \tag{5.3.49}$$

with $\sigma_{\mathrm{T}}=8\pi\alpha^2/(3m_{\mathrm{e}}^2)$ the Thomson cross section and n_{e} the number density of free electrons. The unpolarized Thomson scattering phase function $\omega(\boldsymbol{n},\boldsymbol{n}')$ encodes the angular probability distribution:

$$\omega(\boldsymbol{n},\boldsymbol{n}')=\frac{3}{4}\left[1+(\boldsymbol{n}\cdot\boldsymbol{n}')^2\right]=1+\frac{3}{4}n_{ij}n'^{ij}=1+\frac{1}{2}P_2(\boldsymbol{n}\cdot\boldsymbol{n}'), \tag{5.3.50}$$

where $P_2(\mu)=\frac{1}{2}(3\mu^2-1)$ is the second Legendre polynomial. A crucial property of non-relativistic Thomson scattering is that $\omega(\boldsymbol{n},\boldsymbol{n}')$ and the resulting collision operator are strictly energy-independent. Thomson scattering therefore induces no spectral distortion, justifying our ansatz that the fractional brightness fluctuation $\delta I/(4\bar{I})$ corresponds directly to a temperature perturbation θ [Eqs. (5.3.16), (5.3.32)].

At the background level, the angular symmetry of $\omega(\boldsymbol{n},\boldsymbol{n}')$ ensures that

$$C[\bar{f}]=0, \tag{5.3.51}$$

as required by the spatial isotropy of the unperturbed Friedmann–Lemaître background, where all fluid velocities vanish in comoving coordinates ($u^\mu\propto\delta_0^\mu$).

To first order in cosmological perturbations, transforming from the baryon rest frame to the general coordinate system yields

$$C[\delta f]=\tau'\left[\int\delta f(\eta,x^i,E,\boldsymbol{n}')\frac{\mathrm{d}^2\boldsymbol{n}'}{4\pi}-\delta f(\eta,x^i,E,\boldsymbol{n})+\frac{3}{4}n_{ij}\int\delta f(\eta,x^i,E,\boldsymbol{n}')n'^{ij}\frac{\mathrm{d}^2\boldsymbol{n}'}{4\pi}\right], \tag{5.3.52}$$

where the photon energy E in the electron frame is related to the coordinate energy $\bar{E}$ by $E=-k^\mu u_\mu^{\mathrm{b}}=\bar{E}[1-(B_i+v_i^{\mathrm{b}})n^i]$, giving phase-space measure transformation

$$E^3\,\mathrm{d}E=\left[1+4(B_i+v_i^{\mathrm{b}})n^i\right]\bar{E}^3\mathrm{d}\bar{E}.$$

Integrating Eq. (5.3.52) over energy weighted by $4\pi E^3\,\mathrm{d}E$ yields the collision operator for brightness perturbations:

$$C[\delta I]=\tau'\left[\int\delta I\frac{\mathrm{d}^2\boldsymbol{n}'}{4\pi}-\delta I+4\bar{I}(B_i+v_{\mathrm{b}}^i)n^i+\frac{3}{4}n_{ij}\int\delta I(\eta,x^i,\boldsymbol{n}')n'^{ij}\frac{\mathrm{d}^2\boldsymbol{n}'}{4\pi}\right]. \tag{5.3.53}$$

The first term simplifies to the brightness monopole $\bar{I}\delta_\gamma$, and the quadrupole term evaluates to $3n^{ij}\Pi_{ij} = (4\bar{I})n^i n^j \pi_{ij}/16$. Defining the normalized temperature collision term $C[\theta] = C[\delta I]/(4\bar{I})$, we obtain:

$$\boxed{C[\theta] = \tau' \left[\theta_0 - \theta + (B^i + v_{\rm b}^i)n_i + \frac{1}{16}n^i n^j \pi_{ij}\right],} \tag{5.3.54}$$

where $\theta_0 = \frac{1}{4}\delta_\gamma$ is the temperature monopole [Eq. (5.3.33)]. Since $C[\theta]$ vanishes identically in the unperturbed universe, the Stewart–Walker lemma (Chapter 5) guarantees that $C[\theta]$ is gauge invariant at linear order. Moreover, in the multipole decomposition, the Thomson collision operator couples to the fluid velocity and metric only up to $\ell = 2$; for all higher multipoles ($\ell > 2$), the collision term reduces simply to

$$C[\theta_\ell] = -\tau'\theta_\ell.$$

5.3.2 Gauge invariant expressions

A comprehensive treatment of the gauge transformation properties of relativistic distribution functions is given in Refs. [19, 21].

Gauge invariant distribution function

The split $f = \bar{f} + \delta f$ depends on the choice of background spacetime mapping. In linear perturbation theory, defining δf requires choosing an isomorphism ψ between the physical and background mass shells:

$$\psi : \mathcal{P}_m \to \mathcal{P}_m : (x^\alpha, p^\beta) \mapsto (\bar{x}^\alpha, \bar{p}^\beta),$$

so that

$$f = \bar{f} \circ \psi + \delta f. \tag{5.3.55}$$

Under an infinitesimal coordinate transformation $x^\alpha \to y^\alpha = x^\alpha - \xi^\alpha$, the 4-momentum transforms according to

$$p^\mu \to p^\mu - p^\nu \partial_\nu \xi^\mu. \tag{5.3.56}$$

This phase-space transformation is generated by the tangent vector field

$$T_\xi = (\xi^\mu,\ p^\nu \partial_\nu \xi^\mu) \tag{5.3.57}$$

in the natural coordinate basis $(\partial_\mu, \partial/\partial p^\mu)$. To first order, the distribution function transforms via the Lie derivative along T_ξ:

$$f \to f + \pounds_{T_\xi} f. \tag{5.3.58}$$

Because the background $\bar{f}$ is defined strictly on the mass shell $\mathcal{P}_m$, and the generator T_ξ is not necessarily tangent to $\mathcal{P}_m$, the gauge variation of δf is properly defined by projecting T_ξ along the mass shell ($T_\xi^{\|}$), as shown in Ref. [21]. Under a general gauge transformation, the perturbation δf transforms as

$$\boxed{\delta f \to \delta f + (\mathcal{H}T + n^i \partial_i T)E\frac{\partial \bar{f}}{\partial E},} \tag{5.3.59}$$

where $T \equiv \xi^0$ is the temporal gauge shift parameter. This result can also be understood physically: the photon energy $E = -N_\mu k^\mu = a(1 + A)k^0$ shifts by $E \to E - En^i\partial_i T$, which combined with the background Liouville relation $\bar{f}' = \mathcal{H}E\partial\bar{f}/\partial E$ [Eq. (5.3.13)] reproduces Eq. (5.3.59).

We can therefore construct a gauge invariant distribution function $\mathcal{F}$ by combining δf with scalar metric perturbations:

$$\boxed{\mathcal{F} \equiv \delta f - \left[C + n^i \partial_i (E' - B)\right] E \frac{\partial \bar{f}}{\partial E},} \tag{5.3.60}$$

where E in $(E' - B)$ represents the scalar metric shear perturbation, while outside the brackets E denotes photon energy. Integrating $\mathcal{F}$ over momentum space gives

$$\int \mathcal{F} E^3 \, \mathrm{d}E \, \mathrm{d}^2\boldsymbol{n} = \delta\rho_\gamma + 4C\bar{\rho}_\gamma = \bar{\rho}_\gamma \delta_\gamma^{\mathcal{F}},$$

revealing that $\mathcal{F}$ represents the distribution perturbation in the flat-slicing gauge ($C = 0$).

Alternatively, one can define the gauge invariant distribution function in the conformal Newtonian (longitudinal) gauge:

$$\bar{\mathcal{F}} \equiv \mathcal{F} + \Psi E \frac{\partial \bar{f}}{\partial E} = \mathcal{F} + \bar{f}' \frac{\Psi}{\mathcal{H}}. \tag{5.3.61}$$

Integrating over energy, the corresponding gauge invariant temperature perturbations are denoted by $\mathcal{M}$ (flat-slicing gauge) and Θ (Newtonian gauge). They are related by

$$\mathcal{M} = \Theta - \Psi,$$

and satisfy the respective gauge invariant Boltzmann equations:

$$\mathcal{M}' + n^k \partial_k \mathcal{M} - {}^{(3)}\Gamma^k_{ij} n^i n^j \frac{\partial \mathcal{M}}{\partial n^k} + \left[n^k \partial_k (\Phi + \Psi) + \left(\bar{E}'_{ij} + D_{(i}\bar{\Phi}_{j)}\right) n^i n^j\right] = C[\mathcal{M}], \tag{5.3.62}$$

$$\Theta' + n^k \partial_k (\Theta + \Phi) - {}^{(3)}\Gamma^k_{ij} n^i n^j \frac{\partial \Theta}{\partial n^k} + \left[-\Psi' + \left(\bar{E}'_{ij} + D_{(i}\bar{\Phi}_{j)}\right) n^i n^j\right] = C[\Theta]. \tag{5.3.63}$$

Because the metric perturbation source terms depend on direction only up to quadratic order in n^i, these two equations differ only at the level of the monopole and dipole.

Boltzmann equation for the temperature

Specializing to a spatially flat FLRW background ($K = 0$) and restricting attention to scalar perturbations, the Newtonian-gauge Boltzmann equation (5.3.63) in Fourier space becomes

$$\Theta' + \mathrm{i}k\mu(\Theta + \Phi) = \Psi' + \tau' \left[\Theta_0 - \Theta + \mathrm{i}k\mu V_\mathrm{b} + \frac{1}{16} n_i n_j \Pi^{ij}\right], \tag{5.3.64}$$

where $\mu \equiv \hat{\boldsymbol{k}} \cdot \boldsymbol{n}$ is the cosine of the angle between the wavevector and photon direction. We expand the directional dependence of $\Theta(\boldsymbol{k}, \eta, \mu)$ in Legendre polynomials:

$$\Theta(\boldsymbol{k}, \eta, \mu) = \sum_{\ell=0}^{\infty} (-\mathrm{i})^\ell \Theta_\ell(k, \eta) P_\ell(\mu) \mathrm{e}^{+\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}. \tag{5.3.65}$$

Using the recurrence relation for Legendre polynomials,

$$(2\ell + 1)\mu P_\ell(\mu) = (\ell + 1)P_{\ell+1}(\mu) + \ell P_{\ell-1}(\mu),$$

the streaming term $\mathrm{i}k\mu\Theta$ is transformed into

$$k \sum_{\ell=0}^{\infty} \left[\frac{\ell + 1}{2\ell + 3}\Theta_{\ell+1} - \frac{\ell}{2\ell - 1}\Theta_{\ell-1}\right] (-\mathrm{i})^\ell P_\ell(\mu).$$

Projecting onto individual Legendre multipoles using $\int_{-1}^{1} P_\ell(\mu)P_{\ell'}(\mu)\mathrm{d}\mu = \frac{2}{2\ell+1}\delta_{\ell\ell'}$, and noting that $\Pi_{ij}n^i n^j/16 = (-\mathrm{i})^2 P_2(\mu)\Theta_2/10$, the infinite hierarchy of coupled Boltzmann moment equations emerges:

$$\Theta_0' = -\frac{k}{3}\Theta_1 + \Psi', \tag{5.3.66}$$

$$\Theta_1' = k\left(\Theta_0 - \frac{2}{5}\Theta_2 + \Phi\right) - \tau'(kV_\mathrm{b} + \Theta_1), \tag{5.3.67}$$

$$\Theta_2' = k\left(\frac{2}{3}\Theta_1 - \frac{3}{7}\Theta_3\right) - \frac{9}{10}\tau'\Theta_2, \tag{5.3.68}$$

$$\Theta_\ell' = k\left(\frac{\ell}{2\ell-1}\Theta_{\ell-1} - \frac{\ell+1}{2\ell+3}\Theta_{\ell+1}\right) - \tau'\Theta_\ell \qquad (\ell \geq 3). \tag{5.3.69}$$

As anticipated, metric perturbations and baryonic velocity appear exclusively in the monopole and dipole equations ($\ell = 0, 1$), while the anisotropic shear quadrupole damping enters at $\ell = 2$ with coefficient $\frac{9}{10}\tau'$. For all multipoles $\ell \geq 3$, the collision operator reduces strictly to $-\tau'\Theta_\ell$.

The lowest-order multipoles connect directly to macroscopic fluid variables:

$$4\Theta_0 = \delta_\gamma^\mathrm{N}, \qquad \Theta_1 = -kV_\gamma, \qquad \Theta_2 = \frac{5}{12}k^2\pi_\gamma. \tag{5.3.70}$$

The factor $-k$ between Θ_1 and V_γ reflects the harmonic normalization of Eq. (5.3.44). Thus, the first two equations of the hierarchy (5.3.66)–(5.3.67) are precisely the continuity and Euler equations of the radiation fluid.

For primordial tensor perturbations, the Boltzmann equation simplifies to

$$\Theta' + \mathrm{i}k\mu\Theta = -\bar{E}_{ij}' n^i n^j - \tau'\left(\Theta - \frac{1}{16}n_i n_j \Pi^{ij}\right). \tag{5.3.71}$$

Decomposing $\bar{E}_{ij} = H^{(\pm 2)}Q_{ij}^{(\pm 2)}$ [cf. Eq. (5.3.44)], the tensor hierarchy begins at the quadrupole $\ell = 2$:

$$\boxed{\Theta_2' = -k\frac{\sqrt{5}}{7}\Theta_2 - \frac{9}{10}\tau'\Theta_2 - \dot{H},} \tag{5.3.72}$$

$$\boxed{\Theta_\ell' = k\left(\frac{\sqrt{\ell^2-4}}{2\ell-1}\Theta_{\ell-1} - \frac{\sqrt{(\ell+1)^2-4}}{2\ell+3}\Theta_{\ell+1}\right) - \tau'\Theta_\ell \qquad (\ell \geq 3).} \tag{5.3.73}$$

5.3.3 Polarization

Relativistic kinetic theory and polarization formalisms for the CMB are reviewed in Refs. [22–26], with detailed multipole approaches presented in Refs. [3, 27].

Stokes parameters

A monochromatic electromagnetic wave propagating along unit direction $\boldsymbol{e}_3$ possesses an electric field vector lying in the orthogonal transverse plane $(\boldsymbol{e}_1, \boldsymbol{e}_2)$:

$$\boldsymbol{E}(t, \boldsymbol{x}) = \mathrm{Re}\begin{pmatrix} a_1 \exp[\mathrm{i}\theta_1] \\ a_2 \exp[\mathrm{i}\theta_2] \\ 0 \end{pmatrix}. \tag{5.3.74}$$

When $\theta_1 = \theta_2$, the electric field oscillates in a constant plane at angle $\tan\alpha = a_2/a_1$, describing linear polarization. When the phases differ, the wave is elliptically polarized. A statistical ensemble of radiation is completely described by the four Stokes parameters:

$$I \equiv \langle a_1^2 \rangle + \langle a_2^2 \rangle = I_1 + I_2, \tag{5.3.75}$$

$$Q \equiv \langle a_1^2 \rangle - \langle a_2^2 \rangle = I_1 - I_2, \tag{5.3.76}$$
$$U \equiv 2\langle a_1 a_2 \cos(\theta_1 - \theta_2) \rangle, \tag{5.3.77}$$
$$V \equiv 2\langle a_1 a_2 \sin(\theta_1 - \theta_2) \rangle, \tag{5.3.78}$$

where angle brackets denote time averages over durations much longer than the optical period $2\pi/\omega$. For fully polarized monochromatic radiation, these parameters obey

$$I^2 = Q^2 + U^2 + V^2. \tag{5.3.79}$$

I represents the total intensity; Q quantifies the excess of linear polarization along $\boldsymbol{e}_1$ relative to $\boldsymbol{e}_2$; U measures linear polarization along the diagonal axes at 45°; and V measures the degree of circular polarization (the asymmetry between right- and left-handed helicity states, $V = \langle a_+^2 \rangle - \langle a_-^2 \rangle$). Thomson scattering generates no circular polarization from an initially unpolarized background, so $V = 0$ throughout CMB cosmology.

Under a right-handed rotation of the polarization basis $(\boldsymbol{e}_1, \boldsymbol{e}_2)$ by angle ψ around the propagation axis $\boldsymbol{e}_3$:

$$\boldsymbol{e}_1' = \cos\psi \boldsymbol{e}_1 + \sin\psi \boldsymbol{e}_2, \qquad \boldsymbol{e}_2' = -\sin\psi \boldsymbol{e}_1 + \cos\psi \boldsymbol{e}_2,$$

the intensity I and circular polarization V remain invariant, whereas the linear polarization parameters (Q, U) rotate by 2ψ:

$$\begin{pmatrix} Q' \\ U' \end{pmatrix} = \begin{pmatrix} \cos 2\psi & \sin 2\psi \\ -\sin 2\psi & \cos 2\psi \end{pmatrix} \begin{pmatrix} Q \\ U \end{pmatrix}. \tag{5.3.80}$$

Equivalently, the transformation of the vector $(I_1, I_2, U, V)^{\mathrm{T}}$ takes the matrix form:

$$\begin{pmatrix} I_1' \\ I_2' \\ U' \\ V' \end{pmatrix} = \begin{pmatrix} \cos^2\psi & \sin^2\psi & \frac{1}{2}\sin 2\psi & 0 \\ \sin^2\psi & \cos^2\psi & -\frac{1}{2}\sin 2\psi & 0 \\ -\sin 2\psi & \sin 2\psi & \cos 2\psi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \\ U \\ V \end{pmatrix}. \tag{5.3.81}$$

From Eq. (5.3.80), the complex linear combinations $(Q \pm \mathrm{i}U)$ transform under basis rotation as spin-± 2 quantities:

$$(Q \pm \mathrm{i}U)'(\boldsymbol{e}_3) = \mathrm{e}^{\mp 2\mathrm{i}\psi}(Q \pm \mathrm{i}U)(\boldsymbol{e}_3). \tag{5.3.82}$$

Because the unperturbed universe is strictly unpolarized, Q and U are first-order gauge invariant perturbations that propagate without linear coupling to gravitational metric perturbations. Their kinetic Boltzmann equation takes the form:

$$\mathcal{L}\begin{pmatrix} Q \\ U \end{pmatrix} = \begin{pmatrix} Q \\ U \end{pmatrix}' + n^i \partial_i \begin{pmatrix} Q \\ U \end{pmatrix} - \mathcal{H}E\frac{\partial}{\partial E}\begin{pmatrix} Q \\ U \end{pmatrix} = C\begin{pmatrix} Q \\ U \end{pmatrix}. \tag{5.3.83}$$

Normalizing the Stokes parameters by the background brightness $\bar{I}$ [Eq. (5.3.29)],

$$\begin{pmatrix} Q \\ U \end{pmatrix} \to \frac{1}{4\bar{I}} \int 4\pi E^2 \begin{pmatrix} Q \\ U \end{pmatrix} \mathrm{d}E\,, \tag{5.3.84}$$

we obtain the simplified Liouville equation for linear polarization:

$$\mathcal{L}\begin{pmatrix} Q \\ U \end{pmatrix} = \begin{pmatrix} Q \\ U \end{pmatrix}' + n^i \partial_i \begin{pmatrix} Q \\ U \end{pmatrix} = C\begin{pmatrix} Q \\ U \end{pmatrix}, \tag{5.3.85}$$

with normalized collision integral

$$C\begin{pmatrix} Q \\ U \end{pmatrix} \to \frac{1}{4\bar{I}} \int 4\pi E^2 C\begin{pmatrix} Q \\ U \end{pmatrix} \mathrm{d}E\,. \tag{5.3.86}$$

Spherical harmonics of spin s

For any celestial direction $\boldsymbol{e}_3$ specified by spherical coordinates (θ, ϕ), the tangent space is spanned by the orthonormal basis $(\boldsymbol{e}_\theta, \boldsymbol{e}_\phi)$, determined up to an internal rotation. A function ${}_sf(\theta, \phi)$ on the sphere is said to possess spin weight s if it transforms as ${}_sf'(\theta, \phi) = \mathrm{e}^{-\mathrm{i}s\psi}\,{}_sf(\theta, \phi)$ under a local basis rotation of angle ψ.

Any spin-s field can be expanded in spin-weighted spherical harmonics ${}_sY_{\ell m}(\theta, \phi)$ [Ref. [29]], which satisfy the orthogonality and completeness relations:

$$\int_0^{2\pi} \mathrm{d}\phi \int_{-1}^{1} \mathrm{d}(\cos\theta)\ {}_sY^*_{\ell' m'}(\theta, \phi)\,{}_sY_{\ell m}(\theta, \phi) = \delta_{\ell\ell'}\delta_{mm'}, \tag{5.3.87}$$

$$\sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} {}_sY^*_{\ell m}(\theta, \phi)\,{}_sY_{\ell m}(\theta', \phi') = \delta(\phi - \phi')\delta(\cos\theta - \cos\theta'). \tag{5.3.88}$$

The spin weight can be adjusted using the spin-raising (∂^+) and spin-lowering (∂^-) differential operators:

$$\partial^{\pm}{}_sf(\theta, \phi) = -\sin^{\pm s}\theta \left(\partial_\theta \pm \frac{\mathrm{i}}{\sin\theta}\partial_\phi\right) \sin^{\mp s}\theta\,{}_sf(\theta, \phi). \tag{5.3.89}$$

For spin-2 functions with azimuthal dependence $\partial_\phi[{}_{\pm 2}f(\mu, \phi)] = \mathrm{i}m\,{}_{\pm 2}f(\mu, \phi)$ where $\mu = \cos\theta$, the second-order operators reduce to

$$(\partial^{\mp})^2\,{}_{\pm 2}f(\mu, \phi) = \left(-\partial_\mu \pm \frac{m}{1 - \mu^2}\right)^2 \left[(1 - \mu^2)\,{}_{\pm 2}f(\mu, \phi)\right]. \tag{5.3.90}$$

Acting with these ladder operators on ordinary spherical harmonics ($s = 0$) yields the spin-weighted spherical harmonics:

$${}_sY_{\ell m} \equiv \sqrt{\frac{(\ell - s)!}{(\ell + s)!}}(\partial^+)^s Y_{\ell m} \quad (0 \le s \le \ell), \qquad {}_sY_{\ell m} \equiv \sqrt{\frac{(\ell + s)!}{(\ell - s)!}}(-1)^s(\partial^-)^{-s} Y_{\ell m} \quad (-\ell \le s \le 0). \tag{5.3.91}$$

These harmonics satisfy the conjugation and differential relations:

$${}_sY^*_{\ell m} = (-1)^s\,{}_{-s}Y_{\ell, -m}, \tag{5.3.92}$$

$$\partial^{\pm}{}_sY_{\ell m} = \pm\sqrt{(\ell \mp s)(\ell + 1 \pm s)}\,{}_{s\pm 1}Y_{\ell m}, \tag{5.3.93}$$

$$\partial^-\partial^+{}_sY_{\ell m} = -(\ell - s)(\ell + 1 + s)\,{}_sY_{\ell m}. \tag{5.3.94}$$

E and B modes

In complete analogy with the scalar temperature field expansion,

$$\Theta(\boldsymbol{n}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\boldsymbol{n}), \tag{5.3.95}$$

the spin-± 2 linear polarization fields expand in spin-weighted harmonics as

$$(Q \pm \mathrm{i}U)(\boldsymbol{n}) = \sum_{\ell=2}^{\infty} \sum_{m=-\ell}^{\ell} a^{(\pm 2)}_{\ell m}\,{}_{\pm 2}Y_{\ell m}(\boldsymbol{n}), \tag{5.3.96}$$

with $a^{(-2)*}_{\ell m} = a^{(+2)}_{\ell, -m}$. Applying the differential ladder relations yields

$$(\partial^{\pm})^2(Q \pm \mathrm{i}U)(\boldsymbol{n}) = \sqrt{\frac{(\ell + 2)!}{(\ell - 2)!}} \sum_{\ell=2}^{\infty} \sum_{m=-\ell}^{\ell} a^{(\pm 2)}_{\ell m} Y_{\ell m}(\boldsymbol{n}). \tag{5.3.97}$$

The expansion multipole moments are obtained by projection:

$$a_{\ell m} = \int \mathrm{d}^2\boldsymbol{n}\, Y^*_{\ell m}(\boldsymbol{n})\Theta(\boldsymbol{n}), \tag{5.3.98}$$

and

$$a^{(\pm 2)}_{\ell m} = \int \mathrm{d}^2\boldsymbol{n}\, {}_{\pm 2}Y^*_{\ell m}(\boldsymbol{n})(Q \pm \mathrm{i}U)(\boldsymbol{n}) = \sqrt{\frac{(\ell-2)!}{(\ell+2)!}} \int \mathrm{d}^2\boldsymbol{n}\, Y^*_{\ell m}(\boldsymbol{n})(\eth^{\mp})^2(Q \pm \mathrm{i}U)(\boldsymbol{n}). \tag{5.3.99}$$

Because the Stokes parameters Q and U depend on the coordinate orientation of the polarization basis, it is advantageous to decompose the spin-2 field into coordinate-independent scalar E (electric) and pseudo-scalar B (magnetic) modes:

$$(Q \pm \mathrm{i}U)(\eta, \boldsymbol{x}, \boldsymbol{n}) = -\sum_{\ell=2}^{\infty} \sum_{m=-\ell}^{\ell} (E_{\ell m} \pm \mathrm{i}B_{\ell m})\, {}_{\pm 2}Y_{\ell m}(\boldsymbol{n}), \tag{5.3.100}$$

where the multipole coefficients are defined by

$$E_{\ell m} \equiv -\frac{1}{2}\left(a^{(+2)}_{\ell m} + a^{(-2)}_{\ell m}\right), \qquad B_{\ell m} \equiv -\frac{1}{2\mathrm{i}}\left(a^{(+2)}_{\ell m} - a^{(-2)}_{\ell m}\right). \tag{5.3.101}$$

The designations E and B reflect their definite parity behavior under spatial reflections: under a reflection across the plane containing the propagation vector $\boldsymbol{k}$, $Q \to Q$ and $U \to -U$. The spin-weighted spherical harmonics transform as ${}_sY_{\ell m} \to (-1)^{\ell}\, {}_{-s}Y_{\ell m}$, implying that

$$E_{\ell m} \to (-1)^{\ell} E_{\ell m}, \qquad B_{\ell m} \to (-1)^{\ell+1} B_{\ell m}.$$

Thus, the E-mode behaves as a pure scalar (even parity under spatial inversion), while the B-mode behaves as a pseudo-scalar (odd parity). Linear scalar perturbations possess reflection symmetry and consequently generate pure E-modes with $B_{\ell m} \equiv 0$; primordial tensor gravitational waves break reflection symmetry and generate both E-modes and B-modes.

Relation with the Stokes parameters

The geometric meaning of E and B modes becomes particularly transparent in the flat-sky approximation. Over small angular patches, spherical coordinates (ϑ, ϕ) map onto Cartesian angular coordinates $\boldsymbol{\theta} = (\theta_x, \theta_y)$, with conjugate Fourier wavevector $\boldsymbol{\ell} = (\ell_x, \ell_y) = \ell(\cos\phi_\ell, \sin\phi_\ell)$. In this small-scale limit ($r = 2\sin(\vartheta/2) \approx \vartheta \ll 1$), $Y_{\ell m}(\boldsymbol{n}) \to \exp(\mathrm{i}\boldsymbol{\ell}\cdot\boldsymbol{n})$ and ${}_{\pm 2}Y_{\ell m}(\boldsymbol{n}) \to \exp(\mathrm{i}\boldsymbol{\ell}\cdot\boldsymbol{n} \pm 2\mathrm{i}\phi_\ell)$, so that the fields expand in two-dimensional Fourier modes:

$$\Theta(\boldsymbol{n}) = \int \frac{\mathrm{d}^2\boldsymbol{\ell}}{2\pi} \Theta(\boldsymbol{\ell}) \mathrm{e}^{\mathrm{i}\boldsymbol{\ell}\cdot\boldsymbol{n}}, \tag{5.3.102}$$

$$(Q \pm \mathrm{i}U)(\boldsymbol{n}) = -\int \frac{\mathrm{d}^2\boldsymbol{\ell}}{2\pi} \left[E(\boldsymbol{\ell}) \pm \mathrm{i}B(\boldsymbol{\ell})\right] \frac{1}{\ell^2} (\eth^{\pm})^2 \mathrm{e}^{\mathrm{i}\boldsymbol{\ell}\cdot\boldsymbol{n}}. \tag{5.3.103}$$

Evaluating the action of the spin-differential operators in the flat limit gives

$$(\eth^{\pm})^2 \mathrm{e}^{\mathrm{i}\boldsymbol{\ell}\cdot\boldsymbol{n}} = -\frac{(\ell_x \pm \mathrm{i}\ell_y)^2}{\ell^2} \mathrm{e}^{\mp 2\mathrm{i}\phi} \mathrm{e}^{\mathrm{i}\boldsymbol{\ell}\cdot\boldsymbol{n}} = -\ell^2 \mathrm{e}^{\pm 2\mathrm{i}(\phi_\ell - \phi)} \mathrm{e}^{\mathrm{i}\boldsymbol{\ell}\cdot\boldsymbol{n}}.$$

While the natural spherical basis $(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_\phi)$ varies across the celestial sphere, within a small patch one can fix a Cartesian coordinate grid (x, y), defining a constant polarization reference frame across the field of view.

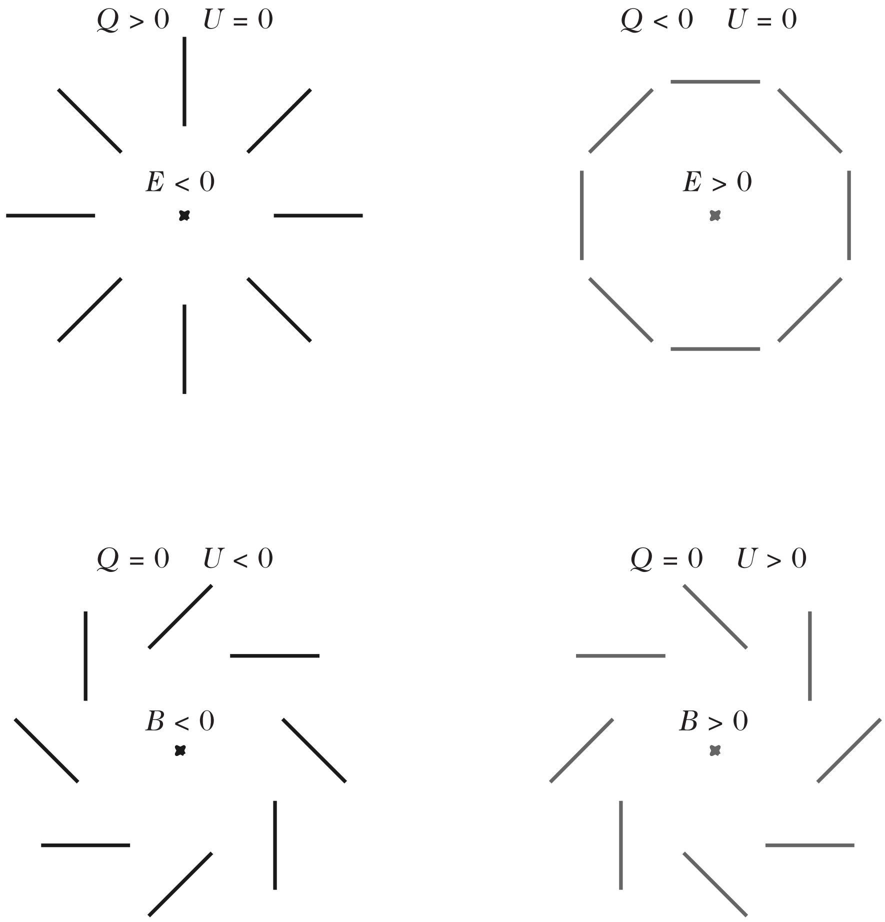


Figure 5.6: Polarization field configurations representing pure E-mode and pure B-mode patterns in the vicinity of a given central point. The line segments indicate the local orientation and amplitude of the linear polarization, directly depicting the Stokes parameters Q and U. These patterns repeat periodically across the plane. Because the scalar E and pseudo-scalar B fields are related to Q and U non-locally, determining their amplitudes at a specific point requires sampling the surrounding polarization texture. Under spatial reflection across a coordinate axis, the two upper configurations (E-modes) are invariant, whereas the two lower configurations (B-modes) are interchanged under reflection, illustrating their opposite parity properties.

Referred to the local vertical axis z, the Stokes parameters in the rotated Cartesian coordinates are related to the celestial basis quantities by $(Q \pm \mathrm{i}U)' = \mathrm{e}^{\mp 2\mathrm{i}\phi}(Q \pm \mathrm{i}U)$. In this Cartesian frame, the Stokes fields take the explicit Fourier representations:

$$Q(\boldsymbol{n}) = \int \frac{\mathrm{d}^2\boldsymbol{\ell}}{2\pi} \left[E(\boldsymbol{\ell}) \cos 2\phi_\ell - B(\boldsymbol{\ell}) \sin 2\phi_\ell\right] \mathrm{e}^{\mathrm{i}\boldsymbol{\ell}\cdot\boldsymbol{n}}, \tag{5.3.104}$$

$$U(\boldsymbol{n}) = \int \frac{\mathrm{d}^2\boldsymbol{\ell}}{2\pi} \left[E(\boldsymbol{\ell}) \sin 2\phi_\ell - B(\boldsymbol{\ell}) \cos 2\phi_\ell\right] \mathrm{e}^{\mathrm{i}\boldsymbol{\ell}\cdot\boldsymbol{n}}. \tag{5.3.105}$$

Using the trigonometric identities connecting the wavevector orientation to its Cartesian components $\boldsymbol{\ell} = (\ell_x, \ell_y)$,

$$\cos 2\phi_\ell = \frac{\ell_x^2 - \ell_y^2}{\ell^2}, \qquad \sin 2\phi_\ell = \frac{2\ell_x\ell_y}{\ell^2}, \tag{5.3.106}$$

one can invert these expressions. In coordinate space, where $\ell_i \to -\mathrm{i}\partial_i$, this yields the spatial relations

$$\boxed{\Delta E = (\partial_x^2 - \partial_y^2)Q + 2\partial_x\partial_y U \qquad \text{and} \qquad \Delta B = (\partial_x^2 - \partial_y^2)U - 2\partial_x\partial_y Q.} \tag{5.3.107}$$

These relations demonstrate that E and B are fundamentally non-local functions of the observable Stokes parameters Q and U. This non-locality is directly inherited from Eq. (5.3.97), reflecting the differential connection between spin-0 and spin-2 spherical harmonics.

Temperature and polarization expansion

The preceding analysis indicates that the radiation temperature perturbation $\Theta(\eta, \boldsymbol{x}, \boldsymbol{n})$ at position $\boldsymbol{x}$ in direction $\boldsymbol{n}$ can be expanded in terms of the scalar basis functions $G_{\ell m}(\boldsymbol{x}, \boldsymbol{n})$ defined by

$$G_{\ell m}(\boldsymbol{x}, \boldsymbol{n}) = (-\mathrm{i})^\ell \sqrt{\frac{4\pi}{2\ell+1}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}} Y_{\ell m}(\boldsymbol{n}), \tag{5.3.108}$$

while the spin-± 2 polarization field is expanded in terms of the spin-weighted basis functions

$${}_{\pm 2}G_{\ell m}(\boldsymbol{x}, \boldsymbol{n}) = (-\mathrm{i})^\ell \sqrt{\frac{4\pi}{2\ell+1}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}} {}_{\pm 2}Y_{\ell m}(\boldsymbol{n}). \tag{5.3.109}$$

Consequently, we decompose the spatial dependence of Θ and $(Q \pm \mathrm{i}U)$ in Fourier plane waves and their directional dependence in spin-weighted spherical harmonics:

$$\Theta(\eta, \boldsymbol{x}, \boldsymbol{n}) = \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^{3/2}} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \Theta_\ell^{(m)}(\eta, \boldsymbol{k}) G_{\ell m}(\boldsymbol{x}, \boldsymbol{n}), \tag{5.3.110}$$

$$(Q \pm \mathrm{i}U)(\eta, \boldsymbol{x}, \boldsymbol{n}) = \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^{3/2}} \sum_{\ell=2}^{\infty} \sum_{m=-\ell}^{\ell} \left[E_\ell^{(m)} \pm \mathrm{i}B_\ell^{(m)}\right](\eta, \boldsymbol{k}) {}_{\pm 2}G_{\ell m}(\boldsymbol{x}, \boldsymbol{n}). \tag{5.3.111}$$

By virtue of the Stewart–Walker lemma, because the background FLRW spacetime is unpolarized, the first-order polarization perturbation variables $E_\ell^{(m)}$ and $B_\ell^{(m)}$ are automatically gauge-invariant.

Scattering cross-section

The differential Thomson scattering cross section for an incident electromagnetic wave with linear polarization vector $\boldsymbol{E}_{\text{in}}$ scattered into an outgoing state with polarization $\boldsymbol{E}_{\text{out}}$ is given by (see Ref. [30])

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{3}{8\pi}\sigma_{\mathrm{T}} |\boldsymbol{E}_{\text{in}}^* \cdot \boldsymbol{E}_{\text{out}}|^2, \tag{5.3.112}$$

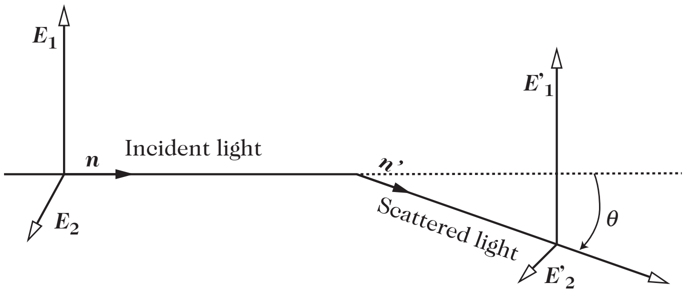


Figure 5.7: Geometry of Thomson scattering in the rest frame of the scattering electron. An incident photon propagating along $\boldsymbol{n}$ is scattered through an angle θ into direction $\boldsymbol{n}'$.

where σ_{T} is the Thomson cross section. It is convenient to introduce the orthogonal partial intensities

$$I_1 \equiv a_1^2 = \frac{I+Q}{2}, \qquad I_2 \equiv a_2^2 = \frac{I-Q}{2}. \tag{5.3.113}$$

For an incoming photon scattered through an angle θ in the plane defined by $(\boldsymbol{n}, \boldsymbol{E}_2)$ (see Fig. 5.7), the outgoing partial intensities are

$$I_1^{\mathrm{out}} = \frac{3\sigma_{\mathrm{T}}}{8\pi} I_1^{\mathrm{in}}, \qquad I_2^{\mathrm{out}} = \frac{3\sigma_{\mathrm{T}}}{8\pi} \cos^2\theta\, I_2^{\mathrm{in}}. \tag{5.3.114}$$

Expressed in terms of the Stokes parameters (I, Q), this scattering relation takes the matrix form

$$\begin{pmatrix} I_{\mathrm{out}} \\ Q_{\mathrm{out}} \end{pmatrix} = \frac{3\sigma_{\mathrm{T}}}{16\pi} \begin{pmatrix} 1+\cos^2\theta & \sin^2\theta \\ \sin^2\theta & 1+\cos^2\theta \end{pmatrix} \begin{pmatrix} I_{\mathrm{in}} \\ Q_{\mathrm{in}} \end{pmatrix}. \tag{5.3.115}$$

Under an arbitrary rotation of the polarization basis, the pair (Q, U) transforms according to Eq. (5.3.80). To evaluate the scattering cross section for an incoming radiation state $(I_{\mathrm{in}}, Q_{\mathrm{in}}, U_{\mathrm{in}})$ along direction $\boldsymbol{n}$ redirected into an outgoing state $(I_{\mathrm{out}}, Q_{\mathrm{out}}, U_{\mathrm{out}})$ along $\boldsymbol{n}'$, we align the coordinates with a fixed reference plane (y, z) (see Fig. 5.8 and Ref. [30]):

1. Rotate about $\boldsymbol{n}$ to bring the scattering plane $(\boldsymbol{n}, \boldsymbol{n}')$ into the plane $(\boldsymbol{n}, \boldsymbol{z})$, corresponding to a rotation (5.3.80) with $\phi = \alpha$.

2. Rotate about the z-axis to bring this plane into coincidence with the reference (y, z) plane; this spatial rotation leaves the Stokes parameters invariant.

3. Apply the scattering transfer matrix of Eq. (5.3.115).

4. Rotate back about the z-axis to restore the scattering plane to $(\boldsymbol{z}, \boldsymbol{n}')$ (again leaving Stokes parameters unaffected).

5. Rotate about the outgoing direction $\boldsymbol{n}'$ through angle $\phi = \alpha'$ to align with the final polarization basis.

Because circular polarization decouples completely and remains zero if initially unexcited, we omit the Stokes parameter V. The preceding transformation sequence yields the linear relation between outgoing and incoming Stokes parameters:

$$\begin{pmatrix} I_1^{\mathrm{out}} \\ I_2^{\mathrm{out}} \\ U^{\mathrm{out}} \end{pmatrix} = \frac{3\sigma_{\mathrm{T}}}{4\pi} \boldsymbol{P} \begin{pmatrix} I_1^{\mathrm{in}} \\ I_2^{\mathrm{in}} \\ U^{\mathrm{in}} \end{pmatrix}. \tag{5.3.116}$$

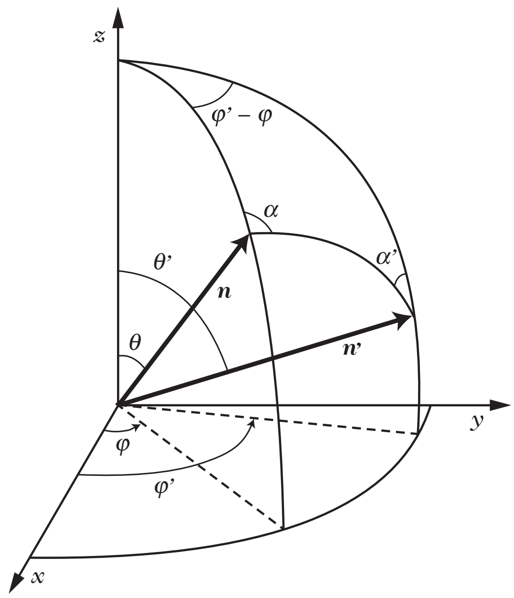


Figure 5.8: Definition of geometric vectors and spherical angles characterizing Thomson scattering between incident direction $\boldsymbol{n}$ and scattered direction $\boldsymbol{n}'$.

Setting $\mu \equiv \cos\theta$ and $\mu' \equiv \cos\theta'$, the scattering matrix $\boldsymbol{P}$ decomposes into three azimuthal harmonic blocks:

$$\boldsymbol{P} = \boldsymbol{P}^{(0)} + \sqrt{(1-\mu^2)(1-\mu'^2)}\boldsymbol{P}^{(1)} + \boldsymbol{P}^{(2)}, \tag{5.3.117}$$

where

$$\boldsymbol{P}^{(0)} = \frac{3}{4}\begin{pmatrix} 2(1-\mu^2)(1-\mu'^2) + \mu^2\mu'^2 & \mu^2 & 0 \\ \mu'^2 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{5.3.118}$$

$$\boldsymbol{P}^{(1)} = \frac{3}{4}\begin{pmatrix} 4\mu\mu'\cos\delta & 0 & 2\mu\sin\delta \\ 0 & 0 & 0 \\ -4\mu'\sin\delta & 0 & 2\cos\delta \end{pmatrix}, \tag{5.3.119}$$

$$\boldsymbol{P}^{(2)} = \frac{3}{4}\begin{pmatrix} \mu^2\mu'^2\cos 2\delta & -\mu^2\cos 2\delta & \mu^2\mu'\sin 2\delta \\ -\mu'^2\cos 2\delta & \cos 2\delta & -\mu'\sin 2\delta \\ -2\mu\mu'^2\sin 2\delta & 2\mu\sin 2\delta & 2\mu\mu'\cos 2\delta \end{pmatrix}, \tag{5.3.120}$$

with the azimuthal difference $\delta \equiv \phi' - \phi$. The standard Stokes parameters (I, Q, U) are recovered via the algebraic transformation

$$\begin{pmatrix} I_1 \\ I_2 \\ U \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}\begin{pmatrix} I \\ Q \\ U \end{pmatrix}. \tag{5.3.121}$$

In the rest frame of the baryon-electron fluid, the collision term governing each polarization state λ is given by

$$C^{(\lambda)} = \sigma_{\mathrm{T}} n_{\mathrm{e}} \left[\int f^{(\lambda')}(E', \boldsymbol{n}') P^{\lambda}{}_{\lambda'}(\boldsymbol{n}, \boldsymbol{n}') \frac{\mathrm{d}^2\boldsymbol{n}'}{4\pi} - f^{(\lambda)}(E', \boldsymbol{n}) \right], \tag{5.3.122}$$

where $P^{\lambda}{}_{\lambda'}(\boldsymbol{n}, \boldsymbol{n}')$ represents the 2×2 submatrix corresponding to (I_1, I_2). This expression provides the full polarized generalization of Eq. (5.3.47).

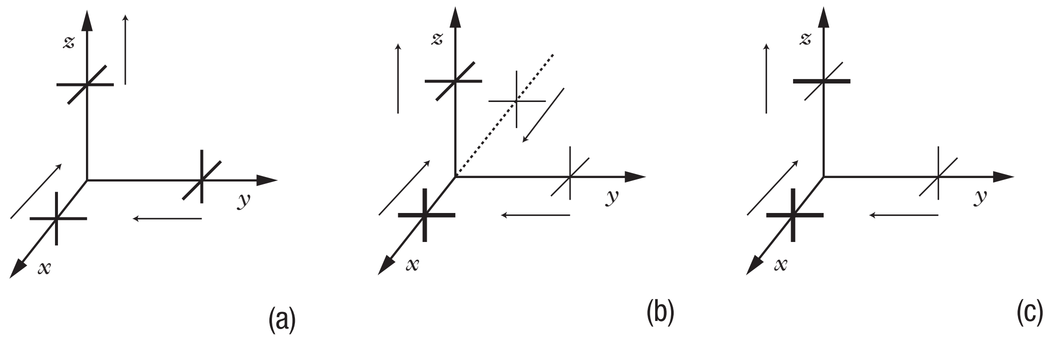


Figure 5.9: Heuristic generation of CMB polarization by Thomson scattering. (a) Under isotropic illumination, the scattered intensities along orthogonal directions balance exactly, producing an unpolarized scattered beam. (b) In the presence of a dipolar temperature anisotropy, opposing hot and cold regions cancel upon angular averaging, again producing zero net linear polarization. (c) A quadrupolar temperature anisotropy illuminates the electron with unequal orthogonal intensities, generating net linear polarization in the scattered radiation.

Expanding the collision integral in the harmonic bases of Eqs. (5.3.108) and (5.3.109), we group the temperature and polarization perturbations into the triplet $\boldsymbol{\mathcal{T}} \equiv (\Theta, Q+\mathrm{i}U, Q-\mathrm{i}U)^T$. One can then establish that the collision operator takes the compact form (see Ref. [31])

$$C[\boldsymbol{\mathcal{T}}] = \tau' \left\{ -\boldsymbol{\mathcal{T}} + \begin{pmatrix} \Theta_0 + n_i V_{\mathrm{b}}^i \\ 0 \\ 0 \end{pmatrix} + \frac{1}{10} \sum_{m=-2}^{2} \int \boldsymbol{\mathcal{P}}^{(m)}(\boldsymbol{n}, \boldsymbol{n}') \boldsymbol{\mathcal{T}}(\boldsymbol{n}') \mathrm{d}^2 \boldsymbol{n}' \right\}, \tag{5.3.123}$$

where the coupling matrix $\boldsymbol{\mathcal{P}}^{(m)}(\boldsymbol{n}, \boldsymbol{n}')$ follows from Eqs. (5.3.117)–(5.3.121) and reads explicitly:

$$\boldsymbol{\mathcal{P}}^{(m)} = \begin{pmatrix} Y_{2m}(\boldsymbol{n}')Y_{2m}(\boldsymbol{n}) & -\sqrt{\frac{3}{2}}\,{}_2Y_{2m}(\boldsymbol{n}')Y_{2m}(\boldsymbol{n}) & -\sqrt{\frac{3}{2}}\,{}_{-2}Y_{2m}(\boldsymbol{n}')Y_{2m}(\boldsymbol{n}) \\ -\sqrt{6}Y_{2m}(\boldsymbol{n}')\,{}_2Y_{2m}(\boldsymbol{n}) & 3\,{}_2Y_{2m}(\boldsymbol{n}')\,{}_2Y_{2m}(\boldsymbol{n}) & 3\,{}_{-2}Y_{2m}(\boldsymbol{n}')\,{}_2Y_{2m}(\boldsymbol{n}) \\ -\sqrt{6}Y_{2m}(\boldsymbol{n}')\,{}_{-2}Y_{2m}(\boldsymbol{n}) & 3\,{}_2Y_{2m}(\boldsymbol{n}')\,{}_{-2}Y_{2m}(\boldsymbol{n}) & 3\,{}_{-2}Y_{2m}(\boldsymbol{n}')\,{}_{-2}Y_{2m}(\boldsymbol{n}) \end{pmatrix}. \tag{5.3.124}$$

When restricted to the unpolarized temperature sector, this general result reduces directly to Eq. (5.3.54).

Origin of the cosmic microwave background polarization

The angular structure of the Thomson scattering cross section provides an intuitive, geometric explanation for the physical origin of CMB polarization (Fig. 5.9).

An electromagnetic wave incident along the x-axis has its electric field polarized in the (y, z) plane. If an unpolarized beam is deflected by 90° into the z-direction, the scattered photon becomes purely linearly polarized along the y-axis. When radiation arrives from all directions with equal intensity (isotropic illumination), statistical isotropy ensures that the y-polarization sourced by waves arriving from the $\pm x$ directions exactly balances the x-polarization sourced by waves arriving from the $\pm y$ directions, resulting in zero net polarization on average (Fig. 5.9a).

If the incoming radiation field exhibits a dipolar temperature anisotropy, the y-directed polarization generated by photons arriving from $+x$ (hot) and $-x$ (cold) averages out to the same mean amplitude as the x-directed polarization from the orthogonal directions. Consequently, a dipole anisotropy also yields an unpolarized scattered beam (Fig. 5.9b).

To produce a net linear polarization in the scattered light, the incident radiation field must possess a non-zero local quadrupole moment (Fig. 5.9c). In this case, orthogonal directions subtend regions of unequal incident flux, so the scattered orthogonal polarizations fail to balance, yielding net linear polarization.

This geometric requirement has immediate cosmological implications:

- Because Thomson scattering can only operate effectively when free electrons are present, CMB polarization is generated almost exclusively during the recombination epoch (and later during reionization).
- Prior to decoupling, tight coupling between photons and baryons enforces near-complete isotropy in the baryon rest frame, keeping the radiation quadrupole very small ($\Theta_2 \ll \Theta_0, \Theta_1$). Consequently, the polarization signal is substantially weaker than the temperature anisotropies (by roughly an order of magnitude).
- For scalar cosmological perturbations, causal velocity gradients only build up quadrupoles inside the Hubble radius. Therefore, scalar-induced polarization is heavily suppressed on super-Hubble scales (at multipoles ℓ below the first acoustic peak). Sub-horizon acoustic oscillations generate polarization peaks that are systematically out of phase with the temperature acoustic peaks.
- In contrast, primordial tensor metric perturbations (gravitational waves) intrinsically produce a quadrupole tidal field even on super-Hubble scales, thereby generating large-angle polarization patterns in both E-modes and B-modes.

Temperature and polarization hierarchy

Aligning the coordinate basis vector $\boldsymbol{e}_3$ with the wavevector $\boldsymbol{k}$, the spatial gradient term along the line of sight becomes

$$n^i \partial_i \to \mathrm{i} n_i k^i = \mathrm{i}\sqrt{\frac{4\pi}{3}} k Y_{10}(\boldsymbol{n}).$$

This directional operator multiplies the angular expansion of the temperature and polarization fields, generating products of spherical harmonics $Y_{10\,s}Y_{\ell m}$. By standard angular momentum coupling, these products expand as

$$\sqrt{\frac{4\pi}{3}} Y_{10\,s}Y_{\ell m} = \frac{{}_s\kappa_\ell^m}{\sqrt{(2\ell+1)(2\ell-1)}}\,{}_sY_{\ell-1,m} - \frac{ms}{\ell(\ell+1)}\,{}_sY_{\ell m} + \frac{{}_s\kappa_{\ell+1}^m}{\sqrt{(2\ell+1)(2\ell+3)}}\,{}_sY_{\ell+1,m}, \tag{5.3.125}$$

where the coupling coefficients are defined by

$${}_s\kappa_\ell^m \equiv \sqrt{\frac{(\ell^2-m^2)(\ell^2-s^2)}{\ell^2}}. \tag{5.3.126}$$

Projecting the linearized Boltzmann equations onto this harmonic basis yields the complete, coupled Boltzmann hierarchy for the multipole moments:

$$\Theta_\ell^{(m)\prime} = k\left[\frac{{}_0\kappa_\ell^m}{2\ell-1}\Theta_{\ell-1}^{(m)} - \frac{{}_0\kappa_{\ell+1}^m}{2\ell+3}\Theta_{\ell+1}^{(m)}\right] - \tau'\Theta_\ell^{(m)} + S_\ell^{(m)}, \tag{5.3.127}$$

$$E_\ell^{(m)\prime} = k\left[\frac{{}_2\kappa_\ell^m}{2\ell-1}E_{\ell-1}^{(m)} - \frac{2m}{\ell(\ell+1)}B_\ell^{(m)} - \frac{{}_2\kappa_{\ell+1}^m}{2\ell+3}E_{\ell+1}^{(m)}\right] - \tau'\left[E_\ell^{(m)} + \sqrt{6}P^{(m)}\delta_{\ell,2}\right], \tag{5.3.128}$$

$$B_\ell^{(m)\prime} = k\left[\frac{{}_2\kappa_\ell^m}{2\ell-1}B_{\ell-1}^{(m)} + \frac{2m}{\ell(\ell+1)}E_\ell^{(m)} - \frac{{}_2\kappa_{\ell+1}^m}{2\ell+3}B_{\ell+1}^{(m)}\right] - \tau' B_\ell^{(m)}. \tag{5.3.129}$$

The gravitational and scattering source terms $S_\ell^{(m)}$ are given by

$$S_0^{(0)} = \tau'\Theta_0^{(0)} + \Psi',$$
$$S_1^{(0)} = -k\tau' V_\mathrm{b} + k\Phi,$$

$$
\begin{aligned}
S_2^{(0)} &= \tau' P^{(0)}, \\
S_1^{(1)} &= -\tau' V_{\mathrm{b}}^{(1)}, \\
S_2^{(1)} &= \tau' P^{(1)} + \frac{k}{\sqrt{3}} \Phi^{(1)}, \\
S_2^{(2)} &= \tau' P^{(2)} - H^{(2)\prime},
\end{aligned}
\tag{5.3.130}
$$

where the vector and tensor metric perturbations are decomposed as

$$
\Phi_i = -\frac{\mathrm{i}}{\sqrt{2}} \sum_{m=\pm 1} e_i^{\pm} \Phi^{(m)}, \qquad E_{ij} = -\sqrt{\frac{3}{8}} \sum_{m=\pm 2} e_i^{\pm} \otimes e_j^{\pm} H^{(m)}. \tag{5.3.131}
$$

The polarization source term $P^{(m)}$, reflecting the anisotropic nature of Compton scattering, is determined by

$$
P^{(m)} = \frac{1}{10} \left[\Theta_2^{(m)} - \sqrt{6} E_2^{(m)} \right], \tag{5.3.132}
$$

which involves solely the quadrupolar moments ($\ell = 2$) of the temperature and of the E-mode polarization.

Significantly, the polarization source $P^{(m)}$ enters only into the evolution equation for the quadrupole of E. The temperature evolution equations should be compared with the unpolarized equations (5.3.66) and (5.3.69) derived previously. Because the system is invariant under azimuthal reflection $m \to -m$, one only needs to solve the hierarchy for positive m. Parity symmetry strictly demands that $B_\ell^{(0)} = 0$ for any scalar perturbation source.

This hierarchy replaces the hydrodynamic fluid equations for the radiation component, while leaving the governing equations for cold dark matter and baryons unaltered. For massless neutrinos, an identical hierarchy applies with $\tau' = 0$ (since neutrinos do not undergo Thomson scattering).

In the tight-coupling regime ($\tau' \gg k$), expanding the scalar evolution equations to leading order yields

$$
E_2^{(0)} = -\sqrt{6} P^{(0)}, \qquad P^{(0)} = \frac{1}{4} \Theta_2^{(0)}, \qquad \sqrt{6} \Theta_2^{(0)} = -4 E_2^{(0)},
$$

which implies the tight-coupling quadrupole relation

$$
\Theta_2^{(0)} = \frac{8}{9} \frac{k}{\tau'} \Theta_1^{(0)}. \tag{5.3.133}
$$

Recalling the shear stress relation of Eq. (5.3.70), we recover

$$
\frac{k^2}{12} \pi_\gamma = \frac{1}{5} \Theta_2^{(0)} = -\frac{8}{45} \frac{k^2}{\tau'} V_\gamma, \tag{5.3.134}
$$

which confirms the shear viscosity relation employed in our earlier analysis of Silk damping in Section 6.2.3.

Power spectra

Given the modal decompositions above, the angular power spectra between two observable fields $X, Z \in \{\Theta, E, B\}$ are given by

$$
(2\ell + 1)^2 C_\ell^{XZ} = \frac{2}{\pi} \int k^2 \, \mathrm{d}k \sum_{m=-2}^{2} X_\ell^{(m)*}(\eta_0, k) Z_\ell^{(m)}(\eta_0, k). \tag{5.3.135}
$$

Figure 5.10 illustrates the four non-vanishing spectra generated by scalar and tensor perturbations. While scalar perturbations generate Θ–Θ, E–E, and Θ–E correlations, they cannot source

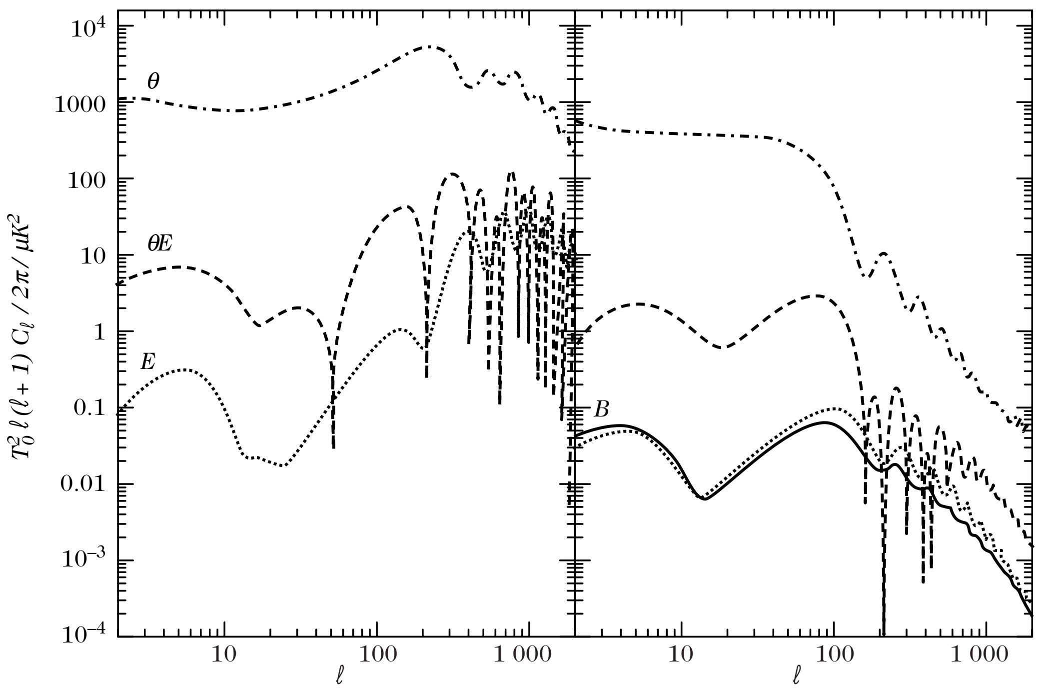


Figure 5.10: The four primary angular power spectra Θ–Θ, Θ–E, E–E, and B–B generated by scalar modes (left panel) and tensor modes (right panel) in an inflationary scenario. Note that the B-mode power spectrum B–B is sourced exclusively by primordial tensor perturbations, while the cross-correlation E–B vanishes identically due to parity conservation.

B-modes. Parity symmetry requires that $C_\ell^{\Theta B} = C_\ell^{EB} = 0$ in the absence of parity-violating physics. Primordial tensor perturbations, however, generate all four non-vanishing power spectra.

In principle, direct numerical integration of the coupled Boltzmann hierarchy can be carried out to obtain these power spectra. Historically, this full-hierarchy approach was the standard methodology, following the evolution of the photon distribution function through the entire cosmic expansion history. However, this brute-force method is computationally demanding: it requires integrating an extensive set of coupled differential equations (typically on the order of $5\ell_{\max}$ equations) for hundreds or thousands of wavenumbers k spanning $k\eta_0 \in [\ell_{\min}/10, 2\ell_{\max}]$. Moreover, careful boundary truncation algorithms are necessary to prevent artificial reflection of power from high to low multipoles.

5.3.4 Numerical integration

Radial modes

A significant computational breakthrough is achieved by separating the radial and angular components of the basis functions ${}_sG_{\ell m}$. Using the Rayleigh expansion for plane waves into spherical harmonics and spherical Bessel functions [Eq. (B.20)],

$$\exp(\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}) = \sum_\ell (-\mathrm{i})^\ell \sqrt{4\pi(2\ell+1)}\, j_\ell(kr) Y_{\ell 0}(\boldsymbol{n}),$$

where $\boldsymbol{e}_3 = \boldsymbol{k}/k$ and $\boldsymbol{x} = -r\boldsymbol{n}$ (with $\boldsymbol{n}$ pointing along the photon propagation direction toward the observer), and applying the recurrence relations for spherical Bessel functions [Eq. (B.46)], we can recast the basis functions as

$$G_{\ell' m} = \sum_\ell (-\mathrm{i})^\ell \sqrt{4\pi(2\ell+1)}\, j_\ell^{(\ell' m)}(kr) Y_{\ell m}(\boldsymbol{n}), \tag{5.3.136}$$

$$ {}_{\pm 2}G_{2m} = \sum_{\ell} (-\mathrm{i})^{\ell} \sqrt{4\pi(2\ell+1)} \left[\epsilon_{\ell}^{(m)}(kr) \pm \mathrm{i}\beta_{\ell}^{(m)}(kr) \right] {}_{\pm 2}Y_{\ell m}(\boldsymbol{n}). \tag{5.3.137} $$

The radial transfer functions $j_{\ell}^{(\ell' m)}(x)$, $\epsilon_{\ell}^{(m)}(x)$, and $\beta_{\ell}^{(m)}(x)$ (with argument $x \equiv kr$) are given explicitly by

$$
\begin{aligned}
j_{\ell}^{(00)} &= j_{\ell}, \\
j_{\ell}^{(10)} &= j_{\ell}', \\
j_{\ell}^{(20)} &= \frac{1}{2}\left(3j_{\ell}'' + j_{\ell}\right), \\
j_{\ell}^{(11)} &= \sqrt{\frac{\ell(\ell+1)}{2}} \frac{j_{\ell}}{x}, \\
j_{\ell}^{(21)} &= \sqrt{\frac{3\ell(\ell+1)}{2}} \left(\frac{j_{\ell}}{x}\right)', \\
j_{\ell}^{(22)} &= \sqrt{\frac{3}{8}\frac{(\ell+2)!}{(\ell-2)!}} \frac{j_{\ell}}{x^2},
\end{aligned} \tag{5.3.138}
$$

alongside the polarization radial functions:

$$ \epsilon_{\ell}^{(0)} = j_{\ell}^{(22)}, \tag{5.3.139} $$

$$ \epsilon_{\ell}^{(1)} = \sqrt{\frac{(\ell-1)(\ell+2)}{2}} \left[\frac{j_{\ell}}{x^2} + \frac{j_{\ell}'}{x}\right], \tag{5.3.140} $$

$$ \epsilon_{\ell}^{(2)} = \frac{1}{4}\left[-j_{\ell} + j_{\ell}'' + 2\frac{j_{\ell}}{x^2} + 4\frac{j_{\ell}'}{x}\right], \tag{5.3.141} $$

$$ \beta_{\ell}^{(0)} = 0, \tag{5.3.142} $$

$$ \beta_{\ell}^{(1)} = \sqrt{\frac{(\ell-1)(\ell+2)}{2}} \frac{j_{\ell}}{x}, \tag{5.3.143} $$

$$ \beta_{\ell}^{(2)} = \frac{1}{2}\left[j_{\ell}' + 2\frac{j_{\ell}}{x}\right]. \tag{5.3.144} $$

Furthermore, these functions satisfy the reflection relations $\epsilon_{\ell}^{(-m)} = \epsilon_{\ell}^{(m)}$ and $\beta_{\ell}^{(-m)} = -\beta_{\ell}^{(m)}$.

Integral solution

The Boltzmann hierarchy of Eqs. (5.3.127)–(5.3.129) admits an exact formal solution in integral form. Projecting onto the radial mode functions of Eqs. (5.3.136) and (5.3.137), the present-day temperature multipoles evaluate to

$$ \Theta_{\ell}^{(m)}(\eta_0, k) = (2\ell+1) \int_0^{\eta_0} \mathrm{d}\eta\, \mathrm{e}^{-\tau} \sum_{\ell'} S_{\ell'}^{(m)}(\eta) j_{\ell}^{(\ell' m)}[k(\eta_0 - \eta)]. \tag{5.3.145} $$

Similarly, the formal solution for the polarization multipoles is given by

$$ \left[E_{\ell}^{(m)} \pm \mathrm{i}B_{\ell}^{(m)}\right](\eta_0, k) = -(2\ell+1)\sqrt{6} \int_0^{\eta_0} \mathrm{d}\eta\, \tau' \mathrm{e}^{-\tau} P^{(m)}(\eta) \left[\epsilon_{\ell}^{(m)} \pm \mathrm{i}\beta_{\ell}^{(m)}\right][k(\eta_0 - \eta)]. \tag{5.3.146} $$

This line-of-sight integration method, developed by Seljak and Zaldarriaga (Refs. [32, 33]), represents a profound generalization of the Sachs–Wolfe formula: it expresses the observed multipoles as line-of-sight integrals over the source functions evaluated throughout cosmic history.

The primary computational advantage of this formulation is that the source terms $S_{\ell}^{(m)}$ and $P^{(m)}$ depend exclusively on moments of order $\ell \leq 2$. As a result, one does not need to solve the full Boltzmann hierarchy up to high $\ell_{\max}$; it is sufficient to solve the coupled differential system truncated at $\ell = 2$ (or conservatively $\ell \sim 8$–10) to determine the source perturbations. The high-ℓ multipoles observed today are then obtained by evaluating the one-dimensional line-of-sight integrals (5.3.145) and (5.3.146), a step that is computationally rapid because the spherical Bessel functions can be precomputed and tabulated on a fine grid.

The structure of Eq. (5.3.145) closely parallels the expressions obtained in Section 5.1.1, which omitted the anisotropic stress of radiation prior to the introduction of Thomson scattering. In particular, the functions $j_\ell^{(00)}$ and $j_\ell^{(10)}$ for scalar modes and $j_\ell^{(11)}$ for vector modes reappear naturally in this harmonic framework. The slight difference in normalization between $j_\ell^{(22)}$ and $\tilde{j}_\ell^{(22)}$ is attributed to the tensor basis conventions noted in Eq. (5.3.46).

Freely available codes

Several publicly available, highly optimized numerical packages implement the line-of-sight integration method to integrate the coupled Einstein–Boltzmann hierarchy for both temperature and polarization:

- `CMBFAST` (Seljak & Zaldarriaga, Ref. [34]), the pioneering code that established the line-of-sight method;
- `CAMB` (Lewis, Challinor & Lasenby, Ref. [35]), based on a covariant approach and written in modern Fortran;
- `CMBEASY` (Doran, Ref. [36]), a C++ object-oriented implementation;
- `COSMICS` (Bertschinger, Ref. [37]), which integrates the complete Boltzmann hierarchy in synchronous or conformal Newtonian gauge.

These packages integrate the perturbation equations within the standard ΛCDM cosmological model, and can be readily extended to encompass dynamical dark energy, non-standard neutrino properties, modified gravity, spatial curvature, and primordial non-Gaussianities.

5.4 Anisotropies of the cosmic microwave background

5.4.1 Observations of the cosmic microwave background anisotropies

We observe the cosmic microwave background on the celestial sphere surrounding our vantage point. Any full-sky temperature anisotropy map $\Theta^{\rm obs}(\boldsymbol{e}) \equiv \Delta T(\boldsymbol{e})/T_0$ measured along line-of-sight directions $\boldsymbol{e}$ can be decomposed in spherical harmonics:

$$\Theta^{\rm obs}(\boldsymbol{e}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m}^{\rm obs} Y_{\ell m}(\boldsymbol{e}). \tag{5.4.1}$$

Under the hypothesis of statistical isotropy, the angular two-point correlation function of the observed temperature field,

$$C^{\rm obs}(\vartheta) \equiv \left\langle \Theta^{\rm obs}(\boldsymbol{e}) \Theta^{\rm obs}(\boldsymbol{e}') \right\rangle_{\boldsymbol{e}\cdot\boldsymbol{e}'=\cos\vartheta}, \tag{5.4.2}$$

depends exclusively on the angular separation $\cos\vartheta = \boldsymbol{e} \cdot \boldsymbol{e}'$ and can be obtained by spatial averaging across the sky. Expanding this correlation function in Legendre polynomials,

$$C^{\rm obs}(\vartheta) = \frac{1}{4\pi} \sum_{\ell=0}^{\infty} (2\ell+1) C_\ell^{\rm obs} P_\ell(\cos\vartheta), \tag{5.4.3}$$

the observed angular power spectrum multipoles are extracted as

$$C_\ell^{\mathrm{obs}} = \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} \left| a_{\ell m}^{\mathrm{obs}} \right|^2 . \tag{5.4.4}$$

Cosmological theoretical models predict only the ensemble statistical properties of primordial perturbations, not the exact fluctuation pattern in our particular Universe. The comparison between theory and observation therefore requires invoking an ergodic hypothesis: we equate the spatial average over the celestial sphere of our unique observable Universe, C_ℓ^{obs}, with the theoretical ensemble average $C_\ell = \langle |a_{\ell m}|^2 \rangle$ evaluated over all possible realizations of the cosmological model. Because our sky provides only a finite statistical sample of $2\ell+1$ independent azimuthal modes for a given multipole ℓ, this comparison is fundamentally limited by cosmic variance.

Estimator of C_ℓ and cosmic variance

In linear cosmological perturbation theory, the temperature fluctuation field is modeled as a homogeneous and isotropic random Gaussian field. The multipole coefficients $a_{\ell m}$ are independent complex random variables of vanishing mean, satisfying

$$\langle a_{\ell m} \rangle = 0, \qquad \langle a_{\ell m} a^*_{\ell' m'} \rangle = \delta_{\ell\ell'} \delta_{mm'} C_\ell .$$

Because the true ensemble power spectrum C_ℓ cannot be measured directly, we construct an unbiased estimator $\widehat{C}_\ell$ by summing over all azimuthal modes m:

$$\widehat{C}_\ell = \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} a_{\ell m} a^*_{\ell m} = \frac{C_\ell}{2\ell+1} V_\ell , \tag{5.4.5}$$

where the normalized variable V_ℓ is defined by

$$V_\ell \equiv \sum_{m=-\ell}^{\ell} \frac{|a_{\ell m}|^2}{C_\ell} .$$

The observed spectrum C_ℓ^{obs} defined in Eq. (5.4.4) is precisely an empirical realization of this estimator $\widehat{C}_\ell$.

In standard single-field slow-roll inflation, the primordial perturbations follow nearly Gaussian statistics, so that each multipole amplitude $a_{\ell m}$ obeys a one-point Gaussian probability density function:

$$\mathcal{P}(a_{\ell m}) = \frac{1}{\sqrt{2\pi C_\ell}} \exp\left(-\frac{a_{\ell m}^2}{2C_\ell} \right) . \tag{5.4.6}$$

Because V_ℓ represents the sum of the squares of $2\ell+1$ independent standard Gaussian random variables, it obeys a chi-square (χ^2) distribution with $2\ell+1$ degrees of freedom:

$$\mathcal{P}_{\chi^2_\ell}(V_\ell) = \frac{V_\ell^{\ell/2-1}}{2^{\ell/2}\Gamma(\ell/2)} \exp\left(-\frac{V_\ell}{2} \right) . \tag{5.4.7}$$

From this result, the probability distribution function of the estimator $\widehat{C}_\ell$ follows directly:

$$\mathcal{P}(\widehat{C}_\ell) = \frac{\ell}{C_\ell} \mathcal{P}_{\chi^2_\ell} \left(\frac{\ell \widehat{C}_\ell}{C_\ell} \right) . \tag{5.4.8}$$

In the high-multipole limit $\ell \to \infty$, the central limit theorem ensures that this distribution approaches a Gaussian, and the estimator converges strictly to the true value:

$$\lim_{\ell\to\infty} \widehat{C}_\ell = C_\ell . \tag{5.4.9}$$

The estimator is therefore consistent. Furthermore, since $\langle \widehat{C}_\ell \rangle = C_\ell$, it is strictly unbiased. The intrinsic variance of the estimator evaluates to

$$\langle \widehat{C}_\ell^2 \rangle - \langle \widehat{C}_\ell \rangle^2 = \frac{2}{2\ell+1} C_\ell^2.$$

By the Cramér–Rao bound, $\widehat{C}_\ell$ is the minimum-variance unbiased estimator of C_ℓ. The relative uncertainty on the power spectrum measurement is fundamentally bounded by the finite number of independent orientations on the sky:

$$\frac{\langle \widehat{C}_\ell^2 \rangle - \langle \widehat{C}_\ell \rangle^2}{C_\ell^2} = \frac{2}{2\ell+1}. \tag{5.4.10}$$

This statistical uncertainty constitutes the cosmic variance. It represents an irreducible physical limitation that dominates the error budget at low multipoles ($\ell \lesssim 10$).

Consequently, no single $a_{\ell m}$ can directly test cosmological models. Scientific inference requires constructing statistical estimators. If primordial fluctuations deviate from Gaussianity (e.g., through non-Gaussian initial conditions or non-linear gravitational evolution), optimal estimator construction becomes significantly more intricate, as two-point correlation functions no longer exhaust the information content of the field.

Status of the observations

Observational cosmology of the CMB has experienced unprecedented progress over recent decades (see Refs. [38–40] for detailed historical surveys).

The foundational breakthrough was achieved in 1992 by the Differential Microwave Radiometer (DMR) instrument aboard NASA's COBE satellite, which detected primordial CMB anisotropies for the first time on large angular scales ($\vartheta \gtrsim 7°$, corresponding to multipoles $\ell \lesssim 20$). As illustrated in Fig. 5.11, observational precision advanced rapidly over the subsequent decade. In 1997, large experimental uncertainties could not decisively distinguish between competing cosmological paradigms. Beginning in 1998, high-altitude balloon-borne experiments—notably BOOMERanG and MAXIMA, followed by Archeops—resolved the first acoustic peak with high significance and provided initial detections of the second peak. By 2003, the Wilkinson Microwave Anisotropy Probe (WMAP) mapped the full sky with sub-degree resolution, measuring the angular power spectrum up to $\ell \sim 1000$.

CMB experimental platforms fall into three primary categories:

1. **Ground-based experiments**: These instruments target small angular scales (high multipoles) using coherent interferometric arrays or focal-plane bolometers (e.g., DASI [41], CAT [42], VLA [43], and CBI). Located at high altitudes or exceptionally dry sites (such as the Atacama plateau or the South Pole), their sensitivity is ultimately limited by atmospheric water vapor fluctuations.

2. **Stratospheric balloon experiments**: Operating at altitudes near 40 km, balloon-borne payloads fly above 99.5% of the Earth's atmosphere, substantially eliminating atmospheric emission and absorption. Flight durations range from overnight flights to circum-polar long-duration ballooning of ten days or more (e.g., BOOMERanG [44], MAXIMA [45], and Archeops [46]).

3. **Space missions**: Space platforms provide full-sky coverage in a thermally stable environment without atmospheric attenuation. Three dedicated space missions have surveyed the CMB: COBE [47], WMAP [48], and Planck [49].

The WMAP satellite mapped the cosmic microwave background in five frequency channels spanning 22 to 90 GHz. Key findings from its multi-year data releases (Ref. [50]) include:

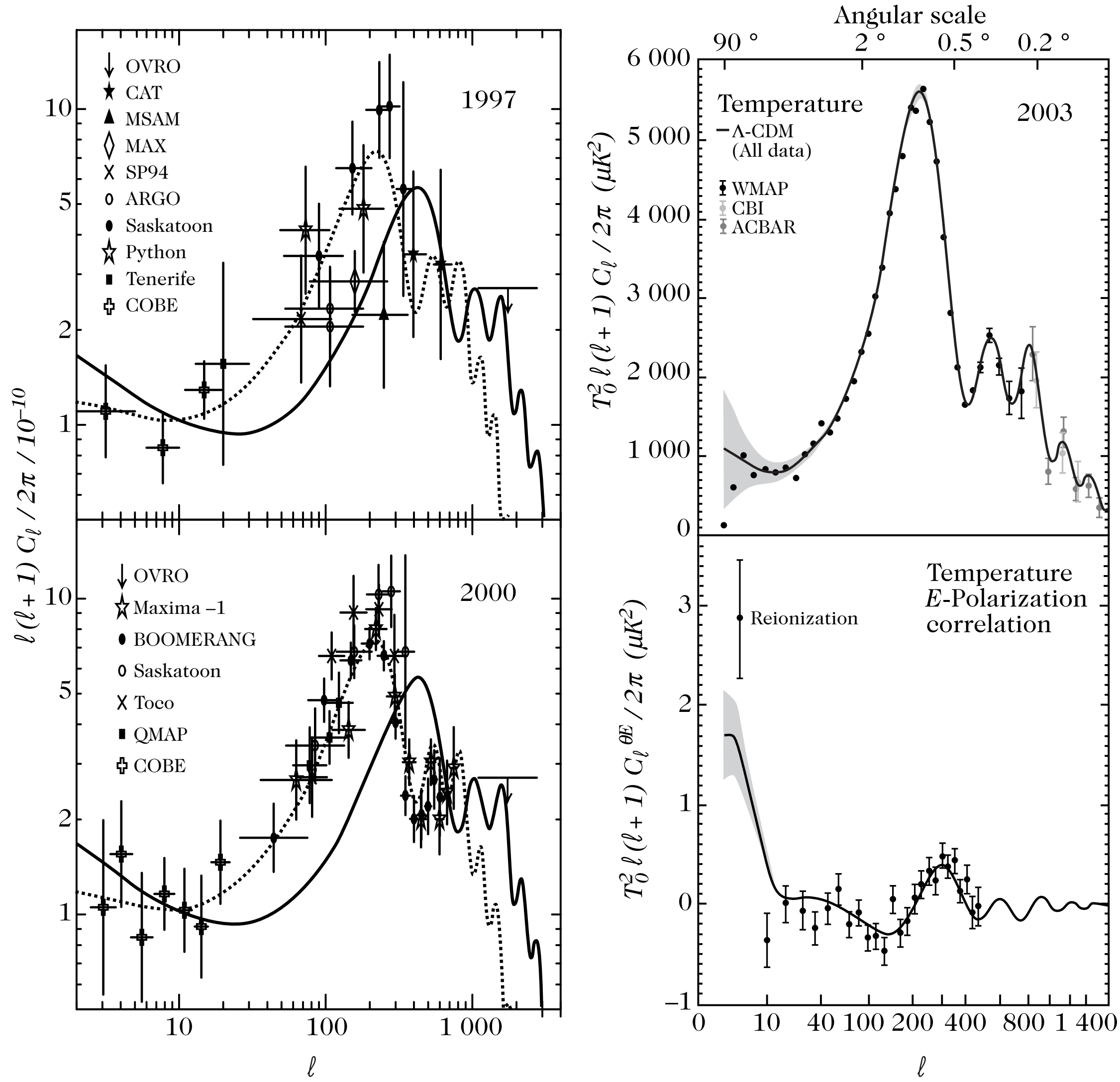


Figure 5.11: Evolution of CMB observational status between 1997 and 2003 (left panel). The solid curve represents the best-fit flat ΛCDM concordance model, while an open model ($\Omega_0 = 0.3$, dotted line) is firmly ruled out by the data. (Right panel): Full-sky angular power spectra for temperature (Θ–Θ) and the temperature–polarization cross-correlation (Θ–E) determined by the WMAP satellite.

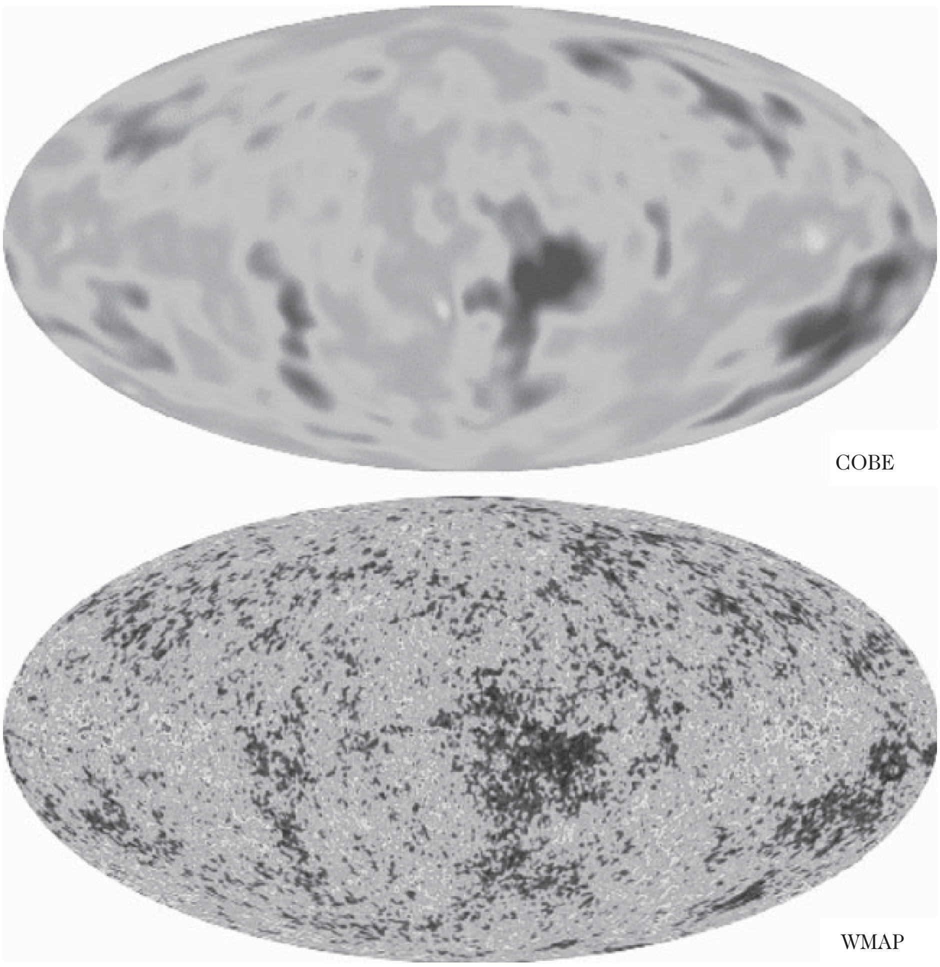


Figure 5.12: Comparison of the angular resolution achieved by the COBE-DMR instrument (top) and the WMAP satellite (bottom) across the microwave sky (WMAP Science Team, 2003).

- Full-sky temperature maps with an angular resolution of 0.3° (Fig. 5.12), measuring the angular power spectrum with high signal-to-noise ratio up to $\ell \sim 1000$.

- High-precision determination of the first acoustic peak position (Ref. [51]):

$$\ell^{(1)} = 220.8 \pm 0.7, \tag{5.4.11}$$

  with an amplitude of $74.7 \pm 0.5\ \mu$K. The second acoustic peak is measured at $\ell^{(2)} = 530.9 \pm 3.8$ with amplitude $48.8 \pm 0.9\ \mu$K. The first and second troughs occur at $\ell^{(-1)} = 412.4 \pm 1.9$ and $\ell^{(-2)} = 675.2 \pm 11.1$, respectively.

- Measurement of the temperature–polarization cross-correlation spectrum $C_\ell^{\Theta E}$ (Fig. 5.11), exhibiting an anti-peak at $\ell = 137 \pm 9$ and a positive peak at $\ell = 329 \pm 19$. The existence of this anti-correlation on angular scales larger than the sound horizon provides definitive confirmation of super-Hubble perturbations at decoupling, ruling out causal post-recombination mechanisms (such as cosmic strings and textures). The relative phase shift between temperature and polarization spectra provides compelling evidence in favor of adiabatic initial conditions.

- An excess in the large-scale Θ–E cross-correlation at $\ell < 20$, indicating an early reionization of the intergalactic medium at redshift $z_{\rm re} = 11.0^{+2.6}_{-2.5}$, with an integrated optical depth $\tau_{\rm re} = 0.089 \pm 0.03$.

- Detection of primordial E-mode polarization:

$$\frac{\ell(\ell+1)}{2\pi} C_\ell^{EE} = (0.086 \pm 0.029)\ \mu\mathrm{K}^2 \qquad (\ell = 2\text{–}6),$$

  interpreted as the Thomson rescattering of CMB photons during reionization.

- Non-detection of primordial tensor B-modes, establishing the upper bound:

$$\frac{\ell(\ell+1)}{2\pi} C_\ell^{BB} = (-0.04 \pm 0.03)\ \mu\mathrm{K}^2 \qquad (\ell = 2\text{–}6).$$

- Strict constraints establishing that the temperature field is consistent with Gaussianity to high precision.

The WMAP observations firmly established the flat ΛCDM concordance model with $\Omega_{\Lambda 0} \sim 0.7$ as the standard framework of modern cosmology. Nevertheless, several subtle anomalies persist at low multipoles:

- A lower quadrupole ($\ell = 2$) amplitude than predicted by the concordance model, first identified by COBE;

- An unexpected planar alignment between the quadrupole and octopole ($\ell = 3$) modes;

- Hemispherical power asymmetries between the northern and southern ecliptic skies.

CMB polarization was first detected experimentally in September 2002 by the DASI interferometer operating at 30 GHz at the South Pole (Ref. [52]), confirming the predicted E-mode polarization at the 5σ confidence level.

The ESA Planck satellite was designed to survey the sky across nine frequency channels (20 GHz to 850 GHz) with angular resolution down to $4'$ ($\ell \sim 2000$), deploying low-frequency radiometers and high-frequency bolometers to map temperature and polarization with cosmic-variance-limited sensitivity.

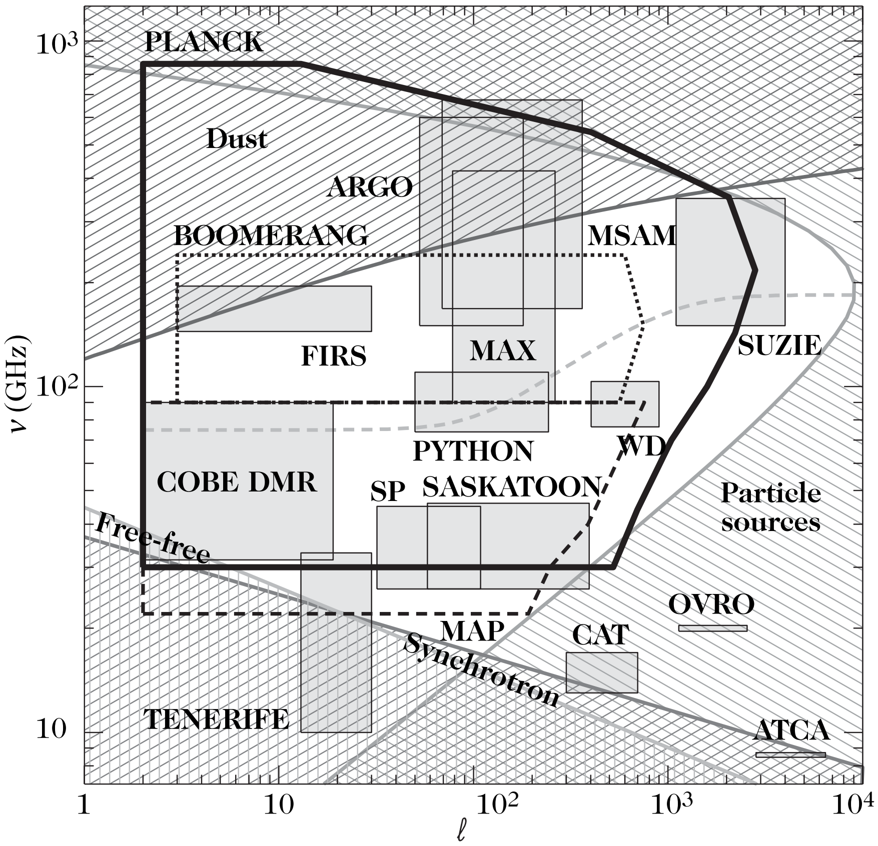


Figure 5.13: Angular and frequency sensitivity ranges for major ground-based, sub-orbital, and space CMB experiments. The shaded regions denote frequency regimes where Galactic foreground emissions (synchrotron, free–free, and thermal dust) exceed 20 μK in the cleanest 20% of the sky. From Ref. [53].

Statistical isotropy and observation on an incomplete sky

Realistic observations cannot measure the primordial CMB across the entire celestial sphere because astrophysical foreground emissions—predominantly Galactic synchrotron, free–free, and thermal dust emission, as well as extragalactic point sources—obscure the cosmological signal. In practice, the Galactic plane and point sources are excised by applying an observational window or mask $W(\boldsymbol{e})$:

$$\widetilde{\Theta}(\boldsymbol{e}) = \Theta(\boldsymbol{e})W(\boldsymbol{e}), \tag{5.4.12}$$

where $W(\boldsymbol{e})$ is a spatial mask equal to zero in masked regions and unity across clean sky areas (or smoothly apodized). Expanding the mask in spherical harmonics,

$$W(\boldsymbol{e}) = \sum_{\ell m} w_{\ell m} Y_{\ell m}(\boldsymbol{e}), \tag{5.4.13}$$

with the reality condition $w^*_{\ell m} = (-1)^m w_{\ell,-m}$, the pseudo-multipole coefficients $\widetilde{a}_{\ell m}$ of the masked map relate to the primordial full-sky coefficients $a_{\ell m}$ via

$$\widetilde{a}_{\ell m} = \sum_{\ell_1 m_1} a_{\ell_1 m_1} \sum_{\ell_2 m_2} w_{\ell_2 m_2} \int \mathrm{d}^2\boldsymbol{e}\, Y_{\ell_1 m_1}(\boldsymbol{e}) Y_{\ell_2 m_2}(\boldsymbol{e}) Y^*_{\ell m}(\boldsymbol{e}). \tag{5.4.14}$$

Using the angular momentum coupling formula for the integral of three spherical harmonics [Eq. (B.23)], this expression evaluates to

$$\widetilde{a}_{\ell m} = \sum_{\ell_1 m_1} a_{\ell_1 m_1} K^{\ell_1 m_1}_{\ell m}, \tag{5.4.15}$$

where the coupling kernel $K^{\ell_1 m_1}_{\ell m}$ is given explicitly in terms of Wigner $3j$-symbols by

$$K^{\ell_1 m_1}_{\ell m} = (-1)^m \sum_{\ell_2 m_2} w_{\ell_2 m_2} \sqrt{\frac{(2\ell_1+1)(2\ell_2+1)(2\ell+1)}{4\pi}} \begin{pmatrix} \ell_1 & \ell_2 & \ell \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_1 & \ell_2 & \ell \\ m_1 & m_2 & -m \end{pmatrix}. \tag{5.4.16}$$

For an unmasked full sky, $W(\boldsymbol{e}) = 1$, corresponding to $w_{\ell m} = \sqrt{4\pi}\,\delta_{\ell 0}\delta_{m0}$, and the kernel collapses to the identity $K^{\ell_1 m_1}_{\ell m} = \delta_{\ell\ell_1}\delta_{mm_1}$. In the presence of a mask, the masked multipoles no longer satisfy the orthogonality condition $\langle \widetilde{a}_{\ell m}\widetilde{a}^*_{\ell' m'}\rangle = \delta_{\ell\ell'}\delta_{mm'}C_\ell$; instead, the covariance matrix becomes

$$\langle \widetilde{a}_{\ell m}\widetilde{a}^*_{\ell' m'}\rangle = \sum_{\ell_1 m_1} C_{\ell_1} K^{\ell_1 m_1}_{\ell m} \left(K^{\ell_1 m_1}_{\ell' m'}\right)^*,$$

reflecting the fact that the cut sky breaks statistical isotropy and couples different multipoles (ℓ, m). Constructing unbiased, minimum-variance pseudo-C_ℓ estimators on partial skies therefore requires inverting the mode-coupling matrix.

5.5 Effects of the parameters on the angular power spectrum

Scalar, vector, and tensor perturbations impart distinct physical signatures onto the temperature anisotropies and polarization fields of the cosmic microwave background. Parity conservation ensures that primordial scalar and tensor perturbations produce only four non-vanishing angular auto- and cross-power spectra: Θ–Θ, E–E, Θ–E, and B–B. The cross-correlations Θ–B and E–B vanish identically by reflection symmetry. Primordial scalar density fluctuations source only the Θ–Θ, E–E, and Θ–E spectra, while tensor gravitational waves contribute to all four observables and provide the unique source for the primordial B–B auto-correlation.

The morphology of these angular power spectra depends sensitively on the cosmological parameters governing the background FLRW dynamics and the perturbation transfer functions, as well as on the primordial initial conditions.

5.5.1 Cosmological parameters

Baryon density: $\Omega_{\mathrm{b}0}h^2$

As established in Section 5.2.2, the principal physical quantity governing acoustic oscillations in the pre-recombination plasma is the baryon-to-photon momentum density ratio $R \equiv 3\rho_{\mathrm{b}}/(4\rho_\gamma)$. At the last-scattering surface, its value evaluates to

$$R_{\mathrm{LSS}} \simeq 27.37\,\Omega_{\mathrm{b}0}h^2.$$

The physical baryon density $\Omega_{\mathrm{b}0}h^2$ exerts four primary effects on the temperature angular power spectrum:

1. **Sound speed and acoustic scale**: Increasing $\Omega_{\mathrm{b}0}$ increases the plasma inertia and reduces the sound speed $c_{\mathrm{s}} = [3(1+R)]^{-1/2}$, decreasing the sound horizon $r_{\mathrm{s}}(\eta_{\mathrm{LSS}})$. This shifts the acoustic peaks toward slightly higher multipoles and compresses peak spacing.

2. **Odd-to-even peak contrast**: Baryons shift the zero-point of acoustic oscillations in the gravitational potential wells. In the adiabatic mode, the oscillation amplitude of odd peaks (compressions) is enhanced by $(1+6R)$ relative to even peaks (rarefactions). A higher baryon density therefore magnifies the first and third peaks while suppressing the second peak.

3. **Acoustic contrast**: The baryon density modulates the relative prominence of the acoustic Doppler velocity term. In the limit of vanishing baryon content ($\Omega_{\mathrm{b}0} \to 0$), the acoustic peaks are heavily smoothed, whereas increasing $\Omega_{\mathrm{b}0}$ sharpens peak contrast.

4. **Silk damping**: Increasing the baryon density increases the free electron density n_{e}, thereby shortening the photon mean free path $\lambda_{\mathrm{mfp}} = (\sigma_{\mathrm{T}} n_{\mathrm{e}})^{-1}$ and shifting Silk diffusion damping toward smaller spatial scales (higher ℓ).

Figure 5.14 demonstrates the systematic deformation of the temperature power spectrum induced by varying the baryon density.

Total matter density: $\Omega_{\mathrm{m}0}h^2$

In the concordance cosmological model, cold dark matter dominates the total matter density ($\Omega_{\mathrm{c}0} \gg \Omega_{\mathrm{b}0}$). Cold dark matter interacts with photons solely through the gravitational potential. The physical matter density $\Omega_{\mathrm{m}0}h^2$ dictates the redshift of matter–radiation equality:

$$1 + z_{\mathrm{eq}} = \frac{\Omega_{\mathrm{m}0}}{\Omega_{\mathrm{r}0}} \simeq 2.4 \times 10^4\,\Omega_{\mathrm{m}0}h^2.$$

Increasing $\Omega_{\mathrm{m}0}h^2$ causes equality to occur earlier (at higher redshift). In radiation-dominated eras, the decay of the Bardeen potentials driving acoustic oscillations induces a substantial early integrated Sachs–Wolfe (eISW) effect, which boosts the amplitude of the first acoustic peak. An earlier equality transition shortens the radiation-dominated era preceding decoupling, thereby attenuating the eISW enhancement and lowering the first peak amplitude relative to the higher multipoles.

Dark-matter–baryon ratio

The baryon-to-dark-matter ratio $\alpha \equiv \Omega_{\mathrm{b}0}/\Omega_{\mathrm{c}0}$ governs the relative source contributions in the perturbed Poisson equation. Because baryonic density perturbations can grow only after decoupling, increasing α diminishes the early gravitational potential well depths established by cold dark matter, lowering the overall amplitude of the acoustic peaks. Furthermore, the acoustic oscillations of the photon–baryon fluid leave a permanent characteristic imprint in the matter power spectrum, known as baryon acoustic oscillations (BAO).

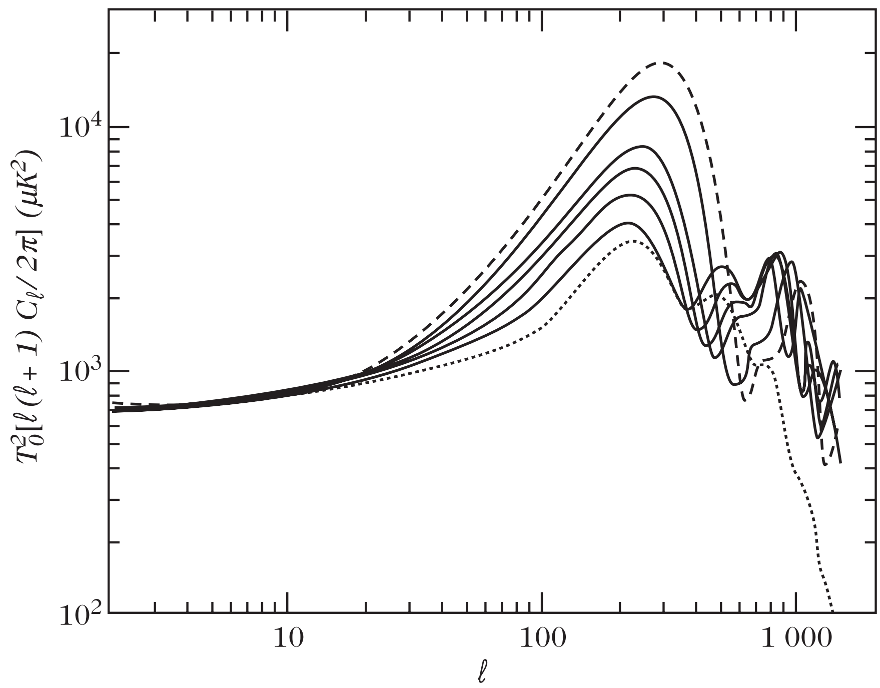


Figure 5.14: Influence of the physical baryon density $\Omega_{b0}h^2$ on the CMB temperature angular power spectrum. Solid curves depict $\Omega_{b0}h^2 = 0.0125, 0.0250, 0.0375, 0.0500$, and 0.0875. The extreme cases correspond to $\Omega_{b0}h^2 = 0.0025$ (dotted line) and 0.1250 (dashed line). From Ref. [3].

Hubble constant: h

The dimensionless Hubble parameter h is constrained by the Friedmann constraint $\Omega_{m0} + \Omega_{\Lambda 0} + \Omega_{K0} = 1$. Its impact on the angular power spectrum depends on which parameters are held fixed during parameter variation. For fixed physical matter density $\omega_m \equiv \Omega_{m0}h^2$, reducing h increases the comoving distance to the last-scattering surface, shifting the entire acoustic peak pattern toward higher multipoles ℓ. Conversely, if Ω_{m0} is fixed while h varies, decreasing h lowers ω_m, delaying matter–radiation equality and enhancing the acoustic peak amplitudes via the early ISW effect (as shown in Fig. 5.15).

Cosmological constant: $\Omega_{\Lambda 0}$

Varying the vacuum energy density $\Omega_{\Lambda 0}$ in a spatially flat universe alters the expansion rate at late times, modifying the comoving angular diameter distance to recombination $D_A(\eta_{LSS})$:

$$D_A = \int_0^{z_{LSS}} \frac{dz}{H(z)}.$$

An increase in $\Omega_{\Lambda 0}$ increases D_A, which shifts the angular projection of the sound horizon toward smaller angular scales (larger multipoles ℓ). Furthermore, dark energy domination at low redshifts causes the gravitational potentials Φ and Ψ to decay at $z \lesssim 1$, producing a late integrated Sachs–Wolfe effect that boosts the temperature power spectrum at low multipoles ($\ell \lesssim 20$).

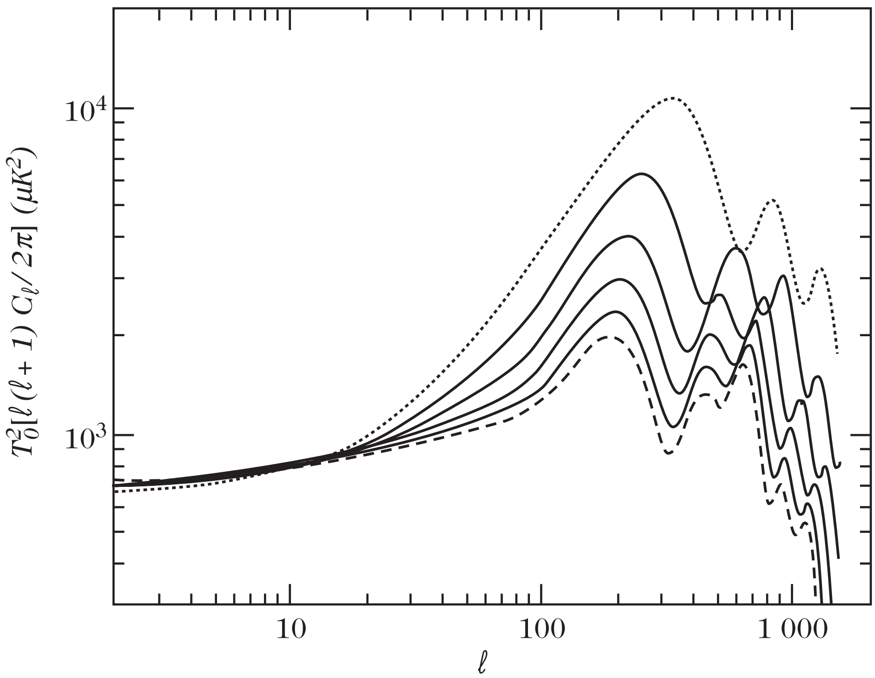


Figure 5.15: Influence of the Hubble constant h on the temperature power spectrum for a fixed baryon fraction $\Omega_{\mathrm{b0}} = 0.0125$. The curves correspond to $h = 0.20$ (dotted line), $0.35, 0.50, 0.65, 0.80$ (solid lines), and 0.95 (dashed line). From Ref. [3].

Curvature: $\Omega_{K0}h^2$

Spatial curvature fundamentally modifies the spatial geometry and the angular diameter distance relation:

$$D_{\mathrm{A}} = \frac{1}{\sqrt{|K|}}\chi_K\left(\sqrt{|K|}\chi\right),$$

where $\chi_K(x) = \sin x$ for closed spaces ($K > 0$) and $\sinh x$ for open spaces ($K < 0$). In an open universe ($\Omega_{K0} > 0$), photon geodesics diverge, reducing the apparent angular size of the sound horizon and shifting the entire acoustic peak spectrum toward substantially higher multipoles ($\ell \propto \Omega_0^{-1/2}$). In a closed universe, geodesics converge, shifting the peaks toward lower multipoles. Additionally, the transition to curvature domination triggers a late decay of the Bardeen potentials, enhancing the low-ℓ power via the late ISW effect.

Notice that we can simultaneously adjust the spatial curvature and the cosmological constant while keeping the resulting angular power spectrum almost unchanged. These compensating effects are illustrated in Fig. 5.17.

Parameter degeneracy

As demonstrated in the preceding figures, varying cosmological parameters simultaneously alters both the positions and relative amplitudes of the acoustic peaks, as well as the shape of the large-scale Sachs–Wolfe plateau. On large angular scales, cosmic variance imposes an unavoidable statistical limitation, allowing substantially different sets of cosmological parameters to produce angular power spectra that are observationally indistinguishable.

These geometric degeneracies arise primarily because the angular locations of the acoustic

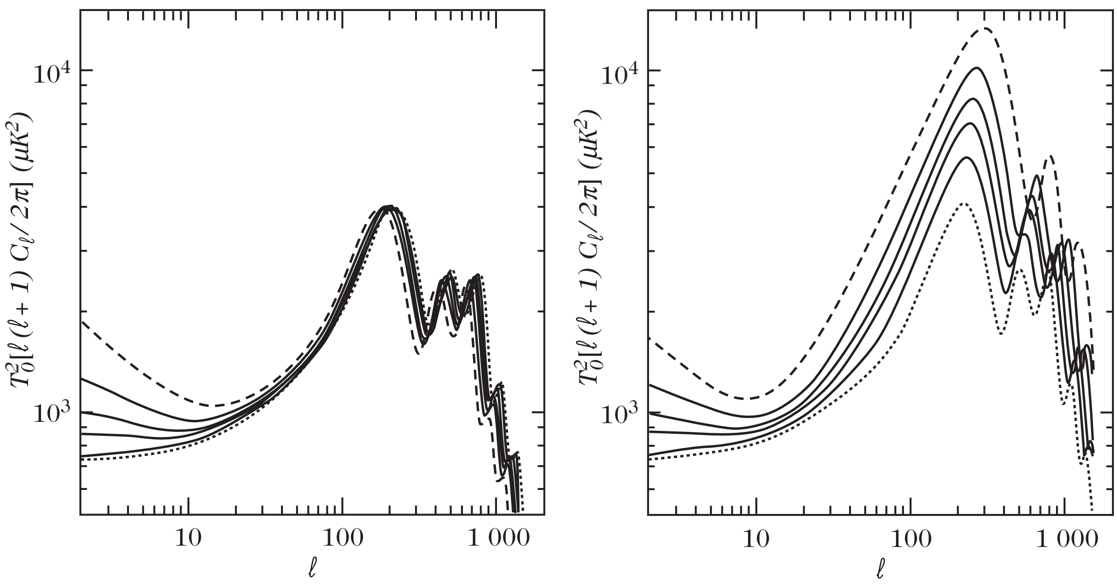


Figure 5.16: Influence of the cosmological constant $\Omega_{\Lambda 0}$ on the temperature angular power spectrum. (Left panel): $\Omega_{\Lambda 0}$ varies while keeping the critical density $\rho_{\mathrm{crit},0}$ and the physical matter density $\bar{\rho}_0$ fixed, which implies that h also varies; $\Omega_{\Lambda 0}$ takes the values 0 (dotted line), $0.4, 0.6, 0.7, 0.8$ (solid lines), and 0.9 (dashed line). (Right panel): $\rho_{\mathrm{b}0}$ and h are held constant so that $\Omega_{\mathrm{m}0}$ varies as $1 - \Omega_{\Lambda 0}$; $\Omega_{\Lambda 0}$ takes the values 0 (dashed line), $0.4, 0.6, 0.7, 0.8$ (solid lines), and 0.9 (dotted line). From Ref. [3].

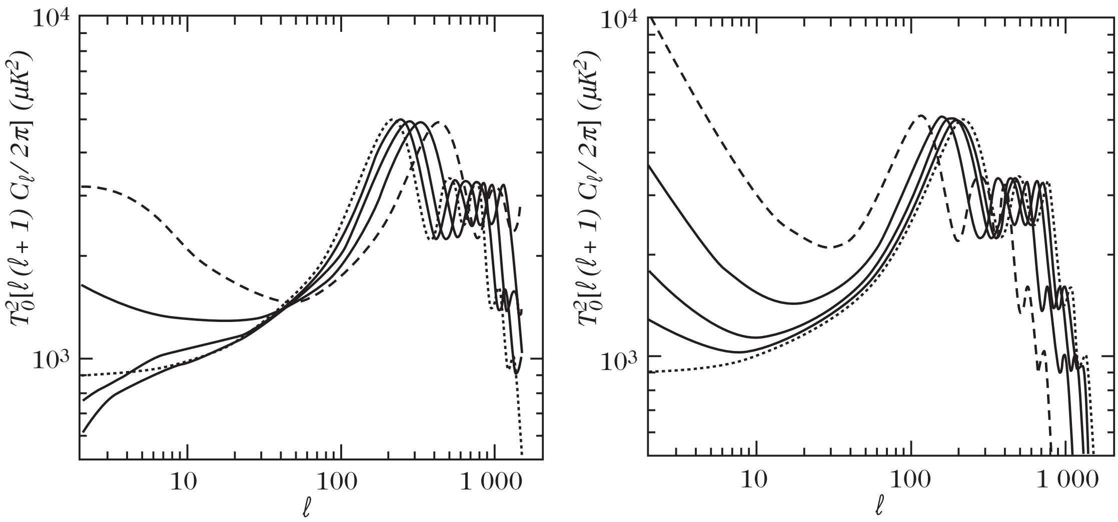


Figure 5.17: Influence of spatial curvature on the temperature angular power spectrum. (Left panel): Spatially hyperbolic (open) universes. The matter density is held constant while Ω_0 takes the values 1 (dotted line), $0.8, 0.6, 0.4$ (solid lines), and 0.2 (dashed line); the acoustic peaks shift toward smaller angular scales (higher ℓ) and potential decay produces a late integrated Sachs–Wolfe effect of opposite sign to the ordinary Sachs–Wolfe term. (Right panel): Spatially spherical (closed) universes. The matter density is held constant while Ω_0 takes the values 1 (dotted line), $1.2, 1.4, 2.0$ (solid lines), and 4.0 (dashed line); the Bardeen potentials grow during the pre-turnaround era, reinforcing the large-scale Sachs–Wolfe plateau. From Ref. [3].

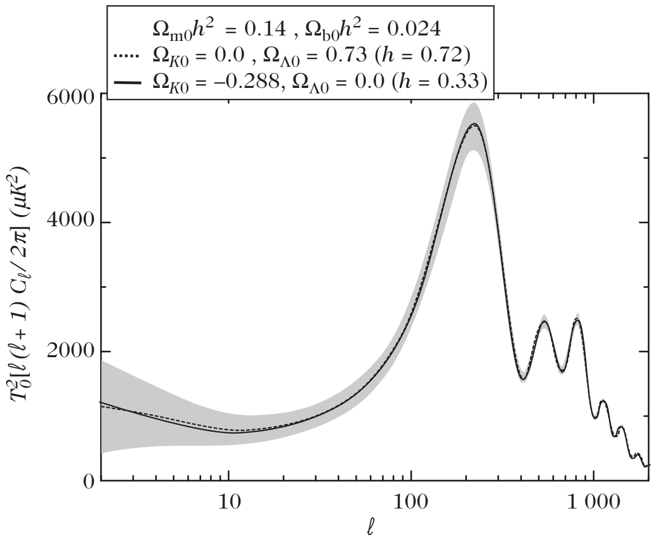


Figure 5.18: Illustration of cosmological parameter degeneracy in the temperature power spectrum. Both cosmological models produce nearly identical angular power spectra within the cosmic variance limit. Both models share identical physical densities $(\Omega_{m0}h^2, \Omega_{b0}h^2) = (0.14, 0.024)$. The first model (dotted line) is the concordance flat ΛCDM model with $(\Omega_{K0}, \Omega_{\Lambda 0}) = (0, 0.73)$ and $h = 0.72$, while the second (solid line) is an open model without dark energy, having $(\Omega_{K0}, \Omega_{\Lambda 0}) = (-0.288, 0)$ and a low Hubble constant $h = 0.33$. From Ref. [3].

peaks are dictated by the acoustic angular scale:

$$\ell_A = \frac{f_K(z_{LSS})}{r_s(z_{LSS})},$$

introduced in Eq. (6.74). Linearizing this angular scale around a flat fiducial concordance model with $(\Omega_0, \Omega_{m0}h^2, \Omega_{\Lambda 0}, w_Q, \Omega_{b0}h^2) = (1, 0.15, 0.65, -1, 0.02)$, where w_Q denotes the dark energy equation of state parameter, the fractional shift of ℓ_A satisfies

$$\frac{\Delta \ell_A}{\ell_A} \simeq -1.1\frac{\Delta\Omega_0}{\Omega_0} - 0.24\frac{\Delta(\Omega_{m0}h^2)}{\Omega_{m0}h^2} - 0.17\frac{\Delta\Omega_{\Lambda 0}}{\Omega_{\Lambda 0}} - 0.11\Delta w_Q - 0.07\frac{\Delta(\Omega_{b0}h^2)}{\Omega_{b0}h^2}.$$

An explicit realization of this cosmological parameter degeneracy is illustrated in Fig. 5.18.

5.5.2 Parameters describing the primordial physics

Primordial physics governs the initial power spectra for both scalar and tensor fluctuations, which are parameterized by their amplitudes, spectral indices, and running of the spectral tilt (Chapter 8). It also determines whether primordial perturbations originate from adiabatic or isocurvature initial conditions.

Spectral index

As established in Eqs. (6.50) and (6.96), the angular power spectrum multipoles on large angular scales scale as $\ell(\ell+1)C_\ell \propto \ell^{n_s-1}$ for scalar modes and $\ell(\ell+1)C_\ell \propto \ell^{n_T}$ for tensor modes.

A red-tilted primordial spectrum ($n_s < 1, n_T < 0$) produces an excess of power at large angular scales (low ℓ), whereas a blue-tilted spectrum ($n_s > 1$) enhances power at small angular scales. A deviation from exact scale invariance ($n_s = 1$) therefore tilts the Sachs–Wolfe plateau and modifies the relative amplitude of the acoustic peaks compared with the low-ℓ plateau. This tilting is illustrated in the left panel of Fig. 5.19.

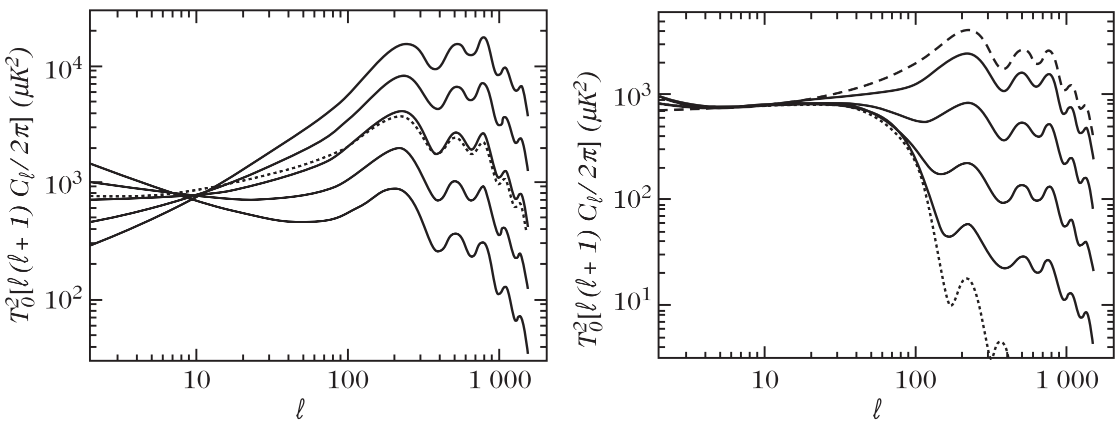


Figure 5.19: (Left panel): Influence of the scalar spectral index n_s on the temperature power spectrum. Curves display $n_s = 0.5, 0.75, 1.0, 1.25$, and 1.5 (solid lines from bottom to top across the acoustic region). The dotted curve denotes $n_s = 1.5$ rescaled by $\ell^{-0.5}$, matching the scale-invariant spectrum $n_s = 1$. (Right panel): Influence of the tensor-to-scalar ratio T/S. The dotted and dashed curves show pure scalar and pure tensor contributions, respectively, with intermediate curves interpolating between $T/S = 0$ and ∞. From Ref. [3].

Primordial gravitational waves

Primordial gravitational waves contribute predominantly to temperature anisotropies at large angular scales ($\ell \lesssim 100$), decaying rapidly once modes enter the horizon during the radiation- or matter-dominated eras. Consequently, tensor perturbations tend to decrease the relative height of the acoustic peaks (which are of purely scalar origin) compared with the Sachs–Wolfe plateau.

When the total power spectrum is normalized to match observational data on large angular scales, a non-zero tensor-to-scalar ratio $r \equiv T/S$ lowers the required scalar amplitude, reducing the overall height of the acoustic peaks (as shown in the right panel of Fig. 5.19).

Adiabatic and isocurvature initial conditions

The choice of initial conditions alters the acoustic peak positions and their relative harmonic spacing. This structural difference stems directly from the oscillation phases described in Eq. (6.74). For isocurvature perturbations, the initial condition is characterized by a vanishing density perturbation at horizon entry, resulting in cosine acoustic oscillations instead of the sine oscillations characteristic of adiabatic modes. Consequently, the position of the first acoustic peak is shifted by a factor of roughly 3/2 toward higher multipoles ($\ell \sim 330$ for isocurvature versus $\ell \sim 220$ for adiabatic perturbations).

A second fundamental distinction appears in the Sachs–Wolfe normalization on super-Hubble scales:

$$\Theta_{\rm SW} = \frac{1}{3}\Phi \quad \text{(adiabatic)}, \qquad \Theta_{\rm SW} = 2\Phi \quad \text{(isocurvature)}.$$

As demonstrated in Fig. 5.20, the observed acoustic peak positions decisively exclude pure isocurvature models, confirming that primordial cosmic perturbations are predominantly adiabatic.

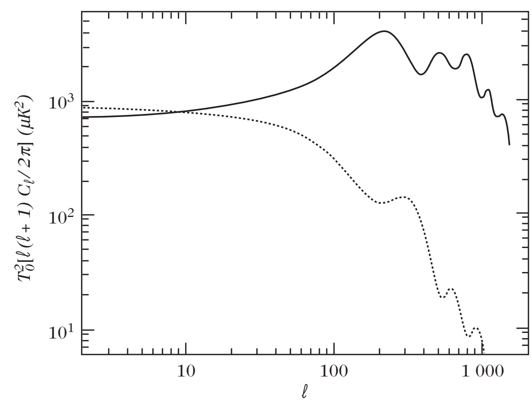


Figure 5.20: Comparison of the temperature angular power spectrum for adiabatic (solid line) and isocurvature (dashed line) initial conditions. The acoustic peaks in the isocurvature spectrum are shifted toward higher multipoles, clearly demonstrating the observational preference for adiabatic perturbations. From Ref. [3].

Chapter 6

Inflation

Inflation is a postulated period of accelerated expansion occurring before the radiation era of the standard Big-Bang model. It was initially proposed as a tentative solution of the problems of the hot Big-Bang model (Chapter 4). In particular, it provides a simple explanation for the homogeneity and flatness of our Universe.

Before the first models of inflation were proposed, precursor works appeared as early as 1965 by Gliner, who postulated a phase of exponential expansion. In 1978, Englert, Brout and Gunzig, in an attempt to resolve the primordial singularity problem and to introduce the particles and the entropy contained in the Universe, proposed a 'fireball' hypothesis, whereby the Universe itself would appear through a quantum effect in a state of negative pressure subject to a phase of exponential expansion. Starobinsky, in 1979, used quantum-gravity ideas to formulate the first semi-realistic rigorous model of an inflation era, although he did not aim to solve the cosmological problems. A simpler model, with transparent physical motivations, was then proposed by Guth in 1981; this model, now called 'old inflation', was the first to use inflation as a means of solving cosmological problems. It was soon followed by Linde's 'new inflation' proposal.

The formulation by Mukhanov and Chibisov of the theory of cosmological perturbations during new inflation, which links the origin of the large-scale structures of the Universe to quantum fluctuations, quickly followed. All these important works were achieved in the early 1980's, which can thus be considered as the 'date of birth' of inflation.

The first models of inflation were actually only incomplete modifications of the Big-Bang theory as they assumed that the Universe was in a state of thermal equilibrium, homogeneous on large enough scales before the period of inflation. This problem was resolved by Linde with the proposition of chaotic inflation. In this model, inflation can start from a Planckian density even if the Universe is not in equilibrium. A new picture of the Universe then appears. The homogeneity and isotropy of our observable Universe would be only local properties, while the Universe is very inhomogeneous on very large scales, with a fractal-like structure.

This chapter describes the first models of inflation, Section 6.1, and explains in general terms how inflation solves the hot Big-Bang problems. We then address, in Section 6.2, the dynamics of single-field inflation and discuss the slow-roll approximation and the chaotic model of inflation. The generation of initial density perturbations during the inflationary epoch is detailed in Section 6.3. To finish, we describe the reheating phase in Section 6.4, which connects inflation to the hot Big-Bang model, the mechanism of eternal inflation, Section 6.5, which can address the question of initial conditions of the Universe, and various possible extensions, Section 6.6.

6.1 Genesis of the paradigm

6.1.1 Original motivations

Guth's argument The flatness problem can be reformulated by noting that so that (4.20) and (4.21) lead to the conclusion that

$$\frac{\Omega_K}{\Omega} = -\frac{3K}{8\pi G_N \rho a^2}, \qquad \left|\frac{\Omega_K}{\Omega}\right| \sim \frac{10^{43}}{S_0^{2/3}}\left(\frac{t}{1\text{s}}\right) \sim \frac{10^{37}}{S_0^{2/3}}\left(\frac{1\text{ GeV}}{T}\right)^2. \tag{6.1.1}$$

where $S_0 = s_0 H_0^{-3}$ is the entropy contained in the Hubble radius today. From (4.140), it is of order $S_0 \sim 10^{87}$. Such a large value of S_0 implies that the Universe should have been very flat in the past. For instance, the curvature parameter must typically be of order $\Omega_K \sim 10^{-16}$ at the time of nucleosynthesis. The flatness problem is therefore related to the fact that the entropy in a comoving volume is conserved. One can then imagine, as proposed by Guth, that it can be resolved if the cosmic expansion is non-adiabatic during a period of time $[t_i, t_f]$ during which

$$S_f = Z^3 S_i, \tag{6.1.2}$$

where Z is a numerical factor to be determined. In Guth's proposal, this entropy production occurs at around $T_{GUT} \sim 10^{17}$ GeV, the characteristic scale of the Grand Unified Theory (GUT) symmetry breaking (see Chapter 9). From (6.1.1), this implies that Z should be of order $Z \sim 10^{28}$ in order for the curvature parameter to be of order unity today, $\Omega_{K0} \sim \mathcal{O}(1)$.

6.1.2 A phase of acceleration

The Friedmann equations (3.25) indicate that $(1-\Omega^{-1})\rho a^2 = 3K/8\pi G_N$ is constant during cosmic evolution. We infer that

$$(1-\Omega_i^{-1})\rho_i a_i^2 = (1-\Omega_f^{-1})\rho_f a_f^2$$

Assuming that the subsequent evolution of the Universe ($t > t_f$) is described by the standard Big-Bang model, and is thus adiabatic, we conclude that

$$(1-\Omega_i^{-1})\rho_i a_i^2 \sim 10^{-56}(1-\Omega_0^{-1})\rho_f a_f^2.$$

The flatness problem can thus be resolved if $\rho_f a_f^2 \gg \rho_i a_i^2$. Moreover, since $H^2 a^2 - 8\pi G_N \rho a^2/3$ is constant in time, we have that

$$\dot a_f > \dot a_i. \tag{6.1.3}$$

So, a necessary condition for inflation is that the expansion of the Universe be accelerated during the period $[t_i, t_f]$, which implies that during this period the strong energy condition must be violated [see (3.26)]:

$$\ddot a(t) > 0 \iff \rho + 3P < 0. \tag{6.1.4}$$

6.1.3 Resolution of the Big-Bang problems

Inflation is therefore defined as a phase of accelerated expansion of the Universe. This property is sufficient to show that the Big-Bang problems can be resolved if this phase lasts sufficiently long. To quantify the length of the inflationary period, we define the quantity

$$N \equiv \ln\left(\frac{a_f}{a_i}\right) \tag{6.1.5}$$

where a_i and a_f are the values of the scale factor at the beginning and at the end of inflation. This number measures the growth in the scale factor during the accelerating phase, expressed in units of the exponential of base 'e', which is the reason why it is usually called the number of 'e-folds'.

Flatness problem The dynamical analysis of the Friedmann-Lemaître equations (Chapter 3) has shown (3.54) that, when $\Omega_\Lambda = 0$,

$$\frac{d\Omega_K}{d\ln a} = (1+3w)\Omega_K(1-\Omega_K).$$

When $-1 < w < -1/3$ the stable fixed point of this dynamical system is on the $\Omega_K = 0$ axis (see Figs. 3.4-3.6). This attractor becomes unstable when w becomes larger than $-1/3$, that is in the matter and radiation eras. So, the flatness problem will be resolved if the attraction toward $\Omega_K = 0$ during the period of inflation is sufficient to compensate its subsequent drift during the hot Big-Bang, i.e. if inflation has lasted sufficiently long.

To give an estimate of the required minimum number of e-folds of inflation, note that, if we assume H to be almost constant during inflation then

$$\left|\frac{\Omega_K(t_f)}{\Omega_K(t_i)}\right| = \left(\frac{a_f}{a_i}\right)^{-2} = e^{-2N}$$

In order to have $|\Omega_K(t_f)| \leq 10^{-60}$ and $\Omega_K(t_i) \sim \mathcal{O}(1)$, we thus need

$$N \geq 70. \tag{6.1.6}$$

Horizon problem The hypothesis (6.1.4) implies that the comoving Hubble radius, $\mathcal{H}^{-1} = (aH)^{-1}$, decreases in time

$$\frac{d}{dt}(aH)^{-1} < 0. \tag{6.1.7}$$

Two points in causal contact at the beginning of inflation can thus be separated by a distance larger than the Hubble radius at the end of inflation (Fig. 8.1). These points are still causally connected, but can seem to be causally disconnected if the inflationary period is omitted. So, inflation allows for the entire observable Universe to emerge out of the same causal region before the onset of inflation.

Figure 6.1: Evolution of the comoving Hubble radius as a function of the scale factor of the Universe assuming a period of inflation (plain line). A given comoving scale (vertical line) exits the Hubble radius during inflation and enters back into the Hubble radius during the hot Big-Bang. All observable scales can thus have been in causal contact at the beginning of inflation. This is not the case without inflation (dashed line) since the comoving Hubble radius is then strictly growing.

For the horizon problem to be solved today, a causal region with scale factor a_i must have at least the size of the current observable Universe, which implies that

$$e^N \sim \frac{a_f}{a_0}\frac{D_H(t_0)}{D_H(t_f)} \sim \frac{T_0}{T_f}\frac{D_H(t_0)}{D_H(t_f)}$$

Assuming that inflation ends at the Grand Unification scale (see Chapter 9), $T_f \sim 10^{16}$ GeV then $D_H(t_f) \sim M_P/T_f^2 \sim 10^{-13}$ GeV^{-1}. Since $T_0 \sim 10^{-4}$ eV and $D_H(t_0) \sim H_0^{-1} \sim 10^{41}$ GeV^{-1}, we infer that

$$N \geq 57 \tag{6.1.8}$$

This value can be smaller if the energy scale at the end of inflation is lower.

Conclusions A primordial period of accelerated expansion allows for the resolution of the hot Big-Bang problems if it lasts sufficiently long. We have estimated that the number of e-folds should typically be at least of the order

$$N \geq 50 - 70, \tag{6.1.9}$$

depending on what the energy scale is at the end of inflation. It can be shown that with such a value, the other problems (monopoles, homogeneity, entropy) are also resolved. It remains to build physical models leading to such an inflationary phase and to understand how it connects with the hot Big-Bang model.

First models of inflation

Every inflationary model relies on one or several scalar fields, called 'inflaton'.

'Old' inflation Models of old inflation rely on a first-order phase-transition mechanism (Fig. 8.2). A scalar field is trapped in a local minimum of its potential, thus imposing a constant energy density, $V(\varphi_i)$, equivalent to the contribution of a cosmological constant. As long as the field remains in this configuration, the evolution of the Universe is exponential and the Universe can be described by a de Sitter space-time. This configuration is metastable and the field can tunnel to its global minimum, $V(\varphi_f) = 0$, hence creating bubbles of true vacuum, which correspond to a non-inflationary Universe (see Fig. 8.2).

Figure 6.2: The model of old inflation (left) is based on a first order phase transition from a local to a true minimum of the potential. The false vacuum is metastable and the field can tunnel to the true vacuum. In the models of new inflation (right) a scalar field slowly relaxes towards its vacuum.

One of the problems with these models arises from the properties of the de Sitter space (Section 8.1.4) that can be considered to be expanding, contracting or static space depending on the choice of coordinates (see below). In the metastable phase, there is a priori no preferred hypersurface and thus no preferred time-like direction to choose a slicing. Without such a direction, the phase transition can occur on any hypersurface. Each bubble of true vacuum will thus have very different physical properties, e.g. spatial curvature, so that the Universe is expected to be very inhomogeneous on large scales. This led to the abandonment of this model.

New inflation In models of new inflation, a scalar field, φ, exits its false vacuum by slowly rolling towards its true vacuum (see Fig. 8.2). This phase can be preceded by a phase of old inflation. The slow-roll phase is essential as it is during this period that density fluctuations, which lead to the currently observed large-scale structures, are generated. This model can only work if the potential has a very flat plateau around $\varphi = 0$, which is artificial. In most versions of this model, the inflaton cannot be in thermal equilibrium with other matter fields. The theory of cosmological phase transitions then does not apply. No satisfying realization of this model has been proposed and it has been progressively abandoned.

Inflation as a de Sitter phase

In the limit of a frozen field, the inflationary phase is described by a de Sitter Universe. This space-time will be useful to understand and illustrate some mechanisms occurring during inflation. In realistic models of inflation, the dynamics of the inflaton induces a deviation of the space-time structure and that of a pure de Sitter space-time. As will also be seen, the limit where the field is completely frozen is usually singular.

Embedding in a five-dimensional space The de Sitter space-time can be represented as a four-dimensional hyperboloid embedded in a five-dimensional Minkowski space,

$$-(z_0)^2 + (z_1)^2 + (z_2)^2 + (z_3)^2 + (z_4)^2 = H^{-2}. \tag{6.1.10}$$

This surface is invariant under five-dimensional Lorentz transformations, so that the de Sitter space is invariant under the 10-parameter symmetry group $O(4,1)$. It is therefore a maximally symmetric four-dimensional space, just as for Minkowski space-time, with curvature

$$R = 12H^2 = \text{const.} \tag{6.1.11}$$

Depending on the choice of the constant-time hypersurfaces, (see Fig. 8.3), the de Sitter space can take the form of a Friedmann-Lemaître space-time with flat, closed or open spatial sections, or a static form similar to the Schwarzschild space-time.

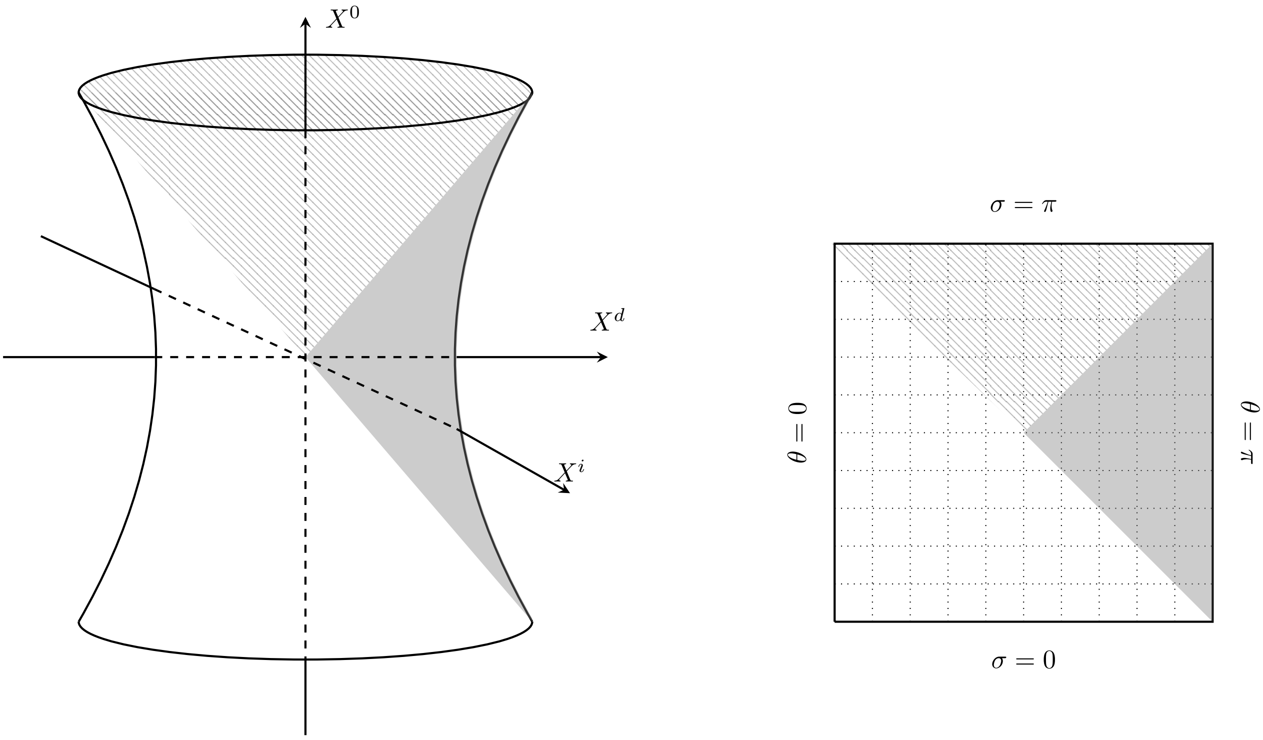


Figure 6.3: Depending on the choice of the constant-time hypersurfaces, the four-dimensional surface (6.1.10) can take various forms. The whole shaded region falls under Euclidean slicing with dark shaded part of it being static slicing.

Euclidean slicing The coordinate choice

$$z_0 = \frac{\sinh Ht}{H} + \frac{H}{2}e^{Ht}x^2, \quad z_4 = \frac{\cosh Ht}{H} - \frac{H}{2}e^{Ht}x^2, \quad z_i = e^{Ht}x_i \tag{6.1.12}$$

leads to the expression for the induced metric

$$ds^2 = -dt^2 + e^{2Ht}\delta_{ij}dx^i dx^j, \tag{6.1.13}$$

with Euclidean spatial sections (see Section 8.4.5 for details on the computation of an induced metric). In conformal time, this space takes the form

$$ds^2 = \frac{1}{(H\eta)^2}(-d\eta^2 + \delta_{ij}dx^i dx^j), \tag{6.1.14}$$

with

$$\eta = -\frac{1}{H}e^{-Ht}. \tag{6.1.15}$$

In this representation, η varies between $-\infty$ and 0^-, with the limit $\eta \to 0^-$ representing 'time-like infinity'. However, it only covers half of the hyperboloid (6.1.10). A static observer in this patch $u^\mu = (1, 0, 0, 0)$ is actually comoving as $a^\mu = \partial_t u^\mu + \Gamma^\mu_{\alpha\beta} u^\alpha u^\beta$ due to $\Gamma^\mu_{tt} = 0$.

Spherical slicing The choice of the system of coordinates (t, χ, θ, ϕ) defined by

$$z_0 = \frac{\sinh Ht}{H}, \quad z_1 = \frac{\cosh Ht}{H}\cos\chi, \quad z_2 = \frac{\cosh Ht}{H}\sin\chi\cos\theta, \quad z_{3\ldots 4} \text{ etc.} \tag{6.1.16}$$

can be used to rewrite the metric in the form of a Friedmann-Lemaître space-time with spherical spatial sections,

$$ds^2 = -dt^2 + \frac{\cosh^2 Ht}{H^2}(d\chi^2 + \sin^2\chi d\Omega^2). \tag{6.1.17}$$

Unlike the Euclidean form (6.1.13), and the hyperbolic version, this slicing covers the entire de Sitter space-time and is thus geodesically complete.

Static slicing To convince ourselves that de Sitter space-time is a static space-time, since it is maximally symmetric, let us derive its static form explicitly. With the choice of coordinates

$$z_0 = \frac{\sinh Ht}{H}\sqrt{1 - r^2H^2}, \quad z_1 = \frac{\cosh Ht}{H}\sqrt{1 - r^2H^2}; \quad z_{2\ldots 4} \text{ etc.} \tag{6.1.18}$$

where $0 \leq rH \leq 1$, the metric takes the Schwarzschild form

$$ds^2 = -(1 - r^2H^2)dt^2 + \frac{dr^2}{(1 - r^2H^2)} + r^2 d\Omega^2. \tag{6.1.19}$$

A static observer ($r =$ const.) in this patch $u^\mu = (\frac{1}{\sqrt{1-r^2H^2}}, 0, 0, 0)$ is actually accelerating as $\Gamma^\mu_{tt} \neq 0$.

6.1.4 Dynamics of single-field inflation

The first models of inflation introduced two ingredients shared by a fair majority of inflation models: the existence of a phase during which the Universe is 'close' to a de Sitter space-time, and a slowly rolling scalar field that rules the expansion of the Universe.

Equations of evolution

The simplest models of inflation are constructed from a single scalar field, φ, evolving in a potential $V(\varphi)$ with an action given by

$$S = -\int \sqrt{-g}\left[\frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi + V(\varphi)\right] d^4x, \tag{6.1.20}$$

so that its energy-momentum tensor takes the form

$$T_{\mu\nu} = \partial_\mu\varphi\partial_\nu\varphi - \left(\frac{1}{2}\partial_\alpha\varphi\partial^\alpha\varphi + V\right) g_{\mu\nu}. \tag{6.1.21}$$

The energy density and pressure of a homogeneous scalar field are, respectively, given by

$$\rho_\varphi = \frac{\varphi'^2}{2a^2} + V(\varphi), \quad P_\varphi = \frac{\varphi'^2}{2a^2} - V(\varphi). \tag{6.1.22}$$

in conformal time, or by

$$\rho_\varphi = \frac{\dot\varphi^2}{2} + V(\varphi), \quad P_\varphi = \frac{\dot\varphi^2}{2} - V(\varphi), \tag{6.1.23}$$

in cosmic time. These expressions show that $\rho_\varphi + 3P_\varphi = 2(\dot\varphi^2 - V)$ and $\dot\varphi^2 = (1+w_\varphi)\rho_\varphi$. Since the Friedmann equations imply $\ddot a/a = -4\pi G_N(\rho+3P)/3$ (see Chapter 3), the Universe can enter a phase of accelerated expansion as soon as $\dot\varphi^2 < V$. The expansion will be quasi-exponential if the scalar field is in a slow-roll regime, i.e. if $\dot\varphi^2 \ll V$.

Equations in cosmic time The Friedmann and Klein-Gordon equation in cosmic time take the form:

$$H^2 = \frac{8\pi G_N}{3}\left(\frac{1}{2}\dot\varphi^2 + V\right) - \frac{K}{a^2}, \tag{6.1.24}$$

$$\frac{\ddot a}{a} = \frac{8\pi G_N}{3}(V - \dot\varphi^2), \tag{6.1.25}$$

$$\ddot\varphi + 3H\dot\varphi + V_{,\varphi} = 0. \tag{6.1.26}$$

Using $\dot H = \ddot a/a - H^2$, we obtain the relation

$$\dot H = -4\pi G_N \dot\varphi^2 + \frac{K}{a^2}, \tag{6.1.27}$$

which will be useful in what follows. As usual, only two of these equations are independent. More importantly, from (6.1.24) for $K = 0$, we have

$$\dot\varphi^2 = 3(1+w_\varphi)H^2 M_{\rm pl}^2. \tag{6.1.28}$$

Thus the background dynamics controls the Effective EoS.

Equations in conformal time These equations can be translated into conformal time. They then take the form

$$\mathcal{H}^2 = \frac{8\pi G_N}{3}\left(\frac{1}{2}\varphi'^2 + Va^2\right) - Ka^2, \tag{6.1.29}$$

$$\mathcal{H}' = -\frac{4\pi G_N}{3}(\varphi'^2 - Va^2), \tag{6.1.30}$$

for the Friedmann equations, and

$$\varphi'' + 2\mathcal{H}\varphi' + a^2 V_{,\varphi} = 0, \tag{6.1.31}$$

for the Klein-Gordon equation.

Hamilton-Jacobi equations As long as the evolution of the scalar field is monotonic, it can be useful to rewrite (6.1.24)-(6.1.26) using φ, instead of t or η, as a time variable. Assuming that $K = 0$, (6.1.27) takes the form

$$H_{,\varphi} \equiv \frac{dH(\varphi)}{d\varphi} = -4\pi G_N \dot\varphi, \tag{6.1.32}$$

which defines the notation $H_{,\varphi}$, so that (6.1.24) becomes

$$[H_{,\varphi}(\varphi)]^2 - 12\pi G_N H^2(\varphi) = -32\pi^2 G_N^2 V(\varphi). \tag{6.1.33}$$

Both (6.1.32) and (6.1.33) are strictly equivalent to (6.1.24)-(6.1.27).

Proof. Consider a system with configuration variable q and Hamiltonian $H(q, p, t)$. The Hamilton-Jacobi equation seeks a *principal function* $S(q, P, t)$ satisfying:

$$H\left(q, \frac{\partial S}{\partial q}, t\right) + \frac{\partial S}{\partial t} = 0 \tag{1}$$

The momenta are given by:

$$p = \frac{\partial S}{\partial q} \tag{2}$$

For time-independent Hamiltonians ($\partial H/\partial t = 0$), we can separate time:

$$S(q, P, t) = W(q, P) - Et \tag{3}$$

where E is constant energy, and W satisfies:

$$H\left(q, \frac{\partial W}{\partial q}\right) = E \tag{4}$$

We now apply this idea to the homogeneous scalar field $\phi(t)$ in a flat FRW universe with metric:

$$ds^2 = -N^2(t)dt^2 + a^2(t)d\vec{x}^2 \tag{5}$$

The total action in this case can be given as follows:

$$S = \frac{1}{16\pi G_N}\int \sqrt{-g}\, R\, d^4x - \int \sqrt{-g}\left[\frac{1}{2}\partial_\mu\phi\partial^\mu\phi + V(\phi)\right] d^4x \tag{6}$$

Under the parametrization of FRW metric, one can write:

$$S = \int dt \left[-\frac{3}{8\pi G_N}\frac{a\dot{a}^2}{N} + \frac{a^3\dot{\phi}^2}{2N} - Na^3V(\phi)\right] \tag{7}$$

Choosing cosmic time gauge $N = 1$:

$$S = \int dt \left[-\frac{3}{8\pi G_N}a\dot{a}^2 + \frac{1}{2}a^3\dot{\phi}^2 - a^3V(\phi)\right] \tag{8}$$

The conjugate momenta of the system with canonical variable $q^1 = a(t)$ and $q^2 = \phi(t, x)$ can be deduced as:

$$\pi_a = \frac{\partial L}{\partial \dot{a}} = -\frac{3}{4\pi G_N}a\dot{a} \tag{9}$$

$$\pi_\phi = \frac{\partial L}{\partial \dot{\phi}} = a^3\dot{\phi} \tag{10}$$

This allows us to write the Hamiltonian:

$$H_{\text{total}} = \pi_a\dot{a} + \pi_\phi\dot{\phi} - L = -\frac{2\pi G_N}{3}\frac{\pi_a^2}{a} + \frac{\pi_\phi^2}{2a^3} + a^3V(\phi) \tag{11}$$

In GR, this is a constrained system. The Hamiltonian constraint (from variation with respect to N) gives:

$$\mathcal{H} \equiv -\frac{2\pi G_N}{3}\frac{\pi_a^2}{a} + \frac{\pi_\phi^2}{2a^3} + a^3V(\phi) = 0 \tag{12}$$

This is equivalent to the Friedmann equation. We seek a principal function $S(a, \phi)$ such that:

$$\pi_a = \frac{\partial S}{\partial a}, \quad \pi_\phi = \frac{\partial S}{\partial \phi} \tag{13}$$

The Hamilton-Jacobi equation becomes the restatement of Hamiltonian constraint as $\mathcal{H}(a, \phi, \partial S/\partial a, \partial S/\partial \phi) = 0$:

$$-\frac{2\pi G_N}{3a}\left(\frac{\partial S}{\partial a}\right)^2 + \frac{1}{2a^3}\left(\frac{\partial S}{\partial \phi}\right)^2 + a^3 V(\phi) = 0 \tag{14}$$

$$-\frac{2\pi G_N}{3}a^2\left(\frac{\partial S}{\partial a}\right)^2 + \frac{1}{2}\left(\frac{\partial S}{\partial \phi}\right)^2 + a^6 V(\phi) = 0 \tag{15}$$

Since the last term contains the product of a and ϕ. We can use the following as separation ansatz:

$$S(a, \phi) = -\frac{1}{4\pi G_N}a^3 H(\phi) \tag{16}$$

where $H(\phi)$ is the Hubble parameter as a function of ϕ. This gives us

$$\frac{\partial S}{\partial a} = -\frac{3}{4\pi G_N}a^2 H(\phi) \tag{17}$$

$$\frac{\partial S}{\partial \phi} = -\frac{1}{4\pi G_N}a^3 H_{,\phi}(\phi) \tag{18}$$

Putting this back in (15), the first and second term can be simplified to:

$$-\frac{2\pi G_N}{3}a^2\left(-\frac{3}{4\pi G_N}a^2 H\right)^2 = -\frac{2\pi G_N}{3}a^2 \cdot \frac{9}{16\pi^2 G_N^2}a^4 H^2 = -\frac{3}{8\pi G_N}a^6 H^2 \tag{19}$$

$$\frac{1}{2}\left(-\frac{1}{4\pi G_N}a^3 H_{,\phi}\right)^2 = \frac{1}{2}\cdot\frac{1}{16\pi^2 G_N^2}a^6 (H_{,\phi})^2 = \frac{1}{32\pi^2 G_N^2}a^6 (H_{,\phi})^2 \tag{20}$$

and hence the equation (15) becomes:

$$a^6\left[-\frac{3}{8\pi G_N}H^2 + \frac{1}{32\pi^2 G_N^2}(H_{,\phi})^2 + V\right] = 0 \tag{21}$$

Which could be rewritten as:

$$-12\pi G_N H^2 + (H_{,\phi})^2 + 32\pi^2 G_N^2 V = 0 \tag{22}$$

or

$$\boxed{(H_{,\phi})^2 - 12\pi G_N H^2 = -32\pi^2 G_N^2 V(\phi)} \tag{23}$$

From $\pi_\phi = \partial S/\partial \phi$ and the ansatz (16):

$$\pi_\phi = -\frac{1}{4\pi G_N}a^3 H_{,\phi} \tag{24}$$

But from the definition: $\pi_\phi = a^3\dot{\phi}$ (with $N = 1$), so:

$$a^3\dot{\phi} = -\frac{1}{4\pi G_N}a^3 H_{,\phi} \implies \boxed{\dot{\phi} = -\frac{1}{4\pi G_N}H_{,\phi}} \tag{25}$$

From (25), we have:

$$\dot{\phi}^2 = \frac{1}{16\pi^2 G_N^2}(H_{,\phi})^2 \tag{26}$$

The Friedmann equation $H^2 = \frac{8\pi G_N}{3}\left(\frac{1}{2}\dot{\phi}^2 + V\right)$ becomes:

$$H^2 = \frac{8\pi G_N}{3}\left[\frac{1}{2}\cdot\frac{1}{16\pi^2 G_N^2}(H_{,\phi})^2 + V\right] \tag{27}$$

$$3H^2 = 8\pi G_N\left[\frac{1}{32\pi^2 G_N^2}(H_{,\phi})^2 + V\right] \tag{28}$$

$$(H_{,\phi})^2 - 12\pi G_N H^2 = -32\pi^2 G_N^2 V \tag{29}$$

which is exactly (23). The second Friedmann equation $\dot{H} = -4\pi G_N\dot{\phi}^2$ follows from differentiating $H(\phi(t))$:

$$\dot{H} = H_{,\phi}\dot{\phi} = H_{,\phi}\left(-\frac{1}{4\pi G_N}H_{,\phi}\right) = -\frac{1}{4\pi G_N}(H_{,\phi})^2 = -4\pi G_N\dot{\phi}^2 \tag{30}$$

using $\dot{\phi}^2 = \frac{1}{16\pi^2 G_N^2}(H_{,\phi})^2$. Here $S(a,\phi) = -\frac{1}{4\pi G_N}a^3 H(\phi)$ is the principal Generating function. S has no explicit time dependence because the system is autonomous (Hamiltonian constraint $\mathcal{H} = 0$ replaces $H + \partial S/\partial t = 0$). In the HJ formalism, slow-roll parameters become geometric:

$$\epsilon_H \equiv -\frac{\dot{H}}{H^2} = \frac{1}{16\pi G_N}\left(\frac{H_{,\phi}}{H}\right)^2 \tag{31}$$

$$\eta_H \equiv -\frac{\ddot{\phi}}{H\dot{\phi}} = \frac{1}{8\pi G_N}\frac{H_{,\phi\phi}}{H} \tag{32}$$

Slow-roll inflation occurs when $\epsilon_H \ll 1$ and $|\eta_H| \ll 1$. □

Slow-roll parameters

The slow-roll formalism is useful in reformulating most models of inflation that do not rely on first-order phase transitions (such as old inflation) in terms of the so-called slow-roll parameters. It makes it possible to compute the predictions of inflation without requiring a specification of the exact form of the inflaton's potential. We assume that $K = 0$ since the curvature term decreases as a^{-2} and becomes negligible in comparison with the contribution from the scalar field. This first implies that $\dot{H} < 0$ [see (6.1.27)], and thus H is always decreasing, and second that $|\dot{H}| < H^2$ if $\ddot{a} > 0$.

Principle During inflation, the scalar field must be in a slow-roll regime. This implies that the scalar field varies little during the inflationary phase and thus satisfies

$$\dot{\varphi}^2 \ll V, \quad |\ddot{\varphi}| \ll 3H|\dot{\varphi}|. \tag{6.1.34}$$

Under these conditions, the evolution equations (6.1.24) and (6.1.26) reduce to

$$H^2 \simeq \frac{8\pi G_N}{3}V, \quad 3H\dot{\varphi} \simeq -V_{,\varphi}, \tag{6.1.35}$$

which implies

$$\frac{|\dot{H}|}{H^2} \ll \frac{3}{2}. \tag{6.1.36}$$

Moreover, since

$$\ddot{\varphi} \simeq -\frac{d}{dt}\left(\frac{V_{,\varphi}}{3H}\right) \simeq -\frac{V_{,\varphi\varphi}\dot{\varphi}}{3H}$$

we infer that the two conditions (6.1.34) imply

$$\left(\frac{V_{,\varphi}}{V}\right)^2 \ll 24\pi G_N, \qquad \frac{|V_{,\varphi\varphi}|}{V} \ll 24\pi G_N. \tag{6.1.37}$$

So, in order for inflation to occur, the potential should be very flat.

Definitions To formalize the previous argument, let us introduce the slow-roll parameters via the relations

$$\epsilon = -\frac{\dot{H}}{H^2}, \tag{6.1.38}$$

$$\delta = \epsilon - \frac{\dot{\epsilon}}{2H\epsilon}, \tag{6.1.39}$$

$$\xi = \frac{\dot{\epsilon} - \dot{\delta}}{H}. \tag{6.1.40}$$

Using the equations of motion we find that the slow-roll parameters can be rewritten as

$$\epsilon = \frac{3}{2}\dot{\varphi}^2 \left[\frac{1}{2}\dot{\varphi}^2 + V(\varphi)\right]^{-1}, \tag{6.1.41}$$

$$\delta = -\frac{\ddot{\varphi}}{H\dot{\varphi}}. \tag{6.1.42}$$

The Friedmann equations imply that the parameter ϵ can be expressed as

$$\epsilon = \frac{1}{4\pi G_N}\left[\frac{H'(\varphi)}{H(\varphi)}\right]^2. \tag{6.1.43}$$

It can be used to rewrite the two Friedmann equations as

$$H^2\left(1 - \frac{1}{3}\epsilon\right) = \frac{8\pi G_N}{3}V, \qquad \frac{\ddot{a}}{a} = H^2(1-\epsilon), \tag{6.1.44}$$

and the effective equation of state of the inflaton as

$$w_\varphi = -1 + \frac{2}{3}\epsilon. \tag{6.1.45}$$

The condition for inflation therefore reduces to

$$\ddot{a} > 0 \iff w < -1/3 \iff \epsilon < 1. \tag{6.1.46}$$

The three parameters (6.1.38)-(6.1.40) are related via the equations of motion as

$$\frac{\dot{\epsilon}}{H} = 2\epsilon(\epsilon - \delta), \qquad \frac{\dot{\delta}}{H} = 2\epsilon(\epsilon - \delta) - \xi. \tag{6.1.47}$$

The slow-roll conditions are thus satisfied as long as ϵ and δ are small compared to 1.

The parameter δ also takes the form

$$\delta = \frac{1}{4\pi G_N}\frac{H_{,\varphi\varphi}}{H}. \tag{6.1.48}$$

It is also useful to express these parameters in terms of conformal time as

$$\epsilon = 1 - \frac{\mathcal{H}'}{\mathcal{H}^2}, \qquad \delta = 1 - \frac{\varphi''}{\mathcal{H}\varphi'} = \epsilon - \frac{\epsilon'}{2\mathcal{H}\epsilon}. \tag{6.1.49}$$

Expansion of the potential and of the expansion rate Two sets of slow-roll parameters can be introduced, based either on the potential V:

$$\epsilon_V = \frac{1}{16\pi G_N}\left(\frac{V_{,\varphi}}{V}\right)^2, \quad \eta_V = \frac{1}{8\pi G_N}\left(\frac{V_{,\varphi\varphi}}{V}\right), \quad \xi_V = \frac{1}{8\pi G_N}\sqrt{\frac{V_{,\varphi}V_{,\varphi\varphi\varphi}}{V^2}}, \tag{6.1.50}$$

or on the expansion rate H,

$$\epsilon_H = \frac{1}{4\pi G_N}\left(\frac{H_{,\varphi}}{H}\right)^2, \quad \delta_H = \frac{1}{4\pi G_N}\left(\frac{H_{,\varphi\varphi}}{H}\right), \quad \xi_H = \frac{1}{4\pi G_N}\sqrt{\frac{H_{,\varphi}H_{,\varphi\varphi\varphi}}{H^2}}. \tag{6.1.51}$$

One can show, using the Jacobi equation (6.1.33), that

$$\epsilon = \epsilon_V = \epsilon_H, \quad \delta = \delta_H = \eta_V - \epsilon_V. \tag{6.1.52}$$

Duration of inflation and number of e-folds The expansion between an arbitrary time t and the end of inflation can be related to N:

$$N(t, t_f) = \int_t^{t_f} H dt, \tag{6.1.53}$$

$$a(t) = a_f e^{-N}. \tag{6.1.54}$$

One can express N in terms of the inflaton as

$$N(\varphi) \equiv N(\varphi, \varphi_f) = \int_{\varphi}^{\varphi_f} \frac{H}{\dot{\varphi}} d\varphi = -4\pi G_N \int_{\varphi}^{\varphi_f} \frac{H}{H'} d\varphi = -\sqrt{4\pi G_N}\int_{\varphi}^{\varphi_f} \frac{d\varphi}{\sqrt{\epsilon}}. \tag{6.1.55}$$

During inflation, a mode k becomes super-Hubble at the time t_k defined by

$$k = a(t_k)H(t_k). \tag{6.1.56}$$

The speed at which scales become super-Hubble is characterized by

$$\frac{d\ln k}{d\varphi} = \sqrt{\frac{4\pi G_N}{\epsilon}}(\epsilon - 1). \tag{6.1.57}$$

Similarly, one can easily show that the relation $k = a_k H_k$ implies

$$\frac{d\ln H_k}{d\ln k} = -\epsilon. \tag{6.1.58}$$

Sensitivity to initial conditions By linearizing (6.1.33), we find that the perturbation δH takes the general form

$$\delta H = \delta H(\varphi_i)\exp\left\{12\pi G_N \int_{\varphi_i}^{\varphi} \frac{H_0(\varphi)}{H_0'(\varphi)} d\varphi\right\}. \tag{6.1.59}$$

The effect of the choice of initial conditions is therefore exponentially erased.

Figure 6.4: Phase portrait in the $(\varphi, \dot{\varphi})$-plane of the dynamics of a scalar field with a potential $V = m^2\varphi^2/2$, assuming $m = 10^{-6}M_P$. This illustrates the mechanism of attraction toward the slow-roll solution.

End of inflation

The end of inflation can be identified with the end of the slow-roll regime. We can therefore define the value of the field at the end of inflation, φ_f, by

$$\varphi + m^2\varphi = 0, \tag{6.1.60}$$

$$H^2 = \frac{4\pi}{3}\left(\frac{m}{M_P}\right)^2\varphi^2. \tag{6.1.61}$$

They can be integrated easily to give

$$\varphi(t) = \varphi_i - \frac{mM_p}{\sqrt{12\pi}}t, \tag{6.1.62}$$

$$a(t) = a_i \exp\left\{\frac{2\pi}{M_p^2}[\varphi_i^2 - \varphi^2(t)]\right\} \tag{6.1.63}$$

where φ_i and a_i are the values of the field and the scale factor at $t_i = 0$. The slow-roll parameters (6.1.38)–(6.1.40) being given by

$$\epsilon = \frac{M_p^2}{4\pi\varphi^2}, \quad \delta = 0, \quad \xi_V = 0, \tag{6.1.64}$$

the slow-roll regime lasts until the field reaches $\varphi_f = M_p/\sqrt{4\pi}$. We infer that the total number of e-folds is

$$N(\varphi_i) = 2\pi\left(\frac{\varphi_i}{M_P}\right)^2 - \frac{1}{2}. \tag{6.1.65}$$

In order to have $N \geq 70$, we need $\varphi_i \geq 3M_P$. If φ_i takes the largest possible value compatible with the classical description adopted here, i.e. fixed by $V(\varphi_i) \leq M_p^4$, we find that $\varphi_i \sim M_p^2/m$. In this case, the maximal accessible number of e-folds would be $N_{\max} \sim 2\pi M_p^2/m^2$. As will be seen later, the theory of cosmological perturbations and observations of the cosmic microwave background impose that the mass is of the order of $m \sim 10^{-6}M_P$, so that $N_{\max} \sim 10^{13}$. The maximal number of e-folds is thus very large compared to the minimum required for solving the cosmological problems.

Consequently, if the Universe is initially composed of regions where the values of the scalar field are randomly distributed, or if we consider different Universes with different field values, then the domains where the initial value of φ is too small never inflate or only for a small number of e-folds. The main contribution to the total physical volume of the Universe at the end of inflation thus comes from regions that have inflated for a long time and that had an initially large value of φ. These domains produce extremely flat and homogeneous zones at the end of inflation with a very large size compared to that of the observable Universe. At larger scales, the Universe has a very inhomogeneous structure. We will come back to this chaotic initial condition idea in Section 8.7.

Figure 6.5: Evolution of the inflaton and scale factor during chaotic inflation for the model (**??**). The scalar field is initially in a slow-roll regime and the expansion of the Universe is accelerated. At the end of this regime, it starts oscillating at the minimum of its potential and it is equivalent to a pressureless fluid.

Extensions This model can be generalized to any polynomial function of the scalar field. As an example, let us consider the family of potentials

$$V(\varphi) = \frac{3M_p^4}{8\pi}\lambda_n \left(\frac{\varphi}{M_p}\right)^n \tag{6.1.66}$$

where λ_n is a dimensionless constant. The slow-roll parameters are given by

$$\epsilon = \frac{M_p^2}{16\pi}\frac{n^2}{\varphi^2}, \qquad \delta = \frac{M_P^2}{16\pi}\frac{n(n-2)}{\varphi^2}, \tag{6.1.67}$$

and the slow-roll regime lasts as long as $\varphi > \varphi_f = nM_P/4\sqrt{\pi}$. We infer that the total number of e-folds is

$$N(\varphi_i) = \frac{4\pi}{n}\left(\frac{\varphi_i}{M_p}\right)^2 - \frac{n}{4}. \tag{6.1.68}$$

So in order for $N > N_{\min}$ we should initially have

$$\varphi_i > M_p\sqrt{\frac{n}{4\pi}\left(N_{\min} + \frac{n}{4}\right)}. \tag{6.1.69}$$

Yet, chaotic initial conditions imply that $V(\varphi_i) \sim M_P^4$, imposing

$$\varphi_i \sim M_p \left(\frac{8\pi}{3\lambda_n}\right)^{1/n}, \tag{6.1.70}$$

a value in general very large compared to M_P. Solving the Klein-Gordon and Friedmann equations in the slow-roll regime approximation gives

$$\varphi(t) = \varphi_i \left[1 + n(n-4)\frac{\sqrt{\lambda_n}}{16\pi}\left(\frac{\varphi_i}{M_p}\right)^{(n-4)/2} M_p t\right]^{2/(4-n)}, \tag{6.1.71}$$

$$a(t) = a_i \exp\left\{\frac{4\pi}{nM_p^2}[\varphi_i^2 - \varphi^2(t)]\right\}. \tag{6.1.72}$$

The expression in square brackets for the solution of the scalar field is positive definite as long as the slow-roll hypothesis is valid. Notice that although expression (6.1.72) is well defined for all n, the scalar field on the other hand is singular for $n = 4$. The direct computation of this specific case shows that (6.1.72) remains valid, with $\varphi(t) = \varphi_i \exp\left(-\frac{1}{2\pi}\sqrt{\lambda_4}M_P t\right)$.

We can reexpress these quantities in terms of the number of e-folds to obtain

$$\varphi(N) = M_p\sqrt{\left(\frac{\varphi_i}{M_p}\right)^2 - \frac{n}{4\pi}N}, \tag{6.1.73}$$

$$H(N) = M_p\sqrt{\lambda_n}\left[\left(\frac{\varphi_i}{M_p}\right)^2 - \frac{n}{4\pi}N\right]^{n/4}; \tag{6.1.74}$$

valid until $N(\varphi_i)$. These expressions show that the Hubble parameter can vary significantly during the slow-roll phase. In particular, and as will be seen later, the amplitude of the observable modes generated roughly 50–70 e-folds before the end of inflation implies that $H/M_P \sim 10^{-5}$. However, this does not imply that the Hubble parameter could not have been much larger before that. Cosmological observations can thus give some inputs on the last 70 e-folds or so of the inflation era but, a priori, put no constraints on the primordial phase of inflation.

Power-law inflation: an exact solution

For most models, the slow-roll approximation is so good that there is no need to go further, even if the end of inflation is not very well taken into account. It is, however, useful to have some exact solutions. Many models exist in the literature but the most attractive one is that of power-law inflation. The advantage of this model is that the perturbation equations can also be solved analytically.

These models are constructed such that the solution of the equations of evolution are

$$a = a_0|\eta|^{1+\beta} \iff a = a_0 t^p, \tag{6.1.75}$$

and

$$1+\beta = \frac{p}{1-p}, \quad \gamma = \frac{\beta+2}{\beta+1} = \frac{1}{p}. \tag{6.1.76}$$

$$\frac{\varphi - \varphi_i}{M_p} = \frac{1}{2}\sqrt{\frac{\gamma}{\pi}}(1+\beta)\ln|\eta|. \tag{6.1.77}$$

The parameter p varies between 1 and ∞ and thus $\beta < -2$. The evolution equations imply that the potential is of the form

$$V(\varphi) = V_i \exp\left[4\sqrt{\frac{\pi}{p}}\frac{(\varphi-\varphi_i)}{M_p}\right], \tag{6.1.78}$$

so that the slow-roll parameters turn out to be constant,

$$\epsilon = \delta = \xi_H = \frac{1}{p}, \quad \xi = 0. \tag{6.1.79}$$

Thus, with such a potential, the slow-roll parameters' inflation never ends. The limit $p \to +\infty$ (i.e. $\beta \to -2$) corresponds to a de Sitter Universe.

Hybrid inflation

Models of hybrid inflation introduce two coupled scalar fields, φ and σ, with the potential

$$V(\varphi,\sigma) = \frac{\lambda}{4}(\sigma^2 - v^2)^2 + \frac{1}{2}g^2\varphi^2\sigma^2 + V(\varphi), \tag{6.1.80}$$

where v is the vacuum expectation value (the VEV, as defined in Chapter 2) of σ and $V(\varphi)$ is the potential of the field φ. The effective mass of the field σ depends on the value of the second field

$$m^2(\sigma) = g^2(\varphi^2 - \varphi_c^2), \quad \varphi_c^2 = \frac{\lambda v^2}{g^2}. \tag{6.1.81}$$

The point $\varphi = \varphi_c$ is the bifurcation point where the effective mass of the field σ changes sign to become negative. As long as $\varphi > \varphi_c$, the potential has a valley along $\sigma = 0$; everything therefore goes as if there was only one field rolling towards the minimum of the potential $V(\varphi) + \lambda v^4/4$. The global minimum is, however, in $\varphi = 0$, $|\sigma| = v$. When the inflaton reaches the value φ_c, the mass of σ changes sign, which results in a spinodal instability triggering the end of inflation. The total number of e-folds is then

$$N = 2\pi\frac{\lambda v^4}{m^2 M_p^2}\ln\frac{\varphi_i}{\varphi_c} \tag{6.1.82}$$

if the potential for the inflaton is of the form $V(\varphi) = m^2\varphi^2/2$.

Figure 6.6: Potential for hybrid inflation with two scalar fields. Inflation occurs when $\sigma = 0$ and stops abruptly when φ reaches the value φ_c of the bifurcation point.

Classification of inflationary models

For single-field inflationary models, the potential can always be characterized by two mass scales, Λ and μ,

$$V(\varphi) = \Lambda^4 f\left(\frac{\varphi}{\mu}\right). \tag{6.1.83}$$

These two scales typically determine the height and width of the potential. In order to discuss the phenomenology of these models, it is useful to classify them as a function of the value of their slow-roll parameters. We distinguish between the four following classes.

A. Inflation with large values of the inflaton: $-\epsilon < \delta \leq \epsilon$. The archetypes of this class are the potentials of chaotic inflation, e.g. (**??**) and (6.1.66) or exponential potentials (6.1.78). In these models, the field is initially away from its minimum towards which it slowly relaxes. These potentials satisfy $V'' > 0$, i.e. $\eta_V > 0$. In the framework of chaotic inflation, the condition $V \sim M_P^4$ implies that $\varphi \gg M_P$, but a value $\varphi \sim M_P$ is, in general, sufficient to obtain the required number of e-folds.

B. Inflation with small values of the inflaton: $\delta < -\epsilon$. These potentials are those that appear naturally in the mechanisms of spontaneous symmetry breaking and were at the origin of the so-called models of new inflation. The field is initially close to an unstable equilibrium position in $\varphi_i = 0$. Inflation occurs when the field is close to the origin and the asymptotic form of the potential is not significant. We therefore have $V'' < 0$, i.e. $\eta_V < 0$ and $\epsilon > 0$ and very small (very flat potential). We infer that $\delta < -\epsilon$. An example of such a potential is

$$V(\varphi) = \Lambda^4 \left[1 - \left(\frac{\varphi}{\mu}\right)^p\right] \tag{6.1.84}$$

that we can consider as a Taylor expansion of the 'true' potential around the origin. In the case of a spontaneous symmetry-breaking potential of the Higgs kind, $p = 2$; μ is then the scale of the symmetry breaking. Another example is the Coleman-Weinberg potential,

$$V = \frac{\lambda}{4}\varphi^4 + \frac{\lambda^2}{44\pi}\varphi^4 \left[\ln\left(\frac{\varphi}{\mu}\right) - \frac{25}{6}\right]. \tag{6.1.85}$$

C. Hybrid models: $0 < \delta < \epsilon$. These potentials often appear in the framework of supersymmetry. The field evolves towards its non-vanishing minimum. Inflation usually ends abruptly due to an instability of the auxiliary field. Another example is given by the potential $V(\varphi) = \Lambda^4[1 + (\varphi/\mu)^p]$.

D. Linear models: $\delta = -\epsilon$. These models, characterized by

$$V(\varphi) \propto \varphi \tag{6.1.86}$$

represent a limit between the models with large and small values of the field.

Of course, such a classification is not exhaustive as it does not incorporate potentials of the type $V(\varphi) \propto \ln\varphi$, or of inverse power laws, $V(\varphi) \propto \varphi^{-n}$ as in intermediate inflation. However, these models require the introduction of auxiliary fields to stop inflation.

Figure 6.7: From left to right, three examples of inflationary potentials with large field values (class A), small field values (class B) and hybrid (class C).

Figure 6.8: The four classes of models of single-field inflation in the space of the two first slow-roll parameters. We have indicated the trajectory of the models of chaotic inflation (**??**) and (6.1.66) and the point represents a model of power-law inflation (6.1.75).

6.2 Quantum fluctuations during inflation

One of the great successes of inflation has been to provide a way to relate the origin of the large-scale structure of the Universe to the quantum fluctuations of the inflaton that are amplified during inflation. To address this central point of primordial cosmology, we start by a warm up and only consider test fields. This study will illustrate the fact that during inflation, any light field (in the sense $m < H$) develops super-Hubble fluctuations with mean amplitude $H/2\pi$.

6.2.1 Massless test scalar field in a de Sitter space-time

The structure of the vacuum in an exponentially expanding space-time is much more complex than that in a Minkowski space-time (Chapter 2). In particular, the wavelength of any fluctuation grows exponentially. When it becomes larger than the Hubble radius, H^{-1}, this fluctuation is said to freeze as it no longer oscillates due to the friction term in the Klein-Gordon equation, and it rapidly converges toward a solution of constant amplitude.

Quantization

To quantize a scalar field in an expanding space-time, we proceed in an analogous way to what is usually done in Minkowski space-time and promote the scalar field to the status of a quantum operator,

$$\hat{\chi}(x,t) = \int \frac{d^3k}{(2\pi)^{3/2}}[\chi_k(t)e^{ik\cdot x}\hat{a}_k + \chi_k^*(t)e^{-ik\cdot x}\hat{a}_k^\dagger] \tag{6.2.1}$$

where the creation and annihilation operators satisfy (2.83). The mode function χ_k is a solution of the classical Klein-Gordon equation

$$\ddot{\chi}_k + 3H\dot{\chi}_k + \frac{k^2}{a^2}\chi_k = 0. \tag{6.2.2}$$

Qualitatively, this equation possesses two regimes. When the mode is sub-Hubble, $k \gg aH$, the friction term is negligible and the equation reduces to that of a harmonic oscillator analogous to that obtained in Minkowski space-time, with the only difference being that the wave number is redshifted by the expansion. When the mode is super-Hubble, $k \ll aH$ and the gradient term is negligible. The equation reduces to $\ddot{\chi}_k + 3H\dot{\chi}_k = 0$ where the dominant solution is a constant. In conclusion, an initially sub-Hubble mode oscillates as its wavelength grows and then freezes when it becomes super-Hubble.

General solution

To solve (6.2.2) it is convenient to work in conformal time and to introduce the auxiliary field

$$v = a\chi. \tag{6.2.3}$$

The Klein-Gordon equation then takes the form

$$v_k'' + \left(k^2 - \frac{a''}{a}\right)v_k = 0, \tag{6.2.4}$$

which introduces an effective time-dependent mass term, $m^2 = -a''/a$. For a de Sitter space, the scale factor is given by (6.1.14) so that v satisfies

$$v_k'' + \left(k^2 - \frac{2}{\eta^2}\right) v_k = 0. \tag{6.2.5}$$

It has the general solution

$$v_k(\eta) = [A(k)H^{(1)}_{3/2}(-k\eta) + B(k)H^{(2)}_{3/2}(-k\eta)]\sqrt{-\eta}, \tag{6.2.6}$$

where $H^{(1,2)}_{3/2}$ are the Hankel functions of first and second kind (Appendix B). Using

$$H^{(2)}_{3/2}(z) = [H^{(1)}_{3/2}(z)]^* = -\sqrt{\frac{2}{\pi z}}e^{-iz}\left(1 + \frac{1}{iz}\right) \tag{6.2.7}$$

we infer the general form

$$v_k(\eta) = A(k)e^{-ik\eta}\left(1 + \frac{1}{ik\eta}\right) + B(k)e^{ik\eta}\left(1 - \frac{1}{ik\eta}\right). \tag{6.2.8}$$

At this stage, we need to postulate a choice of initial conditions to determine the two arbitrary functions $A(k)$ and $B(k)$.

Initial condition

To perform a canonical quantization, we impose the commutation rules on the operators $\hat{v}$ and $\hat{\pi}$, namely $[\hat{v}(x,\eta), \hat{v}(x',\eta)] = [\hat{\pi}(x,\eta), \hat{\pi}(x',\eta)] = 0$ and $[\hat{v}(x,\eta), \hat{\pi}(x',\eta)] = i\delta^{(3)}(x - x')$ on constant time hypersurfaces, $\hat{\pi}$ being the conjugate momentum of $\hat{v}$. From the commutation rules of the annihilation and creation operators, this implies that

$$v_k v_k'^* - v_k^* v_k' = i, \tag{6.2.9}$$

which determines the normalization of the Wronskian. The choice of a specific mode function $v_k(\eta)$ corresponds to the choice of a prescription for the physical vacuum $|0\rangle$, defined by $\hat{a}_k|0\rangle = 0$. To choose the vacuum, let us note that for very high frequency modes, $k\eta \gg 1$, the curvature term is negligible so that the field can be quantized as if it were in a Minkowski space-time (see Chapter 2). We thus pick up the solution that corresponds adiabatically to the usual Minkowski vacuum, keeping only the positive frequencies. We therefore impose

$$v_k \to \frac{e^{-ik\eta}}{\sqrt{2k}}, \quad k\eta \to -\infty. \tag{6.2.10}$$

This choice is referred to as the Bunch-Davies vacuum. We infer that the solution satisfying this initial condition is

$$\chi_k = \frac{H\eta}{\sqrt{2k}}\left(1 + \frac{1}{ik\eta}\right)e^{-ik\eta}. \tag{6.2.11}$$

Let us emphasize that, at this stage, we have not justified why we have to apply the condition (6.2.10) to v_k rather than to χ_k. Such a justification will be made later.

Properties on super-Hubble scales

On super-Hubble scales, the field χ acquires a constant amplitude

$$|\chi_k|(k\eta \ll 1) = \frac{H}{\sqrt{2k^3}} \tag{6.2.12}$$

where we have used $a(\eta) = -1/(H\eta)$ for a pure de Sitter inflationary phase, H being a constant in this case. Let us emphasize that, in this limit, the field actually behaves as

$$\hat{\chi} \to \int \frac{d^3k}{(2\pi)^{3/2}} \hat{\chi}_k e^{ik\cdot x} = \int \frac{d^3k}{(2\pi)^{3/2}} \frac{H}{\sqrt{2k^3}} (\hat{a}_k + \hat{a}^\dagger_{-k}) e^{ik\cdot x}.$$

At this point, all the modes being proportional to $(\hat{a}_k + \hat{a}^\dagger_{-k})$, the quantum operators $\hat{\chi}_k$ commute. Therefore, χ has actually the same statistical properties as a Gaussian classical stochastic field. Effectively, on super-Hubble scales, the quantum operator $\hat{\chi}$ can be replaced by a stochastic field with Gaussian statistics. Introducing a unit Gaussian random variable, $e_v(k)$, which satisfies $\langle e_v(k)\rangle = 0$ and $\langle e_v(k)e_v^*(k')\rangle = \delta^{(3)}(k-k')$, the mode operators are replaced by stochastic variables through $\hat{v}_k \to v_k = v_k(\eta)e_v(k)$ and we can switch from the vacuum quantum average to the classical ensemble average. This explains why the super-Hubble modes have Gaussian statistics.

Power spectrum

The correlation function of v is defined as

$$\xi_v \equiv \langle 0|\hat{v}(x,\eta)\hat{v}(x',\eta)|0\rangle$$

and takes the simple form

$$\xi_v = \int \frac{d^3k}{(2\pi)^3} |v_k|^2 e^{ik\cdot(x-x')} = \int \frac{dk}{k} \frac{k^3}{2\pi^2} |v_k|^2 \frac{\sin kr}{kr}, \tag{6.2.13}$$

where the isotropy of the background space-time has been used to integrate over angles. Because of the symmetries of the background, it is a function of $r = |x - x'|$ only. We thus define the power spectra of v,

$$P_v(k) = |v_k|^2, \quad \mathcal{P}_v(k) = \frac{k^3}{2\pi^2}|v_k|^2. \tag{6.2.14}$$

In the stochastic picture, the correlator of v_k is simply given by $\langle v_k v^*_{k'}\rangle = P_v(k)\delta^{(3)}(k-k')$ from which one easily deduces that the power spectrum of χ is

$$P_\chi(k) = \frac{2\pi^2}{k^3}\mathcal{P}_\chi(k) = |\chi_k|^2 = \frac{|v_k|^2}{a^2}. \tag{6.2.15}$$

It is thus a scale invariant power spectrum,

$$\mathcal{P}_\chi(k) = \left(\frac{H}{2\pi}\right)^2. \tag{6.2.16}$$

WKB approximation

In more general cases, where e.g. the function $a(\eta)$ is not known analytically, we may not have exact solutions for the mode functions. One can generalize the previous construction by relying on a WKB approach in which one introduces the WKB mode functions

$$v_k^{\text{WKB}}(\eta) = \frac{1}{\sqrt{2\omega(k,\eta)}} e^{\pm i\int^\eta \omega(k,\eta')d\eta'} \tag{6.2.17}$$

which, as is easily checked, are solutions of

$$v_k^{\text{WKB}\prime\prime} + [\omega^2(k,\eta) - Q_{\text{WKB}}]v_k^{\text{WKB}} = 0; \tag{6.2.18}$$

where the potential is

$$Q_{\rm WKB} = \frac{3}{4}\left(\frac{\omega'}{\omega}\right)^2 - \frac{1}{2}\frac{\omega''}{\omega}. \tag{6.2.19}$$

The WKB solution is thus a good approximation as long as the WKB condition

$$\left|\frac{Q_{\rm WKB}}{\omega^2}\right| \ll 1 \tag{6.2.20}$$

holds. For a function satisfying an equation such as (6.2.5), $\omega^2(k,\eta) = k^2 - 2/\eta^2$, the WKB condition reduces to $|k\eta| \gg 1$, so that on sub-Hubble scales, the mode function is actually close to its WKB approximation. One might thus reinterpret the quantization procedure by noting the fact that there exists an adiabatic (WKB) solution on sub-Hubble scales.

Interpretation and infra-red divergences

Expression (6.2.11) can be used to express the variance of the field, $\langle\chi^2\rangle$, as a function of the physical momentum, $p = a^{-1}(t)k$,

$$\langle\chi^2\rangle = \int \frac{d^3p}{(2\pi)^3 p}\left(\frac{1}{2} + \frac{H^2}{2p^2}\right). \tag{6.2.21}$$

The first term is simply the zero-point energy of a harmonic oscillator in Minkowski space-time (Chapter 2). This term leads to an ultraviolet divergence that can be eliminated by renormalization. The second term only appears in a curved space-time and is characteristic of de Sitter space-time. It can be interpreted as the existence of

$$n_p = \frac{H^2}{2p^2} \tag{6.2.22}$$

particles of physical momentum p. This term diverges in the infra-red. There are actually two divergences, one when $p \to 0$ and the second one when $\eta \to 0$. The infra-red divergence at $p = 0$ appears because the expansion of the field in creation and annihilation modes do not include the zero mode. For a massless field in Minkowski space in a box of volume V, a complete description of the solutions requires the introduction of the position and momentum operators, $\hat\varphi_0$ and $\hat\pi_{\varphi 0}$, that satisfy the relation $[\hat\varphi_0, \hat\pi_{\varphi 0}] = i$. We can then define the vacuum by imposing $\hat\pi_{\varphi 0}|0\rangle = 0$ and $\hat a_k|0\rangle = 0$, which amounts to describing the state in a space that is the product of a Fock space and a Hilbert space. In a Minkowski space of dimension larger than 2, the continuum limit exists because the phase space volume factor compensates the divergence. However, it persists in a de Sitter space-time. If this divergence is regularised by imposing a frequency cutoff $p > p_0 \sim H$ then

$$\langle\chi^2\rangle \sim \frac{H^2}{4\pi^2}\int_{He^{-Ht}} \frac{dk}{k} \sim \frac{H^3}{4\pi^2}t. \tag{6.2.23}$$

This result is general and the resolution of the infra-red divergence at $k = 0$ does not resolve this second problem. The physical origin of this divergence is found in the eternal expansion that brings modes towards $p = 0$ exponentially. However, these divergences only appear for massless particles in a de Sitter space that characterizes a phase of eternal inflation and naturally disappear in quasi-de Sitter space-times.

Main conclusions

We conclude that during a de Sitter inflationary phase, a test scalar field develops super-Hubble fluctuations with a scale invariant power spectrum and Gaussian statistics. They arise from the parametric amplification of the inescapable vacuum fluctuation of the scalar field on sub-Hubble scales. Their amplitude is dictated by H, that is typically by the energy scale during inflation. As we shall see, most of these conclusions generalize to more sophisticated cases, and in particular to the inflaton.

6.2.2 Massive test field in de Sitter

We can extend the previous case to that of a test field of mass m_χ. Equation (6.2.4) then takes the form

$$v_k'' + \left(k^2 + \frac{m^2/H^2 - 2}{\eta^2}\right) v_k = 0, \tag{6.2.24}$$

when considering a potential $m_\chi^2\chi^2/2$. It can be rewritten as

$$v_k'' + \left[k^2 + \frac{1}{\eta^2}\left(\nu_\chi^2 - \frac{1}{4}\right)\right] v_k = 0, \tag{6.2.25}$$

with

$$\nu_\chi^2 = \frac{9}{4} - \frac{m_\chi^2}{H^2}. \tag{6.2.26}$$

If ν_χ is real, the general solution is a generalization of (6.2.6) and is

$$v_k(\eta) = [A(k)H^{(1)}_{\nu_\chi}(-k\eta) + B(k)H^{(2)}_{\nu_\chi}(-k\eta)]\sqrt{-\eta}, \tag{6.2.27}$$

where $H^{(1,2)}_{\nu_\chi}$ are Hankel functions. The asymptotic form allows us to deduce that the solution that satisfies the initial condition (6.2.10) when $k\eta \to -\infty$ is

$$v_k = \sqrt{\frac{-\pi\eta}{4}} i^{(\nu_\chi + 1/2)} H^{(1)}_{\nu_\chi}(-k\eta) = \sqrt{\frac{-\pi\eta}{4}} i^{(\nu_\chi + 1/2)} H^{(2)}_{\nu_\chi}(k\eta). \tag{6.2.28}$$

On super-Hubble scales, the field freezes to a constant amplitude,

$$|\chi_k|(k\eta \ll 1) = \frac{H}{\sqrt{2k^3}} 2^{\nu_\chi - 3/2} \frac{\Gamma(\nu_\chi)}{\Gamma(3/2)} \left(\frac{k}{aH}\right)^{3/2-\nu_\chi} \tag{6.2.29}$$

and has a power spectrum

$$\mathcal{P}_\chi(k) = 2^{2\nu_\chi - 3}\left[\frac{\Gamma(\nu_\chi)}{\Gamma(3/2)}\right]^2 \left(\frac{H}{2\pi}\right)^2 \left(\frac{k}{aH}\right)^{3-2\nu_\chi}. \tag{6.2.30}$$

If the field has a mass larger than $3H/2$ then $\nu_\chi^2 < 0$. The sub-Hubble solution is not affected and the initial conditions are therefore identical. Nevertheless, the mode function in the super-Hubble limit is also oscillating. After some algebra, we find that the power spectrum is now given by

$$\mathcal{P}_\chi(k) \propto \left(\frac{H}{2\pi}\right)^2 \left(\frac{H}{m_\chi}\right) \left(\frac{k}{aH}\right)^3. \tag{6.2.31}$$

The amplitude of the spectrum is thus reduced by a factor $H/m_\chi > 3/2$ and the spectrum decreases rapidly for long wavelengths. This result is one of the generic predictions of inflation: during inflation, any light test field, $m < H$, develops super-Hubble fluctuations of characteristic amplitude $H/2\pi$, with an almost scale invariant power spectrum.

6.2.3 Massive test field during slow-roll inflation

In the two previous examples, inflation was described as a pure de Sitter phase so that H was constant. Of course, this is only an approximation, and H decreases slowly in time. This time dependence will have an influence on the spectrum of any light field. For a massless scalar field, (6.2.16) means that the spectrum of the super-Hubble modes is given by the value of H at the time where the mode becomes super-Hubble, $\mathcal{P}_\chi(k) \propto H_k^2$. Relation (6.1.58) indicates that in the slow-roll regime, $H_k \propto k^{-\epsilon}$ so that we expect that the spectrum behaves as $\mathcal{P}_\chi \propto k^{-2\epsilon}$. Its spectrum will therefore be red-tilted (since $\epsilon \geq 0$).

Evolution of the scale factor

The expansion of the Universe is now quasi-de Sitter. Integrating by parts, the conformal time is given by

$$\eta = \int \frac{da}{a^2 H} = -\frac{1}{aH} + \int \frac{\epsilon}{a^2 H} da. \tag{6.2.32}$$

Assuming ϵ is constant, we find that the scale factor evolves as

$$a(\eta) = -\frac{1}{H\eta}\frac{1}{1-\epsilon}. \tag{6.2.33}$$

From (6.1.47), this hypothesis amounts to neglecting all terms of 2nd order in the slow-roll parameters since $\dot{\epsilon} = \mathcal{O}(\epsilon^2, \delta\epsilon)$.

Solution of the Klein-Gordon equation

To solve (6.2.24), one should first evaluate

$$\frac{a''}{a} = a^2\left(H^2 + \frac{\dot{a}}{a}\right) = a^2(2H^2 + \dot{H}) = a^2H^2(2-\epsilon) = \frac{2-\epsilon}{(1-\epsilon)^2\eta^2}. \tag{6.2.34}$$

We infer that, to first order in ϵ,

$$\frac{a''}{a} \simeq \frac{2+3\epsilon}{\eta^2}. \tag{6.2.35}$$

The Klein-Gordon equation therefore takes the form (6.2.25) with

$$\nu_\chi^2 = \frac{9}{4} + 3\epsilon - \frac{m_\chi^2}{H^2}. \tag{6.2.36}$$

For a light field, the power spectrum then takes the form (6.2.30), which corresponds to a spectral index

$$n_\chi - 1 \equiv \frac{d\ln \mathcal{P}_\chi}{d\ln k} = 3 - 2\nu_\chi \simeq 2\left(\frac{m_\chi^2}{3H^2} - \epsilon\right). \tag{6.2.37}$$

Conclusion

We recover the expected result. The spectrum can be blue-tilted ($n_\chi > 1$), red-tilted ($n_\chi < 1$) or scale invariant. The spectral index gets a contribution from the mass of the test field and from the parameter ϵ that characterizes the k dependence of H_k. For a massless scalar field, the spectrum is always red-tilted. As an example, consider a model of chaotic inflation with inflaton mass m_φ. The spectral index is then

$$n_\chi - 1 = \frac{2}{3H^2}(m_\chi^2 - m_\varphi^2). \tag{6.2.38}$$

6.3 Quantum fluctuations of the inflaton

The previous section has cleared the ground for the study of primordial fluctuations generated by the inflaton. Since this field dominates the dynamics of the Universe, we cannot neglect the fluctuations of the geometry and have to follow the gauge invariant relativistic perturbation equations.

6.3.1 Overview of the questions to address and expected results

To start with, any fluctuation of the scalar field will generate metric perturbations. We should first study the coupled inflaton-gravity system. We start from 10 perturbations of the metric and 1 for the scalar field, to which we have to substract 4 gauge freedoms and 4 constraint equations. We thus expect to identify 3 independent degrees of freedom to describe the full dynamics.

The second question is that of initial conditions and quantization. In the previous examples, we have decided to quantize the field $v = a\chi$ without justification. Actually, (6.2.4) suggests that the Klein-Gordon equation for v is that of a harmonic oscillator of variable mass. One can conjecture that it is the canonical variable to quantize.

Before we begin, note that if metric perturbations and $\dot{H}$ are neglected compared to H^2, the Klein-Gordon equation for the super-Hubble inflaton perturbations is identical to the time derivative of the background Klein-Gordon equation (6.1.26)

$$\ddot{\varphi} + 3H\dot{\varphi} + V_{,\varphi\varphi}\dot{\varphi} = 0. \tag{6.3.1}$$

Qualitatively, we infer that $\delta\varphi = -\dot{\varphi}_0\delta t$, which implies that $\varphi(x,t) \sim \varphi_0[t - \delta t(x)]$. Due to long wavelength perturbations, the number of e-folds between various super-Hubble zones will be modulated. This will in turn give rise to metric fluctuations of the order of $H\delta t(x) \sim H\delta\varphi/\dot{\varphi}$. We infer that $\delta t \sim H/2\pi\dot{\varphi}$ so that the metric perturbations are approximately of the order of $H^2/2\pi\dot{\varphi}$.

So perturbations are due to the fact the inflaton reaches the bottom of its potential at different times in different points of the Universe. The inflaton also gives rise to primordial gravitational waves. The typical amplitudes are of order

$$\frac{\delta\rho}{\rho} \propto \frac{V^{3/2}}{V_{,\varphi}} \sim \frac{H^2}{2\pi\dot{\varphi}}, \quad h \sim \frac{H}{2\pi M_P}. \tag{6.3.2}$$

The primordial spectra thus contain information on the dynamics and energy scale of inflation.

6.3.2 Perturbed quantities

Stress-energy tensor

The perturbation of the stress-energy tensor (6.1.21) of a scalar field is

$$\begin{aligned}\delta T_{\mu\nu} = 2\partial_{(\nu}\varphi\partial_{\mu)}\delta\varphi - \left(\frac{1}{2}g^{\alpha\beta}\partial_\alpha\varphi\partial_\beta\varphi + V\right)\delta g_{\mu\nu} \\ - g_{\mu\nu}\left(\frac{1}{2}\delta g^{\alpha\beta}\partial_\alpha\varphi\partial_\beta\varphi + g^{\alpha\beta}\partial_\alpha\delta\varphi\partial_\beta\varphi + V'\delta\varphi\right).\end{aligned} \tag{6.3.3}$$

Using the SVT decomposition of the metric (5.52), its components take the form

$$\begin{aligned}\delta T_{00} &= \varphi'\delta\varphi' + 2a^2VA + a^2\frac{dV}{d\varphi}\delta\varphi \\ \delta T_{0i} &= D_i(\varphi'\delta\varphi) + B_i\left(\frac{1}{2}\varphi'^2 - Ba^2V\right) \\ \delta T_{ij} &= [\varphi'\delta\varphi' + (C-A)\varphi'^2 - 2a^2VC - a^2\frac{dV}{d\varphi}\delta\varphi]\gamma_{ij} \\ &\quad + (\varphi'^2 - 2a^2V)(D_iD_jE + D_{(i}\bar{E}_{j)} + \bar{E}_{ij}).\end{aligned} \tag{6.3.4}$$

It is usually more efficient to use the mixed components to obtain the equations,

$$a^2\delta T^0_0 = -\varphi'\delta\varphi' - a^2\frac{dV}{d\varphi}\delta\varphi + A\varphi'^2$$

$$
\begin{aligned}
a^2\delta T^0_i &= -D_i(\varphi'\delta\varphi) \\
a^2\delta T^i_0 &= D^i(\varphi'\delta\varphi + B\varphi'^2) + \bar{B}^i\varphi'^2 \\
a^2\delta T^i_j &= (\varphi'\delta\varphi' - \varphi'^2 A - a^2\frac{dV}{d\varphi}\delta\varphi)\delta^i_j.
\end{aligned}
\tag{6.3.5}
$$

Gauge invariant variables

Only one quantity appears to describe matter perturbations, namely the perturbation of the scalar field itself. Like any scalar quantity, it transforms as $\delta\varphi \to \delta\varphi + \varphi' T$. We can therefore introduce the two following gauge invariant variables

$$\chi = \delta\varphi + \varphi'(B - E'), \quad Q = \delta\varphi - \varphi'\frac{C}{\mathcal{H}}, \tag{6.3.6}$$

related by

$$Q = \chi + \varphi'\frac{\Psi}{\mathcal{H}}. \tag{6.3.7}$$

The variable χ is the perturbation in Newtonian gauge, while Q, the Mukhanov-Sasaki variable, is the one in flat-slicing gauge.

6.3.3 Perturbation equations

Equations in the SVT decomposition

Scalar modes Following the same approach as for a perfect fluid (Chapter 5), we obtain the following three equations, respectively, for the components (00), $(0i)$ and the trace of (ij),

$$(\Delta + 3K)\Psi - 3\mathcal{H}(\Psi' + \mathcal{H}\Phi) = \frac{\kappa}{2}\left(\varphi'\chi' - \varphi'^2\Phi + a^2\frac{dV}{d\varphi}\chi\right) \tag{6.3.8}$$

$$\Psi' + \mathcal{H}\Phi = \frac{\kappa}{2}\varphi'\chi, \tag{6.3.9}$$

$$\Psi'' + 2\mathcal{H}\Psi' - K\Psi + \mathcal{H}\Phi' + (2\mathcal{H}' + \mathcal{H}^2)\Phi + \frac{1}{3}\Delta(\Phi - \Psi) = \frac{\kappa}{2}\left(\varphi'\chi' - \varphi'^2\Phi - a^2\frac{dV}{d\varphi}\chi\right) \tag{6.3.10}$$

The perturbed Klein-Gordon equation then takes the form

$$\chi'' + 2\mathcal{H}\chi' - \Delta\chi + a^2\frac{d^2V}{d\varphi^2}\chi = 2(\varphi'' + 2\mathcal{H}\varphi')\Phi + \varphi'(\Phi' + 3\Psi'). \tag{6.3.11}$$

Only two of the four equations (6.3.8)–(6.3.11) are independent. The Klein-Gordon equation in terms of the variable Q takes the form

$$Q'' + 2\mathcal{H}Q' - \Delta Q + a^2\frac{d^2V}{d\varphi^2}Q = \varphi'\left(X' - \frac{1}{\mathcal{H}}\Delta\Psi\right) \tag{6.3.12}$$

where we recall that $X = \Phi + \Psi + (\Psi/\mathcal{H})'$ [see (5.67)]. Actually, using the Einstein equations, as we shall see later, this equation can be written in a closed form for Q. Moreover, (6.2.35) implies that $\delta T^j_i \propto \delta^j_i$, which directly implies that the two Bardeen potentials must be equal,

$$\Phi = \Psi. \tag{6.3.13}$$

Let us stress one of the properties of these equations in the case of a de Sitter phase where the Universe is dominated by a pure cosmological constant. In this case $\varphi' = 0$ and the Einstein equations imply that

$$\text{de Sitter phase:}\ \Phi = \Psi = 0. \tag{6.3.14}$$

This does not mean that the scalar field does not fluctuate but only that its fluctuations do not couple to the geometry and do not generate any metric fluctuations. This explains why the de Sitter limit is often singular in our equations. In what follows, we assume that $\varphi' \neq 0$.

Vector modes Since a scalar field does not contain any vector sources, there are no vector equations associated with the Klein-Gordon equation and there is only one Einstein equation for these modes

$$\overline{\Phi}_i' + 2\mathcal{H}\overline{\Phi}_i = 0. \tag{6.3.15}$$

The vector modes are therefore rapidly washed out since $\overline{\Phi}_i \propto a^{-2}$. Actually, the constraint (5.113) reduces, in the case of a scalar field, to $(\Delta + 2K)\overline{\Phi}_i = 0$. We can conclude, independently of any model of inflation, that the vector modes are completely absent at the end of inflation.

Tensor modes There is only one tensor equation that can be obtained from the Einstein equations

$$\overline{E}_{ij}'' + 2\mathcal{H}\overline{E}_{ij}' - (\Delta - 2K)\overline{E}_{ij} = 0. \tag{6.3.16}$$

The general study of this equation has been performed in Section 5.3.1 of Chapter 5. As in that chapter, we can expand the gravitational waves into their polarization tensors

$$\overline{E}_{ij}(k,\eta) = \sum_\lambda \overline{E}_\lambda(k,\eta)\epsilon_{ij}^\lambda(k), \tag{6.3.17}$$

where $\lambda = +, \times$ indicates the polarization. This symmetric tensor satisfies

$$\epsilon_{ij}^\lambda(k)\delta^{ij} = 0, \quad k^i\epsilon_{ij}^\lambda(k) = 0, \quad \epsilon_{ij}^\lambda(k)\epsilon_{\lambda'}^{ij}(k) = \delta_{\lambda\lambda'}, \quad \epsilon_{ij}^\lambda(-k) = [\epsilon_{ij}^\lambda(k)]^*. \tag{6.3.18}$$

The last condition is not mandatory but insures that $\overline{E}_\lambda(k,\eta) = \overline{E}_\lambda^*(-k,\eta)$.

Other forms of the scalar equations

Despite the apparent complexity of the scalar equations (6.3.8)–(6.3.11), using the evolution equation for φ, they can be reduced to a second-order equation with time-dependent coefficients for the gravitational potential Φ,

$$\Phi'' + 2\left(\mathcal{H} - \frac{\varphi''}{\varphi'}\right)\Phi' - \left[\Delta - 2\left(\mathcal{H}' - \mathcal{H}\frac{\varphi''}{\varphi'} - 2K\right)\right]\Phi = 0. \tag{6.3.19}$$

This equation is valid only if $\varphi' \neq 0$ and therefore cannot be used for a strict de Sitter phase or during the phase where the scalar field oscillates at the bottom of its potential, as during reheating.

Equation (6.3.19) can take a more suggestive form if we introduce the variables

$$u_s \equiv \frac{2}{\kappa}(\rho + P)^{-1/2}\Phi = \frac{2}{\sqrt{3\kappa}}\frac{a^2\theta}{\mathcal{H}}\Phi, \tag{6.3.20}$$

with

$$\theta = \frac{1}{a}\left(\frac{\rho}{\rho + P}\right)^{1/2}\left(1 - \frac{3K}{\kappa\rho a^2}\right)^{1/2} = \frac{\mathcal{H}}{a}\left[\frac{2}{3}(\mathcal{H}^2 - \mathcal{H}' + K)\right]^{-1/2}; \tag{6.3.21}$$

following what was previously done for a perfect fluid in Section 5.3.2 of Chapter 5. It then reduces to the equation

$$u_s'' - [\Delta + \frac{\theta''}{\theta} - 3K(1-c_s^2)]u_s = 0, \tag{6.3.22}$$

with

$$c_s^2 = \frac{P'}{\rho'} \equiv -\frac{1}{3}\left(1 + 2\frac{\varphi''}{\mathcal{H}\varphi'}\right). \tag{6.3.23}$$

This equation is identical to (5.147) obtained for a fluid. Note that the definition for u_s is the same as that of Ref. [10] for the case of a fluid but differs by a factor $2/\kappa$ for a scalar field since $u_s = (2/\kappa)a\Phi/\varphi'$. We prefer to keep the same definition for both types of matter.

When $K = 0$, this equation can be interpreted as the equation for a harmonic oscillator with a time-dependent frequency $\omega^2 = k^2 - \theta''/\theta$, or as a Schrödinger equation with potential $U = \theta''/\theta$. This potential contains derivatives of the scale factor up to fourth order.

Evolution of the long-wavelength modes

We restrict our analysis to a Universe with Euclidean spatial sections, $K = 0$, in the rest of this section.

Integral solution For long-wavelength modes, $u_s = \theta$ is an obvious solution of (6.3.22), whose solutions are given by the integral form

$$u_s(k,\eta) = \theta(\eta)\left[A(k)\int\frac{d\eta}{\theta^2(\eta)} + B(k)\right] \tag{6.3.24}$$

where $A(k)$ and $B(k)$ are two functions of integration. It can be rewritten as

$$u_S(k,\eta) = \frac{C(k)}{\varphi'}\left[\frac{1}{a}\int^\eta d\eta a^2(\eta)\right]' \tag{6.3.25}$$

where the second integration constant has been absorbed in the integral bounds. This equation allows us to conclude that

$$\Phi(k,\eta) = \frac{\sqrt{3}}{2}C(k)\left[\frac{1}{a}\int^\eta d\eta a^2(\eta)\right]^{\cdot} = \frac{\sqrt{3}}{2}C(k)\left(1 - \frac{H}{a}\int a dt\right). \tag{6.3.26}$$

Conserved quantity In Chapter 5, we introduced the quantity ζ defined by (5.149) as

$$\zeta \equiv \Phi + \frac{2}{3}\frac{\Phi' + \mathcal{H}\Phi}{\mathcal{H}(1+w)} \tag{6.3.27}$$

represents the curvature perturbation in comoving gauge. Equation (6.3.19) is strictly analogous to (5.143) so that ζ satisfies the general equation of conservation (5.150). One can check that, during the period of inflation (assuming $K = 0$), this equation takes the simplified form

$$\frac{3}{2}\mathcal{H}(1+w)\zeta' = \Phi'' + 2\left(\mathcal{H} - \frac{\varphi''}{\varphi'}\right)\Phi' + 2\left(\mathcal{H}' - \mathcal{H}\frac{\varphi''}{\varphi'}\right)\Phi. \tag{6.3.28}$$

Equation (6.3.19) thus implies that

$$\zeta' = \frac{2}{3\mathcal{H}(1+w)}\Delta\Phi. \tag{6.3.29}$$

Curvature perturbation The curvature perturbation of constant density hypersurfaces in flat-slicing gauge (5.153),

$$\zeta_{BST} = -C + \frac{1}{3}\frac{\delta\rho}{\rho + P} \tag{6.3.30}$$

is also a gauge invariant quantity that is also conserved during the evolution. As was shown previously [see (5.156)], this quantity is related to ζ by the relation

$$\zeta = \zeta_{BST} - \frac{1}{3}\frac{\Delta\Phi}{\mathcal{H}' - \mathcal{H}^2} \tag{6.3.31}$$

so that for super-Hubble modes, $\zeta \simeq \zeta_{BST}$. In the literature, it is common to find the quantity

$$\mathcal{R} = C - \mathcal{H}\frac{\delta\varphi}{\varphi'} = -\frac{\mathcal{H}}{\varphi'}Q. \tag{6.3.32}$$

This is the curvature perturbation on a comoving slicing so that $\mathcal{R}$ represents the gravitational potential on constant-φ hypersurfaces. Equation (6.3.9) can be used to establish that $\mathcal{R} = -\zeta$ if $k\eta \ll 1$.

End of inflation Since ζ is constant, one can relate the value of the gravitational potential on super-Hubble scales during inflation to its value during the radiation era, without worrying about the details of the transition between the two epochs. From (5.158), one concludes that

$$\Phi_{RDU}(k\eta \ll 1) = \frac{2}{3}\frac{1+\epsilon}{\epsilon}\Phi_{inf}(k\eta \ll 1). \tag{6.3.33}$$

$\mathcal{R}$ and ζ_{BST} are constant during the transition and the sudden change in the equation of state amplifies the gravitational potential.

In what follows, it will be more convenient to characterize scalar perturbations using one of these conserved quantities as they are not affected by the change in the equation of state, unlike the gravitational potential. ζ is a purely geometric quantity that characterizes density perturbations independently of the exact nature of matter and it allows us to propagate the spectrum of perturbations of the observable modes from inflation to the epoch where we would like to start cosmological computations. In particular, it allows the details of the description of the reheating phase to be avoided.

Junction conditions and their applications

Assuming that the transition between the inflationary phase and the radiation era is instantaneous, one can recover the previous results from junction conditions.

Junction conditions Let us start by determining the junction conditions that shall be satisfied when two space-times are 'glued' along a hypersurface Σ. The space-time embedding of this hypersurface into the (four-dimensional) space-time is defined by

$$x^\mu = \overline{x}^\mu(\sigma^a), \tag{6.3.34}$$

where the σ^a are the intrinsic coordinates and $\overline{x}^\mu$ the functions describing the embedding. The induced metric on the hypersurface is then

$$\gamma_{ab} = g_{\mu\nu}\frac{\partial\overline{x}^\mu}{\partial\sigma^a}\frac{\partial\overline{x}^\nu}{\partial\sigma^b} \tag{6.3.35}$$

This metric can be seen as a tensor defined in the space-time

$$\overline{\gamma}^{\mu\nu} = \gamma^{ab}\frac{\partial\overline{x}^\mu}{\partial\sigma^a}\frac{\partial\overline{x}^\nu}{\partial\sigma^b}, \tag{6.3.36}$$

called the first fundamental form. The Ricci tensor, R_{ab}, of the induced metric allows us to define

$$\overline{R}^{\mu\nu} = R^{ab}\frac{\partial\overline{x}^{\mu}}{\partial\sigma^{a}}\frac{\partial\overline{x}^{\nu}}{\partial\sigma^{b}} \tag{6.3.37}$$

The tensor $\overline{\gamma}^{\mu}_{\nu}$ acts as a projection tensor, projecting tensors at a point of Σ into tangential tensors and can be used to define the orthogonal projection operator by

$$\perp^{\mu}_{\nu} = \delta^{\mu}_{\nu} - \overline{\gamma}^{\mu}_{\nu}. \tag{6.3.38}$$

These definitions actually do not depend on the codimension of Σ. In the case of a hypersurface (i.e. codimension 1), there is a unique perpendicular direction so that $\perp^{\mu}_{\nu} = n^{\mu}n_{\nu}$, where n^{μ} is the unit vector normal to Σ (satisfying $n_{\mu}n^{\mu} = 1$ and $\overline{\gamma}_{\mu\nu}n^{\nu} = 0$) and we have $\gamma_{\mu\nu} = g_{\mu\nu} - n_{\mu}n_{\nu}$. We can then define the extrinsic curvature tensor

$$K_{\mu\nu} = -\overline{\gamma}^{\sigma}_{\nu}\overline{\gamma}^{\alpha}_{\mu}\nabla_{\alpha}n_{\sigma}. \tag{6.3.39}$$

If the vector field n^{μ}, initially defined only on Σ, is geodesically extended in the neighbourhood of the hypersurface (with $n^{\alpha}\nabla_{\alpha}n_{\mu} = 0$) then

$$K_{\mu\nu} = -\nabla_{\mu}n_{\nu}. \tag{6.3.40}$$

One can show in a general way, that two space-times can be glued along Σ if

$$[\overline{\gamma}_{\mu\nu}]_{\pm} = 0, \tag{6.3.41}$$

$$[K_{\mu\nu} - K\overline{\gamma}_{\mu\nu}]_{\pm} = \kappa\tau_{\mu\nu}, \tag{6.3.42}$$

where A_+ and A_- represent the values of A evaluated on each side of the hypersurface and $[A]_{\pm} \equiv A_+ - A_-$ is the jump of A across the hypersurface. The first equation implies that the induced metric is the same whether it is defined from one side or the other of Σ. In the second expression, $\tau_{\mu\nu}$ is the stress-energy tensor of matter fields localized on the hypersurface, including its tension when seen as a membrane.

Application in the context of cosmology For a cosmological transition, Σ is usually defined by

$$\Sigma = \{q = \text{const.}\}, \tag{6.3.43}$$

where q is a scalar function characterizing the hypersurface (i.e. $q = \rho$ for a constant-density hypersurface, or $q = \varphi$ for a constant-field hypersurface, etc.) The unit vector normal to Σ is then given by

$$n_{\mu} = \frac{\partial_{\mu}q}{\sqrt{-\partial_{\alpha}q\partial^{\alpha}q}}, \tag{6.3.44}$$

and we assume that the two space-times have metrics $ds^2_{\pm} = a^2_{\pm}(\eta_{\pm})\{-(1+2\Phi_{\pm})d\eta^2_{\pm} + [(1 - 2\Psi_{\pm})\gamma_{ij} + h^{\pm}_{ij}]dx^i dx^j\}$. Imposing the junction condition, we find:

$$\gamma^a_b = a^2\left\{\left[1 - 2\left(\Psi + \frac{\delta q}{q'}\right)\right]\delta^a_b + h^a_b\right\} \tag{6.3.45}$$

and

$$K^a_b = \frac{1}{a}\left\{-\mathcal{H}\delta^a_b + \left[\mathcal{H}\Phi + \Psi' + (\mathcal{H}' - \mathcal{H}^2)\frac{\delta q}{q'}\right]\delta^a_b + \frac{1}{2}h'^a_b + \partial^a\partial_b\frac{\delta q}{q'}\right\} \tag{6.3.46}$$

where δq is the perturbation of q in Newtonian gauge. The junction conditions (6.3.41)–(6.3.42) then take the form

$$\text{background metric: } [a]_{\pm} = 0, \quad [\mathcal{H}]_{\pm} = 0 \tag{6.3.47}$$

$$\text{scalar modes: } [\Psi]_\pm = 0, \quad \left[\frac{\delta q}{q'}\right]_\pm = 0, \quad \left[\Psi' + \mathcal{H}\Phi + \mathcal{H}'\frac{\delta q}{q'}\right]_\pm = 0 \tag{6.3.48}$$

$$\text{tensor modes: } [h_{ij}]_\pm = 0, \quad [h'_{ij}]_\pm = 0. \tag{6.3.49}$$

The first condition for the background metric tells us that the scale factor and expansion rate must be continuous. But note that the equation of state can be discontinuous, and so $\mathcal{H}'$ as well. To apply these conditions for perturbations, we should assume that the transition is fast compared to the typical oscillation time of the modes considered, which amounts to assuming that these mode are super-Hubble at the time of the transition.

End of inflation Let us apply these junction conditions to relate the end of inflation to the radiation era. Relation (6.3.47) allows us to conclude that the scale factors then take the form

$$a_-(\eta_-) = \frac{C}{(-\eta_-)^{1+\epsilon}}, \quad a_+(\eta_+) = \frac{C}{[(1+\epsilon)\eta_*^{1+\epsilon}]}\left(\frac{\eta_+}{\eta_*}\right) \tag{6.3.50}$$

if the transition happens at $\eta_+ = \eta_* = -(1-\epsilon)\eta_-$.

Assuming that the transition occurs on a constant-density hypersurface, $q = \rho$ and the conditions (6.3.48) imply that $[\zeta_{BST}]_\pm = 0$ and so $[\zeta + \Delta\Phi/3(\mathcal{H}' - \mathcal{H}^2)]_\pm = 0$. Using (5.117), we infer that for super-Hubble modes,

$$\left[\frac{3\mathcal{H}(\Phi' + \mathcal{H}\Phi)}{1+w}\right]_\pm = 0. \tag{6.3.51}$$

Note that if the transition surface had been chosen as a constant-field hypersurface, then we would have concluded $[\mathcal{R}]_\pm = 0$ directly. For super-Hubble modes, it is therefore equivalent to assuming that the end of inflation is a constant-field or a constant-density hypersurface.

In both eras, the gravitational potential is given by $\Phi_- = A_- + B_-(-\eta_-)$ and $\Phi_+ = A_+ + B_+/\eta_+^3$ respectively. The junction conditions imply that

$$A_+ = \frac{2}{3}\frac{1+\epsilon}{\epsilon}A_- \tag{6.3.52}$$

$$B_+ = \frac{1}{3}\frac{\epsilon-2}{\epsilon}A_+ \tag{6.3.53}$$

once the decaying mode during inflation has been neglected. Thus, we recover the previous result: $\Phi_+ \sim [2(1+\epsilon)/2\epsilon]\Phi_-$. However, note that the growing mode inherits a contribution from the gravitational potential during inflation (A_-) and a contribution from the density perturbations at the time of the transition $[(2-\epsilon)\Phi_-/\epsilon]$.

Figure 6.9: Junction between the inflationary phase and the post-inflationary radiation era for the super-Hubble modes. ζ and $\mathcal{R}$ are constant during the transition so that the amplification of the gravitational mode is associated with the appearance of a decaying mode, Φ_d that can then be neglected at late times.

Figure 8.10 illustrates how the growing mode of the gravitational potential can be amplified during inflation while keeping the curvature perturbation constant, at the price of exciting the decaying mode. Let us stress that in the de Sitter limit, $\epsilon \to 0$ and thus $\Phi \to 0$. In this case, perturbations of the post-inflationary era come entirely from perturbations of the surface of transition.

Quantization of the density perturbations

The previous section detailed the evolution of perturbations but did not discuss the choice of the initial state of the fields. To follow the example of the test scalar field, we should determine which are the variables to quantize. This aspect is detailed in Ref. [10] and we only recall the main steps of the construction.

Canonical variable To determine the canonical variables, let us first introduce the variable

$$v = aQ = -z\mathcal{R}. \tag{6.3.54}$$

where Q and $\mathcal{R}$ are the scalar field perturbation in the flat-slicing gauge and the curvature perturbation in comoving gauge, and where z is defined by

$$z \equiv \frac{a\dot{\varphi}}{H} = \frac{a\varphi'}{\mathcal{H}}. \tag{6.3.55}$$

The evolution equations for the perturbations have been obtained by a variational principle from the Einstein-Hilbert action and a minimally coupled scalar field. Since these equations are linear, they can be obtained from the same actions expanded to quadratic order. Starting from the expressions (6.3.54) and (6.3.55), one can show that the total action takes the form

$$\delta^{(2)}S = \frac{1}{2}\int d\eta d^3x \left[(v')^2 - \delta^{ij}\partial_i v \partial_j v + \frac{z''}{z}v^2\right] \equiv \int \mathcal{L} d^4x, \tag{6.3.56}$$

up to terms involving a total derivative and that do not contribute to the equations of motion.

This rather technical calculation is performed explicitly in references. Deriving the evolution equations to first order is strictly equivalent to varying the action expanded to second order, however, the second approach gives us more information on the structure of the theory. This action is that of a scalar field with a time-dependent mass, $m^2 = -z''/z$ in a Minkowski space-time. It is interesting to stress that the natural variable is neither Φ nor χ but v. Note that if we had started from the evolution equations [see, e.g., (6.3.22)], we could have wrongly concluded that u_s was the canonical variable. This equivalence implies that we can quantize this theory as we would quantize a scalar field evolving in a time-dependent exterior field, where here the time dependence would find its origin in the space-time dynamics. Note also the similarity with the case of a test scalar field in an expanding space-time, as considered in Section 8.3. It is therefore the field v that should be promoted to the status of quantum operator in second quantization,

$$\hat{v}(x,\eta) = \int \frac{d^3k}{(2\pi)^{3/2}}[v_k(\eta)e^{ik\cdot x}\hat{a}_k + v_k^*(\eta)e^{-ik\cdot x}\hat{a}_k^\dagger] \tag{6.3.57}$$

In what follows we will work in the Heisenberg representation, in which the time-dependence is carried by the operators.

The first step in performing a canonical quantization is to introduce the conjugate momentum of v,

$$\pi = \frac{\delta\mathcal{L}}{\delta v'} = v', \tag{6.3.58}$$

which is also promoted to the status of operator, $\hat{\pi}$. We can then get the Hamiltonian

$$H = \int (v'\pi - \mathcal{L})d^4x = \frac{1}{2}\int (\pi^2 + \delta^{ij}\partial_i v \partial_j v - \frac{z''}{z}v^2)d^4x. \tag{6.3.59}$$

The operators $\hat{v}$ and $\hat{\pi}$ have to satisfy canonical commutation relations on constant time hypersurfaces

$$[\hat{v}(x,\eta), \hat{v}(y,\eta)] = [\hat{\pi}(x,\eta), \hat{\pi}(y,\eta)] = 0, \quad [\hat{v}(x,\eta), \hat{\pi}(y,\eta)] = i\delta^{(3)}(x-y). \tag{6.3.60}$$

The equation of motion for v obtained from the Lagrangian (6.3.56) provides the field equation for the operator $\hat{v}$, which is similar to the Klein-Gordon equation in flat space-time,

$$\hat{v}'' - (\Delta + \frac{z''}{z})\hat{v} = 0. \tag{6.3.61}$$

Indeed, this is also the classical Klein-Gordon equation that can be derived from (6.3.12) after performing the change of variables (6.3.54) from Q to v. This equation is actually equivalent to the Heisenberg equations

$$\hat{v}' = i[\hat{H}, \hat{v}], \quad \hat{\pi}' = i[\hat{H}, \hat{\pi}], \tag{6.3.62}$$

where $\hat{H}$ is simply the Hamiltonian (6.3.59) in terms of $\hat{v}$ and $\hat{\pi}$.

As for quantization in Minkowski space-time (see Chapter 2), the creation and annihilation operators appearing in the decomposition (6.3.57) satisfy the standard commutation rules

$$[\hat{a}_k, \hat{a}_p] = [\hat{a}_k^\dagger, \hat{a}_p^\dagger] = 0, \quad [\hat{a}_k, \hat{a}_p^\dagger] = \delta^{(3)}(k - p). \tag{6.3.63}$$

These commutation rules are consistent with the commutation rules (6.3.60) only if the field v_k is normalized according to

$$W(k) \equiv v_k v'^*_k - v_k^* v_k' = i, \tag{6.3.64}$$

since

$$[\hat{v}(x,\eta), \hat{\pi}(y,\eta)] = \int \frac{d^3k}{(2\pi)^3} e^{ik\cdot(x'-x)} W(k).$$

This determines the normalization of their Wronskian, W. Then, one needs to construct the Fock representation of the Hilbert space on which the operators $\hat{v}$ and $\hat{\pi}$ act. In Minkowski space-time, the vacuum state is uniquely defined by the condition $\hat{a}_k|0\rangle = 0$ for all k and the other states are obtained by repeated action of the creation operator $\hat{a}_k^\dagger$. This amounts to keeping only negative-frequency modes. The vacuum state $|0\rangle$ is thus defined by the condition that this state is annihilated by all the operators $\hat{a}_k$,

$$\forall k, \quad \hat{a}_k|0\rangle = 0 \tag{6.3.65}$$

The sub-Hubble modes, i.e. the high-frequency modes compared to the expansion of the Universe, must behave as in a flat space-time. So the asymptotic condition to impose for a state in the vacuum is

$$v_k(\eta) \to \frac{1}{\sqrt{2k}} e^{-ik\eta}; \quad k\eta \to -\infty \tag{6.3.66}$$

Let us note that in an expanding space-time, the notion of time is fixed by the background evolution that provides a preferred direction. The notion of positive and negative frequency is not time invariant, which implies that during the evolution time of positive frequency will be generated. Note that this quantization procedure amounts to treating gravity in a quantum way to linear order since the gravitational potential is promoted to the status of operator.

Particle creation If we pick up an arbitrary initial time η_i, then

$$v_k(\eta_i) = \frac{1}{\sqrt{\omega(\eta_i, k)}}, \quad v_k'(\eta_i) = i\sqrt{\omega(\eta_i, k)}, \tag{6.3.67}$$

defines the positive frequency solutions at η_i, indeed for oscillating modes. We can thus define a vacuum state, $|0_i\rangle$ following the previous procedure. At a later time $\eta_1 > \eta_i$, the modes defined by (6.3.67) will no longer be positive frequency and an observer at η_1 will define another vacuum state $|0_1\rangle$ satisfying $\hat{c}_k|0_1\rangle = 0$ for all k, $\hat{c}_k$ being the annihilation operators for the positive-frequency modes at η_1. Since the mode equation is linear, it is clear that the positive-frequency modes at η_1, w_k, are related to mode functions v_k by a linear transformation

$$w_k = \alpha_k v_k + \beta_k v_k^* \tag{6.3.68}$$

with the two complex coefficients satisfying $|\alpha_k|^2 - |\beta_k|^2 = 1$. The number of particles of mode k at η_1 will thus be defined by $N_k^{(1)} = \hat{c}_k^\dagger \hat{c}_k$. Since the two sets of creation and annihilation operators are related by $\hat{a}_k = \alpha_k \hat{c}_k + \beta_k^* \hat{c}_k^\dagger$ implying that $\hat{c}_k = \alpha_k^* \hat{a}_k - \beta_k^* \hat{a}_k^\dagger$, we deduce that the state $|0_i\rangle$ is seen as a state containing $\langle 0_i|N_k^{(1)}|0_i\rangle = |\beta_k|^2$ particles by the observer at η_1. This illustrates the time dependence of the notion of positive frequency and vacuum state. Fortunately the ambiguity of the definition of the vacuum does not make important differences for small wavelength modes for which the WKB approximation holds.

Relation with the perturbation variables The previous definitions allow us to conclude that

$$z = \sqrt{\frac{3}{\kappa}}\theta^{-1} \tag{6.3.69}$$

and $\Delta u_s = -z(v/z)'$. Einstein's equations imply that the gravitational potential is given by

$$\Delta\Phi = \frac{\kappa}{2}\frac{\varphi'^2}{\mathcal{H}}\left(\frac{v}{z}\right)' \quad \text{and} \quad \left(\frac{a^2\Phi}{\mathcal{H}}\right)' = \frac{\kappa}{2}zv, \tag{6.3.70}$$

and we also have the useful relation

$$2\mathcal{H}X = \frac{\kappa}{a}\varphi' v, \tag{6.3.71}$$

where X has been defined in (5.67).

Following the same procedure as for a test scalar field considered in Section 8.3, (6.3.54) implies that the power spectrum of the metric perturbations is

$$\mathcal{P}_{\mathcal{R}} = \mathcal{P}_\zeta = \frac{k^3}{2\pi^2}\left|\frac{v_k}{z}\right|^2 \tag{6.3.72}$$

using the definitions of Section B.5 of Appendix B for the normalization of the power spectrum. The solution of (6.3.61) with initial conditions (6.3.66) allows us to determine the gravitational potential and the power spectrum of $\mathcal{R}$ and ζ. The analysis of Section 8.4.5 then allows us to determine the curvature perturbation and the gravitational potential at the beginning of the post-inflationary phase.

Gravitational waves

Quantization The same procedure can be followed for gravitational waves. Restricting ourselves to tensor degrees of freedom, the action at second order in perturbations takes the form

$$\delta^{(2)}S = \frac{M_P^2}{64\pi}\int a^2(\eta)[(h_{ij}')^2 - (D_k h_{ij})^2 - 2K(h_{ij})^2]\sqrt{\gamma}d^3x d\eta. \tag{6.3.73}$$

Unsurprisingly, the variation of this action gives the equation of propagation (6.3.16). Using the decomposition (6.3.17), and defining

$$\mu_\lambda(k,\eta) = \sqrt{\frac{M_P^2}{8\pi}}a(\eta)\overline{E}_\lambda(k,\eta) = \sqrt{\frac{M_P^2}{32\pi}}a(\eta)h_\lambda(k,\eta) \tag{6.3.74}$$

the action at second order takes the form

$$\delta^{(2)}S = \frac{1}{2}\sum_\lambda\int[(\mu_\lambda')^2 - \gamma^{ij}\partial_i\mu_\lambda\partial_j\mu_\lambda - 2K\mu_\lambda^2 + \frac{a''}{a}\mu_\lambda^2]\sqrt{\gamma}d^3x d\eta. \tag{6.3.75}$$

In this form, the action for the gravitational waves reduces to the action for two independent scalar fields, μ_λ, with time-dependent mass $m^2 = -a''/a$ evolving in a static space-time. We can therefore proceed as for the scalar perturbations and quantize these two degrees of freedom. In the Heisenberg representation, we promote μ_λ to the status of quantum operator

$$\hat{\mu}_\lambda(x,\eta) = \int \frac{d^3k}{(2\pi)^{3/2}}[\mu_{k,\lambda}(\eta)e^{ik\cdot x}\hat{a}_{k,\lambda} + \mu^*_{k,\lambda}(\eta)e^{-ik\cdot x}\hat{a}^\dagger_{k,\lambda}] \tag{6.3.76}$$

If the modes $\mu_{k,\lambda}$ satisfy the normalization (6.3.64), then the creation and annihilation operators satisfy the commutation relations

$$[\hat{a}_{k,\lambda},\hat{a}_{p,\lambda'}] = [\hat{a}^\dagger_{k,\lambda},\hat{a}^\dagger_{p,\lambda'}] = 0, \quad [\hat{a}_{k,\lambda},\hat{a}^\dagger_{p,\lambda'}] = \delta_{\lambda\lambda'}\delta^{(3)}(k-p), \tag{6.3.77}$$

and for each polarization, the vacuum is defined by

$$\forall k, \quad \hat{a}_{k,\lambda}|0\rangle = 0, \tag{6.3.78}$$

and we set the initial conditions in the same way as for scalar perturbations by imposing

$$\mu_{k,\lambda}(\eta) \to \frac{1}{\sqrt{2k}}e^{-ik\eta}; \quad k\eta \to -\infty \tag{6.3.79}$$

The evolution equation for $\mu_{k,\lambda}$, that we will write simply as μ_k when there is no ambiguity, is

$$\mu''_k + (k^2 + 2K - \frac{a''}{a})\mu_k = 0. \tag{6.3.80}$$

As previously, this equation can be solved by using the WKB approximation (6.2.17) with $\omega^2_\lambda(k,\eta) = k^2 + 2K - a''/a$.

Power spectrum The power spectrum of gravitational waves is obtained by solving the evolution equation (6.3.80) with initial conditions (6.3.62). This allows us to define the amplitude of super-Hubble tensor modes, which remains constant. The power spectrum, $\mathcal{P}_T$, of the tensor modes is defined from the variable h_{ij} and is related to the power spectrum $\mathcal{P}_{\overline{E}}$ of one polarization E_λ by

$$\mathcal{P}_T = 8\mathcal{P}_{\overline{E}} \tag{6.3.81}$$

where the factor 8 arises from the sum over polarizations. Because of isotropy, the two polarizations have the same spectrum $\mathcal{P}_{\overline{E}}$ defined by

$$\langle \overline{E}_\lambda(k,\eta_f)\overline{E}^*_{\lambda'}(k',\eta_f)\rangle = \frac{2\pi^2}{k^3}\mathcal{P}_T(k)\delta^{(3)}(k-k')\delta_{\lambda\lambda'}, \tag{6.3.82}$$

with

$$\mathcal{P}_{\overline{E}} = \frac{k^3}{2\pi^2}\frac{8\pi}{M_P^2}\left|\frac{\mu_k}{a}\right|^2 \tag{6.3.83}$$

for each polarization. Thus, we conclude that

$$\mathcal{P}_T = \frac{k^3}{2\pi^2}\frac{64\pi}{M_p^2}\left|\frac{\mu_k}{a}\right|^2 \tag{6.3.84}$$

where each polarization contributes in an identical way.

Gravitational wave density The variation of the action (6.3.73) with respect to the metric allows us to define the stress-energy tensor for the gravitational waves

$$t_{\mu\nu} = \frac{1}{16\pi G_N}\sum_\lambda(\partial_\mu \overline{E}_\lambda \partial_\nu \overline{E}_\lambda - \overline{g}_{\mu\nu}\partial_\alpha \overline{E}_\lambda \partial^\alpha \overline{E}_\lambda). \tag{6.3.85}$$

The gravitational wave amplitude at any time after the inflationary phase can be decomposed as

$$\overline{E}_\lambda(k,\eta) = T(k,\eta)\overline{E}_\lambda(k,\eta_f) \tag{6.3.86}$$

where $T(k,\eta)$ is a transfer function obtained by solving (5.109). The gravitational wave density can then be defined as $\rho_{\overline{GW}} = -\langle t_0^0(x,\eta)\rangle$. As soon as the mode becomes sub-Hubble, it oscillates and one can average over a few periods to extract its amplitude. This procedure is well defined as long as the amplitude varies slowly with respect to the wave period. In practice, this amounts to replacing $T(k,\eta)$ by $\overline{T}(k,\eta)$, which is the average value of T over a few periods. We then obtain the energy density per decade

$$\frac{d\rho_{GW}}{d\ln k} = \frac{1}{8\pi G_N}\left(\frac{k}{a}\right)^2 \mathcal{P}_T(k)\overline{T}(k,\eta), \tag{6.3.87}$$

or equivalently in terms of the density parameter,

$$\frac{d\Omega_{GW}}{d\ln k} = \frac{1}{3}\left(\frac{k}{aH}\right)^2 \mathcal{P}_T(k)\overline{T}(k,\eta). \tag{6.3.88}$$

As long as the mode is super-Hubble, its amplitude is constant (see Chapter 5) and then it enters a damped oscillation regime on sub-Hubble scales. The damping between the time where the mode becomes super-Hubble and today is typically of order $\overline{T}(k,\eta) \simeq (a_k/a_0)^2$ where a_k is the value of the scale factor when $k = aH$. We infer that the gravitational wave spectrum today is

$$\frac{d\Omega_{GW}}{d\ln k} \propto k^2 a_k^2 \mathcal{P}_T(k). \tag{6.3.89}$$

If the mode becomes sub-Hubble during the matter-dominated era, $a_k \propto k^{-2}$, whereas during the radiation era $a_k \propto k^{-1}$. We infer that

$$\frac{d\Omega_{GW}}{d\ln k} \propto \mathcal{P}_T(k) \times \begin{cases} k^{-2} & k < k_{eq} \\ k^0 & k > k_{eq} \end{cases} \tag{6.3.90}$$

This contribution is represented in Fig. 11.15 for a primordial scale invariant spectrum, $\mathcal{P}_T \propto k^0$.

6.3.4 Perturbations in the slow-roll regime

We will now apply the previous formalism to inflation in the slow-roll regime. In this approximation, one only needs to determine the functions $z(\eta)$ and $a(\eta)$ that then allow for the integration of (6.3.61) and (6.3.80). It is useful to note already at this early stage that

$$z = a\frac{M_P}{\sqrt{4\pi}}\sqrt{\epsilon}. \tag{6.3.91}$$

An exact solution: power-law inflation

Before studying the general case, it is useful to consider the case of power-law inflation (6.1.75) for which the slow-roll parameters remain constant, $\epsilon = \delta = 1/p$ and $\xi = 0$. In this case, the expression (6.2.33) is exact, so that

$$aH = \mathcal{H} = -\frac{1}{1-\epsilon}\frac{1}{\eta} \tag{6.3.92}$$

and, as for a de Sitter space-time, η varies between $-\infty$ and 0.

Scalar modes The conformal time evolution of the scale factor (6.1.75) and scalar field (6.1.77) are determined explicitly. The variable z, defined by (6.3.55), is then given by $z = \sqrt{p\pi}\eta^{1+\beta}$, so that

$$\frac{z''}{z} = \frac{\beta(\beta+1)}{\eta^2} \tag{6.3.93}$$

$$\nu = -\beta - \frac{1}{2} = \frac{3}{2} + \frac{1}{p-1} \tag{6.3.94}$$

Equation (6.3.61) then takes the form of a Bessel equation

$$v_k'' + \left(k^2 - \frac{\nu^2 - 1/4}{\eta^2}\right) v_k = 0, \tag{6.3.95}$$

The solution of this equation that satisfies the initial condition (6.3.66) can be expressed in terms of a Hankel function as

$$v_k(\eta) = \frac{\sqrt{\pi}}{2} e^{i(\nu+1/2)\pi/2} \sqrt{-\eta} H_\nu^{(1)}(-k\eta). \tag{6.3.96}$$

For super-Hubble modes ($k/\mathcal{H} \ll 1$), this solution tends towards

$$v_k(\eta) \to 2^{\nu-3/2} \frac{\Gamma(\nu)}{\Gamma(3/2)} e^{i(\nu-1/2)\pi/2} \frac{1}{\sqrt{2k}} (-k\eta)^{-\nu+1/2} \tag{6.3.97}$$

The expression (6.3.72) allows us to find that the power spectrum of curvature perturbations is given by

$$\mathcal{P}_\zeta = \frac{H^2}{\pi M_p^2 \epsilon} \left[2^{\nu-3/2} \frac{\Gamma(\nu)}{\Gamma(3/2)}\right]^2 (\nu - \frac{1}{2})^{-2\nu+1} \left(\frac{k}{aH}\right)^{-2\nu+3}. \tag{6.3.98}$$

The coefficient $(\nu - 1/2)^{-2\nu+1}$ comes from the contribution $(1-\epsilon)^{2\nu-1}$ that appears when replacing $k\eta$ using (6.3.92). If we evaluate this spectrum at $aH = k$, it can then be rewritten as

$$\mathcal{P}_\zeta = \frac{1}{\pi M_p^2} \left[2^{\nu-3/2} \frac{\Gamma(\nu)}{\Gamma(3/2)}\right] (\nu - \frac{1}{2})^{-2\nu+1} \left(\frac{H^2}{\epsilon}\right)_{k=aH}. \tag{6.3.99}$$

which hides the k dependence in the term H^2/ϵ evaluated at the time where the mode k becomes super-Hubble.

Gravitational waves The study of gravitational waves is in this case identical to that of the scalar modes since in this particular model of inflation $z \propto a$. Equation (6.3.80) is thus identical to (6.3.61) and the solution for μ_k is given by (6.3.96). We infer from (6.3.84) that

$$\mathcal{P}_T = \frac{64\pi}{M_p^2} \left(\frac{z}{a}\right)^2 \mathcal{P}_\zeta \tag{6.3.100}$$

The gravitational wave power spectrum is thus given by

$$\mathcal{P}_T = 16\epsilon \mathcal{P}_\zeta. \tag{6.3.101}$$

Spectral indices Note that the amplitude of gravitational waves is determined only by the energy scale, H, during the slow-roll regime, while that of the scalar modes depends on both H and the slow-roll parameter ϵ. The spectral indices, defined by

$$n_s - 1 \equiv \frac{d \ln \mathcal{P}_\zeta}{d \ln k}, \quad n_T \equiv \frac{d \ln \mathcal{P}_T}{d \ln k} \tag{6.3.102}$$

are, respectively, given by

$$n_s = -2\nu + 4 = 2\beta + 5, \quad n_T = -2\nu + 3 = 2\beta + 4. \tag{6.3.103}$$

Note that when $\epsilon \ll 1$, i.e. when $p \gg 1$, then $n_T = -2/(p-1) \sim -2/p = -2\epsilon$ to first order in the slow-roll parameters. We deduce the consistency relation to lowest order in the slow-roll parameters

$$\frac{\mathcal{P}_T}{\mathcal{P}_\zeta} = -8n_T. \tag{6.3.104}$$

General case

In the general case, the slow-roll parameters are time dependent and the evolution equations cannot be integrated exactly. We should thus construct an expansion scheme allowing us to approximate the solution.

Expansion scheme The case of power-law inflation that we considered previously corresponds to constant slow-roll parameters. In an arbitrary model, these parameters vary and their dynamics is dictated, to lowest order, by (6.1.47) and (6.3.61), and (6.3.80) cannot be solved analytically. In order to make an expansion in terms of the slow-roll parameters, we use (6.1.47) to replace the time dependence of ϵ to leading order in all equations, in order to identify the negligible terms. Relation (6.2.32) can then be generalized by performing successive integration by parts and expanding η as a function of $\mathcal{H}$, namely

$$\eta = \int \frac{da}{a\mathcal{H}} = -\frac{1}{\mathcal{H}} + \int \frac{\epsilon da}{a\mathcal{H}} = -\frac{1}{\mathcal{H}} + \epsilon \int \frac{da}{a\mathcal{H}} - \int \frac{\epsilon' da}{a\mathcal{H}^2}, \tag{6.3.105}$$

to obtain

$$\eta = -\frac{1}{\mathcal{H}}\left[\frac{1}{1-\epsilon} - 2\epsilon(\epsilon - \delta)\right] + \dots. \tag{6.3.106}$$

Assuming that ϵ and δ are small parameters, then to first order we can consider them to be constant (since ϵ' is of second order) so that

$$\eta = -\frac{1}{\mathcal{H}}(1+\epsilon) + \mathcal{O}(2). \tag{6.3.107}$$

We will distinguish the leading-order expansion in which only the lowest powers of the slow-roll parameters are conserved from the next-order expansion that contains the next to leading-order corrections.

Expansion of the functions z''/z and a''/a The two functions z''/z and a''/a are essential for solving (6.3.61) and (6.3.80). Starting from (6.3.55), we obtain

$$\frac{z'}{z} = \mathcal{H} + \frac{\varphi''}{\varphi'} - \frac{\mathcal{H}'}{\mathcal{H}} = \mathcal{H}(1 + \epsilon - \delta) \tag{6.3.108}$$

where the second equality uses relations (6.1.49). After differentiating this equation and using relations (6.1.47) and (6.1.49), we find that

$$\begin{aligned}\frac{z''}{z} &= \mathcal{H}^2[2 + 2\epsilon - 3\delta + \delta^2 - \epsilon\delta + \xi] \\ &= \mathcal{H}^2[2 + 2\epsilon - 3\delta + 2\epsilon^2 + \delta^2 - 4\epsilon\delta + \xi_H^2],\end{aligned} \tag{6.3.109}$$

this result being exact. In an identical way, since $a''/a = \mathcal{H}' + \mathcal{H}^2$, we obtain

$$\frac{a''}{a} = (2 - \epsilon)\mathcal{H}^2. \tag{6.3.110}$$

Relation (6.3.107) then allows us to express these quantities in terms of the conformal time. To first order, we obtain

$$\frac{z''}{z} = \frac{2+6\epsilon-3\delta}{\eta^2} + \mathcal{O}(2), \qquad \frac{a''}{a} = \frac{2+3\epsilon}{\eta^2} + \mathcal{O}(2). \tag{6.3.111}$$

Scalar modes As in the case of power-law inflation, (6.3.61) reduces to a Bessel equation of the form (6.3.95) with

$$\nu = \frac{3}{2} + 2\epsilon - \delta. \tag{6.3.112}$$

The solution that satisfies the initial conditions (6.3.66) is given by (6.3.96). For super-Hubble modes, we should then expand this solution, keeping the leading-order term and the next-order ones. Using $\Gamma(a+h)/\Gamma(a) \sim 1 + h\psi(a)$ and $2^h \sim 1 + h\ln 2$ for $h \ll 1$, we obtain

$$\mathcal{P}_\zeta = \frac{1}{\pi}\frac{H^2}{M_p^2\epsilon}[1-2(2C+1)\epsilon+2C\delta]\left(\frac{k}{aH}\right)^{2\delta-4\epsilon} \tag{6.3.113}$$

where $C = \gamma_E + \ln 2 - 2$ and $\gamma_E \sim 0.5772$ the Euler constant. The spectrum in the slow-roll regime takes the two equivalent forms,

$$\begin{aligned}\mathcal{P}_\zeta &= \frac{1}{\pi}\frac{H^2}{M_p^2\epsilon}\left[1-2(2C+1)\epsilon+2C\delta+(2\delta-4\epsilon)\ln\left(\frac{k}{aH}\right)\right] \\ &= \frac{1}{\pi}[1-2(2C+1)\epsilon+2C\delta]\left(\frac{H^2}{M_p^2\epsilon}\right)_{k=aH}.\end{aligned} \tag{6.3.114}$$

Gravitational waves As in the case of power-law inflation, (6.3.80) reduces to a Bessel equation of the form (6.3.95) with

$$\nu = \frac{3}{2} + \epsilon. \tag{6.3.115}$$

The solution that satisfies the initial conditions (6.3.66) is given by (6.3.96). For super-Hubble modes, this solution should then be expanded, keeping the leading and next to leading order terms exactly in the same way as in the previous section. By noticing that $\mathcal{P}_T = 16\epsilon\mathcal{P}_\zeta^{[\delta=\epsilon]}$, we easily obtain

$$\mathcal{P}_T = \frac{16}{\pi}\frac{H^2}{M_p^2}[1-2(C+1)\epsilon]\left(\frac{k}{aH}\right)^{-2\epsilon} \tag{6.3.116}$$

We then infer that the spectrum in the slow-roll regime takes the two equivalent forms,

$$\begin{aligned}\mathcal{P}_T &= \frac{16}{\pi}\frac{H^2}{M_p^2}[1-2(C+1)\epsilon-2\epsilon\ln\left(\frac{k}{aH}\right)] \\ &= \frac{16}{\pi}[1-2(C+1)\epsilon]\left(\frac{H^2}{M_p^2}\right)_{k=aH}.\end{aligned} \tag{6.3.117}$$

Consistency relation at leading order The spectral index of scalar perturbations appears in (6.3.113),

$$n_s - 1 = \frac{d\ln\mathcal{P}_\zeta}{d\ln k} = 2\delta - 4\epsilon. \tag{6.3.118}$$

The spectral index of the tensor modes appears in (6.3.116)

$$n_T = \frac{d\ln\mathcal{P}_T}{d\ln k} = -2\epsilon. \tag{6.3.119}$$

These results summarize one of the generic predictions of inflation. In the slow-roll regime, inflation predicts that density perturbations and gravitational waves develop super-Hubble correlations with nearly scale invariant power spectra. Each spectrum is characterized by its amplitude and its spectral index. Even though the absolute amplitude is arbitrary and is fixed by the energy scale of inflation, H, the relative amplitudes of the scalar and tensor modes must satisfy the relation

$$R \equiv \frac{A_T^2}{A_s^2} = \epsilon. \tag{6.3.120}$$

This implies a consistency relation between the amplitudes of the two spectra and their spectral indices,

$$R = -\frac{n_T}{2} \tag{6.3.121}$$

This relation is incontrovertible and must be satisfied to lowest order by any model of single-field inflation in the slow-roll regime. It is a distinctive signature of inflation and it is difficult to imagine another mechanism that would lead to such an equality.

Consistency relation at next to leading order At next order, the k dependence of the slow-roll parameters should be taken into account. Using (6.1.57) and (6.1.58), we obtain from the logarithmic derivatives of (6.3.114) and (6.3.117)

$$n_s - 1 = 2\delta - 4\epsilon - [8(C+1)\epsilon^2 - (6+10C)\epsilon\delta + 2C\xi_H^2], \tag{6.3.122}$$

$$n_T = -2\epsilon[1 + (3+2C)\epsilon - 2(1+C)\delta]. \tag{6.3.123}$$

By replacing H' by its expression in the ratio between A_s and A_T we find that

$$\epsilon = R[1 - 2C(\epsilon - \delta)], \tag{6.3.124}$$

which corresponds to (6.3.117) at next order. Moreover, from the ratio between (6.3.106) and (6.3.111) we find that

$$n_T = -2R(1 + 3\epsilon - 2\delta). \tag{6.3.125}$$

After replacing ϵ and δ in the brackets by their values at leading order, we obtain the consistency relation at next to leading order

$$n_T = -2R[1 - R^2 + (1 - n_s)]. \tag{6.3.126}$$

It is remarkable that this expression does not involve any derivatives of the spectral indices. Moreover, since the scalar modes are easier to measure than the gravitational waves, it is probable that if n_T is known then so is n_s, so that if the consistency relation can be observationally verified at leading order, one can also verify it at next order without any additional effort.

As shown by relations (6.3.119) and (6.3.120), at this order n_s and n_T depend on the wave number k. We can therefore define

$$\alpha_s \equiv \frac{dn_s}{d\ln k}, \quad \alpha_T \equiv \frac{dn_T}{d\ln k} \tag{6.3.127}$$

which characterize the running of the spectral indices. Using (6.1.47) we find that

$$\frac{d\epsilon}{d\ln k} = 2\epsilon(\delta - \epsilon), \quad \frac{d\delta}{d\ln k} = 2\epsilon(\epsilon - \delta) - \xi. \tag{6.3.128}$$

At lowest order, α_s and α_T are obtained by considering the value of the spectral indices at leading order so that

$$\alpha_s = -4\epsilon(\epsilon - \delta) - 2\xi, \quad \alpha_T = -4\epsilon(\epsilon - \delta). \tag{6.3.129}$$

This implies the existence of a second consistency relation

$$\alpha_T = 2R[2R + (n_s - 1)]. \tag{6.3.130}$$

The measurement of α_T remains for now far from any observational possibilities, and so within the present state of technical possibilities, this relation is not yet very interesting. Table 6.1 summarizes the results obtained for the spectral indices of inflationary spectra in the slow-roll regime. Note also that if these parameters are measured at a given so-called pivot scale, k_*, then the expansion of these spectra takes the form

$$\ln A_s^2(k) = \ln A_s^2(k_*) + [n_s(k_*) - 1]\ln\frac{k}{k_*} + \frac{1}{2}\alpha_s \ln^2\frac{k}{k_*} + ..., \tag{6.3.131}$$

and an analogous version for A_T^2. k_* is usually chosen as the logarithmic mean of the observable frequency band, typically $k_* \sim 0.01h$ Mpc^{-1}.

Table 6.1: Summary of the properties of the scalar modes and gravitational waves power spectra.

	Leading order	Next to leading order
R	ϵ	$R[1 - 2C(\epsilon - \delta)]$
$n_s - 1$	$2\delta - 4\epsilon$	$2\delta - 4\epsilon - [8(C+1)\epsilon^2 - (6+10C)\epsilon\delta + 2C\xi_H^2]$
n_T	-2ϵ	$-2\epsilon[1 + (3+2C)\epsilon - 2(1+C)\delta]$
Consistency	$n_T = -2R$	$n_T = -2R[1 - R^2 + (1 - n_s)]$
α_s	0	$-4\epsilon(\epsilon - \delta) - 2\xi$
α_T	0	$-4\epsilon(\epsilon - \delta)$
Consistency	—	$\alpha_T = 2R[2R + (n_s - 1)]$

6.3.5 Relation to observations

As long as we work in the slow-roll regime, perturbations are characterized by the two primordial spectra A_T and A_s. We briefly review what observables can teach us about them. For super-Hubble modes, ζ is constant during the evolution. Computations of the matter power spectrum and temperature anisotropies of the cosmic microwave background require the knowledge of the spectrum of Φ. Equation (5.149) allows us to obtain

$$\Phi(k\eta \ll 1, \eta < \eta_{eq}) = \frac{2}{3}\zeta, \quad \Phi(k\eta \ll 1, \eta > \eta_{eq}) = \frac{3}{5}\zeta. \tag{6.3.132}$$

Cosmic microwave background

The previous results fix the initial conditions for the computation of the angular power spectrum of the cosmic microwave background anisotropies (Chapter 6). Restricting ourselves to large angular scales, i.e. to small multipoles, the angular spectrum of the scalar modes is given by (6.50), using $\delta T/T \sim \frac{1}{3}\Phi$ for adiabatic perturbations at long wavelengths. Since $\Phi = \frac{3}{5}\zeta$ we infer that

$$C_l^S \simeq \pi A_s^2(k_0)\left[\frac{\pi}{2}\frac{\Gamma(3-n_s)}{2^{3-n_s}}\frac{\Gamma(l+\frac{n_s-1}{2})}{\Gamma^2(2-\frac{n_s}{2})\Gamma(l+\frac{5-n_s}{2})}\right], \tag{6.3.133}$$

if we choose the pivot $k_* = k_0 = 1/\eta_0$. For $n_s \sim 1$, we obtain $l(l+1)C_l^S = \pi A_s^2(k_0)/2$. Observations of the anisotropies by the satellite WMAP impose that

$$Q_{\text{rms-PS}} \equiv \Theta_0\sqrt{\frac{5C_2^S}{4\pi}} = 18 \times 10^{-6}\ \text{K}. \tag{6.3.134}$$

For $\Theta_0 = 2.73$ K we infer that $A_s^2(k_0) = 4.16 \times 10^{-10}$. Using (6.3.91) and (6.1.50) we infer the energy scale of inflation

$$\left(\frac{V}{\epsilon}\right)^{1/4} \simeq 0.0047 M_P \simeq 1.5 \times 10^{17} \text{ GeV}. \tag{6.3.135}$$

This value is actually an upper bound as part of the perturbations can be of tensorial origin. Performing the same computation for gravitational waves, the consistency relation becomes

$$r = \frac{C_2^S}{C_2^T} = f(h, \Omega, \Omega_\Lambda, ...)\epsilon. \tag{6.3.136}$$

The function f depends on a given cosmological model. For the flat standard model with $\Omega_{\Lambda 0} = 0.7$, $f \sim 10$. Analysis from the WMAP data provides the constraint on

$$\epsilon < 0.032, \tag{6.3.137}$$

which implies

$$\left(\frac{H_*}{M_P}\right) \leq 1.4 \times 10^{-5} \iff \left(\frac{V_*}{M_P^4}\right)^{1/4} \leq 2.2 \times 10^{-3}. \tag{6.3.138}$$

The reanalysis of the WMAP data leads to the constraints that $R < 0.0267$ while $16R < 0.0125$ when combined with baryon acoustic oscillations and supernovae data for a tensor-to-scalar ratio at $k = 0.002$ Mpc^{-1}. The scalar spectral index is, respectively, obtained to be $n_s = 0.963^{+0.014}_{-0.015}$ and $n_s = 0.960^{+0.014}_{-0.013}$ at a 68% confidence level.

Link with the number of e-folds

Observations allow us to establish constraints on the pair $(r, n_s - 1)$ that can be related to the slow-roll parameter space. These quantities depend on the form of the potential but also on the number of e-folds, N, between the time when the largest observable scales become super-Hubble and the end of inflation.

To illustrate this, let us consider a model of inflation with large field values (model of type A), for instance, with a power law potential (6.1.65). The expressions (6.1.66) and (6.3.113) allow us to establish that

$$r = \frac{f}{2}\left(\frac{n}{n+2}\right)(1 - n_s). \tag{6.3.139}$$

Inverting (6.1.67) and (6.1.66) for ϵ, we obtain

$$r = f\frac{n}{4(N + n/4)}, \quad 1 - n_s = \frac{n+2}{2(N + n/4)}. \tag{6.3.140}$$

For a given model, for example, $n = 4$, the observable quantities r and $n_s - 1$ depend on the number of e-folds. Figure 6.10 illustrates the position of one of these models in the $(n_s - 1, r)$-plane. As N is increased the model gets closer and closer to the point $(n_s - 1, r) = (0, 0)$. This figure also summarizes the constraints on the two parameters obtained from WMAP data.

Figure 6.10: Constraints on single-field chaotic inflationary models with potentials of the form φ^n with $n = 2$ (dashed), $n = 4$ (solid) and on N-flation with φ^2 potential (dotted). HZ is the prediction for a strictly scale invariant power spectrum. The predictions for $N = 50$ and $N = 60$ e-folds have been plotted. The φ^4 models are excluded at 95% CL.

Given observational constraints in this plane, the viability of a model depends on the number of e-folds. The previous example shows that predictions from a model of chaotic inflation depend crucially on N. The latest analysis of WMAP concluded that a single-field model with $V = \lambda\varphi^4/4$ is far from the 95% confidence level (CL) region for both $N = 50$ and $N = 60$. A massive free-field model is out of the 68% CL region for $N = 50$ and at the boundary of this region for $N = 60$ while being inside the 95% CL region. For a power-law inflation model, $R = 1/p$ and $1 - n_s = 2/p$ so that $p < 60$ is excluded at more than 99% CL.

The symmetry with respect to the end of inflation (for $\eta < \eta_e$, the modes exit the Hubble radius and for $\eta > \eta_e$, the modes enter the Hubble radius, see Fig. 8.5) allows us to put an upper bound on N from H_e and H_0. One can show that

$$N < 60.9 + \frac{1}{2}\ln\left(\frac{H_*}{10^{15}\ \text{GeV}}\right) - \ln\left(\frac{k}{0.02\ \text{Mpc}^{-1}}\right). \tag{6.3.141}$$

Using $H_*/M_P \leq 2 \times 10^{-5}$ and that the largest observable scale is $k = a_0 H_0$ we can then estimate that

$$N < 62.5 + \ln\left(\frac{0.6}{h}\right). \tag{6.3.142}$$

A precise determination of N is important for a good understanding of the end of inflation. An upper bound for the energy scale of inflation can be obtained by assuming that the part of the angular power spectrum that corresponds to the Sachs-Wolfe plateau is due to gravitational waves only. We then obtain $V_*^{1/4} < 3 \times 10^{16}$ GeV.

Consistency relation

In order to check the consistency relation, the detection of primordial gravitational waves is necessary. This is beyond current observations. We summarize here the constraints obtained on the contributions of the gravitational waves. Analysis from WMAP data has established that

$$\frac{C_{10}^T}{C_{10}^S} < 0.32 \tag{6.3.143}$$

so that gravitational waves do not contribute more than 30% of the cosmic microwave background. Relation (6.1.43) allows us to establish that during an interval of ΔN number of e-folds, the field has moved by

$$\frac{\Delta\varphi}{M_P} = \sqrt{\frac{\epsilon}{4\pi}}\Delta N, \tag{6.3.144}$$

which, using (6.3.136), can be re-expressed as

$$\frac{\Delta\varphi}{M_P} = \sqrt{\frac{16\pi r}{f}}\Delta N \sim 2.37\left(\frac{r}{0.07}\right)^{1/2}\left(\frac{f}{10}\right)^{-1/2}\left(\frac{\Delta N}{4}\right). \tag{6.3.145}$$

This estimate allows us to conclude that if $r \sim 0.1$, then the field must have varied by more than M_P. This result has often been used to argue that realistic potentials for inflation imply that r is very small. Due to cosmic variance, the contribution of gravitational waves to temperature anisotropies can only be distinguished if $r > 0.07$. Primordial gravitational waves generate anisotropies in the B polarization (Chapter 6). These B modes are also generated from the E polarization by gravitational lensing by the large-scale structures. This foreground effect limits the detection of primordial gravitational waves to $r > 6 \times 10^{-4}$, which implies that the energy scale of inflation has to be larger than

$$V_*^{1/4} > 3.2 \times 10^{15}\ \text{GeV} \iff \left(\frac{V_*}{M_P^4}\right)^{1/4} \geq 2.62 \times 10^{-4}. \tag{6.3.146}$$

This is only one order of magnitude smaller than the current limit (6.3.138). This limit can be lowered by a factor of 2 if the variation of the spectral index is measured. These considerations show that the consistency relation will be difficult to verify observationally.

Reconstruction of the potential

The measurement of n_s, n_T, R, α_s and α_T must, in principle, make it possible to reconstruct the form of the potential in the neighbourhood of φ_*, the value of φ at the moment where the pivot becomes super-Hubble. To see this, one should establish the link between the expansion (6.3.131) and a Taylor expansion of the potential around φ_*,

$$V = V_* + V'_*\Delta\varphi + \frac{1}{2}V''_*(\Delta\varphi)^2 + \ldots. \tag{6.3.147}$$

The definitions of the slow-roll parameters can be used to express the successive derivatives of the potential as

$$V_* = \frac{M_P^2}{8\pi}H^2(3-\epsilon), \tag{6.3.148}$$

$$V'_* = -\frac{M_P}{\sqrt{4\pi}}H^2\sqrt{\epsilon}(3-\delta), \tag{6.3.149}$$

$$V''_* = H^2(3\epsilon + 3\delta - \delta^2 - \xi_H^2). \tag{6.3.150}$$

As a second step, we express the slow-roll parameters as a function of observables, such as the spectral index. For instance, using expressions (6.3.111) and (6.3.124), we find that V_* is given by

$$V_* = \frac{75}{32}M_P^4 A_T^2(k_*)[1 + (\frac{5}{3} + 2C)R_*]. \tag{6.3.151}$$

The detail of the computation of the other derivatives can be obtained in references. Here, we give just the relations for the two following derivatives:

$$V'_* = -\frac{75}{8}\sqrt{\pi}M_P^3 R_*^{3/2} A_s^2(k_*)[1 + (3C + \frac{4}{3})R_* - \frac{1}{2}(C - \frac{1}{3})(1 - n_{s*})] \tag{6.3.152}$$

$$\begin{aligned} V''_* = \frac{25}{4}\sqrt{\pi}M_P^2 R_* A_s^2(k_*)[9R_* - \frac{3}{2}(1-n_{s*}) + (36C+2)R_*^2 \\ - \frac{1}{4}(1-n_{s*})^2 - (12C-6)R_*(1-n_{s*}) - \frac{3C-1}{2}\alpha_{s*}]. \end{aligned} \tag{6.3.153}$$

Table 6.2 summarizes the different observables that are necessary to reconstruct the potential up to a given order. Note once again that the absolute normalization of the potential requires knowledge of A_T, i.e. the detection of primordial gravitational waves.

Table 6.2: Summary of the slow-roll parameters and observables that are necessary to reconstruct a derivative of given order of the potential.

	Parameters		Observables	
	Leading order	Next order	Leading order	Next order
V	H	H,ϵ	A_T^2	A_T^2, R
V'	H,ϵ	H,ϵ,δ	A_T^2, R	A_T^2, R, n_s
V''	H,ϵ,δ	H,ϵ,δ,ξ	A_T^2, R, n_s	A_T^2, R, n_s, α_s

Flow equations Another way to tackle the problem is to randomly generate potentials and to deduce the observables. For this, we extend the hierarchy of the slow-roll parameters to all orders by defining the quantities

$$^l\lambda_H \equiv \left(\frac{M_P^2}{4\pi}\right)^l \frac{(H')^{l-1}}{H^l}\frac{d^{(l+1)}H}{d\varphi^{(l+1)}}, \tag{6.3.154}$$

which complete the doublet (ϵ_H, δ_H). In particular, $\delta_H = {}^1\lambda_H$ and $\xi_H^2 = {}^2\lambda_H$. Using as a variable the number of e-folds before the end of inflation, we have

$$\frac{d}{dN} = \frac{d}{d\ln a} = \frac{M_P}{\sqrt{4\pi}}\sqrt{\epsilon}\frac{d}{d\varphi}. \tag{6.3.155}$$

The evolution equations (6.1.47) can then be extended into the series of equations

$$\frac{d\epsilon_H}{dN} = 2\epsilon_H(\delta_H - \epsilon_H), \quad \frac{d\,{}^l\lambda_H}{dN} = [(l-1)^1\lambda_H - l\epsilon_H]^l\lambda_H + {}^{l+1}\lambda_H. \tag{6.3.156}$$

This system can be integrated numerically. Figure 6.11 illustrates the distribution of the integration of 10^6 models. In particular, 90% of the potentials have a blue-tilt spectrum with $n_s > 1$ and are thus excluded.

Figure 6.11: Distribution of 10^6 models in the space of observables. Adapted from Ref. [36].

6.4 End of inflation and reheating

The theory of reheating after inflation is one of the most important applications of quantum field theory in curved space-time. Almost all the matter present in our Universe would have been created through a process of particle production during the inflaton decay.

6.4.1 Perturbative reheating

Shortly after the end of inflation, the Universe is cold and 'frozen' in a state of low entropy where the field oscillates around the minimum of its potential (Fig. 8.6). The coherent oscillations of the inflaton can be considered as a collection of independent scalar particles. If they couple to other particles, the inflaton can then decay perturbatively to produce light particles. Reheating would occur after the expansion rate had decreased to a value $H \sim \Gamma_\varphi$.

As illustrated in Fig. 8.6 for $n = 2$, the inflaton behaves as a pressureless fluid during the oscillatory phase and the scale factor evolves as $a \propto t^{2/3}$. To see this, let us rewrite the Klein-Gordon equation as

$$\frac{d}{dt}(a^3\rho_\varphi) = -P_\varphi\frac{da^3}{dt}. \tag{6.4.1}$$

During the oscillatory phase, $H < m$ and for a harmonic oscillator $\langle\dot\varphi^2/2\rangle = (n/2)\langle V(\varphi)\rangle$, so that $\langle P_\varphi\rangle = (n-2)/(n+2)\langle\rho_\varphi\rangle$. The scalar field thus behaves as a dust fluid for $n = 2$ and as a radiation fluid for $n = 4$.

As an example, let us consider couplings described by the Lagrangian

$$-\mathcal{L}_{\rm int} = \sigma v\varphi\chi^2 + h\varphi\bar\psi\psi, \tag{6.4.2}$$

where σ and h are dimensionless coupling constants and v is a mass scale.

Evolution of the inflaton

Including quantum corrections, the Klein-Gordon equation becomes

$$\dot\varphi + 3H\dot\varphi + [m^2 + \Pi(m)]\varphi = 0, \tag{6.4.3}$$

where $\Pi(m)$ is the polarization operator of the inflaton. Even though $\mathrm{Re}[\Pi(m)] \ll m^2$, it has an imaginary part

$$\mathrm{Im}[\Pi(m)] = m\Gamma_\varphi. \tag{6.4.4}$$

We infer that the inflaton evolves as

$$\varphi = \Phi(t)\sin mt, \quad \Phi = \varphi_0 \exp\left[-\frac{1}{2}\int (3H + \Gamma_\varphi)dt\right]. \tag{6.4.5}$$

As long as $3H > \Gamma_\varphi$, the decrease in the inflaton energy caused by the expansion dominates. Since $H = 2/(3t)$, we infer that

$$\Phi = \varphi_f \frac{t_f}{t} = \frac{M_P}{m}\frac{1}{\sqrt{3\pi}t}. \tag{6.4.6}$$

Reheating temperature

Reheating can only occur when $\Gamma_\varphi \geq 3H$. In this regime, we find that

$$\Phi = \frac{M_P}{m}\frac{1}{\sqrt{3\pi}t}e^{-\Gamma_\varphi t/2}. \tag{6.4.7}$$

We define the time of reheating, t_{reh}, by $\Gamma_\varphi = 3H$ so that the energy density at that time is

$$\rho_{reh} = \frac{\Gamma_\varphi^2 M_P^2}{24\pi}. \tag{6.4.8}$$

If this energy is converted into radiation, its temperature is

$$\rho_{reh} = \frac{\pi^2}{30} g_* T_{reh}^4. \tag{6.4.9}$$

The reheating temperature is thus given by

$$T_{reh} = \left(\frac{5}{4\pi^3 g_*}\right)^{1/4} \sqrt{\Gamma_\varphi M_P} \ll 10^{15}\ \text{GeV}. \tag{6.4.10}$$

Decay rate

Γ_φ is the total decay rate into bosons and fermions, given by

$$\Gamma(\varphi \to \chi\chi) = \frac{\sigma^2 v^2}{8\pi m}, \quad \Gamma(\varphi \to \overline{\psi}\psi) = \frac{h^2 m}{8\pi}. \tag{6.4.11}$$

Note that the decay rate for an interaction of the type $\mathcal{L}_{\text{int}} = -g^2\varphi^2\chi^2/2$ is

$$\Gamma(\varphi\varphi \to \chi\chi) = \frac{g^2\varphi^2}{8\pi m}. \tag{6.4.12}$$

Unlike 'three-leg' interactions, this rate decreases quicker than the Hubble rate H. Perturbative reheating can therefore never be produced by 'four-leg' interactions.

Phenomenological formulation

Reheating can then be described phenomenologically as a transfer between two fluids:

$$\dot{\rho}_\varphi + H(\rho_\varphi + P_\varphi) = -\Gamma_\varphi\rho_\varphi, \quad \dot{\rho}_r + 4H\rho_r = \Gamma_\varphi\rho_\varphi. \tag{6.4.13}$$

6.4.2 Theory of preheating

The decay of the inflaton can start much earlier in a phase of preheating where particles are produced by parametric resonance.

Parametric resonance

Consider an interaction of the form

$$V_{\rm int} = \frac{1}{2} g^2 \varphi^2 \chi^2. \tag{6.4.14}$$

The scalar field χ can be decomposed into

$$\hat{\chi} = \int \frac{d^3k}{(2\pi)^{3/2}} [\chi_k(t) \hat{a}_k(t) e^{-ik\cdot x} + h.c.]. \tag{6.4.15}$$

The evolution of each mode is dictated by

$$\ddot{\chi}_k + 3H\dot{\chi}_k + \left(\frac{k^2}{a^2} + g^2\varphi^2\right)\chi_k = 0. \tag{6.4.16}$$

In terms of $X_k = a^{3/2}\chi_k$:

$$\ddot{X}_k + \omega_k^2 X_k = 0, \quad \omega_k^2 = \frac{k^2}{a^2} + g^2\varphi^2 - \frac{3}{4}(2\dot{H} + 3H^2). \tag{6.4.17}$$

Using $z = mt$ and defining

$$q \equiv \frac{g^2\Phi^2(t)}{4m^2}, \quad A_k \equiv 2q + \frac{k^2}{m^2a^2}, \tag{6.4.18}$$

this equation takes the form of a Mathieu equation

$$\frac{d^2}{dz^2} X_k + (A_k - 2q\cos 2z + \Delta) X_k = 0. \tag{6.4.19}$$

Minkowski space-time case

For constant A_k and q, the solution is stable when $\mathrm{Im}(\nu_k) > 0$. In the opposite case, X_k increases exponentially as $X_k \propto \exp(\mu_k z)$. This corresponds to an exponential growth in the occupation number $n_k \propto \exp\left(2\mu_k^{(n)} z\right)$.

Figure 6.12: Stability diagram of the Mathieu equation. The shaded zones are instability zones.

Resonances

When $q \leq 1$, the width is narrow. The number of particles, defined as

$$n_k = \frac{\omega_k}{2}\left(\frac{|\dot{\chi}_k|^2}{\omega_k^2} + |\chi_k|^2\right) - \frac{1}{2}, \tag{6.4.20}$$

grows as $\exp(2\mu_k z) \sim \exp(qz)$. When Φ is large, q can be very large. Production of particles requires that the variation of ω_k is no longer adiabatic:

$$\dot{\omega}_k \geq \omega_k^2. \tag{6.4.21}$$

Expanding space-time

In an expanding space-time, resonances only appear when $q^2 m \geq H$, i.e. when

$$g\Phi \geq 2m\left(\frac{H}{m}\right)^{1/4}. \tag{6.4.22}$$

The expansion modifies the picture by redshifting momenta and damping the amplitude of oscillations ($q \propto t^{-2}$). This leads to stochastic resonance.

Figure 6.13: Evolution of χ_k and n_k as a function of time in Minkowski space-time.

Figure 6.14: Evolution of χ_k and n_k in an expanding space-time. q decreases and the resonance disappears.

Backreaction

The field χ backreacts on the dynamics. The Hubble rate becomes

$$H^2 = \frac{8\pi G_N}{3}\left(\frac{1}{2}\dot{\varphi}^2 + \frac{1}{2}\dot{\chi}^2 + \frac{1}{2}m^2\varphi^2 + \rho_\chi + \rho_\varphi\right). \tag{6.4.23}$$

The description requires coupled equations:

$$\ddot{X}_k + \left(\frac{k^2}{a^2} + g^2\Phi^2\sin^2 mt\right)X_k = -\int X_k(t')\Pi_\chi(t,t';k)dt', \tag{6.4.24}$$

$$\ddot{v}_k + \left(\frac{k^2}{a^2} + g^2\Phi^2\sin^2 mt\right)v_k = -\int v_k(t')\Pi_\varphi(t,t';k)dt'. \tag{6.4.25}$$

In the Hartree approximation:

$$\ddot{\varphi} + 3H\dot{\varphi} + (m^2 + g^2\langle\chi^2\rangle)\varphi = 0, \tag{6.4.26}$$

with

$$\langle\chi^2\rangle = \int \frac{k^2 dk}{2\pi^2 a^3}|X_k(t)|^2. \tag{6.4.27}$$

Discussion

Results are model dependent. With the potential

$$V = \frac{\lambda}{4}\varphi^4 + \frac{1}{2}g^2\varphi^2\chi^2, \tag{6.4.28}$$

the resonance never becomes stochastic. Another case is tachyonic reheating where $V'' < 0$, leading to exponential growth of modes $k < m$. Reheating is then quasi-instantaneous.

6.5 Eternal inflation

Inflation naturally leads to a self-reproducing Universe where there are always inflating regions.

6.5.1 Heuristic argument

Mechanism

Regions separated by H^{-1} evolve independently. In a time $\Delta t \sim H^{-1}$, the field classically decreases by $\Delta\varphi \sim \dot{\varphi}/H \sim -M_P^2/(4\pi\varphi)$. Quantum fluctuations have amplitude $|\delta\varphi| \sim H/2\pi$. They have the same amplitude for

$$|\Delta\varphi| \sim |\delta\varphi| \iff \varphi \sim \varphi_* \equiv \frac{M_P}{2}\sqrt{\frac{M_P}{m}}. \tag{6.5.1}$$

In the regime $\varphi \gg \varphi_*$, the physical volume of regions where the field increases is ten times larger:

$$V_{t+\Delta t}(\varphi > \varphi_*) \sim \frac{1}{2}(\exp H\Delta t)^3 V_t(\varphi > \varphi_*) \sim 10 V_t(\varphi > \varphi_*). \tag{6.5.2}$$

The physical volume inflating grows exponentially.

Figure 6.15: The three phases of evolution of the inflaton (left). Self-reproduction phase (right).

Consequences

The Universe has a fractal structure with continuous production of island Universes. This offers a framework for the anthropic principle.

6.5.2 Stochastic approach

The smoothing scale is

$$R = (\epsilon a H)^{-1}. \tag{6.5.3}$$

The field is decomposed:

$$\varphi(x,t) = \overline{\varphi}(x,t) + \delta\varphi(x,t), \tag{6.5.4}$$

$$\overline{\varphi}(x,t) = \int \frac{d^3k}{(2\pi)^{3/2}} w(kR)[\varphi_k(t)\hat{a}_k e^{ik\cdot x} + h.c.]. \tag{6.5.5}$$

The Langevin equation

In the slow-roll regime:

$$\dot{\overline{\varphi}}(x,t) = -\frac{1}{3H}V'(\overline{\varphi}) + \xi_Q(x,t). \tag{6.5.6}$$

ξ_Q is a quantum noise. With $\xi_Q \simeq -\delta\dot{\varphi}$:

$$\xi_Q(x,t) = -\int \frac{d^3k}{(2\pi)^{3/2}}(kHR)w'(kR)[\varphi_k(t)\hat{a}_k e^{ik\cdot x} + h.c.]. \tag{6.5.7}$$

The correlation function is:

$$\langle \xi_Q(x,t)\xi_Q(x',t')\rangle = \int \frac{k^3 dk}{2\pi^2}\frac{\sin kr}{r}[HR(t)]w'[kR(t)][HR(t')]w'[kR(t')]\varphi_k(t)\varphi_k^*(t'). \tag{6.5.8}$$

The smoothed inflaton evolution is:

$$\dot{\phi}(x,t) = -\frac{V'(\phi)}{3H(\phi)} + \xi(x,t). \tag{6.5.9}$$

For a Heaviside window function in de Sitter space:

$$\langle \xi(x,t)\xi(x',t')\rangle = \frac{H^3}{4\pi^2}\frac{\sin(\epsilon a H r)}{\epsilon a H r}\delta(t-t'). \tag{6.5.10}$$

The Langevin equation becomes:

$$\dot{\phi}(x,t) = -\frac{V'(\phi)}{3H(\phi)} + \frac{H^{3/2}(\phi)}{2\pi}\xi(x,t). \tag{6.5.11}$$

Fokker-Planck equation

The probability distribution function $P(\phi,t)$ satisfies:

$$P(\phi, t+\Delta t) = \int G(\phi, t|\phi', t+\Delta t)P(\phi', t)d\phi'. \tag{6.5.12}$$

It leads to:

$$\frac{\partial P}{\partial t} = \frac{\partial J(\phi)}{\partial \phi}, \quad J(\phi) = -F(\phi)P + \frac{1}{2}\sigma^{2(1-\beta)}\frac{\partial}{\partial\phi}(\sigma^{2\beta}P). \tag{6.5.13}$$

In our case:

$$\frac{\partial P}{\partial t} = \frac{\partial}{\partial \phi}\left\{\frac{V'(\phi)}{3H(\phi)}P(\phi,t) + \frac{H^{3(1-\beta)}(\phi)}{8\pi^2}\frac{\partial}{\partial \phi}[H^{3\beta}(\phi)P(\phi,t)]\right\}. \tag{6.5.14}$$

The mean value is:

$$\langle f(\overline{\varphi})\rangle = \int f(\phi)P(\phi,t)d\phi. \tag{6.5.15}$$

Discussion

The Fokker-Planck equation allows for the study of the dynamics on scales larger than H^{-1} over which the Universe is very inhomogeneous. Note that the Fokker-Planck and Langevin equations are only equivalent when the noise is not correlated. We also stress that we have neglected metric perturbations.

The stationary solution of (6.5.13) is given by

$$P(\phi,t) = N\exp\left[-\frac{8\pi^2V(\phi)}{3H^4}\right]V^{-3\beta/2}, \tag{6.5.16}$$

where N is a normalization factor. The dependence on the parameter β is hence sub-exponential. The solution therefore approaches asymptotically the distribution function

$$P_{eq}(\phi) = N\exp\left[-\frac{8\pi^2V(\phi)}{3H^4}\right]. \tag{6.5.17}$$

As an example, if $V = \lambda\phi^4/4$, then

$$P_{eq}(\phi) = \left(\frac{32\pi^2\lambda}{3}\right)^{1/4}\frac{e^{-2\pi^2\lambda\phi^4/3H^4}}{\Gamma(\frac{1}{4})H}. \tag{6.5.18}$$

The distribution function of ϕ is not Gaussian and satisfies $\langle\phi^2\rangle = \sqrt{3/2\pi^2}[\Gamma(3/4)/\Gamma(1/4)](H^2/\sqrt{\lambda})$ and $\langle\phi^4\rangle/\langle\phi^2\rangle^2 - 3 \sim -0.812$.

It is also interesting to introduce the probability $P(\phi,t|\chi)$ for $\phi(t)$ to have the value ϕ at time t if $\phi = \chi$ initially. This conditional probability distribution satisfies the adjoint equation of (6.5.14) called the inverse Kolmogorov equation,

$$\frac{\partial P(\phi,t|\chi)}{\partial t} = \frac{H^{3(1-\beta)}(\chi)}{8\pi^2}\frac{\partial}{\partial \chi}\left[H^{3\beta}(\chi)\frac{\partial P(\phi,t|\chi)}{\partial \chi}\right] - \frac{V'(\chi)}{3H(\chi)}\frac{\partial P(\phi,t|\chi)}{\partial \chi}. \tag{6.5.19}$$

The stationary solution of this equation is

$$P_{eq}(\phi,\chi) = \exp\left[\frac{3M_p^4}{8V(\phi)}\right]\exp\left[-\frac{3M_p^4}{8V(\chi)}\right]. \tag{6.5.20}$$

Unfortunately, if $V = 0$ at its minimum, this distribution function is not normalizable. This is due to the fact that a stationary state exists only if the stochastic diffusion, which increases ϕ, can compensate the classical drift towards the minimum of the potential. For $\phi < \phi_f$, the stochastic contribution is no longer there. So no physically acceptable solution can be stationary.

To recover stationarity, we should introduce the probability distribution function $P_{phys}(\phi,t|\chi)$, which is the analogue of $P_{eq}(\phi,t|\chi)$ defined above but in terms of physical volumes instead of comoving volumes. During an interval dt, the physical volume of a given region grows by an amount $3H(\phi)dt$ so that the evolution equation of $P_{phys}(\phi,t|\chi)$ is

$$\frac{\partial P_{phys}}{\partial t} = \frac{\partial}{\partial \phi}\left\{\frac{V'(\phi)}{3H(\phi)}P_{phys} + \frac{H^{3(1-\beta)}(\phi)}{8\pi^2}\frac{\partial}{\partial \phi}[H^{3\beta}(\phi)P_{phys}]\right\} + 3H(\phi)P_{phys} \tag{6.5.21}$$

or equivalently,

$$\frac{\partial P_{phys}}{\partial t} = \frac{H^{3(1-\beta)}(\chi)}{8\pi^2}\frac{\partial}{\partial\chi}\left[H^{3\beta}(\chi)\frac{\partial P_{phys}}{\partial\chi}\right] - \frac{V'(\chi)}{3H(\chi)}\frac{\partial P_{phys}}{\partial\chi} + 3H(\chi)P_{phys}. \tag{6.5.22}$$

If we only consider the regions of space for which the density is lower than the Planck density, these equations then have a stationary solution of the form $P_{phys}(\phi, t|\chi) \sim \exp(cM_p t)A(\phi)B(\chi)$. If $V \ll M_p^4$ we recover the result that $B(\chi) \propto \exp\left[-3M_p^4/8V(\chi)\right]$ and the normalized probability to find a given volume in a state with a given value of ϕ is $\tilde{P}_{phys}(\phi, t|\chi) \sim A(\phi)B(\chi)$.

This result has an important consequence. In the heuristic argument, we had only one region of size H^{-1} with $\varphi \gg \varphi_*$ However, if we start with a large number of regions, even with $\varphi < \varphi_*$, since the probability (6.5.20) is non-zero, some of them can enter a phase of eternal inflation. This also implies that any hypothesis on the initial state of the Universe becomes completely disconnected from observations. If a region of the Universe enters a self-reproducing phase, it produces an infinite number of island Universes with properties approaching that of a stationary state, independently of the initial conditions.

6.6 Extensions

6.6.1 Multifield inflation

In high-energy theories, legions of scalar fields exist. If some are light, they develop super-Hubble correlations. Multifield inflation makes it possible to conceive the generation of isocurvature perturbations, to extend the consistency relation, and to generate non-Gaussianities.

Friedmann and Klein-Gordon equations

Friedmann's equations become

$$H^2 = \frac{4\pi G_N}{3}\left[\sum_{i=1}^{N}\dot{\varphi}_i^2 + 2V(\varphi_1, ..., \varphi_N)\right], \tag{6.6.1}$$

$$\dot{H} = -4\pi G_N \sum_{i=1}^{N}\dot{\varphi}_i^2, \tag{6.6.2}$$

and the Klein-Gordon equation for each field takes the form

$$\ddot{\varphi}_i + 3H\dot{\varphi}_i = -\frac{\partial V}{\partial\varphi_i}. \tag{6.6.3}$$

For the case of two fields, we decompose them into a part tangent to the trajectory, σ, and a part perpendicular, s,

$$\begin{pmatrix}\dot{\sigma}\\ \dot{s}\end{pmatrix} = \mathcal{M}(\theta)\begin{pmatrix}\dot{\varphi}_1\\ \dot{\varphi}_2\end{pmatrix}, \quad \mathcal{M}(\theta) = \begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix} \tag{6.6.4}$$

where $\sin\theta = \dot{\varphi}_2/\sqrt{\dot{\varphi}_1^2 + \dot{\varphi}_2^2}$. The equations reduce to

$$\ddot{\sigma} + 3H\dot{\sigma} + U_{,\sigma} = 0, \quad \begin{pmatrix}U_{,\sigma}\\ U_{,s}\end{pmatrix} = \mathcal{M}(\theta)\begin{pmatrix}V_{,1}\\ V_{,2}\end{pmatrix}. \tag{6.6.5}$$

Perturbation equations

The Einstein equations for scalar modes take the form

$$\dot{\Phi} + H\Phi = 4\pi G_N(\dot{\varphi}_1\chi_1 + \dot{\varphi}_2\chi_2), \tag{6.6.6}$$

$$3\dot{\Phi} + (3H^2 + \dot{H})\Phi - \frac{\Delta}{a^2}\Phi = -4\pi G_N(V_{,1}\chi_1 + \dot{\varphi}_1\dot{\chi}_1 + V_{,2}\chi_2 + \dot{\varphi}_2\dot{\chi}_2), \tag{6.6.7}$$

$$\ddot{\chi}_i + 3H\dot{\chi}_i - \frac{\Delta}{a^2}\chi_i + V_{,ij}\chi_j = -2V_{,i}\Phi + 4\dot{\varphi}_i\dot{\Phi}. \tag{6.6.8}$$

Using the decomposition into adiabatic and isocurvature modes, one finds the coupled system

$$\ddot{\chi}_\sigma + 3H\dot{\chi}_\sigma + \left(-\frac{\Delta}{a^2} - \dot{\theta}^2 + U_{,\sigma\sigma}\right)\chi_\sigma = -2\Phi U_{,\sigma} + 4\dot{\sigma}\dot{\Phi} + 2(\dot{\theta}\delta s)^{\cdot} - 2\frac{\dot{\theta}}{\dot{\sigma}}U_{,\sigma}\delta s, \tag{6.6.9}$$

$$\ddot{\delta s} + 3H\dot{\delta s} + \left(-\frac{\Delta}{a^2} - \dot{\theta}^2 + U_{,ss}\right)\delta s = -\frac{\dot{\theta}}{\dot{\sigma}}\frac{\Delta}{2\pi G_N a^2}\Phi. \tag{6.6.10}$$

The evolution equation for the curvature perturbation $\mathcal{R}$ becomes

$$\dot{\mathcal{R}} = -\frac{H}{\dot{H}}\frac{\Delta}{a^2}\Phi + 2c_s^2\frac{H}{\dot{\sigma}}\dot{\theta}\delta s. \tag{6.6.11}$$

As long as $\dot{\theta} = 0$, the modes are decoupled. If the trajectory is curved ($\dot{\theta} \neq 0$), the isocurvature mode δs sources the curvature perturbation $\mathcal{R}$, which is then no longer conserved on super-Hubble scales.

6.6.2 Non-Gaussianity

Since curvature perturbations originate from vacuum fluctuations, they are Gaussian at the linear level. However, general relativity is non-linear. To second order, one can parameterize the non-Gaussianity using the f_{NL} parameter,

$$\Phi_k = \Phi_{lin}(k) + \int \frac{d^3k_1 d^3k_2}{(2\pi)^{3/2}} f_{NL}^{(\Phi)}(k_1, k_2, k)\Phi_{lin}(k_1)\Phi_{lin}(k_2)\delta^{(3)}(k - k_1 - k_2). \tag{6.6.12}$$

In single-field inflation, f_{NL} is typically of the order of the slow-roll parameters:

$$f_{NL}^{(\tilde{\mathcal{R}})}(k_1, k_2, k_3) \sim \frac{1}{4}(n_s - 1). \tag{6.6.13}$$

6.6.3 Trans-Planckian problem

Observable modes today had a sub-Planckian wavelength at the beginning of inflation. To test the robustness of predictions, one can modify the dispersion relation

$$\omega^2(k, \eta) = a^2 F^2\left(\frac{k}{a}\right). \tag{6.6.14}$$

The mode equation becomes

$$v_k'' + \left[\omega^2(k, \eta) - \frac{z''}{z}\right]v_k = 0. \tag{6.6.15}$$

If the adiabaticity condition

$$\left|\frac{\omega_T'}{\omega_T^2}\right| \ll 1 \tag{6.6.16}$$

is violated, modulations appear in the power spectrum, $\mathcal{P} = \mathcal{P}_0\{1 + 2\text{Re}[\beta_k e^{i\phi_k}]\}$. However, such states are constrained by backreaction. The energy density relative to critical density is

$$\frac{\delta\rho_k}{\rho_{crit}} \sim n_k \exp(4N_i)\left(\frac{H_{inf}}{M_p}\right)^2. \tag{6.6.17}$$

Avoiding backreaction requires $n_k \ll 1$, which essentially restores the standard Bunch-Davies vacuum prescription.

6.7 Status of the paradigm

Single-field inflation makes robust, testable predictions: a homogeneous and isotropic Universe, spatial flatness ($\Omega = 1$), Gaussian adiabatic scalar perturbations with a nearly scale-invariant spectrum, and a background of primordial gravitational waves with a specific consistency relation (6.3.118).

The detection of a positive curvature, even small, would be a problem for standard inflation. In a de Sitter phase with $K = +1$, the maximal number of e-folds before the largest scale becomes super-Hubble is

$$\Delta N_{max} = \frac{1}{2}\ln\left(1 + \frac{1}{\delta_0}\right) \sim 2. \tag{6.7.1}$$

Inflation remains a satisfying framework that solves the classical Big-Bang problems and provides a quantum origin for the large-scale structures of the Universe.